ROUTLEDGE INTERNATIONAL HANDBOOKS

Routledge Handbook of Sports Technology and Engineering

Second Edition

Edited by Tom Allen, Andy Alderson, Alan Godfrey, Duane Knudson, James Webster and Kazuya Seo

ROUTLEDGE HANDBOOK OF SPORTS TECHNOLOGY AND ENGINEERING

Since the 1990s, the field of sports technology and engineering has expanded beyond an initial focus on sports equipment and materials to include various topics. These topics span sustainable equipment design and manufacturing, user-centred design, biomechanics and human-equipment interaction, field testing, sensors and instrumentation of sports equipment and clothing, smart textiles, artificial intelligence and big data, and the development of human body surrogates for testing protective equipment. This second edition of *Routledge Handbook of Sports Technology and Engineering* pulls together the full depth and breadth of this field, explores current issues and controversies, and looks to future research directions. Bringing together many of the world's leading experts and scientists, this book emphasises the current understanding of the underlying mechanics associated with sport and physical activity, exercise, training, and athletic performance in relation to sports equipment, clothing, and training and officiating technologies in a broad sense.

This book has five sections:

1 Sports mechanics
2 Sports materials
3 Sports equipment design and manufacture
4 Sports biomechanics and human-equipment interaction
5 Field testing, sensors, and instrumentation

Written by an international team of leading experts, the emphasis throughout this book is on bridging the gap between scientific research and application within sports products and their effect on training and competition. This text is important reading for students, scholars, and others with an interest in engineering related to sport, exercise, and health in general.

Tom Allen is a Reader in the Department of Engineering, Manchester Metropolitan University, UK. He is a Fellow of the Institution of Mechanical Engineers (IMechE) and Editor-in-Chief of *Sports Engineering*.

Andy Alderson is the Director of the Industry and Innovation Research Institute and a Professor of Smart Materials and Structures at Sheffield Hallam University, UK. He is a recognised world authority on auxetic (negative Poisson's ratio) materials.

Alan Godfrey is an Associate Professor (Reader) in the Department of Computer and Information Sciences, Northumbria University, UK. He is a Deputy Editor of *npj/Nature Digital Medicine*, Editor of *Maturitas*, and Associate Editor of *Journal of Neuro Engineering and Rehabilitation*. He is a senior member of the IEEE (SMIEEE).

Duane Knudson is Regents' Professor in the Department of Health and Human Performance, Texas State University, USA. He is a Fellow of the International Society of Biomechanics in Sports, the National Academy of Kinesiology, and the American College of Sports Medicine.

James Webster leads a footwear materials innovation team within the Nike Exploration Team, Nike, USA. He is on the Editorial Board of *Proceedings of the Institution of Mechanical Engineers, Part P: Journal of Sports Engineering and Technology*.

Kazuya Seo is a Professor at Kogakuin University, Japan, and currently serves as Chair of the Sports Engineering and Human Dynamics Division of the Japan Society of Mechanical Engineers (JSME). He is a Fellow of both the International Sports Engineering Association (ISEA) and JSME.

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FOREWORD

In the dynamic world of sports and exercise, technology and engineering have become integral and transformative processes to enhance performance, safety, and the overall experience for athletes, recreational exercise goers, and spectators alike. The sports equipment industry is experiencing robust growth, driven by technological advancements, increased health awareness, and a rising demand for innovative, eco-friendly, and performance-enhancing products across diverse sports disciplines.

This book, *Routledge Handbook on Sports Technology and Engineering*, Second Edition, explores the myriad ways in which technological advancements are reshaping the landscape of sports. The various chapters highlight how technology and engineering principles are used to diversify and solve global demands for addressing better sports population inclusivity, environmental challenges, and pushing the boundaries of human and equipment potential.

Sports can be seen as a reflection of societal values, and today, the emphasis on diversity and inclusion is more pronounced than ever. The rise of adaptive technologies is empowering athletes of all abilities to compete. Technology is enabling access to sports – from advanced prosthetics that enable para-athletes to perform at elite levels to inclusive equipment designs that cater to a broad spectrum of body types and physical conditions.

Environmental sustainability is another critical consideration in the development of sports equipment and infrastructure. The increasing awareness of our ecological footprint has led to innovative solutions that minimise the environmental impact of this high-turnover consumer sector. This book delves into the latest advancements in eco-friendly materials, sustainable manufacturing processes, and the implementation of circular economy principles within the sports industry. By prioritising environmental considerations, we can preserve the natural beauty and integrity of the spaces where we play and compete, ensuring that future generations can enjoy the same sporting opportunities that we do today.

The integration of technology in sports is more comprehensive and sophisticated than ever before. Wearable devices and real-time data analytics are revolutionising how athletes train and recover, how referees determine decisions, how coaches plan and perform, and how medical staff ensures athlete health and manipulates to minimise injury risk. This book examines cutting-edge developments in sports mechanics, sports materials, sports equipment design and manufacturing, sports biomechanics and human-equipment interaction, and field testing. The convergence of these

innovative methods is enabling athletes to achieve new levels of performance while maintaining their health and well-being.

As we embark on this exploration of sports technology, we are reminded that the spirit of competition is inherently human, yet it is through technology that we continue to evolve and redefine the limits of what is possible. Whether you are a sports enthusiast with a passion for technology, a student, a sports equipment developer, or a researcher, this book offers valuable insights into the latest science and transformative power of technology and engineering in sports.

We invite you to join us on this journey, as we celebrate the remarkable achievements in sports technology and envision a future where innovation drives inclusivity, sustainability, and excellence in every athletic endeavour.

I congratulate the authors who contributed to this second edition of *Routledge Handbook on Sports Technology and Engineering*. By embracing diversity, prioritising environmental sustainability, and demonstrating the latest advancements in sports technologies and engineering, we are not just enhancing sports; we are shaping a more inclusive, responsible, and exhilarating future for all who participate and spectate.

Dr Katrine Okholm Kryger

PREFACE

Sports technology and engineering is a dynamic field of research where knowledge is advancing and expanding rapidly. As the field expands, so does the role of the sports engineer. The modern sports engineer must consider how sporting goods interact with the athlete and their environment and how this affects their performance, satisfaction, and injury risk. Sports engineers must also consider how developments in sporting goods could influence competition and whether changes to the associated rules and regulations are needed to ensure fairness and athlete welfare, while maintaining spectator appeal. Environmental sustainability must also be embedded within the design of sporting technologies. As such, the modern sports engineer must have a broad understanding of the field of sports technology and engineering.

The rapid uptake of the sporting goods sector can serve as an ideal test arena and proving ground for new technologies. Examples include advanced materials, like metamaterials, manufacturing methods, like 3D printing, and new ways to collect and analyse infield data, like wearable sensors combined with machine learning. While introducing new technologies within sporting arenas, engineers must consider the broader implications, including those related to environmental sustainability, equality, and athlete welfare. Consider the emergence of so-called "super shoes" that feature a lightweight, resilient foam midsole embedded with a curved stiff plate. These shoes are thought to have helped athletes improve distance running records, which led to a subsequent tightening of the rules around footwear. There is now concern over the environmental impact of footwear that focuses on performance over durability, as well as the potential injury risk of training in such shoes.

There is also concern over the concussion risk associated with contact sports, like American football, rugby, and ice hockey. Engineers are developing technologies for infield monitoring of athletes to assess injury risk and inform the design and testing of headgear. It is important to ensure that sporting technologies meet the needs and work well for a diverse range of users. Sporting goods have traditionally been designed and marketed for males, but this is starting to change. There is increasing awareness of the need to design products specifically for females, with football boots being an example that has helped gain public awareness. The modern sports engineer should embrace an inclusive design approach.

This second edition of *Routledge Handbook of Sports Technology and Engineering* covers current research advances in various areas related to sports technology and engineering. It also

summarises areas where more research is needed, controversies that exist, and where future research directions might be headed. This book has both current and future continuing relevance, not only helping researchers understand current learning but also assisting in formulating research directions and questions for advancing the science and understanding of the diverse relevance of sports technology and engineering in the future. An overview of the dynamic nature of this field and its relevance to a wide range of sport, exercise, and health-related topic areas is provided. There is a comprehensive survey of contemporary sports technology and engineering, providing a complete overview of academic, professional, and industrial knowledge and techniques. This book is divided into five sections spanning mechanics, materials, equipment design and manufacture, human-equipment interaction, and field testing. The sections are structured around key engineering and science topics rather than sports, and the contributors bring in examples from the sports they work on in their chapters.

Mechanics is a fundamental tool in sports engineering. Mechanics in sports may have two roles. One is that mechanics scientifically elucidates mysterious phenomena in sports. This satisfies our curiosity in sports. The other is that mechanics is key to estimating the athlete's performance and the index of performance of sports equipment when it is designed. This is concerned with profit-making businesses. The opening section illustrates sports mechanics with some research examples and methods based on projectiles and fluid dynamics, materials and structural mechanics, and the dynamics of machinery from the viewpoint of experimental, computational, and theoretical studies.

Materials are fundamental to all sports engineering and technology, providing the basis upon which sports equipment is developed to optimise the performance of the athlete. Covering the major classes of natural and man-made materials, the second section reviews the key properties for application in sports technology. The section illustrates how these properties are derived from material microstructure and the processing methods used to achieve the required microstructure and properties. The section includes consideration of the latest developments in materials displaying counterintuitive or smart functionality (responding to external stimuli in desirable and controlled ways) or both, with indications as to where such materials are, or will be, applied in the development of future technologies for the athlete.

Within the sporting goods industry there has been considerable research, design, and development targeted at bringing novel technologies to the market that maximise human performance. Within this quest for optimised performance, new design approaches and manufacturing methods have been developed with a key focus on sustainability and minimising environmental impact. The third section focuses on how design, manufacturing, and sustainability are key factors in the production of sports equipment and discusses what advances have been made within the industry and what considerations are needed when designing for a diverse range of athletes.

The application of scientific knowledge and engineering principles to improving sports equipment has resulted in dramatic improvements in athlete performance and reductions in the risk of injury. This improvement in sports technology, like all technology, can result in unintended consequences. The intelligence and skill of humans perhaps place the greatest challenge to sports engineers in accommodating biomechanical interactions with new sports technologies. The fourth section illustrates how biomechanics can assist in understanding possible interactions of equipment with the complex biology and motor control of human motion. Chapters summarise how footwear and sports equipment interact with the human body and its movements, influencing activity patterns, risk of injuries, and the nature of sports.

Advances in technology have seen barriers to entry reduced, enabling uptake for nuanced use in various sporting contexts. The use of wearable and connected devices is providing new insights into sports-related performance and longitudinal assessment beyond the laboratory. Technology

deployment in the field is providing more habitual data capture to augment traditional approaches. However, the proliferation of novel sports technologies is fraught with inconsistencies and a lack of suitable guidance to ensure technologies are fit for purpose. The closing section illustrates how novel wearable and other ubiquitous technologies are being developed for use in sports to better understand athletes in the wild through the provision of high-resolution and objective data. Hence, chapters begin to explore the digital athlete while considering the need for improved standards to guide the creation and use of technologies in sports.

These section groupings focus on five key overarching areas of current sports technology and engineering research directions and group chapters into areas that explore specific current and emerging themes in sports technology and engineering. As can be seen from the various chapters, these themes combine basic scientific advances and current cutting-edge research areas in sports technology and engineering with applied themes that incorporate relevant engineering, biomechanics, and physics topics as they relate to and are influenced by or explained by research and understanding defined by sports technology and engineering research.

This book is an essential reference for anybody working in sports technology, sports product design, sports engineering, biomechanics, ergonomics, sports business, or applied sport science. The emphasis throughout this book is on bridging the gap between scientific research and application within sporting products and their effect on training and competition. Readers will discover the mechanical principles underpinning sport. They will also learn how developments in materials can improve sporting equipment, allowing athletes to perform better while enhancing their safety and enjoyment, with a particular focus on equipment design, biomechanics, and human-equipment interaction. Readers will also learn about how such equipment is designed and made, including advances in sustainable and environmentally friendly practices. In the final section, readers will learn about the emergence of field testing, sensors, and instrumentation and how they are changing how we train and participate in sport and exercise.

This book is intended for students and researchers in engineering, sports science, materials science, technical clothing and textiles, physics, and mathematics. It should also be beneficial for established researchers seeking a repository of the current state of knowledge, current debates, and relevant literature in sports technology and engineering.

Tom Allen, Andy Alderson, Alan Godfrey, Duane Knudson, James Webster and Kazuya Seo
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CONTRIBUTORS

Cameron Armstrong
Tennis Australia, Melbourne, Victoria, Australia

Antoine Bacha
Babolat VS, Lyon, France

Mikael Bäckström
Mid Sweden University, Östersund, Sweden

Nicolas Bailly
Aix Marseille University, University Gustave Eiffel, Marseille, France

Catherine Barry
University of Massachusetts Lowell, Lowell, Massachusetts

Gill Barry
Northumbria University, Newcastle upon Tyne, UK

Michel Behr
Aix Marseille University, University Gustave Eiffel, Marseille, France

Rebecca Bennett
The Bobbins, UK, and the University of Manchester, Manchester, UK

Filippo Biagi
University of Bologna, Bologna, Italy

Johsan Billingham
Fédération Internationale de Football Association (FIFA), Zurich, Switzerland

Olivia Cant
Tennis Australia, Melbourne, Victoria, Australia, and Institute for Health and Sport, Victoria University, Melbourne, Victoria, Australia

Holger Cebulla
Chemnitz University of Technology, Chemnitz, Germany

Yunus Celik
Northumbria University, Newcastle upon Tyne, UK

Martin Charter
University for the Creative Arts, Farnham UK

Yi Chen
Washington State University, Pullman, WA, USA

Simon Choppin
Sheffield Hallam University, Sheffield, UK

Tom Clark
University for the Creative Arts, Farnham UK

Daniel Colombo
University of Bologna, Bologna, Italy

Martino Colonna
University of Bologna, Bologna, Italy, and RE-SPORT srl, Bologna, Italy

Lorenzo Crosetta
University of Bologna, Bologna, Italy

Jonas Danvind
Mid Sweden University, Östersund, Sweden

Patrick Drane
University of Massachusetts Lowell, Lowell, Massachusetts

Olly Duncan
Manchester Metropolitan University, Manchester, UK

Marcus Dunn
Heriot-Watt University, Edinburgh, UK

Adrian Eassom
Tennis Australia, Melbourne, Victoria, Australia

Nicolas Evans
Fédération Internationale de Football Association (FIFA), Zurich, Switzerland

Armin Farzin
Chemnitz University of Technology, Chemnitz, Germany

Giuseppe La Fauci
University of Bologna, Bologna, Italy

Alan Godfrey
Northumbria University, Newcastle upon Tyne, UK

John Eric Goff
University of Lynchburg, Lynchburg, Virginia, USA

Robyn A Grant
Manchester Metropolitan University, Manchester, UK

John Hart
Sheffield Hallam University, Sheffield, UK

Jade Haycraft
Victoria University, Public University in Melbourne, Australia

Erik M. Henrikson
Ping Golf, Arizona, United States

Andrew Hext
Sheffield Hallam University, Sheffield, UK

Adil Imam
Canterbury Christchurch University, Canterbury, UK

Jörg Kaufmann
Chemnitz University of Technology, Chemnitz, Germany

Pui Wah Kong
Nanyang Technological University, Singapore

Andrey Koptyug
Mid Sweden University, Östersund, Sweden

Kayla Krantz
University of Massachusetts Lowell, Lowell, Massachusetts

Lyndon Krause
Tennis Australia, Melbourne, Victoria Australia

Olga Kravchenko
Rheon Labs Ltd, London, UK, and Imperial College, London, UK

Wing-Kai Lam
Hong Kong Sports Institute, Hong Kong SAR, China

Gemma Leslie
University of Manchester, Manchester, UK

Stefan Litzenberger
University of Applied Sciences Technikum Wien, Vienna, Austria

Bin Lyu
Washington State University, Pullman, WA, USA

Sasho J. MacKenzie
St. Francis Xavier University, Nova Scotia, Canada

Rachel Mason
Northumbria University, Newcastle upon Tyne, UK

Akihiro Matsuda
University of Tsukuba, Tsukuba, Ibaraki, Japan

André Matthes
Chemnitz University of Technology, Chemnitz, Germany

Motomu Nakashima
Institute of Science Tokyo, Tokyo, Japan

Alessandro Nanni
Re-Sport srl, Bologna, Italy

Kati Nispel
Technical University of Munich, Munich, Germany

Marie Lund Ohlsson
Mid Sweden University, Östersund, Sweden, and Swedish School of Sport and Health Sciences, Stockholm, Sweden

Paul Baudach
Chemnitz University of Technology, Chemnitz, Germany

Jing Wen Pan
Nanyang Technological University, Singapore

Maria Federica Parisi
University of Bologna, Bologna, Italy

Thomas Perri
Tennis Australia, Melbourne, Victoria, Australia

Nicola Petrone
University of Padova, Padova, Italy

Machar Reid
Tennis Australia, Melbourne, Victoria, Australia

Sam Robertson
Victoria University, Public University in Melbourne, Australia

Tasneem Sabir
Manchester Metropolitan University, Manchester, UK

Lilian Sanchez-Moreno
University for the Creative Arts, Farnham, UK

Veit Senner
Technical University of Munich, Munich, Germany

James Sherwood
University of Massachusetts Lowell, Lowell, Massachusetts

Lloyd Smith
Washington State University, Pullman, WA, USA

Nicholas Smith
Washington State University, Pullman, WA, USA

Stephen Spurrier
Manchester Metropolitan University, Manchester, UK

Andy Stamm
Rhine-Waal University of Applied Sciences, Kleve, Germany, and Griffith University, Brisbane, Australia

Samuel Stuart
Northumbria University, Newcastle upon Tyne, UK, and Oregon Health and Science University, Portland, Oregon, USA

Phillis Soek Po Teng
Nanyang Technological University, Singapore

Mike Vasquez
3Degrees, LLC, South Bend, Indiana

Janina Wagner
Chemnitz University of Technology, Chemnitz, Germany

Conor Wall
Northumbria University, Newcastle upon Tyne, UK

Jane Wood
University of Manchester, Manchester, UK

Fraser Young
Northumbria University, Newcastle upon Tyne, UK

Giuseppe Zullo
University of Padova, Padova, Italy

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PART I

Sports mechanics

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1

INTRODUCTION TO SPORTS MECHANICS

Kazuya Seo

Introduction

Mechanics is founded upon the fundamental principle of Newton's laws of motion [1] (Figure 1.1). Sports engineering encompasses four main branches of mechanics: strength of materials, fluid dynamics, thermodynamics, and dynamics of machinery. The Sports Mechanics section is covered in seven chapters, from Chapters 1 to 7, which present numerous examples from the world of sports grounded in these four branches of mechanics.

In Chapter 2, Professor John Eric Goff illustrates "Projectiles in sport." Projectiles are ubiquitous in sports, including soccer balls, shuttlecocks, rugby balls, and more. This chapter is educative for beginners in Sports Engineering. For example, Goff introduces an interesting experiment to measure the drag coefficient of a soccer ball, which requires no special or expensive experimental instruments.

In Chapters 3 and 4, Dr John Hart illustrates Computational Fluid Dynamics (CFD). In Chapter 3, titled "Computational fluid dynamics applied in sport: Part 1 – Capabilities and considerations," Hart introduces the application of CFD within sports research and examines how the technique enhances understanding of the physics and performance of equipment. In Chapter 4, titled "Computational fluid dynamics applied in sport: Part 2 – Application in sport," Hart explores the application of CFD in specific elite sports and the development of sports products. Rich and colourful results of CFD are showcased, spanning the 30 years since the technique began to gain traction in sports.

In Chapter 5, titled "Soft materials in sport," Professor Akihiro Matsuda illustrates some soft materials often applied to sports equipment, such as polymer strings and artificial leather for the string bed and grip of a tennis racket, as well as rubber, polymer fabrics, artificial leather, and foams used in running shoes. Finite element analysis (FEA), discussed in detail in Chapter 6, is introduced as a valuable tool in computer-aided engineering.

In Chapter 6, titled "Finite element modelling in sport," Professor James Sherwood and colleagues share numerous examples of FEA applications in sports engineering. All examples utilize LS-PrePost for model construction and subsequent post-processing, with analysis conducted using the LS-DYNA solver. This chapter serves as a valuable resource for beginners in sports engineering as well as those new to FEA.

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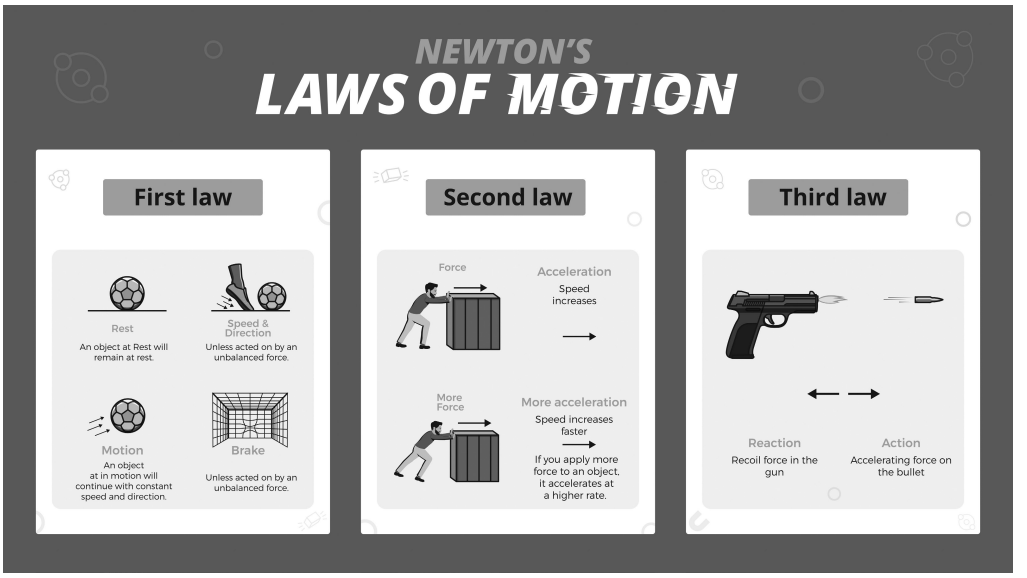

Figure 1.1 Newton's laws of Motion. First Law (Law of Inertia): An object will remain at rest or in uniform motion in a straight line unless acted upon by an external force. It means that objects tend to maintain their state of motion (either staying still or moving at a constant velocity) unless something pushes or pulls on them. Second Law (Law of Acceleration): This law essentially states that the greater the force applied to an object, the greater its acceleration will be, and the greater the mass of the object, the smaller its acceleration will be for the same force. Mathematically, this can be expressed as $\boldsymbol{F} = m\boldsymbol{a}$, where $\boldsymbol{F}$ is the net force applied to the object, m is its mass, and $\boldsymbol{a}$ is its acceleration. Third Law (Action-Reaction): For every action, there is an equal and opposite reaction. This means that when one object exerts a force on a second object, the second object simultaneously exerts a force of equal magnitude but in the opposite direction on the first object.

In Chapter 7, titled "Musculoskeletal simulation in sport," Professor Motomu Nakashima explains the methods of inverse dynamics analysis and forward dynamics analysis. The former involves obtaining forces and torques acting on the human body from kinematic information, while the latter involves solving human motion based on given kinetic information such as joint torques and muscle forces. Finally, this chapter introduces musculoskeletal simulation as an extension of inverse dynamics analysis. Through step-by-step explanations and several examples, readers can gain a comprehensive understanding of musculoskeletal simulation.

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2
PROJECTILES IN SPORT

John Eric Goff

Introduction

Projectiles are ubiquitous in sport. Association footballs fly toward goals; shuttlecocks, tennis balls, and ping-pong balls travel over nets; and baseballs, softballs, darts, and American footballs are thrown toward targets. Air is almost always the fluid through which sport projectiles traverse. Even water sports have projectiles moving through air, such as diving, where the athletes themselves are the projectiles, and water polo, where swimmers tread water, but throw the ball above the water. The aforementioned examples do not come close to comprising an exhaustive list. An alphabetical attempt at such a list would include, in addition to the above sports, basketball, cricket, field hockey, golf, jai alai, lacrosse, pickleball, rugby, sepaktakraw, volleyball, and Wiffle ball. More examples could be listed, but space limitations and taxing the patience of readers are concerns.

A thorough understanding of projectile motion in sport is critical to the continued success of a given sport. The last thing an association football manufacturer, players, fans, and a governing body want is to find the most important piece of equipment – the ball – flying in heretofore unseen ways during the FIFA World Cup, the most viewed sporting event in the world's most popular sport. The same desire applies not only to high-profile sporting events, but also to all levels, even games on sandlots and playgrounds. A sport like baseball has rather specific rules governing the size, shape, surface, and makeup of the ball. Other sports, such as association football and golf, do have rules for the balls, but football panel shapes and dimple counts on golf balls are not standardized. New technological advancements include stronger materials, better waterproofing, and environmentally friendly artistic components. But all the new technologies that lead to new equipment must not alter the well-established and familiar flights of sport projectiles.

Valuable testing of objects to be used as sport projectiles may be performed experimentally with wind tunnels and computationally with sophisticated fluid dynamics software. Fluid dynamics software methods are discussed in Chapters 3 and 4 on Computational Fluid Dynamics applied in sports. This chapter focuses on the physics of sport projectiles and the engineering techniques used to study actual trajectories. The use of wind tunnels requires the suspension of an object in the tunnel, which means the object must be destroyed in some way so that a support rod may be inserted. Computational fluid dynamics (CFD) requires scanned objects for geometrical inputs and

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employs algorithms with intrinsic error. Analyzing actual trajectories has its limitations, too. It is harder to control parameters like launch speed, launch angle, spin rate, and so on with a real sporting object in flight than it is with a wind tunnel or a CFD simulation. Trajectories are often filmed, which requires good lighting and enough cameras to capture the majority of an object's flight. Expensive cameras are needed if the desire is to capture motion that takes place on small time scales. A golf ball struck by a driver might leave the tee with 4000 rpm of backspin. Ascertaining that spin rate at launch requires a camera capable of about 10,000 frames-per-second recording, and the cost of such a camera could easily be a few thousand pounds. Standard photos and videos show two-dimensional images, whereas sport projectiles move in three dimensions. Even in controlled testing environments where two-dimensional motion is created, account must be taken of the three-dimensional aspect of the sport projectile, and care is needed to ensure proper camera focusing in the plane of motion. A minimum of two stereo-calibrated cameras is needed to obtain three-dimensional position data.

For all the stated limitations, analyzing actual sport trajectories has the obvious advantage over wind tunnels and CFD by working in the realm of what really happens when a sport object moves through a fluid. That analysis begins with introductory physics. A projectile's trajectory is the part of its motion from the instant just after it leaves an athlete's hand, stick, bat, etc., up to the instant just before being caught, striking the ground, etc. The only forces on the projectile are due to the gravitational attraction of Earth and the interaction with the fluid through which the projectile moves.

Initial forays into trajectory analysis

Quick trajectory estimates may be made if one assumes the projectile moves through a vacuum. "Ignore air resistance" is a phrase commonly found in introductory physics projectile problem statements. A projectile would have to reach an elevation of approximately 32 km (20 mi) above sea level to experience a magnitude of acceleration due to gravity that is 1% smaller than its value at sea level. Because that elevation is well beyond the maximum heights of projectiles in essentially all terrestrial sports, assuming the magnitude of the acceleration due to gravity to be constant is an excellent approximation. That constant is often written as g and has a value to two-digit accuracy of 9.8 m/s^2 ≈ 32 ft/s^2 ≈ 35 kph/s ≈ 22 mph/s. The macroscopic motion of an object with mass m is determined via Newton's second law equation, given by

$$m\vec{a} = \vec{F}^{(\text{net})}, \tag{2.1}$$

where $\vec{a}$ is the acceleration of the object and $\vec{F}^{(\text{net})}$ is the net external force acting on the object. The above equation is used for point particles because $\vec{a} = d^2\vec{r}(t)/dt^2$, where $\vec{r}(t)$ is the position vector from a previously determined coordinate system origin to the point mass. But, of course, sport projectiles are not point masses. Fortunately, one need not fret over which point should be used in a sport projectile. One can show that the motion of an object's center of mass may be described as if the object's mass were concentrated at the location of its center of mass and moving according to Equation (2.1). Studying the flight of sport projectiles generally involves looking at the motion of the center of mass. If one wishes to investigate the motion of an extended object about its center of mass while in flight, one will need to use more advanced techniques from classical mechanics, namely those that employ principal axes and inertia tensors [1]. The focus in this chapter will be limited to the motion of an object's center of mass.

Neglecting air resistance and assuming that a sport projectile experiences only a constant gravitational force (i.e., weight) $m\vec{g}$, where $\vec{g}$ points downward, leads to a standard introductory

physics problem in kinematics. One result that may be found is the horizontal range of a projectile launched at angle θ_0 from the horizontal with speed v_0. If the initial and final heights are the same, the horizontal range is

$$R = \frac{v_0^2 \sin\left(2\theta_0\right)}{g}. \tag{2.2}$$

The above result is quite accurate as long as the object's weight is the dominant force acting on the object. A runner jumping over a hurdle is such an example because the athlete is not running fast enough for air resistance to be anywhere close to the size of his or her weight. But for a home run hit in a baseball game, Equation (2.2) gives a poor prediction of horizontal range. Suppose a baseball leaves the bat at 30° above the horizontal with an initial speed of 100 mph (45 m/s or 160 kph), typical for a Major League Baseball home run. Equation (2.2) would predict a horizontal range, assuming the same initial and final heights (not usually true for a home run), of approximately 580 ft (180 m, to two-digit accuracy). Such a prediction is roughly 50% longer than an actual home run distance, clearly indicating that vacuum kinematics will neither be sufficient to model the flight of a baseball nor many other sport projectiles.

Going from vacuum to air

The inclusion of air resistance turns an introductory physics problem into a significantly more advanced problem, one that often requires computational techniques for its solution. Air is a fluid with a mass density $\rho \approx 1.2\,\mathrm{kg/m^3}$, more than 800 times less dense than liquid water. The density of air depends on temperature, humidity, and altitude. Air is, to an excellent approximation, an ideal gas, which means the particles comprising air (atoms and molecules) do not interact. Air thus obeys the ideal gas law, given by

$$P\,V = N\,k\,T, \tag{2.3}$$

where P is the air pressure, V is the volume of air, N is the number of particles in the volume, k is Boltzmann's constant, which is $k \approx 1.38 \times 10^{-23}\,\mathrm{J/K}$, and T is the temperature. Equation (2.3) leads to the conclusion that the properties of air are determined by the number of particles in a given volume, not by the identity of those particles. Put another way, it matters not whether the particles are nitrogen molecules, oxygen molecules, or water vapor. The number of particles per volume is $N/V \propto P/T$, which means that for a given pressure and temperature, the number density of particles is fixed, regardless of what the particles are. Humid air contains more water vapor than dry air, but a water molecule (H_2O) is less massive than either dinitrogen (N_2) or dioxygen (O_2), the two types of molecules that make up 99% of air. So despite having the same number of particles in a given volume, humid air has less mass density than dry air, for a given pressure and temperature.

It turns out that humidity is not a large effect on sport trajectories. Water vapor makes up roughly one molecule per 100,000 molecules in air. Even at the dew point, where relative humidity is 100%, at 25°C (77°F), there is only about 20 g of water for each 1 kg of air. At a maximum of just 2% of air's mass, fluctuations in humidity will not make large changes in air's mass density. Leaving humidity aside, focus on $N/V \propto P/T$, which states that for a fixed pressure, the density of particles decreases with increasing temperature. That is the reason balls fly farther on hot summer days than they do on cool autumn days.

It is well known that air's mass density drops with increasing altitude. That is because, for a fixed temperature, $\rho \propto N / V \propto P$, and air pressure drops with increasing altitude. A crude approximation for air density's dependence on altitude z (measured above sea level) is

$$\rho(z) = \rho_0 e^{-z/z_0}, \tag{2.4}$$

where ρ_0 is the density of air at sea level and $z_0 = 8.5\ \text{km}$ [2]. According to the Equation (2.4), a baseball moving through the air in mile-high (1.6-km-high) Denver, Colorado (USA), encounters air that is 17% less dense than the sea-level air in New York City (USA).

Basic fluid dynamics needed for trajectory analyses

Armed with an understanding of how environmental factors influence air density, focus may now shift to a model for the force of air on a sport projectile. Neither this book nor this chapter represents a replacement for the valuable information gained in a course or textbook that focuses on fluid dynamics. But a minimal amount of fluid dynamics is necessary to work in the field of trajectory analysis. An important parameter that an engineer would be wise to estimate before performing experiments is the dimensionless Reynolds number, defined as [3]

$$\text{Re} = \frac{\rho\ v\ L}{\mu} = \frac{v\ L}{\nu}, \tag{2.5}$$

where ρ is fluid density, v is the sport projectile's center-of-mass speed, L is a characteristic length (diameter, if the sport projectile is a spherical ball), μ is the viscosity, and $\nu = \mu / \rho$ is the kinematic viscosity. The Reynolds number is a qualitative measure of the ratio between inertial drag and viscous drag. At very small speeds where fluids creep around objects, and one may imagine a type of friction where two surfaces rub against each other, viscous drag dominates. Such drag is roughly proportional to the projectile's speed. For larger speeds, smooth laminar flow and rough turbulent flow are possible, and inertial drag dominates. One may imagine air molecules smashing into a projectile, exchanging energy and linear momentum with that projectile. For these air flows, air drag is roughly proportional to the square of the projectile's speed. Given the aforementioned speed dependences for viscous and inertial drag, $\text{Re} \propto v$, as Equation (2.5) shows.

For air at 20°C and 1 atm, $\nu \approx 1.50 \times 10^{-5}\ \text{m}^2 / \text{s}$ [3]. Consider a range of sport projectile speeds from 5 m/s (11 mph) to 50 m/s (112 mph), a range that surely encompasses ball speeds in almost every sport. Sport projectiles (not just balls) have characteristic sizes in the approximate range of 30 mm (1.2 in) to 2 m (6.6 ft). The range of possible Reynolds numbers for sport projectiles thus has an order of magnitude 10^4 to 10^7, and because $\text{Re} \gg 1$, inertial drag is the dominant form of air drag on sport projectiles.

Before turning to quadratic inertial drag, it should be noted that the Reynolds number appears in many aspects of fluid dynamics. For example, Re appears as the lone parameter in the dimensionless Navier-Stokes equation. That means if one solves the equation for a given value of Re, one has a solution for any situation matching the given Re. Because $\text{Re} \propto vL$, one could build a geometrically scaled-down model of a sport projectile and investigate its aerodynamics in a wind tunnel. One must scale up the air speed by the same factor to get the identical Re that would be seen in actual flight. This important scaling property is what allows relatively low-cost models to be tested in wind tunnels before high-cost products are manufactured. Dams, automobiles, windmills, buildings, and anything else that experiences forces from fluids are candidates for scaling down.

For all the complicated ways in which air interacts with a sport projectile during its flight, perhaps the simplest way to model the air's force on the projectile is to imagine a single force acting on the projectile's center of mass. Given three spatial dimensions, that force from the air may be split into three orthogonal components. The component opposite the projectile's velocity is the drag force, or air drag, with magnitude [3]

$$F_D = \frac{1}{2} C_D \rho A v^2, \tag{2.6}$$

where A is the projectile's cross-sectional area, and C_D is the dimensionless drag coefficient. The component of the air's force perpendicular to the drag force, but still in the plane formed by the drag force and the projectile's weight, is the lift force, whose magnitude is given by [3]

$$F_L = \frac{1}{2} C_L \rho A v^2, \tag{2.7}$$

where C_L is the dimensionless lift coefficient. Perpendicular to both the drag force and the lift force is the side force, with magnitude [3]

$$F_S = \frac{1}{2} C_S \rho A v^2, \tag{2.8}$$

where C_S is the dimensionless side coefficient. All three component magnitudes contain the dynamic pressure $\frac{1}{2} \rho v^2$, often met for the first time when working with fluid kinetic energy while deriving Bernoulli's equation. That pressure multiplied by the projectile's cross-sectional area, giving the dynamic force, is a great first step in estimating the size of a fluid's force on a projectile. The dimensionless aerodynamic coefficients C_D, C_L, and C_S may thus be thought of as corrections to dynamic forces. Those coefficients contain much of the interesting information about the aerodynamics of sport projectiles because the coefficients themselves may depend on the projectile's speed and spin. Wind tunnels and CFD are common tools used to determine aerodynamic coefficients (see Chapters 3 and 4), but this chapter will explore how they may be found via trajectory analyses.

Before getting into trajectory analyses, it should be pointed out that there is one more force from the air on a sport projectile. Archimedes' principle states that an object displacing a fluid experiences an upward buoyant force on the object equal in magnitude to the weight of the amount of fluid displaced by the object. For mostly hollow sport objects, such as ping pong balls, footballs, and basketballs, the weight of air contained within the ball nearly balances the buoyant force. That means that weighing a hollow ball on a scale and dividing the weight by g essentially gives the ball's mass. For mostly solid objects like baseballs, the buoyant force is not balanced by trapped air. But for a baseball, the buoyant force is less than 0.2% of the baseball's weight. Ignoring the buoyant force in modeling is often done, though its inclusion in Equation (2.1) is trivial.

Starter experiment for trajectory analyses – ball drop

A relatively simple example of trajectory analysis will be presented here. In an effort to control as much of the trajectory as possible, an adidas Telstar 18 football (soccer ball) was dropped with no initial speed from an open window, about 12 m (39 ft) above the ground. The trajectory was filmed with a cell phone; video settings were 60 frames per second and ultra-high definition. Filming trajectories can be challenging. If cameras are too close to the action, objects of interest that

are on the sides of the video image are at too large angles for length calibration to be accurate. Trajectories should be in a plane if only one camera is used, and the camera needs to film from roughly the center of the trajectory to limit lens distortion errors. An item of known length should be in the plane of filming. Figure 2.1 shows the ball-drop experiment.

The ball itself, with its 22-cm (8.7-in) diameter, was used as the length scale in Figure 2.1. Frames of the video and the analysis of the video came from *Tracker*, an easy-to-use, free software [4]. That software allows the user to, among many other things, set a calibration distance, manually note ball position frame by frame in the video, and analyze position data. Simple constant-acceleration kinematics says that if a ball falls in a vacuum from rest at a height h above the ground, the ball will hit the ground with speed $\sqrt{2\ g\ h}$ after falling for a time $\sqrt{2\ h / g}$. According to *Tracker*, the ball fell $h = 11.98$ m, which means if the ball had fallen in a vacuum, it would have hit the ground at 15.32 m/s (55.17 kph or 34.28 mph) after falling for 1.564 s. Note that four significant digits have been kept, though three-digit accuracy is likely what was possible with the video analyses. The aforementioned speed and time represent upper limit and lower limit values, respectively, when air resistance is included. Simple calculations should be done in advance of experiments and analytic work so that one has a rough idea of what physical numbers to expect.

The actual drop time was 1.599 s, which means that at 60 frames per second recording, 97 frames from the video may be used to determine the ball's vertical position as a function of time. Figure 2.2 shows graphical output from *Tracker* for the ball-drop experiment. A vertical axis, labeled y, had its positive direction pointing up, and the ground was very near the zero on that axis. Note in Figure 2.2 that the position data are relatively smooth. The vertical component of velocity appears roughly linear, but there are more noticeable fluctuations than there were in the position data. The vertical component of the acceleration graph shows data that appear to be a mess. One might be able to see, in the acceleration graph an average value in the neighborhood of $-g$, but there is no way that the actual acceleration experienced by the falling Telstar 18 could have had that much fluctuation.

The progressing lack of smoothness seen in the graphs in Figure 2.2 is commonly seen in trajectory analysis. Position data are taken directly from the computer screen, which means locations marked in software by a user are translated, with the help of a calibration distance and a choice of axes' directions, into coordinates. There is uncertainty associated with where the user marks an object's location in a video image. One should mark the same part of an object in each frame, but even trying to do so inevitably leads to uncertainties in position. Velocity and acceleration components are not determined directly. They must be calculated from position data, which means time derivatives are needed. Evaluating those time derivatives numerically means using finite-difference algorithms. Those algorithms come with their own truncation errors. Additional error is introduced for each time derivative needed. That is why the velocity component data in Figure 2.2, which were found via one numerical time derivative, are smoother than the acceleration component data, whose production required two numerical time derivatives.

There are several options one may take with trajectory data if one is interested in extracting aerodynamic information. A couple of those options will be explored here. The first begins by looking at Equation (2.1) in which the net force on the football is its weight pointing down and the drag force pointing up, opposite the ball's velocity. Using Equation (2.6) for the drag force, one finds that the ball's acceleration is

$$a = \frac{C_D\ \rho\ A}{2\ m}\ v^2 - g. \tag{2.9}$$

Figure 2.1 Telstar 18 about to be dropped (left image), approximately halfway to the ground (center image), just hitting the ground (right image).

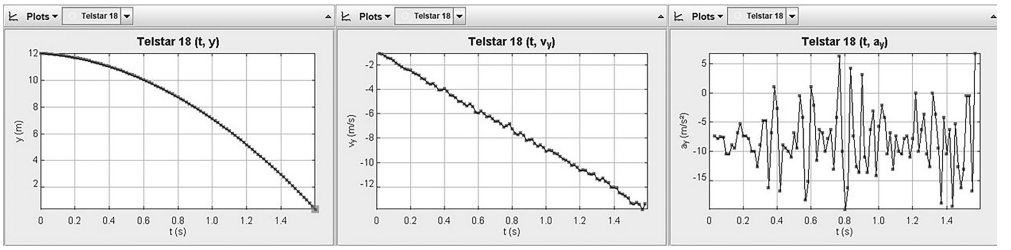

Figure 2.2 Vertical position versus time (left image), vertical velocity component versus time (center image), vertical acceleration component versus time for ball-drop experiment (right image). Graphs produced by *Tracker*.

One could analytically solve Equation (2.9) for the velocity and position via integration if the factor in front of v^2 was constant. But the drag coefficient depends on speed, in general. Still, one could assume that the speeds reached by the falling football did not alter the drag coefficient too much. As a first attempt at extracting a drag coefficient, take the drag coefficient to be constant and let $\alpha = C_D \rho A / 2m$. The velocity as a function of time is

$$v(t) = -v_T \tanh\left(t / \tau\right), \tag{2.10}$$

where $v_T = \sqrt{g / \alpha}$ is the terminal speed and $\tau = 1/\sqrt{\alpha g}$ is the characteristic time. The initial velocity was taken to be zero. Integrating Equation (2.10) gives

$$y(t) = y_0 - \frac{1}{\alpha} \ln[\cosh(t/\tau)], \tag{2.11}$$

where y_0 is the initial position. Differentiating the velocity in Equation (2.10) with respect to time quickly gives

$$a(t) = -g \operatorname{sech}^2(t/\tau), \tag{2.12}$$

which could also have been found by plugging Equation (2.10) into Equation (2.9).

With experimental position data and the analytic expression for position as a function of time given in Equation (2.11), one may find the optimum value of α, and hence C_D, that provides the best fit between the data and Equation (2.11). Carrying out that procedure yields $C_D = 0.30872$ with an adjusted *R*-squared of $R^2 = 0.999926$ (too many significant digits for both parameters are stated, but space does not allow a full discussion of error propagation). Figure 2.3 shows the position data with Equation (2.11).

The agreement between the experimental data and Equation (2.11) in Figure 2.3 is quite good, as also demonstrated by the value of R^2. Only for very small times is the agreement noticeably off. The small-time positions are at the largest angle for the camera, and the drag force is very small just after the ball is released.

A drag coefficient of approximately 0.3 is certainly not out of the realm of what one might expect for an association football. One can do better with trajectory analyses, though, by determining the dependence the drag coefficient has on speed. Equation (2.9) provides what is needed because if the acceleration and speed are known functions of time, the speed-dependent drag coefficient could then be simply determined. The problem with Equation (2.9) is that the numerical

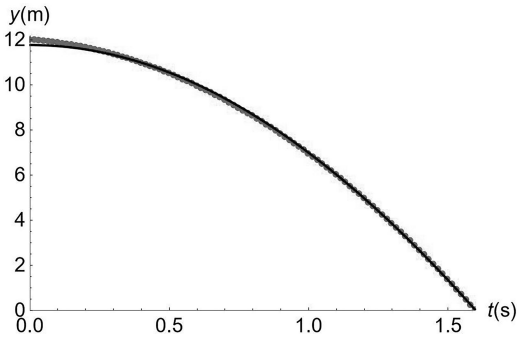

Figure 2.3 Falling football vertical position data (points) and Eq. (11) with optimized C_D .

errors that propagate into the velocity and acceleration components in Figure 2.2 lead to wildly fluctuating values of C_D. The problem may be addressed by smoothing the position data.

There is no perfect way to smooth the position data. Fitting to a quadratic in time will not work because such a fit assumes constant acceleration. Fitting to too large an order in time creates unphysical undulations in the fitted curve. To get a feeling for how smoothing works, just one order past quadratic will be used here. One would hardly notice a difference in the comparison between the data and the fitted function if the order was taken to ten. The position data in Figure 2.2 are easily fit to a third-order polynomial in time using standard algorithms or modern software packages. Even *Tracker* is capable of modest polynomial fitting. Once the polynomial for the position is determined, velocity and acceleration are determined quickly through differentiation.

A parametric plot of the drag coefficient versus speed is shown in Figure 2.4. Several comments need to be made about the graph in Figure 2.4. Quadratic drag is challenging to measure at small speeds, i.e., at speeds where the ball's weight is much larger than the drag force. Extracting the drag coefficient from very small drag forces, meaning very small speeds, means that the drag coefficient can be quite large. The wind-tunnel data shown in Figure 2.4 came from an experiment that did not probe below 7 m/s [5]. The fact that the Telstar 18 ball hit the ground at approximately 14 m/s (50 kph or 31 mph) (see Figure 2.2b) means that, according to the wind-tunnel data [5], much of the ball-drop experiment occurred during the drag crisis, i.e., during the transition from laminar flow to turbulent flow. Trajectory analysis managed to produce a drag coefficient that dropped with increasing speed. The agreement between wind-tunnel data and trajectory analysis is not terrible. Changes in the drag curve happen when the orientation of the non-spinning ball is changed in the wind tunnel [5]. Higher-order polynomial fits yield similar results, though undulations appear in the drag curve. The level of agreement between a trajectory analysis result and a wind-tunnel result becomes much more attractive when one realizes that the former was obtained at significantly lower cost than the latter. Trajectory analyses provide results for objects in flight, without the supporting mechanisms used in wind tunnels, which at least partially reduce the realism associated with an actual sporting event.

The initial assumption, that of a constant drag coefficient, led to a value of C_D roughly in the middle of the curve in Figure 2.4. Such a result is clearly not true but could be quite useful if the level of accuracy needed for a trajectory plot is not too high. Letting $C_D = 0.3$ for all speeds leads to an estimate of 24.8 m/s (89.3 kph or 55.5 mph) for the terminal speed of Telstar 18, a bit small because the drag coefficient is closer to 0.2 for large speeds (which leads to 30.4 m/s ≈ 109 kph ≈ 68.0 mph if $C_D = 0.2$) [5].

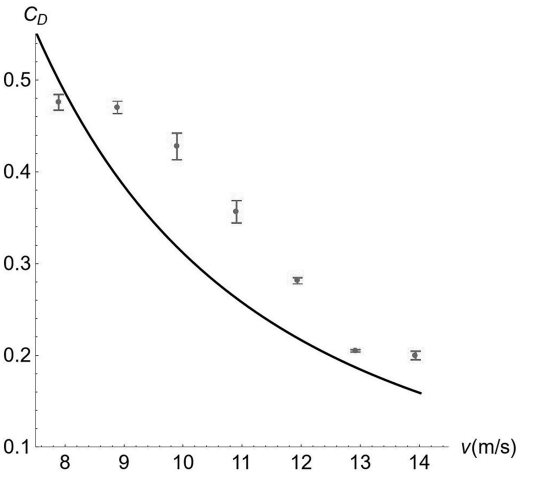

Figure 2.4 Drag coefficient obtained from an $O\left(t^3\right)$ polynomial fit for position. Points and error bars are wind-tunnel data for Telstar 18 [5].

Realistic sport trajectories complicate trajectory analyses

The drop experiment is a simple, low-cost starter project for an investigator looking to extract aerodynamic information via trajectory analyses. But, of course, the world of association football never has balls dropped vertically from 12 m above the pitch. Most football trajectories in real matches have balls with spin, which means two-dimensional vacuum trajectories occupy three spatial dimensions with the inclusion of air. So-called "banana" kicks have a spin axis that is usually neither exactly vertical nor exactly horizontal. Lift and side components of the air's force on the ball are then nonzero, which means Equations (2.7) and (2.8) are required in the trajectory analyses. Extracting aerodynamic information from video of specific trajectories in real football matches, for example, is nearly impossible because the one or two camera angles afforded the viewer show the ball moving toward or away from the camera. Furthermore, cameras filming action for television audiences often move, which negates the ability to have stationary coordinate axes in the video analyses.

The experimenter must create trajectories in a controlled way. Ball launchers are helpful when investigating the aerodynamics of spinning sports balls. Figure 2.5 shows this chapter's author behind a bespoke ball launcher that has four counter-rotating wheels [6]. The ability to set each wheel's rotational speed allows for controlled spin (backspin, topspin, sidespin, etc.). A 2000-frames-per-second camera captured the launch, and an identical camera recorded a portion of the ball's trajectory near its apex. Figure 2.5 also shows the camera image of the football just after being fired from the launcher. Circles show ball locations from the launcher to the position shown in Figure 2.5 (the software used was not *Tracker* but bespoke software at the University of Sheffield).

Cameras must record at a large enough frame rate so that rotational speeds may be determined. Balls (or whatever is spinning) must be marked, if they are not already marked, in such a way as to note how they turn from frame to frame in the video. Breaking symmetry with markings is helpful if balls are turning fast enough to appear not to have turned from one frame to the next. Parameterizing spin is often done with the dimensionless Strouhal number, or spin parameter, defined for a ball as [3]

$$\mathrm{Sp} = \frac{r\,\omega}{v}, \tag{2.13}$$

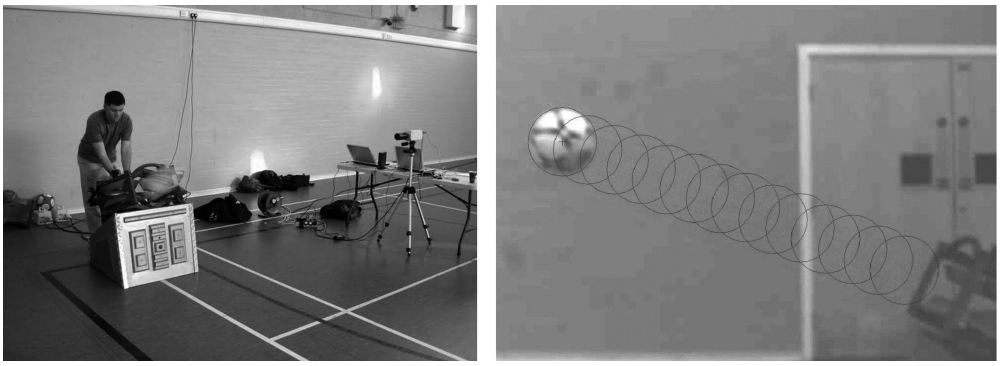

Figure 2.5 Football leaving ball launcher with high-speed camera off to the right (left image); camera's video frame of a particular launch (right image). (Reproduced from [6] with the permission of the American Association of Physics Teachers.)

where the ball's radius, r, and angular speed, ω, multiply to give the tangential speed at the ball's equator. The ratio of that speed to the ball's center-of-mass speed, v, is thus the ratio defined in Equation (2.13). Aerodynamic coefficients depend not only on the Reynolds number but also on the Strouhal number. A hard-kicked football with a lot of sidespin may have $\mathrm{Re} \approx 5 \times 10^5$ and $\mathrm{Sp} \approx 0.2$ [7].

Multi-dimensional trajectory analysis

Trajectory analysis of two- and three-dimensional problems is significantly more complicated than the previously discussed one-dimensional ball-drop example. But the philosophy is the same. The drag coefficient could be extracted from Equation (2.9) if the speed and acceleration were known for all times of the trajectory. Because the velocity changes direction for two- and three-dimensional trajectories, the drag and side forces also change direction. The reader is referred elsewhere [6] for the mathematical derivation of the aerodynamic coefficients in terms of the components of the velocity and acceleration vectors associated with two- and three-dimensional trajectories.

Performing trajectory analyses in three dimensions often requires multiple cameras synced in time and calibrated so that full three-dimensional trajectory data may be extracted from all video taken. Researchers with the funding to obtain several high-speed cameras have the capability of recording numerous trajectories with a myriad of initial launch conditions. Struck sporting objects by athletes or sporting equipment may be used instead of mechanical launchers because numerous trials may be undertaken, knowing that desired trajectories will later be culled from all the video. Interested readers may peruse a study that used four high-speed cameras and 29 separate football trajectories to create drag and lift functions of speed as an example of how trajectory analyses in three dimensions are performed [8].

Flow visualizations via trajectory analyses

Trajectory analyses are not limited to extracting aerodynamic coefficients. Visualizing airflow is also possible with recorded trajectories. Such trajectories might be better created with launchers than with, for example, humans kicking balls. Controlling launch speed, launch spin, and trajectory direction are important if the projectile has to interact with a substance to help visualize

airflow. One low-cost substance that might be used is dust. Simply creating a dust cloud and firing an object through the cloud while a high-speed camera (or cameras) records the action may work as well as what could be produced via a wind tunnel. Using substances to visualize airflow in wind tunnels is tricky because the substances can damage the wind tunnel if they gum up the inner workings of the machine.

Figure 2.6 shows the setup for an experiment involving footballs fired from a bespoke launcher into a dust cloud [9,10]. Trial and error may be involved in getting the best lighting for an experiment. High-speed cameras do not have their shutters open for long as they record frames, which means a lot of light is needed on the object of interest to prevent blurry images of a fast-moving object. The dust cloud for the experiment depicted in Figure 2.6 was created by one researcher standing next to the black screen and clapping dust just before a ball was launched. Many launches were filmed and a subset of the videos obtained led to nice flow visualizations.

Figure 2.7 shows flow visualizations via dust experiments for a non-spinning football moving at two different speeds [9,10]. Though the images are two-dimensional, one must remember that the boundary layer of air separates all around the ball. The images in Figure 2.7 thus provide a cross-sectional view of the full three-dimensional boundary-layer separation. The boundary layer that separates off the ball in the left image of Figure 2.7 did so during laminar flow. The boundary layer separated off the ball roughly halfway around the ball. Boundary-layer separation during turbulent flow is what is visualized in the right image in Figure 2.7. The boundary layer of air in

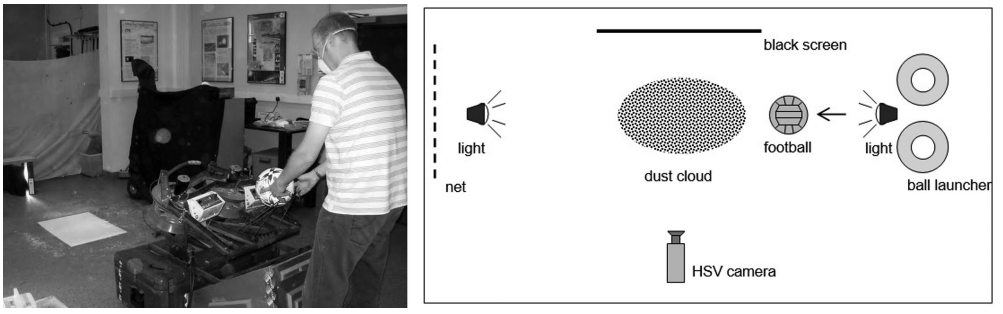

Figure 2.6 Researcher feeding a ball launcher (left image); schematic of the dust-cloud flow-visualization experiment (right image) [9,10]. Copyright Elsevier (2012).

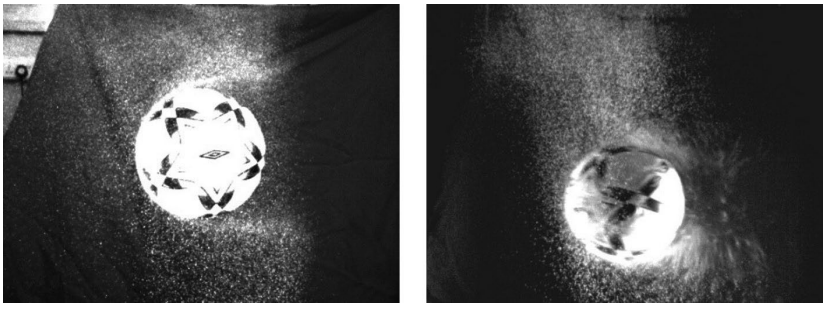

Figure 2.7 Non-spinning football moving to the left through a dust cloud at 7.65 m/s (27.54 kph ≈ 17.11 mph) (left image); non-spinning football moving to the left through a dust cloud at 19.14 m/s (68.90 kph ≈ 42.81 mph) (right image) [9,10]. Copyright Elsevier (2012).

that case separated farther back on the ball compared to the case of laminar separation. By analyzing a myriad of trajectories with various launch speeds, the boundary layer's angle of separation could then be found as a function of speed [9,10]. Spinning balls could be fired into dust clouds to investigate the asymmetric shedding of the boundary layer that leads to nonzero lift and spin coefficients. A finer and perhaps more dense substance might be needed if one wishes to study asymmetric boundary-layer shedding from non-spinning balls. As qualitatively pleasing as the images in Figure 2.7 might be for illustrating the difference between laminar flow and turbulent flow, determining the boundary layer's separation points to high accuracy is quite challenging. The task is made even more challenging if one uses multiple cameras to obtain a three-dimensional view of the boundary-layer separation.

Particle image velocimetry

An approach more sophisticated, and more expensive, than the aforementioned dust experiment to investigate flow patterns around sport objects is particle image velocimetry (PIV). Air, which is almost always the fluid in which sport projectiles are studied, is seeded with tracer particles. The tracer particles act like the dust in the previous section and should follow the air flow around the sport projectile. Space does not allow for a complete and thorough discussion of PIV, but readers interested in this method are encouraged to learn about the Stokes number and note that it must be small compared to unity for proper PIV work [11].

Lasers are often used to illuminate the tracer particles. A sport projectile is fired through the illuminated tracer particles. The PIV system is triggered when the projectile reaches a desired location. A high-speed camera (typically recording a couple of thousand frames per second) captures images, and subsequent image processing reveals fluid flow around the sport projectile. Motion along the axis of the camera cannot be measured, which must be accounted for when drawing conclusions from the planar flow images.

Recent work used PIV techniques to study the velocity field of airflow around golf balls, paying particular attention to the onset of the reverse Magnus effect [12]. The same work determined lift coefficients as a function of spin rate based on the separation points of the boundary layer. Other recent work used PIV methods to investigate the flow patterns around baseballs [13], including the determination of drag and lift coefficients [14].

A look to the future

This chapter has described how trajectory analyses may be used to investigate the aerodynamics of sport projectiles. Though the paradigm employed here was that of football trajectories, essentially any projectile may be investigated via trajectory analyses. Using the techniques on filmed trajectories of humans and other objects lacking near-symmetric geometries can be challenging. As humans traverse through the air with arms and legs moving, their centers of mass move within (or even outside) their bodies. But objects more complicated than balls may still be studied with video if one is interested in, for example, the motion of an athlete's leg while running. Swinging baseball bats or golf clubs through dust clouds could help visualize airflow around those objects. Such experiments are significantly less expensive than using wind tunnels, and they require much less computational skill than using CFD on scanned images.

Technological advancements with cameras, including a reduction in their cost, have made their use in sport nearly omnipresent. Numerous cameras in stadia record pitched and hit balls in baseball. A panoply of data is available for fans and researchers alike [15]. Baseball spin rates

on pitches, plus pitch speed and location, help understand pitching. All Major League Baseball hits are recorded, providing data for launch speed, launch angle, and the ball's spin rate off the bat. Researchers may thus make use of readily available data from an incredible number of hits and pitches. Acquiring that data is extremely cost-effective because researchers simply take data from real-world sporting situations instead of creating those situations in a laboratory setting. A recent study used baseball data available in Japan [16]. Another recent investigation obtained Hawk-Eye data from tennis matches to study the influence of ball wear on the aerodynamics of tennis balls [17].

Football players are tracked with cameras during important matches, like those at the World Cup. Fans may be treated to a great deal of information while watching matches, such as the total distance a player moved on the pitch and that player's top speed. But for all that is currently being done with sport recording, what is still needed is a complete three-dimensional recording of a projectile's flight. That recording should provide center-of-mass coordinates for the entire trajectory, as well as the motion of the object about its center of mass. It would be wonderful to have accurate data associated with the complete trajectory of a hit baseball or golf ball, complete with the time-dependent angular velocity vector.

As recording technology continues its march toward inevitable improvements, physics must be used to convert data into information of use to researchers. A feedback loop exists in this work. A trajectory is recorded, the video is analyzed, and coordinate data are extracted. Physical models are used to find aerodynamic coefficients, and those coefficients may be used with physical models to numerically find new trajectories. These computational trajectories must then be compared with real trajectories. The feedback loop helps improve recording technology and the physical models. Though the essence of classical mechanics has been around for more than three centuries, improving the quality of physical models that employ classical mechanics is an ongoing endeavor. Breakthroughs in trajectory analysis will thus likely arise via improvements in video recording and in creating physical models that more accurately comport with reality than did the models of the past.

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3

COMPUTATIONAL FLUID DYNAMICS APPLIED IN SPORT

Part 1 – Capabilities and considerations

John Hart

Introduction

The influence that fluid dynamics has upon sport has been a focus of intense study for decades. Researchers with interests in sports applications have been driven by a desire to understand and manipulate aero- and hydrodynamics. These can influence both the performance of an item of sporting equipment and the outcome of competition, in subtle or dramatic manners. Through understanding sports fluid dynamics, researchers may identify a subtle alteration to the position of a seam on a cyclist's racing suit that may result in a medal. It is the same flow physics that may dramatically influence the visible motion of sports equipment. For example, the swerve in flight of an association football, which may or may not be an intentional aspect of the equipment design!

The reasons underlying the attention that sports fluid dynamics has received are varied. Research has been driven by commercial factors to develop better sporting products. Sporting goods companies may seek to maintain or grow their market share through the development of products that provide a competitive aerodynamic advantage underpinned by proven research. PING (Arizona, USA) is one such example, having developed their golf drivers and clubs in this manner and published the research in peer-reviewed proceedings [1,2]. Another example would be Speedo International, which partnered with Barry Bixler, who led the way in using CFD to understand swimming [3].

The pursuit of medals in elite sport has driven significant research in recent years, particularly within Olympic sports. Finding a tenth or even a hundredth of a second can be decisive in competitive cycling and winter events such as skeleton bobsled. To this end elite sports and national teams have partnered with both academia and industry to assist with their equipment optimisation. There are examples globally of universities partnering with their national teams to support them in advance of and during Olympic cycles. This is true for the Sports Engineering Research Group at Sheffield Hallam University, which supports elite British teams. The author has worked in the past across summer and winter Olympic cycles optimising bikes and sleds, even modelling air movement on Olympic sailing courses (Figure 3.1). Likewise, Bert Blocken during his time at TU Eindhoven (NL) and in partnership with ANSYS, has supported the Tour de France cycling team, Team Jumbo-Visma.

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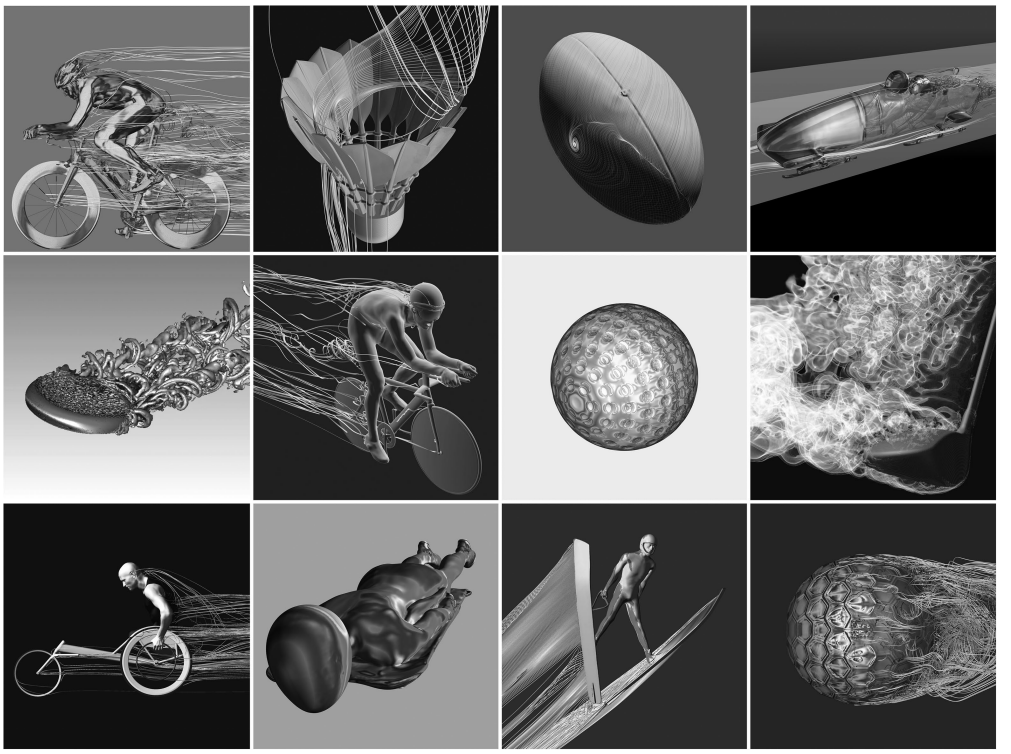

Figure 3.1 For over 20 years, the author has used CFD to understand performance in sport. The images provide example of a small selection of the elite, commercial, and research projects the author has conducted. Top row; time trial cyclist, shuttlecock, rugby ball, two-person bobsled. Middle row; frisbee, elite track cyclist, golf ball, golf driver. Bottom row; wheelchair athlete, skeleton bobsled slider, water-ski jumper, lacrosse ball. (Author's own)

Academics are naturally inquisitive, and many have a desire to educate and engage audiences beyond the walls of institutions. A powerful way to engage the public and future engineers with both physics and science is through the explanation of sporting performances and equipment behaviours that may already have captured their attention. It is common to see engineers engaged in such activity through blogs and articles on news outlets. Whenever a sports team or manufacturer has developed a new piece of equipment that has led to success, or an odd behaviour, news outlets often seek an academic opinion.

Sometimes, however, research that ends up being published about a specific sport or product has arisen for no other initial reason than pure, simple curiosity. Perhaps an engineer's desire to find out if a problem is possible to understand and solve may lead to the discovery of something interesting perhaps significant, that is worthy of publication and could even have implications beyond sport.

Researchers today have a wide range of analytical, experimental, and computational techniques at their disposal. All are actively used in sports engineering research, some of which have been discussed in Chapter 2, Projectiles in Sport. One such technique, Computational Fluid Dynamics (CFD), has become an increasingly popular tool for engineers in their sports research endeavours. In the last 30 years since CFD first started to appear in sports applications, it has become a key tool in the development of both Olympic-medal winning equipment and market-leading sporting

goods. Within academia its use is widely taught, and its application to understanding complex flows is common.

The possibilities for discovery through the utilisation of CFD have grown considerably in the preceding 30 years. New algorithms to accelerate solutions and models with improved accuracy have been developed. The ability to create and handle complex geometric models has improved significantly with automated CFD tools and technological developments in both computing and cameras. Although generic human geometry is still seen in sports simulation, it is now common to scan and include the geometric forms of actual athletes, allowing bespoke development and optimisation of equipment. Twenty years ago, the inclusion of such geometry required expensive laser scanning equipment. Today human and other geometries can be captured using a mobile phone camera, albeit to a level of accuracy.

The inclusion of movement is also increasingly possible and has gone beyond the simple rotational and translational simulations of a spinning ball [4] or bicycle wheel [5]. Geometries are no longer static, and it is now possible to morph a geometric sequence of movement [6]. Such examples in sport include the hydrodynamic analysis of a swimmer's dolphin kick [7] and simulations being used to understand the aerodynamic impact of a cyclist's pedalling action on bicycle component performance [8]. Use of CFD in these applications has only been possible because computing capabilities have increased, as has the accessibility of affordable high-performance computing. What is now economically achievable has brought the possibility of complex simulation within reach of the many rather than the affluent few, increasing research opportunities. The combination of accessibility and advancements in solutions has led to increasingly larger and more complex models. These reveal richer and more detailed flow physics that further increase knowledge and understanding.

However, the fact remains that the level of complexity in a simulation and the speed with which it can be solved is directly proportional to the overall size of the computing resource available. For this simple reason, it is still the most affluent sports and academic institutions that lead the way in the size and complexity of sports simulations being conducted. It is also the reason you will rarely hear a CFD engineer state they are fully satisfied with the solution capabilities they can access. The advent of cloud computing, however, now opens the possibility of further exciting discoveries and should lead to a further and wider increase in user applications.

Basics of CFD?

Computational fluid dynamics is a numerical tool that solves governing equations that describe fluid flow. Put simply, CFD solves a series of equations that calculate the temporal and spatial behavior of fluid flow around or through an object. This allows a user to determine flow variables of interest, such as pressure, velocity, and temperature, at any point on or around that object, as well as the subsequent forces that the object may experience.

CFD codes exist in a variety of formulations, including *Finite Volume* [9], *Lattice Boltzmann* [10], and *Smooth Particle Hydrodynamics* [11]. The approaches differ in the manner they use to solve fluid behavior, but all have been used and all have appeared in peer-reviewed research related to sports applications. Commercial and open-source codes are commonly applied. The specific choice of code may depend on considerations regarding capabilities and functionality, availability, cost, or personal preference. Currently, *Finite Volume* codes are the dominant approach used in sports research. Although interest is growing in the application of *Lattice Boltzmann* and *Smooth Particle Hydrodynamics*, their use within sports is still rare. For this reason, herein we will refer predominantly to the Finite Volume method.

The application of *Finite Volume* CFD requires the creation of a discrete finite grid around or within the object of interest (Figure 3.2). This provides a structural framework for the solution of

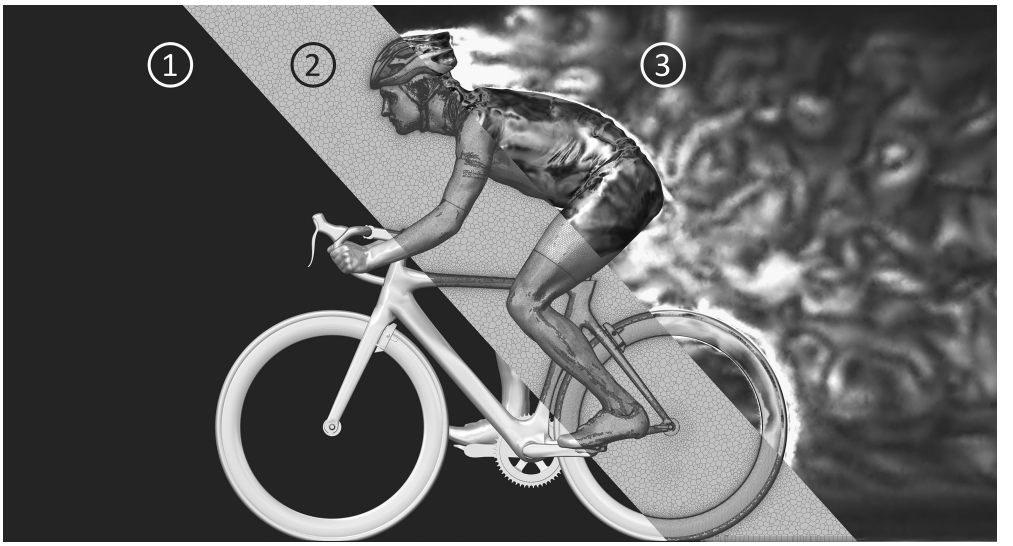

Figure 3.2 The process of CFD: (1) acquire a geometry; (2) create a discrete finite grid around that geometry, specify models and conduct a simulation; (3) analyse results. (Author's own.)

the governing Navier-Stokes (N-S) equations, and additional equations such as turbulence models, at all nodal points within the grid. Values of velocity, pressure, and other variables such as temperature and turbulence are obtained across the grid, building a complex description of the flow field around the object of interest spatially and temporally. This allows flow phenomena and details not readily visible through traditional experiments to be visualised and studied, and the potential performance of equipment to be understood before a physical representation is constructed.

The detail in understanding that can be obtained is directly limited by the spatial resolution of the grids and the temporal resolution of the simulation. However, it is also dependent on the approximations and assumptions within applied models and decisions taken by the user. For this reason, it is important that the user has a solid understanding of fluid behaviour, as well as the limitations and capabilities of the models.

The issue with turbulence

One such area where a user requires a firm understanding of flow is turbulence. Turbulent flow is a mass of instantaneous fluctuations that vary in scale both in space and time. Although the Navier-Stokes equations that *Finite Volume* codes solve provide a description of fluid motion, it is generally accepted that solving all instantaneous fluctuations for all but the simplest of flows is unfeasible. This is due to the wide range of scales typically encountered in real-world applications. The largest fluctuations or eddies in a flow, at an integral scale, have a characteristic size typically as large as the object they pass around or through. These large-scale eddies derive energy from the mean fluid flow and from each other and are subject to vortex stretching. The mechanism of vortex stretching passes turbulent kinetic energy to smaller and smaller eddy scales in an energy cascade until a molecular scale is reached. At these micro-scales, turbulent energy dissipates as heat due to the dominance of viscous effects. Whilst at the smallest scales turbulent eddies can be considered isotropic, the larger integral scales are anisotropic and highly three-dimensional in nature.

To resolve all these scales, with a *Direct Numerical Simulation* (DNS) of the Navier-Stokes equations, would require an extremely fine computational grid and significant computing resources. For sports applications this is not feasible, but that is not to say impossible. Researchers at Arizona State University used a DNS approach to solve the flow around a golf ball [12]. This required a computational mesh of 1.2 billion elements to capture the spatial behaviour and significant computing resources and solution time to capture the time-dependent behaviours.

Billion-element simulations are not unheard of in sports applications, though they are still rare and typically still reserved for Formula 1. However, a researcher, in conjunction with a commercial software provider, used a 3-billion-element simulation to solve the flow of an entire cycling peloton containing 121 cyclists [13]. Consider the size of the simulation that would be necessary to resolve all scales of turbulence in such an application! To make this size of simulation feasible, approximations had to be made. This not only included understandable geometrical simplifications of the cyclist and bicycle but also turbulence was modelled with an averaging approach rather than direct solution.

Modelling turbulence

In sports applications, just as in all other CFD applications, we typically rely on one of three approaches to solving turbulence: RANS (Reynolds Averaged Navier-Stokes), scale-resolving LES (Large Eddy Simulation), or a hybrid approach (a combination of RANS & LES), Figure 3.3 [14].

The simplest approach to resolving turbulence is RANS. This uses a time-averaged form of the Navier-Stokes (N-S) equations as originally proposed by Osborne Reynolds. RANS equations simply average out all but the largest of the eddy structures in a flow with an isotropic turbulence approximation. Time averaging, however, creates additional unknown terms in the N-S equations that need to be solved directly or, more commonly, are dealt with through the application of the Boussinesq eddy-viscosity hypothesis. The Boussinesq eddy-viscosity hypothesis draws an analogy between the Reynolds stresses and the viscous stresses in the flow, using a turbulent viscosity term. RANS turbulence models are then used to obtain a value for this term. These turbulence models can be solved in a manner that assumes flow behaviour is steady-state, a considerable assumption for most sports applications, or in a time-dependent unsteady manner. Unsteady-RANS models, however, simply resolve large-scale single mode vortex structures only, Figure 3.4 [15].

The simplifications of RANS mean the models typically struggle to predict large flow separations, in particular those that may be attributable to an adverse pressure gradient, such as may be experienced in flow around a ball. Additionally, these models do not typically deal with transitional flows; however, advances in this regard have been made in the last decade. The typical trajectories of most sports balls occur at a velocity where the flow in the near surface of the ball transitions from laminar to turbulent behaviour, significantly affecting the aerodynamic forces

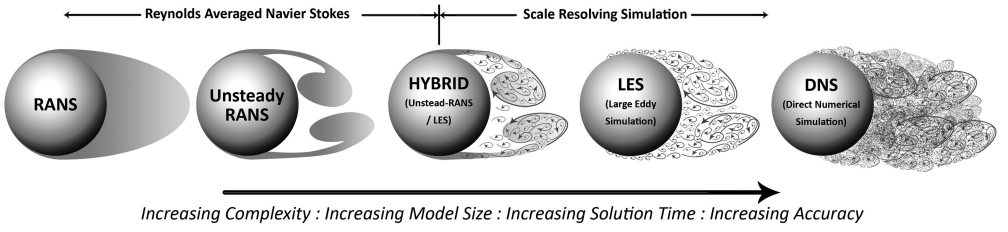

Figure 3.3 Example of turbulent spatial scales resolved by turbulence modelling approach [14].

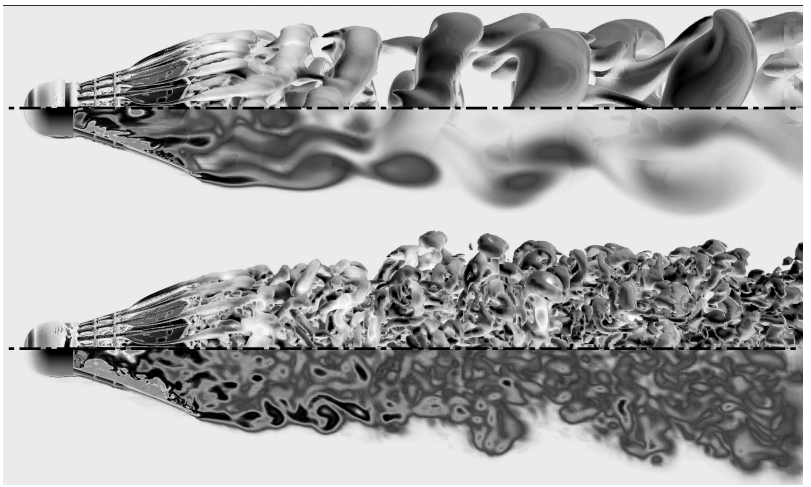

Figure 3.4 Simulation of a feather shuttle revealing in a cut away view the vortex scales resolved internal (below centre line) and external (above centre line) to the shuttle by (a) unsteady-RANS and (b) scale resolving simulation approached [15].

acting upon the body and thus its observed behaviour. As such, RANS is unsuitable for simulating sports balls. This is unfortunate for the analysis of sports balls as RANS solutions can be used to obtain rapid solutions.

At the other end of the feasible spectrum of turbulence modelling approaches is LES. This scale-resolving simulation approach resolves the largest eddy scales directly using the Navier-Stokes equations (as in DNS); however, the smaller scales are filtered and representatively modelled. The point at which filtering commences is dependent on the minimum computational grid element size. It is not possible to resolve an eddy that is smaller in diameter than the grid element in which it resides. LES can cope with transitional flows, and this makes its use in sport attractive. However, large computational grids are required to capture the wide range of turbulent scales. Significant computational runtimes are needed to allow the formation and capture of the time-dependent turbulent behaviours in the simulation. The solution of LES in sports applications is therefore still typically the domain of academia due to the computational power required and the considerable solution times that are often not economically viable for industry.

Currently, the most promising and viable solution for modelling turbulent flows in sports applications are the hybrid family of models that combine LES and RANS. Hybrid models capture the time-dependent behavior of turbulent flows, resolving the largest scales of turbulence with LES, whilst in regions near to a surface, they switch to an unsteady-RANS model. By modelling the smallest scales with an unsteady-RANS approach, considerable computational savings can be made, as resolving these fine structures with LES would require a considerable number of computational grids. An example of this can be seen in the hybrid simulation of an association football in Figure 3.5. The computational grids required for hybrid solutions are more economical and efficient; however, their time-dependent nature can still result in long solution times, and some flow behaviours are still averaged rather than solved.

Ultimately, the choice of turbulence model should be governed by the physics of the problem to be solved and the level of accuracy required in that solution. However, further considerations must be made including the computational resources available and the time-critical nature of the

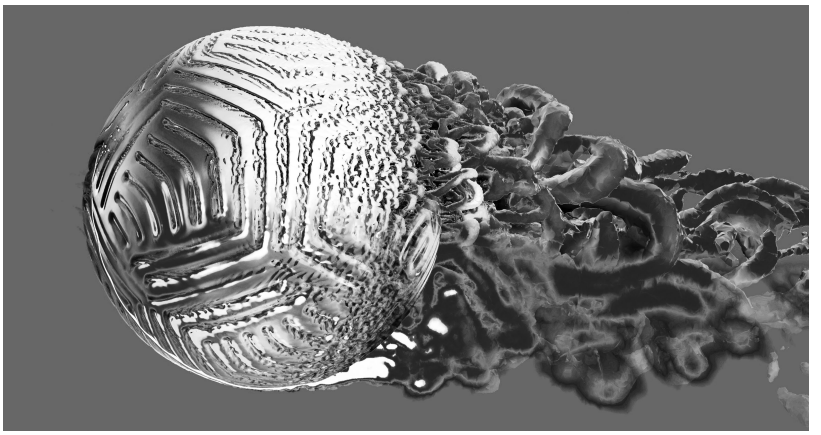

Figure 3.5 A hybrid simulation of a Nike AerowSculpt Flight association football, which models shear flow structure near to the surface of the ball with an unsteady-RANS approximation that averages and essentially smears small-scale structures, away from the surface large-scale vortical wake structures are resolved with LES. (Author's own)

results. A "research" environment may be less time-critical than an industrial application. If time and resources are not available a compromise has to be made. This could include the use of a simplified turbulence model or the use of a non-optimal coarse-sized computational grid. However, even a computational grid that would be considered coarse for LES or hybrid models can often produce reasonable results in comparison to a pure RANS model.

Simplification in simulation need not be a bad thing. Indeed, all simulations should be simple to begin with, and as more is learnt about the problem, the complexity can be increased. From a turbulence standpoint, moving a simulation from unsteady-RANS to hybrid to LES not only can reduce solution time, but if an acceptable level of answer can be found with a simplified model, computational savings can be considerable. Simulations that can be simplified to produce rapid solutions additionally facilitate the use of multiple simulations for the optimisation of a design. Regardless of how a solution is "optimised", the acceptability and credibility of the results should always be carefully considered.

Ensuring credibility

We consider a CFD simulation as being acceptable, or credible, if it reproduces reality accurately [16]. The point of conducting a CFD simulation is to attempt to describe or predict a fluid dynamic reality. However, simulation by its definition is not reality; it is a model or imitation of it. Commercial providers and open-source code developers are continuously increasing the accessibility and ease of use of this tool. Rather than just being a standalone tool, CFD is included concurrently in many CAD packages and is now even being developed as an interactive design tool with the promise of real-time results. The days are gone when users needed ahigher degree to be able to use and apply the codes, which, on one hand, may be no bad thing, but is also perhaps where issues can develop.

Any person with minimal training can conduct a CFD simulation, within reason. Within a matter of a couple of clicks, users can now import complex geometries, benefit from automated grid creation, be guided through the setup of the simulation, and rapidly obtain suggested results. This opens huge possibilities to rapidly understand design, flow situations, and embed CFD in

scenarios previously viewed as too time-consuming. But this tool and its application contain inherent approximations and therefore uncertainties [17]. It is the responsibility of the engineer using the tool to ensure that simulations are credible, and this requires both skill and wider knowledge of fluid flow.

The engineer must have the ability to interpret the obtained results, understand the capabilities of the tool, and therefore be able to identify occasions where results may look plausible but are misleading. This is only possible through possessing a good understanding of both fluid theory and CFD methodology. Software providers have advanced the numerical accuracy of codes, but users should still always verify and validate their results to ensure that a solution provides a credible level of accuracy. It is also why the technique does not sit in isolation as a method for the complete explanation of flow.

Where does CFD sit?

CFD is not the complete answer to understanding sports fluid dynamics. This may, at first appearance, be a bizarre statement from an engineer who has made a career out of using CFD to optimise and understand sports equipment. However, the preceding sections have hopefully provided some indication of why this statement is true. It is, however, a common misconceptionamong many engineering students, who often use simulation without considering the results they obtain.

Instead, CFD is a powerful tool that sits alongside experimental fluid dynamics and analytical fluid dynamics, both of which have a considerable role to play in understanding sports fluid dynamics. The techniques are complementary to each other, and all are required to build a comprehensive understanding of fluid behaviour, ensuring both validity and credibility of analysis. Detailed reviews of research conducted on sports projectiles [18] and cycling [19] have been published, demonstrating how all these techniques are used in combination to build understanding.

It is important to consider which approach is most suitable for the analysis of a problem, considering capabilities and complexity. There is no point in conducting and setting up a CFD simulation where an answer of acceptable accuracy can be obtained using simple analytical equations. However, simple analytical equations are limited in that they can become cumbersome when applied to the analysis of flow in more than one dimension. Experimental techniques are still, perhaps rightly, viewed as the "gold standard" for understanding fluid dynamics. Controlled laboratory experimental fluid studies, utilising wind tunnels and flumes, can provide rich data but can be limited because of the artificial constraining flow boundaries they impose and are complex to set up. However, just as developments in computing have enhanced CFD capabilities, technological developments in sensors and flow visualisation have enhanced the possibilities of experimental studies in sports beyond the constraints of the traditional laboratory. The possibilities of enhanced experimental understanding in real-world settings [20], as discussed in Chapter 2, Projectile in Sport, can create rich validation data sets for a sports CFD engineer.

It is important that a CFD engineer utilises all the validation data available to them, be it the simple analytical calculation of stagnation pressure, force histories, or complex flow visualisation. All can prove invaluable in building credible simulations.

The challenge of using CFD in sports application

The application of CFD in sport can pose a considerable challenge to the user if credible simulation results are to be obtained. This is due to the combination of complex turbulent flows and complex geometric forms that, in turn, control the turbulent flow within sports applications.

Complex turbulent flows

It may be surprising, but perhaps the most difficult geometry to simulate in CFD is a sphere. This simple geometrical form is widespread in sport. Balls are visually differentiated between sports by their specific scale and surface texture. It is the combination of this scale and texture that can create characteristic and notable aerodynamic behaviours at a specific velocity range. This is seen in the golf ball dimples that increase distance, the patterning of seams that can dictate the free-kick swerve of an association football, and the swing delivery of a cricket ball polished on one side and scuffed on the other [21]. The combination of scale, texture, and speed dictates the turbulent flow behaviours around a sphere that in turn govern the forces experienced, which can affect movement. As ball sports are not played at a constant speed, the turbulent behaviours and thus the forces acting on the ball can alter depending on the Reynolds number of the flow, resulting in interesting behaviours [18]. A classic example is the movement seen with the 2010 World Cup football, which, due to its particularly smooth form, at the right speed would move unpredictably in flight. Similar behaviours can be seen in a float volleyball serve where the ball can shift unpredictably at lower speeds, in comparison to a high-velocity spike shot. An example of how the wake of a volleyball varies with velocity is shown in Figure 3.6.

Within CFD, the accurate prediction of transitional flow behaviours is challenging. It is not only difficult to accurately predict the location and strength of transitional flow but also the point at which this will occur. This is why the simulation of a sphere is perhaps the most challenging of problems. With such a simple geometrical form there is nowhere to hide—the simulated physics are either right or wrong. Capturing a flow separation both spatially and temporally, recreating any separation bubble and any subsequent reattachment, is dependent not only on the turbulence modelling but also on the quality of the user input flow conditions and decisions.

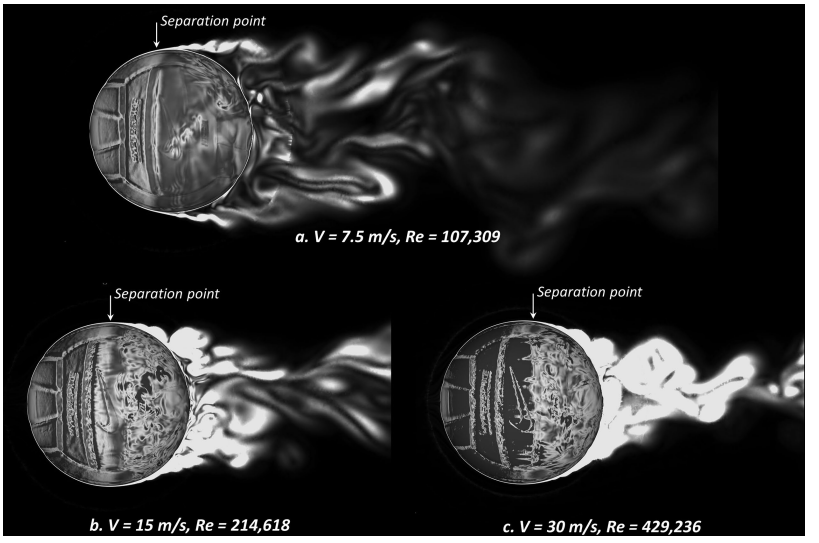

Figure 3.6 A CFD simulation of a Nike Hyperspike Volleyball revealing how the flow patterns around the ball and the skin friction values alter as the flight speed of the ball increases. At lower velocity, flow separates early resulting in a large wake and high drag. With increasing velocity, flow remains attached for longer, the size of the wake decreases and the drag coefficient reduces. (Author's own.)

However, the problems of accurately predicting complex flow behaviours using CFD are not specific to spherical geometries but are common across forms generally categorised as bluff bodies. In sport, there are many such forms, for example, rugby and American footballs (prolate spheroids), golf drivers (flat-faced rounded forms), shuttlecocks (conical), and human anatomy, which is essentially a collection of cylindrical and spherical forms.

Complex forms

A bluff body alone is not a "complex" form; they are often described geometrically by simple equations. Adding surface texture complicates the simulation process. The behaviour of flow around an object can be controlled through the application of texture to that surface. Textures can delay, accelerate, and precisely control the location of flow separation. These textures have been seen to take many forms in sport, ranging from dimples, pimples, chevrons, trip lines, and assorted vortex generators. They are found not just on sports balls but have also featured on golf drivers such as the PING G30, as well as cycling-related equipment and clothing [22]. The textures are also becoming increasingly complicated in form as manufacturing techniques and manufacturers become more adventurous in their application. In recent years, we have seen the introduction of sculpted grooved textures by Nike (Beaverton, USA) on association footballs and raised hexagonal honeycomb textures on Molten volleyballs and even lacrosse balls (Figure 3.7).

Seams in sports balls are another form of texture that directly influences flow behaviour. This can be clearly seen in the evolution of the association football, where the move from 32-panelled stitched balls to thermally bonded moulded designs with as few as eight panels resulted in exhibited erratic aerodynamic behaviours, such as the aforementioned 2010 World Cup football [21,23].

The influence that these textures have on surface flow cannot be readily replicated within CFD through the application of a roughness coefficient. These textures can contain and create distinct

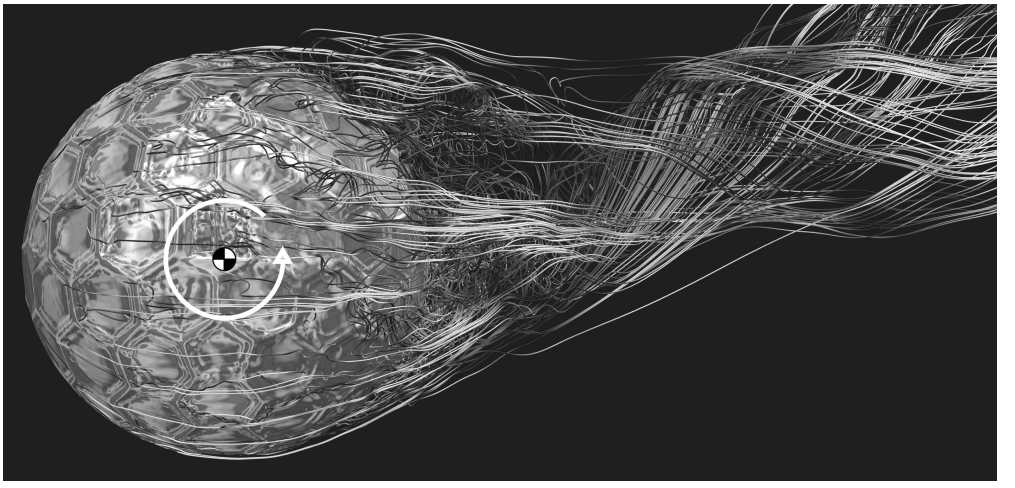

Figure 3.7 Flowlines around a Wolf Athletics lacrosse ball reveal an asymmetric wake behaviour due to top spin being applied to the ball. Wolf has applied a stepped hexagonal surface texture to control the flow physics around the ball that influence its flight behaviour. (Author's own.)

anisotropic turbulent structures. A classic example is the simulation of a golf ball, where it has been shown that dimples have their own unique flow structures, separations, and attachments residing within them [24]. It is these individual structures that need to be resolved in both time and space to replicate the whole. To accomplish this demands large simulations of considerable solution time dependent on the computing power available.

Inclusion of the human anatomy within a simulation could also be considered the addition of a complex form. Although the human form can be simply considered a collection of cylinders and prolate spheroids, human anatomy is organic, with varying curvatures and characteristic scales of length. Limb-specific Reynolds numbers may result in flow behaviours that vary depending on body location.

As was mentioned at the beginning of this discussion, the appearance of generic human form in simulation is decreasing. The ability to capture real-world anatomical form to a high level of accuracy is becoming increasingly accessible, allowing simulations of specific individuals to be conducted. This is important, as beyond the obvious differences that exist between genders, it may sound obvious, but not everyone is the same size and shape! As a result, an item of equipment that may be aerodynamically optimal for one athlete may not perform as well for another. This is sometimes evident, for example, within elite track cycling teams that have optimised their aerodynamic solutions, resulting in the use of a range of helmets. The choice of these may not only be dependent on a specific event but also on what is optimal for the individual's shape, size, and resulting posture.

Finally, the combined simulation of human anatomy with sports equipment can further complicate the solution process. Consider, for example, the cyclist shown in Figure 3.8. The cyclist's anatomy contains a range of length scales that is then further increased by the length scales within the bicycle, considering the cross-sectional forms of the frame, wheels, etc. The aerodynamic interactions between the geometry of the cyclist and the bicycle also need careful consideration, as these will, in turn, influence the flow behaviours and structures that may develop.

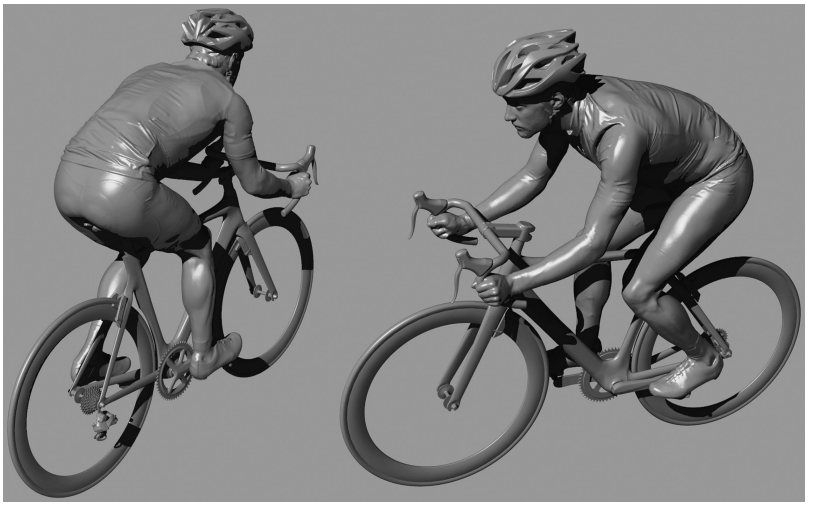

Figure 3.8 Geometry of a cyclist and road bicycle captured using photogrammetric scanning. The geometry contains a range of characteristic scales of length that can result in different turbulent behaviour dependant on location. Interactions between these localised flows further complicate solution, for example, flow around the legs interacting with the frame components. (Author's own.)

However, despite the aforementioned challenges, issues, and considerations that must be made, the application of CFD within sport and wider applications continues to grow. For the simple reason that this powerful technique allows the user to build a detailed, insightful understanding of flow physics that may not be visible by other experimental means, accelerating equipment development and taking it in exciting new directions.

Summary

The use of CFD is today embedded within commercial sports product design, elite sport technical programs, and academic sports research. Just as the application of the technique grows, new codes and numerical models are continuously being developed and optimised, both commercially and as open source. This is with the aim of increasing the credibility of results, simplifying the ease of use, and increasing accessibility of the technology. Although finite volume CFD is still perhaps the dominant method, the range of codes in use, such as Lattice Boltzmann, is increasing. Automated and advanced grid generation technologies and parametric simulation capabilities are now standard. Generic geometries are being replaced with simulations of specific athletes due to advances in geometric capture, such as accessible photogrammetric solutions. Turbulence modelling approaches continue to develop apace with improvements in transitional modelling capabilities.

These advancements have only been possible with improvements in computational hardware, and the accessibility of high-performance computing has created opportunities for increased size and complexity of the computational analysis being created. With cloud computing and the continued evolution of computational hardware, this trend will continue. We have intentionally not touched on the rise and challenges of artificial intelligence (AI) within this chapter. This is a rapidly evolving field, and its application to CFD is still emerging.

With all these increased capabilities, accessibility, and ease of use, we should not fall into the trap of necessarily believing everything we see. CFD is still just a powerful tool, with both distinct advantages and limitations. It does not sit in isolation from experimentation or analytical methods. Indeed, the user should seek to use a range of approaches to verify and validate simulations to ensure credibility.

The application of the technique continues to grow apace. In the 30 years since the application of CFD first started to appear within sport, the extent of that application has grown significantly. This is for the simple reason that the powerful insight and associated benefits to product development and understanding the technique delivers. In the second part of this two-part chapter, we will look at how this application has evolved and is growing.

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4

COMPUTATIONAL FLUID DYNAMICS APPLIED IN SPORT

Part 2 – Application in sport

John Hart

Introduction

In *Part 1 – Capabilities and Considerations* of this review of *Computational Fluid Dynamics Applied in Sport,* we looked at how CFD codes have developed in the preceding 30 years since the use of the technique started to become established in understanding sport. Capabilities have increased significantly; however, application within a sports context can present significant challenges due to the complex flow physics and complex geometric forms typically encountered in sport. Indeed, if you have read *Part 1*, you may now actually be asking, "Why do we bother with applying CFD in Sport?"

CFD is no different from any other method used in understanding sports fluid dynamics; all have their limitations and benefits. However, the limitations within CFD are outweighed by the distinct advantages and what we can learn when careful consideration is given to its application. For example, CFD can identify, isolate, and optimise the aerodynamic performance of a small component on a complete bicycle frame. By using wind tunnels, large aerodynamic gains may be quickly identified; however, as the gains reduce, these can become increasingly difficult to measure and identify. Equally, it can be difficult to understand the aerodynamic interactions of components on the bicycle and the rider using only experimental methods.

CFD can be used to recreate a wide range of environmental conditions, beyond the influence of altitude or temperature. The effects of these can be modelled analytically to predict sports product behaviour, for example trajectory modelling of projectiles. Readers seeking further information on sports projectiles are referred to Chapter 2. CFD goes further; for example, it can be used to recreate meteorological phenomena to discover the influence of a crosswind on a road race or a sea breeze on a sailing course. It is possible to include infrastructure within an analysis. In the winter sports of skeleton and bobsled, the sled sits within a track. Through CFD, it is possible to model the influence that a specific cross-section of track will have on the sled and slider's aerodynamic performance, Figure 4.1. This is not limited to straight tracks; it is possible to simulate the aerodynamics of a curved trajectory of a sled through a bend or a cyclist cornering in a velodrome.

It is for the preceding reasons that CFD proves to be a powerful tool in sports application and why it has been used in both commercial sports product development and in support of successful Olympic medal campaigns. It is important to note at this point that an athlete's sporting success

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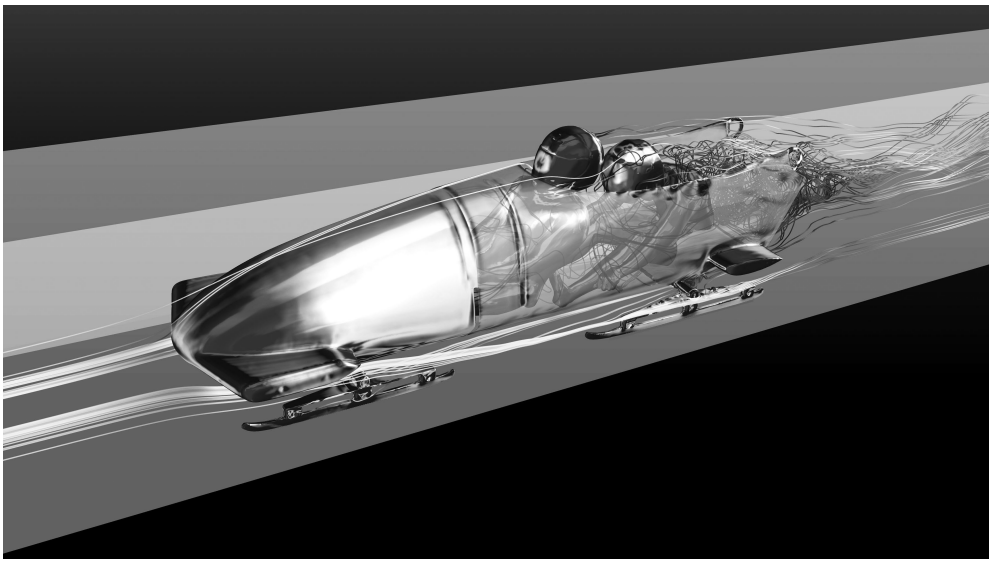

Figure 4.1 Simulation of a two-woman bobsled team within a straight track section; the geometries of the sled and crew were captured using non-contact scanning. The cutaway reveals air movement into the sled body across and around the brake woman. (Author's own.)

can never be declared to be the sole result of the application of CFD. Although CFD can help an athlete or team find the extra tenth or hundredth of a second that may contribute to a medal success, it is always used as part of a wider program of support that underpins an athlete's campaign. Any aerodynamic benefits that can be identified will only ever be effective if every other part of that support program is effective, allowing them to perform to their full extent. Another point of note is that there is/has been more research conducted using CFD in support of elite athletes or commercial ventures than is evident in scientific publication. This is for obvious reasons; whereas academics seek to further knowledge and are expected to publish, commercial companies and elite sports are under no such obligation—information is valuable. In the following section, we will look at some examples of how CFD has been used in support of sport. This is by no means an exhaustive list, and it will not include the classic affluent sports of motor racing or sailing, such as America's Cup yachting. This will be split into two categories: *bluff body sports equipment* and *including the athlete*.

Bluff body sports equipment

In the application of CFD to the analysis of sports equipment, spheres (i.e., sports balls) are perhaps the most simulated bluff geometric form. Of these, the golf ball [1–6] is most common, followed by association footballs [7,8]. Spheres have also received significant interest both experimentally and using CFD outside of sports applications. Experimental studies have sought to visualise and understand the complex vortex structures that evolve with increasing Reynolds number [9,10], alongside studies that present information regarding the transitional drag curve behaviours [11]. It is these complex flow physics that provide an important benchmark for understanding the capabilities of CFD models designed to predict those behaviours [12]. These rich sources of data are valuable for comparison and validation of results from sphere simulations.

Golf is a sport that prides itself on the technology both within and used in the development of its equipment, albeit under the control of the USGA (United States Golf Association) and The R&A (The Royal and Ancient Golf Club). Substantial research has been undertaken into understanding golf balls [13] and developing the design of dimples, both in pattern and form. A key aspect of designing dimples is understanding the flow mechanisms that reside within these structures, and experimental work has gone a long way to revealing this [14]. However, such experimental study is challenging, and therefore researchers have attempted to build further knowledge using CFD. Early CFD studies of golf balls attempted to understand dimple size, depth, and even shape [1,15,16]. Although they used Large Eddy Simulation (LES) modelling of the turbulence structures, the grids were coarse and low in resolution, indicative of computational capabilities at that time. These early studies also only considered flow well within the supercritical fully turbulent region, avoiding the critical flow regime where flow separation moves from laminar to turbulent (Reynolds Number $\geq 4 \times 10^5$ for a smooth sphere) [17]. Indeed, even the first Direct Numerical Simulation (DNS) CFD simulations of a golf ball only considered flow that was either subcritical or supercritical [6]. These simulations were ground-breaking in CFD, using 1.2 billion computational elements for a single item of sporting equipment, outside of what would be considered affluent sport. The effort in understanding the flow mechanisms is placed into context when you consider this level of effort is typically reserved to understand an entire race car or a peloton [17] of 121 cyclists! These simulations revealed in detail the shear layer instabilities that form from the leading edge of dimples, the way they separate and reattach, controlling the ball's overall drag characteristics.

Until recently, most researchers avoided simulations around the critical region of flow transition, for reasons discussed in Chapter 3. With advances in computing and turbulence modelling, this has changed. The DNS simulation of sub/supercritical golf balls [6] was extended to include the critical region [2] and went further by rotating the ball. This required significant computational resources, as DNS is computationally demanding, and the ball at supercritical flow conditions required 260,000 CPU hours per rotation! Spinning golf balls are still a significant simulation challenge; however, advances in turbulence modelling and grid technologies have improved solution times [3].

Indeed, some researchers have sought to use alternative approaches to modelling turbulence depending on whether a ball is static or rotating. For example, in the simulation of association footballs [7], DNS was used in the simulation of static balls, whereas the comparably less computationally intensive LES approach was used for rotating balls. This further demonstrates how DNS is still not a practical tool for everyday use by industry or most of academia. It is for this reason that there is still interest in other, more economic modelling options for the simulation of a sphere and other bluff bodies, Figure 4.2 [18], and why commercial CFD companies continue to optimise Reynolds Averaged Navier-Stokes (RANS) models.

More recent research has been conducted using LES to model golf balls, presented in comparison to smooth spheres. The inclusion of a smooth sphere in the simulation, a widely simulated and simple geometric form with known aerodynamic performance and behaviours, gives confidence in the quality of the applied methodology to the golf ball. These simulations have included both static [5] and rotating [19] balls. The simulations only utilised a 150 million element grid, in comparison to the billion-cell DNS approach, and produced reasonable predictions in comparison to wind tunnel experimentation. In fact, all the scale-resolving simulations mentioned above produced reasonable predictions compared to experiments within their limitations, the main difference being the resolution and detail of resolved flow structures.

Conical bluff bodies, such as shuttlecocks, have also been studied extensively. CFD is an effective tool for developing an understanding of the flow through and around a feather shuttlecock.

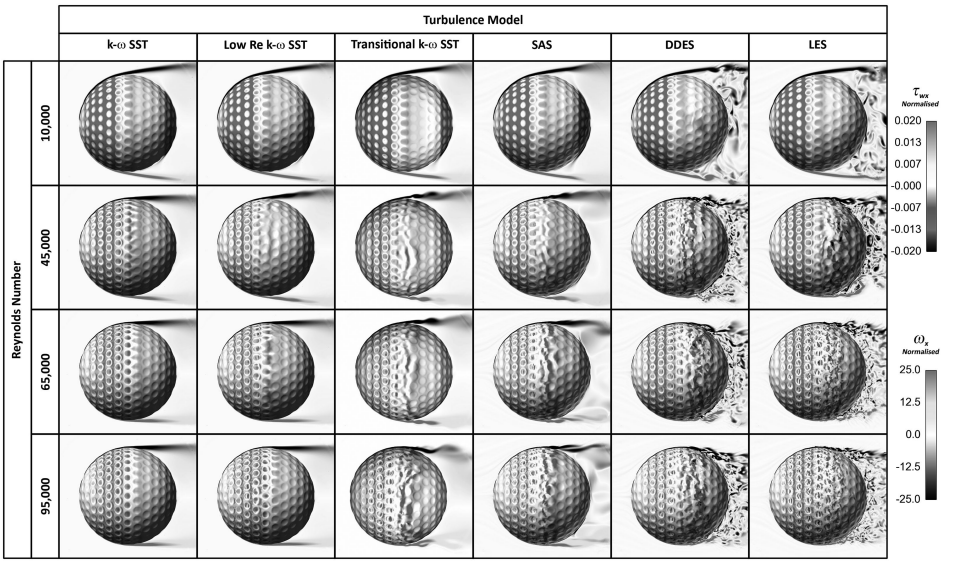

Figure 4.2 Simulation of a golf ball revealing the influence of turbulence model choice on surface shear (τ_{wx}), and flow field spanwise vorticity (ω_x). It can be seen that although pure RANS models struggle to predict large bluff body flow separations, their combined use with Delayed Detached Eddy Simulation (DDES) can produce comparable performance to LES when optimised with consideration [18].

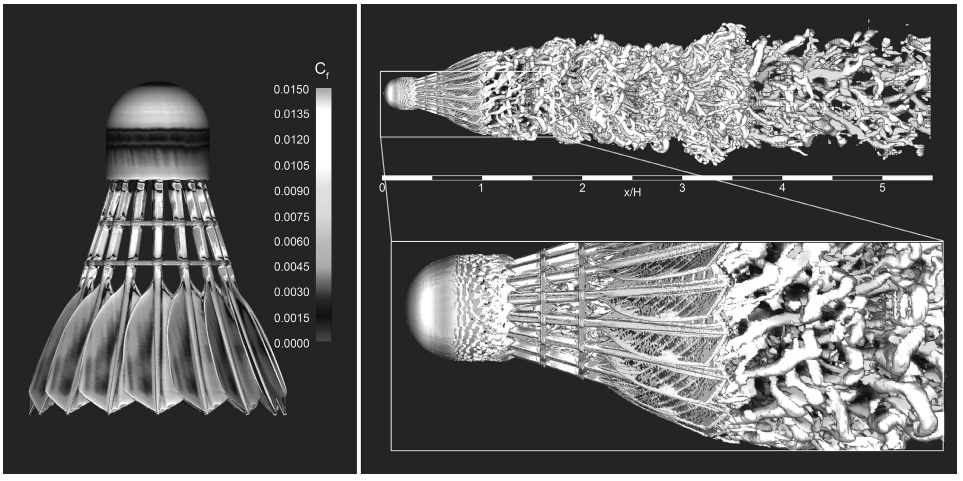

Figure 4.3 Contours of skin friction coefficient of a spinning feather shuttlecock, with visualisation of the vortex structures that form in and around the shuttlecock at Re = 2.7×10^5 [20].

The skirt of a feather shuttlecock is constructed from 16 individual feathers that have their vanes shaped and orientated in an overlapping pattern. Visualising flow experimentally around this structure is difficult, and the experimental wind tunnel equipment required to hold a shuttle in position can potentially compromise the flow structures. With CFD, these structures can be revealed in detail, even whilst the shuttle rotates at high speed [20], Figure 4.3.

The creation of a realistic and accurate geometric model of the shuttlecock for use in CFD is a challenge. Capturing the curvature of a feather vane, its varying thickness, and the complex cross-sectional shape of the feather rachis (feather shaft) requires the use of scanning equipment [21]. Once the geometry has been created, simulation of a shuttle is more straightforward than that of a sphere. Unlike a sphere, it has been shown that an optimised unsteady-RANS simulation of a shuttle can provide comparable results to a scale-resolving simulation [22]. However, it is only through the use of scale-resolving simulations that the full detail of the complex vortex structures created around and through the vanes can be fully revealed.

It is important not to overlook the little details that can have such a dominant influence on aerodynamic performance. Golf drivers are a good example of this, where rules set by the USGA on shape and volume led manufacturers to seek alternative solutions to control undesirable flow behaviour. The USGA stipulates that the width of a club (heel to toe) should be greater than its depth (striking face to back), and, controls the overall volume of the driver head. This combination creates the classic wide and deep bluff body forms associated with this equipment. Coupled with a largely flat leading surface, with which to strike the ball, drivers suffer large-scale flow separations at or around the leading edge of the crown and sole. Manufacturers have sought solutions to control these separations, to create an aerodynamic performance gain leading to increased head speed and ball distance. An example of this was the development of the PING G30 driver.

PING (Arizona, USA) had worked with the Sports Engineering Research Group (SERG) at Sheffield Hallam University for several years investigating the use of CFD, and had implemented the use of CFD within their company [23]. The partnership and knowledge learnt through CFD analysis of golf drivers, and brought from SERG analogous sports research, led to the use of vortex generators (or "*turbulators,*" as PING marketing dubbed them) on the crown of the PING G30 driver [24]. Vortex generators are designed to control or delay the separation of flow around a bluff body, reducing the overall drag experienced, and are common within the aerospace industry. Inclusion and optimisation of these features resulted in a 25% reduction in drag at impact [24], Figure 4.4. The driver sawcommercial success in the US and UK, according to Golf Data-Tech LLC (Kissimmee, USA), who track golf retail sales, and other golf companies subsequently launched their own approach to this issue [25]. The Callaway XR16 utilised a crown ridge trip,

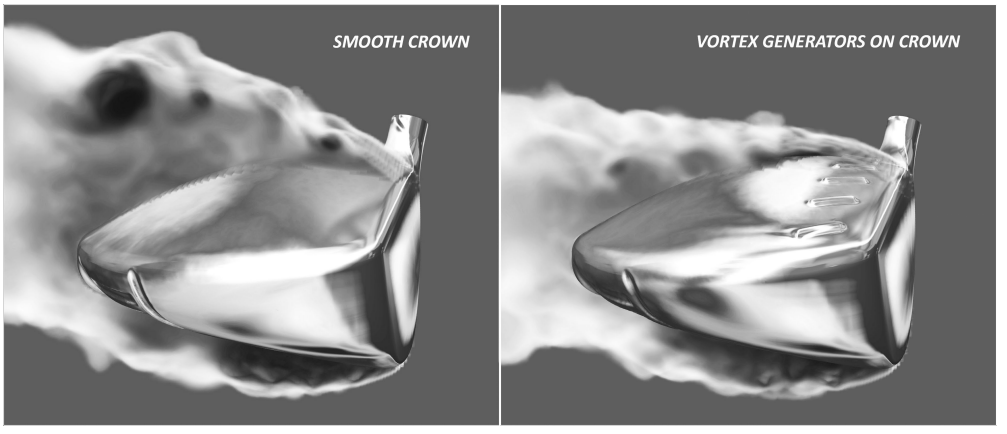

Figure 4.4 Flow analysis around a golf driver with a smooth crown, with and without vortex generators at $Re = 3.29 \times 10^{4.}$ The ability of the vortex generators to reduce the size of the crown wake is clearly visible. (Author's own.)

whilst Taylormade (California, USA) used a raised and textured crown, both designed to help keep flow attached.

The use of vortex generators, textures, and methods of controlling surface-level turbulence are common features of sports apparel. In particular, skin suits used within winter sports such as speed skating [26], [27], and skeleton bobsled, and within cycling both on the track and road [28]. Simulations allow the user to understand the position of probable flow separation and surface frictional coefficients. Thus, CFD can be useful in determining where these features, or specific textures of fabric, should be placed to best control flow behaviour.

The use of vortex-generating features can be controversial within sport. In particular, the opening time trial of the 2017 Tour de France was notable for Team Sky adopting dimple vortex generators on the upper arms of their cycling suits, with four riders placing within the top 10 of the stage. Claims of unfair advantage, a result of alleged time savings up to 25 seconds, were directed at Team Sky. It should be noted they were not the only team to utilise such features at the Tour stage, nor the first in cycling. Using CFD, and predictive analytical modelling using Newton's 2nd Law, it was possible to understand the potential time gains that could be achieved [29].

A series of simplified simulations were conducted of a cylindrical bluff body, of comparable diameter to the upper arm, to determine the drag reductions possible with different designs of features, Figure 4.5. It can be seen in Figure 4.5 how the different surface textures influence the separation location and subsequent width of the trailing wake. The earlier the separation and the wider the wake, the higher the drag value acting upon the cylinder. In the figure, it can be seen how the use of pimples, chevrons, or a trip line creates a later separation and narrower wake than a smooth surface or dimple pattern. Trip lines are the most effective method to control flow separation, but they are particularly sensitive to flow direction. In an external environment, a textured surface of pimples or chevrons provides more consistent behaviours.

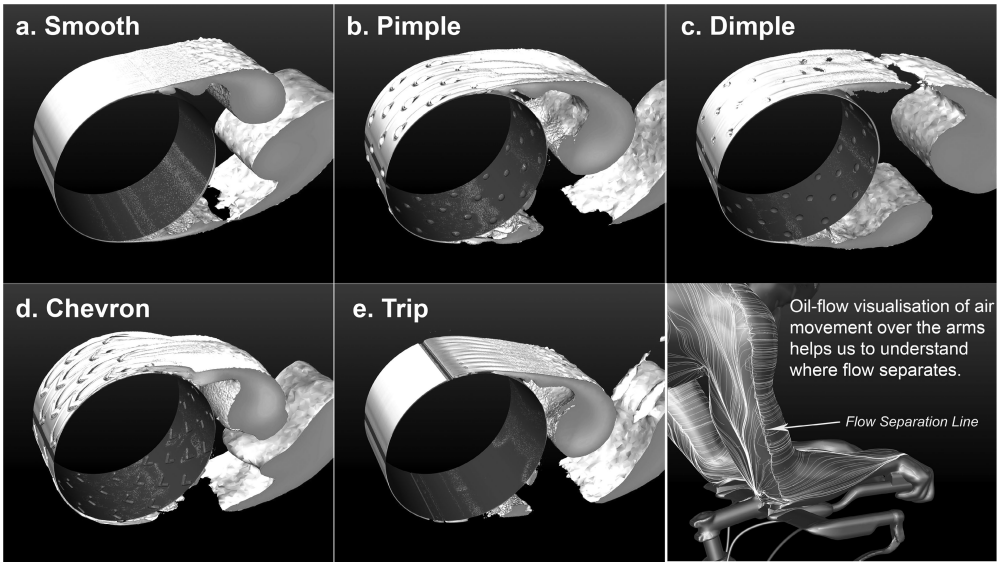

Figure 4.5 Flow analysis of vortex generators and trips around a cylinder. It can be seen that the application of a Chevron texture, or trip line, delays the flow separation, results in a narrower wake, and therefore lowers the drag force acting on the cylinder [29].

The results of the study revealed that the potential time savings were closer to 7 seconds across the stage. So, were the textures worth incorporating? The winner of the stage, Geraint Thomas, finished 5 seconds ahead of second-placed Stefan Kung. Although a bluff body approximation was used to determine potential savings [29], it is always better to conduct such an investigation on a realistic human geometry, so long as time constraints and computing power are not a significant issue.

Including the athlete

Several sports exist where both the equipment and the athlete should be considered in combination. Within cycling, the aerodynamic performance of the bicycle and rider are intrinsically linked; the same is true for winter sport simulations of skeleton bobsled or luge, whereas in swimming, the geometry is the athlete.

Over twenty years ago, when the author first began running CFD sports simulations it was still commonplace to use representative human geometry. The first sports simulation the author was asked to undertake was of a British wheelchair sprinter using a representative humanoid form. The aerodynamics of this particular sprinter were of interest as, due to the athlete's disability, he had to adopt a predominantly upright stance, being unable to tuck forward. This simulation could perhaps be viewed as progressive for the time, whereas research on para-athletes is more common today. However, the geometry was of a manually digitised and highly faceted CAD mannequin. Using representative geometry of this type had limitations. The faceted nature of the surface created unrealistic flow separation and stagnation regions and limited the ability to create a truly tailored aerodynamic optimisation for the participant. Some years later, the author was able to return to this study and use non-contact scanning to capture a more realistic humanoid form, Figure 4.6. This was due to knowledge and skills the author had subsequently acquired while working within cycling.

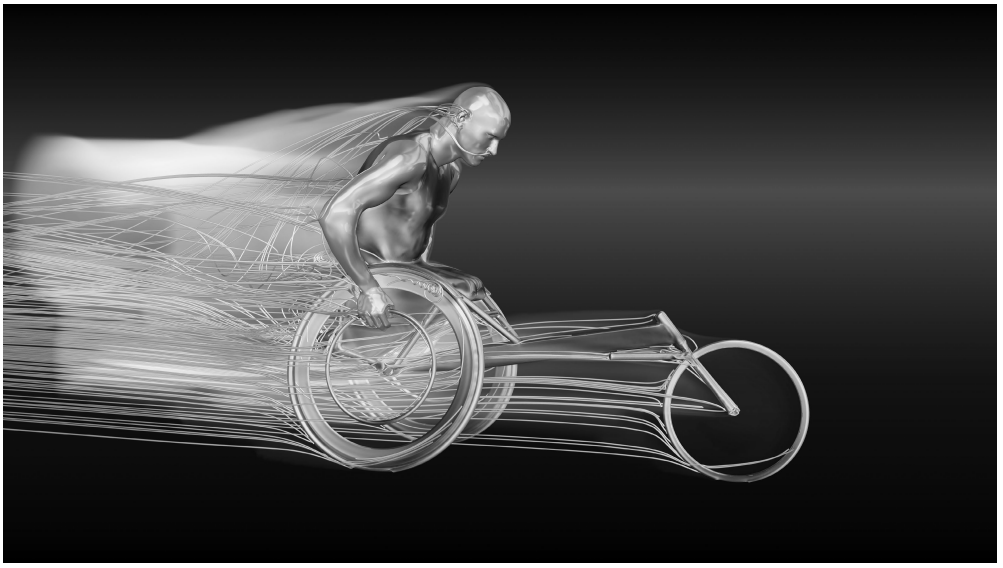

Figure 4.6 Simulation of a para-athlete with an upright body posture. Optimising the aerodynamics of such an athlete is challenging due to the frontal area created by the athlete's stance. (Author's own.)

Cycling

Perhaps it is within cycling that we have seen more simulations with the inclusion of human form than in any other sport. The aerodynamic performance of the bicycle and athlete are explicitly linked; it makes limited sense to simulate one without the other. Cycling is a sport where equipment rules are tightly controlled by its governing body, and it is a technical sport. CFD is ideal, therefore, for developing equipment and finding that extra tenth of a second in performance.

When the author was asked to conduct simulations in support of British Cycling at the Athens 2004 Olympics, it was identified that a more advanced geometric capture ability was required. A "mobile" non-contact laser scanning solution was adopted, allowing the researchers to travel to the athletes. This equipment had a laser scanning head, attached to a tripod-mounted portable metrology arm. Such equipment is still widely used today for precision measurements within the automotive and aerospace industries but was not ideal for capturing human form. The scanning process took over an hour to capture a full cyclist, required breaks for them, and created a subsequent challenge of repositioning them. Other faster scanning solutions were available at that time; however, these were either fixed, could not accommodate a cyclist, or struggled with the line of sight (you can only scan what you can see). To obtain a high-quality human scan in this manner required the development of specific processes and methodologies, which were time-consuming to implement.

Today, handheld photogrammetric and structured light scanners are commonly used for capturing athlete geometry. Lightweight, highly mobile, and with a wide field of view, these allow geometry to be captured in a matter of minutes. They are perhaps no more accurate than the laser scanning solutions; however, their advantage is in the time taken to scan and the automation of the process to manipulate scanned data into a form suitable for use in CFD. This is further enhanced by grid generation technologies within CFD codes that can handle and wrap an imperfect scanned geometry to create a model that will be stable in solution.

Simulations of cycling are the most common sports CFD analysis to contain human geometry. It is where researchers have expended the most effort and where the most notable advances in sports simulation have been evident. This is due to both the high level of technology acceptance within the sport and that cycling is perhaps one of the more affluent sports outside of motor racing.

Full cyclist simulations were being conducted in 2000 by the UK Sports Institute (UKSI), albeit with highly simplified bicycle and cyclist geometry to understand the influence of crosswind on wheels [30]. This is still a current topic of great interest in the development of bicycle wheels. The simplified UKSI geometry was perhaps indicative of both computational capabilities at the time and what was perhaps deemed necessary to produce a simulation of acceptable accuracy.

In fact, our simulations for British Cycling ahead of the 2004 Athens Olympics, which required a detailed analysis of cycle helmet aerodynamics, contained an approximation where only the upper torso and head of the cyclist were considered. This was justified in that the level of accuracy required to build credible understanding of the helmet's performance, with the imposed time and equipment constraints, could not be obtained if a full cyclist was simulated. Four years later, with improved computational capabilities, our simulations for the 2008 Beijing Games considered full and detailed track cyclist models in both pursuit and sprint configurations with multiple static leg positions.

These simulations were conducted using unsteady-RANS turbulence models and validated through wind tunnel testing. By 2010, our commercial cycle simulations were also fully detailed, including small but significant features, such as wheel spokes, brake cables and calipers, gearing and derailleurs, Figure 4.7. These contained grids using thirty-three million elements, but again with an unsteady-RANS turbulence modelling approach. Today, our single cyclist simulations are

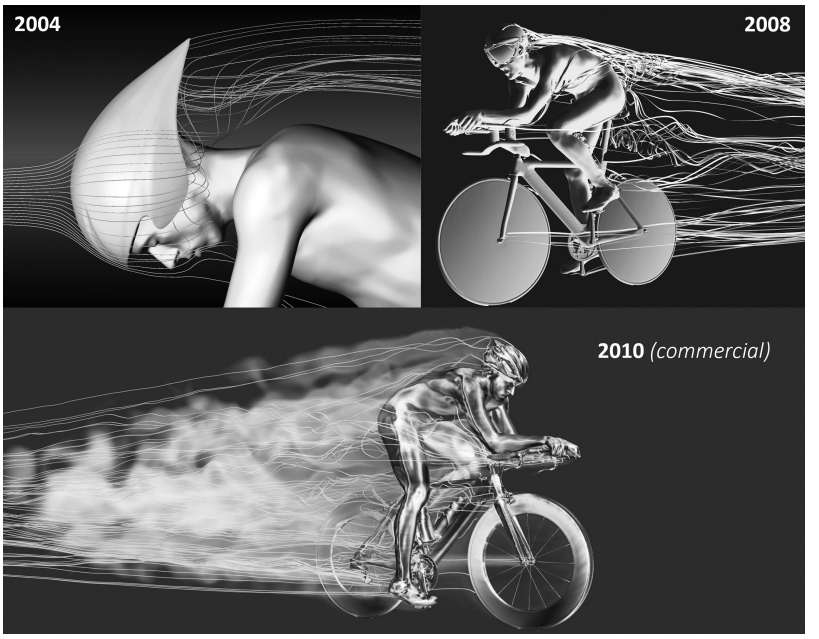

Figure 4.7 Evolution of the author's use of CFD in cycling: 2004 simulations of the upper torso of a cyclist to understand helmet performance was replaced in four years later with full track bicycle analysis. Simulation capabilities rapidly evolved, and full detail high-resolution simulations of cyclists became the norm from 2010 onwards. (Author's own.)

three times the size and use scale-resolving transitional turbulence models. This is possible due to advances in computational codes, models, and the accessibility of computational hardware.

Cycling simulations have moved beyond understanding component and equipment optimisation and are now being used to understand tactical positioning and athlete posture, both static and moving. Simulations have appeared in scientific publications of tactical positioning, ranging from investigations of a pair of drafting cyclists [31], to detailed insights into a cyclist's position within a peloton of 121 riders [17]. The peloton study was validated with an army of quarter-scale prototype riders within the Wind Tunnel Laboratory at the University of Liège in Belgium and remains the largest published cycling study to date, using a 3 billion element grid. This revealed a detailed map of the potential drag reduction that a cyclist could achieve dependent upon their position within the peloton. Other studies have also revealed the influence that nearby motor vehicles, which accompany and film elite cyclists in competition, can have on the aerodynamic forces experienced by the cyclist [32,33].

Other sports

Although we have focused on cycling within this review, human geometry is, and needs to be, included in the analysis of many other sports. Within elite winter sports, CFD has been used to analyse the sliding sports of bobsled [34,35], skeleton bobsled, and luge. Speed skiing [36], speed skating [37], and ski jumping [38] have also featured. Then there are the perhaps even more extreme sports of water ski jumping [39], where ski jumpers hit a launch ramp at 112 kph and

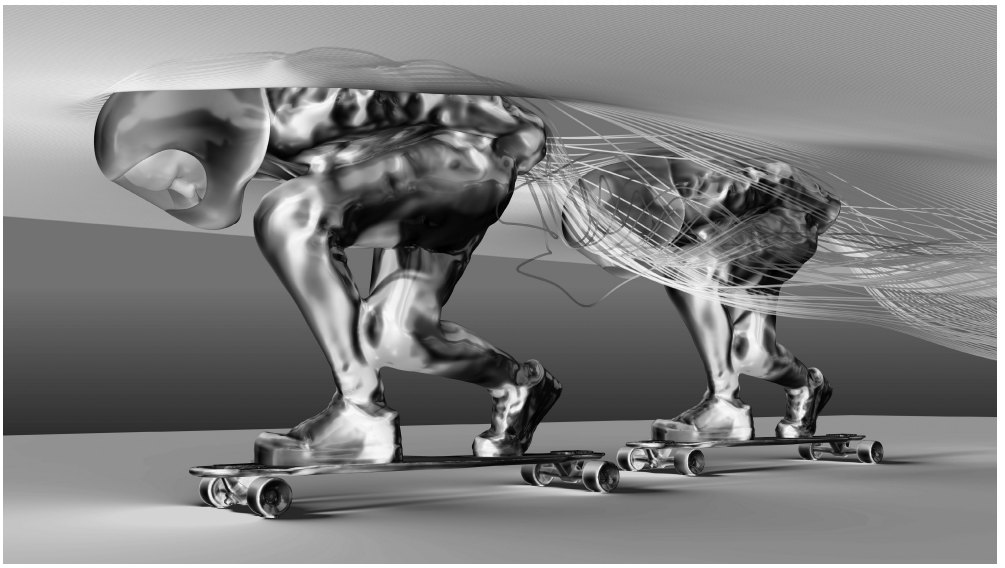

Figure 4.8 Downhill skateboard analysis of drafting skateboarders. The optimum position for the pursuing skateboarder is dictated by the leading leg of the skateboarder in front, as their chosen position results in an asymmetric wake [40].

cover distances in excess of 70 m whilst trying to control and land their jump, and downhill skateboarding. Grab your longboard, a helmet, some motorcycle leathers (advisable!), and head for the nearest mountain with a road to roll down at speeds of up to 90 mph.

In 2009, we were approached by a student who was a competitive downhill skateboarder, developing his own equipment and had an interest in analysing his aerodynamics. We subsequently conducted a study of the then British number one downhill skateboarder using CFD to inform his race tactics. The use of CFD to analyse drafting and position is not just limited to cycling [40]. Through CFD, it was possible to reveal how the optimal drafting position for a pursuing skateboarder could be determined from studying the position of the leading leg of the skateboarder in front, Figure 4.8. The crouched position adopted by the skateboarders twisted their upper torso, resulting in an asymmetric trailing wake. By observing the position of the lead leg, the wake offset, and the optimum position to tuck in behind could be read.

Within swimming, it was the engineer Barry Bixler who initially led the way, studying the propulsive forces generated by a swimmer's hand and forearm [41]. The work showed how CFD could be used to predict significant flow separations, and that simple theory alone could not be used to calculate the forces generated across the stroke. This highlights the importance of not relying on one method to analyse fluid dynamics. Bixler went on to collaborate with Speedo (Nottingham, UK), who, in conjunction with ANSYS (Canonsburg, USA), conducted simulations of swimmers in a glide position to assist in the development of their FastSkin suits [42]. In doing so, they were able to identify zones around the swimmer's torso where the application of specific fabrics or vortex generators would be beneficial. However, swimming is a highly dynamic sport, and inclusion of actual movement within the simulation can enhance understanding further. Dynamic studies are also unsurprisingly of interest to those wishing to understand the pedalling motion of a cyclist.

Movement and manipulation of geometry

Static simulations have been used to understand the influence of the torso position of a cyclist [43] and of leg position during the different stages of a cycling motion [44]. However, a cyclist, as with many athletes, is not static in nature, and dynamic simulations have been conducted that include the pedalling motion of the legs and even the roll of the upper torso that can result. TotalSim Limited (Charwelton, UK), a British commercial CFD consultancy that has worked with elite British sports, released an animation in 2016 of a CFD analysis using open-source codes that captured all these effects. However, the work was not openly published due to its commercial nature. Academic studies have since been published that have detailed dynamic CFD analysis of pedalling motion [45]. The movement relies on the underlying athlete geometry being rigged, creating pivot points at joints that allow the limbs to be repositioned. Due to the way in which the geometry is adjusted, the resultant form is not wholly physically accurate. This is notable around the joints and does not capture changes in muscle shape [46]. In comparison to issues regarding the simulation of the flow physics, this is perhaps not as significant an issue.

Creating a moving human geometry for use in CFD is not a particularly new practice. Researchers have previously used laser-scanned geometry of a swimmer and Autodesk MAYA animation software to create an animated form suitable for simulation of a dolphin kick [47]. This allowed the detailed vortex structures that form from the leg kick to be visualised and understood. However, it must be caveated that animating and simulating a dynamic swimming motion is less complicated than that of a pedalling cyclist interacting with, and physically attached to, their bicycle.

Manipulating geometry of equipment parametrically is also a common technique, varying sizes and curvature to aerodynamically optimise the form. Registration points are distributed across the surface of a geometry, and by manipulating the position of these points, the geometry can be morphed to a desired form. The geometric form is iteratively optimised through a series of geometrically "static" simulations until a threshold performance has been achieved [46].

With the technology that is now available, and the advances in computational codes and models, we can build a detailed geometric model of a specific athlete and their equipment, create a dynamic animation of their movement, and use this within a CFD simulation to optimise and understand their aerodynamic performance in detail. For now, static and parametric analysis are still more common than fully dynamic analysis. A CFD user should always seek to optimise their solution and, as discussed in *Part 1* of this review, should consider what is needed to obtain both acceptable and credible knowledge. Today, full dynamic simulations are still challenging and computationally expensive, but with time, this will no doubt change.

Summary

In the 30 years since CFD first started to appear in human-powered sports and sports equipment development, the extent of its application has grown significantly. It is embedded within elite sport technical programs and in the development of commercial sports products. The potential range of applications within sport continues to grow, in much the same manner as wider engineering applications. CFD provides insight into aerodynamic and hydrodynamic performance, allowing the visualisation of flow phenomena that cannot be viewed by other means. It is possible to easily isolate and study individual components within the athlete-equipment system, further enhancing our understanding of performance and allowing engineers to identify the subtle alterations or improvements that result in an Olympic medal or world-leading sports product.

The size and detail of simulations have increased significantly, which is directly attributable to the advancements that have been made in computational hardware, the accessibility of

high-performance computing, and the advent of cloud computing. Cycling is an obvious example here, where just over 20 years ago, limited single-rider studies with generic geometries were being conducted, whilst today researchers conduct detailed simulations of 121-rider pelotons! It is the same reasons that simulations that are dynamic in nature, such as the analysis of a pedalling cyclist, or a swimmer's kick, have appeared. However, these are still limited in number, but this will surely change in time.

In whatever manner CFD is applied to analyse and understand a sport or item of sports equipment, the user should always consider what they wish to determine, the desired level of accuracy, and their time and computational constraints. It is perhaps for this reason that we still see static and pseudo-static simulations in sport, but with ever-increasing size and resolution. As discussed in *Part 1* of this review turbulence and transitional modelling approaches continue to develop apace, and these will only increase the range of applications to sport as the credibility of such CFD analysis further grows.

It is typical to see at the end of reviews that consider the application of CFD predictions of where the technique may be in a further 20 to 30 years' time. The author intends to make no detailed predictions, apart from saying that CFD will continue to have an important role to play in the design, optimisation, and understanding of sports equipment and competition for the foreseeable future.

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5
SOFT MATERIALS IN SPORT

Akihiro Matsuda

Introduction

Soft materials are materials that can be deformed by stress at room temperature. Soft materials include liquids, polymers, foams, gels, colloids, granular materials, as well as most soft biological materials. On the other hand, hard materials are materials that do not deform easily under stress. Hard materials include steel alloys, ceramics, glass, fiber-reinforced composites, and so on.

Soft materials are applied to many kinds of sports equipment. For running shoes, rubber, polymer fabrics, artificial leather, and foams are used, as shown in the example in Figure 5.1a. Rubber and foam are applied to the outer sole and the midsole. Artificial leather and other synthetic fabrics are applied to the upper part of the shoes. Many parts of baseball gloves are made with natural leather, which is a soft biological material, as shown in the example in Figure 5.1b. Finally, polymer strings and artificial leather are applied to the string bed and grip, respectively, of a tennis racket, as shown in the example in Figure 5.1c. There are many other applications of soft materials in sports.

In this section, some soft materials that are often applied to sports equipment will be introduced. Specifically, the concept and fundamental mechanical behavior of rubber and elastomers are described in the first subsection. The characteristics and manufacturing methods of rubber and elastomers are also described. In the second subsection, the concept and applications of foam materials are discussed, with applications to the midsole of running shoes introduced. In the third subsection, the concept and fundamental mechanical behavior of artificial and natural leather are described. In the fourth subsection, polymer strings used in tennis rackets are introduced. Some constitutive models for soft materials are introduced in the fifth subsection. These constitutive models are useful for designing sports equipment when used in computer-aided engineering (CAE), such as finite element analysis. In the final subsection, some research topics related to soft materials are introduced.

Rubber (elastomer) in sports equipment

Rubber is obtained in the form of latex from the rubber tree. In current usage, the term rubber is not restricted to original natural rubber; rather, it is applied to any material that exhibits mechanical properties similar to those of natural rubber, regardless of its chemical constitution. The modern term elastomer is sometimes employed in relation to synthetic materials with rubber-like mechanical

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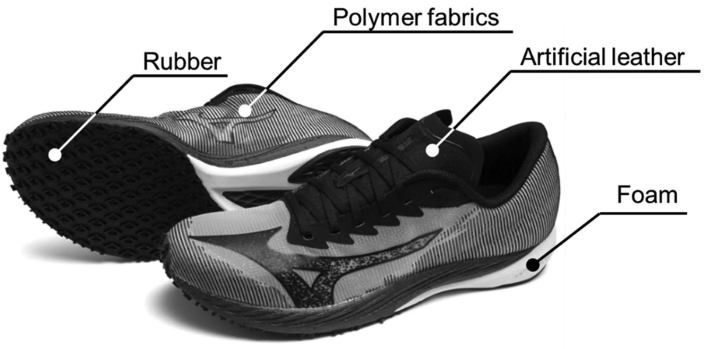
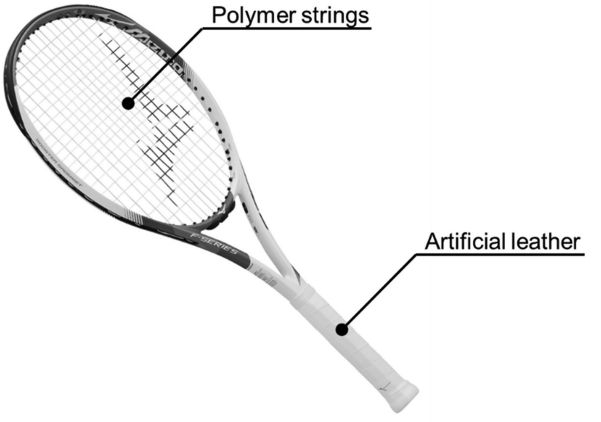


(a) Running shoes (Wave Duel 3 wide by Mizuno Corp.)

(b) Baseball glove (5DNA Technology by Mizuno Corp.)



(c) Tennis racket (F300 by Mizuno Corp.)

Figure 5.1 Soft materials in sports equipment.

properties. Natural rubbers are processed using a technique called vulcanization. Vulcanization is a chemical process during which rubber latex is heated with sulfur, an accelerator, and activation agents. The accelerator of the vulcanization process is defined as a compound that increases the speed of vulcanization. The temperature of the vulcanization is between 140 and 180 degrees Celsius. The process involves the formation of cross-links between long rubber molecules, so vulcanized natural rubber exhibits higher elasticity, viscosity, tensile strength, and water insulation. The extensibility of typical natural rubber normally falls within the range of 500% to 1,000% of simple extension [1].

On the other hand, artificially synthesized rubber-like materials are called synthetic rubbers. While natural rubber is mainly composed of natural rubber latex, synthetic rubber is composed of long synthetic polymer molecules. There are many types of synthetic rubbers due to the varied materials and processing methods. Both natural and synthetic rubber require vulcanization to produce a rubber-like elasticity.

A common application of rubber is in industrial products such as car tires and rubber tubes. Rubbers are also common in the sports industry, where their most important mechanical behaviors are deformability without great stress and strong friction against other surfaces. Bicycle tires and the outsoles of gymnastic shoes, for example, are important applications of rubber in sports. Tennis balls are also a noteworthy application of soft materials: a tennis ball consists of a rubber core wrapped in felt.

The term elastomer is applied to materials that exhibit rubber-like mechanical behaviors with or without the vulcanization process. Some elastomers are processed without the vulcanization process to make cross-links between long polymer molecules. One such class of elastomers typically applied in the sports industry is thermoplastic elastomers (TPE). Thermoplastic elastomers are copolymers or physical mixes of some polymers that deliver thermoplastic and elastomeric behaviors without cross-links. Injection moulding is applicable to the manufacturing process of thermoplastic elastomers.

Both natural and synthetic rubbers exhibit higher extensibility and greater tensile strength than general thermoplastic elastomers, but the latter are suitable for application to sports equipment because they do not have the rubber odor that results from the vulcanization agents. In sports, thermoplastic elastomers are applied to items such as wristwatch straps, swimming goggles, golf grips, shoe soles, and so on.

Foam materials in sports equipment

Foam materials can be made from almost any material: plastics, metals, ceramics, and even composites. The size and shape of a material's microstructural cells determine the mechanical properties of the foam. An important feature of foam is its relative density, which is the density of the foam divided by the density of the solid from which the cell walls are made [2].

Running shoes are well known as an application for polymer foam. For example, foam materials that are applied to the midsole of sports shoes require highly specific stiffness and thermal/water insulation. Polymer foams have been widely used in various applications, such as the midsole of running shoes, protector pads for contact and action sports, bar tape for bicycles, and so on.

Foams are lightweight, soft materials made by foaming the base polymer with a catalyst blowing agent and foam stabilizer. Polyethylene (PE) and polyurethane (PU) are generally applied to a matrix polymer for sports equipment. Both types of foam have three-dimensional, microscopic structures and are classified as either open-cell foams or closed-cell foams. Open-cell foams have better water drainage, and closed-cell foams exhibit water insulation. Generally, the closed-cell foams are stiffer than open-cell foams, due to the air trapped within the cells providing extra support. Ethylene-vinyl acetate (EVA) foams are usually applied to midsoles of running shoes because

polyurethane foam hydrolyzes. The hydrolysis of polyurethane is a chemical reaction with water that chemically and structurally breaks down the urethane linkage. Variations of catalyst blowing agent and foam stabilizer have been studied to produce lighter and stiffer EVA foams. Recently, expanded thermoplastic polyurethane (E-TPU) foams have been applied to midsoles, because the thermoplastic polyurethane which had enough elasticity for the mid soles was developed.

Early polymer foams exhibited large viscous dissipation with cyclic deformation as well as creep deformation after long-time usage. So, it used to be thought that wearing shoes with thinner mid soles would help runners increase their speed. The lower mass of such thin midsole shoes was also advantageous for increasing running speed. Thanks to advances in material development in recent years, dissipation and the viscous effect of polymer foams have decreased considerably. Thick midsoles can now be made lightweight. In track and field events such as marathons, runners from amateurs to top athletes have begun to use running shoes with a thick midsole enclosing a stiff plate. Such stiff plates are typically made from a fiber-reinforced composite material.

Protector pads for contact and action sports is another application for foams. Foams are applied to the inner pad of helmets, baseball catchers' chest protectors, football shin guards, snow sports and mountain bike back protectors, and so on.

Artificial and natural leather

Baseballs are well known as applications of natural and artificial leather. A baseball consists of a rubber or cork core wrapped in yarn and covered by white natural leather or artificial leather. Natural leathers consist of a fiber layer of biological collagen fibers and a surface layer. There are two types of natural leathers: processed and unprocessed. Artificial leathers, also called synthetic composite leathers, consist of a layer of chemical fibers and a surface resin layer. The two-layer structure exhibits good deformability and high durability for cyclic usage. Specifications of artificial and natural leathers are shown in Table 5.1 [3].

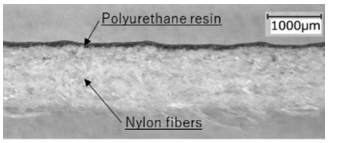
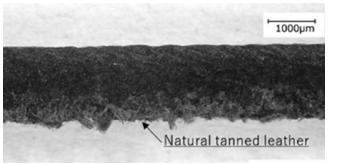
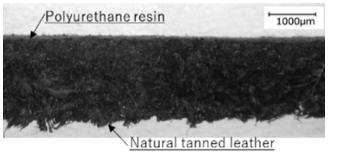
Natural and artificial leathers have similar structures. Both have fiber layers in the body. The dermis layer of animal skin, which consists of a papillary dermis and a reticular dermis, is used to produce natural leather. The many thin fibers in the papillary layer and thick fibers in the reticular layer are tangled complexly, so natural leather is highly durable. On the other hand, chemically produced fibers, such as polyethylene fibers, nylon fibers, and so on, comprise the fiber layer of artificial leather. The complex tangle of chemical fibers in artificial leather is created by reproducing the structure of the fiber layer of natural leather. Due to the fiber layers in leather materials, both natural and artificial leathers exhibit anisotropic mechanical characteristics.

Natural leather is often used for football shoes, basketballs, baseball gloves, and so on. Recently, alternatives to natural leather, such as artificial and synthetic leathers, have been increasingly used to replace natural leathers from the perspective of higher performance. They have some strengths over natural leather in that they are lightweight and affordable, among other factors. However, it is necessary to develop artificial leathers with higher performance than those that already exist to completely replace natural leather in the sports field. Indeed, a constitutive model for natural and artificial leathers has been proposed based on tensile loading test results [3].

Polymer strings for tennis rackets

Tennis was originally called lawn tennis and was introduced in the 1870s. Tennis rackets predating the 1970s were made from wood. From the 1970s, tennis rackets were made from wood, metal, and fiber-polymer composites. Post-1970 tennis rackets were made mainly from fiber-polymer

Table 5.1 Specifications of artificial leather (AL), non-processed natural leather (NL), and processed natural leather (PNL) [3]

Material	*Surface layer*	*Main layer*	*Cross section*
Artificial leather (AL, thickness was 1.06 mm)	Polyurethane resin (80 μm)	Nylon fibers	
Non-processed natural leather (NL, thickness was 1.39 mm)	None	Natural tanned leather	
Processed natural leather (PNL, thickness was 1.16 mm)	Polyurethane resin (30 μm)	Natural tanned leather	

composites [4]. Tennis strings are made of nylon, polyester, or natural gut. Originally, only natural gut strings, which are made from sheep or cow intestines, were used in tennis rackets. Compared with synthetic strings, natural gut strings provide a softer feeling while also generating a lot of power, but they have poor durability, especially in rainy and humid conditions.

Natural gut strings have been used for tennis rackets since the 1870s. Alternative synthetic strings were developed since the 1950s. Synthetic strings have lower extensibility and elasticity than natural gut strings but exhibit greater durability in rainy or humid conditions. Nylon and polyester are the two main materials now used for tennis strings. Monofilament synthetic strings and multifilament synthetic strings are available, and players choose which they prefer based on their own playing style. Substances and surface treatment play an important role here as well.

Constitutive models for soft materials

Constitutive models are mathematical formulations used to describe a material's mechanical behavior. Constitutive models for soft materials are useful for designing high-performance sports equipment, particularly when used in finite element analysis. Finite element analysis is a numerical simulation method used to solve partial differential equations with boundary or initial conditions, or both. Finite element analysis can also be applied to the deformation of solids, heat transfer, fluid flow, and so on. Linear elasticity is the fundamental constitutive model for soft materials; it can be used to consider linear stress-strain relationships. Viscoelasticity may be required for time-dependent properties; it can be used to consider linear stress-strain relationships with the viscous effect. Incompressibility may be required for natural and synthetic rubbers, thermoplastic elastomers, and biological soft materials. Incompressibility of a solid means that its volume does not change when it is deformed. For incompressible elasticity, which occurs in some rubbers and biological materials, hyperelasticity is effective for material modeling. Hyperelasticity is a constitutive model that represents the nonlinear stress-strain relationships of large deformation elasticity.

The stress-strain relationships of hyperelasticity are derived with the strain energy density function W. The Ogden model is a fundamental constitutive model in terms of the principal stretches λ_1, λ_2, and λ_3, as follows [5]:

$$W\left(\lambda_1,\lambda_2,\lambda_3\right)=\sum_{i=1}^{N}\frac{2\mu_i}{\alpha_i}\left(\lambda_1^{\alpha_i}+\lambda_2^{\alpha_i}+\lambda_3^{\alpha_i}-3\right)+\frac{1}{D_1}\left(J-1\right)^2 \tag{5.1}$$

where N, μ_i, D_1, and α_i are material parameters. These parameters can be obtained using the least-squares method by comparing experimental results with theoretical solutions. In the tensile loading test, the stretch λ_t is defined as the ratio between the current length l and the initial length L of the specimens in the tensile direction and is calculated as $\lambda_t = l / L$. The stretch λ_t is approximately related to the nominal tensile strain ϵ_t of the tensile loading specimens by $\epsilon_t = \lambda_t - 1$. To compare with linear elasticity, the initial shear modulus μ, and the bulk modulus κ can be calculated as follows:

$$\mu=\sum_{i=1}^{N}\mu_i \tag{5.2}$$

$$\kappa=\frac{2}{D_1}. \tag{5.3}$$

Here, before proceeding with the description of other constitutive models, some definitions must be introduced. The deformation gradient $\boldsymbol{F}$ is given as follows:

$$\boldsymbol{F}=\frac{\partial \boldsymbol{x}}{\partial \boldsymbol{X}}, \tag{5.4}$$

where $\boldsymbol{x}$ and $\boldsymbol{X}$ are the current and reference positions of a material. The total volume change of a material J is calculated as:

$$J=\det\left(\boldsymbol{F}\right). \tag{5.5}$$

The deformation gradient, when volume change is omitted, is given as:

$$\bar{\boldsymbol{F}}=J^{-\frac{1}{3}}\boldsymbol{F}. \tag{5.6}$$

The right-Cauchy Green deformation tensor $\boldsymbol{C}$ is calculated as:

$$\boldsymbol{C}=\boldsymbol{F}^{\boldsymbol{T}}\cdot\boldsymbol{F}. \tag{5.7}$$

The deviatoric right-Cauchy Green deformation tensor $\bar{\boldsymbol{C}}$ is calculated as:

$$\bar{\boldsymbol{C}}=\bar{\boldsymbol{F}}^{\boldsymbol{T}}\cdot\bar{\boldsymbol{F}}. \tag{5.8}$$

The general Mooney-Rivlin model is another fundamental constitutive model in terms of the first and second principal invariants of deformation $\bar{I}_1$ and $\bar{I}_2$ [6,7,8]:

$$W\left(\bar{I}_1,\bar{I}_2,J\right)=\sum_{i+j=1}^{N}C_{ij}\left(\bar{I}_1-3\right)^i\left(\bar{I}_2-3\right)^j+\sum_{i=1}^{N}\frac{1}{D_i}\left(\frac{J}{J_{th}}-1\right)^{2i}, \tag{5.9}$$

where N, C_{ij}, and D_i are material parameters $\overline{I}_1$ and $\overline{I}_2$ are the first and second principal invariants of the deviatoric right Cauchy-Green deformation tensor $\overline{\boldsymbol{C}}$, respectively. J and J_{th} are the total volume strain and thermal volume strain, respectively. To compare with linear elasticity, the initial μ depends only on the polynomial coefficients of order $N = 1$ as follows:

$$\mu = 2\left(C_{10} + C_{01}\right) \tag{5.10}$$

The bulk modulus κ can be calculated by Equation (5.3). If $N = 1$, $C_{10} = C_1$ and $C_{01} = C_2$ are known, the Mooney-Rivlin model is given as follows [8]:

$$W\left(\overline{I}_1, \overline{I}_2\right) = C_1\left(\overline{I}_1 - 3\right) + C_2\left(\overline{I}_2 - 3\right) + \frac{1}{D_1}\left(\frac{J}{J_{th}} - 1\right)^2. \tag{5.11}$$

Since the Mooney-Rivlin model has only three material parameters C_1, C_2, and D_1, it cannot reproduce complex elastic behavior. As a guideline for the application, the strain range is about 20% or less. If $C_2 = 0$, the neo-Hookean model is obtained as follows [8]:

$$W\left(\overline{I}_1, \overline{I}_2\right) = C_1\left(\overline{I}_1 - 3\right) + \frac{1}{D_1}\left(\frac{J}{J_{th}} - 1\right)^2. \tag{5.12}$$

The neo-Hookean model is a robust model with easy determination of material parameters but is suitable for small strain levels. As a guideline for the application, the strain range is about 20% or less. When the variation of the energy density function with respect to the second invariant is small, the following Yeoh model is proposed, omitting the terms related to the second invariant [8,9]:

$$W\left(\overline{I}_1, \overline{I}_2, J\right) = \sum_{i=1}^{N} C_{i0}\left(\overline{I}_1 - 3\right)^i + \sum_{i=1}^{N} \frac{1}{D_i}\left(\frac{J}{J_{th}} - 1\right)^{2i}. \tag{5.13}$$

If the incompressibility of soft materials can be assumed, C_1 in Equation (5.12) and C_{i0} in Equation (5.13) can be approximated only by the tensile or compression loading test results.

Time-dependent phenomena, such as stress relaxation or strain rate dependency of soft materials, are considerable for finite-strain viscoelasticity. The generalized Maxwell model is considered a strong candidate for the material modeling of the time-dependent behavior of soft materials. In this model, several Maxwell models are arranged in parallel. A generalized Maxwell model consists of a hyperelastic spring and a viscous damper. The free energy function of finite-strain viscoelasticity is derived as follows [10,11]:

$$\psi\left(\boldsymbol{C}, \boldsymbol{Q}_i\right) = W\left(\overline{I}_1, \overline{I}_2, J\right) - \sum_{i=1}^{N} \overline{\boldsymbol{C}} : \boldsymbol{Q}_i + \Xi\left(\sum_{i=1}^{N} \boldsymbol{Q}_i\right). \tag{5.14}$$

Here, Ξ is a certain function of internal variables, but it does not affect the calculation of stress. $\boldsymbol{Q}_i$ represents stresses in the viscous dampers with the equation of evolution, defined as follows:

$$\dot{\boldsymbol{Q}}_i + \frac{1}{\tau_i}\boldsymbol{Q}_i = 2\frac{\gamma_i}{\tau_i}\frac{\partial W\left(\overline{I}_1, \overline{I}_2, J\right)}{\partial \overline{\boldsymbol{C}}}, \tag{5.15}$$

where γ_i represents the strength of the i_{th} Maxwell model and is set from 0 to 1. τ_i is the relaxation time of the i_{th} viscous damper. These models are also available in CAE software [8,12] and play an important role in the design of sports equipment. For biological tissues, such as blood vessels, anisotropic hyperelastic constitutive models are proposed. In the anisotropic hyperelastic constitutive model, additional invariants are added to indicate the direction of anisotropy [13,14].

The stress-strain relationship of the neo-Hookean model under tensile loading is given as follows:

$$\sigma^{\text{eng}} = 2C_1\left(\lambda - \frac{1}{\lambda^2}\right) \tag{5.16}$$

Here, σ^{eng} and λ are the engineering stress and the stretch in the tensile direction, respectively. The stress-strain relationship of the Mooney-Rivlin model ($N = 1$, $C_{10} = C_1$, $C_{01} = C_2$), under tensile loading, is given as follows:

$$\sigma^{\text{eng}} = 2C_1\left(\lambda - \frac{1}{\lambda^2}\right) + 2C_2\left(1 - \frac{1}{\lambda^3}\right) \tag{5.17}$$

These stress-strain relationships are useful for reproducing the nonlinearity of the mechanical properties of polymeric soft materials.

In previous studies, a hyperelastic model with a viscous effect, given in Equation (5.14), was used to model the material of a golf ball [15]. The Yeoh model, shown in Equation (5.13), and the neo-Hookean model, shown in Equation (5.12), were applied to the spring parts of the viscoelasticity. A polynomial strain energy density function of the hyperelastic model, shown in Equation (5.9), was applied to the material modeling of the latex bladder of a soccer ball [16]. In this study, the bladder was modelled as a hyperelastic material with viscoelastic components based on dynamic mechanical analysis (DMA) measurements. The strain energy density function of the hyperelastic material was calibrated against the tensile loading test results of the bladder material.

An anisotropic hyperelasticity was applied to the material of the swimsuit [17]. In this study, the potential of anisotropic hyperelasticity was obtained by identifying it against the tensile test results of rectangular specimens of swimwear material. The generalized Maxwell model, shown in Equation (5.14), was applied to a ball of table tennis and the foam of a table tennis racket [18]. Nine discrete Maxwell elements were identified for the material behavior of the ball and the foam. The Ogden model, shown in Equation (5.1), was applied to a finite element model of a tennis ball [19]. The material definition of the rubber core was based on the Ogden model, and the felt was modelled as low-density foam.

Research topics

Here, some past research topics related to soft materials based on the research conducted by the authors are presented [3,20,21].

1 Tensile loading tests and cyclic tensile loading tests of TPEs [20]

Tensile loading tests and cyclic tensile loading tests were conducted to evaluate the static mechanical properties of TPEs. The specimens were made from a mixture of styrene elastomers, polyolefin, and processed oils. They were processed by injection molding after heat melting. First, tensile loading tests on the TPE specimens were conducted using a uniaxial

tensile loading machine. The deformation of the specimens was measured using a displacement measurement system, which calculates the displacement between target markers using digital information processing. The strain rates of tensile loading tests were set to 0.01%/s, 0.1%/s, 1.0%/s, and 10%/s, respectively. Figure 5.2a shows the relationship between stress and strain given by the tensile loading test. The tensile loading test results revealed that the TPEs could be deformed to over 300% strain. Additionally, cyclic tensile loading tests were conducted; stretches for cyclic deformations were 150%, 200%, and 250%, respectively. Four cycles of tensile deformation were applied to each tensile specimen continuously, and the maximum stretch increased in 50% increments from 150% to 250%. The strain rates of cyclic tensile loading tests were set to 0.1%/s. The cyclic loading test results for the first and third cycles are shown in Figure 5.2b.

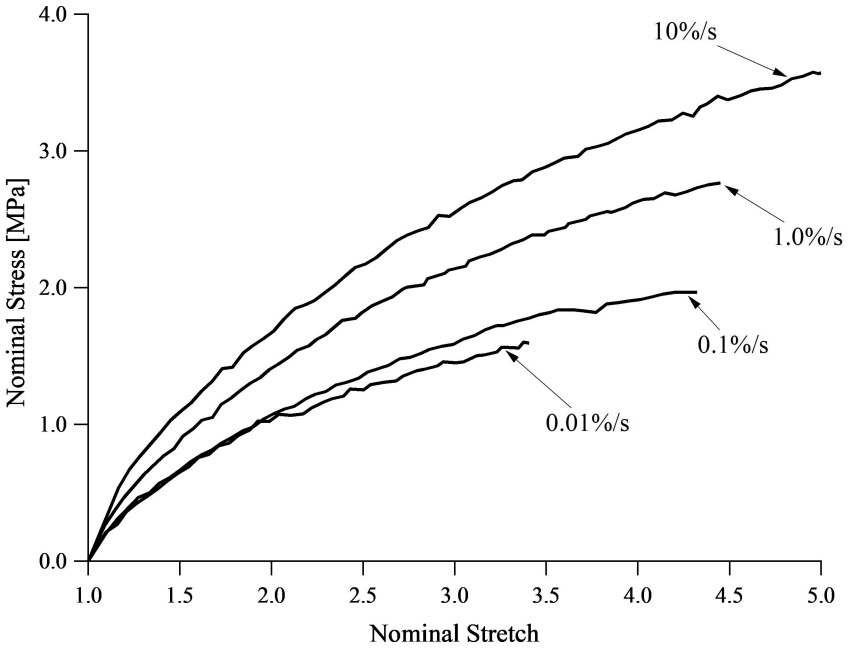
2 Finite element analysis of polyurethane foam based on computed tomography scan (CT scan) data [21]

Finite element analysis with a model based on three-dimensional CT scan data was conducted to consider the effect of microscopic structures on the mechanical properties of polymer foam. The polyurethane foam that was prepared for CT scan measurement is shown in Figure 5.3a. CT scan data for the polyurethane foam were acquired in voxel form. A voxel is a set of three-dimensional cubes reproduced in three-dimensional space. Voxels are frequently used in the visualization of medical data. All three-dimensional scan data for the polyurethane foam are shown in Figure 5.3b. Three rectangular areas were extracted from the CT scan data and are shown in Figures 5.3c, d, and e. The length of one side of each CT model was 2 mm. Finite element analysis of the CT models was conducted under the following conditions: The top and bottom surfaces were completely constrained, and the side surfaces were free. Compression strain was applied to the top surface in increments of 0.015 up to a strain of 0.3 in the vertical direction. The material parameters of the matrix were determined based on the tensile loading test results of the solid polyurethane. Figures 5.3f, g are diagrams of the model at strains of 20% and 30% for CT model-1. Figure 5.4 shows the stress-strain relationships obtained with the simulations. The percentage annotations in the graph indicate the relative density of each model. Depending on the position where the CT scan models were taken, the relative densities of the simulation models were 8.0%, 8.16%, and 9.23%, respectively. The simulation results all roughly agree with each other. Differences in relative density did not affect the calculation results. These results suggest that the sample areas contained representative microstructures of polymer foam. In addition, the results show the bend in the stress-strain relationship at about 10% compression, which represents the plateau region. Therefore, it is implied that applying the voxel finite element analysis with a CT model can predict the stress-strain relationship of polymer foam up to the plateau region.

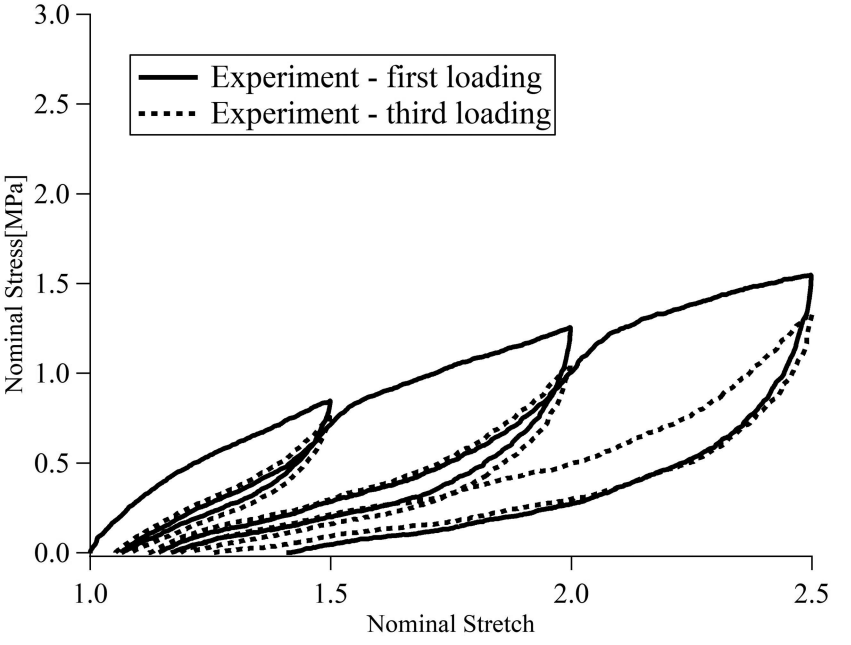
3 Mechanical behavior of natural and artificial leathers [3]

Uniaxial tensile loading tests performed on three types of leather materials were conducted to investigate their mechanical properties. The leather materials examined in this test are commonly used for sports equipment. Sports products made of leather are affected by manufacturing batches and weather conditions during use. Since performance instability of these sports products has been a concern for a long time, it is important to reveal their mechanical properties and control the performance of the equipment. For this purpose, tensile loading tests were conducted to obtain the mechanical characteristics of the leather materials.

Tensile tests were performed on artificial leather, tanned natural leather, and natural leather covered with polyurethane resin on the surface, as shown in Table 5.1. All leather materials for tensile

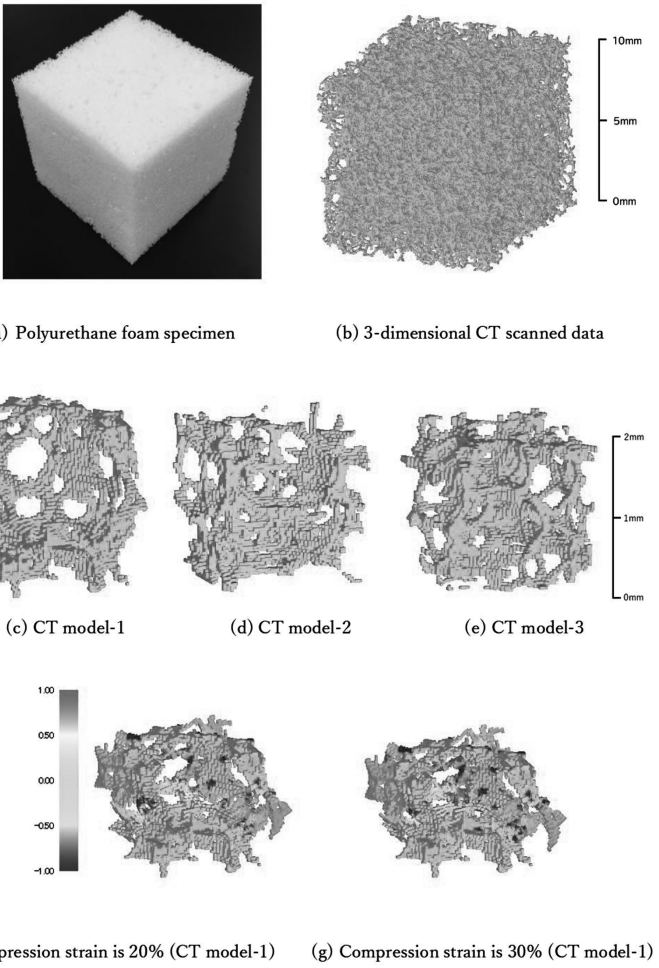


(a) Simple tensile loading test results



(b) 0.1%/sec

Figure 5.2 Single and cyclic tensile loading test results for TPE [20].



(a) Polyurethane foam specimen (b) 3-dimensional CT scanned data

(c) CT model-1 (d) CT model-2 (e) CT model-3

(f) Compression strain is 20% (CT model-1) (g) Compression strain is 30% (CT model-1)

Figure 5.3 Polyurethane foam and CT models for finite element analysis [21].

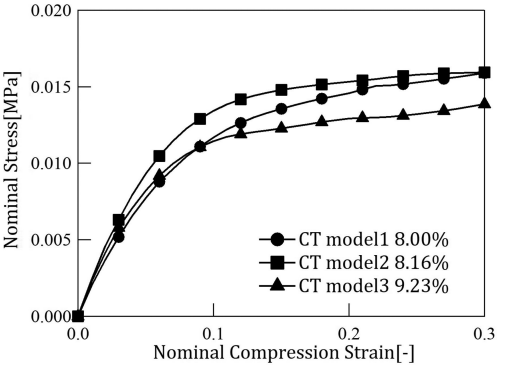
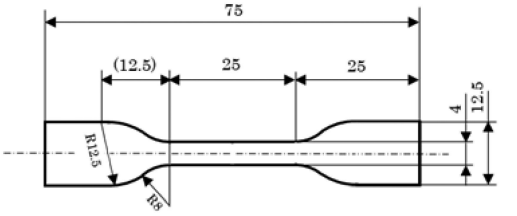


Figure 5.4 Stress-strain relationships obtained with finite element analysis of CT models [21].

tests were supplied by Mizuno Corp (Osaka, Japan). The fiber layer of the artificial leather (AL) was made of nylon fibers injected in a particular direction and tightly tangled. Additionally, the surface was processed with polyurethane resin, so the artificial leather had two layers: the fiber layer and the resin layer. The tanned natural leather was processed only by tanning and is hereinafter referred to as non-processed natural leather (NL). The surface of the processed natural leather (PNL) was artificially covered with a polyurethane resin layer.

To investigate the anisotropy of the AL, specimens were subjected to tensile loading tests. The tensile angles φ of the specimens were defined in $15m/180$ degrees, where $m = 0,1,\ldots,11$ the tensile angle φ was set to 0 in the direction orthogonal to the processing direction of AL, NL, and PNL. The specimens were cut out at every 15 degrees centered on a certain point of a leather sheet. Since the AL was manufactured artificially, the test specimens could be cut out from a wide range of AL sheets. Therefore, three specimens were prepared for each cut-out angle, and the results for each cut-out angle were defined as the mean of the three specimens. For the natural leathers, the tensile angles φ of the specimens were defined in $10n/180$ degrees, where $n = 0,1,\ldots,17$ the specimens were cut out at every 10 degrees centered on a certain point of a sheet of each natural leather material. Since natural leathers are biomaterials derived from animal skins, their fiber orientation distributions were assumed to be more complex than those of artificial leather. In addition, natural leathers had limited areas where test pieces could be cut out. Thus, there was only one specimen for each cut-out angle, and the intervals of cut-out angles between adjacent specimens were set smaller than for the AL. Thirty-six specimens for AL and 18 specimens each for NL and PNL were prepared for the tensile loading tests. The distribution of tensile test specimens for NL sheets is shown in Figure 5.5a.

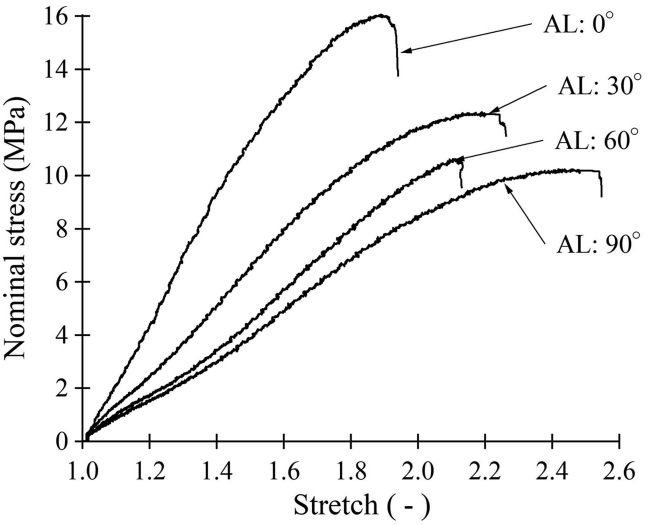
(a) Distribution of tensile test specimens on NL sheet



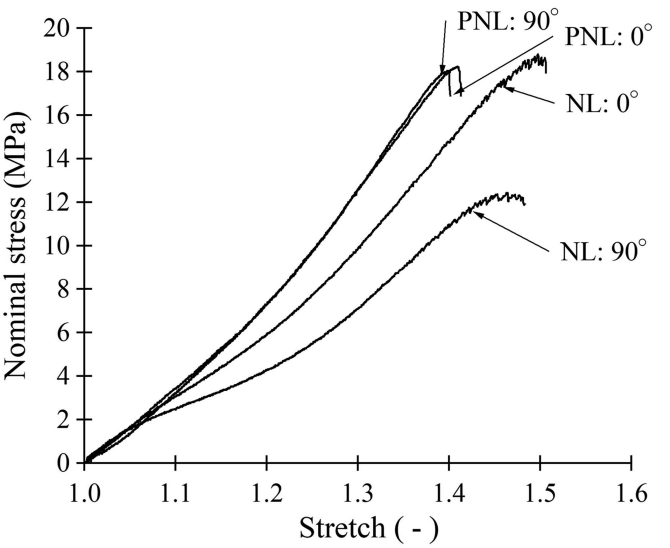
(b) Shape of tensile loading test specimen[22][23]

Figure 5.5 Tensile loading test results of AL, NL, and PNL [3].

In this study, fiber distributions of the three types of leather materials were assumed to have 180° periodicity. The thicknesses of the polyurethane surface layers of AL and PNL were 30μm and 80μm, respectively. Tensile test specimens were cut in the shape of the dumbbell test specimen type No. 6 in the Japanese industrial standard (JIS) K 6251:2017 [22]. The dumbbell test specimen type No. 6 in the JIS K 6251:2017 is the same as the Type 2 dumbbell test specimen in



(a) AL (cut-out angles α = 0°, 30°, 60° and 90°)

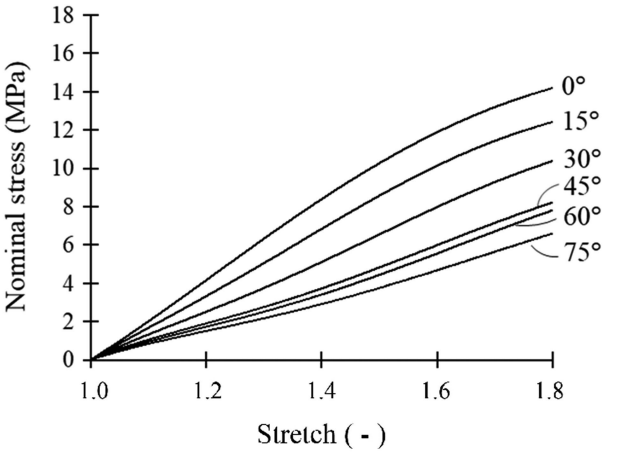


(b) NL and PNL (cut-out angles α = 0° and 90°)

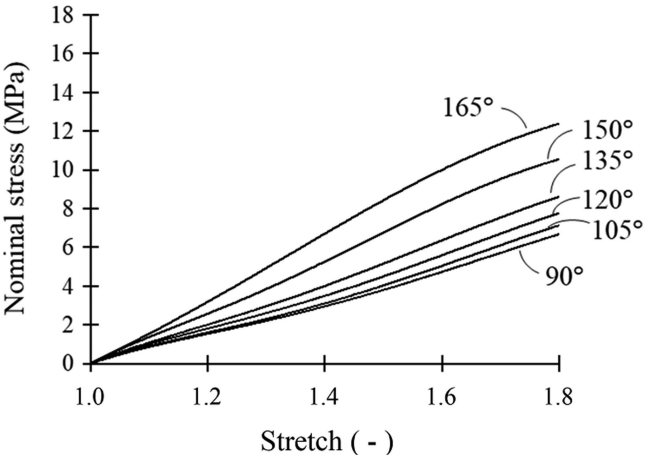
Figure 5.6 Tensile loading test results of AL, NL, and PNL [3].

ISO 37:2017 [23]. The overall length is 75 mm, the width of the ends is 12.5 mm, and the width of the narrow portion is 4 mm, respectively. The shape and dimensions of the tensile test specimens are shown in Figure 5.5a and b. A tensile deformation of 1.0 mm/sec was applied until rupture of the specimens. The tensile deformation of 1.0 mm/sec corresponds to approximately 4%/sec, since the length of the tensile test section is 25 mm. In Figure 5.6, some tensile loading test results for AL, NL, and PNL from start to rupture are shown to demonstrate the overall mechanical properties. The fracture strain of AL specimens ranged from 0.85 to 1.55, and the fracture strains of NL and PNL ranged from 0.4 to 0.55. From the viewpoint of the design of sports equipment, the large elastic deformations of AL, NL, and PNL must be considered.

The relationship between nominal stress and stretch for the AL is shown in Figure 5.7. The results for cut-out angles $\varphi = 0°$, 15°, 30°, 45°, 60°, and 75° are illustrated in Figure 5.7a. In



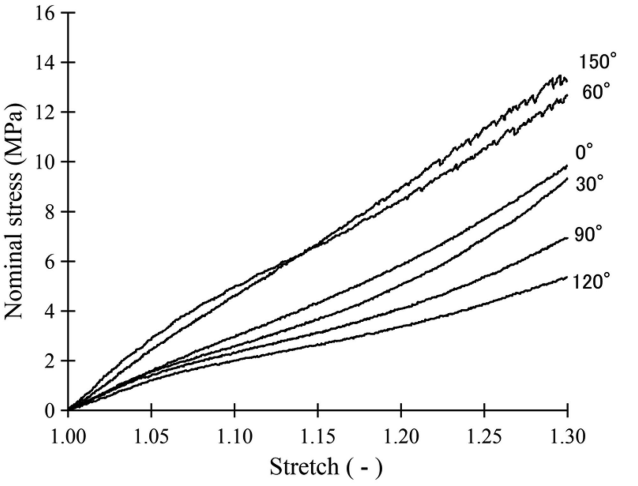
(a) Cut-out angles α = 0°, 15°, 30°, 45°, 60°, and 75°



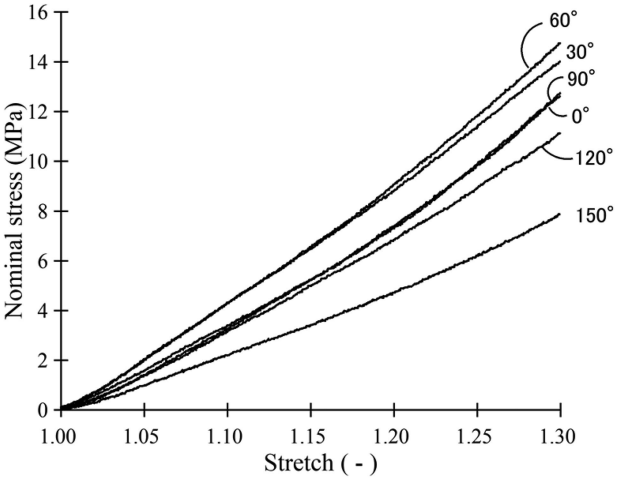
(b) Cut-out angles α = 90°, 105°, 120°, 135°, 150°, and 165°

Figure 5.7 Relationships between nominal stress and stretch for AL [3].

Figure 5.7b, the results for $\varphi = 90°$, 105°, 120°, 135°, 150°, and 165° are shown. Figure 5.8a shows the relationship between nominal stress and stretch for the NL; the results for cut-out angles $\varphi = 0°$, 30°, 60°, 90°, 120°, and 150° are shown. The results for the PNL are shown in Figure 5.8b; the results for cut-out angles $\varphi = 0°$, 30°, 60°, 90°, 120°, and 150° are illustrated. As seen in Figures 5.8a and b, direction dependencies of the relationship between nominal stress and stretch were observed for the NL and PNL. The anisotropic mechanical characteristics of the leather materials were confirmed.



(a) NL

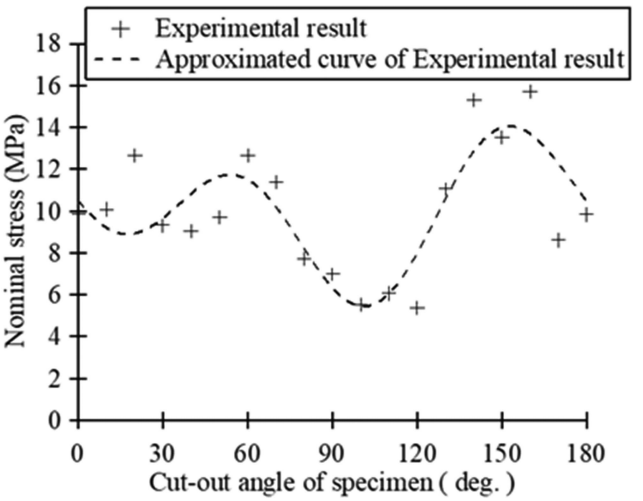


(b) PNL

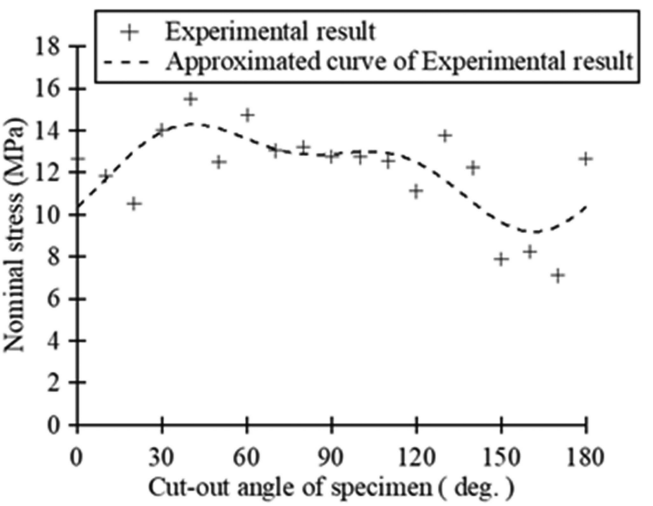
Figure 5.8 Relationships between nominal stress and nominal strain. Results for the natural leather, the surface of which is covered with polyurethane resin layer, are illustrated. Results for cut-out angles $\alpha = 0°, 30°, 60°, 90°, 150°$ are shown [3].

Figures 5.9a and b show the relationships between nominal stress and the cut-out angle of NL and PNL specimens, respectively. The experimental results shown represent nominal stresses when the applied strains reached 0.3. The experimental results are displayed as plotted plus signs, and the approximated curves of the test results are illustrated by the trigonometric function denoted as follows:

$$y = a_1 \cos 2\left(\theta - \theta_1\right) + a_2 \cos 4\left(\theta - \theta_2\right) + b \tag{5.18}$$



(a) NL



(b) PNL

Figure 5.9 Relationships between nominal stress and cut-out angle of specimens from uniaxial tensile loading tests; the approximated curves of the experimental results are illustrated by the dotted lines. Results for a nominal strain of 30% are plotted with black plus signs [3].

The experimental results for the two natural leathers show two respective high peaks of stiffness, so the combination of trigonometric functions of periods π and $\pi / 2$ was applied to approximate the experimental results. The parameters of Equation (5.18) were calculated using the least-square method with the experimental data. Due to the assumption of the π-periodic fiber distribution of leather materials, the trigonometric function was applied to the approximations. Slight differences between the experimental results and approximation curves in Figures 5.9a and b were observed; however, the relationships between nominal stress and the cut-out angle of specimens were approximated.

Conclusion

In this section, some soft materials that are often applied to sports equipment were introduced. Specifically, the concept and fundamental mechanical behavior of rubber, foam, polymer fabrics, natural leather, and artificial leather were investigated. Additionally, some fundamental constitutive models for soft materials were introduced. Some research results on foams, thermoplastic elastomers, artificial leathers, and natural leathers were also investigated. Soft materials are used in many applications for sports equipment. The application of CAE will help the development of sports equipment with higher performance. As the manufacturing development of soft materials advances, higher-performance equipment will become available.

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6

FINITE ELEMENT MODELLING IN SPORT

James Sherwood, Patrick Drane, Catherine Barry, and Kayla Krantz

Introduction

Sports engineering is a multidisciplinary field that blends the principles of mechanical engineering, materials science, and biomechanics to design sports equipment to support the performance, protection, and overall experience of athletes. Finite element analysis (FEA) is a powerful tool for investigating the interactions of these principles and understanding the complex behaviors of sports equipment.

In this chapter, we explore the application of finite elements in sports engineering. Through computational modeling, FEA provides insights into the displacements, stresses, and strains experienced in sports-related scenarios and how such modeling can be used to conduct parametric studies to evaluate how changes in materials and design influence the resulting performance of sports equipment.

This chapter starts with a brief introduction to finite element analysis. This introduction will not provide a deep dive into finite element theory. Numerous texts are available to explore the theory and application of the finite element method, e.g., [1–5]. Rather, this brief introduction to finite element analysis will give an overview of the basics needed to build a model of the sports equipment under consideration. This overview will discuss the common element types and how to choose the right element type(s) for the situation being investigated, discretization techniques, how to define the appropriate boundary conditions, and numerical solving procedures. The rest of the chapter will share examples of the application of FEA in sports engineering. These examples will emphasize practical considerations in model creation, the input of material properties, and the definition of boundary conditions, thereby enabling the user to develop accurate and meaningful simulations of sports-related scenarios. The dynamic and high-impact nature of sports demands special attention to factors such as contact mechanics, fluid-structure interaction, and human biomechanics. By understanding these nuances, the sports engineer will be well-equipped to tackle the complex engineering problems encountered in the realm of sports.

Examples of the application of finite element modeling in sports engineering are shared in tutorials that are available online at https://www.uml.edu/FEM-Tutorials. These examples use LS-PrePost from ANSYS (Canonsburg, Pennsylvania, USA) to build the models and to conduct the subsequent postprocessing, and all the example models in these tutorials are analyzed using the

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LS-DYNA solver. The student version of the LS-DYNA suite can be downloaded online at https://www.ansys.com/academic/students/ansys-ls-dyna-student. After reading this chapter and completing some of the tutorials, you will possess a good understanding of the application of finite element analysis in sports engineering. From designing better sports equipment to optimizing playing surfaces and providing athlete protection, FEA empowers engineers to make well-informed decisions and to drive innovation in this dynamic field. By blending basic finite element concepts with practical examples, this chapter aims to inspire your curiosity and motivate you to explore how FEA can be used to investigate the intersection of engineering and sports.

While all the modeling, analyses and postprocessing steps associated with this chapter are performed using LS-PrePost and LS-DYNA, these models can be done equally well in other commercial finite element packages. For example, FEMAP and its companion solver NX/Nastran are freely available to students and can be downloaded from the Siemens (Munich, Germany) website at https://www.femto.eu/stories/femap-student-edition/. FEMAP is a powerful pre/postprocessor and can be used to translate a finite element input file from one solver to another.

The basics of finite element modeling for sports engineering

A finite element model (FEM) is a computational tool used in engineering to simulate and analyze physical phenomena. Essentially, the model discretizes a real-world structure into small, simple parts called finite elements, which, in turn, are used to develop a set of simultaneous equations that describe the mechanical behavior of the structure. The model can be used to investigate how the structure will behave under various physical conditions, such as applied forces and displacements. The basic components of a finite element model include:

Geometry: This is the first step, where you define the shape and size of the object or structure you want to analyze. The geometry can be as simple as modeling a golf club shaft as a beam or as complex as a running shoe comprised of many layers and different materials. For example, Song et al. [6] used finite element modeling to guide footwear design. This model was comprised of different materials for the shoe and included a model of the foot.

Meshing: After the geometry is defined, the structure is discretized into small, manageable pieces known as elements. This discretization process is called meshing. The quality of the mesh significantly affects the accuracy of the analysis. Elements might be triangular, quadrilateral, tetrahedral, etc., depending on the complexity of the geometry as well as the application under consideration.

Material Properties: For the model to simulate real-world conditions accurately, material properties need to be assigned to the elements. The properties include, at a minimum, the elastic modulus and the Poisson's ratio for the quasistatic analysis of a structure made with an isotropic material, e.g., a hollow aluminum baseball bat. Density is needed for dynamic analysis, e.g., the impact of a wood baseball bat [7]. In any commercial code, there are many different material models, and the number of material properties depends on the material model being used. For example, a rubber ball can use a viscoelastic material model, and a hockey stick may need a composite material model. Burbank and Smith [8] provide an excellent example of the characterization of a rigid foam used in finite element sports ball simulations. Dong et al. used a wood material model to investigate the stick used in Japanese bō [9]. Readers seeking more information on material models for soft materials are referred to Chapter 5, "Soft materials in sport".

Boundary Conditions (BCs) and Loads: Here, the user specifies the conditions under which the model operates. This step in the modeling process includes applying loads (like forces, pressure, or temperature) and constraints (like fixed supports or joints) to simulate real-world conditions.

The advantage of a finite element model is that the same finite element mesh can be used for performing static, dynamic, or thermal analysis. The only change from one type of analysis to another is the modification of the boundary conditions.

Solver: The solver is the computational engine of the FEM. It takes all the information—the mesh, material properties, loads, and boundary conditions—and calculates the structural (or thermal) response. The solver finds the step-by-step solution to the set of simultaneous equations that are developed as a result of building the model. These equations are not explicitly shown to the user.

Results and Post-Processing: The solver will generate one or more output files, and these output files can be used to visualize the structural (or thermal) response. This post-processing can include displacements, stresses, strains, impact forces, temperature distributions and more. Post-processing involves analyzing these results to make engineering decisions.

Elements

The selection of the correct element type is a fundamental aspect of finite element analysis. Element choice is not merely a procedural step; it is a decision that significantly impacts the accuracy, efficiency, and overall success of the FEA. Each element type, e.g., truss, beam, shell, or solid, has its own mathematical formulation and assumptions. These formulations are designed to model specific types of behavior and geometry. For instance, beam elements are good at handling bending and torsional loads in slender structures, while solid elements are more suitable for complex three-dimensional stress states.

Element types fall into three categories: 1D (one-dimensional, e.g., trusses and beams), 2D (two-dimensional, e.g., shells), and 3D (three-dimensional, e.g., solids). If you choose an inappropriate element type for your application, several issues can arise. The most critical is the potential for inaccurate results, which can lead to erroneous conclusions about the structural integrity or performance of your design. Additionally, using a more complex element type than necessary can result in longer computation times and increased demand on computational resources. Thus, it is always necessary to have a good physical understanding of the nature of the problem that you are modeling and then choose the proper element type(s) for your model. Your ability to select the most appropriate element type for each situation will be a key factor in becoming a proficient and effective expert in finite element analyses. The following subsections will walk through the common element types in order of increasing element complexity.

1D elements

Truss elements

Truss elements, sometimes called "rod" elements, are slender members that can only carry axial loads, either in tension or compression. They are useful in simulating structures where axial loads are predominant. A key characteristic of truss elements is their simplicity. They are defined by just two nodes (one at each end), as shown in Figure 6.1 and possess the ability to resist axial forces but not bending moments. The degrees of freedom (DOFs) at each node are the translations in the x, y, and z directions—no rotational DOFs—as denoted by U_i, V_i, and W_i, respectively, where the subscript i denotes the node number. This simplicity makes truss elements computationally efficient. However, this simplicity also comes with limitations. Truss elements cannot model bending or torsional effects, and they assume that the member connections are pinned (or hinged), allowing no moment transfer between members. In real-world applications, these limitations must be carefully

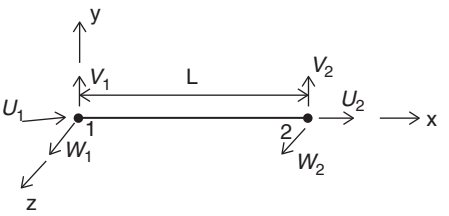

Figure 6.1 Example of a two-noded truss element showing the three DOFs at each node where U_1, V_1, and W_1 denote the translations in the x, y, and z directions, respectively, at Node 1 and U_2, V_2, and W_2 denote the translations in the x, y, and z directions, respectively, at Node 2.

considered to avoid inaccuracies. In sports engineering, truss elements may be a good choice for modeling the strings on a tennis racquet, where the tensile stiffness of the strings dominates the mechanical behavior. However, if bending stiffness cannot be concluded to be insignificant, then a beam element may be the better choice.

Here are some examples of the use of truss elements in sports engineering:

- Stadium and Arena Structures: Many sports stadiums and arenas use truss structures for their roofs or large-span components. Truss elements can be used to model these components, helping to analyze the structural integrity and load-bearing capacity of these large structures.
- Bicycle Frames: For certain types of bicycles, especially those used in competitive cycling, the frame can be modeled using truss elements to optimize the structure for weight, strength, and aerodynamics. However, for more sophisticated analyses of a bicycle frame, beam elements or shell elements may be a better selection.
- Support Structures in Sports Equipment: Support structures in sports such as net systems (e.g., tennis or volleyball) and goal cages (e.g., in ice hockey or soccer) often include components that function similarly to truss frameworks. Truss elements can be used to efficiently model the poles, cables, and structural frames, enabling accurate assessment of how these components respond to loads such as tension, impact forces, or wind. This approach supports the optimization of structural integrity while minimizing material use and weight.
- Adventure and Outdoor Sports Equipment: Equipment used in adventure sports, like climbing frames, zip-line structures, and suspension bridges in adventure parks, can also be analyzed using truss elements. This analysis is vital to study the integrity of the system during in-service conditions.

Beam elements

Beam elements are used for modeling long, thin structural members that are subject to bending, shear, and torsional loads as well as axial loads. Unlike truss elements, which can only handle axial forces, beam elements can resist bending and twisting moments in addition to the axial forces captured by truss elements, thereby making beam elements more versatile. Figure 6.2 shows a two-noded beam element and the associated six DOFs at each node. The mathematical formulation of beam elements allows them to consider bending in one or two directions, shear deformations, and torsional twisting. The DOFs for a beam element are x, y and, z translations and θ_x, θ_y and, θ_z rotations. Beam elements assume a certain level of simplification and work best when the length of the member is significantly greater than its cross-sectional dimensions, and they assume a uniform cross-section along their length. Thus, beam elements are not suited for analyzing details within the cross-section of a beam, such as stress concentrations or complex 3D stress states.

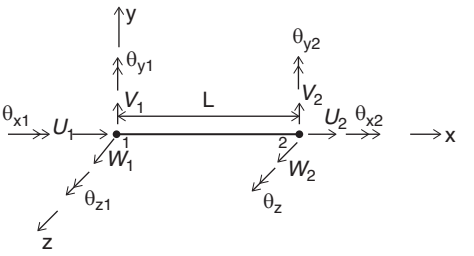

Figure 6.2 Example of a two-noded beam element showing the six DOFs at each node where θ_{x1}, θ_{y1}, and θ_{z1} denote the rotations about the x, y, and z axes, respectively, at Node 1 and where θ_{x2}, θ_{y2}, and θ_{z2} denote the rotations about the x, y, and z axes, respectively, at Node 2.

Here are some examples of the use of beam elements in sports engineering:

- Bicycle and Racing Car Frames: The frame of a bicycle or racing car can be modeled using beam elements to understand how it behaves under different loads. For example, for the bike, the loads could be the rider's weight, dynamic forces during pedaling, or stresses during a race. These analyses help in designing lightweight bicycle frames. For the racing car, such a frame analysis can investigate torsional, longitudinal, and lateral stiffnesses, as reported by Luque et al. [10].
- Golf Clubs: The shaft of a golf club is a perfect candidate for modeling with beam elements. Engineers can analyze how the shaft bends and twists during a swing, influencing the design for control and power transfer.
- Pole Vault Poles: The pole used in pole vaulting experiences significant bending as the athlete uses it to vault over a high bar. Using beam elements to model the pole helps in understanding its flexural behavior, aiding in the design of poles with a good blend of stiffness and flexibility.
- High Jump and Gymnastics Bars: The bars used in high jump and gymnastics are subjected to dynamic bending and vibrational loads. Beam elements can be used to simulate these conditions, helping in the design of bars that have the right balance of rigidity and elasticity.

2D elements

Thin shell elements

Thin shell elements are used where the thickness is small relative to the other two dimensions. Figure 6.3 shows examples of three- and four-noded shell elements. Each node on the shell element has six DOFs. The loads carried in the element include in-plane loads in the local x and y directions as well as bending about the local x and y axes. It is important to note that thin shell elements come with certain limitations and assumptions. They work best when the thickness of the structure does not change significantly under load. The four-noded thin shell is preferred for use over the three-noded thin shell. However, complex geometry may be better captured with a three-noded shell. The user should use the three-node shell sparingly, if possible.

Users should be careful not to confuse the thin shell element with a membrane element. The thin shell and membrane elements look visually the same on the screen; however, the two element types work differently. A membrane element can only carry in-plane loads, i.e., no bending or twisting. Thus, if bending and twisting are not important to the analysis, then the three- or four-noded membrane element may be a good choice.

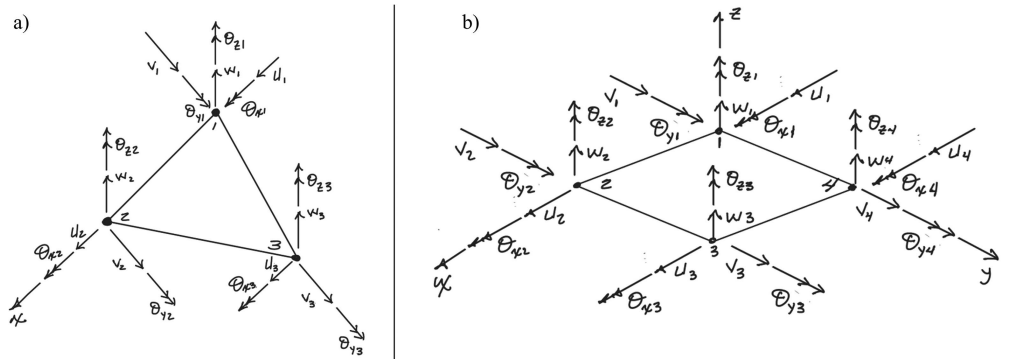

Figure 6.3 Examples of thin shell elements with six DOFs at each node (a) three-noded (triangle) and (b) four-noded where U_i, V_i, and W_i, correspond to the x, y, and z translations, respectively, at Node i, and θ_{xi}, θ_{yi}, and θ_{zi} correspond to the rotations about the x, y, and z axes, respectively, at Node i.

Here are some examples of the use of thin shell elements in sports engineering:

- Sports Helmets: Helmets are important protective equipment for cycling, motor racing, and contact sports, e.g., American football. Thin shell elements can be used to model the helmet outer shell, helping engineers analyze and improve its impact resistance and overall design for protection.
- Skateboards and Snowboards: These sports boards often have complex shapes and are subjected to dynamic loads. Thin shell elements allow for the analysis of their structural integrity and the effects of different materials and geometries on performance and durability.
- Balls: Shell elements can be used for the modeling of hollow sports balls, such as tennis balls, American footballs, and soccer balls. Taha and Hasnun [11] used composite shell elements to model the two layers of a soccer ball: the inner bladder (0.2 mm thick) and the outer panel (2.2 mm thick).
- Racing Car Bodies: In motorsports, the aerodynamic shells of racing cars are key to performance. Using thin shell elements, engineers can simulate the aerodynamic loads and stress distributions on these shells, contributing to designs that optimize speed and stability.
- Tennis Rackets and Golf Clubs: The frames of tennis rackets [12] and the bodies of golf clubs can be effectively modeled using thin shell elements to understand how these pieces of equipment interact with balls and how their design can be tuned for performance. Mase et al. [13] used thin shells in the design optimization of a titanium golf club driver.
- Archery: The vibration of arrows was studied using thick shell elements by Rieckmann et al. [14].
- Skiing: Skis and ski poles can be modeled using shell elements. Schindelwig et al. [15] used shell finite element models to study slalom ski-pole impacts.

3D elements

Solid elements

Solid elements are the most general type of elements used in FEA for modeling three-dimensional structures. They are typically the best choice when modeling a component that has significant

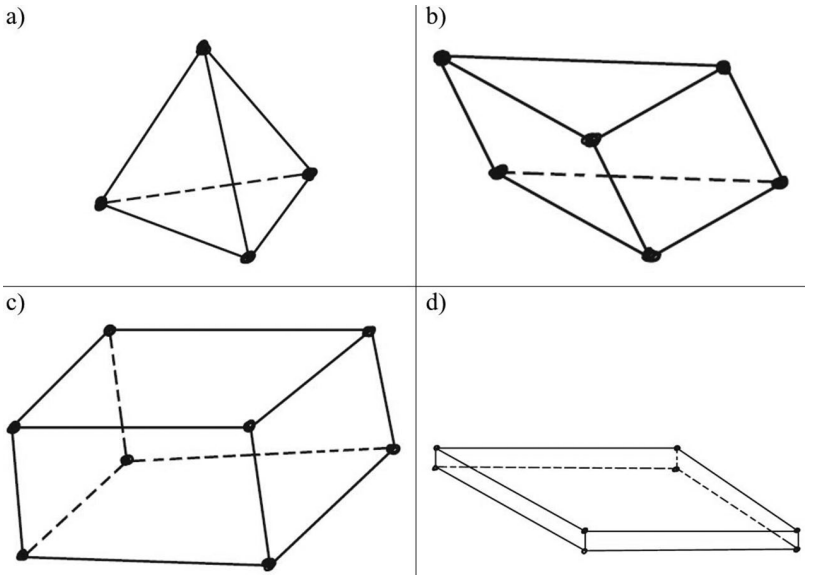

Figure 6.4 Examples of (a) a four-noded tetrahedral element, (b) six-noded wedge element, (c) an eight-noded brick element, and (d) four-noded thick shell element.

depth, width, and height and experiences complex stress states, such as material behavior under multi-axial loading. There are two primary options for linear solid elements: the four-noded tetrahedral element, also known as "tets," and the eight-noded hexahedral element, also known as "bricks." Each of the nodes has three DOFs: translation in the x, y, and z directions, i.e., there are no rotational DOFs. Figure 6.4 shows examples of tet and brick elements. Figure 6.4 also shows a six-noded wedge element. Wedge elements are often used when transitioning from a fine to coarse mesh. (The definitions of fine and coarse meshes are presented in the subsection on Mesh Density.)

One of the key advantages of solid elements is their ability to model complex geometries with varying cross-sections and intricate features. However, a challenge with solid elements is ensuring an adequate mesh density to capture stress gradients accurately, particularly around areas of high stress concentration. This challenge often leads to a mesh that is a trade-off among accuracy, fidelity, and computational efficiency.

Bricks are the preferred choice when using solid elements; however, it is often not feasible to build a solid mesh entirely of eight-noded bricks. Sometimes bricks need to be degenerated to become six-noded wedges. Tets are an excellent choice for capturing the intricate details of a complex geometry and sometimes are the only choice when bricks would need to be extremely small to capture small details in the geometry.

Here are some examples of the use of solid elements in sports engineering:

- Sports Equipment Design: In designing sports equipment like golf clubs, tennis rackets, or baseball bats, solid elements allow for a detailed analysis of the equipment's response to impact forces. For example, the clubhead of a golf club can be modeled to analyze stress distribution upon hitting a golf ball, guiding design choices for materials and geometries to tune performance and durability. Mase et al. [16] used brick elements in combination with the boundary element method to model the sound of a golf club impact. Mase et al. [13] used solid elements

to model a viscoelastic golf ball in their club optimization study. Another example of solid elements for golf club modeling was done by Peterson and McPhee [17]. Tanaka et al. [18] constructed a finite element model of golf balls with hyperelasticity and viscoelasticity to devise a method for constructing ball models by investigating the relationship between the material properties and the behavior of the golf ball during and after impact.

- Sports Balls: Golf balls, tennis balls, and baseballs can be modeled using solid elements. Goodwill et al. [19] used solid elements when modeling the impact of a tennis ball on a rigid surface. Tanaka et al. [20] used solid elements to model the core, mantle, and cover of a golf ball.
- Protective Gear: Solid elements are a good choice to assist in the design of protective sports gear such as shin guards and other types of body armor, which are too thick to be modeled using thin shell elements. The solid elements enable the simulation of how these items absorb and dissipate impact energy, which is vital for enhancing protective features while maintaining comfort and usability. Ferriell et al. [21] used solid elements to study American football faceguards.
- Biomechanics: Parts of the human body can be modeled using solid elements to investigate the relationship between impact intensity and injury. Patton et al. [22] used the finite element method to investigate unhelmeted football head impacts. Razaghi and Biglari [23] used finite element modeling to study ocular injury in boxing, and Razaghi et al. [24] used modeling to investigate the mechanical performance of protective headgear materials to minimize injury to the boxer's head.
- Athletic Footwear: The complex structure of athletic shoes, including soles with varying material properties and geometries, can be effectively modeled using solid elements. A 3D analysis provides insight into the cushioning, stability of support, and overall shoe performance, contributing to the design of footwear that can improve athletic performance and mitigate the risk of injury. Hladnik and Jerman [25] used solid elements to investigate a cross-country ski boot. Li et al. [26] used tet elements in their coupled finite element model of foot and barefoot running footwear. Shimoyama et al. [27] used the finite element method to perform a design optimization of a sport shoe sole structure.
- Sporting Facilities: Components of sporting facilities, like rubberized running tracks or artificial turf systems, can be modeled using solid elements. Such analyses aid in understanding how these surfaces interact with athletes, influencing factors such as energy absorption, traction, and injury prevention. A good example of such modeling was done by Farhang et al. [28], which included a finite element model of a foot, and the viscoelastic behavior of the sport surface was extracted by compression stress-relaxation tests with various strain rates to import into the finite element model.

Thick shell elements

Thick shell elements are used to model structures where the thickness cannot be neglected compared to the other dimensions but where the thickness is still not large enough to require the use of a solid element. Thick shells serve as a bridge between thin-shell and solid elements, combining aspects of both. Unlike thin-shell elements, which assume that the thickness has no effect on the stress distribution, thick shell elements account for transverse shear deformation. This inclusion of transverse shear deformation makes thick shells more accurate than thin shells for moderately thick structures where shear deformation plays a significant role. The degrees of freedom for the nodes of thick shells include translations and rotations.

Here are some examples of the use of thick shell elements in sports engineering:

- Protective Equipment: Thick shell elements can be used to analyze the layered materials often used in helmets and protective pads. These elements provide an accurate representation of behavior under impact.
- Sports Boards: Equipment like surfboards, snowboards, or skateboards can be modeled using thick shell elements. These elements help in understanding how the board flexes and reacts to forces, which is needed for tuning performance and durability.
- Athletic Footwear: The midsole of sports shoes, which is neither very thin nor very thick, can be effectively analyzed using thick shell elements to study cushioning, stability, and overall structural integrity.
- Sports Surfaces: The layers of a football pitch can be modeled by specifying material properties for each layer. Bustamante et al. [29] proposed a finite element model of synthetic turf using a hierarchical approach and validated the model using independent test data. Cole [30] developed a model to describe the behavior of 3G artificial turf systems under vertical loading. Andena et al. [31] built a three-dimensional finite element model of athletic tracks. The model is based on data from quasi-static compression tests performed on small laboratory samples to tune the constitutive parameters.

Implicit vs. explicit solvers

FEA solvers fall into two categories, implicit and explicit solvers. It is important to understand the distinction between these two as each solver has its own set of advantages and ideal applications.

In an Implicit Solver, the equations governing equilibrium are solved at the end of each time step. This approach requires solving a set of simultaneous equations, which can be a computationally intensive process for large systems. However, the implicit method is unconditionally stable, meaning the modeler can use relatively large time steps without the risk of numerical instability. The downside of large time steps is that there can be some loss of fidelity in the output data, i.e., there could be some clipping of the results. Thus, the output may "clip" a maximum or minimum response state. Therefore, the user should be careful in selecting the size of the time steps.

Examples of ideal applications for an implicit solver include:

- Quasi-static Problems: Where the loading is applied slowly relative to the object's natural frequency.
- Linear and Nonlinear Static Analysis: Particularly effective in problems where stability and large deformations are a concern.
- Low-Energy Dynamic Problems: Such as thermal analysis or slow dynamics where small time steps are not required.

An Explicit Solver calculates the state of a system at a future time based on the current state, without solving simultaneous equations. It is computationally less demanding per time step than an implicit solver, making it computationally efficient for certain types of problems. However, it is conditionally stable, meaning that the time step must be sufficiently small—typically smaller than the smallest natural period in the system—to avoid numerical instability.

Examples of ideal applications for an explicit solver include:

- High-Energy Dynamic Events: Such as crashes, explosions, or impact analysis, where the time scales are very short.
- Problems with Complex Contacts and Boundaries: Where the simplicity of the explicit time integration is advantageous.
- Large Deformation and Nonlinear Material Behavior: Explicit solvers can handle these scenarios effectively, albeit with small time steps.

The choice between an implicit and explicit solver in FEA depends on the nature of the problem being addressed. Implicit solvers are better suited for slow or quasi-static scenarios where accuracy and stability over large time steps are critical, while explicit solvers excel in dynamic, high-energy situations where computational efficiency per small time step is essential.

With respect to sports engineering, the explicit solver should be used whenever there is a high-speed impact situation, e.g., a ball impact, and the implicit solver can be used for quasi-static models, e.g., applying a force to a ball being squeezed in a universal testing machine.

Applications in Sports Engineering for an explicit solver:

- Impact Analyses: In the design of protective gear such as helmets or padding used in football, hockey, rugby, or cycling for scenarios that involve high-speed interactions and large deformations. Imam et al. [32] developed a finite element model for assessing the efficacy of rugby body padding in reducing the risk of cuts and abrasions.
- Ball Impact: Impact of a tennis ball with the court surface [33].
- Impact Testing: For baseball or cricket, understanding how a ball reacts upon impact with a bat at high speeds.
- Material Failure Analysis: Simulating scenarios where equipment might fail, such as a wooden baseball bat or composite helmet cracking.
- Dynamic Structural Analysis: Investigating the dynamic behavior of sports equipment during use, such as the flexing of a ski during a slalom race.
- Athlete Biomechanics: Studying the effects of sudden movements or impacts on an athlete's body or understanding the complex interactions between the athlete and equipment or playing surfaces, such as a runner's foot striking the ground.
- Durability Assessment: Testing the durability of sports equipment, such as how a helmet withstands a fall or a bat breaking during a bat/ball impact.

Figure 6.5 shows a finite element model of a baseball and a wooden baseball bat. The finite element models of the ball and bat are each built using solid brick elements. A mesh refinement study (see the next section) was conducted to determine the number of elements needed in the ball and

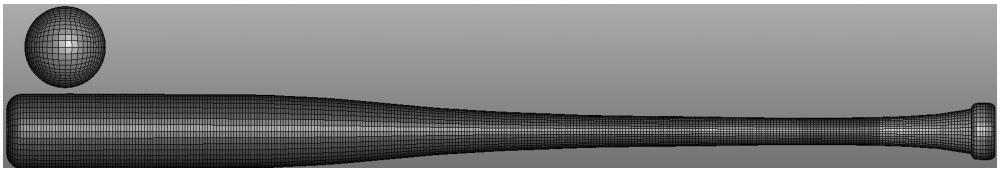

Figure 6.5 Finite element models of a baseball and a wood baseball bat.

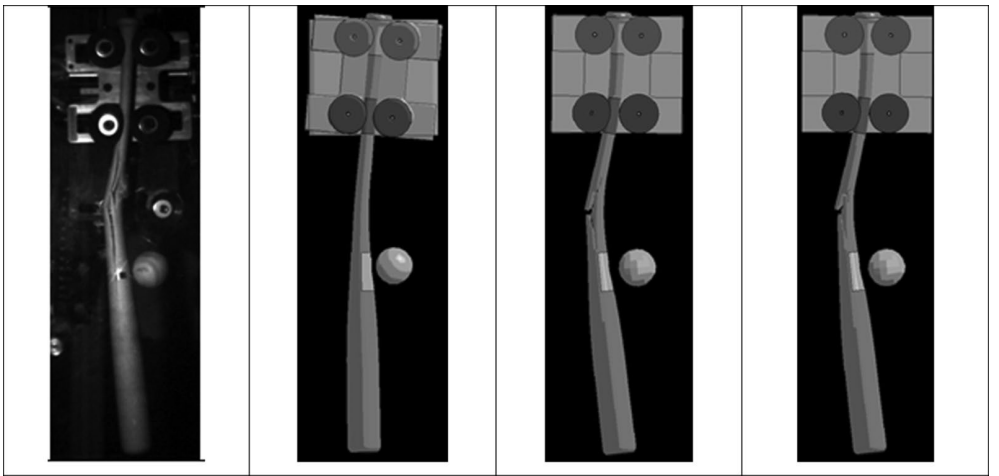

Figure 6.6 Wood bat being broken in the (a) durability testing machine and finite element models of the same test setup at three speeds (b) 135 mph, (c) 140 mph, and (d) 145 mph.

bat to achieve a converged solution, i.e., the stress distribution did not change if the number of elements was increased beyond that shown in Figure 6.5. This model has 12,096 solid elements in the ball and 130,554 solid elements in the bat.

Figure 6.6 shows the finite element model of a yellow birch wood bat being impacted by a baseball as the bat is supported in a durability testing machine. These analyses were completed using the explicit solver in LS-DYNA. The test machine can "throw" a baseball at a prescribed location on the bat at speeds up to 200 mph (322 km/hr). Figure 6.6a shows the bat breaking in the durability testing machine at an impact speed of 147 mph (237 km/hr). In the finite element models depicted in Figures 6.6b–d, the speed of the baseball was incrementally increased from 135 mph (217 km/hr) to 145 mph (233 km/hr) to show the evolution of the bat from not breaking, to cracking, and then to ultimate failure. This kind of modeling can give the sports engineer insight into how the bat breaks and can guide modifications to the variation of the diameter along the length of the bat to mitigate wood bat breakage. For the bats shown in Figure 6.6, the *MAT_WOOD material model in LS-DYNA was used in conjunction with the *MAT_ADD_EROSION option to simulate damage [34]. The *MAT_ADD_EROSION option was also used by Imam et al. for modeling soft tissue damage [32].

Computational efficiency

When building a finite element model, the user should keep these words in mind:

> Make everything as simple as possible, but no simpler.
>
> – Albert Einstein

Building the mesh of nodes and elements for a finite element model is sometimes more of an art than a science. The finite element preprocessor makes it easy to create a mesh with many elements, ensuring that every minute detail of the geometry of the structure is captured. However, such details may not be necessary for the analysis under consideration or may not be computationally reasonable. Thus, it is essential to follow Einstein's wise words: keep the model as simple as possible, but do not make it so simple that the model does not capture all that the sports engineer needs to gain useful insights into the problem being investigated.

Mesh density

The fidelity of the finite element model is a function of the size of the elements, i.e., the smaller the elements, the better the resolution of the stress variation in the structure. However, increased fidelity comes with increasing computer run time and greater resource requirements, such as memory and disk space for data storage and workspace. Thus, it is recommended to start with a coarse mesh (low number of elements, low density) and proceed to increase the number of elements (increase mesh density) to enhance the fidelity if needed. A mesh refinement study is one approach to determine if a higher-density mesh is required. First, build as "coarse" a mesh as is reasonable to capture the geometry sufficiently. Then run the analysis and post-process the data to see the stress or displacement state. Use the option in the preprocessor to split elements; for example, solid elements can be split in each of the three directions ($2 \times 2 \times 2$), thereby increasing the number of elements by a factor of eight. If the "coarse" and "refined" solutions are within a small percentage of difference, then the simpler coarse mesh may be sufficient. However, if there are large differences between the two analyses, then another refinement should be made. This mesh refinement can continue until two consecutive mesh densities give essentially the same result, i.e., within a few percent.

Another option to performing a manual mesh refinement is to use an "adaptive" meshing option in the finite element solver. When this option is turned on, the solver will use the stress gradient, i.e., the rate of stress changes over an area, to refine the mesh automatically in a local region or across the entire structure.

Order of element integration

The finite element method uses Gauss points to perform numerical integration. The details of this numerical integration are beyond the scope of this chapter, but the application of this numerical integration is important for the level of output detail that can be achieved with a finite element model as well as for the computational efficiency of the analysis. In general, single-point (or reduced integration) integration is acceptable for an explicit solver analysis where the mesh is sufficiently refined. However, there may be times in an implicit analysis where using more integration points, e.g., $2 \times 2 \times 2$ integration (or full integration) for a brick element, and fewer brick elements are preferred over a fine mesh of brick elements using single-point integration. The reduced integration will assume a constant strain within the element, while the full integration will provide a parabolic distribution of strain. The advantage of reduced integration over full integration is a shorter run time. The disadvantage is that more elements are required to achieve the same fidelity of the stress contours as can be achieved with full integration. General practice is to use reduced integration for explicit solver analyses and full integration for implicit solver analyses.

Fluid-structure interaction

Fluid-structure interaction (FSI) is the interaction of some movable or deformable structure with an internal or surrounding fluid flow. One commercially available computational fluid dynamics (CFD) solver is the ICFD (Incompressible Computational Fluid Dynamics) solver, an integrated computational fluid dynamics tool developed by LSTC (Livermore Software Technology Corp., now part of ANSYS), used in conjunction with LS-DYNA. Readers seeking more information on CFD are referred to Chapters 3 and 4, "Computational fluid dynamics applied in sport."

The ICFD solver allows for seamless interaction between fluid dynamics and structural mechanics, where there is a mesh for the structure and a mesh for the fluid. Just as the quality of the mesh is important for conducting structural analyses in FEA software such as LS-DYNA, the quality of the mesh for the fluid is crucial for accurate ICFD results. Thus, the modeler needs to define the fluid region and apply an appropriate mesh that captures the essential dynamics of the fluid flow. The default meshing in ICFD is that the volume mesh is automatically generated based on the user-defined mesh boundaries. Thus, the user can control the mesh density by defining the element size in certain regions. The user also defines boundary conditions for the fluid, such as inlet velocity and pressure conditions, and fluid properties, such as viscosity and density, relevant to the specific problem.

The LS-DYNA ICFD solver uses a partitioned solution approach to solve the fluid and structural domains separately and then iteratively exchanges load and response information between them. The partitioned solver approach is suitable for scenarios where the interaction between the fluid and structure is significant but not dominant. An example might be the analysis of wind loads on a sailing yacht [35]. The uncoupling of the solvers has the advantage over a coupled solver methodology in that the uncoupled solvers approach is generally less computationally demanding than the coupled solver.

An FSI model includes a mesh of the fluid model and a mesh of the structural model. The analysis can be two-way or one-way coupling. The FSI model may be one of three options:

- Two-way coupling: Loads and displacements are transferred across the FSI interface, and the full nonlinear problem is solved. It gives weak FSI coupling when the fluid solver is coupled to an explicit mechanical solver and strong FSI coupling when the fluid solver is coupled to an implicit mechanical solver.
- One-way coupling: The solid mechanics solver transfers displacements to the fluid solver.
- One-way coupling: The fluid solver transfers loads to the solid mechanics solver.

The loose coupling analysis is a simplified approach where the interaction between the fluid and structure is weak or incidental. In this method, the fluid and structural domains are solved independently, and their interaction is considered in an approximate manner. This method is used when the structure's influence on the fluid flow, or vice versa, is minor. An example might be the impact of a slow-moving fluid on a large, stiff structure, e.g., the case of a baseball moving through the air, where the deformation of the baseball can be considered negligible compared to the motion of the fluid as the ball travels through the air. The advantage of the loose coupling solver is that it is the least computationally intensive method, making it suitable for quick, preliminary analyses or for large-scale problems where detailed interaction modeling is not feasible. The downside is lower accuracy in capturing the nuances of FSI compared to strong FSI coupling. An example of strong FSI coupling would be a parachute descent, where the interaction between the air and the parachute canopy and suspension lines is significant. The parachute descent situation is a continuous

interaction where the fluid (air) affects the structure (parachute and suspension lines), and the structure affects the flow field. Hanna [36] provides a nice overview of FSI in sports.

Applications in Sports Engineering:

- Rowing: Interaction of the oars with the water (Leroyer et al. [37]).
- Aerodynamics of Vehicles in Motorsports: Analyzing the airflow around racing cars to optimize their shape for reduced drag and increased downforce.
- Swimming Gear Design: Optimizing the design of swimsuits and caps to reduce water resistance.
- Ball Aerodynamics: Studying the behavior of balls in sports like golf, soccer, or baseball to understand their aerodynamic properties.
- Athlete Training and Performance: Simulating the airflow around athletes, for instance, cyclists or skiers, to improve their postures and equipment for better aerodynamic performance.
- Ventilation in Indoor Sports Facilities: Ensuring optimal air circulation for comfort and safety in indoor arenas and training facilities.

Tutorials

This chapter provided a brief introduction to the finite element method with respect to element choices, meshing considerations, solver options, and general topics to be considered when using the finite element method to investigate a sports engineering application. The next step in the process of understanding the value of simulation in sports engineering is to explore the creation, solving, and postprocessing of some basic models. A set of tutorials that serve as a good introduction to the application of the finite element method has been developed to complement this chapter. These tutorials are available online at https://www.uml.edu/FEM-Tutorials.

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7

MUSCULOSKELETAL SIMULATION IN SPORT

Motomu Nakashima

Basics of dynamics of machinery

A human body is subject to various forces from inside and outside the body. A force has both direction and magnitude; therefore, it is represented by a vector. Consider a force vector $\boldsymbol{F}$, which acts on the point P on the body, as shown in Figure 7.1a. The magnitude of the vector can be expressed as:

$$F = \left| \boldsymbol{F} \right| \tag{7.1}$$

The magnitude F is a scalar and has the unit N (Newton) in the SI unit system. An action that rotates about another point Q due to the force $\boldsymbol{F}$ is called the moment of the force $\boldsymbol{F}$ about the point Q. When the position vector from the point Q to the point P is denoted by $\boldsymbol{r}$, the moment $\boldsymbol{M}$ can be expressed using the outer product of the vectors as:

$$\boldsymbol{M} = \boldsymbol{r} \times \boldsymbol{F} \tag{7.2}$$

The length l of the perpendicular line dropped from the point Q to a line on which $\boldsymbol{F}$ exists can be calculated as:

$$l = \frac{\left| \boldsymbol{r} \times \boldsymbol{F} \right|}{\left| \boldsymbol{F} \right|} \tag{7.3}$$

This length l is called the "moment arm." The moment arm represents the length of the lever when the force acts as the moment to rotate the body.

Consider a point mass A that has no size with mass m [kg]. The position vector of the point mass A in the absolute coordinate space is denoted by $\boldsymbol{x}$. When the point mass A is subject to a force $\boldsymbol{F}$ and is in accelerating motion, the following equation of motion holds:

$$\boldsymbol{F} = m\ddot{\boldsymbol{x}} \tag{7.4}$$

The symbol $\ddot{\boldsymbol{x}}$ represents the second derivative of $\boldsymbol{x}$ with respect to time, that is, the so-called acceleration. This equation can be considered either as the cause-and-effect relationship in which the

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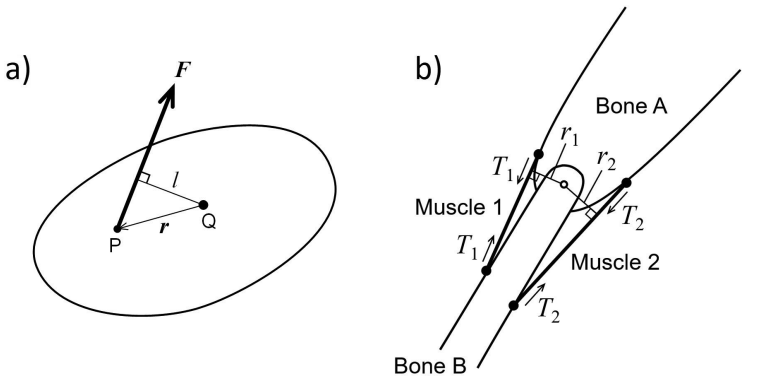

Figure 7.1 (a) A force F acts on the point P on the body. (b) A simplified musculoskeletal model around a joint.

acceleration $\ddot{\boldsymbol{x}}$ occurs as the result of the force $\boldsymbol{F}$, or as the equilibrium relationship of the forces in which the virtual inertial force $\boldsymbol{m\ddot{x}}$ is balanced with the external force $\boldsymbol{F}$.

Next, consider not a point mass but a rigid body that has a size with mass m [kg] and does not deform. When an external force $\boldsymbol{F}$ acts on the center of mass of the rigid body, the equation of motion of the rigid body also becomes Equation (7.4), the same as in the case of the point mass. On the other hand, in the case of the rigid body, the problem becomes more complicated than in the case of a point mass, since it is necessary to consider the rotational motions in three degrees of freedom in addition to the translational motions in three degrees of freedom. For simplicity, first consider a motion in a two-dimensional plane. In this case, the rotational motion has only one component about the axis perpendicular to the plane, resulting in rotation in one degree of freedom. When the rigid body is subject to a moment M [Nm], the following equation of motion of the rigid body for the rotational motion holds:

$$I\dot{\omega} = M \tag{7.5}$$

Note that the symbol ω [rad/s] represents the angular velocity of the rotation about the axis perpendicular to the plane, and $\dot{\omega}$ [rad/s^2] represents the first-order derivative of the angular velocity ω. The symbol I is called the "moment of inertia," which is the quantity corresponding to the mass in translational motion, and its unit is kgm^2. The moment of inertia I can be calculated as:

$$I = \int \rho r^2 dS \tag{7.6}$$

The symbols S, r, and ρ represent the area of the rigid body, the distance from the center of mass and the uniform density of the rigid body, respectively.

In the case of motion in three-dimensional space, the angular velocity becomes a vector and can be denoted by $\boldsymbol{\omega}$. The direction of the vector represents the direction of the rotation axis. The magnitude of the vector represents the magnitude of the angular velocity. The moment that acts on the rigid body becomes a vector, too. If this vector of the moment is denoted by $\boldsymbol{M}$, the equation of motion of the rigid body for the rotational motion can be written as:

$$\boldsymbol{I\dot{\omega}} = \boldsymbol{M} \tag{7.7}$$

The symbol $\boldsymbol{I}$ is a 3 × 3 matrix and is called the inertia tensor.

Application to a human body

Consider applying the dynamics of machinery described above to the human body. A human body has an endoskeleton structure. The skeleton is composed of bones connected by connective tissues, such as ligaments. When a bone can move relative to another bone, that part is regarded as a joint. The joint is moved by skeletal muscles, which are attached to the skeleton. In other words, the muscles are the actuators for joints, just as electric motors are for a robot. Normally, a joint has muscles on both sides to allow movement in both directions. The muscle on one side, when compared to the muscle on the other side, is called the "antagonist muscle."

Consider the musculoskeletal model in which joint parts are simplified, as shown in Figure 7.1b. The muscles 1 and 2, which are respectively attached to bones A and B, exert tensions of T_1 [N] and T_2 [N], respectively. In this case, consider the joint torque T_{joint} [Nm], which rotates the joint and is produced by the muscles. Since the products of the respective muscle tensions (T_1 [N] and T_2 [N]) and the distances between the joint center and the lines of action of the tensions (r_1 [m] and r_2 [m]) become moments, T_{joint} can be represented by:

$$T_{\text{joint}} = T_1 \times r_1 - T_2 \times r_2 \tag{7.8}$$

Here, the torque that rotates bone B against bone A in the clockwise direction is defined as positive. This produced joint torque is used to move the body and to oppose external forces, such as gravity.

To produce a torque that rotates this joint in the clockwise direction, T_2 can be zero. That is, it is sufficient for only muscle 1 to exert a muscle force. Or, if the necessary joint torque is zero, both T_1 and T_2 can be zero. However, in actual human body motions, T_1 and T_2 often do not become zero in such conditions. For example, to oppose an unexpected external disturbance, assume that both muscles 1 and 2 exert muscle forces, and the magnitudes of the first and second terms on the right side of Equation (7.8) become equal, resulting in the total joint torque T_{joint} becoming zero. That is, both muscles pull the joint by the same amount. In this case, the joint seems to become stiff. This action in which muscles on both sides exert muscle forces and the resultant joint torque becomes zero is called "antagonism" of the muscles.

Since the human body does not deform largely except at the joints, it is possible to consider that the body parts, except for the joints, are regarded as rigid bodies, and that the joints are regarded as connecting points of the rigid bodies. In this case, each rigid body is called a "body segment" or simply a "segment," and this model is called the "body segment model." Examples of the body segment model are shown in Figure 7.2. A two-dimensional model in the sagittal plane (the side view for a human) is shown in Figure 7.2a, and a three-dimensional model is shown in Figure 7.2b. The number of divisions for a human body generally changes according to the purpose of analysis. Although the number of divisions for the trunk part often varies, an upper limb is usually divided into the upper arm, forearm, and hand, while a lower limb is usually divided into the thigh, shank, and foot. With respect to the inertial properties of the segments, such as mass, moment of inertia, and location of the center of mass, values measured in experiments are often used [1].

Inverse dynamics analysis

In the analysis of the human body using the body segment model, the joint torques necessary for motion and the produced joint forces can be calculated when the motion of the body segment model is given. This method is called "inverse dynamics analysis." The reason this method is called "inverse" is because of the cause-and-effect relationship between force and motion. It is

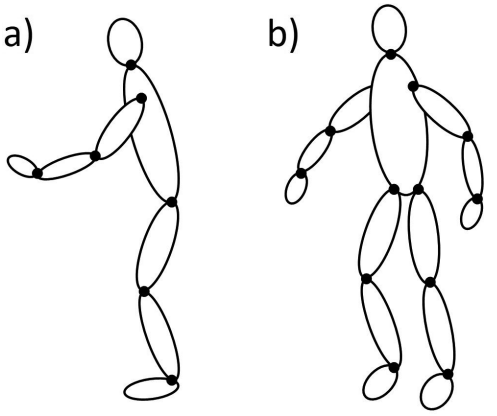

Figure 7.2 (a) Two-dimensional body segment model in sagittal plane. (b) Three-dimensional body segment model.

natural to consider that the order is straightforward when force acts on a human body and motion is produced as the result. From this viewpoint, the order is regarded as "inverse" when motion is given, and force is solved.

In inverse dynamics analysis, loads on a human body (joint torques and forces) are generally calculated from the distal end of the human body. As shown in Figure 7.3a, consider a situation in which a body segment model stands on the ground by the foot. In this situation, the forces acting on the foot segment are the ground reaction force (one force vector representing the distributed reaction forces from the ground), gravitational force, inertial force (due to acceleration), and joint force and torque from the shank segment. Therefore, if the motion of the segment is obtained by experimental measurement, the inertial force and the gravitational force can be calculated. In addition, if the ground reaction force is also obtained, all the forces and moments, except for the joint force and torque from the shank segment, are obtained. Then, from the equilibrium condition of the force and moment, the joint force and torque from the shank segment (labelled (1) in Figure 7.3a) can be obtained. Alternatively, if the joint force and torque from the shank segment

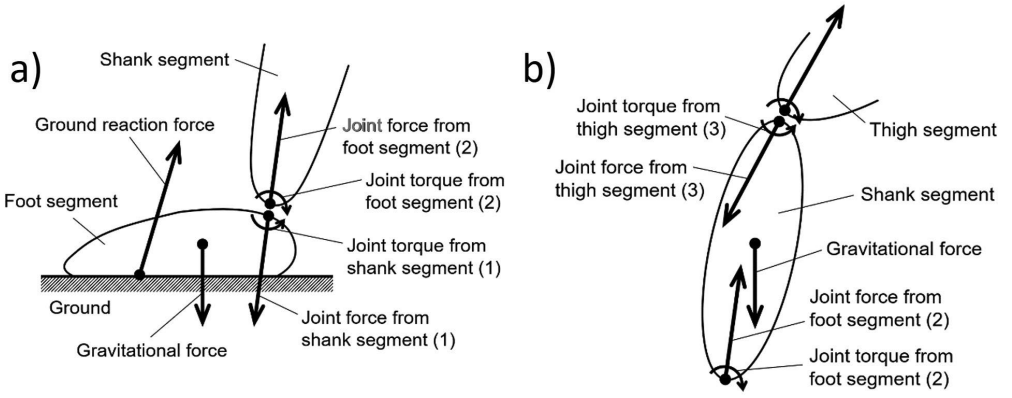

Figure 7.3 Body segment model stands on the ground by the foot. (a) Foot and shank segments. (b) Shank and thigh segments.

are obtained, from the law of action and reaction, the opposite force and torque of those become the joint force and torque of the shank exerted from the foot (labelled (2) in Figures 7.3a and 7.3b). Then, if the gravitational and inertial forces acting on the shank are calculated, the joint force and torque between the shank and thigh at the knee joint (labelled (3) in Figure 7.3b) can be obtained. In this way, the calculation can be done for more proximal segments sequentially.

Recently, the measurement environment for human body motion has made remarkable progress. It is becoming easier to measure simultaneously the motion by a motion capture system and the ground reaction force by a force plate. Some commercially available motion capture systems have the function to directly connect with the force plate and automatically conduct the inverse dynamics analysis, calculating the joint torques immediately after the measurement. Therefore, inverse dynamics analysis is widely used as one of the popular research methods since it has high affinity with the motion measurement system for the human body.

Take a look at an example of a study using inverse dynamics analysis. The golf swing was analyzed with a full-body biomechanical model [2]. In this analysis, the whole golfer's body was modeled as 15 body segments connected by 15 joints in three-dimensional space, as shown in Figure 7.4a. The club shaft was also modeled as 15 segments connected by massless three-dimensional beam elements. The human model was driven by joint motions measured experimentally, and the joint torques and works were calculated, as shown in Figure 7.4b.

Look at another recent example of a study using inverse dynamics analysis. The effect of an arm prosthesis on sprint running for runners with unilateral transradial deficiency was investigated [3]. In sprint races for runners with upper limb amputation/deficiency, it is allowed to use arm prostheses. However, the dynamical effect of an arm prosthesis on sprinting was not directly calculated or discussed for runners with transradial amputation/deficiency. In this research, the arm swing of runners with unilateral transradial deficiency was measured by a motion capture system and analyzed using inverse dynamics calculation. Since the runners targeted in this study had unilateral transradial deficiency, one side of their arm is intact, while the other side is deficient (the forearm is very short and has no hand at all). The targeted runners used conventional pipe-type prostheses attached to their forearms. A flexible-type prosthesis was also proposed

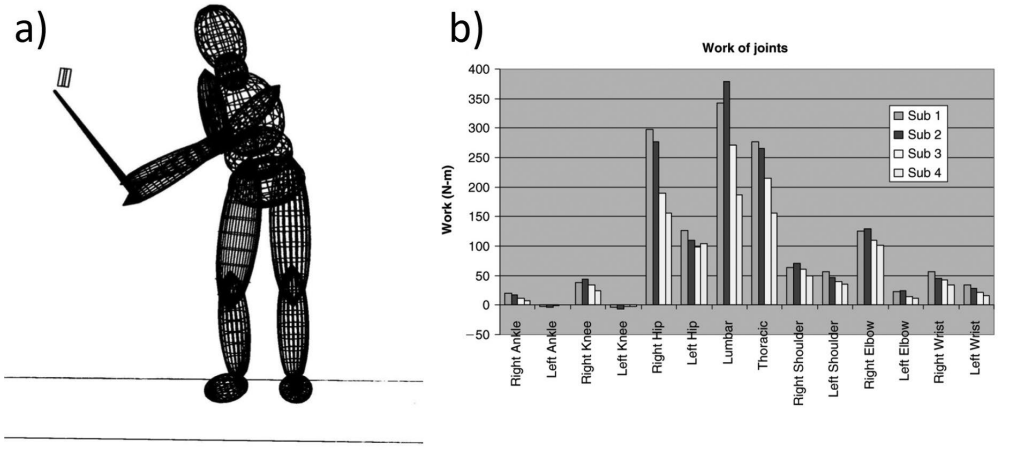

Figure 7.4 An example of study using inverse dynamics analysis [2]. (a) The golfers model consists of 15 body segments connected by 15 joints in the three-dimensional space. (b) The calculation results of work of joints during the down swing.

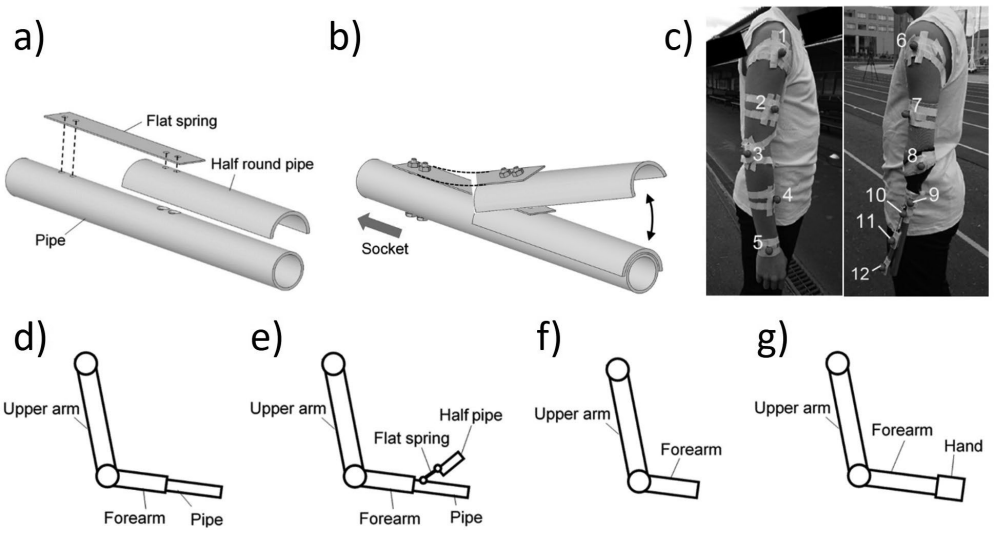

Figure 7.5 A recent example of study using inverse dynamics analysis [3]. (a) A flexible-type arm prosthesis for a runner with unilateral transradial deficiency, which consists of a main pipe, a flexible flat spring, and a half round pipe. (b) The half pipe part moves according to the arm swing motion of the runner. (c) Markers (shade balls) were attached to the runner's upper limb for motion capture system. (d) The body segment model for the upper limb with a conventional pipe-type prosthesis. (e) The model with the flexible prosthesis. (f) The model without a prosthesis. (g) The model for the intact limb.

in this study. The structure of the proposed prosthesis is shown in Figures 7.5a and 7.5b. The marker (shade balls) locations for the motion capture system are shown in Figure 7.5c. The body segment models with the prosthesis are shown in Figures 7.5d–7.5g. The intact upper limb was modeled as a series of three body segments: the upper arm, forearm, and hand, as shown in Figure 7.5g. The deficient limb was modeled as a series of upper arm, forearm, and with/without a prosthesis. A conventional pipe-type prosthesis was modeled as shown in Figure 7.5d. A flexible prosthesis was modeled as shown in Figure 7.5e. The deficient upper limb without prosthesis was modeled as shown in Figure 7.5f. Using these models, the joint forces at the shoulder were calculated by the inverse dynamics analysis. From the analyses, it was found that the arm swing had the effect of increasing the vertical ground reaction force, and that attaching a prosthesis enlarged that effect. With respect to the anteroposterior and lateral directions, it was suggested that the arm swing contributed to maintaining balance and that the prosthesis helped to produce the force which counteracted the force produced by the intact limb. In addition, the effect of the flexibility of the prostheses was found to be very small. Such detailed dynamical analyses can be utilized in the design of the arm prosthesis.

Forward dynamics analysis

In contrast to inverse dynamics analysis, the time variations of joint torques are given and the resultant motion is solved in "forward dynamics analysis." The word "forward" is included because the cause-and-effect relationship is straightforward. Since the joint torques generally cannot be obtained by experiments, forward dynamics analysis is often used for simulations of motion

generation. Since each segment in the body segment model becomes an independent rigid body in forward dynamics analysis, some motion analysis method for a multibody system is necessary to solve the whole-body motion. This problem is called "multibody dynamics," which is quite complicated and itself becomes an academic discipline. There are commercially available analysis software packages to solve the multibody dynamics problem.

Look at a recent example of a study using forward dynamics analysis. How a Paralympic wheelchair curling athlete with tetraplegia controls his body was investigated using forward dynamics simulation [4]. A two-dimensional body segment model of a Paralympic curling athlete with tetraplegia was developed. The body segment parameters were subject-specific, that is, adjusted to the athlete's body size, shape, and mass. Three joints of the hip, shoulder, and elbow could flex/extend to push the stone in the sagittal plane. However, this link mechanism of the athlete's body has redundancy. Indeed, there are numerous combinations of angles at the three joints to move the stone in a certain way. Therefore, the joint torques to drive joint angles throughout pushing the stone were determined by optimizing calculations combined with forward dynamics simulations. In principle, if the joint torques are given, the resultant motion can be obtained by forward dynamics simulation. From the resultant motion, the joint torques, joint power, and angular joint velocities and accelerations throughout the motion can be calculated. If these quantities are used as the objective function to minimize, the optimized motion can be obtained by using some algorithm for the optimization. In this research, GROPS II software [5] was used for the optimization. On the other hand, the joint angles during a wheelchair curling motion were experimentally measured using an inertial measurement unit system. Then, the predicted joint angles from different optimization objective functions were compared with those experimentally measured. It was found that minimizing angular joint accelerations produced the most accurate predictions of the joint angles. Such a method can be utilized in predictive simulations for wheelchair design optimization.

Musculoskeletal simulation

Using the above-mentioned inverse dynamics analysis approach, the joint torques can be obtained for the experimentally measured body motion. Then, consider how the joint torques are realized by forces exerted by the muscles. In the example in Figure 7.1b, the problem is quite simple, since only one muscle contributes to the rotation of the joint in one direction. However, it is rare that only one muscle contributes to the one direction of one joint; rather, multiple muscles normally contribute to it. For example, in the case of the knee joint, the biceps femoris, semimembranosus, and semitendinosus muscles contribute to the flexion, while the quadriceps (rectus femoris, vastus lateralis, vastus medialis and vastus intermedius) and tensor fascia latae muscles contribute to the extension. In such cases, since there are countless combinations of muscle forces that realize a certain joint torque, the distribution of muscle forces cannot be determined from the joint torque condition. In other words, the musculoskeletal system has redundancy. Although the actual distribution of the muscle forces is determined by the cranial nervous system of humans, it is difficult to know the output signals from the cranial nervous system to the muscles. Therefore, in biomechanics, several kinds of algorithm to estimate the distribution of the muscle forces have been devised. The basic idea of these kinds of algorithms is: "in human body, unreasonable distribution in which the load concentrates on one muscle, is not conducted, rather, muscles as many as possible are recruited so that the load does not concentrate on a certain muscle. In addition, total muscle forces should be minimum." Based on this idea, a large muscle should be in charge of a large force, while a small muscle can only be in charge of a small force. Therefore, the load of muscles must be considered according to the size of the muscles. For this purpose, an algorithm to minimize the sum

of the p-th power of muscle stress (muscle force divided by physiological cross-sectional area; PCSA) for the whole body is often used. This algorithm can be expressed as:

$$\text{minimize} \sum_{i=1}^{N} \left(\frac{T_i}{A_i} \right)^p \tag{7.9}$$

The symbols T_i (>0) [N] represent the muscle force of the i-th muscle, N is the number of muscles, and A_i [m^2] is the PCSA of the i-th muscle, respectively. The power p is generally an integer. The PCSA is the cross-sectional area of the muscle, perpendicular to the muscle fibers, and is generally considered to be proportional to the muscular strength that can be exerted. Equation (7.9) is basically an algorithm in which the minimum muscle exertion is obtained so that the sum of muscle stresses of all muscles is minimized. Changing the power p has the effect of adjusting how much protrusion of some muscles is allowed. For example, it becomes a simple sum when p is 1. When p is 2, however, the terms in Equation (7.9) become larger for muscles with larger muscle stresses, while smaller for muscles with smaller muscle stresses, resulting in muscles with larger muscle stresses affecting the total sum more significantly. In addition, if p is ∞, only the maximum muscle stress affects the total sum, meaning that Equation (7.9) becomes a minimization of the maximum (this algorithm is called the "Min/Max Criterion" [6]). For better reproduction of real situations, it is generally suggested that a lower p is recommended for lower physical load intensity, while a higher p is recommended for higher physical load intensity. In many actual analyses, p values of two or three, i.e., minimizing the sum of squares or sum of cubes, are used [7]. Besides minimizing the sum of muscle stress, minimizing muscle activity is used in some cases. In this case, some muscle model, in which muscle force dependency on contraction speed is considered, is used (Hill's model [8] is listed as a representative). In actual calculation, the distribution of muscle force to solve the minimum in Equation (7.9) is determined by an optimizing calculation.

A muscle that moves one joint is called a "mono-articular muscle," while a muscle that moves multiple joints is called a "bi-articular muscle." For example, the biceps femoris on the back side of the thigh contributes not only to the flexion of the knee joint but also to the extension of the hip joint. This is because this muscle starts at the hipbone, which is above the hip joint, and ends at the shank bone, which is below the knee joint. For such muscles, some attention may be needed to estimate the muscle force distribution using the above-mentioned algorithm.

Look at a recent example of a study using musculoskeletal simulation. In the section of inverse dynamics analysis, the study about the effect of arm prosthesis on sprint running for runners with unilateral transradial deficiency [3] was taken as an example. The musculoskeletal simulation was also conducted in that study. The musculoskeletal model is shown in Figure 7.6. The human model with unilateral transradial deficiency is shown in Figure 7.6a. The dark lines in the figure are the muscles modeled as wires. To represent the deficiency, some modifications from the standard model were conducted. First, the length, width, and height of the (non-existent) deficient hand were set to negligibly small values, and its mass was set to zero. The insertions (attaching points) of the muscles in the deficient forearm were moved to the distal end of the shortened radius and ulna (the forearm bones). The model of the prosthesis is shown in Figure 7.6b. The prosthesis was represented as an inertia ellipsoid of a rigid body, whose mass and moment of inertia were the same as those of the actual prosthesis, and which was attached to the distal end of the radius. Regarding the motion, the translational movements of the three degrees of freedom and the rotational movement of one degree of freedom in the yaw direction were given to the pelvis segment. The flexion/

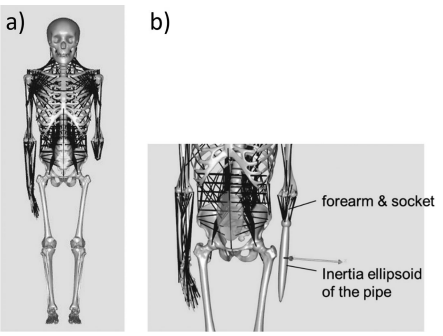

Figure 7.6 A recent example of study using musculoskeletal simulation [3]. (a) The human model with unilateral transradial deficiency. The dark lines are the muscles modelled as wires. (b) The model of prosthesis.

extension, abduction/adduction, and external/internal rotations of the shoulder joint as well as the flexion/extension of the elbow joint were given. The objective function used in the estimation of the muscle forces was to minimize the sum of the cubes of the muscle activities. As a result, it was found that the impulses of the muscle forces (time integrals of the muscle forces) for the deficient limb were basically smaller than those for the intact limb. However, the ratios for the particular muscles were very large. The reason for this was considered to be that the difference in the arm swing between the intact and deficient limbs affected the muscle force characteristics. With respect to the effect of attaching prostheses, it was found that most of the muscle force impulses increased by attaching the prostheses. In particular, the ratios of the brachioradialis muscle were significantly larger than the other muscles for both male and female participants. The brachioradialis muscle is an elbow flexor that starts at the humerus (upper arm bone) and ends at the distal end of the radius (forearm bone). Overall, it was suggested that attaching a prosthesis had the advantage of increasing the vertical ground reaction force, which can be utilized for propulsion, and had the disadvantage of increasing the loads on particular muscles around the shoulder and elbow. Such detailed musculoskeletal analyses can be utilized for runners to determine whether or not to attach a prosthesis and to design a more appropriate prosthesis.

Summary

In this chapter, the dynamics of machinery in sports biomechanics were introduced. The concepts of the body segment model, inverse dynamics analysis, forward dynamics analysis, and musculoskeletal simulation were presented with actual examples in sports.

While these methods are indeed powerful and valuable for analyzing human motion in sports, they assume that the human body can be regarded as a series of rigid body segments connected by joints. Strictly speaking, however, this assumption involves certain simplifications. One simplification is that the effects of motions of soft tissues, such as muscles and fat, are neglected within the body segments. To address this issue, there is an approach to incorporate the effects of motions of the soft tissues [9]. Another simplification is that joint locations are considered fixed in relation to the body segments. In reality, human joints have a highly complex structure, and joint locations can vary depending on the joint angle. Users of the body segment model should be mindful of whether these simplifications could potentially introduce any issues in their specific analysis.

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PART II

Sports materials

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8

INTRODUCTION TO SPORTS MATERIALS

Andy Alderson

Introduction

Advances in materials and their manufacturing processes have driven the development of sports products with enhanced properties for the sport in question. Materials enable optimisation of the physical properties of the equipment for specific sporting activities, and can equally adapt to the dynamically changing conditions and needs of the athlete. The rapid uptake of new materials and technologies characteristic of the sporting goods sector is increasingly being fed by the acceleration of materials design and discovery as part of the digital revolution in this space. Undoubtedly, this materials evolution has contributed to improved athlete performance, to the point of stimulating debate on the fair use of equipment in sports performance and influencing competition rules to take into account the impact of new and improved sports equipment materials [1]. Barriers to the introduction of new materials can include cost and environmental sustainability. The chapters in this section consider all of the above factors for materials in a range of sports equipment. The focus is on cellular solids, fibre-reinforced polymer composites, and advanced textiles, where each of these is underpinned by a microstructural building block (e.g., cell or fibre) approach to the final foam, lattice, composite, or fabric. For each of these, the choice of constituent materials, their geometric arrangements, relative ease of design and simulation, and the ability to host or incorporate sensing, actuation, or smart functionality, has opened a panoramic library of material options and solutions for high-performance sporting goods.

In Chapter 9, Duncan, Bailly, Kravchenko, and Behr introduce the class of cellular solids (honeycombs, foams and other 2D, 3D, and 4D lattices) [2]. They demonstrate how these materials enable tailored, and in some cases counterintuitive, mechanical properties through rational design (including so-called mechanical metamaterials) that have found applications in a variety of sporting impact and vibration damping scenarios (including helmets, running shoes, hockey sticks, and rackets). Consideration is given to manufacturing, testing, and computational design developments in the drive towards sports products that are safer, lighter, and more comfortable for the athlete. The same authors then provide more detail on micromechanical models for the structure and properties of cellular solids in Chapter 10. These models facilitate the selection or design and development of honeycomb, lattice, and foam materials for use in protection and other sports applications discussed in the previous chapter.

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Chen, Lyu, Smith, and Smith point out in Chapter 11 that the combination of light weight and high stiffness of cellular solids finds application as the core material in sandwich panel composites used in, for example, sailboat hulls. However, the main focus of their chapter is predominantly on fibre-reinforced composites [3] which typically form the skins in sandwich panels. Beneficial properties of fibre-reinforced composites, derived from the reinforcing fibre and polymer matrix constituents, include controlled or tailored stiffness, light weight, high strength, impact energy absorption and dissipation, penetration resistance and corrosion resistance. Consequently, alongside advances in manufacturing and the ability to form complex-shaped components, these materials deliver enhanced energy storage and return, vibration reduction, aerodynamic performance and vehicle dynamics, finding application in a variety of sporting products. Specific fibre-reinforced composite sports applications considered by Chen et al. include golf clubs, hockey sticks, footwear, vaulting poles, tennis rackets, Paralympic sport, skis, cycles, racing cars, helmets, shoulder pads, and sailing. The authors acknowledge these materials have sparked significant debate on the fairness of the use of performance-enhancing composite-based sports equipment, with well-known examples relating to running shoes and prosthetic running blades. This is an example where advanced materials development has influenced and, subsequently, led to changes in competition rules. Challenges with composites are also addressed. In particular, the prohibitive cost for some applications, and the critical issue of environmental sustainability due to difficulty in the end-of-life recycling of composite materials, are discussed. The sustainability challenge is accentuated by the fast turnover in sports products, and efforts to address this include the development of biodegradable and recyclable natural fibre composites.

The use of fibre-reinforced polymer composites and cellular solids specifically in board sports equipment is given a detailed treatment by Kaufmann, Farzin, and Baudach in Chapter 12. Common reinforcement and matrix materials, and preform types, are reviewed in terms of their manufacturing and attributes lending themselves to board sports equipment. These and other materials such as various types of wood and, to a lesser degree, some metals are then considered for the specific sports board cases of surfboards, skateboards, wake and kite boards, foilboards, and snowboards. The historical development of these cases is rounded off by a consideration of future prospects, including refinements to properties and performance, and incorporation of advanced materials, new manufacturing processes, and sensors leading to bespoke boards providing performance data for athlete enhancement. Environmental and sustainability considerations are, again, recognised as a key issue driving the use of natural and bio-based materials and end-of-life product management. The authors also foresee potential for further spillover into other sports and recreational equipment, as well as more integration of advanced materials and processes to create new board sport disciplines and an increase in the consumer base at recreational and elite levels.

Of course, not all fibres are reinforcing fibres in the manner of fibre-reinforced composites. In Chapter 13, Sabir, Wood and Bennett give a broader consideration of fibres as the building blocks of textile materials for sports clothing [4]. Here, typically, the requirements are for the clothing to optimise response to heat loss and perspiration, whilst providing sufficient mechanical behaviour (elasticity and strength), feel, appearance, and aesthetics. Natural and synthetic fibres, and fibre blends, are reviewed for enhanced performance in terms of combinations of desirable properties for different sports activities. This includes a survey of assessment approaches and commercial fibres. Here, too, sustainability is a critical issue, and the consideration of natural fibres addresses both their benefits and drawbacks, with a need to consider the carbon footprint of the full processing and transportation routes. The authors then describe the features of woven, knitted, and non-woven fabrics, demonstrating the variety of fibre, yarn, and fabric structures available to tailor

relevant properties for sportswear in woven and knitted structures through case studies. Light and thin woven fabrics can generate sufficient compressive force to streamline the athlete's body, whilst the weave structure together with suitable hydrophobic chemical finishes can control water flow. This combination of properties makes them the material of choice for racing swimsuits. On the other hand, the strength, low-stretch, tear resistance, breathability and moisture management requirements of rugby shirts are met through knitted fabrics incorporating bicomponent fibres, alongside different filament counts, hydrophilic and hydrophobic treatments, and surface textures on the face and reverse sides of the fabric. Potential non-woven sportswear applications are envisaged in the future as the technology develops. Alongside these traditional textile technologies, current developments in biotechnology, 3D printing, and smart textile materials are increasingly providing novel and exciting approaches to performance enhancement, sustainability, increased functionality, and monitoring.

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9

CELLULAR SOLIDS FOR SPORTS APPLICATIONS

Part 1 – Macroscopic properties and applications

Olly Duncan, Nicolas Bailly, Olga Kravchenko, and Michel Behr

Introduction

A cell is a small, enclosed space, and a cellular solid is a solid collection of cells. Cellular solids include man-made and natural lattices (including honeycombs—Figure 9.1a, b), where all the unit cells are similar (Figure 9.1a), and foams (Figure 9.1c), where cells are formed naturally or synthetically and have a degree of randomness [1]. The ability to make lattices or foams gives engineers new combinations of properties, like low density or tunable stiffness. Here, we will not cover every type, use and property of cellular solids, and anyone who would like more detail could read Gibson and Ashby's *Cellular Solids textbook* [1], or Nigel Mills' *Polymer Foams Handbook* [2]. Instead, we will summarise the mechanical properties, design and manufacturing methods, recent developments, and open challenges relating to the application of cellular solids in sporting goods. Part 2 will cover micromechanical models, and cellular solids are also mentioned in Chapter 5 Soft materials in sport.

The basic mechanical properties of cellular solids discussed in this chapter (also in Chapter 5 Soft materials in sport) are [3]: (i) Young's modulus (E)—dimensionless stiffness obtained from

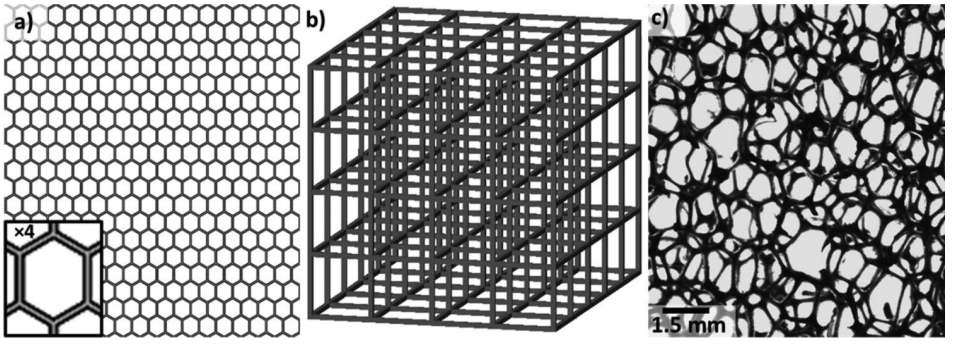

Figure 9.1 (a) A model of an idealised (2D extruded) honeycomb (insert shows single unit cell and ribs extending from junctions, magnified 4×), (b) a lattice with cubic unit cells, and (c) a microscopic image of a ~1-mm thick slice of open cell foam.

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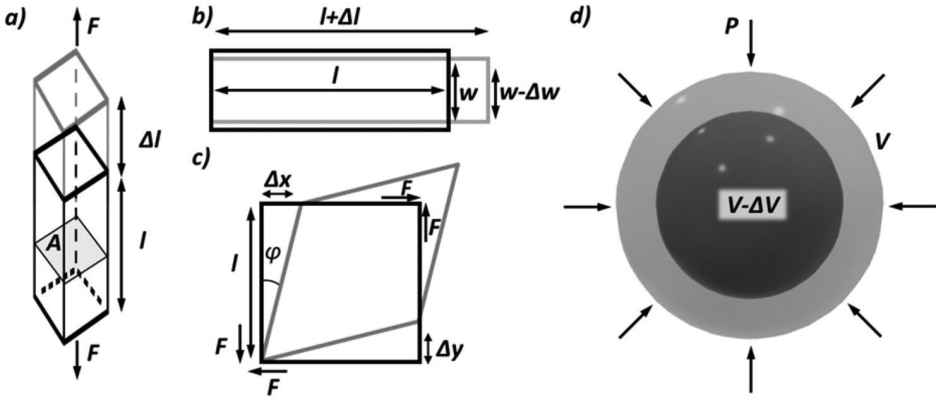

Figure 9.2 Schematic showing (a) a cuboid with length l and cross-sectional area A under applied tensile force F; (b) the transverse contraction (reduction in width, w) of the same cuboid; (c) a shearing cube, deformation $\Delta x = \Delta y$, creating angle ϕ, (d) a solid sphere (volume, V) under hydrostatic pressure (P), shrinking by ΔV.

linear, elastic Cauchy stress (or similar) following applied loading. Young's modulus is often defined as the ratio of engineering stress (σ = load/original cross-sectional area—Figure 9.2a) to engineering strain (ε, resultant deformation/gauge length—Figure 9.2a), over low strain ranges and rates. (ii) Poisson's ratio (v), which is the negative product of the ratio of transverse to axial strain (isotropic limits are $-1 > v > 0.5$—Figure 9.2b). (iii) Shear modulus (G—the ratio of shear stress to shear strain, Figure 9.2c); (iv) Bulk modulus (k, the ratio of hydrostatic pressure (P) to volumetric strain, Figure 9.2d). It is worth noting that engineering stress and strain values can be used at low strains, but tend to provide inaccurate relationships at high strains, where true values are preferred. True strain is the natural logarithm of one plus engineering strain, while true stress is the applied load divided by the cross-sectional area, after accounting for changes in cross-sectional area caused by Poisson's ratio [3].

There are various classes of cellular solids [1,4]. The most distinct divisions in artificial cellular solids are between honeycombs (2D extrusions of repeating unit cells [1,4]—Figure 9.1a), lattices (3D repeating or systematically varying unit cells—Figure 9.1b [4,5], e.g. [6–9]), and foams (Figure 9.1c). Honeycombs, being extrusions of two-dimensional (2D) cells [10–12], are sometimes known as 2.5D [13,14], and can have some small variation in their third axis [12,15,16]. Lattices that change their structure over time or in response to external stimuli are known as 4D [17–19]. Spanning each of these classes of cellular solids are those relating to the base material, which can be made of metals or polymers; elastic, hyperplastic [20–22], viscoelastic [23–25], or brittle—meaning the cellular solid will be crushable [21,26,27]. The boundaries between these classes can be blurred, particularly those between 3D lattices and 2.5D honeycombs with some variation through their thickness, or elastic and viscoelastic polymers. Connected rotating shapes can be cut into solids and are similar to cellular solids [28–31]. Their mechanical properties are achieved by topology (distribution of material [32]) and that of the base material, and indeed some cellular solids include rotating shapes [33]. Rotating shapes are not always cellular, though [28–31].

Lattices can also be classified by the cell shape. These include 2D: hexagonal [1], hexagonal re-entrant [1], chiral (non-symmetric) [6,34,35] and antichiral [6,34] honeycombs, or 3D lattices based on various shapes (Figure 9.3g, h); kelvin (hexagonal extrusions arranged to form squares

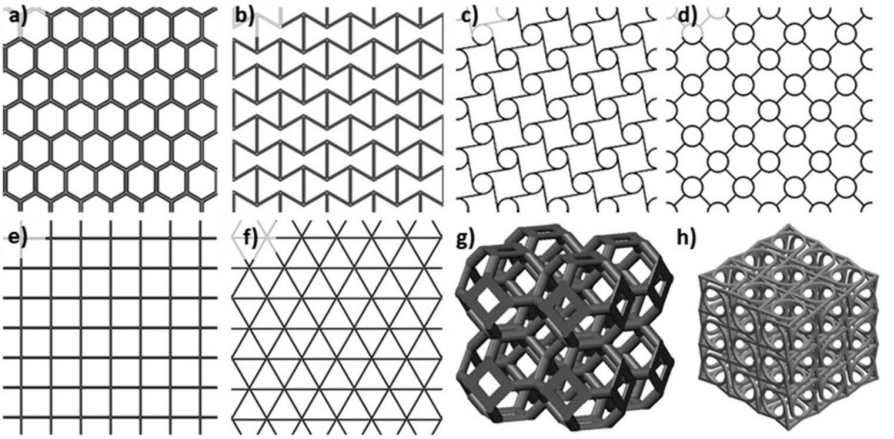

Figure 9.3 Example models of 2D/2.5D honeycombs: (a) regular hexagonal; (b) re-entrant hexagonal; (c) chiral; (d) anti-chiral; (e) square; (f) triangular and 3D lattices; (g) kelvin; and (h) chiral.

between them [4,36], forming a tetrakaidecahedron [37], Figure 9.3g), chiral (Figure 9.3h), or a range of common shapes (cubic, body-centred cubic, triangular, etc. [4]).

Foam (and lattices) can be open-cell, whereby air can flow freely between cells, or closed-cell, meaning the cells are sealed. The solid edges of open-cell foam cells are typically called ribs [1], and these foams are porous (permeable) [38,39], so the cells are sometimes called pores. The faces of closed cells are typically called walls [1], and these foams are not porous.

Readers with an interest in advances in cellular solids may have come across the term(s) metamaterial or mechanical metamaterial (e.g., [40–43]). Mechanical metamaterials are broadly understood to have engineered structures causing combinations of physical properties that cannot usually be achieved in their unstructured constituent (meta) [44], which average out (or can be homogenised) to conform to some type of continuum model (they behave like a "material") [41]. With their properties achieved by topology, certain lattices and foams are mechanical metamaterials—and design concepts used for mechanical metamaterials are key to the development of sports products [35,40,42].

We will first introduce the various sporting applications of cellular solids, giving context to the properties that cellular solids can provide. Then, we will build structural complexity as we describe the topology versus physical property relationships of: (i) simple 2.5D, then 3D, regular hexagonal lattices, (ii) open, closed cell and crushable foams—regular hexagonal lattices with a degree of randomness, and (iii) other lattices, including mechanical metamaterials and 4D lattices. We will end with a summary of testing considerations (specific to cellular solids), analysis methods, and some interesting data-driven advances in cellular solid design and manufacture.

Applications

Cellular structures provide three main sporting benefits: low stiffness, low mass, and tailorable mechanical properties, often based on cell rib or wall buckling in compression [2]. Low stiffness can increase deformation under impact and lower the risk of injury by gradually decelerating colliding bodies. As such, many items of protective equipment use cellular solids for cushioning. Factoring in their low mass, tailorable mechanical properties, and wall/rib buckling (discussed

later), we can wear comfortable, effective personal protective equipment (PPE) with little sacrifice to performance (see examples in the sections on helmets and running shoes). For the same reasons, foams can be used to improve the ergonomics and performance of sports equipment (see the example in the running shoe section). Additionally, their low density means cellular solids have long been applied in the hulls of boats or as liners in various water sports equipment (e.g., surf boards and buoyancy aids) [45].

Impact applications

Cellular solids are used in most sporting protective equipment. They are used in crash mats, traditionally composed of open-cell foams surrounded by a fabric cover [46], and in PPE such as helmets [21], back protectors [47], protective gloves [48], knee [49] and shin guards [50], etc. Two of the most used and studied items of PPE are helmets, as head injury is a key concern for regulatory bodies and sports participants [51], and footwear, the largest sector of the sports equipment market [52]. The rest of this section will focus on these two applications to highlight the typical roles and challenges of cellular solids in PPE.

Helmets

Helmets traditionally have a stiff shell, a liner, low-density foam comfort padding, and a chin strap (Figure 9.4). The rigid shell is usually made of plastic (ABS) or a fiber-polymer composite. The shell resists penetration by concentrated loads while providing structure to avoid liner disintegration during impacts [53]. Helmet liners are typically made of one or several cellular solids that deform under impact to absorb energy while gradually decelerating the head or other colliding bodies [43]. The magnitude of acceleration and the duration of an impact contribute to the risk of brain injury. High accelerations or longer impacts of similar magnitude acceleration pose a greater risk of brain injury [54–57]. Thus, the role of the liner is to minimise peak acceleration and impact duration (e.g., by mitigating rebound). There are various criteria used to predict head injury, summarised in previous reviews (e.g., [54–57]). Historically, the liner of most helmets was made of expanded polystyrene (EPS). EPS provides low cost and density, high energy absorption before

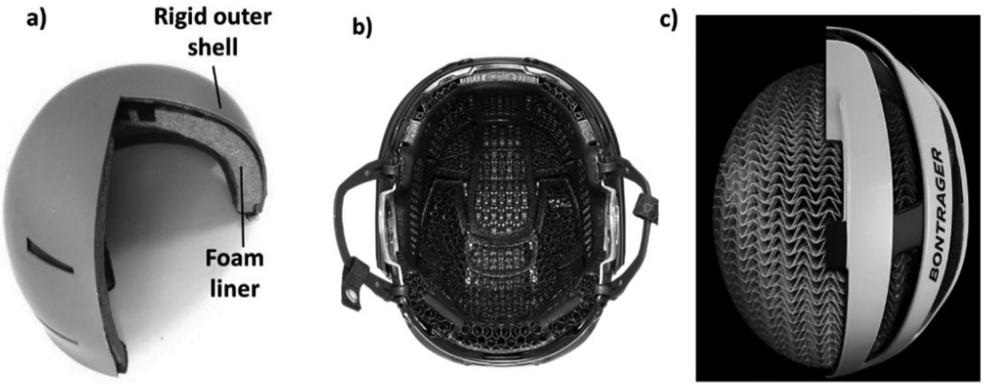

Figure 9.4 Helmet structures: (a) traditional bicycle helmet; (b) an ice hockey helmet with a printed lattice liner (CCM, Super Tacks X); (c) a bicycle helmet with a collapsible cellular liner (Bontrager, WaveCel – image credit, Michael Bottlang).

foam densification (starting at ~80% compression, described in the section on foam manufacture), and a crushable response that reduces rebound and impact duration [21,58,59]. In sports with repeated impacts (i.e., contact sports like American football and ice hockey), EPS is usually replaced by foam that performs well over multiple impacts (e.g., vinyl nitrile, expanded polypropylene) [60–62].

In some of these constructions, in addition to impact mitigation, the cellular solid in the helmet liner also contributes to penetration resistance, provides comfort, or reduces rotational acceleration. Other constructions exist, such as those without an outer shell, 3D-printed lattice liners (Figure 9.4b), collapsible structured liners (Figure 9.4c), or the addition of a device intended to mitigate rotational acceleration [63]. Helmet constructions without cellular solids (e.g., [64]) have been proposed, but these are uncommon and will not be discussed here.

Helmet liner energy absorption mechanisms

When compressed, the stress versus strain relationship of many cellular solids has three phases (Figure 9.5a): (i) linear elastic, (ii) relative stress plateau, and (iii) densification. The energy absorbed per unit volume, or strain energy density (U), corresponds to the area under a stress-strain curve (Equation 9.1).

$$U = \int \sigma \, d\varepsilon \tag{9.1}$$

Energy absorption efficiency is the absorbed energy per unit volume divided by the stress at the point of interest (e.g., foam densification, σ_d) [65–68]. If stress were constant (σ_d) from the start of impact until complete compression was achieved (i.e., when the magnitude of engineering strain is 1), then strain energy density would be the product of σ_d and 1, meaning efficiency would be 1. Thus, efficiency is low during the linear elastic and densification phases and highest during the plateau [1].

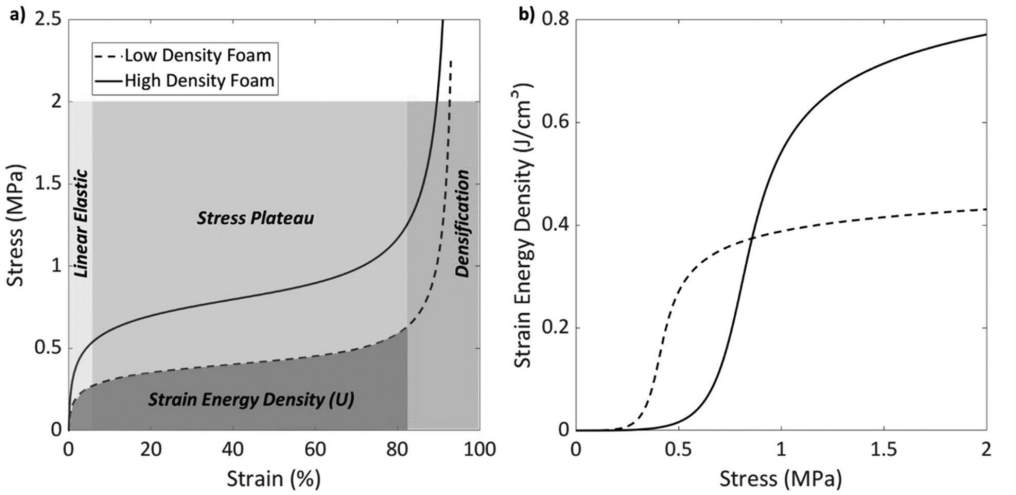

Figure 9.5 (a) Annotated stress-strain curves and (b) energy-absorption diagrams for a dense (stiff) foam, and a less dense (softer) one.

The main factor when selecting helmet foam is the stress at the plateau. The stress at the plateau can be tailored by modifying the cellular solid density, the shape of the cells, and the material properties of the base material [1,69–71]. The stress at the plateau is also usually affected by the impact speed and is often higher during impacts than in slow compression [2,26,65,66], due to the rate dependency of the base material [70,72]. Internal strain gradients caused by local cellular collapse [73] and air contained in closed cells, which requires some time to permeate cell walls [74,75], also affect the stress at the plateau.

Energy absorption diagrams (e.g., Figure 9.5b) can be used to select a piece of foam [1,65,66]. These diagrams present the absorbed energy per unit volume versus the stress (sometimes divided by the modulus of the base material [1]). The best foam for a permitted stress (below the injury threshold) is the foam that absorbs the most energy up to that stress (noting that high stress increases injury risk). Thus, to choose the best cellular material for a helmet, it is necessary to define its energy absorption specifications, along with other limiting factors (e.g., allowable helmet mass and dimensions).

Helmet certification standards

Measures of the minimum level of protection provided by a helmet are typically defined in certification standards. There are many sports helmet certification standards [56], but these are often similar, with a helmeted headform dropped onto an anvil. Pass or fail criteria typically require that the linear headform acceleration remains below a certain threshold. The impact speed and type of anvil (e.g., flat surface, kerbstone, edge, or modular elastic pad, to name a few) vary between sports and certification bodies.

The threshold for maximal linear acceleration of the head is often between 250 and 300 g, based on injury tolerance curves for severe brain injury and skull fracture [76]. For material selection, this acceleration threshold can be translated into a maximal force threshold, where force is the product of the acceleration threshold and the mass of the headform ($F \approx 250\ \text{g} \times 4\ \text{kg} \approx 10\ \text{kN}$). Given that during impact, the compression of the liner (between the headform and the anvil) is distributed over a projected area (A) of about 0.1 m^2 [1], the maximal permitted stress is $\sigma^{max} \approx \text{F} \times \text{A} \approx 1\ \text{MPa}$. Hence, the best materials to pass the certification absorb the most energy up to a stress of around 1 MPa. In other words, the point of interest to calculate energy efficiency is 1 MPa, and the best material has a stress versus strain relationship with a long plateau at ~1 MPa. This desirable stress versus strain response is seen in EPS foam with a density of ~50 kg/m^3, as used in many helmets [21,58,59].

Challenge 1: Helmets to limit rotational acceleration

Historically, protective headgear has been thought of, developed, and evaluated to reduce the head's linear kinematics [56]. This approach was initially based on a correlation between the linear acceleration of the head and the risk of skull fracture or severe brain injury [76,77]. However, there is an increasing volume of evidence that linear kinematics are not the only mechanism of brain injury and that a more refined understanding of brain injury mechanisms may help to evaluate and design protection and other interventions. Since the 1970s, it has been repeatedly shown that rotational accelerations influence brain injury risk [55,78–81]. Rotational acceleration is thought to be the main cause of certain types of brain injury, such as diffuse axonal injury, as the rotational acceleration of the head may cause more shear strain in the brain and connective tissue than linear acceleration [78]. Recent studies, based on finite element modelling of the brain, have allowed

researchers to refine knowledge of injury mechanisms [57]. Suggested risk factors include impact location [82,83], high brain stress, strain, or strain rate [84–87], high axonal strain [88,89], the propagation of shear waves at frequencies harmful to the brain [90], or a combination of brain mechanical and connectivity disruption [91].

The shear stiffness of most helmet foams is considered too high to allow gradual rotational deceleration during impacts. Helmet technologies claimed to reduce rotational head acceleration have been proposed [15,63,92–95]. These technologies usually increase shear deformation in the helmet, either by adding a low-friction layer (not discussed here) or by tailoring the shear behaviour of the cellular material. Indeed, several works have suggested an intertwined relationship between the shear stress level in the cellular structure of the helmet and the rotational acceleration of the head [66,92,96], as seen for compressive stress and linear acceleration (Figure 9.5). Ideally, the cellular structure in a helmet liner would have the correct shear and compressive stiffness at all expected levels of compression or combined shear and compression. As such, the engineering challenge becomes a multi-parameter optimisation, which is not trivial to solve. Several approaches have been proposed to tailor the shear behaviour of the cellular solid in helmets. Developed liners are often anisotropic, varying their properties in different directions to reflect the different direction-dependent requirements. Such anisotropy can be imparted by structuring foam layers [97,98], modifying the cell structure of foams [96,99], designing a lattice or honeycomb structure [11,100–102], taking inspiration from natural protection systems [103,104], or using a cellular structure that changes its deformation mechanism when impacted directly or obliquely [93,105].

Challenge 2: Adapting to different impact conditions

In most sports, the types of impacts can be diverse. For instance, in skiing, the head impact conditions associated with a collision with a tree differ from those during a fall on snow [106]. Similarly, in ice hockey, falls onto the ice differ from collisionsbetween players or the high-speed impact of a puck [43,107–109]. Sports participants expect helmets to reduce the risk of severe head injury in high-energy impacts and the risk of mild TBI (or concussion) in low-energy impacts [87]. Yet, linear and rotational acceleration thresholds for concussion are well below the 250 g tested in most standards, for which helmet cellular materials are usually selected [55,110]. Several tests have been developed to account for the variety of impact conditions in sports, their epidemiological frequency, and their severity [107,108,111–113]. Thus, designing the helmet liner adds another layer of complexity into the multi-objective optimisation process.

Helmets should deform sufficiently during low-energy impacts, which may be frequent, to reduce the risk of concussions or repetitive impact injuries [114]. Helmets should also avoid over-compressing the foam and exceeding severe injury thresholds during high-energy impacts [56]. Two main approaches have been used. The first is to use multi-layered or graded structures in which part of the material is designed for less severe impacts, and part is designed for more severe impacts [115–118]. The second approach is to use rate-sensitive materials in which the stress at the plateau increases with the loading speed [94,119]. As knowledge of head injury develops, engineers face new challenges in helmet design, so a broad understanding of cellular solid deformation mechanisms will play a key role in reducing concussion risks [43].

Performance and cushioning: Running shoe case study

Running shoe design differs from helmet design in that rebound is desirable, and they must withstand thousands of impacts [120,121]. Running is widely practiced, although it carries injury risks.

Almost 30% of people who run regularly have suffered running-related injuries [122]. The most common running injuries affect the lower limbs and can involve soft tissue [123–126]. Runners' injuries can also affect bones, including acute bone fractures, or more often, so-called fatigue or stress fractures [127,128]. The repetition of impacts and the amplitude of impact forces contribute to injury risk [129]. Here, we will examine how foam can mitigate the injurious effects of these impacts. Readers are also referred to Chapter 24, Biomechanical Interactions with Footwear.

Runners' shoes soles, structure and materials

The amplitudes of running impact forces are about three times the weight of a runner [130]. The sole mitigates the effects of these impacts by reducing peak force while absorbing impact energy, a process known as cushioning. The midsole provides cushioning [129,131,132] and also alters the running experience, influencing running patterns. The heel-to-toe drop distance of most running shoes (the difference in midsole thickness between the heel and the ball of the foot) affects running patterns and injury risk [133]. Similarly, bending stiffness may also influence injury risk and performance [134].

Running shoe midsole foam hardness is typically around 30 Shore A, with a modulus between 1 and 5 MPa [121]. The most common materials used in midsoles are poly(ethylene-vinyl acetate) (EVA) and polyurethane (PU) foams. PU foams are considered more durable [135] but less efficient (more viscoelastic). Softer soles were previously reported to reduce injury risk [136,137], although there is little consensus.

Several studies have evaluated midsole performance, degradation, and overuse injuries. Usual test procedures are based on constant-energy drop tests like those in ASTM-F1614 [138], sinusoidal compression [139], or repeated drop tests with force-time curves that mimic a runner's ground reaction force pattern [140]. Constant-energy compression tests appear better adapted to quantifying midsole performance than force-imposed or displacement-imposed compression tests, as midsole properties and a runner's technique change simultaneously. Midsole aging (over 100 miles of mechanically simulated running) can decrease thickness and energy absorption before the onset of foam densification, increasing peak impact force [141,142]. This finding has led to suggestions that the American Association of Podiatric Sports Medicine may be too generous in recommending a total running distance of 350 miles for shoe replacement. Conversely, more recent studies have shown such foams to recover after rest periods during mechanical aging. Some of the degradation was attributed to cell wall ruptures, gas being pushed out of cells [120,143,144], and changes in polymer bonding [141].

In recent years, materials claimed to improve cushioning and spring for runners have been developed. Expanded thermoplastic polyurethanes (E-TPU) and polyether block amide (PEBA) elastomer foams [145] are two examples of these new "super-elastic and lightweight" midsole materials. EVA foams infused with graphene have also been proposed as options to potentially improve cushioning and rebound [146]. The cushioning and rebound of a midsole can be addressed by three parameters: (i) ***Energy Input (J)***: Energy transferred into the midsole heel area by the runner's foot. This energy equals the area under the force/deformation curve during the loading or compression phase. (ii) ***Energy Loss (J)***: Energy absorbed by the foam and dissipated into thermal energy or lost by other forms of hysteresis (e.g., foam plasticity or enthalpy in gases pushed out of closed cells, as shown in Figure 9.6). The energy loss equals the area under the force/deformation curve during the loading phase minus the unloading phase (hysteresis). (iii) ***Relative Energy Loss (%)***: The ratio between energy loss and energy input. Relative energy loss is a common criterion used to define the ability of a shoe midsole to cushion impact.

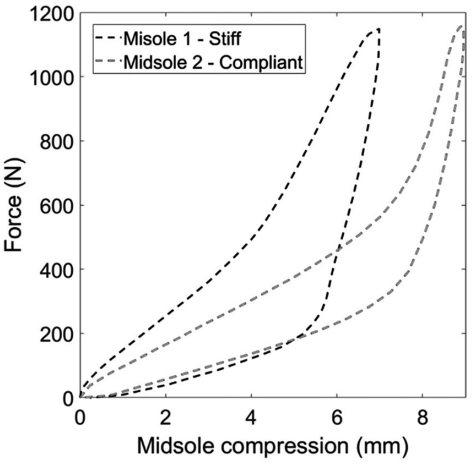

Figure 9.6 Loading/unloading curves for two 10-mm thick midsoles with different cushioning/spring properties.

The loading and unloading curves of two midsoles are shown in Figure 9.6. The two midsoles exhibit different cushioning and spring properties when impacted under constant energy conditions, such as those detailed in ASTM-F1614 [138]. Midsole 1 is stiffer than Midsole 2. At the end of the loading phase for Midsole 2 (above ~7 mm compression), the nonlinear force/displacement is due to foam densification (see Figure 9.5). During unloading, Midsole 1 exerts a notably lower force at each displacement increment than during loading. In these examples, Midsole 2 seems too soft, entering the densification region, while Midsole 1 appears less efficient (evidenced by a larger hysteresis loop).

Challenge 3: Towards user specific midsoles

Running style and variations in fitness, size, gender (see Chapter 24), and other basic differences affect shoe requirements. As a result, manufacturers are increasingly interested in customising shoes. For example, during a study carried out at the Laboratory of Applied Biomechanics (Université Gustave Eiffel, France), on behalf of a shoe brand, user-specific midsoles were developed. Midsoles were customised using 10 mm-thick removable heel pads, made of 1 mm layers of different foam. The personalised heel pads exhibited approximately 40% lower ground reaction force and 20% higher efficiency. Several other attempts to develop user-specific midsoles have been proposed, including the use of 3D-printing. These 3D printed midsoles have also shown promising results, reducing injury risk metrics [147], improving the direction of impact energy transfer [148], and enhancing energy absorption before densification [149].

Cavity/liners

Cellular structures provide support and damping to sandwich structures and reduce risk following cracking when used in water sports equipment (e.g., the hulls of boats, surfboards, and paddle boards). Most sandwich structures have stiff, thin shells with a cavity filled by a cellular solid (Figure 9.7a). Thin layers of stiff materials (such as metals, plastics, or composites) can be designed to have high bending stiffness by using hollow cross-sections (Figure 9.7b). The bending

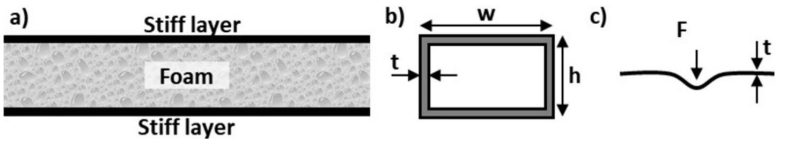

Figure 9.7 Schematic of (a) a sandwich structure, (b) a box section (w = width, h = height, t = uniform wall thickness), and (c) a section of the box section under a locally applied load (F).

stiffness of a beam is proportional to its second moment of area (I) and the material's Young's modulus. For example, if the cross-section in Figure 9.7b were extruded to become a beam, its second moment of area would be:

$$I_x = \frac{wh^3}{12} - \frac{w_I h_I^3}{12} \tag{9.2}$$

where, w and h are width and height, respectively, x is parallel to w, and the subscript *'I'* denotes internal dimensions. Since I_x is proportional to h^3, thin walls (i.e., large values of h/t) provide low mass but high bending stiffness, i.e., high specific stiffness. Hollow structures can be sensitive to localised impacts, particularly through thickness (Figure 9.7c), where the low second moment of area of the thin upper plate defines the bending stiffness. In water sports, failure of hollow sections can range from costly to life-threatening. To address this, closed-cell foams or similar honeycombs or lattices are added to provide support, maintain buoyancy in case of cracking, and increase vibration damping [45].

With their low mass and high stiffness, travelling waves moving through such structures have a high velocity. The speed of a travelling wave (v) can be estimated as [1]:

$$v_{ij} \approx \left(\frac{1}{\rho} \left(C_{ijkl} \right) \right)^{1/2} \tag{9.3}$$

where, ρ is density, and C is an elastic modulus tensor in an axis parallel to the travelling direction of the wave (denoted by the subscripts *i, j, k, or l* = 1, 2, or 3). Energy dissipation of a travelling wave is maximised at low speeds; low stiffness, dense materials provide high damping, while stiff, hollow structures provide low damping. Foam cores with low stiffness, which also add mass to a sandwich structure, can increase the amount of damping [150,151]. The viscoelasticity of these cores may also help dissipate vibrational energy as heat energy [152]. As such, foam cores surrounded by stiff composites are being used in various bats, sticks, and rackets, including hockey sticks [153] and paddle rackets [154], a sport growing in popularity but with limited research to date.

Lattices

Sports engineers interested in developing equipment that includes honeycombs or lattices should understand how honeycombs are made and where limitations are likely. For descriptions of how structuring materials affects their mechanics, see Chapter 10. Lattice manufacturing, like many other forms of manufacturing, is typically subtractive, assembly-based, additive, or formative [155]. Simply put, the voids can be cut using subtractive methods [156–158] (Figures 9.8a–d),

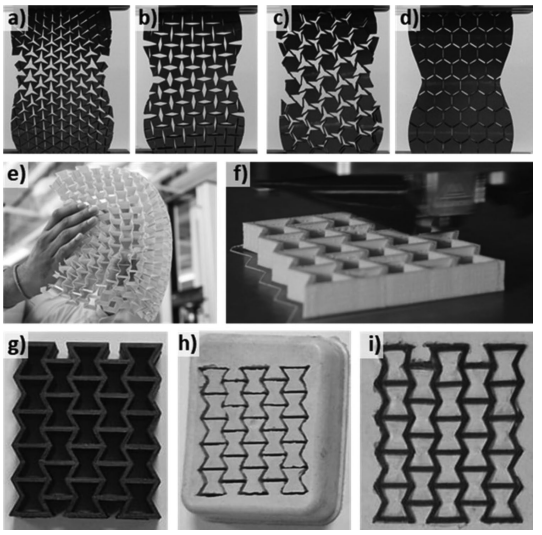

Figure 9.8 Example of (a) rotating triangular; (b) rotating square; (c) chiral; (d) three-point connected rotating hexagons laser cut into foam (image credit – C. Moroney [163]); (e) to (i) 2.5D re-entrant lattices created (e) using kirigarmi (cutting and sticking, image credit – A. Alderson [35]: (f) by 3D printing (image credit: T. Shepherd) and (g) to (i) (image credit – A. Dunleavy) soft moulding, (g) is the hard tool used to make the mould, (h) is the soft mould, and (i) shows a moulded polymer lattice ready for extraction.

the ribs can be prefabricated and assembled [5] (Figure 9.8e), extruded using additive methods [6,10,12,159,160] (Figure 9.8f), or lattices can be formed by molding or casting [94,161,162] (Figures 9.8g–i). With most of these methods, there is a trade-off between cost and manufacturing precision (both in available materials and structural complexity).

While additive methods like 3D printing allow precisely made lattices and simple topology modification during design, these methods are (at the time of writing) often too costly for mass production in sports equipment [164], with exceptions such as Riddell, Adidas, CCM, and Specialised, which now market high-end sportswear with 3D-printed lattices [165]. Additive methods can also limit the selection of base materials to those compatible with a specific 3D printer type, brand and model—although options are increasingly expanding [166]. 3D printing types include fused deposition modelling (layering of typically polymer filaments—low cost but limited by maximum overhang angles, which cause lattice structure limitations), selective laser sintering or melting of powders (higher cost but capable of producing a greater range of printable structures), and liquid binding of powders [167]. Digital light processing methods, such as digital light synthesis, cure and solidify resin in targeted volumes [168,169]. These methods have gained popularity, particularly in sports equipment [165], as they allow accurate printing of 3D structures using soft polymers [168,169].

Traditional, lower-cost formative or subtractive manufacturing methods can be applied to a wider range of materials than 3D printing, but obtaining variation or voids in three dimensions is not always possible. Moulding highly viscoelastic polymers to create structures for impact protection allows some variation in a third axis, as they can be removed from a mould (slowly) but will stiffen during dynamic loading [94]. Assembly methods, like weaving and joining, can also be used to make lattices and are employed industrially [5]. Similarly, Kirigami (similar to origami,

but also including cutting and joining components together) can be used to make lattices [170–172]. Such methods can be low-cost, or automated, and they are used commercially [1], as seen in products like a recyclable (paper) bicycle helmet [173].

Analysing, designing, and making 3D lattices inevitably adds complexity. Subtractive manufacturing methods require cuts along multiple axes, while options to mold 3D lattices are limited. Extrusion or deposition additive manufacturing methods require support material, which must then be removed. There are various academic and commercial examples using resin or laser sintering to make 3D lattices. While the costs of producing 3D lattices are decreasing, these are still typically considered expensiveand suitable only for low-volume production [164]. Assembly methods are also feasible but require multiple operations to join lattice trusses along three axes [5,156].

Foams

Foam manufacture

Foam cells can be open (Figure 9.9a) or closed (Figure 9.9b). Foams made from a stiff, brittle polymer are known as crushable foams (Figure 9.9c). Those interested in more detailed information on foam manufacturing could refer to Nigel Mills' *Polymer Foams Handbook* [2]. For descriptions of how foam cell structures affect their mechanical properties, see Chapter 10.

Flexible open (Figure 9.9a) and closed-cell foams (Figure 9.9b) are made by foaming—typically blowing a gas into a molten polymer before solidifying or curing [174]. Blowing can be used to make foam from most thermoplastic polymers, with blown EVA, polyethylene (PE), and polypropylene (PP) foam commonly used in sports products [174,175]. The viscosity and elasticity of the molten polymer, along with further processing conditions, determine whether the foam is open or closed cell. The size of the foam pores can be tuned by modifying the molten polymer molecules [174]. Foam can also be made by dissolving a gas in a polymer in an autoclave (under pressure) and then allowing the gas to expand as the foam cures [174]. When applied in a mold, products can be foamed and formed simultaneously, as in helmet and shoe midsole manufacturing.

Some foam, like polyurethane foam, is made by mixing reagents that form both a gas and a solid polymer. The pores are created by the gas, while ribs or walls solidify around these. For example, polyurethane is made from a di-isocyanate and a diol [176]. CO_2 can be generated by simultaneously reacting a 1–4% water content with the di-isocyanate monomer to form the pores.

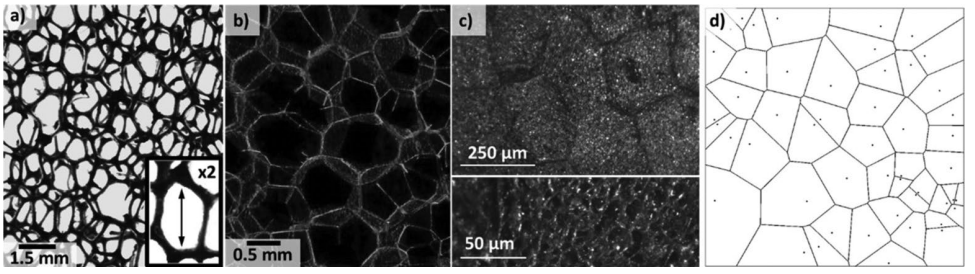

Figure 9.9 (a) Microscopic image of open cell foam structure (insert shows single cell enlarged by two times, and the elongated cell rise direction); (b) micro-computed-tomographic image of closed cell foam structure (from [70]); (c) microscopic image showing EPS helmet foam (top showing bead boundaries, bottom showing bead microporous structure); and (d) Voronoi diagram with dots showing nuclei.

Crushable foams (Figure 9.9c), used as energy-absorbing liners in most helmets, are typically expanded beads of polystyrene (PS) or polypropylene (PP) [177]. Expanded PS (EPS) foam is usually made in two steps: bead expansion followed by fusing. Agents are applied to bead surfaces to prevent them from merging or to control how they merge. Expanded PP (EPP) production is similar, but rods rather than beams are foamed and then pelletised into beads. EPS and EPP beads are steamed, with the steam temperature fusing the beads together (Figure 9.9c – top), while steam absorption into the beads causes expansion to fill a mould (Figure 9.9c – bottom). The shape of the beads becomes similar to the pores in conventional foams. After steaming, the expanded foam shrinks slightly (typically by <1%).

Foam structure often resembles a Voronoi diagram (Figure 9.9d). A Voronoi diagram is a mathematical method to segment space from seed points, with each segment encompassing the space closest to each seed. Cellular solids can form in a similar manner to a Voronoi diagram, with ribs or walls arising at the relative centre between a network of points, or nuclei. The cells are typically hexagonal, with a degree of randomness. The specific reaction, external pressure, and additional blowing agents can affect foam density [176]—or the relationship between rib thickness and length [1]. Nucleating agents can be added to increase the number of points where bubbles form and reduce cell size [174]. The gas can affect the shape of the pores, often producing elongated cells as the gas rises while the mixture is curing (Figure 9.9a) [176]. Other additives or processing conditions (heat, pressure, moisture) can also affect the constituent polymers and copolymers, imparting flame retardancy, thermal stability, or changes in stiffness and viscoelasticity.

Foam mechanics

Open-cell foam

Open-cell foam stiffness is typically low, with Young's modulus ranging from about 10 to 300 kPa [1,22,36,46]. The polymers in open-cell foam are often viscoelastic, and thus the modulus can double under impact when the strain rate increases to >200 s^{-1} [70,178]. To absorb enough energy to bring sporting projectiles or impacting bodies to rest before the foam densifies and peak forces notably increase, open-cell foam pads in isolation must be thick [21,22]. Consequently, open-cell foams are predominantly used in crash pads, as they tend to become too bulky to wear. The gradual deceleration provided by thick open-cell foam pads is well-suited to crash pads, allowing athletes to manage the risk from repeated falls from moderate heights in activities such as gymnastics, indoor bouldering, high jump, and pole vaulting, to name a few [46]. Open-cell foam is also sometimes used as a comfort liner in PPE or other equipment, such as wheelchair seats or prostheses, reducing the likelihood of pressure sores [21,59,99].

Closed cell foam

Flexible closed-cell foam is used in most wearable protective equipment, footwear midsoles, and prosthesis liners [21,35,50,121,142,143,179–182]. Here, the gas trapped in closed cells can affect foam properties. Atmospheric pressure is ~100 kPa, so these foams typically have a Young's modulus above 100 kPa, often exceeding 1 MPa [179]. Gas is a compressible fluid and thus provides support to cell walls, resisting compression while forcing them outwards [183]. Most polymers are viscoelastic, and as with open-cell foams, these flexible closed-cell foams stiffen under high strain rates or impacts. Indeed, the stiffness of open-cell foams typically doubles under strain

rates expected during sporting impacts [66,70]. The contribution of gas also reduces the prevalence of the relative stress plateau in flexible closed-cell foams [50,179,184].

Closed-cell foam typically has lower hysteresis than open-cell foam at typical sporting impact speeds. Hysteresis is usually caused by polymer viscoelasticity rather than energy dissipation by the stored gas [179]. Therefore, while flexible closed-cell foam can reduce peak deceleration or force, impacting bodies may still rebound. (Refer to sections on impact applications and cavity/liners for respective descriptions of when rebound is unwanted or desirable.) For this reason, such foam is used to reduce injury risk in running shoes and from non-life-threatening impacts (e.g., body armor and padding) but is not typically used in helmets. Helmets made for multiple impacts (e.g., American football and ice hockey) often use viscoelastic closed-cell foam [185], such as vinyl nitrate (VN) foams [66,186], which recover slowly after an impact. Helmets designed for single or infrequent impacts typically use crushable foam (see section on impact applications). The modulus of viscoelastic foams can increase dramatically under impact—by approximately tenfold [66,185,186].

In addition to polymer degradation or thermal softening with hysteresis, a secondary degradation mechanism is possible for closed-cell foam. Following cyclic compression (e.g., running), gas can be pushed out of the cells, sometimes rupturing cell walls [120,143,144]. During a run, a season of American football [187], or a cyclic test, foam may soften. It has been shown that closed-cell foam can recover over time as air is reabsorbed into the closed cells [143].

Crushable foam

A critical challenge for sports engineers is reducing concussion risks following head impacts. As discussed in the section on impact applications, various metrics can increase the risk of brain injury [56,57,188]. The key difference from other impact locations is that the brain is suspended inside the skull [57,188,189]. Thus, head acceleration and brain mass determine the boundary forces and internal/surface stresses that define injury risk. The brain is known to be susceptible to damage caused by moderate stresses sustained over time and stresses resulting from rotational acceleration [57,188,189].

Crushable foam recovers and rebounds less than other types of foam after impact, which helps reduce impulse and brain injury risk. As such, crushable expanded foams such as EPS and EPP foams have long been used as energy-absorbing helmet liners in cycling, snow sports, and climbing, among others [56,58,59,190]. This sectionl describes the key deformation mechanisms and differences between each type of expanded foam.

EPS foam exhibits a relative compressive stress plateau, providing high ratios of energy absorption to peak force during impacts [21,26,191–193]. For EPS foam, the plateau is likely caused by plastic hinging/yielding of cell faces [21,184,193], with plastic hinges identified on impacted samples by optical microscopy [21,184]. The relative plateau typically occurs at around 5% compression [21,26,184,191–193]. A secondary failure mechanism, cracking and crack propagation between beads, is also common in EPS foam [177], and depends on the fusion between beads. Higher plastic deformation has also been identified near bead boundaries after compression tests [193], and the bulk material in these boundaries is thought to buckle under compression. As with other foams, densification (beyond 50% strain) causes a sharp increase in stiffness [21,26,184,191–193].

Increasing EPS foam density does not typically increase plateau strain but appears to stiffen the polymer, increasing Young's modulus and plateau stress while decreasing densification strain [26,191–193]. Low-density (~30 kg/m^3) EPS foam has a modulus of ~10 MPa [191,193]; ~70 kg/m^3

can have a modulus of 20 to 30 MPa [26,191,193,194], and ~120 kg/m^3 has a modulus of ~80 MPa [26,191]. Increasing compressive strain rate has a similar effect [26]. Modulus typically increases by two to three times between low (<1 s^{-1}) and high (>100 s^{-1}) strain rates [26,192,195], similar to that of open- and closed-cell foam—likely related to polymer viscoelasticity [70,178].

EPP is less stiff than EPS foam for an equivalent density (e.g., at ~70 kg/m^3, the modulus is <15 MPa [74,194]). The stiffness reduction is similar to that of the base polymer; the Young's modulus of EPP is ~1400 MPa [65,196], while that of EPS is >3000 MPa [197,198]. The relative stress plateau occurs at about 5% compression, and densification begins beyond 50% compression, decreasing to 80% for lower-density foam [65,194]. As with EPS foam, deformation and buckling occur in the cell walls and ribs, as well as in the bulk solid between bead faces [74]. The major difference between the two is that EPP is less brittle and recovers better than EPS after an impact [58], with bead interfaces being less prone to crack propagation [74,177].

Mechanical metamaterials

From the Greek meta, meaning "beyond," metamaterials have unusual combinations of properties derived from their micro/macrostructure rather than their molecular structure. The first metamaterials were discovered by Sir John Pendry [199–201]—fibrous structures with unusual (negative) electromagnetic permeability (resistance to creating an internal electric field). The term has since been applied to other unusual physical phenomena, typically limited to interactions with waves (e.g., electromagnetic [202], acoustic [203], and photonic [204]). To retain their definition as a "material," rather than just an arbitrary shape, the structure of a metamaterial should be much smaller than the wavelength it interacts with, and it should behave according to Maxwell's equations (the foundation of electromagnetism).

The original definitions of metamaterials, which are still retained as the strict definitions, use this relationship between feature size and wavelength. Regarding photonic and mechanical metamaterials, metamaterials may go beyond interaction with waves; photons or continuous/static mechanical loading are not always waves. The UK MetaMaterial Network defines metamaterials as "3D structures made of metals, plastics, and/or ceramics which have properties not possible with only their unstructured constituent materials" [205]. Thus, mechanical metamaterials are broadly understood to have engineered structures causing unusual combinations of physical properties (meta) [44] that average out to conform to some type of continuum mechanics (they behave like a "material") [41]. These include classical Cauchy continuum/Hookean elasticity theory and higher-order theories that consider internal variations in stored energy following the application of a uniform load. In terms of cellular mechanical metamaterials, these can be lattices or foams [41].

Properties

To date, a prolific area of mechanical metamaterials research has been auxetic materials [206]. These negative Poisson's ratio materials (Figure 9.10a) exhibit unusual shape changes and combinations of mechanical properties. The simplest of these is expansion in tension, which is thought to increase the comfort of straps by reducing limiting pressures between the strap and the wearer [35] (Figure 9.10a). Similarly, auxetics form a dome during bending (Figure 9.10b), which has been used to fit a helmet liner, made as a flat sheet, into a helmet [15]. Such behaviour may also enable the development of apparel and PPE that deform with the body [35].

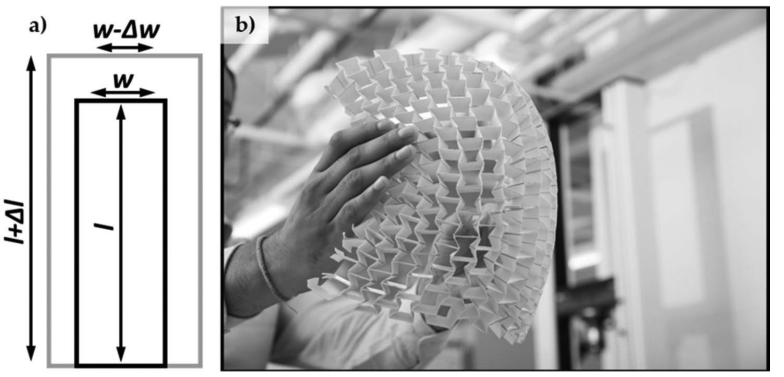

Figure 9.10 Example of an auxetic material (a) fixed underneath side and extended upwards (black lines – undeformed, grey lines – deformed); (b) doming while bent in one direction (photo credit – Prof Alderson, Sheffield Hallam University [35].

Auxetics also exhibit some potentially beneficial mechanical properties. Hertzian indentation force is (F_i) [207–209]:

$$F_i = D\left(\frac{E}{1-\upsilon^2}\right) \tag{9.4}$$

where the term inside the bracket is also the plane stress modulus, and D is a function of the indenter shape and indentation depth. The limits of Poisson's ratio in an isotropic solid are $-1 < \nu < 0.5$ [3]. As Poisson's ratio reduces below -0.5, indentation force increases above that of maximum conventional foam equivalents. The negative Poisson's ratio would also mean auxetics densify underneath an indenter or impactor [210–212] (Figure 9.11), although the benefits of this densification have not been empirically demonstrated. As such, auxetics have been studied and tested under impact [10,99,213]. Readers interested in auxetic materials for sports applications could refer to this review [35].

A related area of interest is materials that exhibit force/torque coupling, such as twist following the application of a uniaxial load [6,214,215] (Figure 9.12a). Twist is a less-studied mechanism for transversely distributing a concentrated load [216]. Interestingly, studying twist is a practical solution for investigating extreme effects of internal material length scales and internal strain gradients—covered in more detail in Chapter 10.

Another negative variant of a usually positive variable, with applications in sports, is negative stiffness. Negative stiffness is typically achieved by snap-through elements (Figure 9.12b), where the buckling of beams causes a reduction in force with subsequent deformation [217]. Such negative stiffness inclusions have been used to create or tune plateau regions in cellular solids, increasing the amount of energy absorbed without increasing peak forces [218]. Similarly, negative stiffness inclusions can also increase damping [217].

A final property of interest is topological protection. Topological protection, usually achieved by materials/metamaterials that break reciprocity, has some potential applications in sporting PPE. Most materials display reciprocity; in Figure 9.12c, if a signal created at point (i) is transmitted through a material (ii) to point (iii), the velocity potential at (iii) would typically have the same magnitude and phase as that at point (i) if the same signal were transmitted from point (iii) [219].

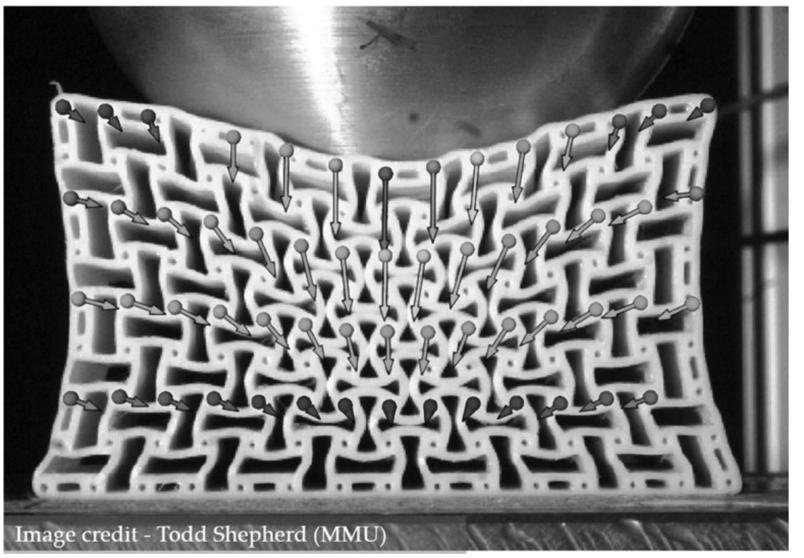

Figure 9.11 Vector plot showing a 2.5D-auxetic lattice flowing to densify underneath a spherical indenter (data and footage collected as part of a published study [10]).

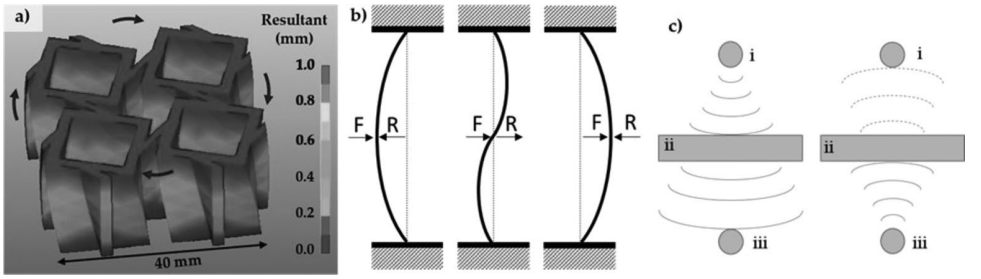

Figure 9.12 (a) A chiral material twisting during through thickness compression; (b) three stages of snap through, whereby *F* is the applied force and *R* is the resultant force; and (c) a non-reciprocal wave response.

Reciprocity has been broken during static loading by creating cellular solids that have different properties when compressed from above or below, typically by activating "floppy modes" on one surface [220]. Such alternate behavior could allow the tuning of layers of PPE—stiffening when indented from the outside while remaining compliant on the inside for comfort.

Multiple anomalous properties can be designed into a metamaterial. By arranging negative stiffness elements, both negative Poisson's ratio and stiffness, occurring concurrently, are possible [221]. Furthermore, tuning when properties occur—such as adaptive stiffness or auxetic behavior being selected only in certain impacts—using electrical control loops [222–224], strain rate-sensitive polymers [94,225], or carefully designed topologies [226], are directions of interest for mechanical metamaterials.

Honeycombs and Lattices

Both 2.5D honeycombs and 3D lattices can be precisely designed to exhibit unusual combinations of properties [41]; they can function as mechanical metamaterials. Indeed, the level of control over topology and base material via additive manufacturing enables the achievement of almost unlimited combinations of mechanical properties. These are designed and produced in a manner similar to lattices, covered in the section on Lattices, although a strong research culture around metamaterials has led to a broader variation of topologies and base materials. These base materials could include biobased or recycled materials, with topology accounting for property limitations associated with such sustainable raw materials [218].

The engineered structures and metamaterial design concepts allow the testing of a wide range of properties, developing fundamental knowledge that can inform sports equipment design. The sporting goods market has been an early adopter of auxetic materials (e.g., [119,227–231]) and negative stiffness inclusions [232]. With growing interest in products that adapt to a user's environment or actions, it is likely we will see commercial interest in 4D-printed materials—those that change over time (i.e., in a fourth dimension) [17–19].

Auxetic foam

Open-cell auxetic foams were first developed in the 1980s [233]. These foams have a re-entrant-like cell structure, causing a negative Poisson's ratio via outward alignment of cell ribs in tension or inward folding of cell ribs in compression (see Figure 9.11a and b). Auxetic foam manufacture typically targets foam cell rib buckling in compression to impart re-entrant-like cell structures. Indeed, Lakes' thermo-mechanical method—compressing foam in three axes to buckle cell ribs, then fixing over time by heating and cooling the compressed foam—is the simplest and most widely used [233]. Developments have increased options to create more types of auxetic foam (e.g., closed-cell [234–237], gradient auxetic foam [183,238–240]), improve manufacturing efficiency [241–243], increase the magnitude of the negative Poisson's ratio [244,245], and reduce foam variability [240,245,246]. The concepts of foam cell structure modification, related to auxetic foam fabrication, have been used to achieve extreme values of foam stiffness/density, adjust the shape of the foam stress plateau, and create direction-dependent Poisson's ratios [70,183,240,247]. The reader is referred to Jiang et al.'s detailed review of auxetic foam manufacture [248]. Auxetic foam is a useful medium for studying mechanical metamaterial concepts (e.g., topology and unusual combinations of properties) [210,239,240,249–252]. However, as the uncontrolled variation of foam cells can be high, it does not strictly fit some definitions of a metamaterial [44,201,205].

Open-cell auxetic foams have been reported to exhibit ~3 to 10 times lower peak forces than their conventional counterparts under low-energy impacts [253–257]. The stark peak force increase, however, is a somewhat binary measure—indicating that the foam has densified (e.g., Figure 9.5). There are some established benefits of auxetic foams and other cellular solids that contribute to the improved performance of auxetic foams under impact. Firstly, auxetic foams can have higher indentation resistance than their equivalent stiffness conventional counterparts [210–212]. Indeed, with a magnitude of Poisson's ratio above 0.3—the typical magnitude for many foams—resistance to indentation increases (Equation 9.4, [210]). Thus, an auxetic pad designed primarily for impacts with a large contact area could offer better protection during impacts with, for example, a stud or a rock, reducing injury risk. As with lattices (Figure 9.11), auxetic foam can densify under an indentation or impact site [210,211], and the negative Poisson's ratio can maintain a non-zero value over a greater strain range than conventional foam [210].

With the higher volumetric deformation due to the negative Poisson's ratio, vibration damping is reported to be higher in auxetic foams than in conventional foams [48,258]. Interestingly, the damping here is not at the polymer level but is related to the interplay between foam structure and polymer viscoelasticity. This higher damping could contribute to the better impact performance of auxetic foams and also makes them suitable for use in seating, gloves, and various types of grips.

While the above benefits are considered positives, other changes to foam properties resulting from the modifications to cellular structure during auxetic foam fabrication also occur. Firstly, the foam density increases (typically by ~3×) following compression during fabrication—potentially increasing the mass of a piece of protective equipment [248]. However, it is possible to select a low-density parent foam for auxetic foam production. There are also some benefits to increasing density, with higher density reducing the speed of traveling waves in a material (Equation 9.2 [151]). The relative stress plateau can also become less pronounced or disappear because of the additional hinge in the foam ribs and the earlier densification strain. Again, this is a potential negative compared to the parent foam, but with potential benefits (e.g., maintaining a non-zero Poisson's ratio over a greater strain range [210]) that should not exclude auxetic foam from use in PPE. For example, it is also well-documented that the plateau is less pronounced in closed-cell than open-cell foams [50,179,184], yet the former are frequently used in PPE.

Auxetic foams have potential benefits for sports applications, particularly related to impact protection and indentation resistance. As with any other set of mechanical properties, designing for a negative Poisson's ratio has broader implications, including resultant changes to manufacturing methods and other foam properties. With new methods to produce auxetic foam and a better understanding of its applications, sports products featuring it are starting to appear on the market [259].

Testing and analysing properties and performance

As sports equipment often undergoes complex loading, such as high strains or rates, sports engineers should undertake material characterisation rather than relying on data sheets. When mechanically testing cellular solids, there are some considerations that differ from those for conventional bulk materials, like polymers or metals. The structural variation and often low stiffness mean product sizing and strain measurement methods need to be carefully selected. Characterisation can be conducted to understand the mechanical behavior of the cellular solid, create material models for finite element modeling (FEM), or validate microstructurally faithful FEMs—and each of which has different considerations. There are already test standards for most types of foam, so we will review these first before discussing lattices.

Testing foam

For flexible foam, both open- and closed-cell, ASTM D3574 covers measurements, tensile testing, compression testing, and indentation testing, to name a few [260]. For all these tests, controls on environmental conditions are specified: 45% to 55% relative humidity and 21°C to 25°C. Polymer foams are known to be affected by moisture [261], and controlling these conditions allows for comparisons between tests. For compliant foams, seemingly basic measurements (e.g., of sample width) can show variation, as the foam is easily compressed by a set of Vernier calipers, so care is advised [260].

To account for Saint-Venant end effects, the standard specifies that tensile dog-bone samples should be cut based on those in ASTM D412 [262], meaning pure tensile stress is expected in the gauge length. To ensure a minimum number of pores is included and that the foam behaves

as a bulk material, thickness should be ≥1.5 mm, length ≥59 mm, and width ≥12 mm [260,262]. Slender samples, with foam gripped by device jaws or with fitted end tabs, can be used to test samples pre-failure [262]. Ideally, sample length should be much longer (>5 times [262]) than the sample width [183,238,240]. Lengths three times the sample width have been used as a pragmatic compromise, with non-contact (e.g., optical) strain measures undertaken in the centre of samples over a gauge length equal to the sample width [239,263,264]. Where present, the elongated cell rise direction should also be specified in all mechanical tests and should be through-thickness to comply with ASTM D3574 [260]. It is common to vary cell rise orientation to obtain properties in all orthogonal axes [248,265].

While device displacement is an adequate measure to calculate tensile strain when testing dog-bone samples [260], some straining close to the grips and Saint-Venant end effects can cause errors in modulus measurement [266]. Flexible foams can be affected by strain gauges or contact extensometers, so non-contact strain measurement methods, such as video extensometers or digital image correlation (DIC), are advised [260]. DIC on open-cell foam can either use the foam pores to provide a speckle pattern [263], add powder coating to enhance it [210], or add a painted speckle pattern (provided it does not affect foam mechanics) [267]. The benefit of using DIC is that Poisson's ratio and measures of strain variation can also be obtained. Readers looking to use DIC should refer to the iDIC good practice guide [268].

To comply with ASTM D3574, compression samples should have a 50 × 50 mm cross-section and be 25 mm thick, with thickness ≤75% of the sample width [260]. Such samples will be subject to frictional end effects, and so different-shaped samples are commonly used, particularly when Poisson's ratio is being measured [248,265]. Selection of compression sample dimensions to prevent such end effects should aim to maximise sample length while avoiding global buckling—the latter being most important and ensured in the standard sample dimensions. Both DIC and video extensometers have been used to measure axial and transverse strain in compression, over the entire sample or in the centre, avoiding areas close to the compression plates [248,265]. For indentation testing, a 380 × 380 mm, 100 mm thick sheet, and the flat end of a cylinder with a diameter of 200 mm are specified [260]. The large sample and indenter dimensions help to ensure continuum effects, with many cells deforming.

The final class of tests to consider is shear testing, which can be measured directly [269–271], by a three-point bend test [260,270], or observed by mixed compression and shear (most relevant to application in a helmet) [66,191,192]. While shear or bending modulus calculation should be possible from classical elasticity theory, both Lakes and Novak noted errors, particularly as foam cell size increased relative to sample size [269,271]. Indeed, Lakes has undertaken studies applying Micropolar Continuum theory, a variation of classical elasticity, to cellular solids that is thought to work better as sample size approaches that of the unit cell [214,272–277] (see Chapter 10).

For FEM, particularly under impact, to design and understand PPE, homogenised properties are needed. Stress versus strain data at high compressive strain and strain rates are required to replicate the impact scenario. The typical approach [21,26,59,66,70] is to apply specifications in ASTM D3547 [260] but vary the strain rate. Dynamic tests can be performed in a device capable of high speeds or under impact, noting that under impact, the applied strain rate will vary [66]. Poisson's ratio measurements at high strain rates require high-speed cameras [66,70,257]. Related to high strain rate testing, stress relaxation tests, typically for FEM material models, can be conducted in compression, calculating shear stress and strain for Prony series fitting [278]. For each of these tests, foam anisotropy should be considered. It is also worth noting the minimum dimensions of the foam test sample or foam sample in the simulated product to account for the limitations of Cauchy continuum mechanics used to replicate bulk solids [214,272–277] (see Chapter 10).

Testing Lattices

Lattice tests are less developed than foam tests, meaning specific unit cell versus sample size ranges to allow direct comparisons between tests are not typically controlled. With large unit cells, there is likely to be some difference in how strain is measured compared to foams. This problem can be resolved quite simply by placing physical or virtual markers around lattices to apply DIC or using video extensometers [10,216,279]. Defining the effect of unit cell versus sample size and lattice homogenisation methods at high strain are open problems (see Chapter 10). Therefore, sports engineers developing lattices should ensure they prototype and test using samples with a similar number of unit cells and similar dimensions to the product they are developing or studying.

Computational Design

"A computer would deserve to be called intelligent if it could deceive a human into believing that it was human" (Alan Turing)

Computational design (CD) and recent developments in generative artificial intelligence (AI) can help facilitate the application of mechanical metamaterials in sports products. Computational design is a global term for the application of computational strategies to the design process [280]. The development of computing power, access to big data and more design-oriented coding environments using biomimicry and algorithmic thinking have changed approaches to design [281].

Unlike traditional design, where designers typically aim to develop one geometry, CD can quickly propose and evolve various complex geometries and structures [282]. Designers input design decisions into an algorithm that can produce near-infinite candidate geometries based on these inputs. Moreover, the various geometric solutions can be analysed, optimised, and evolved with evolutionary algorithms. CD allows designers and engineers to link analysis, design exploration, and evaluation in the same loop.

These CD methods are particularly useful for generating and varying complex topologies with multiple objective and subjective targets, as required when applying metamaterials to protective sports equipment. Generating such topologies using traditional computer-aided design (CAD) is inefficient and impractical. Sports engineers traditionally modeled unit cells that were designed, tested, and iterated individually [10,15,283]. This manual approach can limit the generation of complex geometries and solutions requiring many modifications to a unit cell (e.g., [284]).

Some sports products have been launched using CD. For example, the digital manufacturing company Carbon3D partnered with sports equipment companies, including Riddell [285] and Adidas [286], to generate lattices with varied local properties. Hyperganic's algorithmic engineering approach has been used to develop custom saddles, helmets, and insoles [287]. Rheon Labs structures strain-rate-sensitive polymers for various sports products (Figure 9.13a). An increasing range of CD software enables engineers to solve complex design problems using metamaterials, such as nTopology, Rhinoceros 3D with Grasshopper, and Hyperganic Core [283]. In combination with generative AI, platforms such as ChatGPT (by OpenAI) and Midjourney (by Discord, Figure 9.13b) can also generate data such as code, images, text, simulations, and 3D objects based on prompts and training [288]. These data extend the designer's ability to visualise and generate concepts, including metamaterials. The availability of open, homogenised data sets, such as the meta-genome, could allow both numerical and AI models to be trained and validated [289].

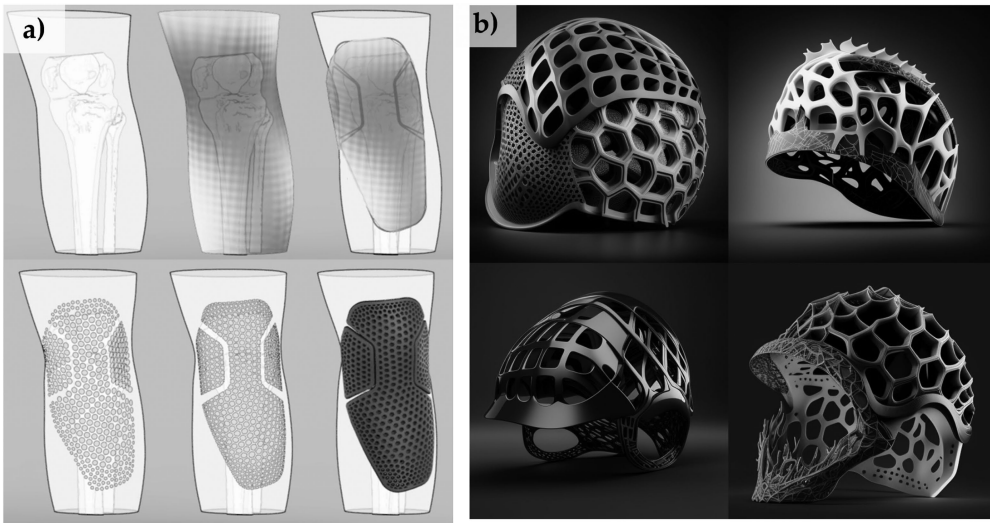

Figure 9.13 (a) Rheon Labs, computational design of a motorcycling knee protector; and (b) Midjourney prompt: helmet with energy absorbing lattice, published under Midjourney's creative commons license [290].

Computational design of tuneable structure in helmets

Mechanical metamaterials for helmets and PPE require properties that not only cover a range of impact conditions but also allow for fit and comfort across various head shapes and sizes. Cellular solid unit cell design should consider the base material properties, how these can be further enhanced, and how varying the unit cell will affect performance and user experience [53,285]. From a CD perspective, helmet liner designs can be generated by combining several data inputs: i) To address fit, the boundary space can be defined by an individual's head scan or derived from manufacturers' existing size splits; ii) Performance can be tuned by changing cell shape, size, and relative density to optimise linear and shear stiffness in specific zones of the helmet, based on data from testing, sensors embedded in helmets during use (smart helmets), or both.

Carbon3D's solution for Riddell and CCM (Figure 9.4b) was created by analysing data from on-field collisions and generating performance targets for conformal lattices. Their solution utilises tetrahedral and rhombic conformal lattices with zoned performance. The goal is to change mechanical properties, particularly lattice stiffness, by changing the thickness of the beams and modifying the shape of the base unit (e.g., from tetra to rhombic). This is an inverse design process whereby desired properties are the input and Carbon's Design Engine software generates a mechanical solution based on a library of validated solutions [53,291]. The Engine's predictive accuracy has not been independently verified, and dynamic or large relative unit cell size solvers are not available, so users should still validate their own designs.

HEXR produces custom 3D-printed cycling helmets based on individual head scans [292]. When the geometry of an individual's head surface is obtained, a honeycomb structure is wrapped between the surface and the manufacturer's shell. The radii of the hexagons change depending on performance requirements, and HEXR claims reductions in rotational and linear acceleration during impacts.

RHEON Labs developed helmet solutions for Fly Racing and Xenith using triangular and hexagonal re-entrant structures, respectively. The structures utilise Rheon's strain-rate-dependent polymers, stiffening to adapt to possible impact angles and intensities [94]. The base units are designed to absorb both high and low energies, while keeping linear and rotational accelerations below a threshold value. Fit is considered by calculating the distance between the headform and the EPS or shell and generating a point cloud (using Rhinoceros 3D with Grasshopper) where parts could be integrated without compromising fit and avoiding the need to produce a new shell. These generated structures are then prototyped, tested, and iteratively improved (experimentally) [293,294]. The performance of the helmets has also been independently checked. For example, they both achieved certification (e.g., [293,294]), and the Xenith helmet topped the 2020 NFL Helmet Tests and was 5-star rated by Virginia Tech [295].

The development process uses an iterative loop, generating geometries in the software, then prototyping, experimentally testing and manually updating the design script. Updates to the design script are based on empirical test data and user feedback. In the design of experiments (DOE), the effects of parameters such as cell size, shape, or height on performance and user comfort can be used to tune and optimise the part. Changes to the topology, and thus the design script, are based on expert knowledge. For example, a designer may suggest changes to zonal cell size with the intention of changing the stiffness of the candidate geometries proposed by the software (rather than simply asking the code to increase stiffness).

In current processes, topology variation is still suggested by the designer, based on their knowledge of part requirements and property/topology relationships. While there are various ways to simulate liner performance, numerical simulation of helmets as systems and understanding the influence of design variables on them, is still a complex matter. The optimal design process could involve entering a collision and use metrics into a design script that would suggest geometries based on numerical or AI predictions of performance, airflow, and fit zoning.

Computational design of structure in body armour

The use of CD to create metamaterials allows for the integration of performance and comfort in a much more direct way. There is a case to explore more metamaterials in body armour than currently seen on the market. Human tolerance to impact depends on anatomical structure and the posture of the impacted area [296]. Mechanical properties of bone and soft tissue differ, and the combined shape and thickness of each tissue influence non-penetrating injury severity [296].

As mentioned in the section on applications, traditional PPE typically uses a combination of EVA or PU foam and a stiff shell to cover a broad area around the part of the body being protected. A minimal protection zone is typically defined by focusing on a main joint or bone and covering morphologically and mechanically different adjacent bones and soft tissue [50,180,297]. Such an approach can result in protectors that cover incorrect areas or provide unsuitable protection to other areas, meaning padding is often bulky and can limit a user's motion. Man, Yue, and Tian [49] explored using CD for a non-specific 3D-printed knee protector based on pressure maps from wearable sensors. Similarly, Rheon Labs produce a range of motorcycling body armour that integrates anatomic data, pressure, mapping, body posture and impact direction.

Rheon's armour utilises weighted centroidal Voronoi tessellation based on anatomic data (Figure 9.13a), changing the geometry of the cells and the distance between them to adjust stiffness rather than simply changing the component thickness (again, using Rhinoceros 3D with Grasshopper). Thus, the impact performance of the armour varies between direction and region. The algorithm to create Rheon's knee pad is initiated by taking CT scans of lower legs and knees

in various positions. These scans are analysed to make measurements of anatomic structures, such as the thickness of bone and soft tissue and the distance between bones and the skin surface. By combining this data with common postures during impacts and impact vectors, a weighted centroidal Voronoi tessellation is created to design the knee pad.

The CD process translates requirements into heat maps and interpolates them into a single encoding for the design script. Generating parts with such intricate topologies is unfeasible with traditional CAD, whereby each cell has a unique structure driven by its location on the body and the required performance and specification. The script allows the designer to first create and parametrically iterate the part, then create the manufacturing tool (in this case, a mould as the inverse of the part) in one step. Rheon again uses a mixed approach, iteratively improving suggested designs (from CD) based on experimental results from impact testing of prototypes. After the design/iteration process, these products achieved certification according to EN 1621-1:2012 [298]. While the process of generating the part is streamlined, integrating design decisions and empirical data, challenges remain in creating a direct link between results and design iterations (relying on expert user input, as described above). As with the helmet case study in the section on Computational Design of Tuneable Structure in Helmets, these methods could be developed further by integrating efficient numerical simulations or AI performance predictions with topology generation.

Summary

There are various forms of cellular solids, providing low density and often tunable mechanical properties. With their stress plateau and similar or marginally lower order of magnitude of stiffness compared to many human tissues, these materials are finding applications in protective equipment. Helmet liners are often designed for single or relatively low-occurrence impacts, and their protective capacity is critical. Rebound should be minimised, while other requirements are evolving with our understanding of brain injury. The one-off performance of running shoes is less critical; instead, running efficiency (rebound, bending stiffness, and mass) and long-term degradation are key parameters.

Designing solutions to these problems requires an understanding of the deformation mechanisms and potential design variables afforded by cellular solids. With a rich history of analytical models, foam and lattice deformation mechanisms are generally well understood. Cellular solids can achieve a wide and often counterintuitive range of properties—including negative Poisson's ratio, negative stiffness, force-torque coupling, and non-reciprocal responses. However, with high computational costs, the optimal performance of cellular solids across various impacts has not yet been achieved. The use of computational design and generative AI for the development of metamaterials in sports engineering has contributed to the creation of safer, lighter, and more comfortable products. However, challenges remain in structuring reliable digital models and improving development speed. The development of bespoke digital tools could help accelerate development processes, enhance accuracy, and ultimately improve the quality of solutions.

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10

CELLULAR SOLIDS FOR SPORTS APPLICATIONS

Part 2 – Micromechanics

Olly Duncan, Nicolas Bailly, Olga Kravchenko, and Michel Behr

Introduction

This chapter supplements Cellular Solids for Sports Applications, Chapter 9. The mechanical properties of cellular solids are caused by three classes of parameters: (i) the base solid(s) they are made of, (ii) the structure(s) these solids form (i.e., their topology), and (iii) the contents of the space inside the cells. This chapter focuses on topology, with exceptions where the base solid or cell contents display interesting or useful interactions with the cellular structures. The analytical models are available as MATLAB codes in [1].

Honeycomb and lattice micromechanics

Regular hexagonal honeycombs – 2D and 2.5D micromechanics

Structures with variants of the regular hexagonal cell appear throughout natural and man-made materials (biological cells, honeycombs, and foams); these remain the most used and studied cellular solids. As a result, there is an established micromechanics literature base (e.g., [2]) explaining the links between the topology of hexagonal cells and their mechanical properties. For sports engineers, understanding the property/topology relationships allows the design of safer protective equipment. Specifically, there are three key concepts of interest: low-strain stiffness, rib buckling and the relative stress plateau (see Figure 9.5 in Chapter 9), and Poisson's ratio (contributing to shear modulus or rotational acceleration in helmets [3–5] and to resistance to indentation [6–8]).

In porous honeycombs, lattices, and foams, rib deformation contributes to the mechanical properties. For honeycombs, there are four key unit cell parameters (Figure 10.1a): (i) the thickness of the ribs (b), (ii) the length of the diagonal (l), (iii) the vertical (h) cell ribs, and (iv) the angle of the diagonal cell ribs (θ). We will describe the key deformation mechanisms of these ribs and how they contribute to mechanical properties, first considering 2.5D honeycombs, and then how these concepts can be extrapolated for 3D lattices and foams.

There are three dominant deformation mechanisms: stretching of the cell ribs (Figure 10.1b), hinging around the junctions between the ribs (Figure 10.1c), and flexing of the cell ribs (Figure 10.1d). Hinging around the junctions of cell ribs is caused by shearing within the joint, local

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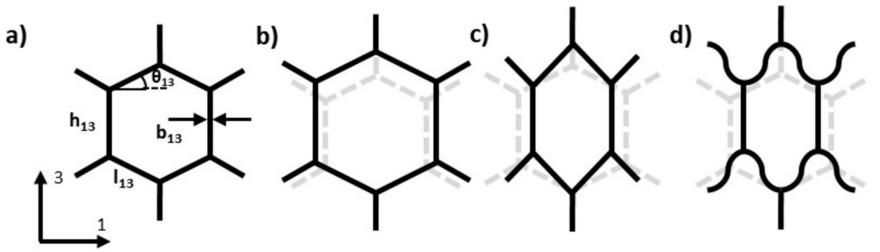

Figure 10.1 Schematics showing (a) a regular hexagonal unit cell, with key parameters (vertical rib length, h, diagonal rib length, l, rib thickness, b, and diagonal rib angle, θ, in the 3-1 plane), and 20% tension in the 3-axis caused by pure (b) cell rib stretching (no change to θ, equal change to h and l); (c) cell rib hinging (θ changing, with no change to h or l); (d) cell rib flexing (no change to h, l, or the cell rib angle close to the junction).

bending of the ribs, or both [9]. These deformation mechanisms cause shape change in each (1 and 3) axis in response to an applied load. If all the ribs were to stretch, without hinging or flexing, an applied tensile force would cause transverse expansion (e.g., Figure 10.1b) or a negative Poisson's ratio (see Chapter 9 for more information about auxetic (negative Poisson's ratio) materials). Note: axes are labelled 1 to 3 to denote internal mechanisms rather than Cartesian coordinates (x, y, and z).

Increasing the ratio of rib thickness to length (b in Equation 10.1), either by increasing rib thickness while maintaining cell size (and rib length) or by decreasing cell size while maintaining rib thickness, increases stiffness [2]. Provided cell dimensions do not change, the stiffness of a hexagonal honeycomb increases with density. Indeed, for a regular hexagonal honeycomb (h = l, $\theta = 30°$, which is transversely isotropic between the 1 and 3 axes), subject to small strains, Young's moduli in the 1 and 3 axes (E_1 and E_3) are [2]:

$$\frac{E_1}{E_s} = \frac{E_3}{E_s} = 2.3\left(\frac{b_{13}}{l_{13}}\right)^3 \tag{10.1}$$

where E_s is Young's modulus of the rib material. For 2.5D honeycombs, density (ρ) is a function of b and l; $b/l \propto \rho$. Thickening (or shortening) cell ribs to increase b/l increases density and stiffness by the ratio of $E \propto \rho^3$ [2].

Increasing the rib angle, θ, decreases the bending and hinging moment in the 3-axis (Figure 10.1), which increases E_1 and decreases E_3, making the honeycomb anisotropic. Within classical elasticity theory, there are various conditions of symmetry [6]. The symmetric compliance condition means the product of E_j and v_{ij} should equal that of E_i and v_{ji}. If a material has a stiff axis, the energy transfer following deformation in that axis will increase the magnitude of deformation in a perpendicular compliant axis, or vice versa. To maintain the symmetric compliance condition, $E_1 v_{31} = E_3 v_{13}$, the magnitude of v_{13} (Poisson's ratio due to expansion in the 3-axis caused by loading in the 1-axis) must increase as E_3 decreases, while v_{31} should decrease in magnitude. Such trends have been shown experimentally [2,9].

By balancing forces (and moments) on diagonal and horizontal ribs (e.g., Figure 10.2), it is possible to create force constants—relating changes in rib angle and length to each deformation [9].

To understand how changing constituent material parameters and cell geometries affect cell deformation, we must first introduce equations for the force constants relating to cell rib stretching

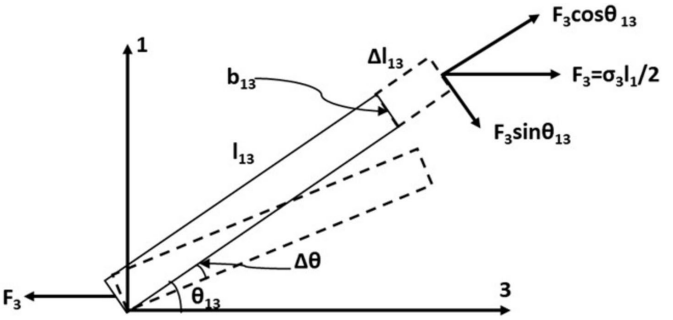

Figure 10.2 Forces acting on a diagonal rib caused by loading in the 3-axis (F_3) (l_1 is length in the 1-axis and σ_3 is stress in the 3-axis). These cause length change and re-orientation (dashed lines).

(K_s for the diagonal rib and K_s^h for the vertical rib), hinging (K_h), and flexing (K_f). K_{hf} is K_h and K_f combined. Introducing a term (w) for rib thickness in the 3-axis, the equations, developed and experimentally validated by Masters and Evans [9], are:

$$K_s = \frac{E_s w b}{l} \tag{10.2}$$

$$K_s^h = K_s \frac{l}{h} \tag{10.3}$$

$$K_f = \frac{E_s w b^3}{l^3} \tag{10.4}$$

$$K_h = \frac{G_s w b}{l}\left(\text{Shearing of junction}\right) \tag{10.5}$$

$$K_h = \frac{E_s w b^3}{6 l^2 q}\left(\text{Local bending at junction}\right) \tag{10.6}$$

$$K_{hf} = \frac{K_h K_f}{K_h + K_f} \tag{10.7}$$

whereby q is the axial length of the curved hinge, and subscripts for the cell axis were removed for simplification. The shear modulus of the rib material is:

$$G = \frac{E}{2\left(1+\nu\right)} \tag{10.8}$$

From Equations 10.5, 10.6, and 10.8—for a rubbery isotropic material (ν_s = 0.5, $G_s \approx E_s/3$) [6] and a curve length $q = l/10$, the dominant hinging mechanism (causing lower K_h) is local bending when $b/l < 0.45$ and shearing when $b/l > 0.45$ [9]. Based on these force constants, uniform changes to the base material properties should have little effect on deformation mechanisms. Clearly, increasing Young's modulus (E) would increase all forms of stiffness equally, provided Poisson's ratio, and thus G/E, remain unchanged. Reducing the Poisson's ratio of the base material from 0.5 could increase the ratio of b/l associated with the transition from the dominant hinging mechanism from local bending to shearing (e.g., at ν = 0.2, the transition would be at b/l = 0.5). In practice, honeycombs with such thick ribs are uncommon and should not be modelled using the

approach above, as the ribs are too thick to be approximated as beams. The boundary between local bending and shearing is closer to $b/l = 0.2$ when local bending is constrained to $q = l/100$. Here, the Poisson's ratio of the rib material could affect the force constant k_h, according to Equations 10.5 and 10.8.

Changes in the deformation mechanism are also likely when the constituent material is non-uniform. For example, honeycombs with softer materials around the joints will deform by hinging [10–12]. Depending on the base material, some softening of the hinge is expected after cyclic loading, due to stress concentration and fatigue, reducing K_h (Equation 10.5, [9]). Such effects are evident in the cyclic loading of running shoes, for example. Viscoelastic effects (typically shear strain rate-dependent stiffening) occur in the base material. Thus, foam, honeycombs, or lattices made from viscoelastic base materials could change their deformation mechanism with loading rate, effectively stiffening at the shearing junctions when loaded quickly. Such fast loading is common in sports equipment.

Finally, to demonstrate how changing cell parameters and force constants affect honeycomb properties, we introduce analytical equations to calculate the 3-axis and 1-axis mechanical properties. Recall Figure 10.1—the 3-axis is parallel to the straight, vertical sides (h) while the 1-axis is horizontal. The reader is referred to the work of Masters and Evans [9], or the adapted version (supplementary materials of [13]) that uses the combined flexing and hinging force constant presented here, for the derivations.

$$\nu_{13} = \frac{\sin\theta\cos^2\theta\left(\frac{1}{K_{hf}} - \frac{1}{K_s}\right)}{\left(\frac{h}{l} + \sin\theta\right)\left[\frac{\sin^2\theta}{K_{hf}} + \frac{\cos^2\theta}{K_s}\right]} \tag{10.9}$$

$$\nu_{31} = \frac{\sin\theta\left(\frac{h}{l} + \sin\theta\right)\left(\frac{1}{K_{hf}} - \frac{1}{K_s}\right)}{\left[\frac{\cos^2\theta}{K_{hf}} + \frac{\frac{2h}{l} + \sin^2\theta}{K_s}\right]} \tag{10.10}$$

$$E_1 = \frac{\cos\theta}{b\left(\frac{h}{l} + \sin\theta\right)\left[\frac{\sin^2\theta}{K_{hf}} + \frac{\cos^2\theta}{K_s}\right]} \tag{10.11}$$

$$E_3 = \frac{\left(\frac{h}{l} + \sin\theta\right)}{b\cos\theta\left[\frac{\cos^2\theta}{K_{hf}} + \frac{\frac{2h}{l} + \sin^2\theta}{K_s}\right]} \tag{10.12}$$

Varying θ from −90° to +90°, and arbitrarily using $b/l = 0.2$, $h = l$, and choosing ratios of $K_{hf}/K_s \approx 0$ (only flexing/hinging), $K_{hf}/K_s = 0.005$ and 0.04 (combined), and $K_{hf}/K_s \approx \infty$ (only stretching), we can plot the elastic constants. The K_{hf}/K_s value 0.04 is from Equations 10.4, 10.5, 10.7,

and 10.8 (v_s = 0.5), while the lower limit (0.005) is from Equations 10.4, 10.6, and 10.7, as $q \to l$. For completeness, the shear modulus (G_{13}) is:

$$G_{13} = \left[\frac{b\left(\frac{h}{l}\right)^2 \left(1+\frac{2h}{l}\right)\cos\theta}{K_{hf}\left(\frac{h}{l}+\sin\theta\right)} + \frac{b\left(1+\frac{h}{l}\sin\theta\right)^2}{K_S\cos\theta\left[\frac{h}{l}+\sin\theta\right]} \right]^{-1} \tag{10.13}$$

Honeycombs with positive diagonal rib angles have a positive Poisson's ratio—they contract laterally in tension or expand in compression [2]. The counterintuitive property of a negative Poisson's ratio (or auxetic behaviour [14]) is also possible and was first shown (using foams) by Rod Lakes in the 1980s [15]. Negative cell rib angles will typically cause negative Poisson's ratios. These trends are evident for the flexing and hinging cases ($K_{hf}/K_s \neq \infty$) in Figures 10.3a and 10.3b. As shown via the schematic (Figure 10.1c), pure stretching ($K_{hf}/K_s \approx \infty$) produces the opposite effect (Figures 10.3c and 10.3d)—although it requires telescopic ribs or similar. Thus, as K_{hf}/K_s tends towards 0, the magnitude of the positive or negative Poisson's ratio typically increases (Figures 10.3a and 10.3b).

Considering both Young's moduli and Poisson's ratio, these hexagonal honeycombs are typically anisotropic—but can have the same properties in the 1 and 3 axes at crossover points at $\theta \approx +/-30°$ [2,9]. For magnitudes of θ below zero, E_1 is typically higher than E_3, and vice versa (Figures 10.3c and 10.3d). Young's modulus (E_1) is typically higher when θ is negative, corresponding to a reduction in cell size without changing the rib thickness (i.e., *a* density increase) [9]. Conversely, E_3 then typically reduces with θ as the gauge length for applied strain reduces, but the amount of deformation per applied load due to hinging and flexing of the cell ribs does not (Figure 10.3c). As θ tends toward zero, stretching of the vertical ribs becomes dominant (Figure 10.3c). Using elongated, anisotropic foams in helmet liners, with high through-thickness stiffness and low transverse stiffness (causing low shear modulus), can reduce peak headform rotational acceleration during oblique impacts [5]. Honeycombs with similar behavior could be designed using Equations 10.11 and 10.12.

In regular hexagonal honeycombs, the trends in Figure 10.3 can be expected in tension or low levels of compression—often up to 5% compressive strain [2]. Rib buckling (which can be elastic) is then expected, effectively creating an extra hinge that reduces the compressive modulus and changes the force for further applied displacement to close to zero [2]. As discussed in Part 1 and other works (e.g., [4,7,16,17]), this buckling is critical to efficient protective equipment, allowing a region of energy absorption with minimal additional injury risk (i.e., higher force; see Figure 9.5, Chapter 9). This relative plateau region is typically from the point of buckling until the point of densification—when cell walls come into contact and stiffness increases toward that of the base material [2,7]. Cell walls behave like an end-loaded column, and so buckling can be predicted as F_{crit}, the point at which the load exceeds the Euler buckling load [2,6]:

$$F_{\text{crit}} = \frac{n^2\pi^2 E_s I}{h^2} \tag{10.14}$$

whereby *n* describes the rotational stiffness of the cell junction and *h* is (again) the vertical rib length. Thus, slender ribs will buckle at a lower force. As an example, for loading in the 3-axis, each column carries a load related to the applied stress (σ_3) of [2]:

$$F = 2\sigma_3 lb\cos\theta \tag{10.15}$$

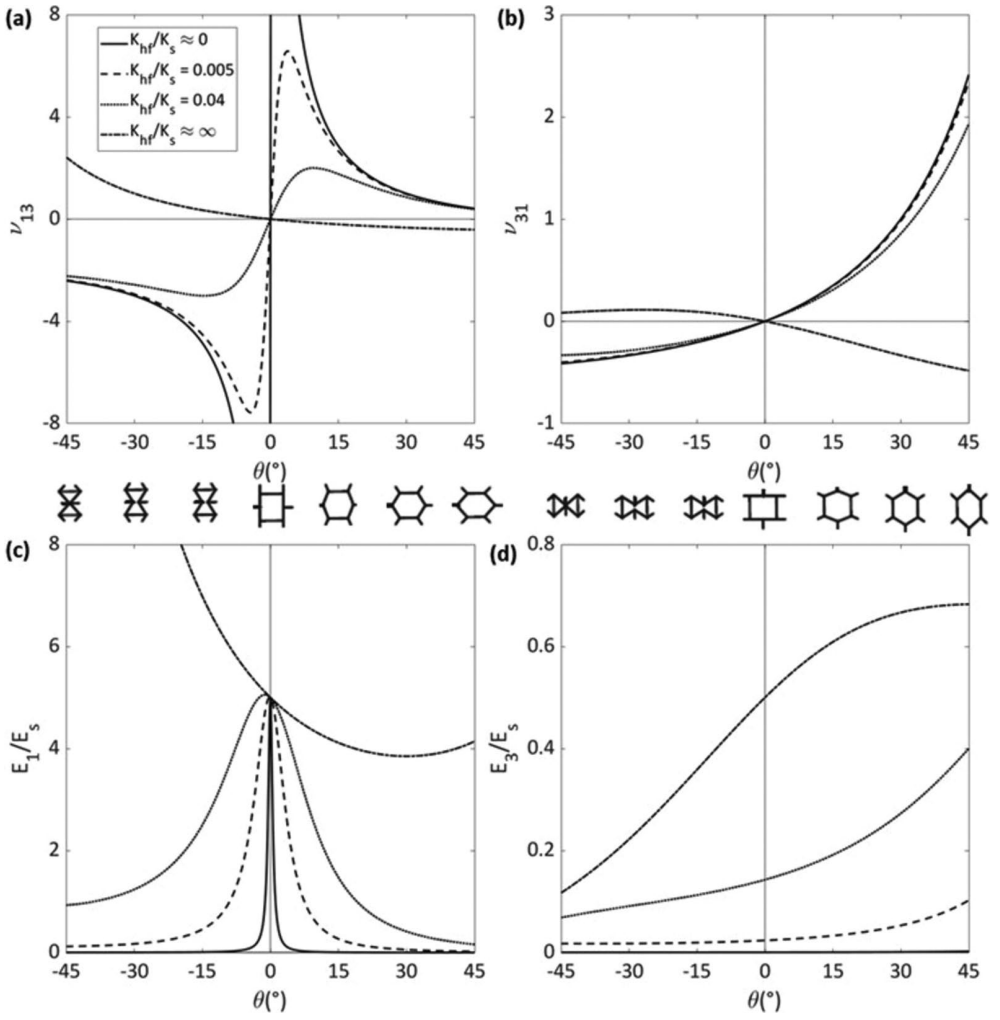

Figure 10.3 Calculated Poisson's ratios (a) from Equation 10.9, (b) Equation 10.10 and Young's moduli (c) from Equation 10.11, (d) Equation 10.12 vs. diagonal cell rib angle (with schematics showing cell shapes either below (a) and (b) or above (c) and (d)), the respective plot – with the direction of loading relative to the schematics aligned vertically, and transverse strain horizontally on the page. Note that, in (a) and (c), values that tend towards infinity have been cropped from the *y*-axis. The same legend (a) for all.

So, buckling will occur when the applied stress is [2]:

$$\sigma_3 = E_s \frac{n^2\pi^2}{24} \frac{b^3}{lh^2} \frac{1}{\cos\theta} \tag{10.16}$$

Calculating cell junction rotational stiffness is challenging [2]. If rotation around the junction is freely allowed, *n* will be 0.5, while if no rotation is allowed, *n* will be 2 [2]. Partially constrained rotations will fall somewhere in between. For 2.5D regular hexagonal honeycombs, *n* is 0.67.

Readers are referred to lookup tables and detailed calculation methods in Appendix 4c of Gibson and Ashby's textbook [2] for other cases.

As well as analytical models, microstructurally faithful finite element models can estimate honeycomb bulk properties [18–20] or application-specific performance [21–23]. There are two sets of variables: the properties of the base material and the honeycomb topology. Complex loading, such as fast or cyclic loading, affects the constituent polymer. For example, consider a honeycomb compressed at a strain rate of 100 s^{-1} to a compressive strain of 80%. The constituent material—deforming predominantly by rib flexure and hinging—would have sections in tension, shear, and compression, deforming at different rates. Ideally, to estimate honeycomb properties, the constituent polymer should be characterised across the range of expected strains and strain rates [21–23], which could be estimated from a finite element model or experimentally (e.g., by digital image correlation). Honeycomb ribs are typically slender, meaning the applied strain in much of the material remains low until the point of buckling, when the deformation mechanism changes. Thus, models assuming linear stress versus strain relationships in the constituent material can provide reasonable approximations [18–20,24].

Regular hexagonal lattices – 3D micromechanics

The typical analytical approach to understanding and analysing the deformation of 3D lattices is to consider 2D projections of cell shapes, aligned along each orthogonal axis. Continuing with the hexagonal unit cell, a 3D variant is an elongated dodecahedron (12 flat faces) (Figure 10.4 [25,26]). The 2D projections, with the 3-axis aligned vertically (Figure 10.4a), are similar to those previously presented. The 2–1 plane (Figure 10.4b) forms a square. Projection deformation mechanisms are typically considered separately during analytical modelling [25–27], defining interactions between planes considered at the bulk level using concepts of orthogonal symmetry. The combined angle of the diagonal rib between (say) the 2–1 and 2–3 planes (example highlighted in Figure 10.4c) causes some errors in predictions of the force constants, which have been adjusted to fit experimental data [13]. An alternative approach is to numerically characterise unit cells and then form a homogenised (spatially averaged) set of material models based on the arrangement of unit cells in two or three dimensions [28].

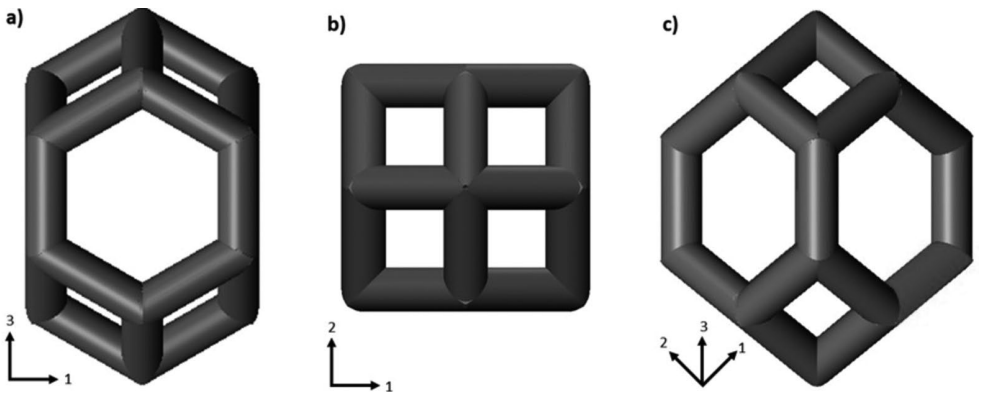

Figure 10.4 Elongated dodecahedron (a) frontal view, (b) top view, (c) frontal view rotated 45° around the 3-axis, with a diagonal rib between the 1-3 and 2-3. Axis orientation is show for each subfigure.

The key mechanical difference between 2D or 2.5D lattices and 3D ones is the presence of porosity in all three axes. These 3D lattices can be isotropic, unlike their 2.5D counterparts. As such, while the complexity of micromechanical models increases, trends between macroscopic mechanical properties (Young's, shear, and bulk moduli) can be clearer and simpler. The reduction in directional sensitivity can be important in the design of sports equipment. For example, a honeycomb helmet liner that undergoes more shear deformation than typical helmet foam during an oblique impact could reduce injury risk [29], as described in Chapter 9.

The changing trends between various lattice-specific relationships differ between 2.5D and 3D lattices. With the added potential for ribs to flex, hinge, or buckle in a third axis, 3D lattices are often softer than their 2D counterparts. Thus, while trends in 2D cell projections can be used to understand topology/property relationships (e.g., Figure 10.3), reducing design iterations, confirmation by experiment or microstructurally faithful finite element models is still needed.

Foam mechanics

Open-cell foam

Open-cell foam micromechanics are closer to those of typical lattices than closed-cell foam. Lattice cells are more regular than foam cells. While foam cells have a degree of randomness, these can be spatially averaged, meaning homogenised foam properties can be explained using analytical equations like those in Section 3 [2,13,25]. With the reduced order in foam cells, alignment is variable, so differences in mechanical properties between orientations (anisotropy) can be reduced in comparison to idealised lattices [13,30]. The relationship between structural order and anisotropy is also observed in conventional ordered/crystalline (often anisotropic) and disordered/amorphous (often isotropic) materials. Similarly, buckling in open-cell foam occurs over a wider strain range than in a typical lattice, and the subsequent region is only a relative plateau; force will typically gradually increase during the plateau before a clear rise at densification [30].

Analytical homogenisation of foam properties, estimating bulk properties from those of one unit cell, has been achieved by applying a mean cellular rotation (ϕ—approximately 10°) [13]:

$$E_1(\phi) = \left[\frac{\cos^4\phi}{E_1} + \cos^2\phi \sin^2\phi \left(\frac{1}{G_{13}} - \frac{2\upsilon_{13}}{E_1} \right) + \frac{\sin^4\phi}{E_3} \right]^{-1} \tag{10.17}$$

$$\upsilon_{13}(\phi) = E_1(\phi) \left[\frac{\left(\cos^4\phi + \sin^4\phi\right)\upsilon_{13}}{E_1} - \cos^2\phi \sin^2\phi \left(\frac{1}{E_1} + \frac{1}{E_3} - \frac{1}{G_{13}} \right) \right] \tag{10.18}$$

whereby all on-axis (unrotated) moduli are as calculated in Section 2.1 (Equations 10.9 to 10.12). With the mean rotation applied, the local minima or maxima for various moduli, with respect to cell parameters, become less pronounced (Figure 10.5). Note that if $\phi = 0°$, $E_1(\phi)$ and $\nu_{13}(\phi) = E_1$ and ν_{13}, respectively, while if $\phi = 90°$, $E_1(\phi)$ and $\nu_{13}(\phi) = E_3$ and ν_{31}, respectively. Foams stiffen as they enter the densification region, much like lattices. Densification often occurs at approximately 70 to 80% compression [2,30] (Figure 10.5).

There are analytical models capable of more closely replicating the properties of open-cell foam than those presented here, such as those based on Voronoi's method to reflect foam randomness [31]. We have not covered these here, choosing to use previously introduced models to intuitively

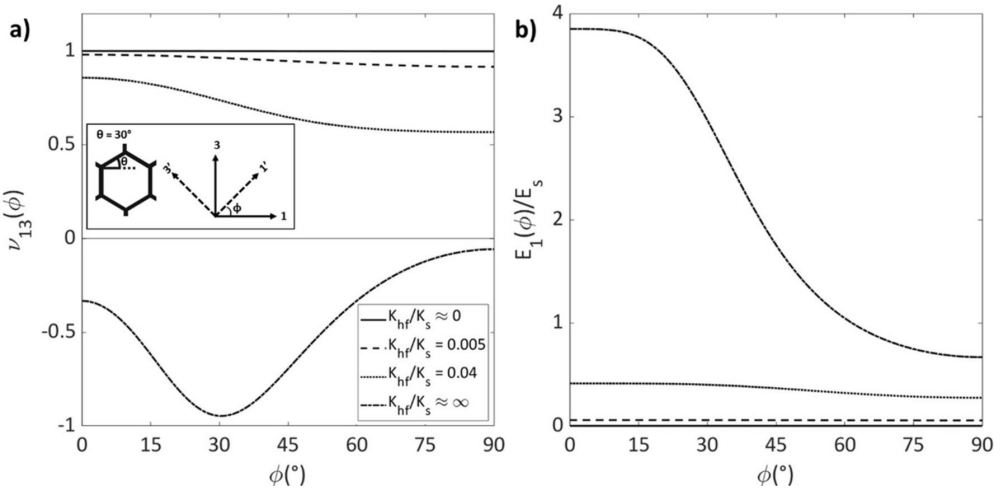

Figure 10.5 (a) Poisson's ratio and (b) Young's modulus, from Equations 10.17 and 10.18, with the applied offset angle and the diagonal rib at an angle of $\theta = 30°$ and all other parameters equal to those used in the section on regular honeycombs. The same legend for (a) and (b), while schematic in (a) shows the offset angle.

demonstrate the effects of changing cell parameters on foam properties. The reader is referred to Chapter 7 of Mills' *Polymer Foam Handbook* should they require different analytical models [30]. Finite element models based on microcomputed tomography have also been conducted for open-cell foam, providing rich information about structure/topology relationships [32–35].

Closed-cell foam

Closed-cell foam is often used in sports equipment (see Chapter 9 and Chapter 5: Soft materials in sport). It has been analytically modelled to high compressive strains [36], considering the ribs as beams, cell walls as membranes, and the effect of gas pressure—all acting in parallel (Figure 10.6). While it may be expected that the gas pressure inside the foam would be equal to atmospheric pressure, that is not always the case [37], possibly due to the foam's loading history. Thus, if the foam volume fraction is R, the initial gas pressure p_0, and the polymer stress is σ_0, then the foam stress at an applied strain (ε) is [37]:

$$\sigma = \sigma_0 + p_0\left(\frac{1-R}{\left(1-\varepsilon\right)\left(1+\upsilon\varepsilon\right)^2 - R} - 1\right) \tag{10.19}$$

Broadly, the changes described in Section 2 still apply to the polymer in closed-cell foam. Transferable, low-strain models based on those in Section 2, which reflect changes in topology, have been developed for various closed-cell foam structures [38]. These models cannot be applied to high strains as they do not account for changing cell wall flexure with increasing air pressure. Similarly, analytical models have been developed for closed-cell foam that can reflect their high-strain response but do not accommodate irregular (e.g., re-entrant-like) cellular structures [36]. The

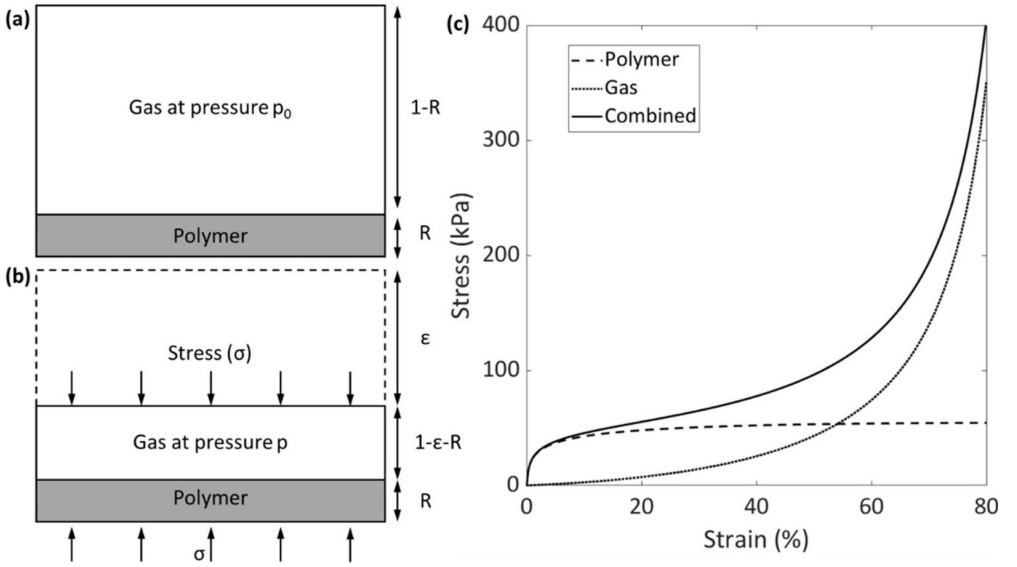

Figure 10.6 Schematics showing (a) gas and polymer fractions before loading, (b) gas and polymer fractions in compression (with $\nu = 0$, i.e., compressive strain = volumetric strain), and (c) polymer and gas contributions for a simulated closed-cell foam in compression (Equation 10.19), with constants $\nu = 0.4$, $p_0 = 80$ kPa, and σ_0 set by an arbitrary function, assuming densification is not reached.

contribution of the gas is higher for closed-cell foams made of soft polymers, whereas for closed-cell foams with rigid polymers, the gas contribution becomes negligible [37]. An analytical model for such foams, based on one for flexible closed-cell foam, has been presented [36] but does not closely reflect experimental results, so it is not included here. Indeed, the specific deformation mechanisms of such foams are complex.

Homogenisation

Cellular solids are heterogeneous; there is an obvious difference in material properties between solid ribs/walls and hollow voids. Indeed, any material will become heterogeneous at some scale: metals display heterogeneity at the atomic scale (and often within their microstructure), polymers at the molecular scale, and fibre-polymer composites between fibres and the polymer matrix. Cellular solids are often heterogeneous at the mesoscale. The mathematical method of homogenisation allows averaged, bulk properties to be obtained based on the analysis of one unit cell, providing computationally efficient analysis. Constraining unit cell edges change the response (e.g., uniaxial and shear stiffness). Thus, for periodic structures, such as lattices and honeycombs deforming under small strains, bulk properties can be obtained from finite element analysis of a single unit cell with periodic boundary conditions [39–41]. For the honeycomb models presented in this chapter (Figures 10.3 and 10.5), homogenisation can be applied by adjusting the estimated force constants (Equations 10.2 to 10.7) to fit experimental/microstructurally faithful finite element models [9,13]. Foams are not periodic, so further modification of unit cell relationships is needed (see the subsection on open-cell foam).

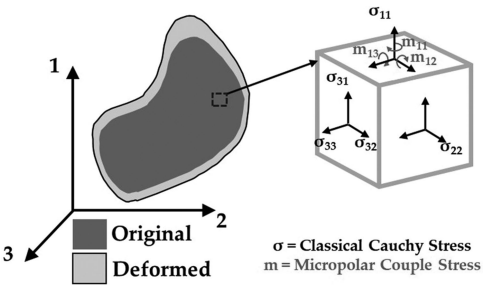

Figure 10.7 Schematic showing example Cauchy stress (black) and micropolar stress acting on the top face only (grey) for a deforming solid.

For foams, honeycombs, and lattices, as with other solids, the simplest homogenisation approach is based on Cauchy continuum mechanics, where the bulk elastic response is described by Young's modulus and Poisson's ratio [6]. Linear mapping can be applied between local and global responses. For those wishing to avoid using higher-order micro-continuum theories, which can be challenging to characterise and apply [42], one approach is to model relatively large structures with small unit cells. If unit cells are large, or if localised bending or low-area indentation is expected, Cauchy mechanics assumptions (that internal stresses cause negligible external change) will not be met [43–49], so results may be inaccurate. For such cases, microstructurally faithful finite element models, experiments, or alternative micro-continuum theories are needed.

There are three main types of micro-continuum theories, covered in detail by Eringen [50]. The generalised theory is micromorphic continuum theory, where internal features have six degrees of freedom (i.e., three rotations and three translational axes). A special case variant, microstretch, allows internal volume change while features retain their shape, much like the cells of breathing lungs. Micropolar continuum theories, such as the Cosserat continuum, are more common, having been developed since the late 19th century [46]. These allow only rotation of internal features, providing more degrees of freedom than the Cauchy continuum (Figure 10.7). Micropolar continuum theories include internal rotations of crystals (or unit cells [51–54]), localised bending [44], and variable friction or attraction between locally deforming crystals, granules, or unit cells [46]. For lattice/cellular solid design and optimisation, such mechanics allow more degrees of external freedom—specifically twist/compression coupling [53,54]—and more accurate simulations [55–58]. These micropolar continuum theories are limited to small strains [50]. Under high strains, the length scale over which the micro-rotations occur may change, so microstretch or micromorphic continuums are needed. An awareness of the possibilities of such internal deformations is useful when designing or modelling cellular solids.

Summary

Analytical models are presented to describe the effect of cell microstructure on the mechanical properties of cellular solids. Those relating to honeycombs are included in detail. Approaches to account for out-of-plane variation in lattices are then described. A foam homogenisation approach, accounting for the somewhat random distribution of cells, is also included. These relationships could be used to select candidate cellular structures for application in protective equipment (e.g., footwear, helmets, and crash pads) or foam layers in sandwich structures (e.g., surfboards), as described in Chapter 9.

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11

FIBRE-REINFORCED COMPOSITES FOR SPORTS APPLICATIONS

Yi Chen, Bin Lyu, Lloyd Smith, and Nicholas Smith

Composite materials

A "composite material" refers to any material made from multiple constituents. Here, we will limit the discussion to carbon-fiber composites. The carbon fibers themselves have high stiffness and strength relative to other materials and are bound together using some form of matrix material, often a thermoset epoxy or a thermoplastic polymer, which is significantly less stiff. A table of some common materials and their mechanical properties is shown in Table 11.1. Of particular interest in many sporting applications are the specific strength and stiffness, which are the strength or stiffness divided by the density of the material. This shows why carbon-fiber composites are used so often to reduce weight.

Mechanics of composite materials

High-performance carbon fiber composites are typically made from continuous carbon fibers that are either woven together or arranged with unidirectional strands. Cylindrical structures are often formed by winding thin strands around a mandrel, while more plate-like structures are formed by stacking sheets on top of a tooling shape and compressing them using vacuum and external pressure. In either case, the properties of the final structure need to be calculated in two steps: the first is to determine the properties of the individual composite layer (which, unlike metals will depend

Table 11.1 Tensile properties of common engineering materials

Material	*Tensile Strength (MPa)*	*Tensile Modulus (GPa)*	*Specific Strength (MPa/[g/cm³])*	*Specific Modulus (GPa/[g/cm³])*
6061 T6 Aluminum	310	69	114	25.5
4340 Steel	1030	200	132	25.5
Nylon 6/6	75	2.8	66	2.5
Silicon Carbide	0.4	400	0.1	118
E-glass	3448	72	1358	28
IM7 Carbon Fiber	5175	276	2907	155
Epoxy Resin	100	3.5	82	2.9

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on the direction, with one set of properties in the fiber direction and another acting transverse to the fiber). The second step is only needed when multiple layers of composites with fibers oriented at different angles are used. It involves assembling the various layer stiffnesses into a laminate stiffness of the structure using classic lamination theory.

As a simple example, consider the case of a quasi-isotropic composite laminate with a stacking sequence of $[0,45,-45,90]_S$. In laminate notation this indicates that there are four unique layers, with fibers oriented at 0 degrees, 45 degrees, −45 degrees, and 90 degrees from the test axis. The subscript s indicates that this stacking sequence is repeated once symmetrically. In a high-performance composite, it is typical to see a fiber volume fraction of 60%, with the remaining 40% as epoxy. Under these conditions, the in-plane composite stiffness would be 58 GPa. While this is notably lower than even aluminum in Table 11.1, it has a specific stiffness of 36.3 GPa/[g/cm^3], which is nearly a 50% improvement over both steel and aluminum.

This anisotropy in stiffness allows composite materials to have tailored stiffness [1]. Some regions can be made stiffer while others are more compliant without changing the thickness or weight of the structure. This property of composites is used in many sports to give precisely tailored mechanical properties.

Despite their advantages, composites fail in different ways compared to more traditional materials [2]. In a composite material, internal damage can be difficult to detect and affects the performance of the material significantly [3]. Impact damage is particularly important in composites for this reason, as transverse impacts can affect the adhesion between layers and change the mechanical properties.

Failure modes

There are many different failure modes that composite materials can experience. They can be grouped into three categories: translaminar, intralaminar and interlaminar [4]. Translaminar failures occur when the reinforcing fibers themselves fail. Intralaminar failures occur when the matrix material between fibers fails. Interlaminar failure, also known as delamination, occurs when failure occurs between composite layers.

Many researchers have studied the fracture surfaces of composites under tension [5–8], and in most cases, forms of intralaminar failure are observed for high-performance fibers. When good bonding between the matrix and fiber is present and lower-strength fibers are used, fiber breakage can also be observed. But in most cases, matrix cracking and debonding of the fiber/matrix interface are the most common under this loading scenario.

Unlike most metallic materials, composites can have significantly different strengths under compression compared to tension. Researchers studying the fracture surfaces of composites under compression have categorized many specific failure modes [9,10]. For the sake of this review, we can group them into two broader categories: translaminar failure (often some form of fiber buckling) and intralaminar failure (matrix failure).

Composite materials are frequently subjected to bending loads. Since bending features both tension and compression on opposite surfaces, it is not surprising that researchers have found similar features from both tensile and compressive failures on bending fracture surfaces [11].

Composites in sport

Composite materials are used for their mechanical performance and specific strength, and they often have good performance in fatigue. In some sports, these features are used to gain an advantage

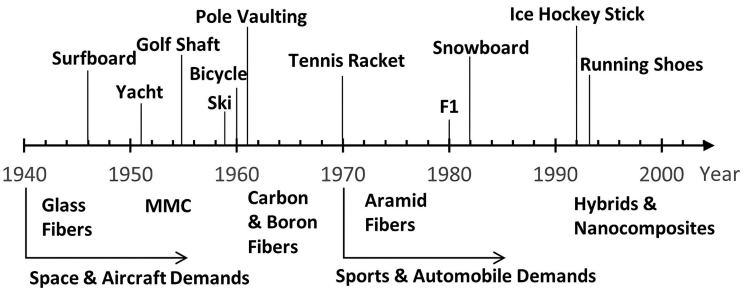

Figure 11.1 Timeline of composites introduced in sports [12]–[17].

in energy storage and transfer, while others benefit from weight reduction. Fiber-reinforced composites first appeared in sports equipment with the introduction of fiberglass around the 1950s and gained momentum with the rise of carbon fibers in the late 1960s. The growing markets of sports and automobiles in the 1970s promoted the development of composites, as shown in Figure 11.1. In this work, we will discuss the effects of composites on several sports.

Ball-equipment impact

The following considers how composites affect performance in sports involving ball-equipment impact. To understand how composites contribute to performance, we must first understand some basic properties of sports balls.

Upon impact, all balls dissipate energy. Ball energy dissipation is typically measured from a rigid wall impact and reported as a coefficient of restitution, or $\underline{e}$. In most cases, e is measured at a lower speed than occurs in play. Tennis uses a speed of 7 m/s, while play speeds reach 60 m/s [18]. Soccer uses a speed of 6 m/s, while play speeds reach 30 m/s [19]. The difference in testing and play speeds is important since the coefficient of restitution is not constant but decreases with increasing speed. A softball, for instance, with $e = 0.42$ at 30 m/s will have $e = 0.35$ at 50 m/s, where 88% of the impact energy is dissipated [20]. There is some subtlety with speed, since impact forces against a rigid wall are greater than impact forces at the same speed against a recoiling object [20]. In most cases, nevertheless, laboratory impact forces (and energy dissipation) are lower than those that occur in play. Understanding and quantifying the nuances of ball energy dissipation is important in sport since materials in general, and composites in particular can be designed to recover some of this lost energy.

When a sports ball impacts another object, both deform and dissipate energy. With few exceptions, balls are softer and less elastic than the objects they impact. This results in higher deformation and energy dissipation of the ball compared to the object it impacts. In most cases, impacting objects (bats, clubs, rackets) are also more elastic than balls. Impact energy dissipation will be reduced, therefore, if the impacting object is made softer, reducing ball deformation in exchange for increased impacting object deformation. This is often referred to as the "trampoline effect". This phenomenon was observed by accident 50 years ago when baseball bat manufacturers began making hollow metal bats as a more durable alternative to solid wood bats [21]. In addition to improved durability, the bats were lighter, easier to swing, and tended to hit the ball further. An arms race quickly ensued between manufacturers seeking thinner walls, stronger alloys, new designs, and new materials. The trampoline effect transferred to other sports, including golf, where all drivers are now hollow designs and increasingly composite.

The magnitude of the trampoline effect is limited by the strength of the material of the bat or club. If the wall is too thin, it will dent or crack upon impact. If the wall is too thick, the trampoline effect is negligible. Ten years after hollow baseball bats were introduced, another design innovation increased the trampoline effect beyond what materials of the day could achieve: the double-wall bat. A double-wall bat has a barrel made of two concentric shells that are in contact but not bonded together. As a simple example, consider a simply supported beam, impacted at its center, as shown in Figure 11.2a. Now compare this to a system of two stacked beams in Figure 11.2b with the same total thickness as the beam in Figure 11.2a. The beam stiffnesses, k, differ by a factor of four, or

$$(k)_{\text{single}} = \frac{4Ebt^3}{L^3} \text{ and } (k)_{\text{double}} = \frac{Ebt^3}{L^3} \tag{11.1}$$

for the solid and stacked beams, respectively, where E is the modulus, b is the plate width, and L is the span. By equating the kinetic energy of the ball with the stored elastic energy in the plates, we can develop an expression for the maximum plate stress, which is the same for the two cases as

$$(\sigma)_{\text{single}} = (\sigma)_{\text{double}} = 3v\sqrt{\frac{Em}{bLt}} \tag{11.2}$$

where m and v are the ball's mass and speed, respectively. The bending stress depends on the bending moment, beam thickness, and beam inertia. The stacked beams in Figure 11.2b have lower stiffness than the single beam in Figure 11.2a, resulting in a reduced bending moment. Each stacked beam in Figure 11.2b carries half the moment, but has half the thickness and one-eigthth the inertia as the single beam in Figure 11.2a. These competing factors cancel out, producing the same stress for the thicker single beam and the stacked beams. This simple idea had a dramatic effect on the equipment performance of the day and was quickly adopted by all manufacturers.

About 20 years ago, the toughness of composite materials improved to the point that they could be used in ball impact sports. Composite baseball bats and golf clubs were lighter than their metal counterparts, and the orthotropic materials allowed unprecedented stiffness control. Composite stiffness and strength increase with the fiber volume fraction. The fiber volume fraction, in turn, tends to be highest in composites made from prepreg. (Prepreg is a thin layer of highly aligned fiber pre-impregnated with partially cured resin.) Composite structures are commonly made of multiple plies of prepreg, which are stacked and cured under pressure and heat. In the case of cylindrical structures such as baseball bats, ski poles, golf clubs, and fishing rods, pressure is applied using a heat-sensitive polymeric tape that shrinks when heated.

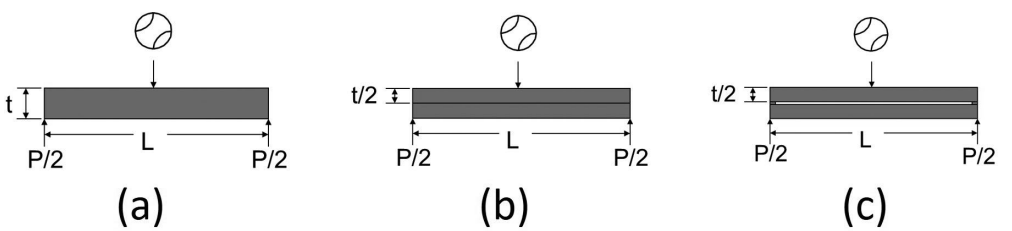

Figure 11.2 Idealized models to compare the mechanics of single wall (a), double wall (b), and double wall gap (c) bat barrels.

The shrink tape is coated with a release agent to allow removal after cure. Composite multiwall baseball bats are readily manufactured by leaving the shrink tape intact and laying up additional shells over it. The process is so efficient that composite bats with three walls are commonplace, and bats with four walls are also commercially available. Thus, composite materials not only reduce weight and allow increased stiffness control but can also be more easily produced with performance-enhancing multiwall designs.

The observation that multiple walls improve performance and durability has created an interesting feature for composite baseball and softball bats. Composite materials have high in-plane stiffness and strength and relatively low interlaminar strength. As composite structures age, damage in the form of matrix cracks can occur. Impact damage often develops in the form of delaminations, which tend to lower stiffness. It is not uncommon, for instance, for the performance of composite baseball and softball bats to increase with age. Some will intentionally damage new bats to take advantage of this effect. This is in contrast to the perception of metal bats, which some players will complain having "lost their pop" as they age.

About 10 years ago, the multi-wall baseball and softball bat design was further improved. The concept involves designing a gap between the concentric walls of the bat. It is depicted in Figure 11.2c using our beam example. A player with limited ability will not generate high swing speeds or large impact forces. A thin-walled bat with a large trampoline effect would be suitable for such a player. Multi-wall bats with gaps between their walls were designed to allow a single bat to be used by batters of differing abilities. A strong batter will generate high bat speed, large impact forces, and engage both walls of the bat. While multi-wall bats with gaps tend to lower the difference between batters, swing speed is still the dominant factor in ball exit speed. Even with this innovation, batters with the high swing speed will still hit the ball farther.

The stiffness control afforded by composite materials has a unique application for baseball and softball bats. Recall that the trampoline effect requires a softer barrel to increase performance. This can be achieved by aligning the fibers parallel to the length of the bat. Most sporting equipment, including bats, dissipate energy through vibration. Baseball bats, tennis rackets, and golf clubs have a so-called "sweet spot", which, when impacted, produces a high ball exit speed with relatively little energy transferred to the athlete. Balls impacting the sweet spot travel far, and the impact feels good to the athlete. which is indeed sweet. The penalty for impacts away from the sweet spot can be minimized by increasing the bat's flexural stiffness. Thus, in the case of baseball bats, aligning fibers parallel to their length optimizes the trampoline effect and lengthens the size of the sweet spot. As with thinning the walls of metal bats, however, the fiber alignment of composite bats has limits. If the fibers were all parallel to the bat's length, for instance, the barrel would not have sufficient strength to withstand ball impacts. The fiber angle is typically a compromise between maximizing performance and achieving the desired durability.

Golf shafts

When modern golf was invented in the 13th century, golf shafts were made of wood. In the 1920s, the first steel-shafted golf clubs were made in the United States. Fiber-reinforced composite materials started to be used to make golf shafts around 1970 [22].

Composite golf shafts are lighter than steel and can have similar stiffness without reducing shaft length or increasing weight. It was found that 95 g carbon fiber shafts can be as stiff as 120 g steel shafts, for instance. Studies show that shafts with high stiffness have higher angular velocity at the grip but lower clubhead speed [23,24]. Composite materials also absorb and damp vibrations more effectively than steel.

Some high-level players feel that the performance of composite shafts is unpredictable and requires more practice to learn to control [25]. Reduced weight in the shaft gives a composite club a different weight balance and feel. Carbon fiber shafts are nevertheless favored by most golfers for their reduced weight and tailored stiffness.

Ice hockey

Hockey sticks have changed from wood to aluminum to composite over the last 30 years. Composite hockey sticks are the most popular today and are commonly made from prepreg. Compared with wood and aluminum, carbon fiber hockey sticks can be lightweight and have tailored stiffness.

The mechanism of a hockey shot requires the stick to store and return energy to the puck. This is known as loading and is done by striking the ice with the stick prior to impacting the puck. Studies have found that more flexible sticks store more energy [25]. Similar to a golf shaft, hockey sticks are manufactured to have tailored stiffness. One study showed that the effect of stiffness on puck speed depends on the shot type [26]. For shots involving large stick loading, a more flexible stick will give a higher puck speed, while for shots involving low stick loading, the trend reverses.

Athletic shoes

Sports footwear can utilize a carbon fiber insole or midsole to improve energy storage and return it to the runner [27]. The Nike VaporFly, for instance, generated controversy by claiming to improve running efficiency by as much as 4% [28]. The shoes contain a curved carbon fiber plate embedded in the midsole. Studies have shown that those claims are credible [29,30]. One study found that the carbon fiber plate increases the longitudinal bending stiffness of the shoe, reducing running economy by 1% [31].

The use of composites in shoes initiated a debate on fairness to athletes. In 2020, World Athletics amended its rules governing competition shoes for elite athletes [32]. The rule limits the maximum sole thickness to 40 mm and allows no more than one carbon fiber plate to be used. The VaporFly has a 39.5 mm sole, just within the limits of the regulation.

Carbon fiber plates also play an important role in basketball shoes. This strong and lightweight material provides superior stiffness with minimal added weight. Composite midsoles improve the torsional stiffness of basketball shoes and help smooth the transition from heel to toe, helping athletes in jumping higher. The first application of carbon fiber in basketball shoes was the Air Jordan 11 [33]. Today, most professional basketball shoes are designed with carbon fiber midsoles. Studies have found, nevertheless, that the effects of carbon fiber on performance and injury prevention are small [34,35].

Pole vaulting

In contrast to most other sports, equipment is not regulated in pole vaulting. According to the IAAF regulations, the pole may be made of any material or combination of materials and can have any length or diameter [36]. Traditionally, poles were made of solid ash, hickory, or cedar and had relatively high stiffness. In 1904, a major improvement in vaulting performance was achieved with the introduction of the bamboo pole. Bamboo is a naturally hollow material and is much lighter than a solid pole, with desirable stiffness and strength. In the 1960s, glass fiber composites were introduced and brought revolutionary changes to pole vaulting. The pole vaulting world record by year and the associated pole material is shown in Figure 11.3. The efficiency of the pole was

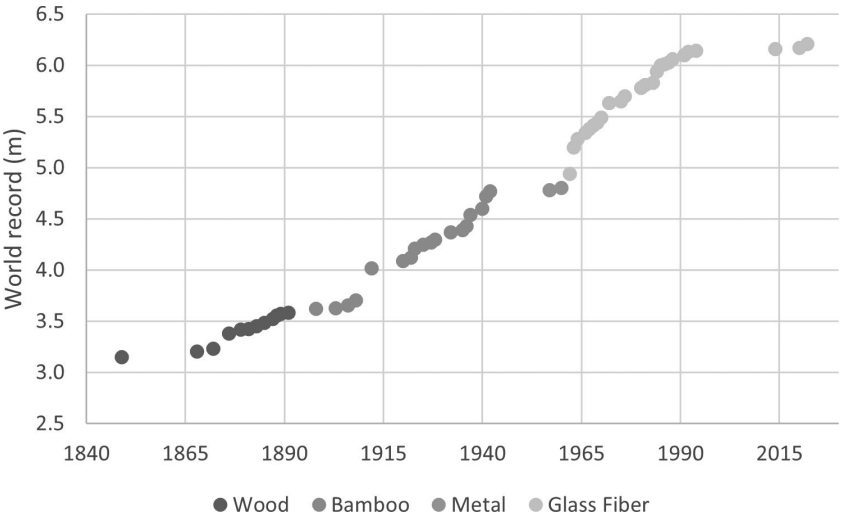

Figure 11.3 Progress of world record by year and associated vaulting pole material changes.

further improved with the introduction of carbon fiber composites in the 1980s, which led to a further reduction in pole weight [37,38]. Carbon fiber poles have not been widely adopted by the world's best pole vaulters, however, and have not yet broken a world record [39].

Pole stiffness and strength must be carefully tuned to the athlete's size and ability to maximize vault height. While carbon fiber is stiffer and stronger than glass fiber, it has a lower failure strain. Novel pole designs tailored to carbon fiber will likely be required for this material to outperform glass fiber poles.

Tennis rackets

Tennis rackets today are mostly made of carbon fiber composite materials. The application of carbon fiber to tennis rackets began in the 1970s [15]. Composite materials have higher specific stiffness than other materials, allowing composite tennis rackets to be lighter and stiffer than traditional wood rackets [40].

The stiffness of carbon fiber composites allows for greater power transfer from the racket to the ball, resulting in more powerful shots. Using stiffer rackets can help reduce vibration and increase ball speed [41].

Composite rackets are also known for their excellent control, as the stiffness and responsiveness of the material allow players to accurately place shots. This can be particularly beneficial for players who rely on precision and placement rather than power to win points [42].

Paralympic sport

Carbon fiber composites play an important role in Paralympic sports. The application includes both upper and lower limb prosthetics. The evolution of prosthetics has a long and storied history, with an artificial leg dating back to about 300 B.C., made of bronze and iron [43]. After thousands of years of development, today's devices are much lighter, mostly made of fiber-reinforced composite materials, providing users with the most functional devices.

Similar to other sports, one of the main breakthroughs of composite materials in Paralympic sports is energy storage and release. One example is prosthetic running blades. Thanks to their curved shape and carbon fiber construction, they are light and elastic, in some cases allowing runners to reach higher speeds than traditional prosthetics. Oscar Pistorius competed in sprint events using the Flex-Foot Cheetah transtibial prosthetic [44]. He also competed in non-disabled international competitions, including the Olympic Games. Some studies have shown that the running blades he used gave an unfair advantage [45,46]. The IAAF reports that prosthetic blades use 25% less energy than an able-bodied leg [47].

Upper limb prosthetics are used in sports such as baseball, cycling, golf, kayaking, etc. [48]. The purpose of upper limb prosthetics is to grip and maneuver apparatuses, and sometimes to help support the athlete. As a result, the main advantages of using carbon fiber in upper limb prosthetics are their lighter weight and higher strength.

Skiing

Skis are constructed from multiple layers. The basic construction of a ski has a laminated wood core at its center. The wood core is surrounded by composite layers. A top sheet is attached and a base is attached to the bottom [49]. The composite layers are designed to carry the ski's torsional stress. These layers protect the ski's core and contribute to the flexural and torsional stiffness of the ski. The composite layers are thin but provide a ski with 50–80% of its overall stiffness [50].

The most common composite used in ski construction is fiberglass. While fiberglass has dominated the market for years, the use of carbon fiber has increased over the last decade. The weight reduction compared to fiberglass makes carbon a popular material in backcountry and touring skis, where weight is important.

The longitudinal and torsional stiffness are important factors in a ski's performance. High stiffness provides a more stable platform with increased speed. Lower stiffness makes turning easier but results in a less stable feel [51,52]. Ski stiffness often varies along its length. A large difference in flexural stiffness and a small difference in torsional stiffness leads to better ski performance [53]. Ski manufacturers use different combinations of the wood core along with the composite layers to provide tailored stiffness to the ski.

Cycling

There are many design requirements for road racing bike frames, but the foremost are weight and lateral stiffness. Lighter weight decreases energy consumption uphill or during acceleration and provides a more responsive feel to the rider's movements [12]. The use of a 7 kg bicycle will decrease the time required for a 20 km climb by 29 s to 3 min compared to a 10 kg bicycle [54].

Since the first all-composite bike frame (fiberglass) was invented in 1960 [12], bicycles in the Tour de France have changed dramatically as frames made of traditional steel have evolved to carbon fiber. Figure 11.4 shows the average speed and bike weight for a number of Tour winners [55–58]. In 1986, the first fiber-reinforced composite frame (Look KG86) was raced in the Tour de France and helped Greg LeMond win the title. The tubes were made of Kevlar with layers of woven carbon fiber and bonded with aluminum lugs. Thanks to the carbon tubes, the KG86 weighed 7.2 kg, 1 kg lighter than its metal opponents. For the following decade, metal frames continued to dominate the Tour, while carbon bikes suffered quality problems due to poor lamination methods and voids. However, the push for lighter weight bikes had begun. For example,

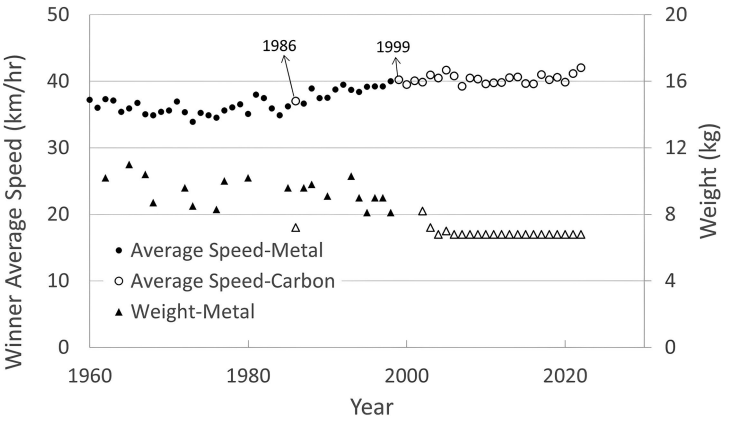

Figure 11.4 Average speed and bike weight for some winners of the recent Tour de France.

the last aluminum frame bike (Bianchi Mega Pro XL Reparto Corse) to win the Tour in 1998 weighed 8.1 kg, compared to the last winning steel bike (Pinarello) in 1994 (9 kg) and the ceramic-reinforced aluminum matrix composite bike (Pinarello Keral Lite) in 1996 (9 kg).

As fabrication technology for carbon epoxy prepreg, implemented from the aerospace industry, matured, carbon fiber bikes grew in popularity and began to dominate the Tour de France with the success of Lance Armstrong and his carbon fiber bikes [59]. The success of Trek manifested how carbon fiber's lightweight and higher stiffness assisted riders in competitions. The Trek 5500 (1999) was the first bike with a 100% carbon frame, where the pure carbon lugs and tubes were bonded together to win the Tour. The frame (not the entire bicycle) weighed only 1.1 kg, which was the lightest at that time [59]. From 1998 to 2006, the weight of winning bicycles dropped from 8.1 kg to 6.8 kg, while the average speed increased by 0.8 km/h, which could cause a rider to be an hour faster over a tour.

Carbon fiber parts may be shaped into complex aerodynamic structures and their directional properties contribute to the growing use of carbon fiber in every major component. A lightweight wheel helps minimize energy consumption during acceleration and uphill races. Carbon fiber composites have allowed the manufacturing of ultra-light disc and spoked wheels, including wheels with three to five spokes, compared to the traditional 16 spokes of a wire-spoked wheel [60]. Carbon/epoxy handlebars are 60% and 80% lighter than aluminum alloy and steel handlebars with the same dimensions, respectively [61]. The stiffness of a bicycle's frame can be accessed at different locations: steering, pedaling, and vertical [62]. In general, a bicycle with high pedaling stiffness is faster due to less energy lost to the frame. However, a stiff bicycle also gives the rider an uncomfortable feeling and can affect braking performance and traction [63]. Carbon and aluminum frames are about 20% stiffer than titanium or steel. Besides being stiffer and lighter than other materials, carbon fiber allows for tailored stiffness.

In 2000 the International Cycling Union (UCI) introduced a minimum weight limit of 6.8 kg for racing bikes. However, the pursuit of lightweight bikes has not stopped, as lighter materials allow for aerodynamic structures. A study showed that the combination of boron and graphite fibers in an epoxy prepreg can achieve a bike frame with a 10% weight reduction and comparable damage tolerance [64]. In 2016, Dassi introduced the first graphene bike, the Interceptor, to the market. Compared with a carbon frame, the frame made of carbon fiber enhanced with graphene is 30% lighter and twice as strong [65].

Formula one

First introduced in 1981 by the McLaren Formula One (F1) team, carbon fiber composites now make up almost 85% of the volume of a contemporary F1 car while accounting for less than 30% of its mass [66]. The latest Formula One regulation issued by the Fédération Internationale de l'Automobile (FIA) requires a minimum weight of 796 kg (including the driver) [67]. As with cycling, the weight reduction from carbon fiber improves vehicle dynamics [68], allowing a ballast to lower the center of gravity for improved acceleration, cornering, and braking [69].

Since the gearbox is located at the rear of the car in a relatively high position, weight reduction is important for gearbox design. In 2004, the BAR-Honda Formula One design team produced the first carbon fiber-reinforced gearbox. They integrated the gearbox into the design of the rear of the car and reduced its weight by 30% compared to a traditional metal version [70]. Formula One cars use disc brakes consisting of a rotor and caliper at each wheel. The Brabham team replaced steel or cast-iron rotors with carbon-carbon composite rotors in 1976 [71]. The corner weight (car weight on each wheel) of a full carbon brake system was 6 kg, 70% lighter than an iron brake system [72].

Responding to the safety requirements proposed by the FIA in 1963, a variety of methods and equipment to protect drivers from accidents have been used in F1. Composite materials played an important role in dramatically decreasing race fatalities. Figure 11.5 shows the number of fatalities during FIA World Championship events and other accidents with Formula One cars [73,74], with safety innovations and regulations involving composites [66,75]. The first carbon fiber composite chassis were met with public skepticism in 1980. The doubts were quickly erased after a driver walked away from the debris of a car that smashed into the barriers during the 1981 Italian Grand Prix. The carbon chassis absorbed and dissipated a large portion of the impact energy, while its high stiffness limited the load transferred from the impact area to the monocoque and driver since the chassis did not involve plastic deformation [66]. The specific energy absorption of a steel tube is between 11.5 and 38 kJ/kg [76], an aluminum tube ranges from 22 to 43 kJ/kg [77], while a carbon fiber/epoxy tube is around 60 kJ/kg [78]. Between 1980 and 1992, the rate of fatal and severe injury in Formula One was less than 1/250, significantly reduced from the rate in 1980 (1/40) [79].

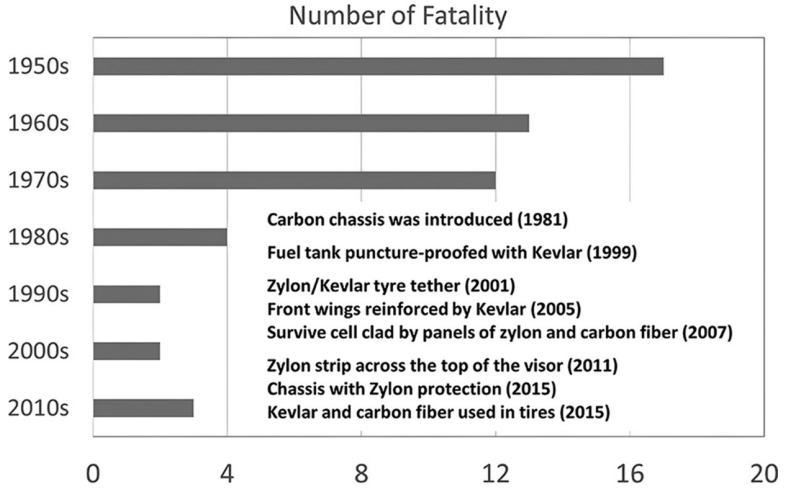

Figure 11.5 The number of fatalities related to F1 cars and safety innovations and regulations involving composites.

Kevlar fibers provide good impact and penetration resistance and are mandatory within some structures at the front of the car [66]. For example, the front wing endplates must be covered by a layer of Kevlar to keep the carbon fiber together and prevent sharp fragments from breaking the wings in a collision. The survival cell must include Kevlar, Zylon, and carbon fiber to prevent front suspension components from penetrating the monocoque [80]. Kevlar combined with carbon fibers is widely used in tires to prevent penetration damage. Another vital safety regulation is the Zylon strip across the top of the helmet visor. This evolution prevented F1 driver Max Chilton from being injured by another car's debris striking the Zylon strip during the German Grand Prix in 2013.

Helmets

Head injury in sports can be fatal, ranging from structural damage to the skull, mandible, and soft tissues to traumatic brain injury [81]. Head protection in sports was first introduced over 100 years ago due to severe injuries and deaths in American football [81] and is now extensively used in a range of sports. Helmets have been effective in reducing moderate to severe head injuries and have become mandatory in football, biking, ice hockey, and snowboarding [82].

The protection system of a modern sports helmet generally consists of an outer shell, an inner liner, and sometimes face protection. The rigid outer shell provides the initial defense against impact, absorbs a portion of the impact energy, prevents penetration by sharp objects, and holds the liner components in place. The relatively soft liner prolongs the impulse duration and absorbs impact energy, reducing force and acceleration to the head.

The shell is usually made of conventional thermoplastics like acrylonitrile butadiene styrene (ABS), polyurethane, polycarbonate, or composites [83,84]. While composite shells have higher material and processing costs, they absorb more impact energy than thermoplastics [83]. Bhudolia et al. developed a thermoplastic and a hybrid composite bicycle helmet shell [85]. These composite shells absorb 38% to 66% of the impact energy, compared to 13% for polycarbonate shells. The composite shells reduced the probability of critical and fatal injury by around 40% and 60%, respectively.

Some have found that the composite shell stiffness affects the helmet impact response. Kostopoulos et al. provided a simulation of a motorcycle safety helmet, including damage and delamination mechanisms [86]. A Kevlar shell achieved longer impact duration, lower peak acceleration, and higher absorbed energy than carbon-fiber and glass-fiber shells. They also found that composite shells with high stiffness provided good protection under high impact energy levels, where fiber damage was activated and dominated. Thus, fiber-reinforced composites are increasingly used to form the outer shell of high-energy impact sports helmets [87].

Shoulder pads

Shoulder pads are used in American football, ice hockey, and lacrosse. Among the players of the National Collegiate Athletic Association (NCAA), shoulder injuries are the second most common injury for ice hockey [88] and football players [89], and the fifth for lacrosse players [90]. Shoulder pads are essential to protect players from shoulder separation injuries while not restricting their motion. The design of shoulder pads, including shape and size, may differ from sport to sport. The primary shoulder pad structures are similar, consisting of soft foam padding and plastic caps covering the chest and shoulders. Similar to the protection mechanism of a helmet, foam in the shoulder pads disperses and absorbs impact energy on the shoulders and mitigates shock by

increasing the space between the impact surface and acromioclavicular joint, while the plastic caps are strong and rigid to withstand point loads [91].

Unlike helmets with relatively mature fiber-reinforced methods and materials, the applications and benefits of composite materials in shoulder pads are still under discussion. Since ice hockey shoulder pads are required for mobility, carbon fiber-reinforced composites are under consideration to replace the plastic caps due to their high strength-to-weight ratio. Compared with a polyethylene shoulder cap, there was a slight improvement in the carbon fiber cap in energy-absorbing tests [91]. Some lacrosse shoulder pads employ a carbon fiber-reinforced thermoplastic for the shell of sternum and spine protection and shoulder cap [92]. During a pendulum test, where a pendulum struck the shoulder of a test dummy, a carbon fiber pad was the most effective in reducing accelerations in low- and medium-impact testing. The carbon pad also had the least effect on the vertical range of motion, with participants losing only 0.3% of their uninhibited range. One of the first commercial football shoulder pads with carbon fiber caps reduced the maximum applied force by 81% at impact with a 10% reduction in weight [93]. Despite these findings, high costs and limited evaluation have hindered the use of composite materials in shoulder protection.

Sailing

The first fiber-reinforced boat was built in 1937, but it was not developed further due to the brittle nature of the matrix and immature manufacturing methods [94]. In addition to the high strength-to-weight ratio, glass-reinforced plastic shows excellent resistance to corrosion and environmental degradation compared to steel, aluminum, and wood. Today, over 90% of all marine composite structures are made with glass/polyester and glass/vinyl ester materials [95]. The Laser Olympic dinghy is one of the most popular sail racing competitions. One study showed that the effect of weight on upright resistance of a Laser dinghy reduced resistance between 2.7% and 6.2% for a weight change of 10 kg from a normal total weight of 160 kg [96].

Many lightweight racing yachts are built using carbon-reinforced composites, which have higher stiffness than glass/polyester and glass/vinyl ester laminates. While sailing vessels have developed with the help of composite materials, yacht rigging has also taken advantage of the favorable material properties of composites. Kevlar, Polybenzobisoxazole, and carbon fibers have replaced steel as the dominant materials in rigging, as they provide significant weight reduction with comparable stiffness and improved fatigue and corrosion resistance [97].

Boat designers and manufacturers have used honeycomb structures for many years to achieve weight-saving and improved performance. Mostafa et al. developed a composite sandwich panel for sailboat hulls using glass fiber/epoxy facing and a polyurethane foam core [98]. A kayak called Flyak made extensive use of epoxy/carbon laminates and DIAB sandwich cores, offering excellent lightness and strength [99]. Leading market manufacturers focus on the development of foam core materials, offering excellent strength-to-weight performance for all types of marine applications [100–102].

Environmental sustainability

With the growing demand for high-performance, lightweight, and improved materials in sports, the market for composites in global sporting goods is rapidly developing and is expected to reach $1.1 billion by 2027 with a compound annual growth rate of 6.3% from 2021 [103]. However, the fast turnover of sports products results in increased composite waste that can be difficult to recycle

[104]. There are three methods for fiber-reinforced composite waste disposal: landfill, incineration, co-incineration, and recycling. Landfill has been the primary method for many years but is now limited by government legislation [105]. Incineration with partial energy recovery from the combustion of organic parts leaves 50% of the combusted waste to be landfilled [106], and the residual ash can cause air and underground water pollution. Co-incineration of glass fiber-reinforced composites in a cement kiln can be used to produce clinker but has restrictions, including waste size, ingredients, and calorific value [107]. In contrast, recycling (mechanical [108,109], thermal [110], and chemical [111]) is the most desirable disposal method, though it faces the challenges of high cost, low processing efficiency, and concerns about the recyclates' quality.

To decrease the adverse environmental impacts caused by the disposal of composite products, researchers and designers are shifting their attention to bio-composites using natural fibers as reinforcement materials, which are biodegradable and recyclable. Flax fibers can replace glass fibers in a racing sailboat with less density and comparable strength and modulus [112]. With high damping capacity, flax fibers are often combined with carbon fibers in tennis rackets [113], skis [114], and bicycle frames [115] to improve impact toughness, effectively reduce vibrations, increase athletes' control over their equipment, and reduce injury risks. Bamboo fibers are suitable for sports apparel due to their high moisture absorption capacity, breathability, and fast drying behavior [116,117]. The high strength and stiffness of hemp fibers make them suitable for snowboards [118] and surfboards [119]. Other natural fibers, such as coir fibers [120], banana fibers [121], and jute fibers [122,123], are seeking opportunities in the sports industry. Further research on the diversity of natural fibers, characterization and modification of bio-composites properties, the contributions of natural fibers to systematic mechanical properties, and optimization of manufacturing methods are required to expand the promising market with environmental sustainability and economic benefits.

Summary

Advances in composite materials have played a significant role in sports, reducing equipment weight and increasing athlete performance and safety. The progress has primarily been due to improved materials, from the development of new toughened polymers to novel manufacturing methods. Improved material design has further strengthened composite sport applications, where their directional properties provide additional performance advantages. While some sport federations limit the benefits of composite materials, composites have made the sport more enjoyable and accessible to both professional and amateur athletes alike. Composite materials will be a driving factor affecting sport performance and safety in the future. Regulation will continue to play a key role, helping to ensure technological advances are managed in a way that ensures fair, safe, and enjoyable competition. New uses of stiff, lightweight composite materials will continue to evolve, where novel approaches to energy management will further enhance athlete performance and safety. Cost and end-of-life use constitute difficult barriers for composites in sport. Equipment cost, rather than athletic ability, may become a barrier for athletes from developing areas. On the other hand, broad acceptance of composite sporting equipment will present challenges in disposing these materials, which are difficult to recycle or reuse. Bio-composites are expected to contribute to the promising market.

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12

ADVANCED MATERIALS FOR BOARD SPORTS

Jörg Kaufmann, Armin Farzin, and Paul Baudach

Introduction

Nowadays, board sports involve various sports, and there is sports equipment for almost every surface on this planet. There are boards made for water and snow, as well as for practicing on concrete, tarred roads, or sand. To meet the requirements of the different surfaces, various materials are used in the board sports sector. The most commonly used materials are wood, plastics, and material combinations such as fiber-reinforced plastics (FRP). Metals are also used in some areas. Almost every board sport began with simple sports equipment made of wood (the first surfboard, skateboard, and snowboard) and became more complex and optimized in terms of its field of use. Wood is widely available and can be processed with simple tools. However, its lightweight construction potential is limited. Therefore, most wooden constructions were supplemented or even replaced by alternative materials. Composites have been introduced for almost every kind of board sport to produce equipment that is as light and durable as possible. Board sport equipment is made with varied thicknesses, forms, and constructions to accommodate different surfaces and environmental factors that board sport athletes perform on. Regardless of the sports activity, the boards must be able to withstand bending and impact stresses, as well as have the proper ratio of strength, flexibility, and stiffness [1].

As well as the technical demands on sports equipment, the sustainability requirements have also increased in recent years, and there is a growing desire among customers for more environmentally friendly alternatives (see Chapter 17 “Sustainable design and manufacturing of sports equipment”).

This chapter will cover an introduction to the most common materials used for board sports construction and how their selective application can improve the performance of sports equipment. Furthermore, processes used to manufacture the preforms are discussed, which offer further optimization potential through the more targeted use of materials. As sandwich construction (see schematic drawing in Figure 12.1) is used in most board sports products, the description of the materials is divided into core materials, which are subject to high demands in terms of lightweight construction and shear strength, and materials for the reinforcement layers, which must withstand the various stresses associated with use. To ensure that the products have a good graphic design and that the core and fibers are protected from environmental influences such as moisture and ultraviolet (UV) light, they are usually coated with printed polymer films, gel coat layers, veneer, or similar.

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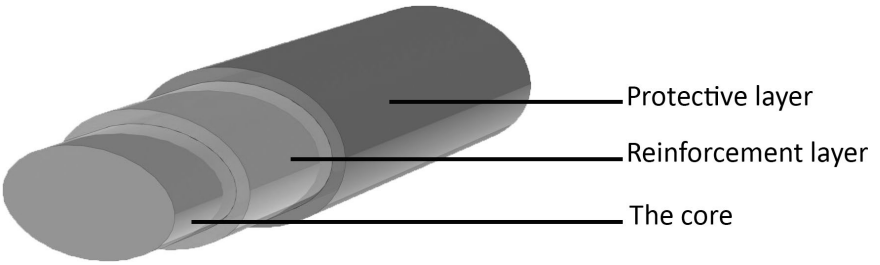

Figure 12.1 Board sports' composite structure.

Type of material

Reinforcement fibers and textiles

As well as solid materials such as wood, plastics, or metals, a combination of reinforcement fibers and matrix materials, known as FRP or fiber-reinforced composites, are used in the board sports sector. These combinations enable the sports equipment to be more durable and lightweight than would be possible with solid materials alone. This first section will focus on the fiber materials.

Fiber is a linear, flexible, and elementary structure. Fibers are the basic material for yarns, non-wovens, or FRP. Fibers can be broadly classified into two types. On the one hand, man-made fibers are produced artificially in the form of continuous filaments. On the other hand, natural fibers are taken from plant or animal material and are only available in shorter lengths. These must be processed into longer or continuous yarns using a spinning process. Depending on the fiber material used, the properties of the end product can be influenced and thus adapted to the requirements of the sports equipment.

Due to their high flexibility, fibers buckle under axial compression. However, they can withstand comparatively high tensile loads. Therefore, in addition to the fiber type, attention should be paid to their aligned orientation when developing sports equipment. The most common fibers in board sports will be explained in more detail in the following section.

Synthetic fibers

Man-made fibers are those in which the base material is converted, usually thermally or chemically, to melt the material and then brought into the form of a fiber. The fluid material is then processed through a fine nozzle, and a so-called filament is formed when the material cools down in a medium directly after the nozzle [2]. Most materials that can be melted and hardened again can be used to produce fibers. Fibers with lower tensile strengths, such as polyester fibers, are mainly used in the manufacture of clothing and accessories. Fibers with excellent mechanical properties, known as reinforcing fibers, are used for the lightweight construction of board sport goods.

Reinforcement fibers offer excellent functional characteristics alongside superior mechanical properties at low density. Fibers with exceptionally high levels of physical and chemical characteristics are known as high-performance fibers. Glass (G), carbon (C), and aramid (AR) fibers are examples of high-performance fibers [3].

GLASS

Glass refers to the class of inorganic, non-metallic materials. Depending on the type and the application of glass, it consists of finely powdered, iron-free, and pure quartz sand, soda (sodium

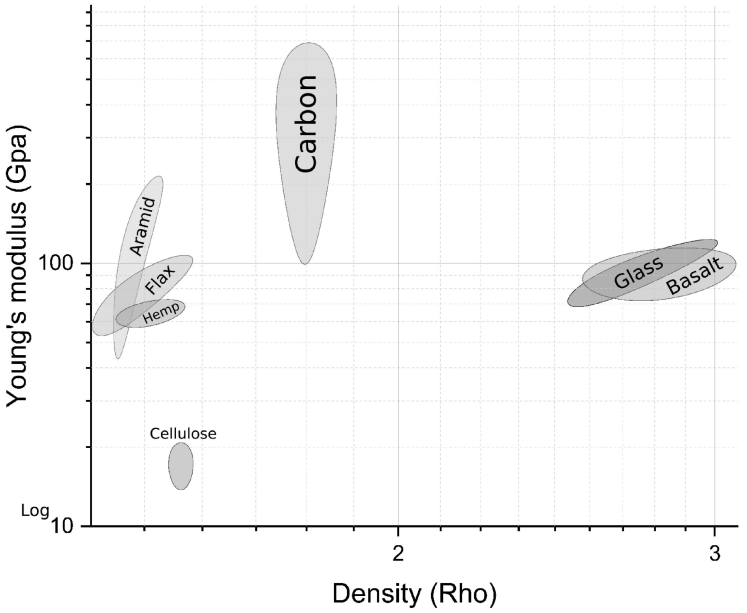

Figure 12.2 Young's modulus and density range of different fibrous materials [3, 11, 21, 22, 25, 26].

carbonate) or potassium carbonate as a flow agent, and other additives. Technical glass is made from raw materials like quartz powder (silicon dioxide), limestone (calcium carbonate), colemanite (borate mineral), kaolin (kaolinite), and other finely dosed oxides (such as boron and aluminum oxides, calcium, and magnesium oxides). These materials combine with tailored chemical and physical properties to create glass that is ideal for processing into glass fibers [3]. The high permeability of glass fiber promises a homogeneous polymer infusion, resulting in a uniform composite material. Furthermore, reliability, ease of production, and comparatively lower price make glass fibers (especially in the form of non-crimped fabric or woven fabric) an ideal material for manufacturing board sport sandwich composites. Typically, the Young's modulus for glass fiber ranges between 72 and 115 GPa, with a density of 2.52 to 2.89 g/cm^3 [3] (Figure 12.2).

ARAMID

Aramids (a well-known commercial term, Kevlar®) are synthetic fibers created from synthetic polymers formed of aromatic long-chain polyamides, where at least 85% of the mass is linked directly with two aromatic rings, and up to 60% of the amide bonds can be replaced by "imide" bonds [3]. Aramid fibers, used in composite materials, are primarily made from three polymers: poly-m-phenyleneisophthalamide (MPIA), poly-p-phenylene terephthalamide (PPTA), and poly-p-phenylene 3,4-diphenylene. Among these, para-aramid fibers (from PPTA) are prioritized for composite production. The unique structure of para-aramid fibers, with high crystallinity and orientation, can be manipulated during spinning to achieve specific fiber profiles. Notably, drawing enhances crystallinity, producing high-modulus fibers. Aramid fibers exhibit high strength and stiffness at low density, and their tenacity allows plastic deformation before breaking. The fibrillar structure limits rotation around the longitudinal axis. These fibers excel in energy absorption and impact resistance. However, they are susceptible to compressive loads in the axial direction due to

their structure [3]. Using aramid materials in woven form enhances the composite's impact resistance, making them desirable for applications such as board sports equipment in situations like rock collisions in surfing or in foiling equipment connectors with high damping properties in regions subject to high stresses. Typical Young's modulus values for aramid fibers range from 58 to 186 GPa, with a density span from 1.39 to 1.45 g/cm^3 [3] (Figure 12.2).

CARBON

Fibers or whiskers (monocrystals) of carbon are products of pyrolysis from organic carbon compounds. Based on the fiber structures and carbon crystal alignment, two kinds of carbon fiber filaments are distinguished. Isotropic carbon fibers are those without identifiable preferred orientations of carbon crystals. Though these fibers are excellent for fillers, strings, and packing for thermal insulation, they are not well suited for manufacturing composite materials because of their comparatively low strength. On the other hand, when carbon fiber layer planes are aligned with the fiber's axis, the product is called anisotropic carbon fiber which shows good mechanical performance in composites [3]. Anisotropic unidirectional carbon fiber filaments are the main products used in high-end board sports products and are the focus of this chapter. Compared to other fibers, carbon has the following key advantages: high tensile strength, high stiffness, low density, and strong chemical resistance. When paired with a suitable matrix material (polymer resin), all these benefits can result in composite components with exceptional mechanical properties [4]. Usual Young's modulus values for carbon fiber range from 240 to 650 GPa, with a density of 1.73 to 2.15 g/cm^3 [3] (Figure 12.2).

Natural fibers

Natural fibers originate from plants, animals, or minerals, creating new types of fibers that show promising results as reinforcing materials, specifically in polymer-matrix composites [5]. Their lightweight nature, high specific strength and modulus, nontoxicity, biodegradability, recyclability, and renewability are the factors that make them enticing for use in polymer matrix composites (PMC) [5]. Another important feature influencing the usage of natural fiber composites (NFC) specific to sports and leisure products (snowboards, surfboards, etc.) is their good vibration-damping characteristics [6]. Furthermore, in NFCs, natural fibers can be combined with glass and carbon fibers or used alone to enhance the mechanical properties [7]. Although issues such as inconsistent raw material properties, unpredictable permeability, proper sizing material, and low fiber volume content remain partially unresolved in natural fibers, they can still be advantageous in manufacturing sports boards. The most utilized natural fibers and their applications in board sports will now be described.

FLAX

Flax is a natural fiber derived from the linen plant. Two methods are used to remove the fiber from the stem: mechanical and bacterial. After fiber extraction and the combing process, the material is incorporated into untwisted yarns (roving). A chemical bonding agent is then added to the material to ensure cohesiveness between fibers [7]. Flax fiber composites have good mechanical properties, low density, and minimal environmental impact, making them great options for lightweight applications. Furthermore, flax composites show higher specific stiffness than glass fiber composites in both tension and plate bending, and only slightly lower values than carbon fiber composites in plate bending. These fibers also possess a distinguished high vibration damping capacity, making them exceptional candidates for sporting goods applications [8]. A product line of Notox surfboards (based in France) [9] and Magine snowboards (located in Canada) [10] are reinforced with flax fibers

as examples of applied cases in board sports. Typical Young's modulus values for flax fiber range from 60 to 100 GPa in the longitudinal direction at a density of 1.4 to 1.5 g/cm^3 [11] (Figure 12.2).

HEMP

Hemp is the second most widely used natural fiber (after Sisal, which is not common in board sport composites) as reinforcement in composites. Hemp is naturally one of the most ecologically friendly fibers [12]. Cannabis sativa L., the scientific term for the production-worthy hemp plant, has become one of the most fashionable plants for composite usage [13]. Various techniques are used to extract fiber from hemp stems, such as steam explosion, osmotic degumming, enzymatic retting, dew retting, water retting, and mechanical decortication, which break down pectin, lignin, and hemicellulose [13]. According to a 2021 study [14] comparing hemp and flax fiber, hemp has a high production of long-line fiber, and as hemp fields typically produce more biomass than flax fields, more long-line fibers might be produced. The tensile properties and fiber division of the produced hemp fibers are suitable for use in load-bearing composite materials. As such, they might be seen as a high-potential supply of natural fiber to be used in composites [14]. As examples of use cases in board sports, Solid Surf (in the USA), manufactures their surfboards using hemp fiber laminated composites [15], and Silbaerg Snowboards (in Germany) [16] uses a combination of recycled carbon fiber and hemp in one of their product lines [17]. Typically, the Young's modulus values for hemp fiber range from 58 to 70 GPa at a density of 1.4 to 1.5 g/cm^3 [11] (Figure 12.2).

CELLULOSE

Cellulose is the most common organic polymer on Earth [18] and the main structural component of green plants' cell walls. Fibers made from esters, ethers, or cellulose, which can be extracted from the bark, wood, leaves, or plant stems, or other plant-based materials, are usable as bio-based reinforcement fibers. In addition to cellulose, the extracted fibers may also contain hemicellulose and lignin, and different percentages of these components will alter the mechanical properties of the fibers. Cobra (based in Thailand) [19] has utilized this material in a range of their surfboards since 2009. Furthermore, rayon (or viscose) is the regenerated form of cellulose. Dissolving cellulose (usually from wood pulps) and then reconstructing it as insoluble fibrous cellulose creates rayon. With a proper and modified process, this kind of cellulose fiber can make strong continuous rovings, usable in lightweight structures. Manufacturing rayon in traditional processes generates lots of contaminated water and reduces the desirability of this semi-synthetic natural fiber. However, developing new manufacturing processes and considering the life cycle perspective of its composites can open new use cases for rayon, such as in board sports. An example of a supplier of technical continuous rayon fibers is Cordenka (based in Germany) [20]. Typical Young's modulus along the fiber direction of rayon ranges from 14.5 to 20 GPa [21] at a density of 1.53 g/cm^3 [22] (Figure 12.2).

BASALT

Unlike the three aforementioned fibers, basalt is a mineral-originated natural fiber. Basalt is a glassy lava rock, mostly composed of the oxides of silicon, aluminum, iron, calcium, and magnesium. For the manufacturing of basalt filaments, only rocks with a silicon dioxide level exceeding 46% are usable [23]. The melt-spinning process to produce basalt fibers requires restricted viscosity, crystallization, surface tension, and thermal and chemical homogeneity of the molten material [3]. The produced unidirectional filaments usually have diameter ranges lower than glass fibers and

slightly higher than carbon fibers, resulting in relatively good mechanical performance for use in board sports production (as thinner fibers result in stronger composites). Also, compared to carbon fiber, basalt fibers possess more elasticity. Though the energy requirements for producing basalt fibers are more favorable than those for carbon fibers, sourcing the raw material in nature and providing a constant and controlled production line is challenging and requires further advancements to lower their price. As an example of a use case, Rossignol (based in France) [24] uses basalt fiber preforms in some of their ski and snowboard product ranges. Usually, the Young's modulus values for basalt fiber are from 79.3 to 110 GPa and a density of 2.63 to 3.05 g/cm^3 [25] (Figure 12.2).

Matrix systems

In composite materials, the matrix acts as a binder, preserving and shielding reinforcement fibers from abrasion, chemical degradation, or other environmental factors. Polymers, metals, and ceramics are the three categories of matrix materials. The characteristics of these three matrix categories differ considerably, which impacts the properties of the composite end product [26]. Due to their lightweight nature, applicability in molds, and ease of use for manufacturing sandwich board structures, PMCs are the main type of composites applied in board sports. The two main types of polymer matrices, thermoplastics and thermosets, will now be described. Unlike the previous descriptions of reinforcement fibers and textiles, the typical material properties of polymer matrices are not stated for three reasons. First, the strength of a composite relies mostly on the properties of the reinforcement fibers, not the matrix. Second, within board sports applications, the properties of the polymer used for the matrix vary considerably, depending on the specific application. Third, each manufacturer uses a specific polymer for their product based on the combination of the preforms and the manufacturing process. Furthermore, manufacturers tend to keep the properties of these polymers confidential.

Thermoplastics

Thermoplastics define any polymeric material that becomes malleable or moldable at a certain elevated temperature and hardens upon cooling [27]. Thermoplastics obtain the simplest molecular arrangement among chemically independent macromolecule materials [28]. These polymers show potential as low-cost raw materials with long shelf lives and short manufacturing times. They offer good impact qualities and can be recycled and reshaped by heating. Thermoplastic polymer melts processing yields inexpensive intermediate composite products, which are later assembled into intricate, high-performing structures through flow molding and other cost-effective manufacturing processes. Composites made from these polymers are appealing as structural elements due to their exceptional impact resistance and capacity for large-scale manufacturing [29]. On the other hand, constant energy consumption during their processing, relatively high creep deformation, and mostly low resistance to UV light and other environmental factors are some of the main reasons these materials are less desirable for board sport applications. Although these materials have already been utilized in board manufacturing—snowboards from the manufacturer Nairoby [30] serve as an example—they are mostly exploited for board sport accessories with more complex geometries, better suited for injection molding processes. In these scenarios, fiber-reinforced thermoplastics tend to be a good solution. For instance, snowboard bindings [31] and surfboard fins [32] are examples of such commercialized accessories. Another use case for these materials is as a decorative cap (foil) or protective gliding bottom (polyethylene) surface in snowboards, which is also utilized in other board sports [33].

Thermosets

Thermosets are polymers made by the chemical reaction of two polymeric components under ambient conditions or induced by radiation or heat, forming a highly cross-linked network solid. The formation of these materials is an irreversible process [34]. Like thermoplastics, thermosets are independent macromolecules before hardening. However, in their final form, upon solidification, they form a three-dimensional structure achieved through chemical crosslinking, which occurs either during (e.g., compression or injection molding) or after (e.g., spray-up molding or filament winding) the process [28]. Although the recyclability of thermoplastics makes them a more sustainable solution than thermosets in most cases, these polymers possess many advantages that make them the more desirable material in board sport applications. Qualities such as consistency in production, customization and repairability, less complicated molds, greater design freedom, and lower energy consumption in composite manufacturing are some of their advantages over thermoplastic matrix systems in board sports. Furthermore, although they are not as reusable as thermoplastics, advancements in bio-based epoxy resins and other thermoset matrices are expected to reduce their ecological concerns in the future.

Types of preform

The qualities of the finished composite or product are significantly influenced by the textile reinforcement of the composite facesheets, technically referred to as the "preform." Using an appropriate textile production technique, the preform formation process provides precise control over the fiber design and orientation [35]. The alignment and architecture of reinforcement fibers are important in board sport composites. Preforms often consist of different layers of fiber reinforcement. The parameters of a layer include material (glass, carbon, etc.), fiber orientation, textile weave (the architecture of the textile), and area weight. In the following section, based on their weave type, the most commonly utilized types of preforms in board sports will be introduced.

Non-woven fabrics

Non-woven textiles are created by arranging long fibers (short fibers 0.1 to 1.0 mm, long fibers 1 to 50 mm, and endless fibers >50 mm long) into sheets, thereby increasing the manageability of these otherwise unstructured fibers. In some cases, rows of needles may be temporarily deployed to puncture the fiber sheet, repositioning some of the fibers to establish a mechanical bond between them, resulting in needle-punched fabrics. Another approach is to bond the long fibers together with adhesive [36]. Regardless of the technique used to make such material, the result will be an anisotropic textile [37], with certain drawbacks and capabilities for use in different composite structures. To mention some of their potential uses in board sports, they can be used as local reinforcement or aesthetic lightweight finishing coats. Furthermore, producing such materials from recycled composites or leftover textiles from other processes can help move toward a circular economy in board sports (see Chapter 17: "Sustainable design and manufacturing of sport equipment").

Woven fabrics

Warp (0-degree) and weft (90-degree) threads are interlaced to create woven fabrics in a regular pattern or weave style (a biaxial fiber direction textile). The mechanical interlocking of the

strands preserves the fabric's integrity. The weaving style of a cloth primarily determines its stability, surface smoothness, and drape (the capacity to conform to a complex surface). As they combine 0 and 90-degree fiber orientations, they are beneficial for composite manufacturing applications where more than one fiber orientation is required [3]. Being common in the market, available in various materials (carbon fiber, glass fiber, aramid fiber, and natural fibers) and different styles, woven textiles tend to be usable in board sports. Besides the waste of materials in the cutting process of the preforms, in the case of continuous unidirectional fibers such as carbon and glass, weaving increases the undulation of each fiber, which decreases the tensile and shear strength of the final composite [38,39]. On the other hand, the increased undulation can improve regional impact resistance [39], which can be beneficial in some use cases of sandwich board constructions.

Non-crimp fabrics

Non-crimp fabrics are fabrics made of numerous layers of unidirectional (UD) fibers that are layered on top of one another and bonded together, usually by stitching with a thin thread of (often) polyester. Recent research has shown non-crimp fabrics to be a particularly beneficial addition to the spectrum of textiles accessible for composite material applications [3]. Fabrics with various parallel, straight yarn bands are called multi-non-crimp fabrics. Different angles can be applied to stack these bands on top of one another. Due to the straight orientation of the yarns and the ability to arrange them in various narrowly adjustable directions, multi-non-crimp textiles are mostly used for composite applications rather than garments. Diverse types of multi-non-crimp fabrics exist depending on the application. The arrangement and orientation of the fibers determine the various textiles. There are three main kinds of non-crimp fabrics: multiaxial (fibers are oriented in more than two directions), biaxial (fibers are oriented in two directions), and uniaxial (fibers are oriented in one direction alone) [35]. As a semi-finished product, their usage in the board sports industry is easy and accessible. Non-crimp fabrics made of glass fiber, carbon fiber, and natural fibers are available in the market with various properties. On the other hand, depending on the sport and the design of the board sport goods, they are limited to the pre-made fiber direction angles and layer arrangements. Additionally, waste is substantial in their preparation and preform manufacturing stages.

Tailored fiber placement

Embroidery is a well-known textile technique that has been used for centuries to apply yarns to textile surfaces, usually for decorative purposes. The way and direction in which the yarn material is inserted, as well as how much of it is used, can vary widely. Modern drive and calculation technology allows embroidery machines to produce patterns with a high level of productivity and reproducibility. Tailored Fiber Placement (TFP) is a technological advancement in embroidery. This method enables the creation of textile preforms with the requisite reinforcing yarn arrangement for the manufacture of fiber composite components as well as the targeted local reinforcement or functionalization of textile semi-finished goods [3]. In the use case of unidirectional fibers, TFP can reduce the preform strength by increasing the twist and undulation of the rovings [40]. Tailored Fiber Placement has, however, been shown to be an effective solution to reinforce stress-concentrated weak regions of the preform, such as holes and edges [41]. In preforms used in composite boards, such use cases can include mounting bolt holes on skateboards to install trucks, the trucks themselves [42], or parts of snowboard bindings.

Dry Fiber Placement

In this method, spread and pre-bonded unidirectional (usually carbon) fiber rovings are automatically converted into dry textile preforms using 2-D and 3-D laying systems. The goal is to automate current manual lay-up operations to make the process simpler, more reliable, and easier to handle while making preforms. The resulting process promises more dependability, less waste, and flexible manufacturing [43]. The semi-finished fiber products manufactured on traditional textile lines have a width of several inches. The most popular width is 1.27 m, or 50 inches, which results in considerable production waste (up to 30%). In contrast, the fabrication of preforms for a split board (a subgroup of snowboards) using the dry fiber placement technique results in a roughly 60% reduction in carbon fiber waste during the production of fiber semi-finished products, as well as a notable improvement in economic efficiency [44]. Complicated layer structures that can only be generated manually may also be produced automatically enabled by the dry fiber placement technology's variable tape positioning [3].

Applications in board sport goods

After describing the main materials used in board sports equipment, this chapter describes selected examples of their use. Surfboards, skateboards, wakeboards, kiteboards, foilboards, and snowboards are covered.

Surfboard

Surfing has a long history dating back to the 18th century when European explorers first documented the art of surfing in Hawaii. In the 1950s and 1960s, music, movies, and a growing surf culture helped spread the sport worldwide. Technical innovations in board design made the sport more accessible. Surfing is now a recognized sport, celebrated in both professional competitions and as a recreational activity. Its inclusion in the 2020 Olympic Games underscores surfing's global recognition and popularity, honoring its cultural roots as it evolves and expands.

Core material and structure

Early surfboards did not have a separate core. The original shape consisted of a solid piece of wood, with three typical Hawaiian woods being used. The use of balsa wood and the first frame construction methods in the 1920s made it possible to reduce weight for the first time. With the increasing popularity of FRP, sandwich construction was introduced to surfing in the 1950s. Since then, the construction of a lightweight foam core and fiber-reinforced shell has become established.

For a long time, polyurethane (PU) foam was the main material for the core [45]. As this is available in different densities, the right density can be selected for the different fields of use (longboard, shortboard and big wave). To increase the bending stiffness and prevent the surfboard from breaking easily, the foam core is reinforced with a wooden stringer (see Figure 12.3). A stringer is a thin strip of wood or plywood that runs from nose to tail down the center of the surfboard blank. The most common woods for stringers are balsa, birch, basswood, red cedar, redwood, and spruce. Here too, the kind of wood and the thickness of the stringer are adapted to the field of use. Occasionally, boards have also been built with expanded polystyrene (EPS) foam. However, epoxy resin had to be used, as polyester resin dissolves the EPS foam.

It was only when the Clark Foam factory (PU foam provider) closed in 2005 that other construction methods gradually gained popularity. Due to the shortage of surfboard blanks, alternatives to PU foam had to be found. Some companies used this situation to experiment with new construction

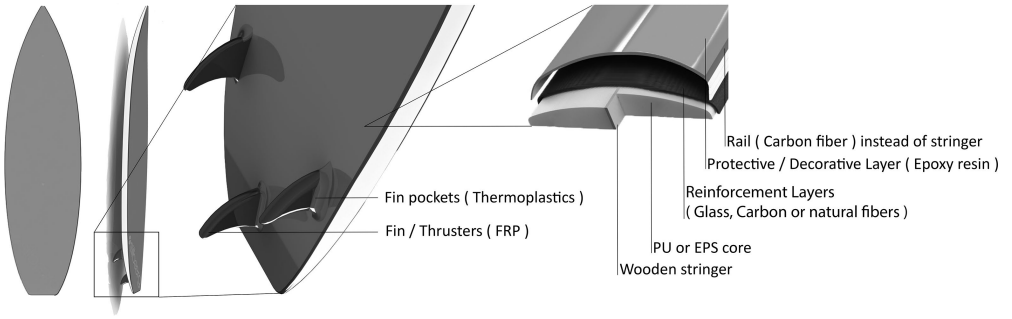

Figure 12.3 Advanced material combination in a surfboard.

methods and rethink the general structure of the boards. This led to the development of "Future Shape Technology" by Firewire (based in the USA) and the "Future Flex" construction method by Haydenshapes (based in Australia) [46]. There are now major companies that offer their range purely in EPS construction (Firewire, Stretch [based in the USA]), and most well-known manufacturers allow the customer to choose between PU construction and EPS construction. Other core materials that are used in smaller quantities include extruded polystyrene (XPS) and, to meet the increased demand for bio-based materials, agave wood (Grow Blanks, based in Kenya), BioFoam® (BEWi, based in Norway), and PU and EPS cores with a certain recycled content (Polyola, based in France; Marko Foam, based in the USA). The use of stringer-reinforced PU foam cores is based on many years of experience with this construction method, shorter process times, lower material costs, and a lower CO_2 footprint of around 170 kg per surfboard [47]. The core is responsible for around 50 kg of CO_2 emissions. The flex behavior of PU boards is also described by some surfers as "more familiar," as they are more accustomed to the properties of the material. The advantage of construction with an EPS core lies in the closed-cell nature of the foam. This means that water cannot penetrate the core if it is punctured, and the damage is easier to repair. EPS also has better pressure stability and can be installed without needing stringers. This means that surfboards using this construction method can be built lighter. This more durable construction method also extends the average service life of a surfboard from 380 hours with PU to 680 hours with EPS construction [47].

Usual reinforcement fibers

With the introduction of sandwich construction to surfboard manufacturing in the 1950s, fiberglass became one of the most important materials [45]. As the load cases in surfing, unlike in many other sports, are not well-defined, there were no real scientific approaches to determining the ideal fiber thickness for the shell. Instead, the optimum grammage of the glass fiber was approximated by trial and error. Factors such as the wave height and strength in the various areas played just as important a role, as did the ability of the producers to achieve the best possible fiber volume content in the hand lay-up process. The primary objective was always to achieve a good compromise between lightweight construction and durability. Over the years, the composition of one layer of woven glass fiber fabric with 135.6 g/m² on the bottom and two layers with 135.6 g/m² on the top became established. This construction can still be found on most surfboards on the market today. For surfboards that are expected to withstand greater forces (big wave boards, longboards) a reinforcement by a thicker composition, which can consist of two layers on both the bottom (135.6 g/m²) and the top (203.4 g/m²), will be used. Contest boards for professionals, on the other

hand, are only glassed with one layer (135.6 g/m²) on each side to achieve optimum lightweight performance. However, sometimes a board breaks after just a few hours' use [45,46].

With the introduction of carbon fiber into the sports industry, attempts were also made to further increase the lightweight construction potential of surfboards. However, the savings potential is much lower than in other fields, such as cycling. Depending on the size of the board, the amount of glass fiber only accounts for around 300 g of a surfboard's total weight of 3,000 g. Therefore, only a small amount of weight can be saved by using carbon fiber. Another trend is the use of natural fibers, often in combination with more environmentally friendly epoxy resins. The founding of the "Ecoboard" project by Sustainable Surf (based in the USA) in 2012 was a key milestone [48]. This independent organization awards various labels to surfboards depending on their sustainability in production. The newly introduced label and increasing customer demand for more sustainable products led to an increase in alternative construction methods and materials. The most used fibers are flax and hemp fibers. Attempts have also been made with sisal fibers. The disadvantages here are the varying range of properties of natural fibers and the more difficult handling in production. As some natural fibers can absorb up to 8% of their weight in water, suitable drying must precede the lamination process. Recently, the first surfboards based on cellulose fiber have been brought to market. Compared to other natural fibers, cellulose does not absorb humidity and is easier to integrate into existing production processes. However, there is currently no data available on how the durability of these surfboards compares to the classic fiberglass construction method.

Skateboard

Skateboarding originated in the late 1940s when surfers in California were looking for a way to "surf" the streets when the ocean waves were flat. This "sidewalk surfing" began with wooden boxes or boards with roller skate wheels attached to the bottom. Over the decades, skateboarding evolved, both in terms of the skateboards themselves and the culture surrounding them. By the 1980s, skateboarding had cemented its place in youth culture, with skate parks proliferating and the sport taking a prominent place in the media. Innovations in tricks, board design, and the rise of professional skateboarding competitions propelled skateboarding from a niche hobby to a globally recognized sport and lifestyle, culminating in its inclusion in the 2020 Tokyo Olympics.

Core material and structure

A skateboard is a downsized surfboard-shaped sheet with usually four wheels attached underneath (see Figure 12.4) to glide on a track [49]. Nowadays, skateboards come in different styles and designs based on preference, the type of track surface, and the activity. The two main classes of design are freestyle boards and longboards. The former is used for "street" and "park" (both are Olympic categories) and for "vert" skating (halfpipe), while longboards are used for cruising or downhill skating. Freestyle or common skateboards usually consist of seven to nine layers of plywood. Canadian maple is almost exclusively used as the wood type, as it offers the best performance characteristics. Since this is particularly a hardwood, maple can withstand the stresses and strains of use. It also produces the best "pop" during the "ollie" which is essential for the jump height of most tricks performed. Various materials can be found in longboard or cruiser board construction. Depending on the field of use (downhill, carving, dancing), the deck needs to be stiffer or springier, utilizing different core materials. Downhill boards need to be stiff, and their weight is less of a concern. Due to its high resistance to wear, plywood made of Canadian maple is used for these boards. For carving and dancing, skaters need a certain flex in the deck. The spring of the

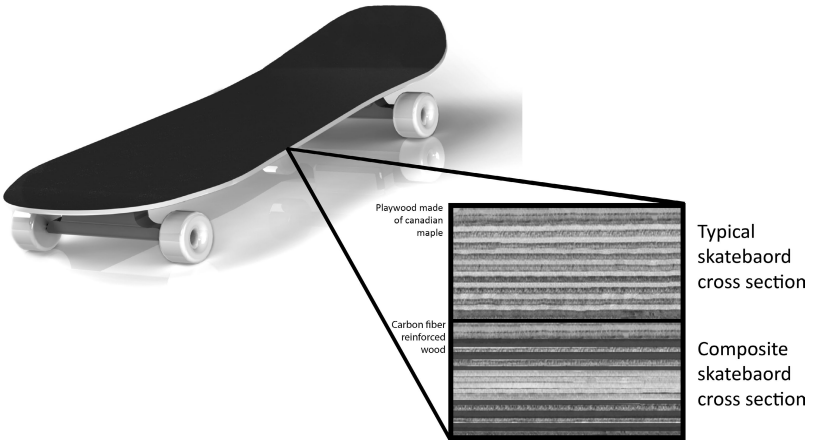

Figure 12.4 Skateboard deck materials.

deck allows the rider to generate extra speed when skating on flat terrain and cushions the jump when they land on the board after performing a trick. For these boards, the best performance can be achieved by building the decks out of bamboo plywood. However, there are also combinations of wood (maple, poplar) and glass fiber available on the market. A cruiser is a shorter type of board mostly used for cruising around and as transportation. Due to the short wheelbase, the deck does not need to withstand high force and can be made from a wide range of materials. Most common- are still plywood decks, as the production process is familiar to the companies, along with decks made of thermoplastics like polypropylene (PP), which are inexpensive for mass production.

Usual reinforcement fibers

For a long time, a typical freestyle skateboard consisted only of several layers of maple plywood. Recently, some companies have started adding extra layers of carbon fiber to their decks (for instance, Titus Skateboards based in Germany [50]). By adding carbon fiber between the plywood layers, the board is designed to last longer and resist breaking as easily. Furthermore, the reinforcement ensures that the riding characteristics, especially the pop, do not degrade and that the wood does not become softer over time. Still, carbon fiber does not help against the usual tear (resulting from impacts) at the tips of the board. This includes the wear of the wood when using the deck to brake and the chipping of the edges upon impact. Fiberglass reinforcements are often used for cruiser and dancer boards. Due to the larger wheelbase, the deck must withstand higher bending stresses. The advantage of fiberglass in this case is that it can achieve the desired stability with less weight gain and thickness increase compared to a construction using extra layers of plywood. Furthermore, substituting wood layers with fiberglass reinforcements ensures that the board is springier. This particularly enhances the riding characteristics when cruising and dancing.

Wake and kiteboard

Wakeboarding, a dynamic blend of water skiing, snowboarding, and surfing, was conceived in the 1980s and evolved into its own sport in the early 1990s. Originally, the sport involved riders being towed behind a boat on a small board while performing tricks and jumps from the wake of the boat.

The invention of foot bindings and a concave shape allowed for greater control and aerial maneuvers. Wakeboarding's growth was further fueled by the establishment of professional competitions and dedicated cable parks, making it a globally recognized water sport celebrated for its creativity, community, and exhilarating athleticism.

Kiteboarding developed during the same time span and merges elements of windsurfing, surfing, and paragliding. The first notable experiments in the 1980s involved using kites for propulsion on water. It was not until the 1990s, however, that kiteboarding began to solidify as a sport, due to advancements in kite design and safety. The inflatable kite design became the foundation for modern kiteboarding, making kites safer and more controllable. This led to the sport's rapid expansion in the early 2000s. Today, kiteboarding is a thriving global sport, known for its spectacular jumps, freestyle maneuvers, and speed, attracting a dedicated following of enthusiasts and professionals. It was also part of the Olympic Games 2024 in Paris.

Core material and structure

Kiteboarding is fundamentally the same activity as wakeboarding, with the caveat of having a kite instead of a motorboat as a thruster. The essential difference between a kiteboard and a wakeboard is in the "rocker" (the curve of the board from the tail to its nose [51]) and the fins (see Figure 12.5). As wakeboarders reach higher speeds, the wakeboard's rocker is more pronounced to avoid risky wipeouts (falls). On the other hand, excessive speed is not usually an issue for kiteboarders. In kiteboards, the main concern is harnessing the power of the wind and accelerating as much as possible [52]. Based on this, kiteboards usually possess less rocker and are lighter than wakeboards. In the structural design, the main objective of these boards is to overcome the maximum hydroelastic loads during the so-called "slamming" phenomenon that occurs when the board enters the water after a jump. A plastic foam core or a fiber-reinforced wood core is used to create wake and kiteboards [53] (see Figure 12.5). For better flow on the

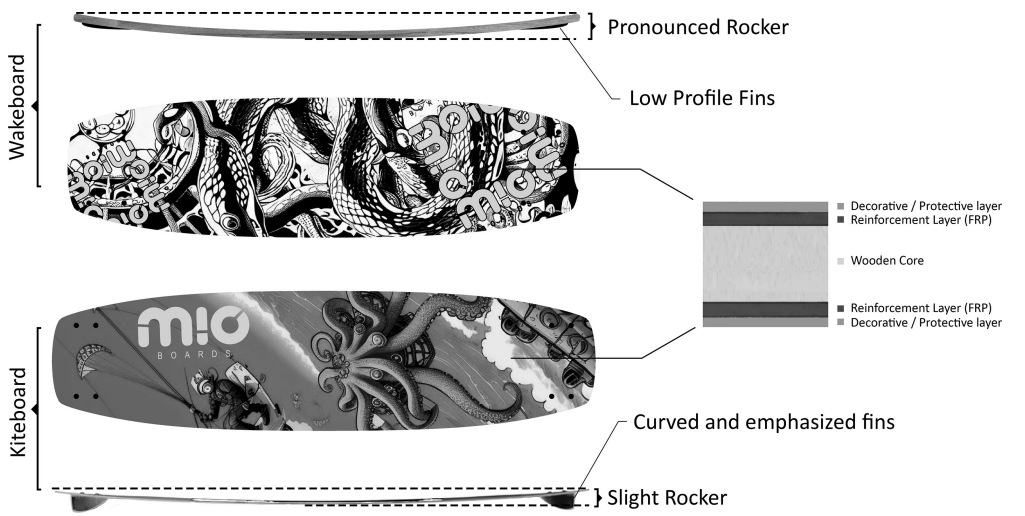

Figure 12.5 Generic design of Kiteboard and Wakeboard. Although the reinforcement and the core material are the same, the structure and shape of the fins are different between the two. Graphics from Mioboards/Germany [55].

water, as in surfing and foiling, low density is necessary for wakeboards. Due to this requirement, the core material is usually a rigid, closed-cell polymer material such as PU or EPS. Although different wooden structural frames can sometimes be added to the core, most of the surface area of a wakeboard's core is made of polymeric material.

Usual reinforcement fibers

In designing a wakeboard, one might assume that producing the faces from a more durable material enables them to be thinner. However, this is untrue, as the minimum face thickness required to prevent buckling phenomena (resulting from slamming the board) is determined solely by the board's radius of curvature, the thickness of the core, and the length of the delamination. It is not affected by the face sheet's elastic modulus [53]. Bearing this in mind, glass fibers (which are not the thinnest of technical fibers) are the most common material used as reinforcement in wakeboards. On the other hand, due to the thinner profile and weight constraints, carbon fiber is more commonly utilized in manufacturing kiteboards.

Foilboard

Foiling, a revolutionary advancement in water sports, uses hydrofoil technology to provide a surreal experience of flying over water. A hydrofoil, mounted beneath a board or craft, works by lifting the vessel above the surface as speed increases, drastically reducing water resistance. This innovation, which emerged in the early 2000s, has been applied to various disciplines, including sailing, surfing, and kiteboarding. Nowadays, foil surfers are increasingly using a wing held in the hand instead of a kite directly connected to the rider. Foiling not only raises the performance ceiling by allowing for higher speeds and smoother rides but also transforms the rider's experience by providing a unique sensation, like gliding through the air. Its introduction represents a leap forward in water sports, pushing the boundaries of what is possible on the water and inviting enthusiasts into a new era of speed, efficiency, and exhilaration. Foiling was part of kiteboarding at the 2024 Olympic Games.

Core material and structure

A hydrofoil is described as a lifting surface strut with a solid contour that functions in water [54]. In other words, this refers to a single wing or a composition of wings that are attached to a watercraft underneath the water surface [55]. Foil boards, unlike surfboards, are equipped with a hydrofoil instead of fins underneath the board. The whole setup consists of two main parts. First, the board itself, which, for competent users, can be a compact, thin wooden or polymer frame filled with polystyrene and reinforced with fibers and a matrix system. For beginners, it is a thicker, larger wooden or polymer structure filled with EPS foam and enveloped in an outer polymer shell. There are also computer numerical control (CNC) milled bionic wood core structures on the market that combine sustainability with lightweight construction [56]. The second part, the hydrofoil, contains a connector plate, the mast, a fuselage, and the front and rear wings. The hydrofoil is connected to the board's bottom by the connector, which must withstand the concentrated stress on its edges generated by the movement of the board. The connector plate can be made of aluminum or titanium alloys, but most recently, fiber-reinforced components made in a resin transfer mold (RTM) have been manufactured and utilized. Next, the mast, a long, thin blade with the cross-section of an airfoil, is typically composed of extruded aluminum or PMC and keeps the unit submerged in the water to produce lift force. The fuselage connects the wings to the mast and is usually made of

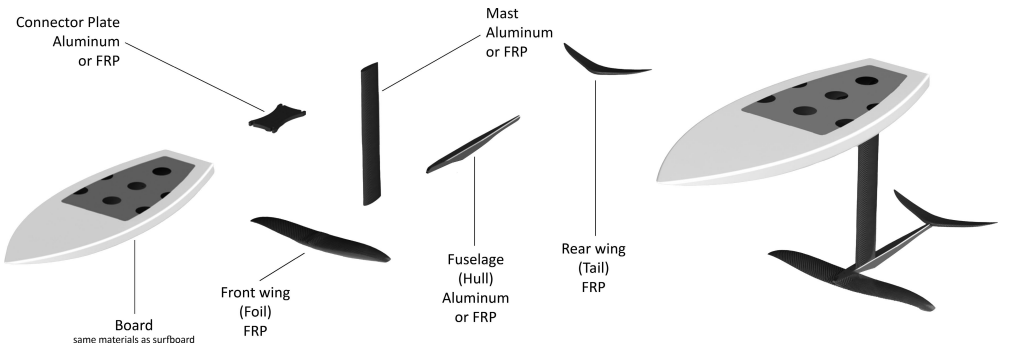

Figure 12.6 Typical foilboard design and explosion view of its different parts.

milled aluminum parts or a brittle metallic alloy. Finally, lift is produced by the front wing, while stability and maneuverability are sustained by the rear wing. These two parts are mostly made of FRP in an RTM [57] (see Figure 12.6).

Usual reinforcement fibers

For the foilboard, the usual reinforcement fibers and matrix system are the same as those used for surfboards or kiteboards. Aramid, carbon, or glass fibers in woven forms, or non-crimped preforms, are commonly used in producing the boards. The composite connector plates are typically made from woven carbon or glass fibers. This is because the concentrated stress on the edge points of the plate benefits from the interlocking strength of the woven fabrics and their high structural integrity, which are helpful for generic designs. Fabricating a composite mast requires materials capable of withstanding the dynamic loads resulting from the interaction between the board and the water streams beneath the surface. These masts are commonly made from woven carbon fiber prepregs. The wings are typically manufactured from woven carbon fiber to maintain the high performance required and achieve low thickness in the airfoil cross-sections.

Snowboard

Snowboarding, a sport that blends elements of surfing, skiing, and skateboarding, originated in the 1960s, bringing the spirit of surfing to traditional winter sports. Its inception is often credited to Sherman Poppen, who, in 1965, created the "Snurfer," a rudimentary snowboard for his children by attaching two skis together. This invention sparked interest and led to the development of the modern snowboard. Throughout the 1970s and 1980s, pioneers such as Jake Burton Carpenter and Tom Sims innovated and refined snowboard designs, added bindings, and promoted the sport. Despite initial resistance from ski resorts, snowboarding grew in popularity, leading to its inclusion in the 1998 Winter Olympics. Today, it is celebrated worldwide as a vibrant and integral part of winter sports culture, showcasing a spirit of freedom and creativity [58].

Core material and structure

Typically, the focus in designing a sandwich structure for a snowboard is to optimize the vibration qualities and the elements that quiet and dampen the snowboard on the snow [33]. Weight, as well

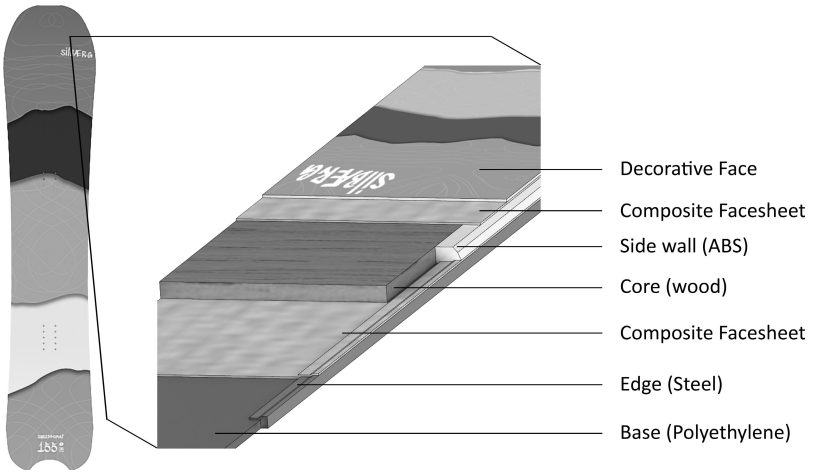

Figure 12.7 An advanced snowboard composite structure cross-section. Graphics from silbaerg GmbH/ Germany.

as bending and torsional stiffness, are determined primarily by the setup of the core [59]. The core is mostly a few laminated wood strips that have been bonded together, and the rest of the board will be constructed upon them. Species such as poplar, aspen, beech, maple, ash, birch, bamboo, abachi, balsa, and paulownia are common for these strips. Additionally, honeycomb cores, empty or filled with foam polymers, pure polyvinyl chloride (PVC), or wood chip fiber boards, are other alternatives for wooden cores in snowboards.

The most typical is a wooden core (see Figure 12.7) made of light wood (usually poplar), which is reinforced by two strips of hardwood in the longitudinal axis of the insert's region (holes for fastening the binding). Due to the diverse preferences, applications, and different body types and demographics of the athletes, the thickness and layout of the wood core change between models and sizes. A typical wood core has a thickness between 5 to 7 mm between the bindings. The core thickness reduces toward the tips of the board.

Usual reinforcement fibers

In snowboards, the reinforcement fibers not only differ in material (commonly glass, carbon, basalt, aramid fibers, and most recently natural fibers like flax and hemp), but also in the angle arrangements, stacking sequences, and manufacturing methods of the preforms. "Flex," "pop," and "edge hold" are criteria that customers are concerned about and can be modified through different reinforcement approaches. Flex refers to the bending behavior between the bindings in the longitudinal axis of the board and the energy transport during turns. Due to the typical outline (sidecut and radius) of the board and the centrifugal force from the user, the whole snowboard will bend in the longitudinal axis during a turn when it is on its edge. When the board is placed back on the base after the turn, the stored energy will be released. This spring-loaded energy gives the rider a speed boost after each turn and provides a lively driving experience. Flex is achieved by adding unidirectional fibers to the wood core between the bindings. The selection and positioning of different fiber materials (e.g., glass and carbon fiber) allow the board to be built according to set specifications and needs.

Pop also refers to the bending behavior, but mostly in the area between the binding and the board tips. Compared to flex properties, which are important for every snowboarder, pop is more of a concern for performing freestyle. In this subcategory, the user executes tricks like jumps and slides, most of them initiated by an "ollie," where the front foot will lift, and the board will bend like a bow. The energy stored between the rear binding and the tip behaves like a spring, enabling a higher jump. By choosing the right reinforcement fibers, pop can be maximized, and more technical tricks can be performed. Edge hold is defined by the torsional stiffness of the board. During turns, all the force will be funneled onto the board's edges. The distance between the user's center of gravity (centrifugal force during turns) and the board's edge generates a torsional twist of the board. If the torsional stiffness of a snowboard is not coherent with the riding speed, the board can twist, and the edge will lose grip. This will make the athlete drift away during a turn or when initiating a jump with rotation and can lead to falls. Therefore, most advanced snowboards are equipped with some torsional reinforcement. This is done by adding a layer of triaxial fibers or placing some patches in the crucial areas of the board. Obtaining proper reinforcement comes from experience and trial-and-error development processes, which are held as a company secret for each snowboard manufacturer. There are few approaches known to computer-aided design (CAD) processes, and finite element analysis can be applied to develop the preform layup [60].

Summary and future perspectives

This chapter focuses on the use of fiber-reinforced plastics and core components for sandwich construction parts as part of board sports equipment. The first sections focused on the textile preforms' architecture and materials utilized in composite sports boards, respectively. Further sections went into more detail about different board sports goods and their distinctive applications of the materials mentioned. As a key user of FRP, the sports industry is likely to apply these materials more in the future, and, widely dependent on their gear, board sports are not distinguished from this trend. For instance, the following factors are some of the future potential perspectives for advanced composite materials in these sports:

1 Development of high-performance FRP: FRPs' lightweight and high-strength properties are well-suited for weight-sensitive board sports goods. Future developments may emphasize refining these composite material properties to enhance the strength, flexibility, and overall performance of the board sport products.
2 Advanced design and manufacturing techniques: The development of new advanced materials can contribute to the ongoing research in biomechanics and sports engineering. This can result in designing board sport goods with intricate shapes and structures. Furthermore, the development of new manufacturing processes may enable greater customization, allowing athletes to have boards tailored to their morphology and preferences.
3 Integration of smart technologies: The integration of smart technologies into boards may become more common. As seen in other sports sectors, such as bicycle frames [61], FRP sporting goods could also be designed to accommodate sensors, providing performance data such as speed, balance, and technique. Development of technologies such as dry fiber and tailored fiber placement may make the integration of sensors in reinforcement facesheets achievable.
4 Environmental considerations: A growing emphasis on the environmental impact of materials and manufacturing processes in board sports may drive the development of eco-friendly FRP or the exploration of sustainable alternatives. Natural reinforcement fibers and utilizing bio-based

resins as composite matrices may become inevitable. Also, further developments in managing boards at the end-of-life stage are expected [62].

5 Innovating new sports equipment: FRP might extend beyond traditional board sport goods, finding new applications in other recreational equipment or innovative uses in adaptive sports. Collaborations between material scientists, engineers, and athletes could lead to innovative uses of FRP in board sports products. Looking back, hydrofoils, which were only used for high-performance motorboats, transformed into a new board sport category. It is not far-fetched to see advanced materials enabling the invention of other new board sport disciplines.

6 Consumer accessibility: Refined and cost-effective manufacturing processes could increase access to FRP sporting goods for a broader range of consumers, including both professional athletes and recreational enthusiasts.

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13
TEXTILES FOR SPORTS CLOTHING

Tasneem Sabir, Jane Wood, and Rebecca Bennett

Introduction to textile fibres

The human body goes through enormous changes during physical activities [1]. It is vitally important that clothing choices withstand environmental conditions and maintain a balance between thermal and fluid factors. Sports clothing differs from casual clothing in that it offers enhanced performance and functionality [2]. Traditional textiles have undergone process changes to create highly functional alternatives. Polyester, polyamide, acrylics, and elastane still dominate much of the sporting apparel market [3]. The construction of materials for sportswear is generally based on comfort, stretchability, and ease of movement, in addition to performance characteristics such as water resistance/repellency, aerodynamics, durability, and breathability. Smart materials that can sense and react to external stimuli are being developed to gather information from the environment and the body. These innovative technologies are now finding a place in sportswear. The rise in ongoing health and well-being interest is a primary driver in the consumption of sporting apparel [4]. From recreational activities to high-intensity training schedules, fabrics are chosen based on comfort (keeping cool in the summer and warm in the winter), easy care properties, and lightweight design. This chapter presents an overview of textile fibres, along with a discussion of their characteristics, properties, and applications specific to sportswear. Further, the chapter explains the differences in fabric structures and the evolution of smart materials for sportswear using case studies as examples.

Characteristics of textiles fibres: overview

Textile fibre growth is projected to reach 127.3 million tonnes by 2027, with a projected growth of 2.2% CAGR (Compound Annual Growth Rate) between 2020 and 2027 [5]. The sportswear industry, a sub-section of the textiles industry, is now projected to increase its sales by 5.6% CAGR. Textile fibres are fundamental to the creation of cleverly engineered sports materials and garments. The growth in fibre preference is based on affordability, comfort, and sustainability [6]. The selection of fibres brings together material science, technology, and engineers to develop textiles that are not only aesthetically pleasing but also offer enhanced functionality and innovation in the form of wearable electronics.

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Textile fibres are chosen based on their properties, whether to protect the body from environmental causes, provide protection from the elements, or assist in health and well-being. The properties of all fibres are either inherent or induced through the manufacturing process. Sports garments are used by elite athletes to enhance performance, provide recovery, and assist in the prevention of injuries. The selection of components to begin the journey into sports materials or apparel starts with the basic unit – textile fibres. These are cleverly constructed into yarns, which are then interlaced or interlooped into fabrics, which are further assembled into garments. Examples of different natural and manufactured fibres can be found in Table 13.1.

The origin processing route of all fibres can categorise them into either natural or manufactured fibres (man-made). Both are considered relevant in the production of sports materials, offering unique properties to benefit the individual or sport. A textile fibre is a unit of matter characterised by flexibility, fineness, and a high length-to-width ratio [11]. The classifications or groups of fibres can be defined by their origin and length [12]. To determine a fibre's suitability in sportswear, they must meet certain functional requirements for different sports, as shown in Table 13.2. General fibre properties are presented in Table 13.3.

Natural fibres are obtained from natural sources including plants and animals, and their textile fibre properties are inherent in their genetic structure. The most common plant fibres (also known as cellulosic) are cotton and flax. Animal (protein) fibres include wool and silk. In addition to

Table 13.1 Examples of textile fibre classifications [7–10]

Natural fibres (short staple fibres)		*Manufactured fibres (long continuous filaments)*	
Protein	*Cellulosic*	*Regenerated cellulosic*	*Synthetic*
Fur	Cotton	Viscose Rayon	Polyester
Hair (alpaca, mohair, cashmere)	Flax	Bamboo	Polyamide (Nylon)
Wool (Merino Wool)	Jute	Acetate	Aramid
Silk (only natural fibre considered a filament due to its long continuous length)	Hemp		Acrylic Elastomers Olefins
Inorganic: Carbon, Ceramic, Glass, Metallic			

Table 13.2 Requirements for sportswear

Different sports	*Functional requirement*
Shirts for tennis, volleyball, golf, football, rugby, baseball uniform, athletic attire, tracksuits	Sweat absorbing, quick-drying, cooling, UV resistant, tactile sensation, sweat regulation
Skiwear, wind-breakers, rainwear	Moisture vapour permeability, waterproofness, sunlight absorption, thermal retention, and low fluid resistance
Swimming race and skating costume, ski jump and downhill skiing suits, cycling costume	Aerodynamic/hydrodynamic through water and air, body mapping,
Swimwear, leotards, skating costume	Stretchability, opacity, moisture and vapour management, flexibility
Snowboard wear, baseball uniform, football uniform	High tenacity, resistance to abrasion, stretchability, windproof, waterproof
Helmets, gloves, padding	Impact protection, breathable, durable, thermal balance, and thermal regulation

Source: Adapted from Chowdhury et al. [18].

Table 13.3 Textile properties and relevant test methods

Properties	*Natural*				*Cellulosic*			*Synthetic*					*Standard test methods*
	Cotton	*Flax*	*Wool*	*Silk*	*Viscose rayon*	*Lyocell*	*Acetate/ Triacetate*	*Nylon*	*Polyester*	*Acrylic*	*Olefin*	*Elastomeric*	
Touch, Handle Appearance	Soft, cool, comfortable	Fresh, cool, stiff, hard, smooth, lustrous	Warm/cool, hard/ smooth, fine/bulky (depends on fibre)	Very soft, smooth, lustrous, cool, drapes well	Soft, fine, comfortable, lustrous Good drape	Soft, fine, comfortable	Good drape, smooth	Depends on cross section. Flat filament: smooth, firm, lustrous Staple yarn or textured filament: warm, soft					N/A
Moisture Absorbance	Good	Good	V. Good	Good	V. Good	V. Good	Fair	Poor	Poor	Poor	Poor Wicking	Poor	ISO 20158:2018
Crease Resistance	Poor	Poor	V. Good	Good	Poor	Poor	Poor	V. Good	V. Good	Good	V. Good	V. Good	BS EN 22313, ISO 2313
Resiliency							Tri (good)						
Durability, Easy to Wash	Good	Good	Poor	Good	Poor	Fair	Poor	Good	Good	Good	Good	Good	BS EN 26330, ISO 6330
Abrasion	Good	Good	Good	Good	Fair	Good	Poor	V. Good	V. Good	Poor	V. Good	Poor	BS EN ISO 12947-1
Elasticity	Poor	V. Poor	Good	V. Good	Poor	Fair	Good	V. Good	V. Good	V. Good	Good	Highest	BS 4294
Pilling	Fair	Fair	Poor	Fair	Good	Good	Good	Poor	Poor	Fair	Good	V. Good	BS EN ISO 12945-1
Electrostatic Charge	Good	Good	Good	Good	V. Good	V. Good	Fair	Poor	Poor	Poor	Good	V. Good	ISO 18080-1:2015(en)
Colourfastness	Good	Fair	Good	Good	Fair	Fair	Fair Tri (poor)	Poor	Good	V. Good	Good	Fair	BS EN 20105 BS EN ISO 105 ISO 105

genetic influences, the properties of these fibres are also governed by environmental conditions and harvesting routes. These fibres are elaborately structured products of living organisms, and their very complexity endows them with wonderfully versatile and powerful properties. Natural fibres were used in sporting materials for clothing before the introduction of synthetic fibres.

The use of cellulosic fibres in sports materials can offer breathability, absorbency, next-to-skin comfort, and easy care. They are extensively blended with manufactured fibres to create better and longer-lasting materials. Protein fibres can offer warmth and are highly absorbent with good elasticity due to their crimp configuration. Merino wool (fine and soft from Merino sheep) is used in materials, as this unique fibre is quick-drying, breathable, and has excellent water vapour permeability [13].

Manufactured fibres are created by converting solids to liquid (referred to as dope) and forcing them through a spinneret (similar in appearance to a shower head) to create strands known as filaments. Three classifications of manufactured fibres include: regenerated (materials which cannot be used in their original form – viscose rayon and acetate), inorganic (glass, ceramic, and metallic-based fibres which have seen a rise in sports apparel), and finally, synthetic fibres (polyester, nylon, and spandex) [14]. Due to the ease of creating almost perfect fibres with specific properties, the manufacturing process can create filaments with different cross-sectional shapes. Differences in shapes play a pivotal role in fabric properties. From trilobal, dog-shape, to hollow, each engineered cross-section can promote a range of properties, from aesthetic properties to high performance [15]. Each shape is designed and formed by the spinneret, creating properties that can be used for sportswear, as seen in Figure 13.1. Round formations offer dull fibres, hollow centres provide thermal insulation, and serrated designs are used for liquid transportation [16].

Of the manufactured fibres, polyester is regarded as the preferred choice for sports garments and is often blended with elastane, wool, acrylic, and nylon. Polyester is known for its durability, easy-care properties, moisture resistance, shape retention, wicking, and quick drying. This makes polyester a very versatile fibre for sports apparel. Generally, synthetic high-performance fibres provide high tensile strength, water resistance, elasticity, comfort, and breathability and are often favoured for sportswear.

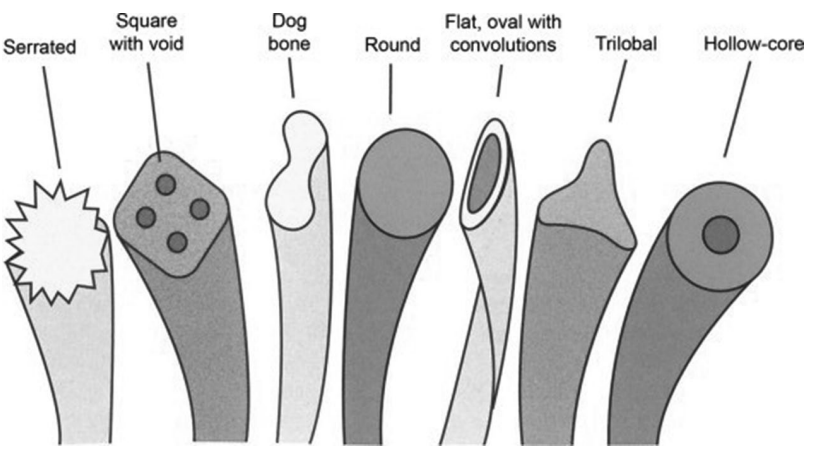


Figure 13.1 Cross-sectional shapes of textile filament fibres designed through a spinneret (a show-like device).

Fibres properties for sportswear

The most important aspect to consider is managing all levels of fabric design and construction as this can have a large effect on the overall performance of functional clothing. Fibres are the building blocks of all textile materials; therefore, each fibre is considered for its selection in the development of sporting apparel. Blending is often considered when creating textile fabrics with enhanced performance. The blending process can comprise natural and natural fibres or natural and synthetic fibres. The purpose of blending is usually to provide enhanced properties that one fibre may lack or to encourage enhanced functionality. Blending can also occur when different textile materials are bonded (usually comprising various fibre types). Different sporting activities require a combination of desirable properties. Sports garments can be designed for different weather conditions or specific types of sports [17]. Below summarises some examples of functional requirements for various sports (Table 13.3):

The performance of functional clothing is achieved through the selection of appropriate fibres that can provide the wearer with a balance between heat loss and body perspiration, strength, elasticity, aesthetics, appearance, and feel [3,19]. Clothing comfort, regarded as the most desirable attribute, is essential in sports garments. Expected clothing comfort contributes to the following [20]:

1 Transfer of sweat from the skin,
2 Quick drying,
3 Breathability,
4 Balancing body heat by moving away moisture and ventilating the body's critical heat areas,
5 Not charging electrostatically,
6 High strength and abrasion resistance,
7 Non-restrictive body movements,
8 Being soft and durable,
9 Joining techniques that cause discomfort or irritation,
10 Pleasant feel/handle,
11 Aesthetic appeal,
12 Providing easy care.

Assessing fibres for performance

To select fibres for their performance in sports garments, textiles can be assessed for their suitability. Essentially, properties and materials are based on the user's needs and sporting requirements. Many of the fibre properties discussed in the chapter can be assessed through textile testing, which allows manufacturers to test materials for their fitness for purpose. For consideration in sports applications, the following performance characteristics are assessed, as listed below [2,13,21]:

Comfort: The human body temperature is 37°C. Deviation from this can result in devastating effects. Comfort enables the body to cool down after exercise. Materials are tested for their moisture management and thermal insulation properties, as well as fit (BS EN ISO 9073-15).

Absorbency (moisture regain): This property refers to the ability to take in water from atmospheric conditions (ISO 20158:2018).

Durability: The strength and longevity of materials through wear and laundering are tested through mechanical and physical methods (bending, tensile, rubbing, flexing) (BS EN ISO 13934, BS EN ISO 13937-3, and ISO 13938).

Wicking: The ability of materials to transfer moisture from one space to another through capillary actions (AATCC 79, AATCC 197, AATCC 198, and AATCC 195).

Moisture Management: The ability of textiles to absorb humidity from the skin, transport it to their outer surface and release it into the surrounding air (ISO 13029:2012(en)).

Stretch and Recovery: A material's ability to extend and recover without deformation (BS 4294).

Thermo-physiological: Thermal balance occurs when the rate of heat loss from the body equals the rate at which it is being generated by physiological processes (ASTM D7984).

Water Repellent: The material's ability to repel water molecules from the surface, making it hydrophobic (BS EN 20811 ISO 811, BS EN 24920 EN 24920 ISO 4920).

Breathability: An attribute of comfort that facilitates the evaporation of moisture from the skin.

Windproof: The fabric's ability to prevent air from passing through the fabric (BS 5636, ASTM D737).

Commercial fibre development

With the growth of sports apparel, the textile industry is now engineering the fibre development process to induce properties based on consumer preferences. With science and technology, these exciting fibre developments are now being used commercially on a national and international scale. Listed below are some commercial fibre developments (adapted from [22]). Properties detailed below are in accordance with brand literature on company websites:

TENCEL™ Active (TENCEL™): TENCEL cellulosic fibre keeps the body cool with natural comfort, versatility, and freedom of movement. Due to its renewable wood sourcing, its naturally structured channels transport moisture away from the body, keeping it cool and dry.

LYCRA® SPORT technology (©The LYCRA Company): This fibre offers comfort and performance with precise fit and shape. It provides optimum fabric performance, enhanced wearer experience, and is washable.

drirelease® (© 2022 drirelease): This fibre optimises wicking, drying, soft feel, and comfort through naturally occurring properties.

X-STATIC® Technology (Noble Biomaterials): Fibre is infused with silver to inhibit the growth of bacteria that live on fabric, eliminating human odour.

Advansa (ADVANSA): ADVANSA's innovations include Coolmax®, Thermolite® and Thermocool®. Coolmax® is designed to allow moisture to flow away from the body. THERMOLITE®, due to its hollow structure, provides thermal insulation. THERMCOOL® enables evaporative cooling when wearers perspire, plus thermo-buffering to maintain a comfortable body temperature.

HYGRA (© 2022 UNITIKA TRADING CO., LTD): Unitika Limited has launched Hygra. The properties of this fibre are that it absorbs moisture such as sweat and water vapour emitted from the body, and releases absorbed moisture to the outside air to suppress stuffiness and stickiness.

CELLIANT® (CELLIANT®): This technology transforms your body's heat into full-spectrum infrared energy, promoting local circulation and cell oxygenation while keeping you drier and at the perfect temperature.

NanoFuse (© VALOUR SPORT): Bamboo charcoal fibre with natural antibacterial properties, odour resistance and anti-irritant fibres. it has excellent absorbency with a natural infrared radiator. NanoFuse also activates human cells to increase cellular energy and speed up circulation and metabolism.

There is a demand in the textile industry to look at sustainability as consumers are now becoming more environmentally conscious [23]. Therefore, there is a drive towards plastic-free fibres. Today, technologists are experimenting with naturally occurring materials. The green movement has given rise to sustainable sportswear. Organic fibres, derivatives of organic matter, and recycled fibres are now being considered. Although they may not present the superior properties exhibited by synthetic fibres, these fibres can offer comfort, breathability, and inherent medicinal properties as well as being lightweight, soft, and versatile. Some of the listed properties associated with sustainable textiles are either claims made by manufacturers or through aggressive marketing. Those that consider a sustainable approach to their practice should test materials for their performance. Many of the listed sustainable fibres are sourced and processed globally [24]. Geographical locations include India, Bangladesh, China, and Europe. The processing route generally entails cultivation, harvesting, and retting, followed by transportation from the field to the processing unit. Fibre processing occurs after this, followed by transportation to either Asia or Europe or within Europe [25]. It is therefore essential to consider the carbon footprint and understand the ecological and environmental categories to assess the climate impact of products. Fibres that are generally considered sustainable are listed below:

Organic Cotton: Grown without pesticides, these fibres offer durability, biodegradability, recyclability, softness, and lightweight properties. Applications: Sweatshirts, hoodies, shorts, and socks.

Bamboo: A lightweight material with moisture-wicking capabilities, inherent medicinal properties, odour resistance, and strong fibres. Applications: Sports leggings, socks, undergarments, and yoga wear.

Hemp: One of the most versatile cellulosic fibres with antibacterial properties, very resilient and durable. Applications: Sports bras, leggings, biker shorts, and T-shirts.

Nettle Fibre: A versatile fabric with dual properties, keeping you cool in summer and warm in winter. Applications: Potential use in sportswear for different climatic conditions.

Coffee Ground Fibre: An excellent choice for sporting garments with natural anti-odour qualities, UV ray protection, and quick drying time. Applications: Outdoor and sports performance wear.

Pineapple Fibre: Similar to leather, offering strength, versatility, breathability, softness, and flexibility. Applications: Footwear as an alternative to leather and fashionable goods.

Banana (Abaca) Fibre: High strength, biodegradability, and good moisture absorption quality. Applications: Bags, but it can be easily applied to sportswear.

Polyester is the most consumed fibre in both fashion and sportswear [26]. Reliance on fossil-based materials is damaging the environment. For this reason, the textile industry has now been challenged to increase the market share of polyester that comes from recycled resources from 14% in 2019 to 45% in 2025 [26]. Polyester is made from polyethylene terephthalate (PET), whereas Nylon is based on petroleum oil. Both fibres dominate the sportswear market but are made from non-renewable sources and high-energy-intensive processes and eventually end up in landfills [27,28]. It is the responsibility of each sports brand to act on sustainable and ethical actions, driven by consumers becoming more environmentally conscious. With the continued use of polyester, nylon and elastane derived from petroleum, this is causing significant carbon emissions and harming biodiversity [29]. Well-known brands like Nike and Adidas are now producing ranges with recycled polyester and nylon. As of 2020, Nike is investing in reusable plastics, yarns, and textiles to significantly reduce emissions [30]. Adidas plans to achieve climate neutrality by 2050. By

2024, Adidas has stated that they will only use recycled polyester where possible [31]. These two giants of the sports brand industry are setting the benchmark for others brands to follow.

There are benefits and drawbacks to using recycled content, including

+ Reduction in water, energy, and dye usage	– Can reduce some properties (e.g., strength)
+ Properties similar to virgin fibres	– Degradation of quality
+ Less pollution	– High levels of re-dyeing, using chemicals, water, and energy
+ Reduced amounts ending up in landfills/oceans	– Release of microplastics

Fabric properties for sportswear clothing

Textile structures are created by using a method of interlocking yarns or fibres and can be fundamentally broken down into three categories: woven, knitted, and non-woven. These categories, alongside the selection of yarns, fibres, and finishing techniques, offer a certain set of complex performance criteria that determine the choice of textile for specific end uses. Additionally, the complexity of the structural units often makes the behaviour of the textile structure more difficult to predict than that of other engineering materials [32].

Woven fabrics

This is the oldest method of fabric construction, thought to originate from the interlocking of reeds in basket making. Woven fabrics are constructed from the interlocking of vertical (warp) and horizontal (weft) yarns, usually at right angles, to form a textile sheet. Woven fabrics are created on a weaving loom, where the complexity of the yarn interlacing can be manipulated to form a variety of structures to alter either aesthetic or performance properties. With an appropriate selection of fibre, yarn, and weave structure, properties such as strength, porosity, durability, moisture vapour permeability, and extensibility can be manipulated to meet the requirements of the end user [32,33]. Table 13.4 provides an overview of the most common weave structures and their properties.

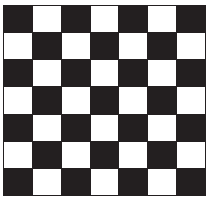
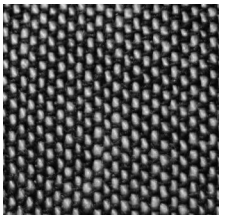
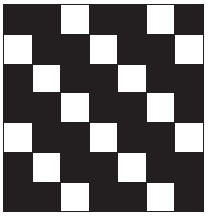
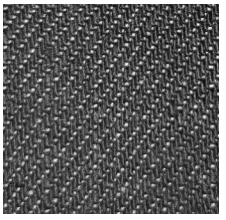
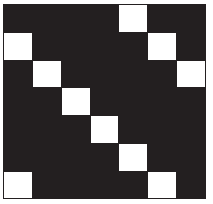
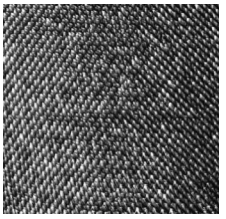
As a result of its manufacturing technique, a woven fabric has distinct ‘grain lines.’ Due to the directional differences in woven fabric properties, it is critical that special attention is given to grain lines when developing garment designs.

The warp grain runs vertically along the length of the fabric and displays the best stability in terms of tensile strength, propensity to distortion, and wash shrinkage. It is usual in garment creation for the warp grain to run from top to bottom of the wearer’s body. The weft grain line (also known as the crosswise grain) runs across the width of the fabric. As weft yarns are often weaker and more textured than warp yarns, the crosswise grain usually displays more stretch but less tensile strength and wash stability. Weft grain usually runs across the garment, i.e., horizontally across the body.

The bias grain runs diagonally (at 45°) across the fabric and allows the most stretch in a woven structure (in the absence of stretch yarns). Cutting a garment on the bias grain allows for fluidity and drape in a garment structure, enabling ease of movement in certain apparel designs [33].

In sportswear applications, parameters such as air permeability, moisture management, elasticity, thermoregulation, and comfort are critical to the performance of the wearer and weave structure can have an impact on all these areas. For example, researchers have shown that moisture

Table 13.4 Common weave structures and their properties [1,2]

Weave name	*Technical representation*	*Appearance*	*Properties*[a]
Plain			Even, flat surfaces are useful for print applications. Good abrasion resistance, tensile strength, dimensional stability, and snag resistance. Poor tearing strength, elasticity, and drape. Considered the cheapest and quickest method of woven fabric manufacture.
Twill			Defined diagonal lines on the fabric surface, lines running upwards to the right are 'Z' twill structures, lines running upwards to the left are 'S' twill. Superior abrasion resistance and stability compared to plain weave. The most common twill is denim. Considered the 'strongest' weave and therefore synonymous with workwear and durable apparel.
Satin/sateen			Lustrous, fluid, drapey fabrics. Snags easily due to floating yarns on the surface. Traditionally produced using silk yarns, now widely used with synthetics.

[a] *Comments on properties are a generalisation for the structure and can be altered depending on fibre/yarn selection, weave density and subsequent fabric processing.*

absorption is reduced with an increase in the density of a woven structure, and it is accepted that a decrease in weave structure density allows for improved air permeability [32]. However, it is always critical to remember that many factors (as discussed above) contribute to overall textile performance.

Knitted fabrics

Knitted fabrics are constructed by the interlocking of yarn loops, with each loop referred to as a stitch. Stitches are formed in rows, with each row interlacing with the next. In an equivalent way

Case Study 1 Woven stretch textiles in swimwear applications

Many materials used in swim and beachwear garments are made from warp-knitted textiles. A warp-knitted material provides excellent stretch and recovery properties, ideal for these tight-fitting, compressive garments. Much of the warp-knitted textiles are made from polyamide and elastane, with 80% polyamide and 20% elastane being the most common composition. However, there are many garments on the swim market made from 100% polyester. These textiles contain a bicomponent polyester fibre, as discussed in Case Study 2, which creates the required stretch and recovery. Whilst this physical characteristic is not as ideal as that created by a material containing elastane, it is a much more stable property since chlorine in swimming pools breaks down elastane over time, something that does not happen to polyester. Swimmers who train regularly tend to use these 100% polyester suits as they are far more durable than those containing elastane.

Whilst swimmers train in suits made from warp-knitted textiles, they professionally race in garments made from woven textiles. Racing swimsuits need to be very compressive; they must streamline the athlete's body, stabilise their muscles and remain anchored tightly against the body as they move through the water at high speed. By using a stretch-woven textile, as opposed to a warp-knitted one, a far higher compressive force can be achieved from a much lighter, thinner textile.

The stretch wovens used in racing suits tend to be very high-density plain weaves, with elastane in both the warp and the weft directions. The warp and weft yarns have an elastane core, which is covered with a high-twist polyamide outer, and a total elastane content in the finished fabric of around 25–30%.

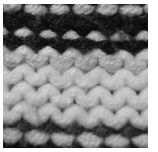
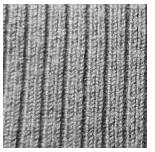
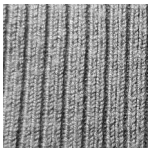
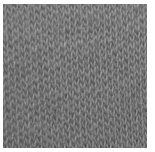
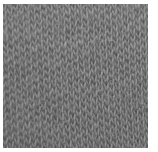
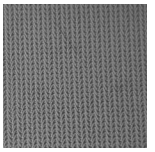
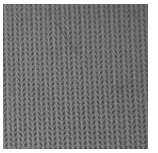
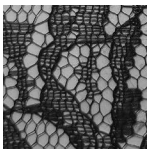
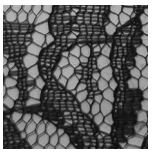
Different woven structures are being introduced into racing suits specifically to direct the flow of water over the swimmer's body. These textured textiles are woven on Jacquard looms (looms with extensive patterning capabilities). The surface design is engineered with the guidance of either aero- or hydrodynamicists to create a textile with specific surface drag characteristics.

All the textiles used in professional racing swimsuits need to pass the rules imposed by the governing body of swimming, FINA requirements for swimwear approval (FRSA) 4.1, 2017 [34]. These rules centre around the thickness and air permeability of the textile. As the woven textile is very lightweight and thin, it often needs to be used in two layers for reasons of modesty. The same FINA rules apply regardless of whether the textile is being used in one or a maximum of two layers.

For the textile to remain as light as possible for the duration of a race it is often treated with a hydrophobic chemical. This chemical repels water from the surface and delays the speed at which water absorbs into the textile structure. As a woven textile in this case is lighter and thinner than a warp-knitted one, the capacity for water absorption is lower, and therefore the garment remains relatively light. When racing suits were first introduced, the hydrophobic chemicals used were long-chain (C8) PFC (fluorocarbon)-based chemistries. As global legislation has been introduced to restrict the use of these chemicals due to their bioaccumulative nature (Annex 1 of the EU's Persistent Organic Pollutants (POPs) Regulation), extensive research has been conducted to create finishes with equal performance but a lower environmental impact.

Whilst a racing swimsuit will never be as durable as a training suit, attention should be given to the tensile and tear strength of the textile used. Fabric tears and seam ruptures are the most common sources of failure in a racing swimsuit, occurring predominantly when the garment is being put on. This is due to the extremely tight garment fit and the high force required to don (put on) the suit. Whilst a woven textile does not ladder (unlike a weft-knitted structure), it has lower tear strength, and attention should be given to these properties, with the provision of more durable areas to aid in the donning of the suit.

Table 13.5 Common knit structures and their properties

Knit name	*Appearance*		*Properties*
	Face	*Reverse*	
Single jersey / stockinette (weft knit			Simplest weft knit structure. Produced on circular or flat knit machines. Unbalanced structure; different face and reverse. Prone to unravelling, edge curling, and distortion. Limited recovery after stretching.
Rib (weft knit)			Produced on weft knit 'v' bed or double jersey machine. Balanced structure, more stable than single jersey. Can be stretched and distorted and displays a degree of recovery.
Interlock (weft knit)			Produced on weft knit double needle bed machines. Appearance is the same on both sides of the fabric. Lies flat and is a stable structure, which is difficult to distort but displays relatively little stretch compared to rib and single jersey fabrics.
Tricot (warp knit)			Produced on warp knit tricot machine. A different face and reverse; loops appear to run vertically on the face and horizontally on the reverse. A stable structure with little stretch capability compared to weft knitted fabrics.
Raschel (warp knit)			Produced on warp knit raschel machines. Usually very similar face and reverse but can differ due to the versatility of this knitting method and the ability of the machine to use novelty and fancy yarns.

to the warp and weft notation of woven fabrics, the horizontal lines of loops in a knitted structure are known as courses, and the vertical lines are called wales (Table 13.5).

Knitted fabrics were traditionally created by hand using thick and heavy woollen yarns and two knitting needles. The first mechanical knitting machine, created around the end of the 16th century, by William Lee, produced tubular fabrics for hosiery using cotton, wool, and silk. The development of fine, strong synthetic yarns caused a revolution in knitted fabric manufacture and led to the evolution of machinery capable of creating fine, intricate fabrics at high speeds.

Modern knitted fabric production can be split into two types of structure: warp and weft knits. Weft knitting involves loops being formed across the width of the fabric structure, whereas warp knit structures comprise loops formed along the length of the fabric.

Weft knitted structures are stretchy (without the need for stretch yarns) in all directions. However, the degree to which they recover their shape after stretching is highly dependent on the specific knit structure and the types of yarn present. The inherent stretch means weft knitted

fabrics are highly desirable for a myriad of garments such as sportswear, hosiery, and intimate apparel. However, as recovery is critical in these applications, the presence of a stretch yarn is also required. Nevertheless, the speed and ease with which basic weft knitted structures such as single jersey (also referred to as stockinette), rib, and interlock can be produced make this type of fabric popular for most apparel applications.

Compression is one of the most important properties of weft knitted structures pertinent to sportswear. Due to the structure and placement of the loops in a weft knitted fabric, this structure is found to be much easier to compress than its woven counterparts, particularly in the presence of stretch yarns.

Warp knitted fabric production is considered extremely versatile and can be further categorised, depending on the mode of manufacture, into tricot and raschel structures. In contrast to weft knits, warp knitted structures are usually created on machines fed by warp beams of yarn, in a similar way to a weaving loom. Tricot fabrics are usually associated with sports and active wear, creating distinctive mesh-like structures. Raschel fabrics are slower to produce and are generally stiffer in structure, normally creating intricate structures such as lace.

The development of knitting technologies has led to a revolution in the way knitted garments are designed and created. The manufacture of garments from flat pieces of knitted fabrics, with garment panels cut and sewn together, is still a popular method. However, knitting technology innovation has opened the door to ‘seamless’ garments and machines that can knit a garment in its entirety in one operation. The advantages of this technique include the elimination of seams (and the associated comfort and seam failure issues), better fit, and enhanced aesthetics. Additionally, this type of machine has the capability to engineer garments with specific areas of performance, such as compression and moisture management. However, this type of garment manufacture, although not completely uncommon, has not been adopted as widely as anticipated and therefore is not being used to its full commercial potential. It is thought this is due, in part, to the skill set required to bring this product to market, such as designers with detailed technical expertise in this type of product, as well as highly skilled software engineers and machine mechanics. Nonetheless, this is still a critical area for consideration in functional apparel [35].

Non-wovens

Non-woven fabrics are those considered to be made directly from fibres (i.e., omitting the yarn creation stage). Non-wovens are created by ‘sticking together’ the fibres in some way to form a mesh or web-like structure. Common construction techniques include bonding the fibres using chemical, mechanical, or thermal methods. As with the other techniques of fabric creation already discussed, the selection of fibres and the subsequent finishing of the textile can greatly affect the performance attributes of the non-woven textile. Whilst non-woven fabrics are not traditionally noted for their strength, they are used extensively for filtration and absorbance applications, usually as single-use textiles in the field of healthcare. However, non-woven composites are becoming more prevalent in construction applications as their technology develops and thus may have a place in sportswear apparel in the future.

Alternative manufacturing approaches

Innovations in manufacturing techniques have led the textile industry to adopt novel approaches to textile creation.

Case Study 2 Weft Knit textiles in a teamwear application - Rugby

When selecting or developing a textile for the application of a rugby jersey, there are certain characteristics that cannot be compromised. The textile needs to be strong, with very high tear resistance. The chosen textile must also contribute to the creation of strong seams and arm, neck, and leg openings.

The textile needs to allow the wearer to remain comfortable during high aerobic activity and, therefore, must be breathable and capable of moving sweat away from the body. Due to the contact nature of the game of rugby, it is advisable to minimise the stretch of the fabric to prevent other players from holding or pulling an opponents clothing.

A double jersey weft knit structure is the ideal choice to address the needs of the rugby player. In this case, the double jersey structure is created from yarns comprising different filament counts on the face and reverse sides of the fabric. This 'denier gradient' construction creates a structure that can physically draw moisture away from the body to the outer surface of the fabric, from which it can readily evaporate. This process is referred to as the fabric's wicking capability. This physical wicking process is often boosted by the addition of a hydrophilic chemical, which speeds up the movement of sweat away from the body. In some cases, a slightly more complex chemical system is used, whereby a hydrophilic chemical is applied to the reverse of the textile, which sits next to the skin, while a hydrophobic chemical is applied to the face of the fabric, reducing the absorption of moisture from the environment (e.g., rain or moisture on the ground).

Double jersey textiles often have a texture on the side closest to the body, with a smooth flat surface facing outwards. The texture contributes to the moisture management of the textile, whilst the smooth face provides an ideal surface for printing.

Currently, materials used in teamwear applications are made from polyester, as this allows them to be printed using sublimation methods. This printing technique is ideal because stocks of finished textiles can be held and printed with seasonal designs and team sponsors' logos.

In terms of fibre composition, most rugby jersey textiles do not contain any elastane fibres. Whilst the addition of elastane allows for greater freedom of movement and ease of donning and doffing, it has a detrimental effect on the material's strength. By switching from an elastane stretch fibre to a bicomponent stretch fibre (one with two different properties in the same composite), a textile with comparable strength at a lighter weight can be created, which is beneficial in such a fast-paced sport. In the case of textiles for rugby jerseys, the bicomponent fibre is made of two polymers with different shrinkage characteristics placed side by side. When the fibre is dyed, each side of the fibre shrinks to differing degrees, resulting in a helix-type structure. The helix has an inherent stretch, which is then used to provide the stretch and recovery to the knitted material.

The material needs only enough stretch to allow the player to don and doff the jersey, which can ultimately be done with the help of team members. Therefore, most jerseys will have a composition of 100% polyester.

To check that a textile is suitable for use in rugby, the textile material is tested for its bursting strength. The bursting strength test is ideal because it closely replicates a grab or pull on the material, albeit at a much slower speed. To test for bursting strength, the textile is mounted over a diaphragm that is inflated to subject the material to 360-degree pressure. The pressure is increased until the material bursts. If a textile with a higher degree of stretch is to be used, the ball burst test method is preferable, where a metal ball is forced down onto the clamped textile. It is imperative to also test the strength of seams and neck openings, as these must be as strong as the textile itself.

Biotechnologies

Inspiration from nature has gained popularity in the textile and apparel [36] arena, with many 'natural' sources being investigated, alongside synthetically generated materials designed to mimic those found in the natural world [37]. This is still a new field of exploration, and therefore many developments are in their infancy and at a trial stage. However, these materials could provide interesting alternatives in sportswear apparel, such as spider silk for its superior strength properties; crab shells and bird beaks for their impact protection; sharkskin and bird feathers for their aqua and aerodynamics, respectively; pinecones, leaf respiration, and animal skins for environmental management; and naturally occurring microbes for their potential to generate novel textiles with a wealth of enhanced properties [38]. Alongside the enhanced and novel approaches to performance properties that nature has to offer, many researchers are taking the opportunity to use the biomimetic approach as a way of addressing the sustainability challenges faced by the textile and apparel industries.

3D printing

Developments in 3D printing technologies have led to the exploration of such methods in apparel and textile applications. Whilst this technology is still in its infancy for mainstream apparel, it has been widely used for garment components (such as buttons and zips), making it valuable for the customisation of apparel. Some work has been conducted to use 3D printing technology to 'mimic' traditional woven [39,40] and knitted [41] textile structures, but this requires further development. Instead, researchers are exploring the possibilities of using 3D printing to develop textiles for apparel that do not adhere to the constraints of traditional woven or knitted structures [42,43]. This is a field that shows exciting potential for the future but has yet to be realised in the mass market. However, for certain high-performance garments requiring additional functionality, 3D printing has been used to produce items such as sensors and electrodes that can be intergrated into the textile and garment structure. The enhanced functionality of garments is the most rapidly growing field within the realms of textiles for sportswear apparel.

Smart textile materials

Smart textiles can be defined as those that sense their surroundings, react to, and adapt to their environment, thus serving the needs of the end user [44,45].

Several types of materials fall into the category of smart materials:

Shape memory

These textiles are temporarily deformed by external stimuli (such as pressure, light, magnetic fields, or temperature), returning to their original form once the stimulus is removed. In sportswear applications, this type of material is useful for thermoregulation, with companies such as Schoeller Textiles developing responsive materials for apparel.

Phase change

Phase change materials absorb energy to facilitate a change in physical state [46]. Technologies such as Outlast use the principle of wax encapsulated into fibres, which change from a solid to a liquid state when absorbing heat from the body (thus eliciting a 'cooling' effect). Conversely, the

change from liquid to solid wax emits heat when the environmental temperature falls (creating a 'warming' effect). This type of technology has been successfully incorporated into clothing, with Outlast claiming reduced sweating and improved thermal balance in activewear products.

Dilatant materials such as RHEON, Poron XRD and D3O® can also be considered in this category. When not under stress, the material is flexible and pliable, but upon impact, the force is absorbed as the structure interlocks and becomes rigid, thus absorbing the energy of impact and offering protection to the body. These materials have been successfully used in many products in the field of sportswear apparel, predominantly in the category of body armour.

Chromic materials

The application of colour in textiles is an important aesthetic consideration. Developments in textile dye chemistry mean that colour can be engineered to respond to the external environment. The categorisation of these types of chromic materials is outlined below:

Light = Photo
Electrical current = Electro
Heat = Thermo
Chemical = Chemo
pH = Iono
Mechanical = Piezo
Moisture = Solva

The application of this type of technology is used in the form of sensors. For example, photochromic textiles have been used to alert the wearer to overexposure to sunlight (UV rays), and thermochromes have been used to alert the wearer to changes in body temperature [47].

Electronic textiles

Many garments have been considered 'electronic' or 'smart' (as seen in Figure 13.2), but these have mostly been due to the clever incorporation of componentry and wiring into the garment structure, rather than the actual electronic componentry being the textile itself [48–50]. Sports garments currently on the market that could be considered 'wearable electronics', with some degree of amalgamation of textile and electronic componentry, include the Nadi Yoga pant and the Antelope EMS suit and calf guards. However, true 'electronic' textiles are an exciting field of development, particularly as they are starting to explore the possibilities of linking with the Internet of Things (IoT) [49].

The barrier to incorporating electronic componentry in textiles and apparel has been the power source; often, these are bulky and cumbersome, difficult to conceal within the garment, lead to a degree of wearer discomfort, and usually need to be removed prior to laundering [49]. Some research studies have explored alternative approaches, such as textile-based supercapacitors [51,52] and energy harvesting [53], as alternatives to the more traditional battery pack, but these are still in the early stages.

The focus of research in performance apparel, aside from sensors used for physiological monitoring, has been on the use of textile actuators (also known as textuators). Initially employed in soft robotics, these actuators are increasingly used as 'textile muscles' for mobility assistance,

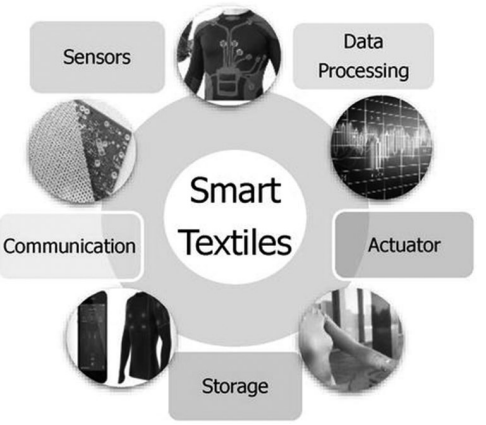


Figure 13.2 The diagram displays various elements that make up electronic/smart textiles from sensors, data processing, communication, storage, and actuator.

high-performance compression garments [54], adaptive clothing [55], and haptic feedback – all of which have great potential to drive sportswear apparel into the future.

Summary and future trends

The discussion in this chapter has demonstrated that the unique characteristics of fibres, their formation into fabrics, and their application in sporting materials and garments provide users with properties to enhance performance, monitor biometric signals, and assist in recovery, health, and well-being.

The chapter provides the reader with knowledge on the selection of textiles for sportswear. The development of smart materials and a growing interest in new fibres with sustainable features offer the sportswear industry an opportunity to develop sports apparel that connects with the digital world. The current geometry of textile materials is being engineered with uniquely designed fibres, complex structures, and smart technology, which will offer the sportswear industry the required comfort levels and functionality, as well as the potential to transform an athlete's experience during sporting activities. Garments are now being developed to communicate, transform, conduct energy, and grow. Future innovations in textiles will involve how these materials interact with our bodies and the environment. The next generation of textiles will not only employ wearable technology but also meet the demands of the new market, green textiles, and environmental sustainability.

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PART III

Sports equipment design and manufacture

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14

INTRODUCTION TO SPORTS EQUIPMENT DESIGN AND MANUFACTURE

James Webster

Introduction

Over the past century, there have been incredible advances in technology, spanning materials, manufacturing, and design capabilities, among others. The sports industry has benefited from these advancements. For example, within footwear, the industry has moved from handmade products limited by the ability of humans working with naturally occurring materials to automated production with incredible precision and repeatability, utilizing tailored engineered materials specifically developed for the end use (Figure 14.1). Alongside the gain in capabilities from a product creation and development viewpoint, there has also been an overwhelming increase in sports research, from biomechanics, anthropometrics, physiology, and psychology (Figure 14.2). These increased learnings have helped guide the design and development of new sports equipment, apparel, footwear, and sports surfaces aimed at maximizing performance and reducing injury risk. One example of these advancements is the development of the so-called "super shoes," where the integration of high-performance foams and carbon fiber composite plates, among other things, has

Figure 14.1 Evolution of footwear from traditional materials and cobbling to super shoes with complex construction and high-performance materials.

DOI: 10.4324/9781003303626-17

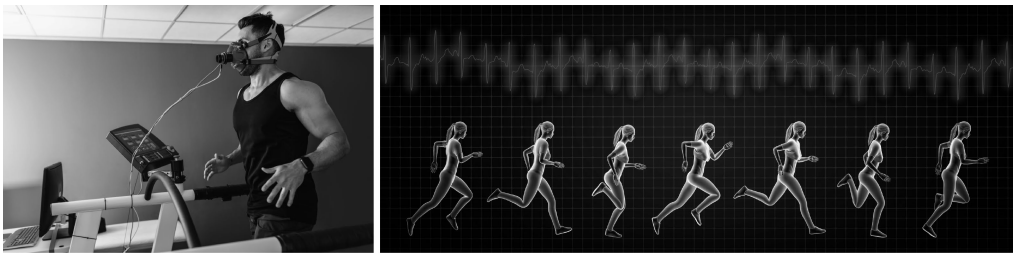

Figure 14.2 Human performance evaluation to inform future product development.

seen measurable increases in performance [1–3]. Several of these advancements have been well documented in the literature, e.g., [4–20].

These developments have created an ever-evolving landscape within the sports industry. Brands are continually looking to integrate the athlete learnings with the unlocking of new manufacturing principles and novel high-performance materials to optimize the elite athlete as well as increase the desirability of the product to the general population, bridging the gap between elite-level sport and lifestyle products. Although these developments have enabled increased product capabilities in terms of performance and unique aesthetics, they do come with their own limitations, which designers, developers, and manufacturers must be aware of and consider before embarking on the product creation journey.

This section of the book covers several aspects of the design and manufacture of sports equipment. Although not exhaustive, it will provide a good understanding of the topic and enable readers to further develop their understanding through continued research. Chapter 15, "Design and manufacturing of textile sports equipment," is the first chapter within this section, where Wagner et al. discuss the technological revolutions across a broad spectrum of materials and the integration of these materials in various products from ropes to footwear. Within this work, they identify emerging trends and highlight the influence of design and manufacturing on these textiles. In Chapter 16, "3D Printing in sports," Vasquez highlights the development of 3D printing technologies and how these technologies are utilizing different materials to enable novel sports equipment to be developed or how the technology can be used to enable faster and more economical development through prototyping. In Chapter 17, "Sustainable design and manufacture of sports equipment," Colonna et al. present the complexity of sustainability, discussing different approaches to reduce the negative impact of sports products on the environment, including information on materials, manufacturing, and recyclability. They highlight the importance of design in attaining more sustainable products. In Chapter 18, "Cricket gear and sustainability," Charter et al. present a unique view of how impactful climate change can be on sport and what is currently being done to reduce the carbon footprint of cricket. They outline how this needs to be a top-to-bottom effort, including governments and governing bodies, as well as equipment manufacturers and participants.

Chapter 19, "Design considerations for body padding in rugby," Imam discusses the current regulations within rugby. Imam also discusses current materials and test methods utilized for protective equipment. The detailed review of current requirements, materials, and athlete needs is used to outline design considerations and ideologies to inform potential pathways for future research. In Chapter 20, "Ergonomics and user-centered design," Nispel and Senner discuss how the combination of athlete measurement, computer-aided design and manufacturing, and material developments, such as those seen in 3D printing, can be integrated to create novel products

centered around the user and their performance. Their worked example focusing on bike saddles shows a detailed process to enable advanced customized products to be created. In Chapter 21, "The application of biological engineering to sports technology," Grant and Spurrier discuss how biology can inspire the design of sports equipment, taking learnings from nature to create unique design concepts. They also discuss how biological engineering can be used in the testing and analysis of the designs. In the final chapter of this section, Chapter 22 "Design for athletes," Koptyug and Bäckström discuss the design process and how a multi-faceted approach is required, with consideration being given to performance, functionality, comfort, ethical issues, legislation, affordability, manufacturability, safety, reliability, sustainability, emerging technologies and materials, and athlete feedback to enable the development of a successful piece of equipment. The chapters within this section cover a large array, but not exhaustively, important factors and areas of consideration within the design and development of sports equipment, apparel, or footwear, aimed at furthering the understanding of the design and manufacturing process.

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15

DESIGN AND MANUFACTURING OF TEXTILE SPORTS EQUIPMENT

Janina Wagner, André Matthes, and Holger Cebulla

Emerging trends in sportswear equipment

Textile sporting equipment comprises any sports gear or equipment made using fabrics and fibers, and it can include various product types such as footwear—ranging from athletic shoes to football boots and gym shoes—seamless compression garments, damping elements, tents, ropes, nets, balls, sails, rackets, and protective headgear [1]. The interest in sports equipment has risen continuously over time due to its ability to provide various athletes with greater comfort, mobility, and protection during their physical performance in indoor and outdoor surroundings. Equipment for sportswear can also be furnished with specific features and functions, such as moisture-wicking ability, force and energy transfer, antimicrobial properties, and thermoregulating advantages to improve athletic performance and overall comfort in application [1, 2]. In many cases, the combination of materials and structural design can cause synergies to achieve the posed requirements, but it is also possible that, for example, equipment that benefits from aerodynamic aspects can interfere with the thermoregulation of the product. This is due to the missing thermal radiation and evaporation and can be found in cycling, speed-skating, or skiing garments, which shows the need for new smart fabrics that can combine a variety of technical requirements [1].

Due to the rising demand for sporting gear, the segment continues to grow steadily and is expected to play an ongoing important role in the sportswear market share [2]. In general, the global sporting equipment and sportswear market size was estimated to reach a value of USD 542 billion in the year 2021 and will increasingly grow to a level of about USD 1030 billion by 2030 [2]. This expansion is mainly driven by the increasing popularity of sports and physical activities caused by growing concern about possible health issues and the advertising of prominent sporting events, such as the Olympics and the FIFA World Cup [2, 3]. However, the sportswear industry also has a rising innovation potential regarding material and fiber development and advancements [6]. Some future trends are presented in Table 15.1.

Emerging trends in the sector of sportswear equipment include accessibility for all types of users and sustainability in terms of recycling and the usage of environmentally friendly fibers [2, 7]. This sustainable trend is partly due to the changing and fast-emerging regulations related to sustainable textiles in the EU, such as the EU strategy for sustainable and circular textiles. An important factor is to ensure the possibility of end-of-life recycling, which can be more difficult when fibers

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Table 15.1 Examples of trends in sports gear [4–6]

Textile sports equipment	*Trends*
Ropes	Replacement of hard fiber ropes with synthetic ones
Nets	Smart nets equipped with sensors for real time feedback and performance, multi-sport nets
Tents	Siliconized tents with taped seams, super lightweight materials, integrated technology like built-in LED
Bandages and damping elements	Spacer fabrics, tri-laminated flame bonded structures, socks with tapered toe area to match parabolic shape of the foot
Sport shoes	Traction and grip technologies, integrated sensors, single or dual material shoes
Fitness mats	Smart mats (e.g., electronic mats)

made from different materials are mixed or combined. Closely connected to the sustainability of a product is also its durability, which makes the gear suitable for all types of climate conditions or contributes to the extended lifecycle of a product. An example here is the construction of a tent, which can be used in heavy rainfall and summer heat. Apart from that, multipurpose gear also adds to the extended usage of sporting equipment as it enables a product to be used while performing the preferred sporting activity but also in regular urban life [7]. Moreover, reactive equipment, for example, in the case of thermal materials and moisture-wicking fabrics, enables the user to detect changes in body temperature [2]. Lastly, consumers continue to demand new technologies and wearable smart devices to provide insights into their fitness performance. Different kinds of sensor technology that can be incorporated into many kinds of equipment, such as tennis rackets, fishing rods, and fitness mats, are used [2]. These sensor-equipped gears can show biometric data as well as motion control and improvement and will probably become indispensable in the future [8]. However, the difficulties in establishing recyclable products will be one of the biggest challenges for this product category in the future. Readers seeking more detail on the topic of sustainability are referred to "Chapter 17: Sustainable design and manufacturing of sport equipment."

Due to the diversity of applications, this chapter will focus exclusively on ropes, nets, tents, bandages, sports shoes, and fitness mats as product categories of textile sports equipment. Readers interested in textiles in relation to clothing are referred to "Chapter 13: Textiles for sports clothing".

Ropes and nets

Ropes and nets made of textile materials have been used for various practical applications for centuries, such as fishing, transportation, and construction work. However, due to the rising awareness of health issues and well-being in general, ropes and nets have found their way into the world of sports equipment in recent years. Nowadays, textile ropes and nets are commonly used as essential components in various sporting applications, such as rock climbing, gymnastics, aerial arts, and ball sports like volleyball and tennis. Both textile ropes and nets are utilized to guarantee safety, support, and performance due to their properties. These include strength, flexibility, and elasticity, which make them ideal for various physical activities involving sports equipment.

The following chapter will explore the different types of textile ropes and nets commonly used in sports, their manufacturing, and their ability to enhance athletic performance and safety through unique qualities.

Ropes

A textile rope is a compilation of yarns and fibers that are mainly manufactured through either twisting or braiding and has various possibilities for application. Examples in sports gear include rock-climbing ropes, gymnastic ropes, and jump ropes, but laces, tent lines, and trouser cords also belong to the category of ropes. There are also metallic ropes, which are used in various non-sports applications, such as bridge cables and hoists. This different application area is mainly because they have a higher density than technical textile fibers [9].

The peculiarity of a rope lies in its resistance to large axial loads in contrast to bending and torsional loads [10]. Other qualities include high durability, a high strength-to-weight ratio, high strength in general, and sturdy spinning capabilities [11]. With advancements in manufacturing technology, textile ropes are constructed with more and more performance abilities, which enable synthetic ropes to evolve into light, strong materials with low corrosion susceptibility and floatability [11]. Nevertheless, natural fibers are also used in the construction and manufacturing of ropes because of their good gripping surface, convenient usage, non-melting property, and low recoil energy when breaking [11]. A blend of both natural and synthetic fibers is also possible.

Materials

Synthetic fibers and specialty fibers are either used as staples or continuous filaments in rope construction. Different kinds of synthetic fibers show varying properties in fineness, extension at break, and tenacity. For example, aramid is a relatively fine fiber with lower extension and higher tenacity than polyamide. Looking closer at the atomic structure, polyamide consists of polymers with amide links holding together the repeating units. The high strength of aramid, derived from the words "aromatic polyamide," is therefore a result of its molecular structure. The paraposition, achieved by the aromatic rings, results in a high orientation of the polymer strands, which are kept in position by hydrogen bonds and result in long-chain, straight, rod-like structures [12, 13].

Even though polyester also shows a higher extension than high-performance fibers, such as aramid and thermotropic liquid crystalline polymer (TLCP), it has lower tenacity and strength. High modulus polyethylene (HMPE), on the other hand, is known for its high tensile strength but rather poor creep behavior and low melting point [14]. When looking at the molecular structure of HMPE, it has long molecular chains that are stretched, oriented, and crystallized in the specific direction of the fiber to achieve a strong material [13]. A comparison among the most widely used synthetic fibers is made in Table 15.2.

Table 15.2 Common synthetic fibers used in rope construction [10, 14–18]

Fiber	*Fineness (dtex)*	*Extension at failure (%)*	*Tenacity (mN/tex)*	*Strength (MPa)*
Kevlar (Aramid)	<2	<5	2,000	2,900
Polyamide	>6	20	840	960
Polyester	>6	15	530	500
Polypropylene	>5	20	820	560
Polyethylene	>5	20	620	1,130
HMPE	>55	3,5	3,500	3,400
TLCP (Vectran)	>1,700	3,5	2,200	3,100
PBO	>2	3	3,700	5,700

To determine the given parameters, different standards can be used. For fineness, DIN 53830-3 is typically used, which evaluates the linear density of single and plied yarns through a measuring device with a specified weight load on the fibers [19]. As for extension, ISO 13934-1 is a common standard for evaluating tensile elongation by the strip tensile test, which tests for extension by clamping the test specimen in its full width in the clamping device of the test equipment. For tensile strength, ISO 527 is utilized, which calculates the maximum tensile force achieved and the specimen cross-sectional area at the start of the tensile test [20].

The selection of material, therefore, hinges mostly on the performance requirements, such as strength, durability, flexibility, softness, hardness, and economic availability [15]. For example, polyamide is an appropriate material for climbing ropes, as the high energy absorption capacity provides protection against falling, while polyester ropes are adaptable for mooring applications for ships, as they have the needed amount of working stretch and can absorb shock loads [10].

Construction

To construct a rope, the chosen fibers must be straightened out, mostly by carding or combing. The subsequent twist is necessary to transfer the single fibers into a yarn, which leads to a self-locking structure [15]. Natural fiber ropes require a tighter twist to prevent the fibers from slipping, as they only get their strength from the friction between the fibers. Synthetic fibers can be used as continuous filament yarns. In this case, they are non-slip, and twisting is only required to keep them joined together [9]. The strands can be either twisted or laid, plaited, or braided, which automatically leads to different kinds of ropes being available. Other constructions include wire ropes, parallel ropes, and pultruded ropes [10]. Besides braiding circular knitting is another alternative construction method.

A laid or so-called twisted rope is made by twisting the fibers together. At first, the fibers are twisted into yarns, which are then twisted together in the opposite direction to form strands. These strands can then be twisted into the final product in the form of a laid rope. The twist varies

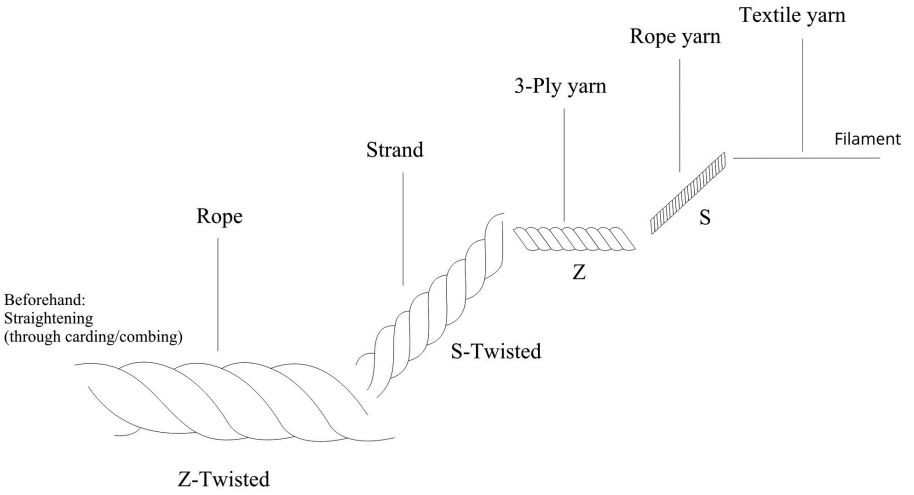


Figure 15.1 Rope construction. Authors' own image, based on [10].

between the S and Z directions to minimize torque [9]. The two most common types of laid ropes are three-strand and eight-strand ropes. While three-strand ropes consist, as the name suggests, of three strands, the eight-strand laid rope is manufactured by combining two right-twisted and two left-twisted pairs of strands together. Laid ropes are typically more flexible than other constructions but can also unwind and rotate with continuous usage [15]. Examples of twisted ropes include battle ropes used in fitness studios. The different types and construction methods are presented in Figure 15.1.

A braided rope consists of single strands that are interlaced together. Single braids include 8 to 12 strands that are plaited together in circular motions, either clockwise or counterclockwise. Double braided rope consists of plaited strands forming a core, with even more braided strands around the core resulting in a sheath, which makes it stronger and more durable [15]. Apart from that, the core is resistant to ultraviolet (UV) damage and abrasion [21]. Examples of braided ropes in sports equipment include climbing ropes.

The combination of different rope constructions and materials can provide different properties that are needed for various uses in sports equipment.

Nets

Another type of sporting equipment made from technical textiles is nets, which consist of a coarse-meshed structure with regular rhombic, square, diamond, or hexagonal openings, achieved by twisting, interlocking, or knitting the yarn [9]. They vary in their end use and can withstand varying climate conditions depending on the material used, leading to applications such as fishing nets, goal nets for ball sports, screens for sporting facilities, and nets for dividing sporting facilities [22]. The structure ranges from light fabric, as in mosquito nets, to stronger fabrics, as in fishing nets.

Construction

To achieve the visual appearance of an open fabric with more holes than yarns, the product is manufactured through gauze weaving, knitting, or knotting [9]. This can be done by hand or on a lace machine or bar raschel machines, for example [9]. Whatever method of manufacturing is chosen, it is important to ensure that the crossovers that form the openings are firmly locked [9]. During the manufacturing process, the parameters can be adjusted to achieve different levels of shade, wind permeability, opacity, elasticity, and stability for the expected end use [22]. These parameters include machine gauge, construction, type of yarn, lapping, and stitch density, for example [22]. The most common technology for nets is warp knitting because it produces flexible yet dimensionally stable textile nets with various possible structures [22]. When using circular warp knitting as a construction method, netted tubular goods, such as packaging nets, can also be developed.

One type of net, for example, is the knotless construction, which is produced on either single- or double-bar raschel machines. Knotless nets are made by processing all yarns vertically into the attachment points and connecting the sections through the guide bars in specified time intervals [22]. The produced nets are known for their equal thickness, slip resistance, good abrasion resistance, high energy absorption capacity, dent-free structure, and sleek surface with a uniform visual appearance of the openings [22]. Most of the nets used in sports today are knotless nets.

Another type of net is the knotted net, which is produced by twisting yarns together to shape net openings that can be varied in size through lapping [22]. This demonstrates flexibility in adjustment and size.

Table 15.3 Material properties of commonly used fibers in net construction [27]

Fiber	*UV-Stability*	*Abrasion resistance*	*Moisture absorption (%)*
Polyamide	Good	Very good	1–7
Polyester	Very good	Very good	0.5–2
Polypropylen	Satisfactory	Good	0
Polyethylen	Good	Good	2–5

Materials

According to the various types of sports, different materials are used. For example, for fishing nets, polyamide twine is used since it shows good elastic recovery and wet resistance [23]. Polyester and polypropylene are also options for sport fishing nets [22]. The same types of materials, like polyester, polypropylene, and polyethylene, are also used for nets in sports like volleyball and tennis [24, 25]. All fibers used have different advantages in netting, such as the low moisture absorption and high UV stability of polyester and the high abrasion resistance of polyamide, for example [26]. Some properties of the various fibers used in net construction are listed in Table 15.3.

As for testing methods, the most common standard for testing UV stability for textiles is ISO 105-B02, which tests for color fastness to artificial light, while ISO 12947-2, ISO 12947-3, and ISO 12947-4, describing the Martindale procedure, are commonly used to test for abrasion resistance [28]. The Martindale method is the most frequently used method for testing textile abrasion resistance and works by rubbing a sample of the material being tested against a specified standard woven fabric under the pressure of a certain weight load. The number of rubbing cycles is determined to evaluate the abrasion resistance of the fabric. For testing moisture absorption, DIN 53923 is used, which specifies the water absorption time and capacity of a textile [29].

Future trends

As for future trends in nets, multi-sport nets that can be used for multiple kinds of sports and smart nets with integrated electronics are targeted. Smart nets could benefit athletes by enabling them to measure their performance with interrelated feedback.

Fitness mats

Another type of sporting equipment used in various athletic disciplines includes mats and pads of all kinds. They can be found in combat sports, such as wrestling and boxing, as well as in gymnastics, dance, camping, and yoga. In general, mats are used to ensure a cushioning effect between the athlete and the ground, reducing injuries from hard floors by providing safety, support, and comfort [30]. This is usually achieved through high energy absorption and suitable friction while keeping the maximum force produced below a specific level, which can vary according to the specific purpose [31]. For example, sleeping pads need to guarantee resting comfort, while gymnastics mats must be slip-resistant. Mats are also used in weightlifting to provide shock absorption and high energy absorption, such as when dropping heavy weights, and in martial arts, where the mat must be stiff enough to prevent the athlete's foot from sinking into it, thereby avoiding injuries to the ankle [31].

Materials

Different fitness mats are made from materials such as polyester, polyvinyl chloride, ethylene-vinyl acetate copolymer, polyethylene, and polyurethane. While mats made from polyethylene exhibit less flexibility but greater durability and allow for lightweight usage, nitrile rubber and polyvinyl chloride materials demonstrate significant irreversible deformation [31]. Polyurethane, on the other hand, is a thermoplastic polymer, meaning that increased heat resistance can only be achieved through crosslinking the molecular chains [31]. This crosslinking also enhances the material's ability to recover after static loads. Additionally, there are air and foam mattresses, which can vary in height.

Construction

The process of manufacturing foam mats involves distributing the foaming agent in the polymer matrix, while bubbles begin to form and cells multiply. Once the volume of the mat increases to a stable stage, the stabilization of the texture is achieved by cooling [31]. Mats made of foam, such as polymer foam, can be categorized according to their porosity into closed, blind, and through pores, and can also be inflatable. The different methods of manufacturing foam mats provide various benefits. Closed-pore, also called closed-cell foam mats, show a high insulation effect due to their lower permeability, high density, and long lifespan, although they are relatively high in cost [31]. Open-pore mats are available at a lower cost but are generally known to have reduced durability and a higher water absorption capacity [31]. Both types provide good insulating and impact properties based on the deformation mechanism of the cells and are known for their excellent energy absorption and impact damping ability [31]. Still, foam mats can cause problems concerning comfort and recycling [32]. These problems are now solved by using waste from cutting polyurethane foam blocks during mat production [31]. The functionality of the foam mat is primarily tested through the falling weight impact test, which measures the force and retardation during the impact of an object falling onto the mat. Readers seeking more details on foams are referred to Chapters 9 and 10, "Cellular solids for sports applications."

Performance requirements

The different requirements categorized by the varying physical activities needed for mats are listed in Table 15.4.

These performance requirements can be ensured by certain standardizations. For example, wrestling is covered by the standard MSZ 20333:2015, while requirements for gymnastics, judo, and rod- and high-jumping are listed in EN 12503 [31]. The EN 12503 includes a measurement method to ensure that the athlete's foot does not submerge into the mat by evaluating static stiffness [31]. It is also important that slipping is prevented by evaluating the coefficient of friction. This is also covered in the EN 12503. A commonly used standard for motorsports and skiing, which demands protective clothing, is EN 1621–2, which states the protection of the textile against mechanical load [31].

For some applications, there may be a need to provide air circulation and moisture transfer to prevent sweating during sporting activity. In this case, warp-knitted spacer fabrics can be used in mats instead of foam to provide high pressure resistance, high air permeability, efficient elasticity, thermoregulation, moisture absorption, chemical resistance, and high bending performance for good draping properties [30, 34]. A spacer fabric features a sandwich structure, consisting of

Table 15.4 Requirements of mats categorized by sporting activity [31, 33, 35]

Sport	*Requirements*	*Material*
Mixed martial art, striking combat sports (e.g., karate, taekwondo)	Slip resistance, avoidance of sinking, landing, striking – high energy absorption without stress level increase achieved by thickness of the mat, secure stance – higher density	Rebonded polyurethane foam, ethylene-vinyl acetate (EVA) or cross-linked polyethylene (XPE)
High jump, pole vault	Shock absorption - low density, high thickness	Polyurethane foam
Gymnastics	Landing, striking, stiffness – high energy absorption without stress level increase achieved by thickness of mat, adequate friction	EVA or rebonded polyurethane foam
Wrestling	Landing, striking	Polyvinyl chloride, nitrile rubber or rebonded polyurethane foam
Judo	Landing, striking – thinner, firmer mats	Polyethylene foam

Figure 15.2 Example of a spacer fabric with top and bottom layer in black and spacer layer illustrated through white spacer monofilaments.

a top, spacer, and bottom layer. The top and bottom layers are connected through the spacer layer with spacer monofilaments, which leads to a three-dimensional (3D) structure [34]. This three-dimensional structure provides the protection, buffering, and force limitation needed to decrease the collision between the body and the floor when using a fitness mat [34]. To achieve these structures, two needle-beds and six yarn guide bars are used to produce symmetrical lapping movements on warp knitting machines [32]. The upper and bottom layers are usually knitted either with plain structures or with meshes [32]. Sandwich structures like spacer fabrics are used when the requirements of a mat are too complex to meet with the basic construction of a foam mat. Sandwich structures allow the mat to integrate different layers, resulting in different combinations of requirements. The construction of a spacer fabric is exemplarily presented in Figure 15.2.

Future trends

There are some new inventions allowing mats to evolve into more innovative fitness products. For example, auxetic materials can potentially be used for better impact protection. Auxetic materials

have a negative Poisson's ratio, meaning they become thicker when stretched and thinner when compressed [36]. This negative Poisson's ratio behavior differentiates auxetics from most other common materials, which typically become thinner when stretched and thicker when compressed [36]. When applying auxetic foam to crash mats, such as those used in gymnastics and for the protection of skiers and snowboarders, it could lead to improved energy absorption and penetration resistance [36]. Currently, there are ongoing experiments involving the use of auxetic materials in crash mats to potentially utilize them in the future. Further details on auxetic materials can be found in Chapters 9 and 10 "Cellular solids for sports applications."

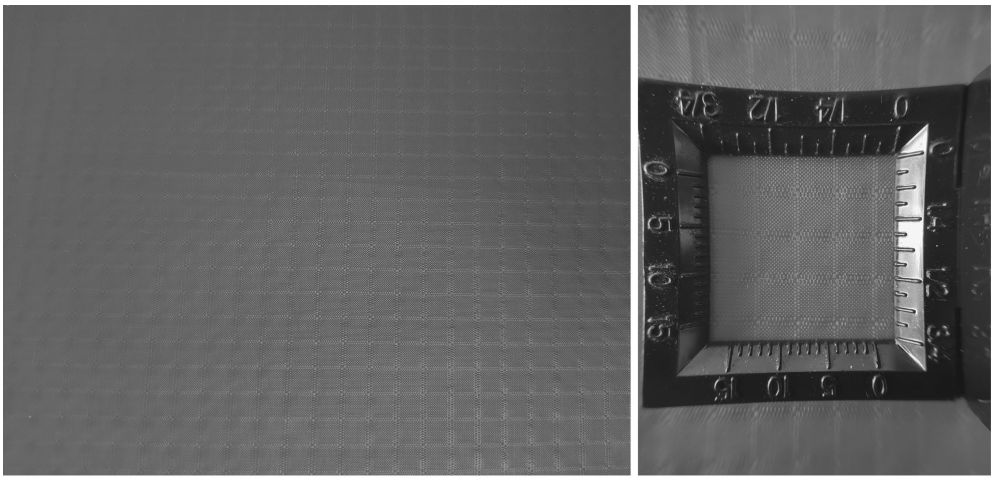
There are also new developments regarding smart textile mats with pressure sensor technology, which allow athletes to count and detect the exercises they perform [37]. This structure is made of conductive polymer fibers layered into a sheet, which is placed between conductive foil [37]. The integrated foil electrodes in the mat can recognize exercises by evaluating the volume resistance when pressed. These are just some examples of innovations in mats as sports equipment.

Tents

Textile tents can be utilized in diverse sports for different climate conditions and usages, as, for example, in camping and hiking. They are mostly used to protect against natural conditions and harms such as sunlight, wind, moisture, and heat, and they can provide shelter outdoors [38]. These functions directly relate to the required properties of the material of a tent, which can include flame retardancy, tensile strength, water resistance, breathability, weight, and moisture permeability, for example [38]. According to the application, tents can also be categorized by their UV resistance, color fastness, dirt repellency, weatherability, infrared reflectance durability, uniaxial and biaxial elongation, insulation, and their defense against insects and bacteria [38–40].

Components and functions of tent layers

Generally, a tent consists of sheets of fabric in a hemispheric order and the bar scaffold, which makes up the basic structure of the tent and provides support. The fabric in the form of a hemisphere also provides stability and minimizes the surface area [40]. Still, a tent can also be complemented by using ropes and stakes [38]. The bar scaffold usually consists of poles made from wood, synthetic material, metal, or fiberglass for either flexibility or rigidity. Synthetic materials are typically used in low-cost models, while aluminum and carbon fiber composites are used for designs of much higher quality. The fabric, on the other hand can be used to fulfill the established requirements by creating synergies through an inner and outer part. The outer layer is usually attached to the inner layer and provides water resistance through coating and film-generating substances to close any leftover space. Commonly used materials for the outer layer are strong and functional textiles such as polyamide or polyester [38]. Cotton canvas, once widely utilized but now scarcely employed due to its heaviness, can serve as an external fabric. In contemporary products, the introduction of "technical cotton" offers the advantage of enhanced breathability. This is attributed to the distinctive swelling characteristics exhibited by cotton fibers upon being wetted, resulting in self-closure of the material during rainfall. Ripstop structures, often used in outdoor applications, are also recommended. These structures have thicker yarns in both directions, resulting in a unique reinforcing technique that prevents further tearing and ripping and gives a high strength-to-weight ratio [41]. These yarns build a cross-hatch design in ripstop textiles as they are intertwined at periodic intervals [41]. An example of a ripstop structure can be seen in Figure 15.3.



(a) (b)

Figure 15.3 Example of a ripstop structure with crosshatch design. Authors' own image.

The inner layer of a tent does not require the same level of weather resistance as the outer one. This is because the inner layer is not directly exposed to climatic weather conditions. While the outer layer acts as a protective shield against rain, wind, and sunlight, the inner one serves primarily for comfort and privacy within the tent. Since the inner layer is not exposed to harsh weather conditions, it can be made of materials that focus on breathability, lightness, and comfort. Common materials for the inner layer include lightweight polyamide or polyester fabrics that allow for proper air circulation and moisture management. These materials are made into mesh-like structures that keep unwanted insects out of the interior while providing a more comfortable sleeping environment by reducing condensation and increasing breathability. The focus of the inner layer is to create a cozy and insulated space for occupants, providing a retreat from the external environment. It typically features mesh panels or windows for ventilation, allowing fresh air to flow while keeping insects out. The inner layer often also includes storage pockets and attachment points for gear organization and convenience. By differentiating the requirements for the outer and inner layers of the tent, manufacturers can optimize its performance and comfort. The outer layer withstands the elements, while the inner one provides a comfortable and protected space for users to rest and relax during their outdoor adventures [39].

Materials

The various materials typically used for both the outer and inner layers can prevent unwanted performance deficits. In general, the fabric needs to be lightweight to be easily transportable and strong enough to withstand weather conditions [39]. To achieve buckling resistance, coatings, laminations, and top coatings are usually utilized to provide fabrics with qualities that ensure protection [40]. Coatings are applied as liquids to the woven fabric. The most commonly used ones for tents are polyvinyl chloride, polytetrafluoroethylene, polyurethane, and silicone [40]. The adhesion values between the coating and base influence the tear strength properties. This is because tearing is a common reason for the rupturing of a fabric. However, since flame retardance, UV, mildew, and water resistance are also desired qualities in tents, chemical compounds are needed

to achieve these properties. To avert aging of the material, for example, polytetrafluoroethylene (PTFE) is commonly used instead of polyurethane, as the performance is better for preventing the bending of the material [38, 40]. Fluorocarbons are, however, increasingly being phased out because of their contribution to global warming and ozone depletion. This has led to challenges in finding suitable alternatives with lower environmental impact. Materials like hydrofluoroolefins and natural refrigerants are currently being researched and tested to replace traditional fluorocarbons. Furthermore, water resistance is achieved by coating the fabric with polymers such as polyurethane, polyvinyl chloride, and polyacrylonitrile [42]. Apart from providing water resistance, polyurethane, polyvinyl chloride, and polyacrylonitrile also provide UV resistance with low flexibility [39]. In this selection, polyurethane might be the best fit for achieving resistance to wet-flexing and cold-cracking while being lightweight [39]. Another option is silicone elastomer, as it shows efficient water repellency, elasticity, and UV and temperature resistance [39]. Disadvantages of silicone elastomer include its low fire-retardancy and the fact that it cannot be repaired with adhesive tapes since it coats the fabric on both sides. Another possibility apart from taping is sealing with solvents or thermoplastic silicones, which also works with silicone materials, or seam sealing with swelling yarn. Swelling yarns work by utilizing cotton yarn, which swells up when wet and closes the stitch hole in this way. Using core-spun yarns, made of both polyester and cotton, is also possible with this technique.

Future trends

New trends include siliconized tents with taped seams and hybrid coatings made of silicon and polyurethane, where no impregnation is needed, and the seams can be welded from the inside. Super lightweight materials and built-in light-emitting diodes (LEDs) to light the inside of the tent are other examples of future trends in tents.

Shoes

Sports shoes are manufactured to deliver support, performance, protection, stability, shock absorption, and comfort during physical activity and to protect against any kind of injury, making them an essential element in sports equipment [43, 44]. This is achieved by securing the feet in a way that is appropriate for different kinds of sports, as footwear for basketball, dancing, and running differs immensely in its construction.

General requirements

In general, sport footwear can be divided by the specific demands categorized in athletic sports, like hiking and running; court sports, like basketball; field sports, such as football; winter sports, like ice hockey; track and field sports; and outdoor sports, such as hunting, fishing, and parachuting [43]. The construction differs by the part of the shoe, like the upper and lower part and the insole, and is influenced by the anatomy, biomechanics, and functional demands of the athlete [43]. Readers wishing to learn more about how athletes interact with their footwear are referred to "Chapter 24 Biomechanical interactions with footwear". Functionality is also differentiated by the kind of sport. For example, for a runner, the shoe needs to provide support through shock absorption and a light and flexible feel, while a hiker needs support and stability through a thick and stiff outsole [43]. Moreover, court sports like basketball and volleyball, characterized by frequent lateral movements, place a significant emphasis on the need for footwear to provide optimal lateral

Table 15.5 Requirements in specific footwear [5, 43]

Sport	*Footwear requirements*
Long distance runner	Lightweight, flexibility, comfort, support, shock absorption, breathability
Outdoor activities	Support, non-water resorption, tight feel
Sprinting	Tight feel with maximum performance, breathability
Walking	Shock absorption, comfort, breathability, support
Winter sports	Ankle protection, thermal stability
Soccer	Balance, traction control, stability
Dance	Movement, stability
Court Sports	Lateral stability, torsional flexibility, and traction control, ankle support

stability, torsional flexibility, and reliable traction control [5]. A way to fulfill those needs is by providing a stable shank. The stable shank is achieved, for instance, by a high-top design in basketball shoes to provide ankle support. In contrast, research has also shown that the high-top design could increase shock transmission and reduce performance when running or jumping [5]. Volleyball shoes, on the other hand, are usually low-cut to enable the ability for repeated lunging and diving [5]. Another form of sports footwear can be found in dancing. Ballet dancers, for example, need at least the first three toes to be of the same length to prevent fractures and injuries [5]. Inside the toe box of the shoe, lamb wool is commonly used to provide protection because silicone and rubber hamper movement and fit [5]. Baseball and softball shoes, on the other hand, must provide many functional abilities since side-to-side movements, running, and fielding are needed when playing. This again shows the need for stability, even though the right foot may be more demanded in these sports [5]. When playing soccer, all movements in all kinds of directions are conceivable, with additional needed features for dribbling and ball control [5]. Soccer shoes, therefore, need to provide balance, traction, and stability, which is often achieved by using detachable cleats made of plastic or metal [5]. Fencing is another sporting discipline that requires special functions in its footwear. Since it is an asymmetrical discipline, different shoes for the right and left foot are even offered [45]. Some requirements for specific footwear can be found in Table 15.5.

To achieve these specific requirements, materials and construction are combined for maximum results. Flexibility is, for example, achieved by focusing on the joint area, while a light shoe is mostly influenced by the upper materials and less cushioning [43]. A good rolling motion is reached by the curved last, and water-resistant shoes are focused on the right seam allowances, closing techniques and materials [43].

Construction and functionalities

The construction of the different parts of the shoe allows for different functionalities. The upper, consisting of various individual components like the vamp, toe box, and foxing, is responsible for moisture management and the fit of the shoe [5]. The bottom, consisting of the midsole, wedge, and outsole, is responsible for shock attenuation and absorption, torsion rigidity, flexibility, dampening, cushioning, and stability. For the midsole, foam rubber with encapsulated air is often used due to its ability to absorb shock efficiently [43]. This is done by a chemical blowing agent that reacts with chemicals in the material, which eventually leads to gas production, forming small bubbles of trapped air that act as shock absorbers when the air cell walls deform for energy absorption [5]. Another possibility is the use of a physical blowing agent, which, contrary to chemical blowing agents, offers more environmental advantages, a consistent cellular structure, adjustable

material properties, and high machining efficiency [46]. They work by using thermoplastic polymers, which show increased flexibility and simple fabrication because of the micro-phase separation structure [46]. A physical crosslink network is established through crystalline hard segments, which are destructed and reconstructed under thermal treatment [46]. This results in thermoplastic polymers being able to foam just like plastics [46]. Instead of thermoplastic polymers, vulcanized elastomers with a low melting point can also be used. For this method, a pre-vulcanization process is needed before the actual physical foaming process, followed by a post-vulcanization step to further improve thermal stability and mechanical features [46]. Midsoles can also be of dual density to highlight support areas [5]. Wedges, on the other hand, are used for a natural roll motion of the foot to increase stability and reduce overpronation, while the outsole can influence stability and motion control drastically [5]. The design of the outsole also affects the function, such as waffle outsoles, which are used when running on dirt, while ripple soles can be utilized when running on asphalt [5]. The insole provides stability and flexibility as a solid base and is also responsible for impeding bacterial or fungal growth [5]. Furthermore, the sock liner, located between the foot and insole, can contribute to sweat transport, energy absorption, and comfort [5].

Materials

Materials also play an important role in preventing pilling and enabling resistance to various climate conditions. The variation of material in toe bumpers, backparts, tongues, and upper materials allows for stability, foot movement, and durability [43]. Examples of materials include Ethylene-Vinyl Acetate foams, which provide impact resistance and balance [44]. For the upper, either special open-pored synthetic fabrics because of their breathability, heat transfer, and washability, or leather, due to its water transmission ability, abrasion and tension resistance, and flexibility to the form of the foot, are often used [5].

Since new biogenic materials are gaining interest among the public, leather materials are slowly but continuously being replaced by vegan alternatives [47]. Examples include natural, animal-free fibrous materials like Muskin®, produced from mushroom fruit bodies [47]. The soft feel could make Muskin® a good substitute for leather, but since it has restricted availability, it will probably not be able to fully replace it. Another way to develop alternatives is by utilizing biotechnical procedures to manufacture fiber-based materials [47]. While micro-cellulosic structures are generated by bacteria, mycelium fiber structures of fungi hyphae are usually composed of chitin, cellulose, and proteoglycans growing on organic waste [47]. Another possibility for substituting leather is by supplanting synthetic constituents with agricultural waste, such as grain, apple pomace, or milled cactus leaves [47]. An example of another substitute is Pinatex®, where renewable fibers of pineapple leaves are transformed into non-woven materials that are laminated with polylactic acid derived from corn starch [47]. Some materials with their desired application fields are listed in Table 15.6.

Future trends

In the realm of sports shoe manufacturing, textiles play a pivotal role in shaping future trends. These trends include:

1 Sustainable textiles: With a growing emphasis on sustainability, manufacturers are turning to eco-friendly textile materials. Recycled fibers, organic cotton, and bio-based synthetics are being utilized to create more sustainable sports shoes [48].

Table 15.6 Materials used for different parts of an athletic shoe [5, 43–45]

Material for athletic footwear	*Part of the shoe*	*Property*
Polyamide	Upper	Resistant to permanent deformation, rapid drying capability, low air permeability
Polyamide mesh	Upper	Strength regardless of tightness of weave
Leather	Upper/reinforcement in upper	Formfitting, perspiration storing, water vapor transmission, tension withstand, abrasion resistance
EVA	Midsole	Cushioning, shock absorption, decreased shearing
Polyurethane	Midsole	Compression resistance, durability
Thermal plastic	Wedges	Stiffness, lightness
Blown rubber	Outsole	Lightness, softness, flexibility, cushioning
Carbon rubber	Outsole	Durability, lightness
EVA	Sock liner	Formfitting

2 Customized textile solutions: The demand for personalized sports shoes has led to advancements in custom textile solutions. Technologies such as 3D knitting, laser cutting, and robotics enable the production of customized shoe uppers, ensuring a better fit for individual athletes [49].
3 Lightweight textile innovations: Lightweight textiles are at the forefront of sports shoe design. Manufacturers are exploring the use of innovative textile materials, such as engineered mesh structures and lightweight synthetic fibers, to reduce the mass of the shoes without compromising performance [50].
4 Smart textiles integration: The integration of smart textiles into sports shoes is a burgeoning trend. Fabrics with embedded sensors can capture data on foot strike patterns, pressure distribution, and temperature, providing valuable insights for performance optimization and injury prevention [51].
5 Breathable and moisture-wicking fabrics: Textiles with enhanced breathability and moisture-wicking properties are highly sought after in sports shoe manufacturing. Mesh fabrics and moisture-wicking coatings are employed to ensure comfort and regulate foot temperature during physical activities [48].
6 Seamless construction techniques: Seamless construction techniques are gaining popularity in sports shoe manufacturing. Seamless knitting or bonding methods eliminate the need for stitching, reducing friction and enhancing overall comfort [3].
7 Barefoot-inspired textile designs: Inspired by the concept of barefoot running, textile designs for sports shoes aim to provide a natural and unrestricted feel. Flexible and breathable materials, combined with minimalist cushioning, simulate the sensation of running barefoot while still offering some protection and support [45].

These textile-driven trends demonstrate the importance of innovative and sustainable textile solutions in sports shoe manufacturing. As technology and material science continue to evolve, textiles will play a crucial role in creating high-performance, comfortable, and eco-friendly sports shoes for athletes.

Bandages and damping elements

Bandages and damping elements comprise a wide field of different parts of equipment with varied functions and versatile applications. Bandages can provide support for pain relief or compression

or be used to fixate joints. They can be described as aids that wrap around parts of the body or fit close to parts of the body and are usually ready-made [52]. Bandages are mostly used as orthopedic aids for soft tissue diseases and are therefore constructed out of flexible textile materials, which contrasts with orthoses made of elastic materials with stabilizing elements [52]. Types of bandages and damping elements include compression bandages, athletic tape made of adhesive fabric to secure joints and ligaments, kinesiology tape (similar to athletic tape but with the function of supporting muscles and improving circulation and recovery), gel pads, insoles and inserts to disperse impact loads using silicone or foam, and lastly, athletic socks. Due to the range of possible examples, this chapter will solely focus on the most researched items, which are compression bandages.

Performance requirements

Compression bandages are used for multiple performance requirements, such as the enhancement of the return of venous and arterial blood flow [53]. This is achieved by applying external pressure on the extremities when the muscle expands against the compression fabric, which eventually leads to a reduction in the diameter of the veins [35]. This reduction initiates a decrease in fluid leakage and an increase in the reabsorption of lymphatic fluid and the blood flow speed, which benefits intravascular pressure caused by certain vascular diseases, as well as physical sporting activity and performance through vasodilation [35, 4].

Construction

As for the construction and fabrication process, compression garments can be found as stockings, pneumatic devices, or bandages and are either multi-layered, overlapped, or stretched [4]. The material used also plays an important role in the performance of the compression garment. A heavier fabric with a higher fiber force of the elastic yarns, for instance, can exert more pressure than a lighter material [4]. The structure is influenced by the method of fabrication, such as weaving, circular or flat knitting, or non-woven production. All methods have different advantages and disadvantages. For example, a circular-knitted fabric can be lightweight and more elastic than a flat-knitted compression garment, which may be more fitted and stiffer, making it also less elastic [5]. This is because, in circular knitting much finer needles are used, which develop finer structures. Knitting, in general, allows for a smooth surface, extensibility, thermal conductivity, water-vapor permeability, flexibility, moisture absorption, and good recovery because of the structural stretch of knitted fabrics in combination with elastic yarns [4]. A woven compression bandage is more suitable for inelastic garments since it is stiff and has lower expandability and recovery function. This is due to the stretched lying threads, which only allow for expandability through material elongation. Non-woven compression bandages are mostly used as a cushioning element in a multi-layer compression product [4].

As well as the materials used, which influence almost all the properties of a textile, there are many structural parameters that enable the specific adjustment of properties in textiles. The thread count and density, for example, are the most important parameters for adjusting the mass per unit area of a textile. Surface formation processes and weave structures influence the strength and elasticity of textiles. Oven porosity affects a textile's moisture and air permeability, to name a few. It is then the task of textile engineers to adjust a desired property by selecting materials, manufacturing technologies, and parameters to fulfill the required function [4]. For bandages, the parameters of density, puncture resistance, elasticity, water vapor and air permeability, and strength are decisive. For special applications, multi-layer constructions are also used. An example of this is spacer

fabrics, which are used for the treatment of edema. This makes it possible to treat swellings of the body tissue due to fluid accumulation in the vascular system with compression bandages [4]. To achieve maximum elasticity, a combination of a base yarn for stiffness and thickness and an interlining yarn for tension can be used [4].

Materials

Synthetic materials are most often used to make compression bandages. These materials include polyamide, elastane, polypropylene, and Lycra [35]. A coating can also be applied to provide antibacterial and anti-odor properties, as well as functional wear properties, such as improved durability and comfort by reducing friction. Examples of coatings include zinc oxide and polyacrylic for adhesion [4].

Summary

This chapter delves into the innovative realm of textile sports equipment, exploring the technological advancements and applications of textile components in different sporting disciplines. It also highlights how textiles have revolutionized the design and safety of sports equipment across different sectors, enabling athletes to demonstrate their best performance. The chapter starts with a brief market overview of the sector, showing the rising demand for textile sports equipment and some emerging trends in the market, such as wearable sensors, smart technology integration, and the substitution of natural fibers with synthetic ones. With the presentation of emerging trends, it is shown how the advancements are poised to shape the future of sports equipment, enhancing athletes' performance, safety, and overall sporting experience. Moving on, the chapter continues to explore the integration of textiles in specific sports equipment categories. By outlining the usage of selected pieces of sporting equipment, such as footwear, fitness mats and ropes, the chapter gives a concise overview of the benefits of textiles in sports equipment, the various usage possibilities, and miscellaneous examples. The focus lies on the ability to not only enhance athletes' performance with textile sports equipment but also aid in injury prevention and post-workout recovery. This is achieved through requirements such as shock absorbability, impact protection, and damping.

Still, the field shows the ongoing need for continuous research to solve problems such as durability, safety, maintenance, and cost-effectiveness, to unlock the full potential of textiles in sports equipment. The challenge of durability can be seen in the difficulty of developing products that are both durable and long-lasting. This is mostly based on the materials used, and ongoing research focuses on investigating how textiles can withstand the demands of rigorous physical activity, abrasion, and impact. It is also challenging to achieve an acceptable balance between performance-enhancing properties and durability, as some high-performance textiles can sacrifice longevity. Proper maintenance in the form of cleaning and special care can impact longevity, performance, and durability. Cost-effectiveness also plays an important role in the challenges faced in textile sports equipment, as the various requirements can be rather costly. High-performance textiles, such as HMPE used in ropes, and advances in manufacturing techniques can also increase production costs. The trend of smart technology integration also shows the need for continuous research in this field, as seamless integration, reliable data collection, and user-friendly interfaces require interdisciplinary collaborations. This can be achieved by introducing collaboration between textile engineers and sports professionals and through ongoing rigorous testing. Addressing all these challenges requires continuous research and development, collaboration, and testing, which may result in the ability to push the boundaries of textile sports equipment.

The literature also shows the transformative impact of textile sports equipment on athletic performance, safety, and comfort. By embracing textile sports equipment, manufacturers can continue to push the boundaries of innovation and create a new era of cutting-edge gear for athletes worldwide.

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16
3D PRINTING IN SPORTS

Mike Vasquez

Introduction

3D printing, also known as additive manufacturing, is a process of creating a physical object from a digital design by building it up layer by layer (see Figure 16.1). This technology has the potential to revolutionize the way sporting goods are designed, manufactured, used, and potentially recycled [1]. There are various book chapters, review papers, and reports on 3D printing, including those related to sporting goods and consumer products [2–12], and this chapter summarizes the themes of these while drawing on the author's experience in the sector.

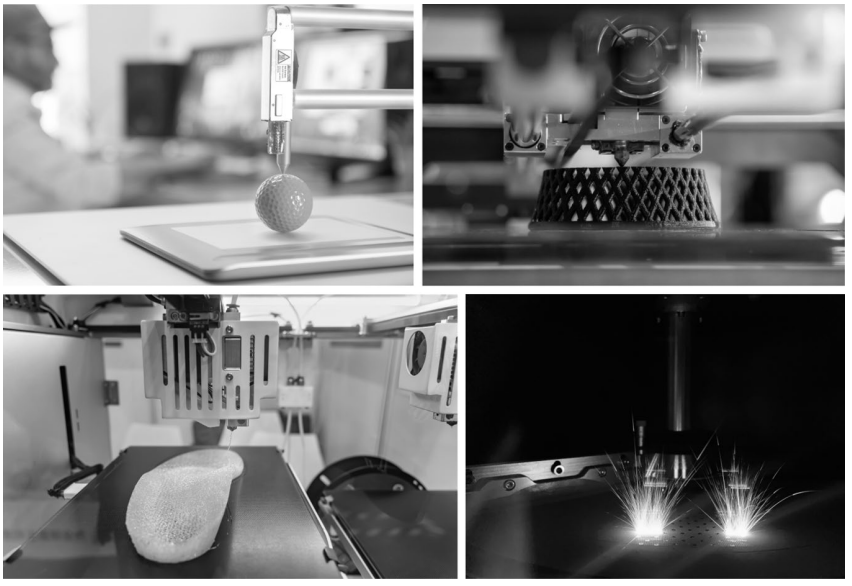

Figure 16.1 Examples of 3D printing, whereby an object is built up layer by layer.

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There are several unique types of 3D printing modalities; however, for the purposes of this chapter, we will focus primarily on the three listed below:

1 Material Extrusion: Material extrusion is a popular 3D printing technology in which a plastic filament is heated and extruded through a nozzle to build up layers of a 3D object. The process is like squeezing toothpaste through a tube or piping frosting on a cake. Material extrusion 3D printers are relatively affordable and are often used for prototyping and creating low-cost, low-strength parts.
2 Vat Polymerization: Vat polymerization 3D printers use a laser to cure a photopolymer resin layer by layer to form a 3D object. The process starts by creating a digital model of the object and then slicing it into thin layers. The printer then traces the shape of each layer onto a bed of liquid resin using a laser, hardening the resin in the desired pattern. These systems can produce high-resolution, accurate parts with a smooth surface finish. However, a drawback of this technology is that the parts can slowly degrade when exposed to ultraviolet (UV) light.
3 Powder Bed Fusion (PBF): Powder bed fusion 3D printing is a process in which a laser is used to fuse together small particles of metal, plastic, or ceramic powder to create a 3D object. The printer builds the object layer by layer by spreading a thin layer of powder and using the laser to fuse the particles together in the desired pattern. Selective laser sintering (SLS) 3D printers are often used for producing functional prototypes and low-volume production parts. This is the most common type of printing system used to construct engineering-grade materials such as Polyamide 11 (PA-11), Polyamide 12 (PA-12), Thermoplastic Polyurethane (TPU), Titanium, Stainless Steel, Aluminum, and Inconel.

Customization is a key advantage offered by 3D printing, allowing the creation of tailored sporting equipment intended to meet individual athletes' unique needs and preferences. Whether it is personalized sports shoes, mouthguards [13–16], or custom cleat plates for professional athletes or "weekend warriors," 3D printing could offer an edge in the industry.

Rapid prototyping is another area where 3D printing excels. This technology can reduce the time and cost associated with traditional prototyping methods, enabling designers to quickly iterate and test new sporting equipment designs. For such an approach to be useful, the material properties of the printed prototypes must be comparable to those obtained from the process that will be used to make the final product. Rapid prototyping has the potential to accelerate product development cycles, aligning with the fast-paced nature of the sporting goods market. The typical sporting goods company has two to four design/launch seasons, millions of products or stock-keeping units (SKUs), and innovation cycles that occur roughly every four years to coincide with major sporting events like the Olympics or the football World Cup.

Sustainability is a focus in modern manufacturing [17], and 3D printing aligns well with this ethos. With minimal waste generation and the potential for localized manufacturing, it can offer an eco-friendlier alternative. The exploration of sustainable materials derived from natural sources or post-consumer waste adds to the appeal. However, the research and development (R&D) required to develop sustainable materials is often lengthy. If a suitable sustainable material is identified that could effectively work in a 3D printing system, the costs to procure, convert, and stabilize it often lead to a less favorable economic situation.

Performance enhancement is a critical factor in sports, and 3D printing can play a role in creating sporting goods that outperform their traditionally manufactured counterparts. Golf clubs with precise weight distribution, bike frames [18] that are both lightweight and strong, and advanced prosthetics for athletes with disabilities [4,19] exemplify the technology's capabilities. Increased

accessibility to sports for individuals with disabilities is also a noble application of 3D printing. Customized prosthetics and adaptive equipment have opened doors for participation and competition, enhancing inclusivity in the sporting world. Finally, the repair and maintenance aspects demonstrate 3D printing's cost-effective and time-efficient ability to produce replacement parts for damaged equipment. It could reduce downtime and give sporting goods a longer life cycle, which can reduce the overall environmental impact.

3D printing has been used to create sporting goods for many years. One of the earliest examples is the use of 3D printing to create customized inline skates in the 1990s. Since then, the use of 3D printing in the industry has expanded to include various products, such as golf clubs, footwear, tennis rackets, bicycle frames, and more. One of the main reasons for using 3D printing in the sporting goods industry is the ability to create customized products tailored to the specific needs and preferences of individual athletes. For example, 3D printing can be used to create customized golf clubs with specific weight distributions or tennis rackets with specific stiffness characteristics. This level of customization is unfeasible with traditional manufacturing methods, which typically require the use of molds and dies that are difficult and expensive to modify. Not only can customization enhance capabilities, but designers can have better control of geometry (where material can go) with 3D printing.

The development of 3D printing, or additive manufacturing, is not unique to the sporting goods space. Starting in the early 1980s, the technology was developed to speed up the process of designing and manufacturing prototypes. Since then, several industries have accelerated their use of the technology. The biggest 3D printing end-use products by revenue are likely to vary depending on the specific market being considered. However, some of the most common end-use products for 3D printing are likely to include:

1 Medical devices and implants: The use of 3D printing in the healthcare industry has grown in recent years, with 3D-printed medical devices and implants being used for various applications [20], including prosthetics, hearing aids, and dental crowns [21,22].
2 Aerospace and defense: 3D printing is being used to create various products for the aerospace and defense industry, including aircraft and spacecraft parts, weapons components, and more.
3 Automotive: The automotive industry is increasingly using 3D printing to create various products, including parts, prototypes, and tooling.
4 Consumer products: 3D printing is being used to create various consumer products, including jewelry, toys, and home goods.
5 Industrial products: The use of 3D printing in the industrial sector is growing, with 3D printing being used to create various products, including machine parts, tools, and fixtures.

Figure 16.2 illustrates the evolution of 3D printing in sports. 3D printing technology emerged in the 1980s primarily for industrial prototyping purposes. While not directly applied to sports, the groundwork for 3D printing was laid during this period. In the 1990s, sporting goods manufacturers

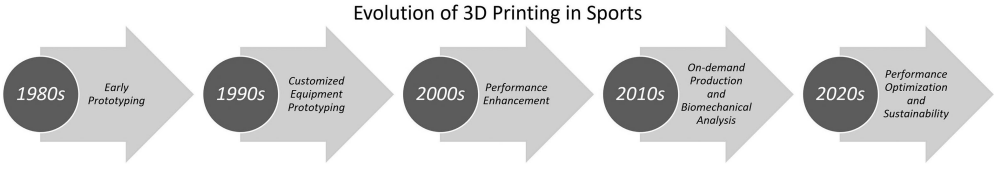

Figure 16.2 Diagram to illustrate the evolution of 3D printing in sports.

began to experiment with 3D printing for rapid prototyping of customized equipment such as helmets and shoe soles. Companies explored the potential for personalized gear tailored to individual athletes' needs. In the 2000s, advancements in 3D printing technology allowed the production of more complex and lightweight designs. Sports teams and athletes increasingly utilized 3D printing for creating performance-enhancing equipment, such as aerodynamic bike components for cyclists and customized orthotics for runners. 3D printing became more accessible in the 2010s, enabling on-demand production of sporting goods. Sports medicine professionals utilized 3D printing to create custom braces, splints, and orthopedic implants, improving athlete rehabilitation and injury prevention. Biomechanical analysis tools integrated with 3D printing enabled the creation of customized equipment optimized for individual athletes' biomechanics, such as ergonomic grips for tennis racquets or golf clubs.

Nowadays, in the 2020s, 3D printing plays a crucial role in pushing the boundaries of performance optimization in sports equipment. New 3D printing materials are leading to the creation of lightweight yet durable gear for various sports, including football helmets, racing car parts, and tennis racquets. The sports industry is increasingly adopting 3D printing for sustainable manufacturing practices, utilizing recycled materials and minimizing waste in the production process. Athletes are benefitting from highly personalized gear designed using advanced scanning and modeling techniques, leading to enhanced performance and injury prevention.

As well as customization, 3D printing has also been used to create sporting goods that are lighter, stronger, and more durable than their traditionally manufactured counterparts. For example, 3D-printed bicycle frames can be designed with complex internal structures that provide strength and stiffness where it is needed while minimizing weight. This has the potential to lead to better performance and a more enjoyable riding experience. Another benefit of using 3D printing for sporting goods is the ability to create prototypes and small batches of products quickly at a low cost. This can be especially useful for small businesses and startups that may not have the resources to invest in expensive tooling and equipment.

This chapter explores some of the ways in which 3D printing is being used in the sports industry. This includes customized equipment design, rapid prototyping, sustainability, performance improvement, increased accessibility, and repair and maintenance. Presented case studies demonstrate the current industrial direction of this technology and new materials.

As 3D printing has matured as a viable manufacturing technology, athletes and sports have been early adopters. Over the past several decades, new materials and manufacturing methods have emerged from industries such as aerospace and defense. Materials like titanium and carbon fiber composite manufacturing followed this pattern. These industries have large budgets and are looking to optimize for high strength and low weight, both of which are useful in sports performance. Unlike aerospace, where safety is critical, sporting goods industries typically have less stringent requirements and standards in this regard. This set of circumstances has enabled many sports technology companies, research institutions, and athletes to be highly innovative with 3D printing systems. The sporting goods industry makes an ideal test arena for new technologies due to the ease, accessibility, relatively loose requirements, and limited risk of testing [23].

Catering to the athlete: Customization using 3D printing

One of the main benefits of 3D printing is the ability to create customized products tailored to the specific needs and preferences of an individual. This can be especially useful in the sports industry, where a piece of equipment tailored to a particular athlete can give them a competitive edge. For example, 3D printing has the potential to create customized sports shoes designed to fit an athlete's

feet perfectly or to create customized mouthguards tailored to the shape of an athlete's mouth. 3D printing can also be used to make sports and sporting goods more accessible to people with disabilities. It could be used to create customized prosthetics for athletes with amputations or to develop specialized equipment for athletes with disabilities, making it easier for them to participate in sports..

The globalization of manufacturing over the last 30 years has favored mass production over customization. Mass-producing a part using conventional tools like machining or injection molding has been optimized for profits. 3D printing and associated advanced production techniques are allowing companies to rethink how they may reach new markets or create enhanced value for their consumer base.

There are many factors to consider when examining the potential benefits of customizing equipment for users. To do so, it is useful to split the customer or user into two categories. First is the high-performance or professional athlete who is sponsored and looking to gain an edge over other professionals. The other end of the spectrum is the "weekend warrior" or the average person who utilizes sporting goods and generates revenue for most equipment brands through their purchases of footwear, equipment, or apparel.

An example of this dichotomy comes from the footwear space. The ability to print custom insoles or midsoles to improve performance and limit overall waste has been a target for 3D printing almost since its inception. In 2013, Nike (Beaverton, USA) did a short-run release of the Vapor Laser Talon American football cleat. The entire cleat plate of the shoe was 3D printed. The geometry of the design meant it could not be produced by conventional injection molding approaches. The design of the cleats was specifically tailored to the artificial turf surface where the athletes were competing. Factors such as weight, cleat depth, position, and stiffness were designed to yield the best results. However, while this approach showed feasibility, the method used to make the parts was slow, labor-intensive, and utilized high-cost materials. Specifically, for these shoes, the SLS technology was used, which takes engineering polymers like PA-12, PA-11, polypropylene, or TPU in powder form and consolidates parts through melting with a laser. For small volumes of parts, this approach can be cost-effective (~1,000 parts depending on size). However, producing parts in the tens or hundreds of thousands with SLS requires a highly specialized use case. Figure 16.3 illustrates economic factors that limit the use of 3D printing technologies.

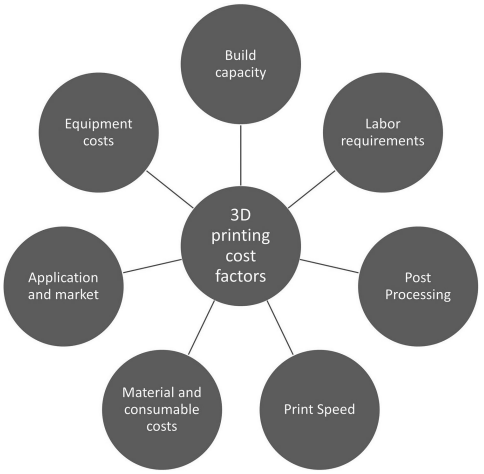

Figure 16.3 Economic factors that influence 3D printing. Build capacity is defined as the number of parts that can be fitted in a machine.

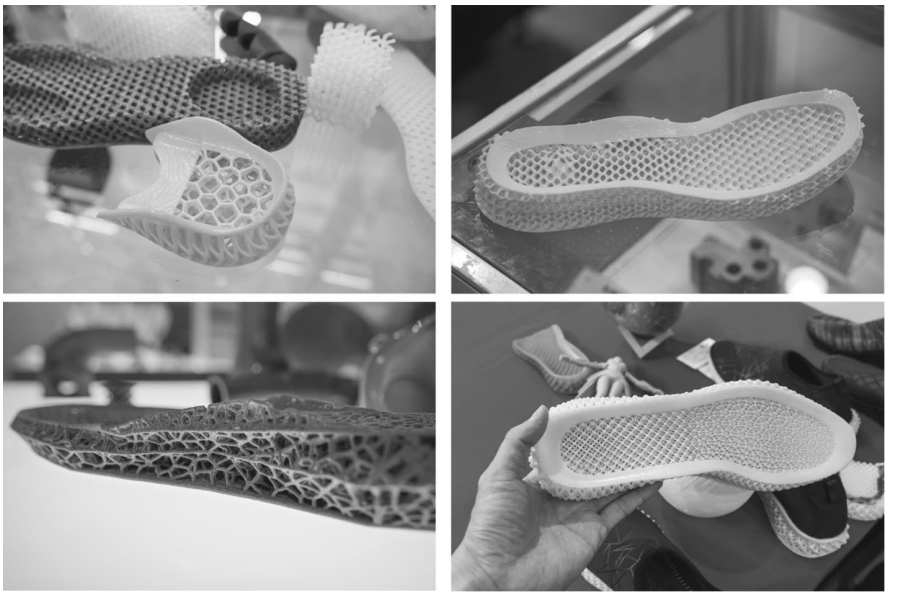

Figure 16.4 Examples of 3D printed shoe midsoles with lattice structures.

Nevertheless, there are many aspects to footwear. While custom cleat plates may not have scaled to a mass market due to costs and investment requirements, other manufacturers have leveraged 3D printing for consumer-grade products. The most notable of these is Adidas' (Herzogenaurach, Germany) use of 3D printing with the Carbon Inc. (Redwood City, CA, USA) system to produce shoe midsoles. Carbon Inc. uses an offshoot of vat polymerization technology to produce flexible components. These midsole parts have complicated lattice structures (e.g., Figure 16.4) that could not be manufactured through conventional tooling-based approaches like injection molding. The company demonstrated the ability to produce shoes using 3D printing and included the technology in their end-to-end manufacturing workflow to construct the entire product.

The business case for this approach was different from what Nike was exploring with their cleat plates. For Adidas, they were using 3D printing to create complex geometries that could not be produced with conventional methods at higher production volumes (not necessarily customized). The process and learnings involved would ultimately lead to future capabilities showcasing customization leveraging the digital underpinnings of 3D printing. As one of the first widely produced 3D-printed products, the price (~$300) was higher than most conventional shoes due to materials, processing, and post-processing costs.

Today, many footwear brands are looking for methods to integrate 3D printing into both elite and high-street applications. Significant challenges remain that constrain the technology's adoption and, more importantly, the business justification for these types of manufacturing methods. Critical on this path is the development of materials that must perform, pass quality requirements, and be economical. The materials currently available for 3D printing applications are about a tenth of what is available in conventional manufacturing processes. The properties of printed materials can also be inferior to those made via conventional manufacturing, depending on the printing quality.

Rapid prototyping: De-risking the manufacturing process

Another benefit of 3D printing is the speed at which it allows for the creation of prototypes [24,25]. In the sports industry, this can be used to quickly test new designs for equipment and make any necessary adjustments before moving into mass production. This can help reduce the time and cost associated with traditional prototyping methods and allow for more rapid innovation and experimentation. The availability of more sophisticated materials at lower costs can also help engineers closely mimic form, feel, and fit to a degree that is realistic compared to what is likely to be produced with a traditional manufacturing process such as injection molding.

An advantage of using 3D printing for prototyping is its speed and efficiency. Traditional prototyping methods, such as machining or molding, can be slow and require specialized equipment and materials. In contrast, 3D printing allows users to go from digital design to physical prototype in a matter of hours or even minutes, depending on the size and complexity of the part. This fast turnaround time allows designers to iterate and test their designs more quickly, potentially leading to faster product development cycles.

Specific to the sporting goods industry, most large brands produce new products on an annual basis to keep up with trends, align with global sporting events, and stay ahead of their competition. The planning for a new product launch often takes place two to three years in anticipation of the projected launch date. If a company can cut down the iteration time for product development using 3D printing versus a typically longer lead-time technology, like molding or machining, products can come to market faster and with more of the ”bugs” worked out.

With traditional prototyping methods, designers are often limited by the materials and processes available. In contrast, 3D printing allows designers to choose from a wide range of materials, including plastics, metals, and ceramics, and to create complex geometries that may not be possible with other manufacturing methods [26], e.g., Figures 16.4 and 16.5. This flexibility allows designers to create prototypes that can be useful for testing designs. It is important for designers and developers who use 3D printing to prototype to maintain awareness of the limitations of traditional manufacturing methods in their designs. For example, while features like undercuts can be printed, issues can arise when these designs are transferred to mass production with traditional methods.

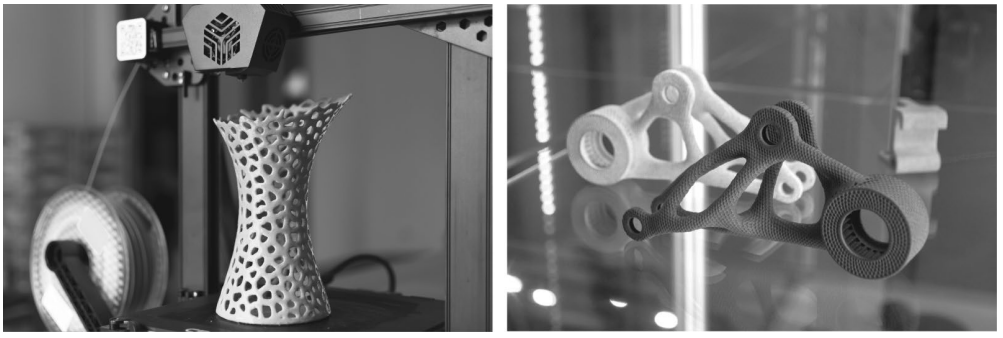

Figure 16.5 Examples of 3D printed geometries that may not be possible with other manufacturing methods.

For most products, 3D printing is not going to be cost-effective (see Figure 16.3) or have adequate scalability to be the solution for final manufacturing. Nevertheless, the variety of materials, textures, and design freedoms can mimic traditional methods. Conventional prototyping methods can be expensive, especially for small quantities or one-off prototypes. In contrast, 3D printing can allow designers to create prototypes at lower costs, making it an attractive option for small businesses and startups with limited budgets.

Overall, 3D printing has revolutionized the prototyping process by offering designers and engineers a fast, flexible, and cost-effective way to create physical models of their designs. It has also been useful for universities in developing testing fixtures, new equipment rigs, and human body surrogates for testing protective devices, e.g., [27–32]. Its speed, versatility, low cost, and sustainability make it an attractive option for prototyping various products, from consumer goods to aerospace and medical devices. As the technology continues to evolve and improve, it is likely to play an increasingly important role in the product development process.

Environmental sustainability

One of the major advantages of 3D printing is the ability to create products with minimal waste. Traditional manufacturing methods often require large volumes of raw materials, which can result in significant waste during production [17]. 3D printing, on the other hand, allows for the creation of products with minimal waste, as the material is added layer by layer and only the necessary amount is used. This can help reduce the environmental impact of sporting goods manufacturing and make the process more sustainable.

Many companies, including sporting goods manufacturers, have realized there are profitable business models associated with the idea of environmental sustainability [17]. This has come alongside regulations in major business areas that require more stringent processes to assess the environmental impact of a product from the start of its production to its ultimate end-of-life. Readers seeking information on the sustainability of 3D printing should explore research published by the Additive Manufacturing Green Trade Association [33].

The distributed manufacturing model that 3D printing enables allows companies to rethink how and where they want to produce their products. Globalization over the past 30 years has resulted in a race to the bottom for low-cost manufacturing labor, with stops in China, Vietnam, and the Philippines, as sporting goods manufacturers seek to optimize business objectives. For the right product, 3D printing can be a more localized manufacturing solution that reduces the amount of labor required and brings production closer to the customer, thereby saving on transportation emissions. Companies such as Wilson Sporting Goods (Illinois, US) are experimenting with this approach, using Azul 3D's (Illinois, US) vat polymerization technology to produce pickleball rackets.

Some of the most exciting work associated with 3D printing is the exploration of new materials that can unlock enhanced properties and lower the carbon footprint. Several material companies, such as Arkema (Colombes, France), have developed raw 3D printing materials from natural sources like castor oil and have used post-consumer waste as a precursor to new feedstock. However, 3D printing is not without limitations when it comes to sustainability. While the layer by layer process does limit the amount of waste compared to subtractive technologies, there are still waste streams from unsintered powders or liquid resin residues. When working with natural, biobased or recycled materials, it can also be challenging to maintain consistency.

Repair and maintenance

3D printing technology could also be used to repair and maintain sporting goods. If a piece of equipment becomes damaged, 3D printing could be used to create a replacement part matching the original. This could help reduce the cost and time associated with repairing or replacing equipment and extend the product's life. This application has most often been seen in vehicle-based sports such as cycling or motor vehicle racing. For example, if a bike pedal becomes damaged, a 3D-printed replacement part could be created to match the original. In most cases, the end-user would not likely print their replacement parts at home (although low-cost printers are becoming more capable of producing finished products). Instead, local hubs, printer farms, or maintenance offshoots of the originating company would provide the 3D-printed replacement.

Several organizations are investigating the ability to digitize entire product replacement outfits. The current business model for industries like automotive involves physically producing and warehousing spare parts until needed. This process could take years to manifest as a sale, but in the meantime, the associated costs are liabilities. Instead, companies could house a computer-aided design (CAD) file of the geometry on the cloud or a local server and print the part as needed.

Industry examples

There are several examples of sports equipment created using 3D printing technology. Here are a few:

Golf Clubs: 3D printing has been used to create customized golf clubs with more precise weight distribution and improved aerodynamics. These clubs are designed to help golfers hit the ball farther and more accurately. For example, Snarr3D Inc. (Texas, USA) specializes in 3D-printed golf club shafts, and Cobra Golf (California, USA) offers 3D-printed putters.

Bicycle Frames: 3D printing has also been used to create lightweight and strong bike frames designed to improve performance (e.g., [18,34]). These frames are often made using composite materials that can be printed in various shapes and sizes to meet the rider's specific needs.

Prosthetics: 3D printing has been used to create customized prosthetics for athletes with amputations, allowing them to participate in sports at the highest level. These prosthetics are designed to be lightweight, strong, and tailored to the athlete's specific needs.

Mouthguards: 3D printing has been used to create customized mouthguards tailored to the shape of an athlete's mouth. These mouthguards can provide better protection and comfort than traditional ones.

Protective Equipment: Beyond mouthguards, 3D printing has been used to create other types of protective equipment, such as shin guards [24] and shoulder pads. These items can be designed to fit the athlete's specific needs, potentially offerring improved protection and comfort. For example, EOS GmbH, a 3D printing manufacturer based in Germany, developed Digital Foam technology (printed material with foam-like properties) and partnered with companies such as Bauer (manufacturer of ice hockey equipment based in the USA) to make ice hockey helmets. Similarly, Carbon Inc. has worked with Riddell Sports Group (Illinois, USA) to produce American football helmets.

While many of these examples are moving towards industrialization at larger production levels, many brands have used 3D printing technology as an opportunity to further their marketing by making promotional products or highly specific components to gain "buzz" or establish a perception of their use of "cutting-edge" technology.

Future of 3D printing in sporting goods

The 3D printing industry has grown significantly over the last decade into a multi-billion-dollar enterprise. This growth is being driven by several industries, including sporting goods. Today, most of the industry remains focused on applications to enhance prototyping, and continuous developments are pushing the technology towards production applications. New materials, lower costs, and faster systems will enable sporting goods designers and manufacturers to supplement their existing manufacturing portfolios with these new technologies, delivering increased benefits to their customers and athletes. Emerging technologies, such as Composite-Based Additive Manufacturing (CBAM) from Impossible Objects (Illinois, USA), are poised to decrease the overall production time of printed components. Once significant gains in cost, speed, and overall ease of production are achieved, the market and opportunities for 3D-printed components will continue to grow.

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SUSTAINABLE DESIGN AND MANUFACTURING OF SPORTS EQUIPMENT

Martino Colonna, Filippo Biagi, Alessandro Nanni, Maria Federica Parisi, Lorenzo Crosetta, Giuseppe La Fauci, and Daniel Colombo

Introduction

Before going into detail on sports equipment sustainable manufacturing and design, it is important to define what sustainability is and how it affects the sports industry. A general definition has been provided by the United Nations Commission in the Brundtland Report in 1987 [1], which defines sustainable development as development that:

> *meets the needs of the present without compromising the ability of future generations to meet their own needs.*

Going into more detail on material and process sustainability, an important definition given by the Oxford English Dictionary regards the depletion of natural resources and defines a sustainable process as one that:

> *is able to be maintained or continued while avoiding the long-term depletion of natural resources.*

More broadly, sustainability can be defined as:

> *the capacity to maintain or improve the state and availability of desirable materials or conditions over the long term.*

not just focusing on natural resources [2]. However, not only environmental sustainability should be taken into consideration, but also other factors (i.e., social and economic) that affect people's lives.

The principle of the *Three Pillars of Sustainability* defines that for complete sustainability, all three pillars must be sustainable [3]. The three pillars are social sustainability, environmental sustainability, and economic sustainability, also known in a simplified way as people, planet, and profit. The three pillars are all important to achieve complete sustainability and are all interconnected.

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Thus, sustainable production should take into consideration not only the use of natural resources, but also the amount of energy used, the well-being of humankind, and should not create social and economic problems.

These aspects do not only apply to sports equipment manufacturing but to all types of manufacturing. However, one important point to consider is the market in which sports equipment, a mass product, is aimed. Indeed, most of the users of sports equipment perform their sports to have a great experience, often in an open and wild environment. This likely makes them more sensitive to the effect of their actions on the environment. A clear example is people performing winter sports, in which the effect of pollution and climate change has a great effect on the possibility of performing their sports activities. For this reason, in several countries, non-profit organizations such as Protect Our Winters (POW) [4] are growing, and people are looking to purchase more sustainable products, even if they are more expensive than non-sustainable ones [5]. On the other hand, governments are creating new regulations to push producers to take sustainability into consideration, particularly the end-of-life of products. This approach was first applied to packaging but now is applied to other sectors such as textiles and shoes. One important example is the new European Community rule regarding extended producer responsibility (EPR) that defines the producers' responsibility for the end-of-life of their products [6]. According to the EPR approach, producers must demonstrate that their products can be recycled at the end of their use and propose feasible processes for recycling.

Going into more detail on the aspects that affect sports equipment sustainability, we can divide them into three main categories related to the materials used, production processes, and the end-of-life scenario.

Regarding the effect of materials, the most important actions to improve sustainability are:

- Lower the amount of material used: reduced material in the product and waste throughout the manufacturing process; less material equates to higher sustainability.
- Use of natural resources to decrease the production of greenhouse gases.
- Improve the durability of the equipment and the possibility of repair or reuse: the longer the use, the less virgin materials must be employed.
- Use of recycled materials that do not require the same amount of energy to be made as virgin materials.

The use of recycled materials is strongly connected to their end-of-life. Indeed, all materials, and in particular the plastic materials that are often used in sports equipment, have a limited lifetime due to degradation reactions caused by use and environmental aging [7]. Therefore, the materials reach a point where they no longer provide the performance and safety characteristics that permit their use [8]. However, sports equipment has not traditionally been designed to be recycled; rather, the main considerations have been cost, performance, and safety. Therefore, even if each material that composes the equipment could, in principle, be recycled, most equipment cannot be recycled since the different materials are glued or molded together and cannot be separated. For this reason, sports equipment is nearly always disposed of in landfills or incinerated [9]. Another important issue to be considered is that plastics, at their end-of-life, especially if not correctly disposed of, generate microplastics during the degradation that occurs after their use due to environmental aging [10].

Regarding the sustainability of the production processes, several points must be taken into consideration:

- Amount of energy and water required during the process.
- Amount of waste generated during the process and not re-used.

- Use of safe and environmentally benign substances, including solvents.
- Production location, in terms of distance from the location of selling/use.
- Workers' conditions.

In the following paragraphs the aspects that affect the sustainability of sports equipment will be described in detail, along with examples of best practices and potential solutions. Since plastics are the most common materials for producing sports equipment, the focus on sustainability will be mainly on this type of material. However, the same approach can be applied to other materials such as metals, ceramics, wood, etc. This chapter covers the sustainability of sports equipment in general, and readers seeking a more specific example are referred to Chapter 18, Cricket Gear and Sustainability.

Sustainable materials

As previously stated, raw materials and energy are necessary to produce sports equipment. Therefore, using less material is an important first step to be taken by sport equipment producers seeking to improve the sustainability of their products. This can be achieved by product optimization, through means such as finite element analysis and topological optimization [11]. Another important aspect to be considered during the design of sports equipment is the selection of plastics that have a low environmental impact in their production. This depends on the manufacturing process of the plastic, but also on the distance between the manufacturing site of the sports equipment and the production plant of the plastic.

Another important way to improve the sustainability of polymeric materials is the use of renewable resources as raw materials for their production. First, it is important to stress the difference between the terms *bioplastic* and *biobased plastic*. Bioplastics are defined by the European Bioplastic Association as plastics either derived from renewable resources or that are biodegradable (and of course, those that feature both properties) [12] (Figure 17.1). Therefore, not only biodegradable plastic can be considered bioplastics, but also biobased and non-biodegradable ones. Biobased materials are defined by the fraction of their carbon content that is sourced from biological materials or agricultural resources, rather than from fossil carbon [13]. Biobased content is measured following the procedures set by ASTM D6866 and EN 16785-1:2015 using the ^{14}C radiocarbon analysis [14,15]. However, the norms do not define the minimum biobased content. For this reason, it is important that sports equipment manufacturers do not just claim the biobased source but also the percentage of biobased material.

A biodegradable plastic is a plastic that undergoes biodegradation in the environment due to the action of naturally occurring microorganisms such as bacteria, fungi, and algae [16]. Composting is the biodegradation of a polymer in a controlled environment that degrades the material, converting it into at least 90% CO_2 and disintegration into pieces smaller than 2 mm in a defined amount of time, thus producing compost that can be used as a fertilizer to grow plants. Composting can be divided into home composting (temperature of 20–30°C and degradation time of 90% degradation of 12 months) and industrial composting (temperature of 58°C and 6 months to reach 90% degradation). The accepted industry standard specifications are ASTM D6400, ASTM D6868, ASTM D7081, and EN 13432 [17,18]. Composting should not be considered a recycling path for plastic materials, since the compost generated by the degradation of plastic is unlikely to be used to grow plants that will then be used to produce bioplastics. It must also be stressed that compostable materials are not always biodegradable under conditions different from those occurring in home and industrial composting, since the degradation process depends on the environment

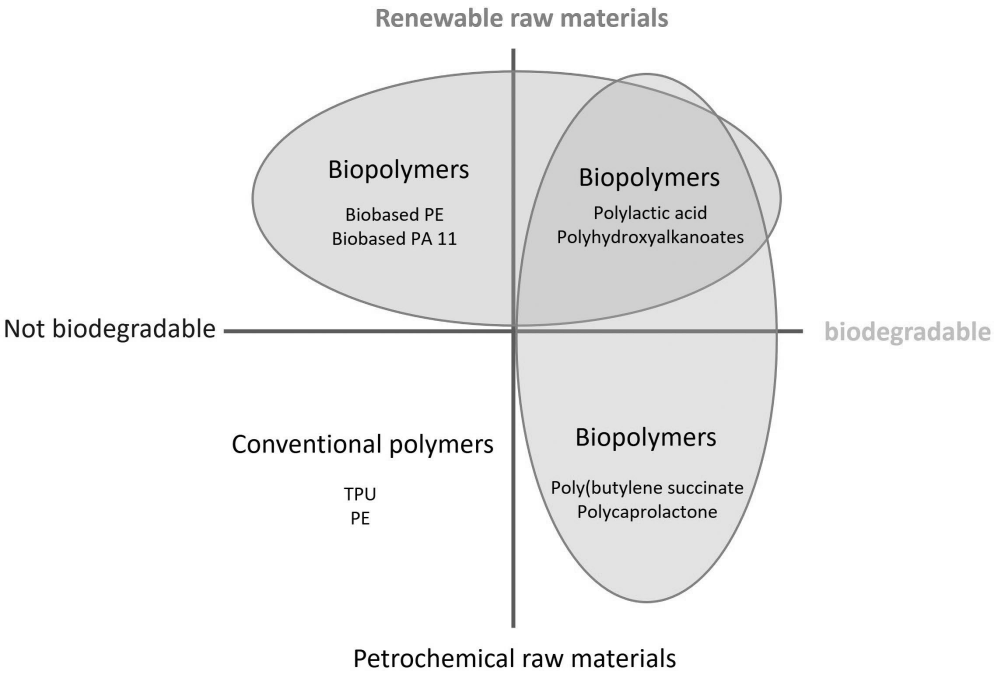

Figure 17.1 Classes of biopolymers according to the European Bioplastic Association.

in which it occurs. As an example, biodegradation is much slower in marine and freshwater environments, and in most cases, compostable plastics require a very long time (up to 500 years) to reach disintegration and degradation in natural environments [19]. It must also be considered that during degradation, the additives and catalysts present in the plastic are released into the environment, thus creating potential problems.

In general, biodegradable polymers have a lower durability than non-biodegradable ones [20]. This is because their chemical structure can be more easily degraded, not only after their use in composting facilities but also in less harsh conditions, such as those encountered during the use and storage of sports equipment [16]. Moreover, biodegradable polymers are more prone than non-biodegradable ones to produce microplastics during their end-of-life disposal [21]. In addition, biodegradable materials in several cases cannot be mechanically recycled due to degradation reactions that occur during the recycling process [16] and in some cases, they create problems in the recycling of non-biodegradable plastics, as in the case of the presence of poly(lactic acid) (PLA) in poly(ethylene terephthalate) (PET) recycling streams [22].

For these reasons, biobased non-biodegradable polymers are the preferred choice as biopolymers for producing durable sports equipment and, in general, for durable goods. This is also indicated by their higher production (five times, according to the European Bioplastic Association [23]) compared to biodegradable plastics. The use of biobased polymers helps to reduce the production of greenhouse gases since the plants used as biomass to produce the biopolymers have captured the CO_2 in recent times and converted it into organic matter through photosynthesis [13]. However, the production of biobased polymers is not carbon neutral, since the conversion of the biomass into intermediates and monomers, the polymerization process, and the transportation of the final materials all require energy consumption [24]. Nevertheless, in most cases, the use of

renewable resources helps to reduce the carbon footprint. An important example is Polyamide 11, which can be obtained from castor oil [25]. Polyamide 11 and its co-polymers (e.g., Pebax®) are widely used in sports equipment manufacturing due to the combination of lightweight, stiffness, impact resistance, and elastic rebound, making them the material of choice for the outer parts of high-end ski-mountaineering boots [8] and for the foam of running shoe midsoles [26]. 11-aminoundecanoic acid (the monomer used for the synthesis of polyamide 11) is obtained from castor oil by cracking reactions [27]. The process was developed over 80 years ago in France, but it has gained industrial importance in the last decades due to the increasing awareness of sustainability. Life-cycle assessment (LCA) studies have shown that the use of castor oil reduces the energy required to produce the plastic by 29% and CO_2 emissions by 32% compared to the same polymer obtained from fossil resources [28].

In the case of biobased Polyethylene (PE), considering the complete supply chain, LCA analysis shows that the use of renewable resources as feedstock (in this case, sugar cane) removes up to 2.15 metric tons of CO_2 from the atmosphere for each ton produced [29]. Calculations of the amount of land needed to produce biobased PE evaluated that with a hectare of land, it is possible to produce 82.5 tons of sugarcane, which can be converted into 7,200 liters of ethanol, which in turn affords three tons of ethylene and therefore three tons of biobased PE [29].

Even if the actual production of biopolymers currently represents under 2% of the overall plastic production [30], a significant surface area of land is clearly needed to produce the chemical substances for producing bio-based plastics. However, it has been calculated that only 2% of the global agricultural area has been used to produce materials (e.g., wood, natural fibers, etc.) and 1% for biofuels [30]. Going into more detail, the percentage of arable land used for bioplastic production is only 0.02% [30]. Nevertheless, there has been criticism toward the use of arable land to produce bioplastics because of its potential competition with food and animal feed [30]. Indeed, the first-generation feedstocks used to produce bioplastics were crops and plants rich in carbohydrates that could be consumed by humans and animals. Nevertheless, first-generation feedstocks are the most efficient to produce bioplastics as they require less land to grow and have a higher yield and efficiency than other feedstock generations. Examples of first-generation feedstocks are corn, wheat, sugarcane, potato, sugar beet, rice, and plant oil. Even if the criticism was more aimed at the biofuels sector than the bioplastics sector, the bioplastics market was also affected. For this reason, second-generation feedstocks have been used to replace the first-generation ones. The second-generation feedstocks are crops and plants unsuitable for human (food) or animal (feed) consumption and can be either non-food crops (cellulosic feedstock) or waste materials from first-generation feedstocks (e.g., waste vegetable oil) [30]. One practical example is the Pragati project that has been implemented in India for the use of non-arable land to grow castor plants used to produce the castor oil, which is then employed to produce Polyamide 11 [27]. The use of agricultural wastes is becoming increasingly important to produce building blocks and monomers to produce polymers and composite materials. An example is the use of wine production wastes for producing bio-fillers to produce biobased-polymer composites [31]. Incentive plans, such as the one developed by Truterra, LLC (Minnesota, United States) to help farmers switch to sustainable processes, should be supported by governments.

Another important approach to reducing the carbon footprint and energy consumption of polymeric materials is the use of recycled materials. Indeed, in the case of recycled materials, the conversion of petrochemical sources or natural feedstocks into chemicals, monomers, and then polymers is unnecessary. However, it must be taken into consideration that waste collection, separation of different materials, and re-processing require significant energy consumption. Therefore,

as described in detail in the following paragraphs, the recycling process (chemical, physical, mechanical, etc.) makes a significant difference in the sustainability of recycled materials.

Production processes

The first simple and important consideration about the sustainability of the production process regards the amount of material that ends up directly in the final product. Of course, as will be reported in the following, the scraps of the process can be mechanically recycled during the same process. However, the recycling process (grinding, drying, injection molding, etc.) requires energy, and thus, even if the use of recycled material is a way to improve sustainability, the best approach is to minimize the production of processing scraps. This can be done by optimizing the process setup and decreasing the number of final models in terms of color and type. Indeed, a change in materials, colors, and molds requires a transient period before getting to the steady state needed for consistent production. Generally, all the materials produced in the transient period must be discarded since they have intermediate properties, thus increasing the waste of the overall process. When scraps are produced, they should be recycled directly in the process through primary recycling. In primary recycling, the waste generated is ground in the same plant where the final items are produced, and then, after drying, re-processed by injection or compression molding. The advantage of this type of recycling is that the materials do not need to be transported and are well-known in terms of characteristics, compositions, and processing conditions. For this reason, this method provides lower environmental impact and lower costs than other recycling methods. However, during the process (particularly injection and compression molding), degradation reactions can occur due to thermo-oxidative processes [32], causing a decrease in the polymer chain length and thus of the thermo-mechanical properties of the material. This makes the polymers more brittle thus giving rise to mechanical failures of the sports equipment. For this reason, for most polymers, recycled material is used in combination with virgin materials. The ratio between recycled and virgin materials depends on the thermal stability and mechanical properties of the polymer. Nevertheless, approximately 20% of recycled material can be used for most polymers [8]. In some cases, all recycled material can be used. As an example, the ski-boot producer SCARPA® (Asolo, Italy) has commercialized over 2,000 ski boots made entirely of pre-consumer scraps of Polyamide 11 produced during the standard injection molding processes [33]. Mechanical and safety tests (according to ISO 527 tensile tests and ISO 179 impact tests) have shown these boots made of recycled material have the properties necessary for their commercialization. The LCA analysis conducted according to the ISO 14004 norm has shown that primary recycling can decrease the carbon footprint of the final product by 27% compared to the same boot made of virgin materials [33].

One limitation of primary recycling concerns the color of the final product. If the recycled material is not separated by color, or it does not have the same color as the virgin material, or both, the final product can have a non-constant appearance. For this reason, during primary recycling, a black pigment is often added to obtain a homogeneous color that covers all the other colors present in the material to be recycled. This limits the aesthetic possibilities of new products containing primarily recycled material. For this reason, it is suggested to educate consumers about the potential environmental benefits of choosing products with fewer color options and with darker colors.

Partially (e.g., foams) or fully (rubbers and composites) crosslinked materials cannot be mechanically recycled and re-entered into the production process since they do not completely melt and therefore cannot be directly re-processed [34] by standard processes such as injection and compression molding. For this reason, they must be recycled in other ways (e.g., by

devulcanization for rubber parts), typically outside the manufacturing plant. The same problem exists for multi-material products that have already been assembled as will be described in the following paragraph on design for recycling.

The energy used during the production process also has an important impact on overall sustainability. For this reason, it is important to perform processes that require low temperatures and, in general, low energy use. This can be achieved by using materials with lower processing (e.g., melting, pressing, curing, etc.) temperatures and systems that provide the best use of energy (e.g., by insulating the processing machines). Another important aspect to be considered is how the energy is produced. As an example, several ski producers use renewable resources (e.g., wood) to produce the heat and the steam necessary for the compression molding presses used for ski manufacturing [35].

The type of processes and their degree of automation also influences the sustainability of the overall process, not only in terms of environmental sustainability but also in consideration of the other two pillars (social and economic). As an example, processes that require a high level of manpower (e.g., the stitching of soccer balls or the sewing of shoes) are generally performed in third-world countries [36], where the cost of workers is lower. However, this creates problems with transportation costs and with working conditions, which should be ethical, responsible, and safe.

Another important aspect that affects the sustainability of sports equipment is the use of solvents or toxic chemicals in the production process. As an example, the production of bisphenol polycarbonate, a material widely used to produce helmets and eyewear [37], has been performed in the past using chloroform as a (toxic) solvent for the interfacial polymerization and phosgene as a (toxic) reagent [38]. The polymerization is now performed in a melt-polymerization process without the use of solvents and toxic reagents [39]. Toxic monomers (e.g., isocyanates to produce polyurethane foams) [40] or expanding gases (e.g., Azodicarbonamide used as a blowing agent, which is now considered toxic in some countries [39]) have been widely used to produce foams. New rules have been introduced in recent years to decrease the use of toxic chemicals, for example, the solvents (particularly dimethylformamide) used for the preparation of elastane fibers (Lycra®), which have been progressively banned in European countries [41].

Recycling processes

The current most important environmental challenge for sports equipment producers is the enormous amount of waste generated at the end-of-life phase [5], with most equipment being disposed of in landfills or incinerated. While for some industrial and manufacturing sectors the product recycling process is already in an advanced state [42], for sports equipment there is still a degree of uncertainty and lack of knowledge. In particular, the disassembly and recycling of products is difficult and expensive due to the multi-material assembly generally used for sports equipment, which has not typically been designed with consideration for its end-of-life. There is a well-defined hierarchy in the sustainability of waste that must be followed to decrease the environmental impact of the end-of-life of a product (Figure 17.2). The best practices to reduce the environmental impact are, of course, prevention (i.e., keep using the product) and reuse. However, this is not always possible, since equipment can degrade and become damaged with use, as is the case with running shoes. Therefore, other options are needed.

The best approach for the recovery of end-of-life products is mechanical recycling, which consists of the grinding, drying, and re-processing of the material. However, as already stated, during the re-processing, the molecular weight of the polymer can decrease due to thermo-oxidative

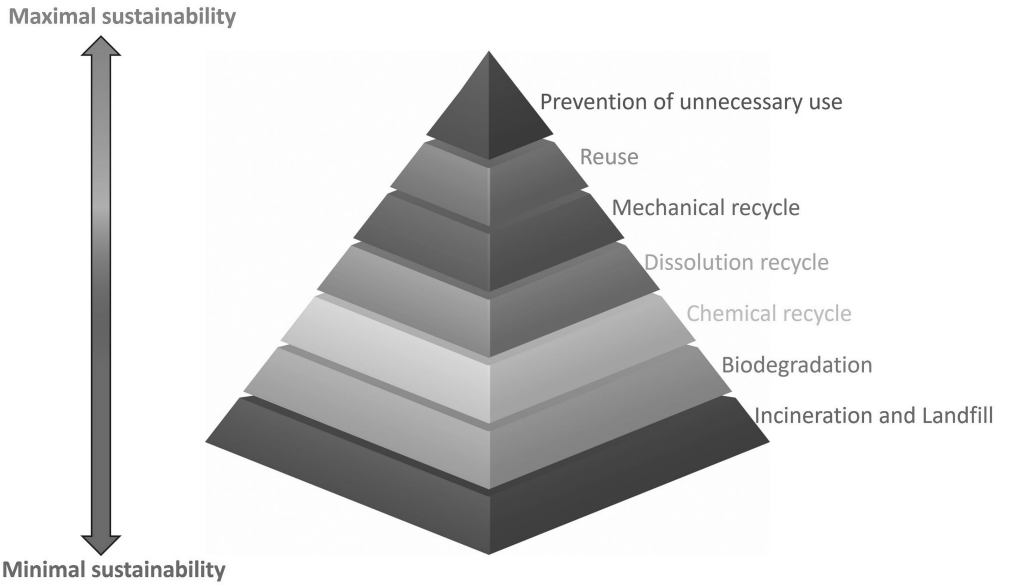

Figure 17.2 End-of-life hierarchy.

side-reactions [32], thus decreasing the mechanical properties of the final material. We have recently evaluated the effect of mechanical recycling on the mechanical properties of thermoplastic polyurethanes (TPU) used to produce sports equipment [43]. We found that materials coming from new equipment can be recycled up to three times without losing performance and safety characteristics. On the other hand, when TPUs from end-of-life equipment have been recycled, the properties tend to worsen after the second or third recycling cycle, depending on the time of use of the equipment [43]. This difference has been ascribed to degradation reactions that occur in the thermoplastic materials during the use of the equipment [43]. For these reasons, the properties of mechanically recycled polymers depend on the conditions of the material to be recycled and its time of use. Adding some virgin material can partially mitigate this effect [43].

In cases where the properties of the recycled material do not fulfill the requirements in terms of mechanical and safety properties necessary for the commercialization of the equipment, it can be used for other applications that have less stringent requirements. In this case, recycling is more correctly called down-cycling, since the end-use has a lower technical or economic value or both [44].

Mechanical recycling is often unfeasible due to the presence of different materials that cannot be disassembled and recycled together, due to the non-miscibility of the polymers in the melt or the presence of components that are crosslinked and cannot be melted. This problem can be mitigated by using sorting technologies that separate materials based on chemical composition, color, and density by means of UV-NIR, triboelectric, and gravimetric separations [45].

If the sorting of the different parts is unfeasible by separation techniques (e.g., in the case of multi-injected materials or glued materials) the best option is dissolution recycling. In this type of recycling, a solvent is used to selectively dissolve one or more materials that are separated from the others by filtration [46]. This approach differs substantially from the chemical recycling described below, since the macromolecular structure of the material is unaffected and therefore

the polymer can be simply recovered by solvent evaporation or precipitation in a non-solvent [44]. Therefore, the mechanical properties of the polymers are unaffected, and no re-polymerization is needed. One example of this type of approach is the recycling of helmets by solvolysis using a natural solvent called limonene [47]. This molecule is the main component of citrus fruit skin. It can selectively dissolve the expanded polystyrene (EPS) of helmets, which can be recovered by precipitation in ethanol and re-used for the preparation of new EPS for helmets [47]. The acrylonitrile butadiene styrene (ABS) polymer that often composes the external part of the helmet can be selectively dissolved in another bio-based solvent (ethyl acetate), which can be recovered by solvent evaporation [47]. This approach cannot be applied to polymers that cannot be dissolved in any solvent and has a drawback related to the boiling temperature of the solvent. The higher the boiling point, the more energy is required for its evaporation. When the boiling temperature is high, it is possible to use a non-solvent that causes polymer precipitation. The resulting mixture of solvents (solvent and non-solvent) can be separated by distillation.

Chemical recycling consists of the chemical degradation of the polymer structure (e.g., by hydrolysis) into smaller molecules (monomers or intermediates) that can then be purified and re-polymerized into new materials [48]. However, this type of process requires much more energy than mechanical and dissolution recycling, as both the de-polymerization and re-polymerization of the material are energy-consuming processes that occur at high temperatures (often exceeding 200°C) [49]. Therefore, this type of process should be used only in cases where the previously described recycling processes cannot be applied. One case is the recycling of partially cross-linked foams that cannot be recycled by mechanical or dissolution recycling, as it is impossible to re-melt and re-form the materials due to the crosslinks [49].

Another option that can be followed for the recycling of foams or crosslinked material is the agglomeration process [50]. This process consists of adding a binding agent to foams that have been ground. Binding agents can be liquid polyurethane pre-polymers or bi-component fibers. In the first case, the ground material is mixed with about 10% liquid pre-polymer and then heated and pressed in a cubic mold to obtain new foams with a minimum density of 70 kg/m^3, which is higher than that of the starting materials [50]. Moreover, the mechanical properties (stiffness, compression set, etc.) are lower than those of the virgin materials. This approach has been used, for example, for the recycling of foams of ski boot liners, to produce new liners [51] or protection mattresses [52]. This type of approach has also been widely used for the recycling of shoe soles and rubber parts of sports equipment with a down-cycling approach, to produce sport grounds and playground surfaces [53].

Rubber parts have also been recycled by grinding as infill for artificial turfs [54]. However, this approach generates a high amount of microplastics, since during the use of the turf, a significant quantity of ground rubber is lost in the environment because the particles are not bonded together or to the turf [54]. A more efficient approach for the recycling of vulcanized rubbers is the devulcanization process. This process consists of the partial degradation of sulfur cross-links by thermo-mechanical or chemical reactions [55]. The obtained material can be mixed with virgin rubber (up to 40%) and re-vulcanized to obtain new rubber parts, for example, for sport shoe outsoles [56].

In the case of sport equipment containing composite materials, the recycling path depends on the type of polymer matrix used for the composite. Readers seeking more detail on the use of composite materials in sports equipment are referred to Chapter 11, Fibre-Reinforced Materials for Sports Applications. In the case of thermoplastic matrices, the composite can be recycled by efficient and simple mechanical recycling in the form of grinding followed by injection molding [55]. The main drawback of this type of recycling is the decrease in the length of the fibers due to the

mechanical action of the grinder [57]. In the case of thermoset resins, the main routes for recycling are the combustion of the polymer matrix for the recovery of the fibers or the chemical recycling of the matrix [57]. The most used process is now the recovery of the fibers by combustion of the matrix, which is completely lost [57].

The least sustainable options for end-of-life sports equipment are energy recovery by incineration and disposal in landfills (which is the worst-case scenario), both from economic and environmental viewpoints, even if they are performed according to strict requirements that define the amount of released toxic substances during incineration and landfill.

As a final comment on recycling, biodegradation cannot be considered a form of recycling. This is because the biodegradation of material ends in the production of compost, which is unlikely to be used for cultivating crops that are used for producing chemicals that, in turn, are used for producing plastics. Moreover, the complete cycle to recreate the polymeric material requires lots of water and energy for growing the crops, for converting the biomass into chemicals, and for the polymerization of the chemicals to produce the new polymers. This makes this approach less sustainable compared to mechanical and even chemical recycling routes [16].

Design for recycling

Most sports equipment is currently designed to achieve high performance, safety, lightness, good aesthetic appearance, and low cost, without taking into consideration the recycling process at its end-of-life. Moreover, durability and the possibility of repairing are, in most cases, not key factors considered by producers. This approach, of course, does not favor the recycling and repair of the equipment. A change in mindset is necessary to improve the sustainability of sports equipment, making it possible to recycle at end-of-life and enabling the possibility of repair to extend the lifetime of products.

The first step to achieving easier recycling is to decrease the amount and number of materials employed. Indeed, more material means more energy needed for recycling. Moreover, according to previous paragraphs, the higher the number of materials and parts, the more difficult recycling becomes. Recycling is more complex for materials that cannot be recycled together. However, even when using materials that can be recycled together (e.g., polymers with the same chemical composition, but with different mechanical properties, or miscible polymers with different chemical compositions), the mixed material obtained will likely have properties that are worse than those of the individual materials. To mitigate this problem, the design should aim at decreasing the number of parts (especially if their primary purpose is aesthetic), with the final aim being a mono-material product. In cases where different materials must be used to achieve performance and safety, their choice should be made with consideration for the possibility of recycling them together. If this is unachievable, the design should take into consideration the possibility of easily disassembling the different parts. This can be achieved by connecting parts with removable connections (e.g., screws) or thermo-reversible glues that allow them to be separated. This also has an important effect on the lifetime of products, since broken or worn parts can be substituted, permitting the re-use of the equipment. This approach is also called design for repair. Mechanical metamaterials could be used to achieve the required properties by modifying the 3D structure in different zones of the equipment, thus achieving different properties while using one material. Readers seeking more information on mechanical metamaterials are referred to Chapters 9 and 10 on Cellular Solids for Sports Applications.

In general, thermoplastic materials should be preferred over thermoset or partially crosslinked materials due to their easier and more efficient recycling. However, when thermoset materials

(e.g., vulcanized rubber and thermoset composites) or partially cross-linked materials (e.g., polyurethane (PU) and ethylene-vinyl acetate (EVA) foams) must be used, they should be easily separated from thermoplastic materials.

Conclusions

Sustainability is a major topic in almost all industrial and social sectors and is becoming increasingly important in the field of sports equipment. The situation is rapidly changing due to the increase in end-user's sensitivity and new rules that push producers to consider sustainability and the end-of-life scenario. Therefore, producers are trying to explore different ways to improve their sustainability by using more sustainable materials (bio-based or recycled, or both), more sustainable processes, and developing end-of-life recycling processes. One of the main problems that producers and recycling companies face is related to the multi-material assembly of equipment that has not been designed to be recycled. Therefore, efforts are now being made to design new equipment that can be recycled at its end-of-life. However, the recycling and separation technologies have not yet been fully developed or exploited, and significant advances are expected in the next few years. To aid separation and disassembly, it would be important for equipment producers to provide the composition of the parts that make up the equipment, following the example of the Tecnica group (Giavera del Montello, Italy), which has put a QR code on their ski boots with information on the composition and recycling potential of its different parts. The presence of this type of information can also help with tracking and logistics, which are difficult tasks in the overall recycling process. Indeed, the collection and recovery of end-of-life products still create several issues due to the complexity of the permits needed to move the end-of-life products, which are considered waste, and the associated costs of transporting the products to recycling facilities. A more efficient logistic process and a simpler way to obtain permits would improve the overall efficiency of the recycling process.

Another main problem is connected to the cost of more sustainable materials and processes, which are generally more expensive than traditional ones. For example, the use of bio-based materials is generally more expensive than petrol-based materials. However, due to the increase in the production scale of bio-based materials, the optimization of the production process, and the rising costs of petrol-based materials, a decrease in the price difference between bio-based and non-bio-based materials is expected.

It would be important for governing bodies to support the use of recycled or bio-based materials through laws, as they are doing in other sectors such as automotive and packaging, by increasing taxes for companies that do not use recycled and bio-based materials. Moreover, an extended producer responsibility (EPR) scheme, holding the producer responsible for the end-of-life of products, should be implemented, as in other industrial sectors.

As a final comment, many producers lack existing knowledge of sustainability, and several cases of "greenwashing" claims have been observed in advertisements of "sustainable" products without any scientific evidence of the real advantages in terms of sustainability. In some cases, sustainability logos have not been supported by precise scientific methods and are often double-checked and verified by the same bodies providing the sustainability logos. For this reason, a more scientific approach should be enforced to avoid claims that have no real effect [58]. This can be achieved by providing results and making claims only if fully supported by LCA analysis using standardized norms. However, more clarity and rules should be developed to perform more reliable LCA analyses that are based on real data from production and recycling processes and not on secondary data taken from databases, which can lead to different results if performed using different databases.

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18

CRICKET GEAR AND SUSTAINABILITY

Martin Charter, Tom Clark, and Lilian Sanchez-Moreno

Introduction

As the world's second most popular sport, with over 300 million people playing the game at some level [1], there is little published research into the sustainability of cricket gear (bats, balls and protective equipment), considerable volumes of which are needed to support global growth in participation. The global growth of cricket is particularly related to its status as the number one sport in India and other Asian countries, as well as efforts by the International Cricket Council (ICC) and others to promote the game in non-traditional countries and increase female participation. The Centre for Sustainable Design® (CfSD) at the University for the Creative Arts (UCA) has addressed this gap through the establishment of the Platform for the Acceleration of Sustainable Innovation in Cricket (PASIC) [2], founded in 2021 to conduct research and innovation projects covering sustainability and cricket gear, as well as cricket clothing and apparel. Underpinned by the research undertaken within PASIC, which highlighted important sustainability issues for the cricket gear sector, the following chapter aims to highlight opportunities for sustainable innovation, including the development of circular cricket gear. Readers interested in sustainability considerations in general are referred to Chapter 17, Sustainable Design and Manufacturing of Sport Equipment. The chapter focuses on England and Wales while considering the global context. To understand the sector and sustainability context, the chapter provides an overview of cricket participation, the structure and governance of the game at professional and recreational levels, standards for cricket gear, the market for gear, and the production and supply chain. Participation is the primary driver of demand for cricket gear, while the game's structure and governance influence its use and the potential for innovation, and markets and supply influence sustainability issues in consumption and production.

While not directly within the scope of this chapter, social issues are a consideration in cricket sustainability. The focus of cricket organisations worldwide has been on increasing participation and inclusivity, considered a vital part of the game's development. In its landmark report, Holding up a Mirror to Cricket: A Report by the Independent Commission for Equity in Cricket (2023) [3], the Independent Commission for Equity in Cricket (ICEC) highlighted affordability and access to gear as important considerations for a gear-intensive sport and a significant barrier to playing cricket in the UK. Such issues have been highlighted in other reports disseminated via PASIC, including discussions on the 'golden triangle' that explores the links between environmental,

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economic, and social sustainability issues [2]. Lastly, the chapter also offers insight into the lack of data on cricket participation, cricket gear markets, production and consumption, and life cycle sustainability impacts, including materials use and waste.

Contextual issues

Changes in the game

At the higher level, cricket has changed from being a traditional, heritage-based game to one that is highly commercialised and professionalised, with new management and business structures. In 2005, cricket's institutional focus moved from London to Dubai with the establishment of the ICC headquarters there. India is now the key centre of influence through the Indian Cricket Board (BCCI), with the Indian Premier League (IPL) emerging as arguably the world's most important cricket tournament. New formats, especially T20, have emerged (in 2003 in England and 2007 internationally) to attract spectators, leading to global growth in participation outside traditional playing countries. Perhaps the biggest single change in recent years has been the growth in female participation. While these developments show that cricket is open to change, it remains conservative in many respects, particularly in relation to cricket gear design and development which, compared to other sports, has seen little product innovation throughout the game's history. To contextualise this, the lawn tennis racket, for example, transitioned from the use of wood when it originated in the 1870s to the use of metal frames in the late 1960s, and by the 1970s, manufacturers were experimenting with fibre-reinforced composites, aluminium, and steel [4]. Since the 1980s, high-end tennis rackets have been made from fibre-reinforced composite materials such as fibreglass and carbon fibre, while the shape and materials used to create the cricket bat have remained unchanged since the mid-eighteenth century [4]. As discussed in more detail below, one reason for this is the Marylebone Cricket Club (MCC) Laws and other British Standards relating to gear and gameplay, which present challenges for change and innovation in a game where the willow/leather and bat/ball dynamic is part of its essence. However, as this chapter shows, while innovation for cricket bats and balls was found to be constrained by MCC Laws, opportunities for innovation were identified for other gear, specifically for batting pads. With regard to batting gloves, while the Laws do not constrain the materials used, suitable alternative materials were not identified due to performance and functional requirements. Recommendations for further research and material development for gloves are offered throughout.

Structure and governance

Structure and governance can impact participation, the use of cricket gear, sustainability performance [5], and potential openness to sustainable innovation.

The following highlights key organisations and groups related to professional and recreational cricket globally and in England and Wales.

ICC

The ICC plays a prominent role in governing and developing the international game, which is highly commercialised, for example, through sponsorship, broadcasting rights, and the organisation and governance of cricket's major international tournaments, most notably the Cricket World Cup, T20 World Cup, and ICC World Test Championship.

MCC

Since 1788, the MCC has been the keeper of the Laws of the Game, including laws on cricket gear specifications (see below), which, for example, specify wood for bats and leather for balls. These requirements apply to the professional game and to recreational cricket played within the Laws. The Laws do not apply to recreational cricket played outside the Laws, such as junior cricket, where plastic bats and soft balls, generally produced from a combination of polyester and rubber or a synthetic compound, are often used. In practice, most recreational cricket is played within the Laws, and most gear produced is aligned with the Laws.

ECB

The England and Wales Cricket Board (ECB) is an influential national member of the ICC and governs all aspects of cricket played in England and Wales, both professional (players paid to play) and recreational, through 39 County Cricket Club Boards (18 First-Class Counties and 21 National, previously Minor Counties). There are an estimated 6,500 ECB-affiliated clubs, to which the ECB provides support. About 500 clubs in England and Wales are not affiliated with the ECB.

Apart from an ECB estimate of around 6,500 affiliated clubs, no data was obtained on the number of clubs, teams, and adult players at these various levels. Table 18.1 shows the general structure and composition of cricket in England and Wales.

Table 18.1 Structure and composition of cricket in England and Wales

Level of play	*Structure and composition*
Adult	
Professional/semi-professional	County and National Counties clubs, leagues, and championships T20 The Hundred and other limited over competitions Professional and semi-professional players Men's and women's cricket
Recreational – upper level	ECB Premier League ECB and non-ECB-affiliated clubs Saturday leagues and competitions
Recreational – friendly, formal	ECB and non-ECB-affiliated clubs Sunday leagues
Recreational – friendly, informal	'Park' cricket Indoor cricket
Junior	
Upper level – non-school	County and National County junior teams and competitions ECB-affiliated and other club junior teams and competitions. Boys and girls
Private schools	Over 12 and under 12 (prep) schools Inter and intra school competition Boys and girls
State schools	Over 12 and under 12 schools Inter and intra school competition Boys and girls
Informal	'Park' cricket

Regulations and standards

Various regulations and standards apply to cricket gear. In the context of this chapter, the main regulatory body that influences the potential opportunities for sustainable innovation in relation to materials and dimensions of cricket gear is the MCC Laws that govern cricket gear [6], as well as the rules of play. Law 4 specifies the size and mass of the ball. Law 5 and Appendix B specify the size and composition of the bat: the blade of the bat must be made of wood, and the handle must be principally cane or wood. Appendix E sets restrictions on the size and design of the gloves used by the wicketkeeper. However, within MCC Laws, there is potential scope for change. An example of this is the discussions generated at MCC over the use of a bamboo cricket bat developed by researchers at the University of Cambridge [7]. Bamboo is a grass and not a wood, and therefore, at present, is not permitted for bat making. However, MCC stated in 2021 that they would discuss the topic at the next Laws sub-committee meeting [8]. Additionally, other standards and regulations that should be considered are the ICC regulations on clothing and equipment, as well as those set by the British Standards Institution (BSI), the British Safety Industry Federation (BSIF), and industry-led standards.

In this sense, the ICC Regulations mainly specify the use of colours and logos on clothing and equipment in televised ICC international matches [9]. The ICC permits the use of cricket gear as a vehicle for advertising (e.g., adding manufacturers' logos on bats); however, it specifies that 50% or more of gear and clothing in white-ball internationals 'must be white, or the same colour as the base colour of the relevant team's playing shirt' [10].

In 2019, Gray-Nicolls (East Sussex, England) made a first attempt at addressing sustainability issues related to the production of cricket gear by developing a pair of batting gloves made from off-cuts from the manufacturing process. The gloves were worn by England's T20 vice-captain, Sam Billings, in a televised game. However, the gloves were banned because they contravened ICC regulations by using multiple colours [11]. This means that while these off-cut batting gloves were banned from international use, they can still be used within the recreational game. Procuring all-white off-cuts could enable the use of pre-consumer waste for cricket gear that adheres to ICC Regulations for televised international games, thereby reducing the gloves' overall environmental impact (e.g., reduction in waste and material usage).

There is a range of British Standards related to cricket helmets, balls, gloves, and pads, but not bats. The ICC defers to MCC on cricket gear specifications but requires helmets for international matches to conform to BS7928:2013. The BSI has published a series of technical standards related to the quality and performance of cricket balls and personal protective equipment (PPE) [12]: BS5993:1994 is the specification for leather-covered cricket balls; BS6183:2001 is a series of specifications on protective equipment for cricketers, including pads and gloves; and BS 7928:2013+A1:2019 (Amendment) is the specification for head protectors for cricketers.

In addition to the British Standards, on 21 April 2018, the BSIF highlighted in a press release that under Regulation (EU) 2016/425—commonly referred to as the PPE Regulation [13]—all cricket PPE (including helmets, gloves, pads and other guards) is subject to Class II Regulations, requiring independent testing under UK Certification Authority requirements [14]. Under previous regulations, cricket gear 'softs' (e.g., gloves and pads) were classed as Class I and were subject to self-declaration on testing and other requirements. In this context, there are implications for the cost and availability of third-party cricket gear placed on the market, as well as for repaired and refurbished cricket PPE if resold, as these may need to be retested to comply with existing testing requirements [14]. Currently, enforcement of adherence to Class ll Regulations is through local trading standards bodies.

The main industry-led regulation relates to the quality of willow used for cricket bats, developed from the craft of bat-making. Various companies have developed their own standards for bats and other gear. For example, JS Wright (Chelmsford, UK), which supplies 95% of English bat willow [15], identifies five main grades for both English and Kashmir willow used in cricket bat blades according to grain width, quantity, and other features such as the appearance of redwood, blemishes, pin knots, or visible 'specks'. Top-end cricket bats (Grade 1) offer minimal appearance of redwood, the grain on the face is straight, and they contain a minimum of six visible grains. Meanwhile, Grade 4 blades contain only four grains, and the willow will include imperfections such as 'butterfly stain', attributed to pruning and frost damage [16]. While Kashmir willow was originally planted from English willow, different climatic and growing conditions result in differences in the wood, making it harder and drier, thus affecting its coefficient of restitution (COR), defined as the ratio of relative speed of the objects after and before collision [17]. A previous study has shown that the COR of the bat decreased as the bat stiffness increased [18].

The cricket gear market and industry

The cricket gear market appears to be fragmented with many suppliers, including traditional cricket specialists such as Gunn and Moore (GM) (Nottingham, England), major Indian and other specialists such as Sanspareils Greenland (SG) (Uttar Pradesh), global sports brands such as Puma (Herzogenaurach, Germany), and many smaller suppliers of bats, other gear, and clothing.

Obtaining data and information on the cricket gear industry and market is challenging, reflecting a general lack of transparency in the sector. There is no cricket gear industry organisation or other cricket body compiling such information in England and Wales, and there is a general lack of import statistics for cricket gear. Equally, access to data related to the volume of sporting goods being produced and disposed of is a challenge across the entire sports sector [19].

Customer markets

The cricket gear market in England and Wales can be segmented in various ways: by gear type, by quality (which is material dependent), and by age/size and sex, as indicated in Table 18.2. No data was obtained on the market mix.

Gear type

As previously mentioned, the research [2,5,25,27,28,33,43,45,47] that underpins this chapter focused on cricket gear used by individuals to play the game, e.g., bats, balls, gloves, pads, and

Table 18.2 Cricket gear markets in England and Wales

Gear market segment	*Categories*
Product type	Bats, balls, batting and wicket-keeping gloves, batting, and wicket keeping gloves, helmets
Grade/quality	Higher quality brands/products versus cheaper, especially in the case of bats. Many suppliers offer multiple grades
Customer types	Individuals (professionals, club recreational, leisure/social), associations, clubs, schools
Age/size	Adults, junior
Sex	Male, female

helmets, the main types of gear used by individual players, while recognising that there are other items of individual gear, such as arm and thigh guards, footwear, and other cricket equipment used more broadly to practise and play games, such as practice nets and scoring boards.

Grade/quality/price

As noted in the section on standards above, there are various grades of cricket gear rated by quality/price. This is especially true for bats, with premium pricing for Grade 1+ English Willow bats (referred to as players' grade) used by high-profile brands. In general, higher-quality gear is used at higher levels of the game, although the lifespan of the bat will vary across the five grades. For example, softer willow is considered to have excellent performance properties but a shorter lifespan. Product lifespan also depends on frequency of use, style of play (in the case of bats), and product care. As for other gear, there are various grades of balls and gloves that reflect leather quality. Long-established firms such as Dukes (London, UK) continue to supply the premium end of the cricket ball market.

Age and sex

Adults and juniors are distinct markets, reflected in the size and weight of gear, and, in the case of juniors, the frequency of replacement as players grow. The junior market is especially important since this is where people are most likely to take up or give up cricket. The focus of the chapter is on hard-ball cricket, while recognising that soft balls and plastic bats are widely used at junior and some adult leisure levels, and specifically in women's teams playing softball cricket.

Cricket has traditionally been a male activity, but with increased female participation, women and girls represent distinct markets, accommodated mainly by junior sizes rather than product redesigns specifically for them. While issues related to inclusivity have been recognised across the wider sporting sector, in which there has been significant technological development to enable participation from disabled athletes and address sports-related injuries, sporting goods designed specifically for women are uncommon [19,20]. The Institution of Mechanical Engineers (IMechE) report "Sustainable, Inclusive, Innovative: The Role of Engineering in Sport" also highlights how most sporting goods marketed at females are often a scaled-down version of those offered to men that do not consider female ergonomics [19].

Suppliers

In the past, the industry and market were dominated by well-established English cricket specialist manufacturers, but today there are many suppliers. As with many industries, cricket gear production is predominantly globalised and outsourced to India and, to a lesser extent, Pakistan, complemented by specialist cricket and sporting goods manufacturers supplying the large home as well as export markets, effectively competing against their own brand products (e.g., SG).

In recent years, global sports brands have come to command a large share of the market, as the market outside of India, the UK, Australia, and other countries where the game is played is relatively small. Furthermore, much of the Indian market itself, while large, is at the lower price and profitability end of the cricket gear spectrum. Well-established English companies such as Gunn and Moore (GM), Gray-Nicholls (GN), and Slazenger (Derbyshire, England, UK) continue to compete at the premium-grade end of the market but contract out all or most production to India. These companies have themselves tended to be taken over by larger sporting goods firms

Table 18.3 Cricket gear supply in England and Wales

Supplier type	*Sub-categories*
Manufacturers and brands	
Global sports brands	Global, e.g., Puma, New Balance
	Indian, e.g., MRF
	Other
Major cricket gear specialists	UK: Gunn and Moore (GM), Gray-Nicholls (GN)
	Australian, e.g., Kookaburra
	Indian, e.g., SG, SS, BDM
	Other
Other cricket gear specialists	Balls, e.g., Dukes
	Helmets, e.g., Masuri
UK craft bat makers and gear suppliers	Bats, e.g., B3
	Gear supplier, e.g., SM
Labelled brands	Retailers' brands
	'Bedroom' brands[a]
Willow for cricket bats	Mainly JS Wright
Distribution	
Online	Sports and cricket retailers
	Brand sites
	Factory direct
	Club shops
Physical shops	Sports chains
	Cricket specialists
	Club shops
	Bat makers
	Coaches

[a] Small distributors importing unbranded cricket gear, adding their own label and selling online, e.g., Village Cricket.

(e.g., GM by Unicorn, Nottingham, England). As in other markets, traditional retailing has been substantially displaced by online retailing.

Further research conducted by CfSD indicated that, as most cricket gear used in the UK is produced overseas (e.g., in India and Pakistan), the skills to produce gear have been lost locally [21]. This research also showed there appears to be an emerging market of 'bedroom brands' which import gear to label in the UK. Sustainability considerations appear to be a low priority for the cricket gear industry, as evidenced by the little information on the topic available on brands' websites [22].

Table 18.3 provides a summary of the main supplier types. This does not include a full value chain of primary and intermediary manufacturing and materials supply.

Bat-making

Bat-making is distinct from many other industries in involving considerable manual and craft skills, especially for higher-end bats. The nature of the process for willow bats, including cutting clefts from willow timber, then shaping and pressing, does not readily lend itself to automation.

Using a bat press is the defining characteristic of bat manufacturing, and having a 'feel' for the wood is important since every piece is subtly different. It is for this reason that bat-making in India is concentrated in Jalandhar (Punjab) and Meerut (Uttar Pradesh), which are craft hubs [23].

Traditional cricket bat-making in England and Wales has been in danger of dying out due to costs and other disadvantages, but in recent years there has been a resurgence in craft-level manufacturing, with an estimated 50 small firms and individual bat-makers in business serving niche markets [24]. An important stakeholder in the bat supply chain is willow suppliers. As noted above, JS Wright provides around 95% of the world's supply of English willow used in cricket bats. It supplies stocks and clefts of various grades to bat makers; however, JS Wright has also indicated that there is not enough top-grade willow available to meet demand, and prices have been rising steeply. In turn, this presents a gap in the market for the development of alternative materials that emulate the characteristics of willow and preserve the bat-to-ball balance.[1] Moreover, this could present an opportunity for sustainable innovation and changes to the Laws of the game driven by external factors such as supply chain issues. Further detail regarding the material characteristics of willow is provided in Section "Sustainability Responses by the Sport of Cricket," of this chapter.

Balls

The production of cricket balls is mainly in India and Pakistan but has also spread to Bangladesh, China, and elsewhere, where similar processes such as cutting and stitching are well-established in clothing and leather industries.

Cricket gear sustainability issues

Background

There are various sustainability issues relevant to cricket gear which could impact the development of the game. The focus here is on environmental issues while recognising relevant social and economic issues as referenced in Section 1.

The main issues relate to life cycle considerations, risks, and impacts of materials used in cricket gear. As one of the most gear-intensive sports, significant quantities of materials are used, including animal hide leathers, polyurethane leather, high-density foams, and synthetic fabrics [25].

Moreover, as mentioned above, with production being outsourced primarily to India and Pakistan, the carbon footprint of shipping cricket gear from Asia and the shipping of materials and components through global supply chains is a high contributing factor to the environmental impact of cricket gear. For example, willow is sent from England to India for bat manufacture/assembly and returned to England for finishing. Using standard shipping distances and emission factors, a simple calculation shows that shipping willow to India and returning it to Britain adds about 0.4 kg of CO_2- e per bat. Growing participation will demand more material and generate more waste if circular economy aspects are not considered. The recent 2023 sustainability plan published by the ECB primarily focuses on the impact of climate change on the game rather than on resources and the waste generated by cricket gear [26].

There appears to be little published information on the overall unit sales of cricket gear or its materials content and production processes. Equally, no life cycle assessment (LCA) results appear to be published for cricket gear. As such, research completed by CfSD within PASIC identified materials in balls and cricket PPE [27], and an LCA for batting pads was conducted [28].

The main materials identified were willow for bats; cane and rubber for bat handles; leather, cork, petrochemically derived materials, and fabrics for balls; leather, petrochemically derived components, and fabric in batting gloves and pads; and plastics, fabrics, and metals in helmets. The section below highlights the sustainability considerations of the cricket gear materials identified and potential sustainable alternatives.

Willow

The use of *Salix alba* 'Caerulea' for cricket bats is at the heart of the game because willow has unique qualities and performance that maintain the balance between bat and ball, as required by Law 6 of the game. Despite efforts to find alternatives to willow, such as the use of aluminium for the blade [29], or the design of carbon fibre reinforced composite handles [30], no legal alternatives have been found, and especially top-grade willow continues to be highly prized. Willow growing does not appear to be inherently harmful or unsustainable overall and can have environmental benefits, growing in and helping to stabilise areas prone to flooding. However, the game's dependence on it (and globally also on Kashmir willow) presents a strategic risk since supply is concentrated in two relatively small areas of the world: English willow is almost entirely sourced from plantations in England, and Kashmir willow is mostly supplied from Kashmir, India. Attempts to grow English willow elsewhere, e.g., in Tasmania, have been unsuccessful.

Furthermore, English willow trees take up to 15 years to mature, and supply cannot be easily adjusted to demand unless suppliers cut immature trees (it is thought that some contract farmers have been doing this, cashing in on demand and high prices, but adversely affecting the quality of wood available). To meet demand, European willow is being used in significant volumes, but it is considered poor quality for cricket bat manufacture due to having lots of narrow grains, which make the bat brittle [31]. No data appears to be available on the volumes involved. Alternatives to willow are considered below under opportunities for sustainable innovation.

Cane for bat handles

Rattan cane for bat handles is understood to come from wild sources – mainly Sarawak and elsewhere in Malaysia and Indonesia. It is unknown whether sourcing is sustainable.

Leather

The main use of leather in cricket gear is in balls, where the use of leather and performance characteristics are specified in standards and Laws. Furthermore, the number of new balls per match is stated under the Laws of the game, so large numbers of new balls are needed to fulfil these requirements – an estimated 1.8 million a year in England and Wales alone (see Table 18.4).

Leather production is well known as a potentially highly polluting industry, including greenhouse emissions and other impacts of cattle rearing. Most of the leather production for cricket gear is in India, and it is unknown whether any cleaner production practices are followed in manufacture. Furthermore, there are existing challenges in standardising the tanning processes within the leather industry. Dukes witnessed this first-hand during the summer of 2022 when its cricket balls softened more quickly than other batches and then went 'out of shape'. Dukes attributed this to the grass being fed to the livestock, the environment (climate change), and other issues related to the COVID-19 pandemic [32].

Table 18.4 Estimated annual waste in England and Wales for selected cricket gear

Gear type	*Number of items (000s)*	*Annual waste (tonnes)*
Bats	364	410
Balls	1,805	284
Batting gloves – pairs	496	170
Wicket keeping gloves – pairs	50	59
Pads (batting and wicket keeping) – pairs	351	533
Helmets	319	190
Total	3,385	1646

Assumptions include:

1. *Numbers of players*: 300,000 adults playing recreational cricket; 2000 professional and semi-professional players; 200,000 private school pupils over 12 and 100,000 under 12; 100,000 state school pupils playing at significant levels.
2. *Gear ownership*: All players own at least one bat and set of PPE. Professional and higher echelon adult recreational and junior players own several bats and more than one set of PPE. Recreational clubs and schools, private and state, provide some gear for non-owners as well as balls. It is assumed that 10% of players own wicket keeping gloves and pads.
3. *Average life of bats*: An average life of four years is assumed for adult bats (less for professionals). Juniors will need to replace bats as they grow, and the average life is shorter assumed to be three years.
4. *Average life of PPE:* Average lives for adult gear is assumed to be three years for batting gloves, three years for wicket-keeping gloves, and five years for pads and helmets. Average lives for junior gear, is assumed to be shorter to allow for growth at two years for both batting and wicket-keeping gloves, three years for pads and two years for helmets. For club and school-owned gear it has been assumed that the average life is five years.
5. *Weights*: Average weights are assumed for adult and junior gear, allowing for older heavier and modern lightweight PPE.
6. *Numbers of matches and balls*: 1000 professional matches a year using an average of four balls; 6500 ECB-affiliated clubs with an average of three Saturday league teams (including juniors) and 20 fixtures per season two new balls per match.; 500 non ECB-affiliated clubs playing in Saturday Leagues with three teams and 20 fixtures per season and two new balls per match; 6500 ECB-affiliated clubs with an average of two Sunday league teams and 12 fixtures per season with two new balls per match; 10,000 informal recreational teams playing 12 matches per season, 50% with new balls; 2000 private schools with an average of four teams playing 12 inter-school matches per season, as well as intra-school house matches, all using two new balls; 2000 state schools (out of 3500 secondary schools in total) playing cricket significantly, with an average of three teams playing six matches per summer term; 50 junior teams playing in county and national championships, involving 1000 matches and two new balls per match.

Fossil fuel-based materials

Most other materials used in cricket gear are petrochemically derived. For example, Polypropylene (PP) is used for the finger/thumb inserts in gloves, Polypropylene (PP)/Polystyrene (PS) for the connectors for straps on pads, Polystyrene (PS) for the knee guard in pads, and high-density Polyurethane (PU) foam for the internal padding of both gloves and pads. Readers wishing for more information on foams are referred to Chapter 5, Soft materials in sport, and Chapters 9 and 10 on Cellular Solids for Sports Applications. The use of synthetic materials derived from non-renewable natural resources (e.g., oil) has inherent sustainability issues that contribute to

global warming and have limited end-of-life disposal routes. The volatility in prices of oil-based polymers also influences their pricing [27]. Furthermore, production of such polymeric materials comes under REACH regulations; for example, isocyanates (critical to the manufacture of PUs) are toxic, classified as CMR (Carcinogen, Mutagen, and Reprotoxic), and are already regulated by Environment Protection Agencies and the Occupational Safety and Health Administration; this may tighten further [27].

Sustainable material alternatives

CfSD at UCA and the Centre for Natural Material Innovation at the University of Cambridge have reviewed opportunities for alternative materials in all types of cricket gear [27], including willow, cane, cork, fabrics, plastics, and rubber, and have explored leather alternatives [33].

As part of the investigation into leather alternatives for cricket balls and gloves, two testing methodologies were employed by UoC to evaluate the strength and stiffness of four samples of plant-based vegan leather alternatives (quasistatic tensile testing) and to measure their wettability and absorption behaviour (contact angle analysis). While the study produced a database [33] of over 100 vegan leather alternatives, samples were only obtained for four of these: Piñatex® (Ananas Anam Ltd, London, UK), BarkTex® (Germany), HideBio (Cambridge, UK), and Bananatex® (The Netherlands). While none of the sample materials proved to be suitable alternatives to animal hide leather for cricket balls, due to being too thick, not absorbing enough moisture, and having insufficient strength, the tests enabled a basic comparison between the four samples, which in turn highlighted opportunities for the further development of testing methods for plant-based vegan leathers. Furthermore, while the tested vegan leather materials did not emulate the performance of premium leather in cricket balls, a further development from one of the four candidate vegan leathers, Piñatex® Light (Ananas Anam Ltd, London, UK), was identified to replace the existing PU leather 'outer skin' of cricket batting pads. Based on this, a prototype incorporating other circular design innovations was developed (Figure 18.1). Readers

Figure 18.1 Circular cricket batting pad prototype using Piñatex® Light, second-life internal high-density foams, and Fishy Filaments® 3D printed knee guard.

seeking more detail on the properties of artificial and natural leathers are referred to Chapter 5, Soft materials in sport.

The prototype design innovations focused on the reuse of internal components (high-density foam and cane) from end-of-life gear, the repair of the straps, and 3D printing of the internal knee guards using Polylactic Acid (PLA) and a recycled nylon Porthcurno filament made from end-of-life fishing gear (Fishy Filaments®, Cornwall, UK) to replace existing polyurethane (PE) knee guards (Figure 18.1) [34]. The development of the first circular cricket batting pad prototype concluded that while it is possible to produce a more circular batting pad, there are a range of issues that still need to be tackled before similar products can be placed on the market, including addressing the gap in 'making' skills for batting pads in England and Wales, economic feasibility (labour and materials costs), scalability, and testing of the pad to comply with PPE standards.

Waste

As noted above, no reliable data was found on production, consumption, product life, and waste arising from cricket gear, globally or in England and Wales; but the quantities are likely to be globally significant, with over 300 million people playing the game. Post-consumer waste from the use of cricket gear will arise from the disposal of damaged gear at 'end of life', users seeking performance/quality improvement and replacing gear, teams changing colours and logos, juniors outgrowing gear, players (especially juniors) dropping out, and unused equipment stored in homes or clubs. While there will be some reuse, most is often stored in households and ends up in landfill or waste-to-energy. Ideally, waste could be estimated from actual sales, average product life, and replacement statistics. Since data and information are not available, the following two methods were applied to various levels of play, from professional to junior, to individuals and clubs:

- Method 1 (for bats, gloves, pads, and helmets): Waste is estimated from numbers of players, average ownership, average product life, and average weight per unit.
- Method 2 (for balls): Waste is estimated based on numbers of clubs/schools, teams, matches, and balls per match. This method is applied to balls since there are specific rules and requirements relating to the number of new balls used in matches.

In both cases, an annual estimate of waste was made for a baseline of 2022, when there was some data available on: numbers of participants in England and Wales (around 300,000) [1]; numbers of professional teams and ECB-affiliated clubs (6,500); and average weights of gear and new balls per match. All other numbers are estimates based on expert judgement that have been incorporated into a simple model that can be refined as more information becomes available.

Estimated annual tonnages of redundant gear going to waste in England and Wales are shown in Table 18.4, based on the assumptions below. These estimates cover both adults and school (private and state) pupils playing the game.

Sustainability responses by the sport of cricket

Governing bodies

Cricket's governing bodies have an important role in showing leadership, but to date, sustainability considerations have focused on environmental factors (e.g., climate change) affecting the game and carbon emissions related to the built environment (venues, facilities, and grounds) [35], with little consideration of impacts related to cricket gear.

The ICC has not so far addressed sustainability issues. A review of its website (in 2023) did not show any sustainability policy, although the website stated it had worked with the BCCI to develop a sustainability plan for the 2023 World Cup and is highly active in increasing participation through policies on social issues. ICC policies on the use of gear as a vehicle for advertising in international matches do not consider sustainability. The ECB implements policies agreed with the ICC in England and Wales, as well as having its own policies on ethics and governance, including those covering child labour, human trafficking, and combating racism and discrimination. In 2021, the ECB employed a sustainability manager and recognised sustainability and climate change challenges for the game in its 'Sustainability Ambitions' published in April 2021 [35]. However, the ECB's latest report, 'An Environmental Plan for Cricket' launched in November 2023, highlights that its prime focus remains on reducing the environmental impact of buildings, venues, and events. This is evidenced further by the report's limited focus on the topic of cricket gear, with the prime references to cricket gear being to schemes run by other organisations, e.g., Lord's Taverners Cricket Kit Recycling Scheme and Cric-Kit by Yorkshire Cricket Foundation [26]. In addition, there does not appear to have been significant consideration of sustainability in the procurement of cricket gear by the ECB.

MCC policies on sustainability are closely aligned with those of the ECB. In September 2023, the MCC launched their 'Plan to Net Zero Carbon', which limits its focus on materials to the manufacturing of steel, cement, and glass for buildings and venues [36]. However, as the home of Lord's Taverners, a charity that includes the Lord's Taverners Recycling Unit (LTRU), which facilitated the reuse of about 25,000 items of cricket gear and clothing in 2023 [37], it also has a role in the reuse of cricket gear and clothing overseas.

Sustainability initiatives by county cricket clubs in England and Wales

There appears to be recognition by some county cricket clubs of climate change, especially the risks to grounds affected by flooding. Few examples of sustainability policies and initiatives were identified. However, some clubs have recently created internal sustainability roles, e.g., Surrey County Cricket Club and Warwickshire County Cricket Club [38], while others have been involved in cricket gear reuse and recycling schemes.

Sustainability initiatives by suppliers

A review of the websites of 83 specialist suppliers of cricket bats, balls, and protective gear between August 2022 and March 2023 revealed that sustainability considerations were not perceived as a priority by most of the companies placing cricket gear on the market [22]. There are some examples of sustainability policies, including bat maker Bradbury Cricket (Leppington, Australia), which covers sustainability and circular economy principles by using off-cuts from cutting the raw willow clefts to regulation width to construct birdhouses and shelters for native bees; bat makers Millichamp and Hall (Taunton, UK) have a 'green policy', and Gray-Nicholls recently launched its 'Envyro' sustainability programme that focuses on sustainable production and packaging, which sits alongside its Tree Legacy programme and pioneering development of the first pair of upcycled batting gloves [39]. Masuri, UK-based suppliers of cricket helmets, has an environmental policy and Platinum Grade compliance with WRAP (Worldwide Responsible Accredited Production), including environmental and responsible sourcing performance [40]; and JS Wright [41] states that it plants three willow trees for every one harvested, while various craft bat makers source their willow locally. Major global brand suppliers of sports clothing and footwear have sustainability policies and initiatives as part of their long-standing response to broader

sustainability issues, including working conditions in supplier factories. However, it is not clear whether such companies have been applying any sustainability considerations in the manufacturing of cricket gear and procurement of related materials and components.

Repair initiatives

Under MCC Laws, cricket bat repair is permitted. More specifically, Law 5 indicates what bat repair is permissible for cricket played within the Laws.

It should be noted that, from discussions, cricket bat specialists provide guidance and support for the care of bats to extend their life as repair services. This is a contribution to sustainability by enabling product-life extension. Tests to assess the performance of cricket bats have been developed, with the main test being the 'tapping' test, in which the bat is impacted with a hardwood mallet, or a cricket ball is bounced on the face of the blade. A parameter typically considered to measure the performance of a bat is the ratio of rebound to inbound ball velocity [42]. However, no literature was identified on tests to assess potential deterioration in repaired bats, thus presenting an opportunity in test development. For 'softs' and other PPE (gloves, pads, helmets), no examples were found in England and Wales of significant levels of repair or refurbishment, although Masuri provides replacement chin guards for its helmets [40], and two specialists offer refacing of wicketkeeper gloves (Beckenham Cricket Specialists Ltd and Classic Keeper, Solihull, UK) [43]. Additionally, glove repair was identified in India where labour costs are lower than in England and Wales and skills and manufacturing capability are well established, suggesting that repair is technically possible [44]. This was confirmed by a repair exercise by CfSD in 2023 that replaced end-of-life leather on the palms with vegan chamois leather [45]. However, pilot player testing indicated that the gloves did not perform as well as bovine leather (e.g., flexibility, breathability, durability), and more work is required to develop performance vegan leathers. Further research is also required to identify alternatives to sewing or speeding up the sewing process to make repair economically viable, as well as addressing hygiene issues related to sweat and odour [45].

Reuse initiatives

As noted above, LTRU established in 2012 a cricket gear and clothing collection and reuse scheme to enable underprivileged people to play the game [46]. LTRU was the most developed cricket gear reuse scheme in the UK [43]. An estimated 95% of cricket gear collected for reuse by LTRU was distributed overseas [47]. However, as of Spring 2025, LTRU has ceased all activities due to operational costs. Other schemes have been established, but the scale of these is small, and they appear to relate to suppliers offloading surplus stock to LTRU. As such, the reuse of cricket gear in England and Wales appears to be driven by seven identified schemes[2] and small-scale initiatives being led by individual clubs. While only a fraction of cricket gear reuse is happening in the UK, the reuse of PPE presents challenges related to the safety of second-life gear and its compliance with PPE regulations if placed back on the market. Further studies will thus be required to determine the loss of protective properties of materials used for PPE to assess their suitability for reuse. Aligned with concerns over the limited use phase of protective properties of PPE, while there are ongoing studies to improve helmet design to reduce the risk of concussion using alternative technologies and materials such as mechanical metamaterials [48], sustainability considerations to assess if these new technologies could extend the lifespan of helmets present an opportunity for further research in the field. Readers wishing more information on mechanical metamaterials are referred to Chapters 9 and 10 on Cellular Solids for Sports Applications.

Conclusions and recommendations

This chapter has illustrated that there has been a lack of attention to and data availability on the sustainability aspects of cricket gear in research and policy, and there is a need to address these issues as participation and demand for gear increase. Concerns include the supply of willow for cricket bats, the general life cycle impacts of gear materials, the amount of end-of-life waste, and the low implementation of circularity initiatives.

Research into alternative materials is continuing, but the performance requirements for bats and balls have not been fully emulated in materials that have been researched and tested so far by CfSD at UCA and the Centre for Natural Material Innovation (CNMI) at the University of Cambridge. Opportunities related to extending the life of gear through reuse, repair, and refurbishment require further research.

Drawing on the findings presented in this chapter, the following recommendations are made. There is a need to:

- Raise awareness and understanding of sustainability issues within the cricket industry.
- Enhance clarity on the volume/amount of cricket gear and clothing waste.
- Develop a more robust UK reuse infrastructure for cricket gear and clothing.
- Extend product life through repair services, including reuse of internal components in batting gloves and pads that still appear to have structural integrity. This, in turn, has the potential to address environmental issues and affordability barriers to participation.
- Continue to explore the use of less environmentally impactful materials, building on materials, technical testing, and an LCA recently conducted on a pair of batting pads [28].
- Conduct testing on the sustainable material alternatives identified to replace existing high-density foams, synthetic textiles, and animal hide leathers.
- Test second-life cricket gear products to assess their compatibility with PPE standards if they are to be placed back in the market. For example, test the durability, stiffness, and impact performance of second-life foams.
- Explore opportunities for the local production of cricket gear to reduce the carbon footprint and other impacts of present supply chains.
- Develop improved guidance from governing bodies working with cricket gear companies and other stakeholders to explore and foster innovation and sustainability as participation grows globally.
- Establish sustainability standards for cricket gear, for example: requirements for specific gear to have a specific percentage of recycled or biobased materials; the development of an extended producer responsibility (EPR) scheme for sports gear with a category for cricket gear; the introduction of a baseline for minimum repairability, disassembly, and recyclability of gear.

By addressing these recommendations, stakeholders in the cricket gear industry can take the first steps towards a more sustainable, equitable, and circular future.

Notes

1 Bat balance is a fundamental aspect of cricket that has a significant impact on a batsman's performance. Bat balance refers to the weight distribution along the length of a cricket bat. It involves finding equilibrium between the weight of the handle and the blade.
2 Yorkshire Cricket Foundation's Cric-Kit programmes in Shefield, Leeds and Bradford; the Cricket Kindness Project (incorporated into No Boundaries Cricket Club as of late 2022; Bat for a Chance; Cricket Asylum Foundation Kit Library, and the Kit Drop Swap programme organised by Gloucestershire County Cricket Club.

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19

DESIGN CONSIDERATIONS FOR BODY PADDING IN RUGBY

Adil Imam

Introduction

Rugby's origins can be traced to various football games and sports from around the globe, including Ki-o-Rahi (New Zealand), Marn Grook (Australia), and Caid (Ireland). The first recorded game of rugby took place in 1823 at Rugby School, with William Webb Ellis, a 16-year-old student, picking up a football and running with it. Today, the Rugby World Cup trophy bears his name [1]. From the establishment of the Rugby Football Union in 1871 to the inception of the Six Nations Championship in 2000, rugby has grown considerably. Global rugby participation increased by 11% to 8.4 million between 2022 and 2023 [2].

The World Rugby Strategic Plan [3] for 2021–2025 outlines goals for the sport, including:

- "Leading targeted research informing and advancing injury prevention and mental well-being" and
- "Advancing player welfare law review and trials to further protect players"

The global market for rugby merchandise was valued at USD 18.5 billion in Japan, the host nation for the 2019 Rugby World Cup [4]. The expansion plans of World Rugby, coupled with the increasing number of players, are likely to drive further growth in the market value of rugby equipment. Such growth in equipment sales will also bring to the spotlight the associated environmental sustainability challenges [5]. Readers seeking more information on this topic are referred to Chapter 17, Sustainable Design and Manufacturing of Sport Equipment.

World Rugby reportedly invested ~£575 million in the development of the sport between 2020 and 2023 [2]. As the player base expands, so does the spectrum of skill levels, and diversity in terms of sizes, ethnicity, and competition. At the semi-professional and professional levels, incentives and monetary rewards have heightened the game's competitiveness while increasing the risk of injury [6]. To mitigate this risk, there has been a growing trend among players to wear padded clothing. Over 66% of the rugby players surveyed in one study had used body padding at some point [7]. Currently, 295 former rugby players are suing World Rugby, the Rugby Football Union (RFU), and the Welsh Rugby Union (WRU), alleging a failure to implement adequate player safety measures related to brain injuries [8]. While rugby players can wear World Rugby-approved headgear, this chapter focuses on the design considerations of body padding.

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While research on rugby personal protective equipment (PPE) has increased, there remains uncertainty regarding its effectiveness in reducing injury risk [6]. To address this, World Rugby had established a set of specifications (dated 2007 [6]) and directives governing padded clothing in its Handbook, specifically Regulation-12. This regulation was further divided into three schedules; this chapter will focus on the author's research (starting in 2017) related to what was Schedule 1: Specifications relating to Players' Dress-Shoulder Pads. Schedule 1 provided guidelines for the design, construction, material specifications, performance requirements, and test methods for padded clothing. Given the continuous advancements in technology and materials, it is crucial to regularly review and update these regulations to ensure the effectiveness of padded clothing in protecting players. Moreover, World Rugby's intention for padded clothing is only to reduce the risk of soft tissue injuries, like cuts and abrasions (which are also not tested for), and not to protect against severe injuries like fractures and dislocations.

This chapter summarises the author's research to review the regulation for rugby padded clothing, identify gaps and areas for improvement, and update it to make it appropriate for testing against cuts and abrasions. To identify areas of ambiguity, the existing Regulation-12 Schedule 1 documentation (as of 2017, when the project started) was textually critiqued. This critique was followed by testing of manufacturers' samples (approved and available on the market at that time) using Regulation-12 Schedule-1 instructions. The areas of focus identified were:

i **Materials used for padding and the testing of the material property**: Only homogeneous padding materials were allowed, i.e., no sandwich constructions. There was also a density limit (60 $kg \cdot m^{-3}$), presumably intended to prevent the use of hard materials as padding. While the limitation of density to a maximum of 60 $kg \cdot m^{-3}$ effectively limited the possible padding materials to foams, research has shown that the hardness of a material can be varied without changing the density (e.g., Auxetics/Bio-Inspired, Biomimetic Materials - See the next section on Materials used in personal protective equipment).

ii **Impact testing methodology**: The impact test was defined as a rigid impactor striking the padding when placed on a fixed rigid anvil, which was not representative of an impact during the game of rugby, i.e., collisions between compliant bodies. Moreover, the regulation limited the protection allowed. The limiting criteria were a minimum impact acceleration of 150 g for a 14.7 J impact (5 kg dropped from 0.3 m).

iii **Location of the padding**: A diagram specified the permitted location of shoulder padding on a jersey (see the section, Case Study: Study of Padding Variations). However, ambiguous wording in the regulation meant it was possible for manufacturers to place padding outside the shoulder region without it needing to meet the impact test limitations (see ii.). This led to ambiguity about how to test any padding located outside the shoulder areas.

This chapter presents the research behind each of the areas of focus identified and the reasoning behind the recommended changes to Regulation-12 Schedule 1, which led to the publication of the new Body Padding Performance Specifications [9]. The chapter will also provide insight into some of the technological developments in materials and analysis of other test methodologies that informed the new regulatory framework. The following section will provide insight into some of the materials used in sporting PPE in general. It will show how continuous development in materials necessitates the need to review sporting regulations and ensure that they are adequate to assess and govern the use of PPE.

Materials used in personal protective equipment

PPE designs can vary between sports depending on the intent of the protection (Figure 19.1). Sporting protective equipment is intended to reduce the risk of injuries and can be a part of the playing environment (i.e., landing mats or barriers [10]) or be a part of the athletes' attire. PPE used by athletes can be embedded into their garment/attire (i.e., padded clothing in rugby, padded gear in cycling, motor racing, or fencing) or worn as an extra item (i.e., helmets, body protection in cricket, shin pads).

PPE is typically designed to provide impact protection and reduce the associated forces and accelerations that can cause injury. However, the intention of World Rugby is that rugby shoulder padding (expanded to body padding in the new specifications) (example shown in Figure 19.2) is only meant to reduce the risk of superficial injuries to the skin tissue in the form of cuts and abrasions.

Sporting PPE often contains layers of different materials (Figure 19.3), i.e., sandwich construction, which was not allowed for rugby padding according to Regulation-12. The outer layer of PPE is often a stiff material or shell, to help distribute the impact force and reduce the pressure acting on the body. PPE also usually has one or more layers of compliant material, typically closed-cell foam [11,12], for absorbing impact energy and reducing forces and accelerations [13]. The specific reduction of forces is dependent on the nature of the padding material and how it deforms under impact. Gradual deformation of foam under the impact (without reaching densification and

Figure 19.1 Collection of sporting PPE showing the variations in design and usability.

Figure 19.2 Rugby PPE example, showing XP1000 body armour and junior and protective shorts. Image courtesy Gilbert website.

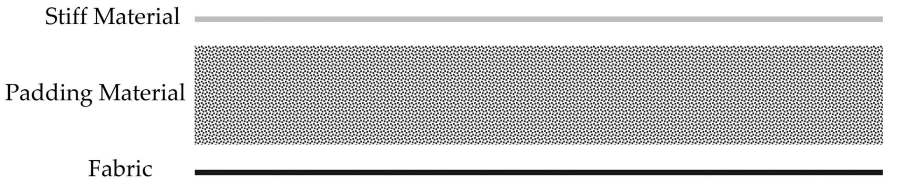

Figure 19.3 Schematic of a typical cross-section of a PPE showing the different materials.

"bottoming out" at >~80% compression) can increase the impact duration, lower the impulse, and hence increase the level of protection.

With improvements in manufacturing and associated technologies, like 3D printing (see Chapter 16, 3D Printing in Sports) and moulding, researchers can make new materials or structures [14] with varied mechanical properties (see Chapters 9 and 10 on Cellular Solids for Sports Applications). Such materials can be made to perform differently in different situations or scenarios. Researchers are also able to ensure material properties remain consistent while manipulating the density of padding material (raising questions about the efficacy of the identified area of focus i.). To assess the suitability of new and emerging materials for use in rugby padding, it is important to understand their properties. The following sub-sections summarise materials that could be applied to padded clothing for rugby, including those that are emerging and could be used in the future.

Auxetic materials

Auxetic materials (or structures) fatten when stretched (i.e., negative Poisson's ratio), unlike conventional materials that contract sideways when stretched [13] (Figure 19.4). This unique property makes auxetic materials useful for various applications, including sporting PPE like helmets [14], shoulder pads [15], and running shoes.

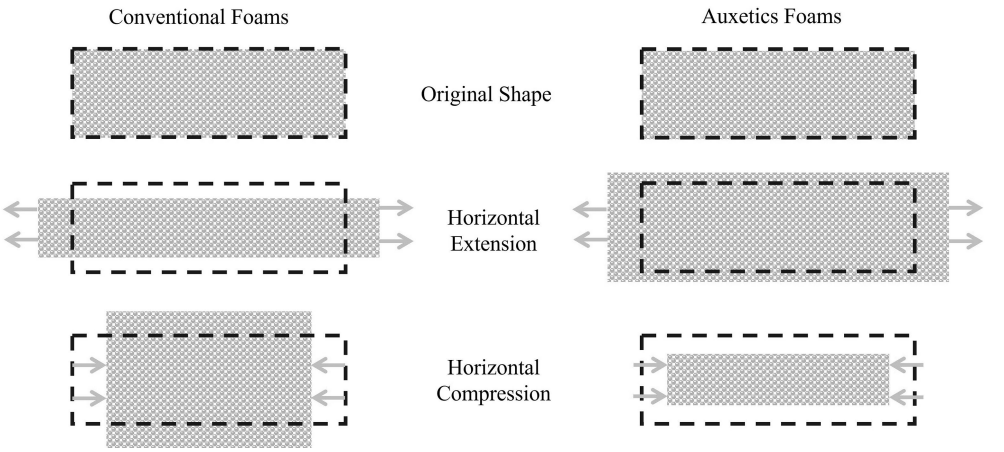

Figure 19.4 Schematic showing a comparison between conventional and auxetic material under deformation due to axial (horizontal) loading. Arrows show the load applied and the dotted outline depicts the original shape for reference. Author's own image, based on Figure 1 from Duncan et al. [13].

Auxetic materials have potential benefits over conventional materials, including:

- Improved indentation resistance: Auxetic materials have shown better indentation resistance and could better protect the wearer from injury [16,17].
- Enhanced synclastic curvature: Auxetic materials could conform more closely to the shape of the body, providing a more comfortable and secure fit [15].

Chan and Evans [18] showed auxetic foams (density range: 69–95 $kg \cdot m^{-3}$) to have improved indentation resilience over conventional foams. Scarpa et al. [19] converted polyurethane (PU) foam (density: 27 $kg \cdot m^{-3}$) into auxetic foam (via a compression process), with densities ranging from 75 to 254 $kg \cdot m^{-3}$. 2-D finite element (FE) simulations of these foams under indentation predicted lower stresses in the auxetic foam than in the conventional foam. Testing of auxetic foam under impact, at energies between 2 and 15 J (comparable to the 14.7 J impact in Regulation-12 Schedule-1), has shown between ~3 and ~8 times lower peak forces than conventional foams [16,20].

While most of the reported auxetic foams had densities above the Regulation-12 Schedule-1 limit of 60 $kg \cdot m^{-3}$, the density (42–52 $kg \cdot m^{-3}$) of the auxetic foam of Allen et al. [20] was within this limit. Suggestions by Allen et al. [20] and Duncan et al. [13], backed by evidence of improved properties, such as peak impact force reduction and indentation resistance, suggest that auxetic foams may soon be used in rugby padding. Continued research in this area may result in a wider variety of auxetic foams that would meet the Regulation-12 Schedule-1 density requirement. With further research and development, auxetic materials have the potential to improve the design of sporting PPE and provide athletes with greater levels of protection and comfort.

Dilatant materials

For many fluids, the viscosity is independent of the strain rate. Fluids that show increased viscosity under higher strain rates are termed dilatant fluids. Since the commercialisation of the dilatant material D3O® (energy-absorbing material, US Patent 7794827), other brands, such

as RHEON Labs Ltd (Patent no: US20120021167A1) and Dow Corning-Deflexion (Patent no: US20110039087A1), have used similar technologies, with claims of improved impact protection in products for sports [21] like American football (e.g., Schutt Sports D3O® Liner, Shadow XR), boxing (e.g., OPRO mouthguards), lacrosse (GameBreaker headgear), and field hockey. When impact testing rugby headgear incorporating dilatants (Poron XRD® and D3O® AERO and DECELL), Kajtaz and Subic [22] reported a 10% reduction in peak force in comparison to traditional (ethylene-vinyl acetate) foams. Recent work has looked at combining auxetic foams and dilatants with similar findings [23].

Though these dilatant materials provided improved impact force attenuation properties, their use in rugby was restricted by Regulation-12 Schedule-1 because they are too dense. Commercially available dilatant materials have densities starting from 140 $kg \cdot m^{-3}$ (Poron-XRD®) or 500 $kg \cdot m^{-3}$ (D3O®), which is much higher than the 60 $kg \cdot m^{-3}$ limit in Regulation-12 Schedule-1. Most of the brands marketing dilatants (D3O® products, in particular) claim that their products are abrasion resistant, which could make them effective in reducing injuries such as cuts and abrasions, although this has not been shown nor tested specifically in the scientific literature. While there is potential for dilatant materials to bring improvements to rugby padding, they were not allowed when Regulation-12 Schedule-1 was being reviewed.

Polymer foams are widely used in applications where the mechanical function is important [24]. Due to concern over climate change, and our impact on the environment in general, there is a strong motivation to replace petroleum-based polymers with alternatives from renewable resources [5] (see Chapter 17, Sustainable Design and Manufacturing of Sport Equipment). For example, starches like polysaccharide are available from wheat, corn, potato, or other plant sources and are of interest as they are widely available and potentially low-cost [24,25]. The properties of starch-based foams have been shown to improve through reinforcement by cellulosic fibres [25]. Regular polymer foams can be modified by adding metal nanoparticles into the cell walls, and hence by varying the density of the foam, properties can be tailored to meet specific applications [24]. Adding functional nanometre-sized elements to biopolymer foams may enhance the mechanical and physical properties while reducing the mass and gas diffusivity of the polymer film [26].

Svagan et al. [24] studied the mechanical properties of biomimetic nanocomposite foams by varying the micro-fibrillated cellulose (MFC) content. They were able to change the compressive stress vs. strain response of the foam by varying the MFC content. Sehaqui et al. [27] developed a plant-inspired biopolymer foam by adding MFC found in plant cell walls during the foam fabrication process. They were able to change the compressive stress vs. strain response of the foam by varying the cellulose content while maintaining a constant density of 21 ± 1 $kg \cdot m^{-3}$ (by varying the foam content). They also state that the energy absorption of the foam under quasi-static compression is higher than expanded polystyrene (EPS) foam. Foam density values reported by Svagan et al. (2008) ranged from 87 to 103 $kg \cdot m^{-3}$. Sehaqui et al. (2010) claim that their biopolymer foams can be tailored to have densities ranging from 7 to 103 $kg \cdot m^{-3}$, with capabilities to alter the stress vs. strain response at a given density. For conventional materials, it was assumed that density is related to stiffness, as shown in Ashby plots [28]. This assumption, along with the research presented, suggests that if a bio-inspired/biomimetic foam were used as shoulder padding in rugby, measuring its density would unlikely provide any indication, or other useful information, about its hardness. Researchers seeking more information on bio-inspired design are referred to Chapter 21, The Application of Biological Engineering to Sports Technology.

While World Rugby intends to restrict the impact force attenuation capabilities of shoulder padding, the above-mentioned (especially dilatant and auxetic) materials have been reported to have high indentation resistance, which may help reduce the risk of cuts and abrasions. Aside from the

protective padding materials typically used in sporting PPE, there are those specifically designed to provide indentation or penetration resistance, such as stab-proof materials, as discussed in the next subsection.

Stab proof materials

Body armour has substantially advanced due to improvements in manufacturing techniques and materials. Modern body armour often utilises ceramic, polycarbonate, or aramid-based fibres, like Cordura® or Kevlar®, to protect against blunt force impacts. These fibres effectively capture the impactor and dissipate the impact energy, while additional layers absorb energy to further halt its momentum. Ceramic layers shatter upon impact, dispersing the impact energy but reducing the armour's future effectiveness.

Research has led to the development of Ultra-High Molecular Weight Polyethylene (UHMWPE) fibres, such as Dyneema® and Spectra®, which offer high performance. These lightweight, high-strength fibres exhibit exceptional abrasion resistance and are 10–50 times stronger than steel and Kevlar®, respectively. Studies [29,30] have shown their high resistance to cuts, abrasions, and punctures, making them suitable for protective clothing applications, including body armour, gloves, chainsaw protection, and fencing suits.

Johnson [31] predicted the future incorporation of materials like Carbon Nanotubes and MagnetoRheological fluids into body armour, and studies [32,33] have shown promising results. Carbon Nanotubes, with their nanoscale dimensions and higher stiffness than Kevlar®, offer enhanced protection. Meanwhile, MagnetoRheological fluid-based armour benefits from improved shear resistance and impact absorption. Studies [34,35] have shown that combining Carbon Nanotube coatings with UHMWPE fibres enhances compressive and bending strength.

As well as impact protection, body armour research focuses on stab resistance and injury prevention from cuts and abrasions. As Regulation-12 Schedule-1 limited the impact force attenuation of rugby padding, alternative approaches to reducing injury risk were considered. Johnson et al. [31] identified two primary methods of reducing injury risk:

- Absorbing and dissipating impact energy: This involves using materials that deform or fracture to prevent the impactor from penetrating the protective layer.
- Deflecting the striking implement: This involves using materials that are stiffer than the impactor to redirect its force. However, using extremely stiff materials in PPE for contact sports may pose injury risks to opponents.

Materials like UHMWPE in combination with dilatants have shown potential for reducing the risk of cuts and abrasions. Their incorporation into protective materials can enhance resistance to penetration and abrasions. The development of body armour continues, driven by advances in materials, manufacturing techniques, and a deeper understanding of injury mechanisms (see Chapter 27 Sports Injury and Trauma Biomechanics). The integration of new materials like UHMWPE, CNTs, and MR fluids holds promise for enhanced protection against various threats. As research progresses, the focus remains on optimising protection while maintaining comfort and flexibility for the wearer.

Modern materials, such as metamaterials, dilatants, and bio-inspired materials [36], have the potential to improve rugby padding by enhancing protection against soft tissue injuries. Auxetics and dilatants can also offer better indentation resistance and impact force attenuation than the foams (e.g., EVA) traditionally used in PPE. However, the modern and stab-resistant materials

described in this chapter tend to be denser (>100 kg·m^{-3}) than the limit (60 kg·m^{-3}) specified in World Rugby Regulation-12 Schedule-1. Research undertaken within the project described here [37] showed that measuring padding density did not provide any information about the hardness of the material and was erroneous (i.e., hard to accurately cut and measure foam samples). Hardness testing of the padding, if carried out using the correct method/strain, would help eliminate the use of hard paddings through a simple non-destructive test methodology [37]. The next section reviews established methods for testing materials designed to reduce the risk of soft tissue injuries. Various test methods are summarised, including stab impact, abrasion tests, cut resistance, and stud impact onto skin simulants.

Tests for personal protective equipment performance against cuts

The United Kingdom Home Office has defined a *Body Armour Standard* [38] that outlines the minimum requirements and testing methodologies for various body armour intended for UK law enforcement agencies. There are three classifications based on protection levels: (i) ballistics, (ii) knife, and (iii) knife with spike. The setup described in the standard is a drop tower assembly with a knife or spike impactor (Figure 19.5). Testing is conducted at four energy levels (24, 33, 36, and 50 J), with different penetration limits allowed at each one.

Another standard intended to test the ability of PPE to protect against cuts and abrasions is BS EN ISO 13997: Protective Clothing – Mechanical Properties – Determination of Resistance to Cutting by Sharp Objects. The standard complements BS EN 388: Protective Gloves Against Mechanical Risks. BS EN 388 defines some tests to determine whether the PPE can protect against sharp and abrasive objects. The standard defines several testing methodologies, including abrasion, blade cut, cut resistance, tear resistance, and puncture resistance.

Abrasion testing, specified in BS EN ISO 13997, is based on EN ISO 12947-1: *Textiles — Determination of the Abrasion Resistance of Fabrics by the Martindale Method*. The method requires a specialist testing machine – the Martindale Abrasion Tester (Figure 19.6) – consisting of a mass pressing a specimen against an abrasive surface. The machine applies randomised cyclic loads that cause the sample to rub against the abradant to mimic wear. Assessment is based on either a visual comparison of the material before and after the test or the number of cycles until the material tears.

Cut resistance testing in BS EN ISO 13997 involves moving a flat blade (with a defined preload) slowly over a specimen in a linear path. The length of the blade path is varied, and the

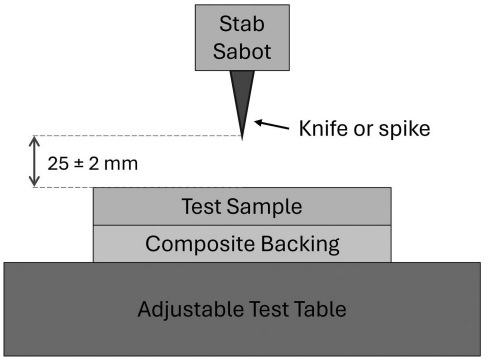

Figure 19.5 Schematic of a typical stab test configuration described by the Home Office Body Armour Standard (2017). Author's own image based on Figure 22 in [36].

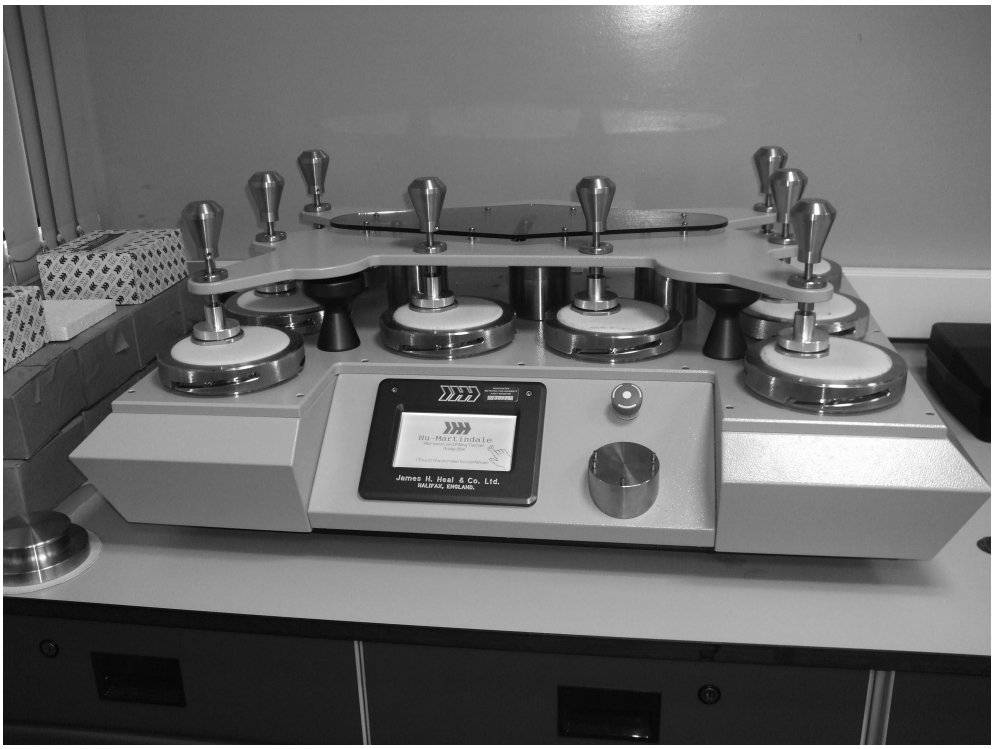

Figure 19.6 Martindale Abrasion Tester, image courtesy of Manchester Fashion Institute, Manchester Metropolitan University.

length of any cuts induced in the specimen is measured. The blade cut resistance test is similar, but the blade is circular and rotating, and the measurement noted is the number of load applications before a cut occurs. In addition to body armour testing and BS EN ISO 13997, there are other standards for testing protective equipment concerning the prevention of cuts and abrasions. BS EN 13594: *Protective Gloves for Motorcycle Riders – Requirements and Test Methods* outlines the test method for determining impact abrasion resistance. The standard states: "*The test machine is designed such as to allow the test piece to be dropped from a set height with defined energy onto an abrasive belt moving at a fixed speed over a rigid horizontal surface.*" This test is intended to replicate the palmar region of the glove sliding on the road or asphalt during a motorcycle fall/crash.

The standard testing methodologies for protection against cuts or stabs use sharp objects like knives or blades. While these tests should provide useful insights into the cut resistance of materials, they do not represent rugby scenarios. The Home Office *Body Armour Standard* (2017) states that the continued development of technology provides the ability to produce test methods that are more representative of the operational scenarios faced by the end user. Hence, when devising a methodology to test rugby padding against cuts and abrasions, it is important to use modern technology to help create scenarios representative of those in the sport.

While there is a gap in our knowledge of the mechanics of cut- or laceration-inducing scenarios in rugby, a test methodology existed in World Rugby™ Regulation-12 Schedule-2 for stud

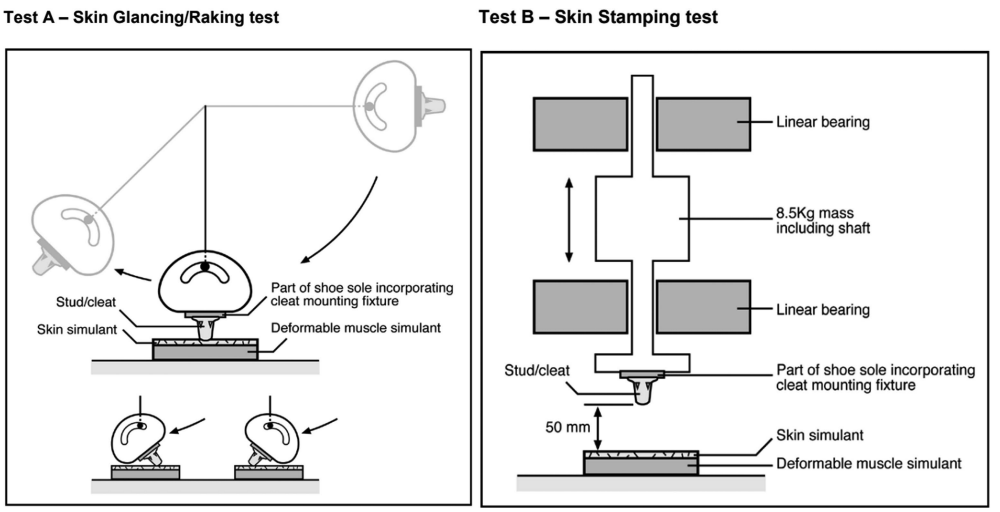

Figure 19.7 Schematic of (a) skin glancing/raking test and (b) skin stamping test in World Rugby™ Regulation-12 Schedule-2 (2017). Permission to use the figure granted by World Rugby.

design, which later became the Studs and Outsoles Specifications in 2019 [39]. Schedule-2 had five optional tests:

- Test A – Skin glancing/raking test
- Test B – Skin stamping test
- Test C – Stud/cleat impact test
- Test D – Fitting of replaceable studs/cleats
- Test E – Wear simulation

The regulation stated that Tests A and B provide a direct method of assessing a stud/cleat's potential to cause injury through glancing, raking, or stamping mechanisms. Test A described a pendulum impactor with a stud impacting a muscle and skin simulant (Figure 19.7). Test B comprised a vertical drop of an impactor (8.5 kg) with a standard stud design from 50 mm (4.17 J impact energy).

For both Tests A and B, Regulation-12 Schedule-2 stated: "A suitable human flesh simulant is required, comprising an artificial skin backed with a thick, deformable muscle-like material." (World Rugby™ Handbook, 2015:218). The Handbook suggests using a 1.5 mm thick poromeric shoe upper material (i.e., a leather substitute) with relatively low abrasion resistance as an artificial skin and gelatine moulded into a suitable form as the deformable muscle stimulant. No other material specifications or data sheets were mentioned for this simulant. The descriptions of the test methods did not include references to literature or other reasoning to support or justify the specified impact energy or flesh simulant.

The Regulation-12 Schedule-2 test methods were comparative. They were used to assess the damage caused to the skin simulant by stud designs in comparison to the standard one defined in the regulation. As such, the tests did not quantify the extent of stud-induced damage to the skin simulant. The test methods used compliant materials that may be more similar to human muscle and skin tissue than metal anvils (i.e., more biofidelic), and mimic scenarios likely to occur in rugby.

Another test, defined by Fédération Internationale de Football Association (FIFA) in their Handbook for Test Methods [40], uses Silicon Skin (Article Code-13110952, 1 mm thickness, ERIKS SAS) as a skin simulant to test abrasion against artificial surfaces (turf). Within this test, the skin simulant is used only to determine skin abrasion and not injuries from blunt impact scenarios. The test method involves sliding a test foot with the silicone skin attached to it across the turf and noting the force readings. Hurtado et al. [41] compared the silicone skin for the tribological performance of mechanical skin equivalents against ex vivo human skin samples. Their results showed that the elastic properties of the silicone skin differed from those of human skin.

Summary of the critique of regulation-12

As the most likely cause of skin injuries in rugby is stud impacts [42], these methods were deemed an ideal starting point for testing the efficacy of padding materials/designs in reducing cuts and abrasions. The impact test setup with the metal anvil in Regulation-12 Schedule-1 was well suited for test houses, where test equipment must be durable and highly repeatable, but it does not closely reflect in-game interaction. The test does not accurately represent rugby tackles or other collisions, especially those that cause cuts and abrasions. Using cadaveric or post-mortem human specimens is not ideal for assessing the effect of protective equipment design on performance in relation to cuts and abrasions [43–45]. The reasons include natural biological and anthropometric variation between individuals, ethical considerations, differences between live and post-mortem tissue, and degradation/damage from storage and repeated testing. Using a more biofidelic anvil [44,46,47] could help improve our understanding of how rugby padding might perform during an impact when worn by a player.

Hughes et al. [48] developed a relaxed muscle simulant (by mixing silicones) to assess the ability of rugby body padding to prevent soft tissue injury. When investigating raking injuries from studs, Hughes et al. [49] covered the muscle simulant with a layer of synthetic chamois leather intended to represent skin [50,51]. Placing such a synthetic tissue or silicone over a metal anvil, whilst not completely mimicking shoulder anatomy, would provide a more biofidelic anvil than the rigid metallic anvil specified in World Rugby™ Regulation-12 Schedule-1. More recent work has tested for friction injuries to the skin due to interaction with turf using compliant components [52,53].

Biofidelic anvils incorporating compliant materials are less durable than metal anvils and can degrade when impacted repeatedly [54,55]. To overcome this issue, an FE model that simulates an impact test with a biofidelic anvil was developed [56]. Using an FE model allows for testing the effect of various parameters, such as anvil size as well as shape, and changes to padding design, in an efficient and repeatable manner. Readers seeking more information on the design of biofidelic test surrogates are referred to Chapter 27: Sports Injury and Trauma Biomechanics.

While the critique highlighted areas of concern in Regulation-12 Schedule-1, examining the approved designs on the market also provided insight into potential conformance issues. The next section presents a case study carried out to highlight the areas of concern identified during the research and how it helped inform the regulation update.

Case study: Study of padding variations

One of the areas of concern (iii.) raised in the review of World Rugby™ Regulation-12 Schedule-1 was the ambiguous definition of the area of padding on a jersey and the associated testing. This section presents a case study on how this issue was identified and addressed. Regulation-12 Schedule-1 was written for shoulder padding, with areas on a jersey where padding was permitted demarcated as shown in Figure 19.8.

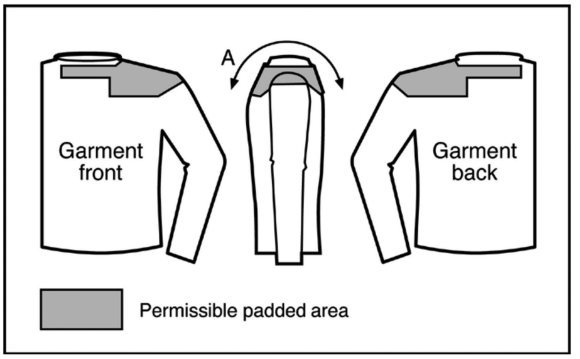

Figure 19.8 Permissible areas of padding as per Regulation-12 Schedule-1 (2017). Permission to use the figure granted by World Rugby.

A Visual inspection of various padded clothing specimens marketed for rugby showed that the areas of padding differed between them. There was visible padding outside the permissible areas highlighted (i.e., Figure 19.8). A list of approved padded clothing was obtained from the World Rugby™ website in 2018 (World Rugby™ Approved Equipment List– Shoulder Padding), which contained a plethora of padding and padding equipment. The list was filtered to account only for padded jerseys (eliminating detachable shoulder pads or those not incorporated into clothing). Corresponding images (front and back) for 24 different jersey designs were obtained from manufacturer/retailer websites. These 24 male padding designs were studied, and the front and rear images were analysed to identify areas of padding (Figure 19.9).

A numeric grid design was overlaid on a jersey template (Figure 19.10-A) and then on each design to recognise areas of padding presence/concentration (Figure 19.10-B). The grids were populated by cumulating the number of designs that had padding in each section. This method was applied to analyse areas of padding on both the front and rear of the 24 jersey designs.

The areas were then grouped into two sections: (i) Areas where padding was present (at least one design) (ii) Areas with a high frequency of padding (i.e., at least 14 different designs (60%) having padding in an area). All but one of the 24 designs (~96%) of padded jerseys analysed had padding outside the permissible padded area, highlighting the point raised in the critique of Regulation-12 Schedule-1 that padding appears outside the described areas. Areas of high concentration of padding on the analysed jerseys are shown in Figure 19.11.

The results presented above were shared at the 2018 World Rugby™ Manufacturers' Meeting. Following discussions with the manufacturers and World Rugby™, an updated regulation, *Performance Specification for Body Padding* (expanded to include all padding on a jersey), was introduced in 2019.

Summary of the research presented

Based on the findings presented in this chapter, a recommendation to World Rugby™ was made to update its regulations for body padding. The updated regulations were named 'The Body Padding Performance Specifications' and included all the padding that could be added to the fabric of a jersey [9]. Under the new requirements, any padding or extra material going over a jersey fabric would be considered padding and would then have to meet the performance test criteria.

Figure 19.9 Examples of approved designs (2017) with varied padding areas outside the shoulder region highlighted in ellipses. The top row shows the front and the bottom row shows the corresponding design's rear paddings.

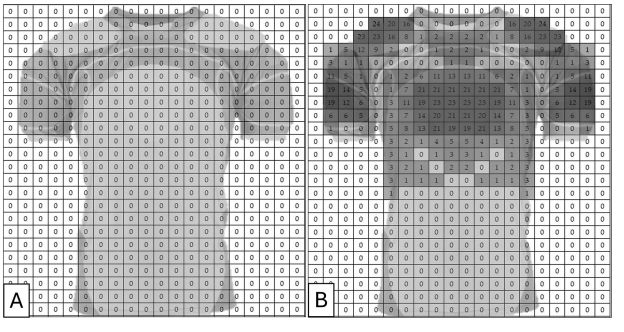

Figure 19.10 Grid used to identify areas of padding on the front of a padded jersey: (a) blank grid over template and (b) grid after being used to analyse 24 designs. Numbers represent the count of different designs having padding in the respective grid.

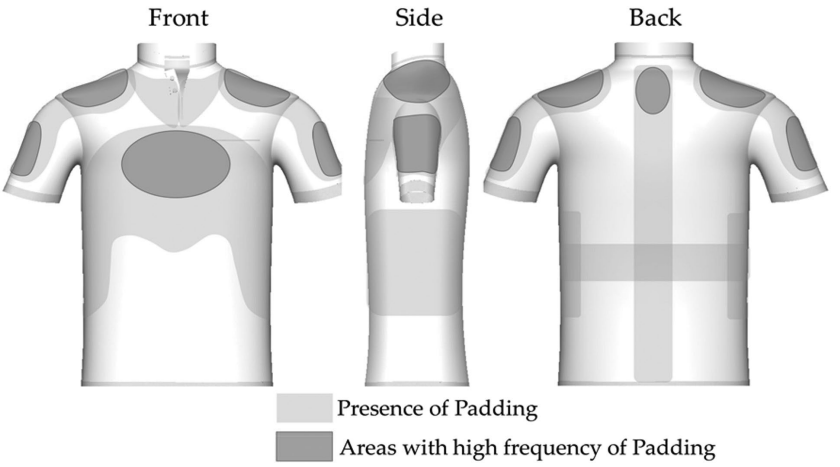

Figure 19.11 Areas of concentration of padding in Regulation-12 Scheduled-1 approved designed as of 2017, the high-frequency area highlighted if padding is present in more than 60% of designs.

The efficacy of padding could be tested using three tests: (i) a quasi-static hardness test (adapted from ISO 2439) presented in reference [37] to ensure the material is not too hard, (ii) impact attenuation testing to ensure excessive impact protection is not provided, and (iii) an impact test using a biofidelic anvil to gauge the ability to reduce the risk of cuts and abrasions. Under the new performance requirements, all padding must now undergo a slow compression test to ensure its hardness and an impact test to measure its impact force attenuation capabilities (in both directions). The density requirement and the limitation of the design being homogenous have been removed, allowing for potential multi-material design or sandwich construction. As before, in Regulation-12 Schedule-1, the impact tests are carried out to ensure that the padding does not provide too much protection, i.e. there is an upper limit to the protection provided. This is to ensure that the padding in place is not over-protecting as the intention is only to reduce the risk of cuts and abrasions and not severe injuries such as fractures and dislocations. Figure 19.12 summarises the research project described here.

Potential design considerations for the future

While this chapter focused on the research and reasoning behind the proposed changes to World Rugby Regulation-12 Schedule-1 and the subsequent introduction of Body Padding Performance Specifications, it provides information that could help engineers review and regulate PPE in the future.

There are three basic considerations that a designer of PPE for rugby must keep in mind:

i The intention of World Rugby is for body padding not to provide excessive impact force attenuation. This ensures that the nature of the game stays the same and that body padding does not provide too much protection, which may alter the psychology of the player and, in turn, lead to injuries from more severe impacts, i.e. players colliding harder. However, future research could investigate whether devices that offer more protection could be introduced without changing the nature of the game and increasing injury risk.

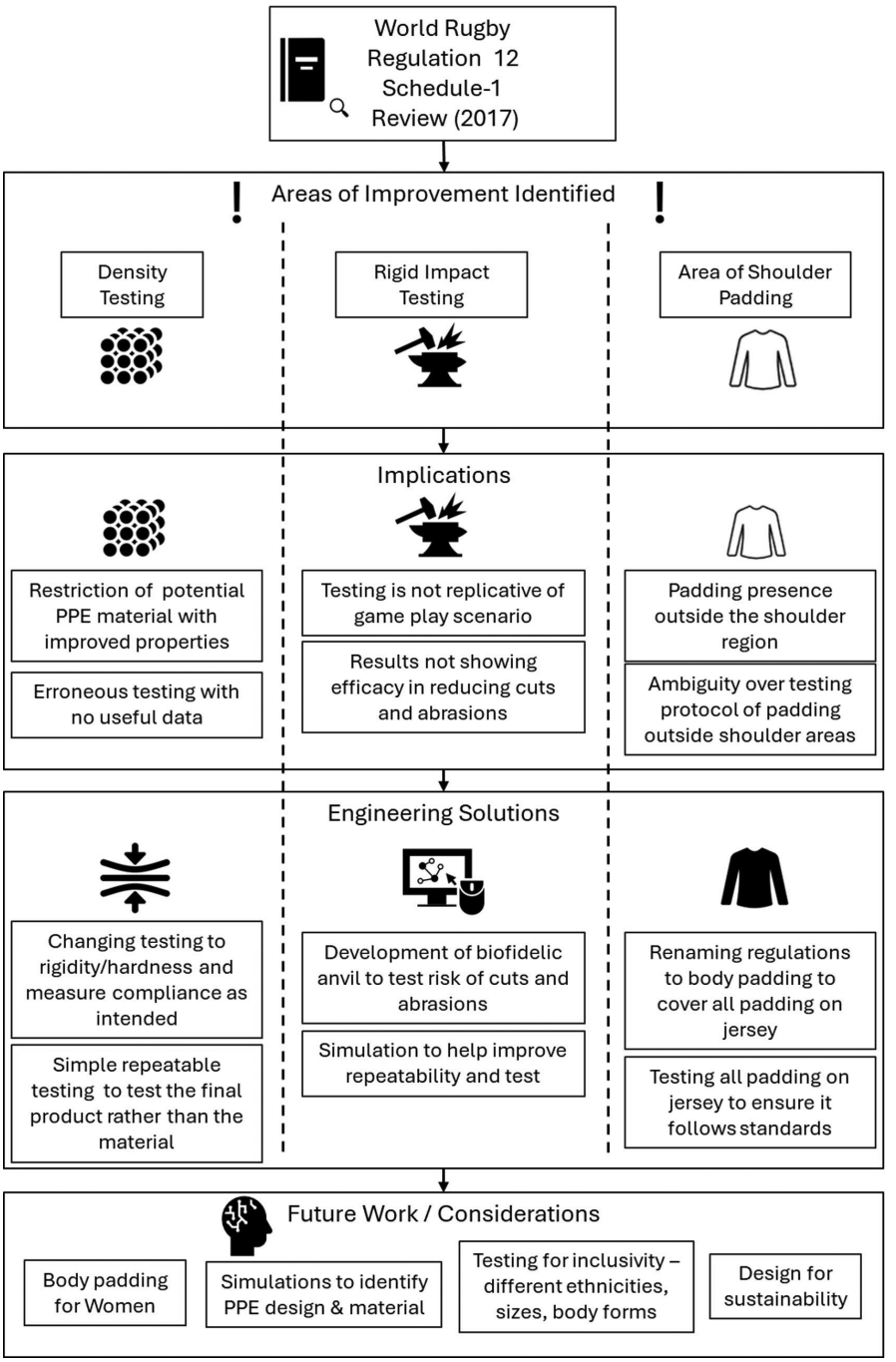

Figure 19.12 Summary of the considerations for the redesign of Regulation-12.

ii The hardness of the material. Incorporating hard shells into padding could enhance impact protection and reduce the risk of cuts and abrasions to the wearer. However, while this modification could mitigate soft tissue injuries for the wearer, it would also deviate from the padding's original design intent. Adopting hard-shell PPE might reduce soft tissue injuries but could also alter playing techniques, potentially leading to different types of injuries. A hard shell could also present an injury risk to the opponent, such as a cut from soft tissue coming into contact with a sharp (i.e. worn, degraded, or lacerated) edge or corner.

iii The final consideration for PPE designers must be the efficacy of materials against cuts and abrasions. To adequately assess protection against cuts and abrasions, testing methods should accurately replicate scenarios that could lead to such injuries. Utilising a rigid anvil would not provide meaningful insights into the padding's ability to mitigate cuts and abrasions. While this is outside the scope of the Body Padding Performance Specifications, as it only limits impact force attenuation performance, designers could assess this and ensure that products meet this requirement. As shown by researchers [49], employing a biofidelic anvil or a surrogate during testing for soft tissue injuries yields data that would provide information about the risk of cuts and abrasions. While surrogate-based testing may lack durability due to degradation during impact, utilising a validated FE model [56] could provide valuable insights into a product's performance capabilities. Such a model also has the benefit of allowing product concepts to be tested virtually before taking the candidates with the best potential forward for prototyping and physical testing (see Chapter 6: Finite element modelling in sport).

While considerable work has gone into updating Regulation-12 Schedule-1 into the Body Padding Performance Specifications, many design considerations remain for designing rugby PPE. The work presented in this chapter was done from a governance perspective while trying to ensure that the nature of the sport remains unchanged. This approach to reviewing the regulation did not focus on improving the design of the PPE itself. Engineers designing future PPE in rugby should explore different body paddings options for the women's game and ways to protect against soft tissue injuries, particularly concerning breast protection.

Issues with the design and fit of women's PPE have been highlighted recently in sport [57,58] and in general [59,60]. With rugby growing and attracting more young and female participants—57% of players in the post-COVID-19 era are pre-teen, and 24% of them are girls [2]—the design of equipment needs to be inclusive of different ages, sizes, genders, and ethnicities, rather than just resized versions of existing male-centric designs [5]. There could also be a focus on ensuring that the fit and design target the user and their ability/skill set, such as considering padding options for specific scrummaging positions. From a testing perspective, more work can be done to improve FE models and ensure testing accounts for all skin types and different environmental conditions (humid, arid, hot, high-altitude, cold, and windy conditions) wherever the sport may be played around the globe. Making designs conform to different anatomies and offering environmental protection would improve the inclusivity of the sport and make it more accessible globally.

With growth in the game and increased sales of equipment, there are additional design considerations for rugby padding that must be taken into account. The biggest question will always be its efficacy in reducing the risk of injuries, especially with the expanded coverage over the jersey. Research could also examine the relationship between padding design and injury locations to better understand padding placement on the jersey and its correlation with injury risk and occurrence. There also needs to be continuous monitoring of the rules and regulations to ensure that they evolve with technology and that any new designs are environmentally sustainable.

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20

ERGONOMICS AND USER-CENTERED DESIGN OF SPORTS EQUIPMENT

Kati Nispel and Veit Senner

How the research got here?

In competitive sports, mankind has always been testing the limits of what is physically possible to achieve. Limitations and boundaries are shifted. This shift is fostered by the exponential rise of technology, which offers new fields of research and development, extending these boundaries even further. With this shift in performance, the risk of suffering acute, as well as overuse, injuries began to rise. This deviates from one of our major goals as sports engineers: supporting sustainable sport and health through technology.

The buzzword "sustainable" in this context not only incorporates the idea of providing technology that is well-adjusted to the user's lifespan but also addresses the need for highly customized solutions, considering the individual health and fitness level, the person's motives for being physically active, and their desired goals. If we developers are able to create such technology, long-lasting enjoyment with these products is guaranteed, which in turn leads to longer product life cycles and more sustainable waste management. However, individualization requires effort from the first step. For instance, we need to gain a full understanding of what the customer needs and under which boundary conditions they will be using the product. Individualization also requires flexible production technologies, processes that are even able to deal with user-specific materials and design features. All this costs time and money—rising severe economic challenges when it comes to individualized products. Yet, there are three trends in our society and in technology that have made customizing feasible:

I Rising wealth has triggered a focus on a healthier lifestyle in society. More people started to get physically active and thus spend money on corresponding leisure time activities and well-being services.
II A broader range of society has started infiltrating the world of sports and physical activities. Many sports that historically have been the domain of male athletes are now being widely practiced by women as well. Further, sport has become an important contributor to inclusion and integration [1].
III With the rise of digital technologies and additive manufacturing, numerous options have evolved for individualizing sports equipment.

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After a short review of the current state of user-centered design, this book chapter will then take a closer look at an exemplary product—a seat pad for women's bicycle pants. Research on females has been less frequent in the past decades, and much sports equipment still focuses on male measurements. Meanwhile, women are increasingly represented in ambitious road cycling. A seat pad designed for the specific needs of women is therefore a representative example of the need for a user-centered product. We will illustrate its design process step by step to clarify and underline the specific characteristics of a user-centered design process.

On the cutting-edge understanding of the topic

Many of us have experienced the discomfort of poorly fitting ski boots, blisters on our feet during long hikes, or the soreness that arises from using an improper bicycle saddle. Conversely, how satisfying is functional, user-friendly sports equipment? The kind of equipment that makes sport a pleasure and makes you feel safe and seen as an athlete. Consider the bliss of a well-fitting backpack for individuals of smaller stature, the assurance of staying dry on a snowy powder day, or the convenience of navigating the cold mountains using a satellite-based radio navigation system without freezing your fingers.

As per ISO 9241-210:2019 [2], user-centered design aims to enhance satisfaction and improve user well-being throughout the product design process. Improving effectiveness, accessibility, efficiency, and sustainability while reducing potential negative effects of the product on user safety, health, and performance is a key principle of the approach, the latter being particularly relevant in sports engineering. That being said, a closer look at the term "design" reveals its interdisciplinary character, with definitions ranging from purely aesthetic aspects to especially functional ones.

The function of a product can be divided into two scopes: the product language and the technical scope. Design, as a concept, goes beyond the purely technical function and introduces aspects that communicate with the user and enable them to interact with the environment [3]. Both the product language and the technical scope can be subject to user-centered efforts.

The product language can be diversified by choices that the user can make during the process of buying a product—a classic application is a car configurator. Fueled not only by new developments in the automotive industry, such as battery-electric cars and complete software centers but also by the wide range of possible product language features like colors and materials available for configuration when purchasing a new car, the number of potential hardware configurations can reach up to ten billion [3]. Refer to Chapter 17 on sustainability to learn about the impact of such vast product configuration options on the environment. Apart from the automotive industry, many sports companies have introduced customization steps in their marketing concepts. For example, the American bicycle company Trek Bicycle Corporation launched a project in which "thousands of colors and design schemes" are available in the bicycle configuration process to express the exact character of the cyclist [4]. Additionally, personalized lettering can be added to the frame (Figure 20.1).

The product language, as the first of two functions of a product, is mainly the playground of designers. The technical scope of the product, in contrast, introduces individualization options that include aspects addressing the individual anthropometrics of the users. To realize both alongside each other, the development team ideally consists of various professionals with diverse skills and expertise.

The interdisciplinary development team continuously shapes and refines the product features within an iterative process—the design process. This process mirrors a cycle, wherein the product's design parameters are subject to constant adjustments (Figure 20.2). The design cycle is

Figure 20.1 Personalized lettering on a Trek Émonda that has been created as part of the "Project-One" customization concept of the American bicycle company (Authors' own image).

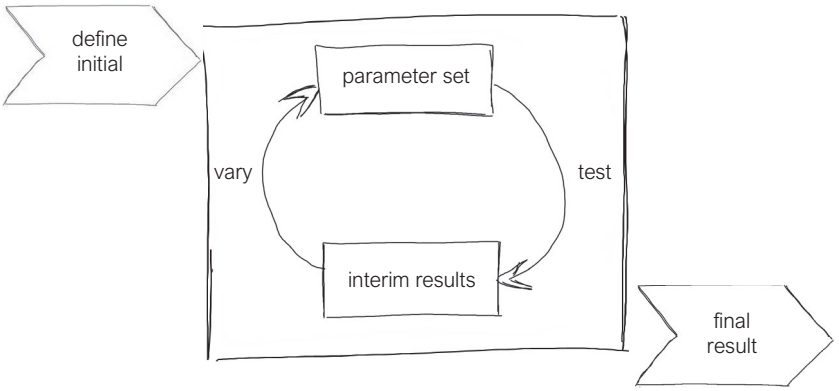


Figure 20.2 The design process of a user-centered product includes an iteration circle which is entered through a set of initial parameters and characterized by constant testing and subsequent parameter variations. The intermediate result is eventually replaced by the final result, which marks the exit of the circle.

entered by setting an initial parameter set. Often, the final product differs substantially from its initial form due to the extensive adjustments made during the design process. This transformation can be so substantial that the final product is barely recognized as an advancement of the initial one. A famous example is the Apple iPhone®, which originally included a prototype called "Extrudo," designed as an aluminum extrusion with plastic caps on its top and bottom ends. The caps were necessary to ensure the functioning of the radio antennas, but they would concurrently ruin the clean and Apple-specific extrusion design. After more iterations, the extruded version was replaced with the "sandwich" design we all know today.

Emphasizing the importance of product linguistic design features, this chapter will now focus on technical product aspects due to their relevance for sports engineering and technology.

Levels of individualization

Before starting the design process itself, it is crucial to identify the targeted user group size for the product under development. Therefore, the preferences, needs, and desires of the user group need to be determined. Note that with respect to the economic aspect of product development, the user group can also be described as the customer group. However, customers include individuals who purchase a product but discontinue its use after a limited number of times, while users keep using products on a regular basis. There are broadly three levels of individualization that can be distinguished: Level 1 (L1), which is uniform; Level 2 (L2), tailored to specific user groups; and Level 3 (L3), customized. In contrast to customization with regard to product language aspects, as mentioned above, customization here describes functional aspects based on anthropometric data.

Figure 20.3 shows the relation between the levels of individualization and the size of the user group that the product addresses. The smallest user group size is n = 1, consisting of one user only. The climax of larger user group sizes is a uniform product intended to be used by everyone. Between these two extremes, user group sizes range from large groups such as "women", or "men" to smaller, more specific groups like "professional left-footed soccer players". The larger the user group size, the more challenging it becomes to include user-centered specifications due to the vast diversity of individuals they have to address. Uniform products, here referred to as L1 products, usually undergo no individualization steps and are designed for general use. In contrast, L3 products intended for single users often include athlete-specific customization features. Significant design effort is required due to the persistently high number of manual steps, which leads to increased costs for the customer but also offers a highly rewarding user experience.

For large companies with high production volumes and highly automated production processes, the required manual production steps for customized L3 products may not be realizable. For small companies or start-ups, lower overhead costs enable the cost-effective realization of customized products. In addition, they often benefit from direct contact with their customers due to easy accessibility, which allows for valuable first-hand feedback on the products.

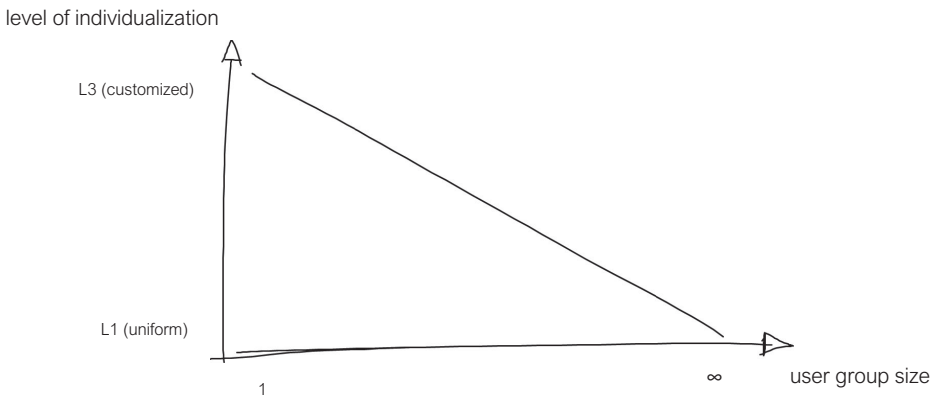


Figure 20.3 The smaller the user group size, the more individualized a product can be. Products for single users are customized (L3), while uniform products (L1) often do not include any individualization features.

Knowing the user

Once the decision on the level of individualization is made, various methods can be applied to understand the user or the user group. Broadly, these methods can be divided into qualitative and quantitative measurements. Qualitative statements describe the subjective needs of the user or the user group and can be gained from interviews with potential users and experts in the field, as well as by researching literature for previously conducted studies. In contrast, quantitative information describes the objective requirements of the user or user group, which can, for example, be measured in experiments.

For qualitative information gathering, one of the least standardized but helpful and frequently used methods to commence the process is through free interviews, where the potential user's opinions are sought through an open dialogue. One major disadvantage of this approach is the unknown loss and change of information content due to misunderstanding and misinterpretation between the interviewer and the interviewee.

More standardization is guaranteed by using predefined questions, still letting the interviewee answer freely, but restricting the interviewer's double-checks. Both the free interview and the interview with open-ended questions represent qualitative research, which aims to gather non-numerical, descriptive data.

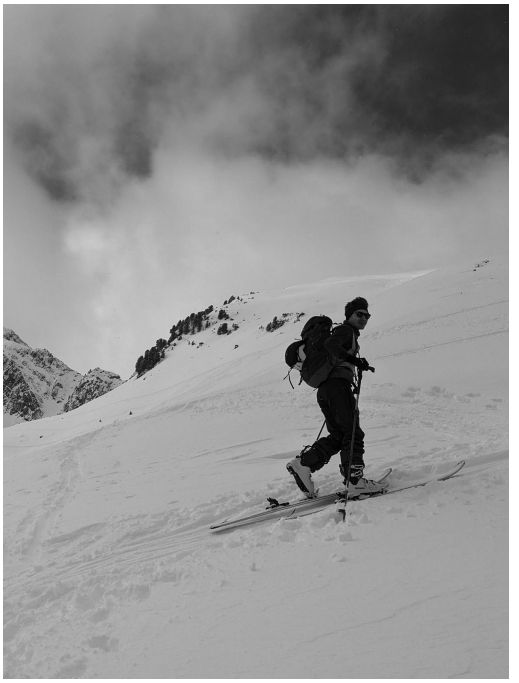
While often associated with quantitative methods, closed-ended questions, when strategically employed in qualitative research, can offer additional insights by providing structured responses that can highlight specific attitudes, preferences, or opinions within a focused framework. Hence, it requires careful consideration not only in formulating meaningful questions but also in selecting the appropriate rating scale that corresponds well with the chosen attributes describing users' rating possibilities. The classical Likert scale, for instance, is bipolar, giving two agreement levels to the negative ("disagree" and "strongly disagree") and two agreement levels to the positive ("agree" and "strongly agree"), symmetrically arranged around the indifferent judging option "neither agree nor disagree" [5]. When planning a questionnaire, it needs to be discussed and decided—among many other aspects—whether such an indifferent answer option should be allowed or, instead, make the questioned person decide either to the positive or negative.

In the effort of getting to know the user, it can be helpful to define a persona, a representative user who further specifies the user's needs in a tangible and understandable way and helps to keep the focus on the user group. An example of a persona in the imaginary context of a ski touring binding development is given in Table 20.1.

The qualitative information collected should give a first insight into the sensitivities of the user group, their challenges, frustrations, problems, and wishes. However, for engineers having to select materials, calculate dimensions, plan manufacturing processes, and determine costs, quantitative data specifying the user's needs are much appreciated, and often necessary. This requires instruments to collect user-specific data objectively, reliably, and—since real-life sport is difficult to transfer to a lab environment—with as much external validity as possible. Well-known methods include, for example, ground reaction force measurements, marker-based motion capturing, and electromyographic (EMG) measurements. In addition to established methods, the digital transformation and scientific advancements of the recent century have introduced innovative approaches to derive quantitative information about the users, such as 3D scanning and biosensor technologies.

3D scanners facilitate the digitalization of the geometry of physical objects through techniques such as laser scanning, structured light scanning, and photogrammetry. In sports engineering, 3D scanning has been instrumental in creating precisely fitting sports gear, including apparel such as functional shirts, running shoes, and skiing boots, as well as protective gear like helmets. 3D

Table 20.1 The exemplary persona "Joseph Light" represents a typical user of a specific user group.



Profession:
Joseph is an environmental researcher, focusing on the impact of human activities on alpine ecosystems. His work aims to research sustainable practices in ski touring, pointing out the need for responsible exploration and conservation in high-altitude areas.

Free Time and Hobbies:
During his free time, Joseph is often found on snowy peaks, equipped with his ski touring gear. With a deep passion for the outdoors, he recharges in the tranquility of the backcountry, where he can escape the chaos of city life and immerse himself in the raw beauty of nature. Besides ski touring, he enjoys capturing mysterious landscapes through his photography, documenting the beauty of the wilderness.

Goals, Wishes, Expectations:
Joseph's ultimate goal is to foster a deeper understanding of the complex relationship between human activities and the fragile alpine ecosystems. He wants to contribute significantly to the global discourse on sustainable practices, hoping to inspire a collective commitment to preserving these environments for future generations. With a strong wish to live in a world where environmental consciousness is strongly present in societal values, Joseph remains consequent in his pursuit of advocating for responsible ecotourism and climate conservation.

scanning can be applied for L2 or L3 products. While in L3 products, the scan of a single person is used to design a product, larger studies with multiple participants can result in average models for a specific user group (L2). For a more detailed exploration of 3D scanning in human subjects, readers are encouraged to consult Bartol et al.'s comprehensive review article [6].

Biosensors, on the other hand, encompass devices that convert biological processes into electrical signals. Commonly applied sensors for measuring human biomechanics include inertial measurement units (IMUs). Biosensors offer a deeper understanding of sports performance by capturing athletes' movements, resultant forces, and physiological parameters like hydration and fatigue. They have been employed in various sports applications, such as analyzing motion patterns in swimming, providing feedback to prevent injuries in baseball, monitoring physiological parameters in road cycling, and assessing fatigue levels in soccer players. It, however, is important and has become one of the major tasks of sports engineers to question these sensors' measurement quality by comparing them to existing "gold-standard" reference systems. Various studies have done this and several of these have revealed major limitations in the devices being assessed [7–9].

Design process

Recent developments have influenced not only measurement methods but also the way we design and manufacture. Once the design cycle shown in Figure 20.2 is entered with an initial set of parameters that appeared suitable based on the collected data and boundary conditions, and the iterative design process is initiated. Frequent testing of product versions may include simulations, physical prototypes, or field studies at a later point in time. The innovative methods in simulation and prototype production include CAE and additive manufacturing.

Computer-aided engineering (CAE) can be viewed as the extension of computer-aided design (CAD) by enabling numerical simulations. Individualized numerical simulations can be especially interesting for customized L3 product categories. They involve techniques such as multi-body simulations (MBS), computational fluid dynamics (CFD), and finite element methods (FEM), which can model various aspects of sports engineering, including athlete-specific muscle forces [10], aerodynamics, and object stresses. Coupled simulations are gaining attention as a holistic tool to represent the athlete-equipment interface, applied in areas such as swimming hydrodynamics, snowboard design optimization, and customized prosthetic socket fitting for Paralympic athletes.

Note that on the product-language side, CAD allows discovering aspects beyond the purely technical function of the product. This is demonstrated in the Apple iPhone® example mentioned above, where design features have been visualized using digital CAD models of the iPhone. In this sense, their application in the development of user group-specific products can be effective for larger-scale manufacturing. Meanwhile, recent technical advancements have significantly contributed to the evolution of digital twins, which extend beyond CAE by creating a dynamic, real-time connection between the physical object or system and its virtual representation [11].

Additive manufacturing, with its layered material accumulation process, allows for the production of complex and lightweight geometries based on CAD models, with minimal material waste. Techniques such as Fused Deposition Modeling (FDM), Stereolithography (SLA), Selective Laser Sintering (SLS), and Selective Laser Melting (SLM) have advanced over the last decade, enabling the manufacturing of various products, ranging from customized trail running shoe insoles to products intended for a broader user group, such as lightweight cycling frames or racing horse saddles. While additive manufacturing is frequently used in prototyping, another main advantage of it is the possibility to realize complex geometries like lattice structures with relatively small effort and great detail [12]. Due to this, the process is not only valuable in developing L3 products but is also frequently utilized in L2 products. Bourell comprehensively summarized the recent status of additive manufacturing and provided perspectives on future developments [13]. Readers are also referred to Chapter 16 on 3D printing in sports.

In summary, the mentioned promising technological developments for enhancing user-centered products are 3D scanning, biosensors, CAE, and additive manufacturing. The next section of the chapter presents an example that illustrates the practical application of the aforementioned theory.

An example: Seat pad for females

During the last decades, sports equipment development has strongly focused on male athletes. In cycling, the most relevant interface between the athlete and the sports device lies in the saddle region. Anatomical differences between male and female athletes are particularly pronounced in this region, yet only minimal progress has been made in the development of female-specific saddles or seat pads. Here, we illustrate a development process of seat pads for female road cycling athletes, which was part of a research project with the company SHER S.r.l. from Italy. In this

study, we documented the cycling experiences of female road cycling athletes, focusing on comfort levels, pain, and the equipment used. We developed a new pressure distribution measurement device based on impressions and used this data to design a lattice-based seat pad prototype tailored to meet the specific needs of women engaged in long-distance road cycling.

Levels of individualization

The user group that is addressed consists of female road cycling athletes who experience pain, injuries, or discomfort in the saddle region. We are therefore talking about an L2 level of individualization that targets a specific user group. However, since pressure distribution can be individually measured for any athlete at a later point in time, the seat pad could also be marketed as a customized product (L3). Incorporating personal measurements would involve more time and lead to higher costs, factors that must be carefully considered in the marketing strategy of each respective company.

Knowing the user

To understand the needs of the users we conducted literature research before carrying out additional qualitative and quantitative measurements. Our research delved into the challenges faced by women in road cycling. Discomfort was found to be widespread, with nearly half of all female road cycling athletes reporting its prevalence [14]. Common issues included genital numbness and sensory dysfunctions [15], which could potentially result in reduced arousal and orgasms [16,17]. One possible explanation for this can be found in the loads occurring in the athlete-saddle interface region, with peak pressure values likely affecting blood perfusion. In general, high pressure in the genital area at the anterior end of the saddle and in the region of the ischial tuberosities in the left and right posterior region of the saddle was found in previous studies [18–20]. Athletes were able to share useful information, such as the fact that long road cycling trips are commonly linked to vaginal yeast infections. In this literature research, we can once again distinguish between qualitative measurements, such as the subjective feeling of discomfort or pain, and quantitative measurements, such as pressure distribution.

We subsequently assembled a pool of 29 athletes who would participate in a study that, as a first step, collected further qualitative data using a questionnaire that examined athletes' historical and recent pain experiences. The need identified was to relieve pain in the saddle area. The results of the questionnaire showed that complaints in the genital or pelvic area are the most common: 38% of athletes often or always suffer from them. The posterior area of the ischial tuberosities was less affected: only 10% said they often or always felt discomfort. Most athletes (66%) reported suffering from skin irritation in the genital area at least some of the time, making this the second most common problem in the survey results.

To further understand the cause of pain and pave the way for the development of a seat pad that increases comfort, quantitative, physical measurements were carried out using biosensory methods. Our hypothesis proposes that pressure plays a crucial role in pain management, both in its peaks and in its overall distribution. Mats of digital pressure sensors determine pressure distributions. However, these mats have limited resolution and may not be able to detect local pressure fluctuations. In addition, they are subject to wear and tear and can malfunction, which can lead to measurement errors.

To overcome these limitations, we decided to develop a new pressure measurement device that is based on impressions (Figure 20.4). It consists of a saddle-shaped, stiff plastic base layer with

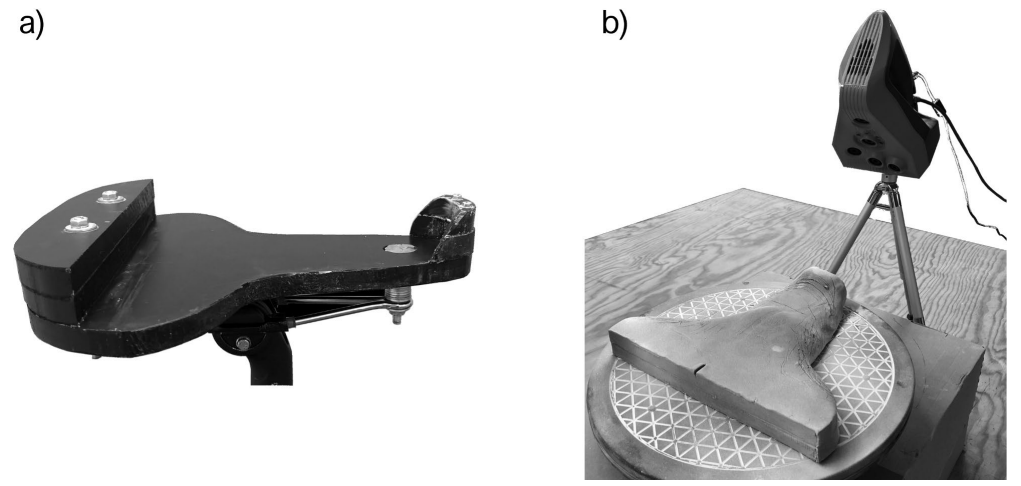


Figure 20.4 (a) The base of a measurement tool is mounted to a common saddle rail and filled with plasticine to create an imprint of the crotch area during cycling. (b) 3D scanning set up with an imprint representing the pressure distribution of an athlete on the bicycle.

fixture components at the anterior and posterior ends (Figure 20.4a) and contains soft plasticine as a molding compound. The device contains a standard saddle rail and clamp to attach it to any seat tube. We used the idmatch bikefitting ergometer in our study. After the most ergonomic cycling position of the respective athlete was determined by the idmatch system (Selle Italia, Casella d'Asolo, Italy), the pressure distribution was measured in two different load cases – static and dynamic. During the experiments, the plasticine was covered with cling film.

Using this analog sensor configuration, we attained a high-resolution indication of the pressure distribution. To optimize data processing, we aimed at the digitalization of our data. For this purpose, we used laser-based 3D scanning. The scans of the impressions were taken from multiple angles, and the resulting 3D models were aligned in CAD software using reference marks in the mold. To evaluate the scans and determine whether there are correlations between them and the data on, e.g., pain history that was collected in the survey, the height of the prints was sampled at equidistant grid points to create a discretized height profile. To simplify matters, we assumed that the depth of the impression was equal to the respective force acting at the same point.

When statistically evaluating the results, we were unable to find a significant correlation between the depth of the impression and the pain level of the athletes that we obtained from the questionnaires. Since we hypothesized a relationship between the load on the saddle and the athlete's stride, this was unexpected. A possible reason for this is the small sample size. However, we remained with the inclusion of the pressure distribution measurements in the pad design process. The qualitative and quantitative data collection forms the basis for the development process and can be found on the far left of the flow chart in Figure 20.5, which summarizes the complete process.

Design process

After having intensively studied the users with respect to conversations, literature research, a survey, and physical measurements, we were able to enter the design circle shown in Figures 20.2 and

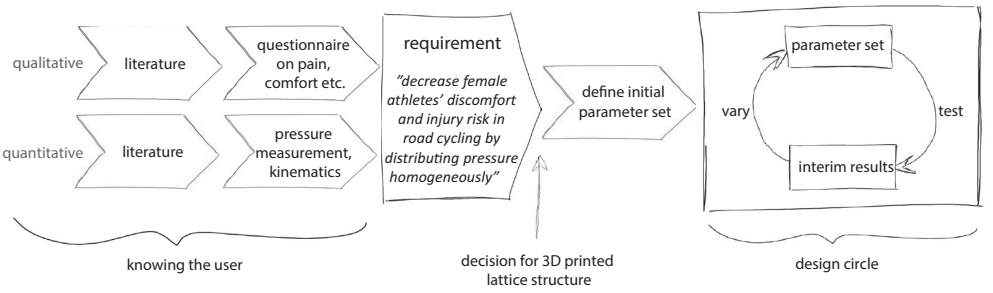


Figure 20.5 Flow chart of the seat pad development process. Qualitative and quantitative measurements were executed first to get to know the user. Based on their results, we defined the requirements. When the decision for a pad design was made and 3D printing was chosen as a manufacturing method, the design circle could be entered with a carefully determined initial parameter set.

20.5. For the group of users this product addresses, namely ambitious female road cycling athletes (L2), we set the following design requirement for the seat pad:

> We want to design a seat pad that prevents pressure peaks by distributing pressure more evenly, such that female athletes' discomfort and injury risk in road cycling is decreased.

Transitioning the level of individualization to a customized product (L3) can be achieved by measuring an athlete's pressure distribution and tailoring the seat pad to her specific needs. For this group of users, though, we averaged the height profiles across all participants gained from the measurements, both for the dynamic and static cases, and noted the resulting profile as an input for our pad design.

According to the pressure distribution, the stiffness of the pad should vary to distribute the pressure more evenly. A lattice structure could allow for local stiffness variations by varying the beam diameter of the lattice cells. More information on the design of lattice cells can be found in Chapters 9 and 10 on cellular solids. 3D printing is predestined to manufacture lattice structures due to its additive character. It is also adaptable to design changes within a certain range, which enables the switch to a customized product by adapting the lattice beam diameters in the CAD model based on another pressure distribution profile. In addition, lattice structures are characterized by the potential for increased air circulation. Combined with the use of a medical class II photopolymer (Bloxx® splint, Megadenta Dentalprodukte GmbH, Germany) material, infections could be prevented more effectively. 3D printing of lattice structures is often presented in the media as a straightforward technology. In reality, it is a complex process that requires careful calibration, appropriate beam geometries, and appropriate material selection.

Having decided on a 3D-printed lattice structure as the main concept for the pad design, new parameters were introduced into the design process: cell type, cell size, and beam diameter range. To manage the vast variations of only these three parameter combinations, we conducted a preliminary process. In multiple iterations, samples with different parameter combinations were designed and 3D printed (G2, Genera Printer GmbH, Austria) to determine appropriate ranges for beam diameter and cell type. The approach was similar to the design process shown in Figure 20.2, but on a smaller scale. This included parameter selection, 3D printing of a sample, and adjusting parameters in an iterative process. This preliminary process led to promising parameters for the initial set in the overall pad design process.

After the inside of the seat pad had been determined, the overall geometry should be defined. We derived the maximum length of the pad from the anteroposterior length of the collected pressure data [19,20]. The pad width was set to the maximum value of the distance between the ischial tuberosities in women. To ensure that the pad has enough flexibility to adapt to the crotch when walking, we added a longitudinal gap centrally at the back end.

From the preprocessing for the lattice parameters, literature research, conversations with athletes and experts, a self-conducted study with a peer group, and physical measurements of pressure distribution, we derived the following initial parameter set for the design (Figure 20.6).

During the iterative design process, parameters will be continuously adjusted to identify the optimal design. It is beneficial to initially determine which parameters will be modified at different stages of the design process. In our example, we decided to maintain the defined cell size, cell type, and beam diameter range for as long as possible, as these values were found to be particularly suitable for our requirements in the preprocessing and were in the best agreement with the given boundary conditions, as dictated, for example, by the 3D printer. To vary stiffness, we chose to adjust local beam diameters before considering a material change due to its easier applicability. Consequently, the overall pad dimensions and the local beam diameters were selected as the primary varied parameters, particularly in the first iteration steps.

With this first set of parameters, we designed a first version of a seat pad, as shown in the image below (Figure 20.7). To predict its performance, we implemented a FEM simulation of the loaded pad and 3D printed an initial prototype. By combining the physical and digital implementation of the pad, we have integrated CAE into our design process. The advantage of this approach is that a first impression of the behavior of the pad under load can be obtained before extensive field studies or destructive material tests have to be carried out. The load in the FEM simulation can be defined based on an athlete's measured pressure distribution, enabling a user-centered analysis of padding behavior during actual cycling. Once validated, the FEM model could also be used to predict the outcome of a specific design or material change.

There is potential to take the CAE simulation approach a step further. When the athletes' anthropometry and cycling kinematics, along with the appearing loads, are known, a biomechanical simulation can be implemented. MBS is suitable for representing the dynamics of a system

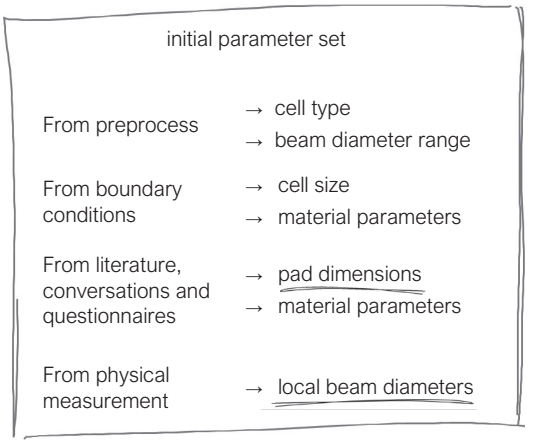


Figure 20.6 The initial parameter set as part of the design circle on the left. The parameters that are favored for adjustment in the first iterations of the design circle are underlined.

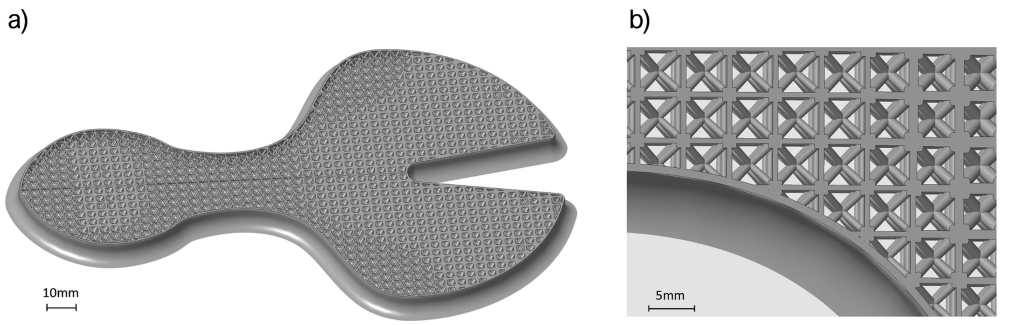


Figure 20.7 (a) First prototype of a 3D printed seat pad with the lattice showing locally adapted beam diameters according to the measured pressure distribution. (b) Detail of the beam diameter variations in the pad.

of rigid bodies. With increasing popularity, MBS simulations are supplemented by FEM bodies, resulting in a coupled simulation that is able to predict deformations and inner stresses in non-rigid bodies. Through this method, the dynamic loads between the athlete and the bicycle can be calculated, leading to more detailed load predictions and suggestions on the individual pad design. However, the complexity of simulations is directly proportional to their susceptibility to errors and must be carefully considered.

Upon evaluating the results of our initial testing, we found that the chosen design parameters worked well in the FEM simulation, meaning that we found sufficiently low compression of the surface of the pad and non-critical material stresses within the lattice structure. The printed prototype showed low durability of the material under bending loads, indicated by material failure in single lattice beams. Low flexibility was observed, especially around the anterior-posterior axis, which would most likely cause discomfort when wearing the pad while walking. To overcome the observed limitations, we incorporated a larger gap in the posterior end to achieve more flexibility around the anterior-posterior axis. Ultimately, we had no choice but to adjust the material composition to prevent material failure during bending. Optimizing a medical material for additive manufacturing can be challenging, especially when complex tasks such as 3D printing of lattice structures are intended as a use case. We recommend careful consideration of available materials before resorting to material development.

Adapting the material and changing the geometry marked one iteration in the design circle of the seat pad. However, notable shortcomings persisted, underscoring the tedious character of the process. After more iterations in the design circle, testing of the prototypes will meet the defined requirements. To further validate the function of the seat pad, the user group must be included as per field studies. Additional questionnaires will give insights into the seat pad's effectiveness in reducing discomfort and potential avenues for improvement. This ongoing process is not part of this chapter, but an article with more insights has been submitted to the Journal of Sports Engineering and Technology [21].

Controversies and future directions:

The example we provided is just one of many reasonable products that have been realized through a user-centered design approach. However, while customization in the technical function scope of a product holds great potential to solve issues for athletes, it always comes with some sort of data

acquisition. At a time of increasing interest in personal data, ethical concerns regarding consent, ownership, and privacy are increasing, highlighting the need for careful data protection measures in user-centered design processes.

Moreover, just as digital cameras are seen as a significant improvement over cameras but hardly ever reach the resolution of an analog camera, digitalization comes with its limitations. As already mentioned above, discretization always comes with a loss of accuracy. A study on the accuracy of wrist-worn activity trackers in 2019, for example, revealed significant errors in the estimation of maximum oxygen uptake and energy expenditure in running exercises [8]. In addition to the accuracy, digital products are more susceptible to integrated component failures simply due to the number of components they contain.

Note as well that the sheer availability of highly digitalized, innovative methods makes it tempting to incorporate them into user-centered sports products. However, products like this have had difficulty establishing themselves in the market in the past. One example is the adidas_1 shoe by Adidas AG, which was released to the market in 2005 and contained a microprocessor that would adjust the cushioning system based on the wearer's gait profile [22]. With a retail price of around $250, few athletes considered it worth investing in, and the shoe was eventually discontinued. The lesson learned from this example is that while using innovative methods in the design process can be advantageous, the focus of the user should remain clear throughout the process.

Looking to the future, improving experience and technology will further advance additive manufacturing and lead to more robust printing processes, with more complex geometries being achievable. The code of the human body is further encoded, and biosensors are supposed to recognize the corresponding signals. The rapid progress of artificial intelligence could eventually replace conservative measurements such as motion capture, which in turn would lead to more time-efficient, tailored implementations of numerical simulations. As this progress unfolds, it remains one of the biggest tasks of our generation to synchronize innovation with human needs in a responsible way.

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21

THE APPLICATION OF BIOLOGICAL ENGINEERING TO SPORTS TECHNOLOGY

Robyn A Grant and Stephen Spurrier

Introduction

The field of sports technology and engineering is inherently multidisciplinary. Just looking at those words, they span aspects of applied physics, maths, machinery, and equipment, all applied to sport. However, even outside of maths, physics, and engineering, other disciplines are also embedded in sports technology and engineering. For example, computer science. Computational imaging techniques, for example, are being adopted for goal-line technologies, and aspects of machine learning are being applied to large sports datasets. Biology is also useful in the field of sports technology and engineering. Biomechanics, for instance, or sensors to measure athlete physiology, are examples that easily spring to mind. Of course, to design sporting goods, there must also be some biological understanding of human shape (ergonomics) as well as movement (see Part 4: Sports biomechanics and human-equipment interaction). However, in this chapter, we move beyond the idea of simple ergonomics and biomechanics and start to explore how the tools and principles of biology can be used to further improve the design and testing of sporting goods. We define sporting goods here as products used for sport, including equipment and protective devices (e.g., clothing, footwear, running blades).

Useful definitions

The approach of applying the principles and processes of biology to applications in engineering is termed *Biological Engineering* or *Bioengineering*. A biological engineer is defined by the Institute of Biological Engineering as someone who studies "biological processes and integrate(s) them with engineering principles to develop solutions for a wide variety of technical problems… biological engineers should possess the scientific knowledge of biology, including its philosophical views, be proficient in the principles and practices of engineering, and be capable of integrating discoveries from multiple disciplines to design solutions for problems." Institute of Biological Engineering [1].

A biological engineer, therefore, is someone who takes aspects of biology and applies them together with engineering principles. It is considered a science-based rather than an engineering-based subject, aligned with the perspectives and foundations of biology. Biological

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engineering has a much broader remit than *engineering biology*, which is more strictly defined as synthetic biology applied to engineering solutions (UKRI, [2]), and can include making vaccines and synthetic blood cells, as well as new biomaterials such as biosilk. An engineering biologist will take an aspect of biology that is synthetically reproduced and apply it to solve a problem.

There are other terms synonymous with the biological influence of nature on engineering. Biomimicry, biomimetics, bionics, and bioinspiration are often used interchangeably, although there are some subtle differences between them. The terms biomimicry and biomimetics can operate on three levels: (i) morphology and structure, (ii) molecular, and (iii) process [3]. In biomimicry, there tends to be a direct link from the biological example to the synthetic product, where the biological example is explicitly "mimicked" [4,5]. This approach fits well with engineering biology, where the biology is synthetically produced.

In bionics, artificial systems are constructed that have some of the characteristics of living systems. It is a discipline that seeks to replicate, enhance, or replace biological functions with electronic or mechanical equivalents [6]. Therefore, bionics has more of a functional focus rather than a focus on form or process. In sports technology and engineering, artificial limbs, test surrogates, or testing machines, such as golf swing robots, are all bionic examples.

The term bioinspiration has largely been considered the overarching term for incorporating anything biological into engineering and, in this way, seems to align best with biological engineering. Just like biological engineering includes aspects of engineering biology, bioinspiration is an umbrella term that includes biomimetics, biomimicry, and bionics. With biological engineering and bioinspiration defined in the broadest possible way that biology can inspire engineering, these are the terms we will use for the rest of this chapter.

Sports technology and engineering design

If we are considering how biology can inspire sporting goods, it might be best to first define the process of how sporting goods are designed so that we can really start to think about where biological engineering might fit within this process. Design process models are maps or diagrams that designers follow to give themselves a plan during the process. Given the variation in the field of product design, there is no single clearly adopted design process used throughout the field. This has led to the development of multiple design processes and no clear definition of what a standard design process model should involve. One of the earliest concepts for the design process was created by Mihajlo Mesarovic in 1964. Known as the spiral model, it reveals a movement from a wider problem to a specific solution and offers three main concepts for each stage: Analysis, Synthesis, and Evaluation (ASE). It involves the collection of information related to the design problem, the formulation of solutions to solve the problem from the analysis stage, and arriving at a solution using evaluative techniques [7].

Indeed, it is important to adopt an iterative design process, where concepts and prototypes are tested and evaluated at each stage. Mehalik et al. [8] presented a review of design processes and suggested that most design processes include the construction of requirements (or specifications), a preliminary design stage, and then a more detailed design stage that can include coding, as well as testing and evaluation. Evaluation and usability were also common in many other studies. Wilson et al. [9] presented the first design process model for sporting goods design and emphasised the need to also involve users throughout the process. We present here in Figure 21.1, a generalised design process that is applicable to sports engineering (loosely based on suggestions by [9]

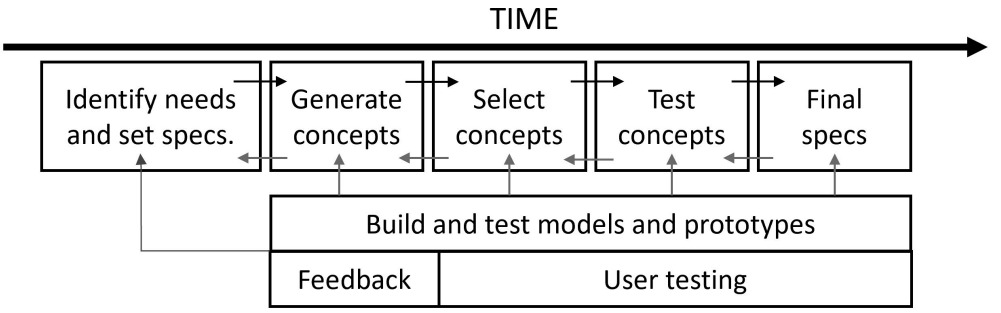

Figure 21.1 Example of sports technology and engineering design process. Black lines show a standard timeline, and grey lines indicate required feedback loops that need to be considered at the end of each stage.

(their Figure 9) and [8] (their Figure 1). The iterative nature of the process can be seen here, and many more iterations may even exist between the latter three stages. Evaluation is embedded within the prototype and user testing elements of the process, so the design can be continually updated. The model could also be expanded to include production, installation, maintenance, and end-of-life considerations, which are all key features of a design specification and are used to evaluate the success of a product. We will use this generalised design process model in Figure 21.1 to suggest where biological engineering feeds into sports technology and engineering design.

Current applications of biological engineering

Armed with sufficient knowledge of biological engineering definitions and design process models, let us now marry these two ideas together and consider how biological engineering can inform the design of sporting goods. Reflecting on this, we have three suggestions for how biological engineering is being applied in sports technology and engineering currently, above and beyond the areas of ergonomics and biomechanics. Firstly, to generate bio-inspired designs in the research stage; secondly, for generating and selecting concepts; and finally, for further testing and data analysis techniques.

Researching concepts: Bio-inspired design

Bioinspired design takes inspiration from millions of years of natural evolution and is thought to have the potential to improve products [10]. Out of all the examples we consider in this chapter, it might well be the most obvious application of biology to product design. Overall, two main approaches are taken to bio-inspired design. One of these approaches is to identify a natural solution first and use it to inspire a new concept, whereas the other is to start with the problem and look for inspiration in nature [10]. The latter is the most usual to see in the literature, probably because this is the more natural approach for an engineer to take.

The rapid-uptake sporting goods sector makes it an ideal test arena for new technologies and processes, lending itself well to experimentation with bio-inspired design. The most famous examples to date are probably swimsuits that are loosely inspired by shark skin [11,12]. The skin of sharks and rays has small denticles, or teeth-like structures (Figure 21.2b), which reduce hydrodynamic drag during locomotion [11]. Swimsuits have been made with reduced seams and

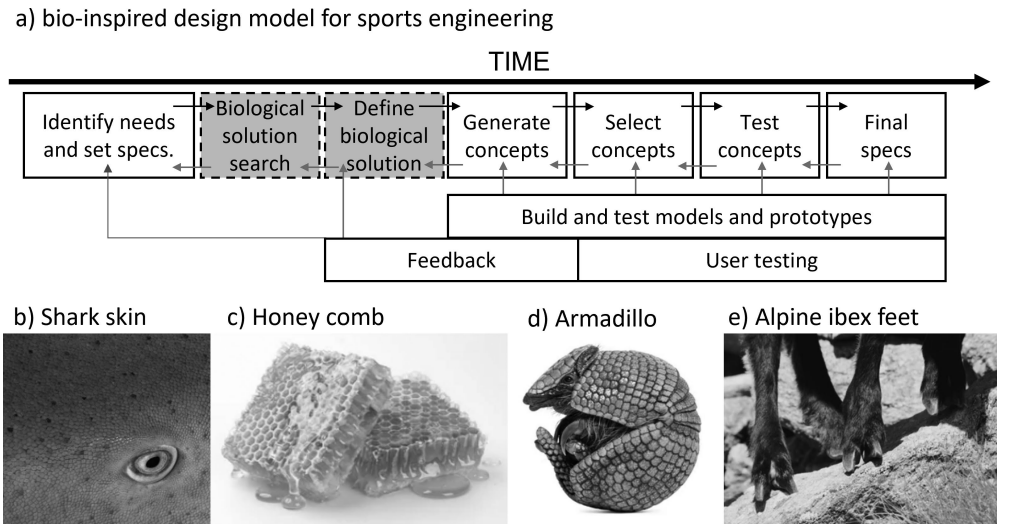

Figure 21.2 (a) Example of bio-inspired sports technology and engineering design process model. Black lines show a standard timeline, and grey lines indicate required feedback loops that need to be considered at the end of each stage. The biological engineering steps are identified in grey. Photographs of biological examples used in bio-inspired sporting goods studies, including: (b) shark skin; (c) honeycomb; (d) armadillo; (e) Alpine ibex. The biological engineering steps are identified in grey.

strategically positioned panels with various ridges and dents to enhance swimmer performance [11,12]. These textures do not strictly mimic the shark skin denticles; therefore, the suits are only bioinspired and not biomimetic. Studies of animal feet have also inspired sports engineering designers, who have looked to pad arrangements and shapes to reduce shock absorption in running shoes using ostrich feet for inspiration [13], and increasing the grip of shoe soles based on leopard feet [14].

Biological structures, or lattices, can often be seen reproduced in sporting goods and may represent honeycomb (Figure 21.2c), cellular matrices, spider webs, pinecones, or carbon structures in nature (Readers wishing further information on such structures are referred to Chapters 9 and 10 on Cellular solids for sports applications). These structures have been used in water sports boards [15], scooter decks [16], and helmet liners [17,18]. They can offer high stiffness-to-weight ratios [15,16], which can enhance the performance and safety of such sporting goods.

Animal exoskeletons are flexible but protective and have been an inspiration for some sporting goods. For example, seahorse spines have inspired backpacks [19], and armadillo shells (Figure 21.2d) have inspired knee protectors [20]. These shapes have good impact protection and support but also offer flexibility and comfort. In most of these examples, the properties of the biological idea loosely coincide with those of the engineering problem; however, in most examples, it is unclear how the biological example itself is selected. There is little evidence in the literature to suggest that comprehensive and systematic biological example selection often occurs. Indeed, sometimes only one natural example is given, with limited justification as to why that one is chosen specifically.

Gorea et al. [21] propose a multi-stage process to formally consider how to systematically search for a biological solution. These stages include (i) defining the problem, (ii) searching for an analogous biological solution, and (iii) designing a solution. They used sports bras as a case study and wanted to solve the problem of moisture wicking. They conducted literature reviews and brainstorming to identify biological solutions, which included biological terms such as "passive actuation," "plant biomechanics," "hydrophilic fibres and surfaces," and "cell structure." They then adopted natural hydrophilic plant fibres into sports bras and conducted extensive testing and evaluation to assess wicking. This really showcases biology integrating into the engineering design process, as it is the natural fibres themselves in the product rather than a synthetic alternative. The development of bio-inspired synthetic materials is also increasing, especially in fashion. One example is synthetic spider silk, which can make products biodegradable and waterproof, although it has not yet appeared in the commercial marketplace.

Spurrier et al. [22] also systematically searched for a biological solution to inspire climbing shoe designs. However, rather than looking in the literature, since the climbing feet of very few animals have been formally described, Spurrier et al. systematically described the feet of 166 mammal species from museum specimens. They identified textures and foot pad arrangements associated with climbing mammals (e.g., the Alpine ibex, Figure 21.2e), and the next stage of their work is to (i) describe textures and pad positions in more detail in order to understand their role in increasing friction; (ii) identify candidate designs; and (iii) test candidate designs. Indeed, Helms et al. [23] also suggested that the bio-inspired design process involves multiple stages. It needs a stage to (i) search for the biological solution (which may involve describing the outcomes too, as per [22]) and then to (ii) define the biological solution before moving on to candidate designs. We have incorporated this into Figure 21.2a, which showcases how bio-inspired design can be used to search for and then define or describe biological solutions (grey boxes in Figure 21.2a) before generating concepts.

Generating and selecting test concepts

Bioinspired design mainly looks to natural structures for inspiration. The fields of evolutionary biology and palaeontology have generated many ways to measure and analyse natural shapes and structures. Geometric morphometrics (statistical shape modelling) is one such established technique [24–26]. It can take into account variations in shape (geometric properties) or form (shape and size) and help relate performance to form [26]. Geometric morphometrics can be applied in two or three dimensions [24–28]. Grant et al. [29] used two-dimensional shape analyses, termed morphometrics, to quantify how tennis racket shape has changed over time. Specifically, they extracted the two-dimensional outlines of 515 tennis rackets and modelled them using elliptic harmonics—a mathematical formula that can be used to capture the shape of each racket (e.g., Figure 21.3ci). Using principal component analysis (PCA) to capture the variation in their data, they showed that innovations in materials, especially in fibre-polymer composites, have overcome the mechanical constraints of wood, allowing modern rackets to have larger and more oval (or egg-shaped) heads than their wooden predecessors. Morphometric analyses enabled the racket shape to be described, quantified, and explored in detail. Using this technique, the authors were able to generate "mean shapes" of rackets (Figure 21.3cii), which can be used to select representative racket shapes for further computational testing, such as finite element analysis. The principal component data (e.g., from PCA 1) could also be used to select a median racket example (Figure 21.3ciii) that can be tested further, either experimentally or computationally. The PCA space

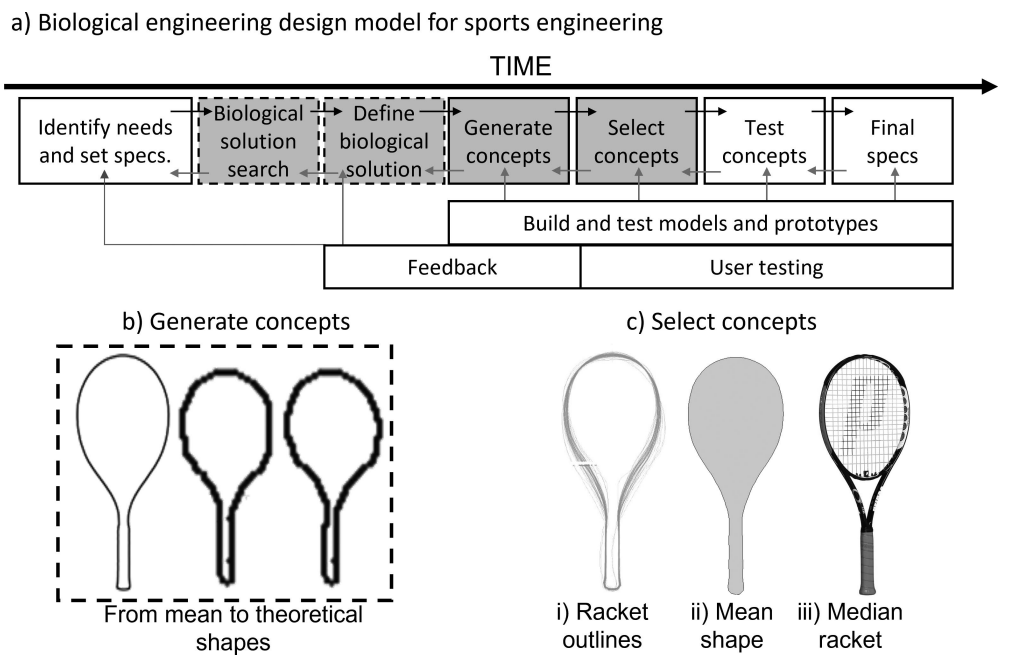

Figure 21.3 (a) Example of biological engineering design process model for sports technology and engineering feeding into biological solution searches and solution, as well as the generation and selection of concepts (all in grey boxes). Black lines show a standard timeline, and grey lines indicate required feedback loops that need to be considered at the end of each stage. Generating concepts can be through theoretical morphospaces (b) or generative design, and selecting concepts could be selecting specific summary measures (mean or median; photograph shows a Prince Ozone 7 racket) (c) or using genetic models for selection.

(e.g., plotting PCA1 against PCA2) creates a shape space, or *morphospace*, which illustrates the distribution of all the racket shapes tested. Because racket shape can be captured using an equation, simple adjustments to the variables can transform the shape of the racket. Therefore, a representative racket shape can be created for any point on the morphospace, even if there is no data from that area. Furthermore, the whole area can be extended into a theoretical zone called a *theoretical morphospace* [30]. This is an emerging technique in evolutionary biology used to explore all possibilities of a given form. Using this technique, we can go beyond the bounds of previous designs and extend into a theoretical space [30] to explore future design concepts that have not been used before (e.g., Figure 21.3b). So far, morphometric analysis has only been applied to tennis rackets in two dimensions, but there are many applications across sports technology and engineering.

Another area of design that often relies on prior designs to inform new concepts is *Generative Design*. It too draws inspiration from evolutionary biology. Evolution is the change in traits of biological populations over successive generations. These traits are therefore heritable (i.e., caused by the passing on of genetic material to the next generation). The most popular generative design tools are loosely based on evolution, using genetic models and artificial intelligence to iteratively create concepts that are usually constrained to certain design specifications [31,32]. Commonly, these are produced in computer-aided design (CAD) software to produce three-dimensional shapes and designs [31] and have so far been applied to developing trainers (e.g., Under Armour Architect) and artificial

limbs. However, image datasets can also be used to train AI models to generate design concepts. This has been trialled on a dataset of trainer images to create new trainer design concepts [33]. Once these design concepts have been generated, it is imperative to shortlist them to the most suitable candidates for testing. This can be done with user input in feedback sessions [9] but also more formally using design specifications and quantitative techniques, such as the Pareto genetic algorithm [32], which ranks the concepts according to their fit to the specification and its priorities. Once the concepts are shortlisted and produced—either physically or in CAD software—they are then ready for testing.

Testing concepts

Testing is applied in sports technology and engineering to measure product wear or degradation, as well as performance and safety. Many of these tests need to be inspired by human form and movement to properly replicate how the product is used (Readers seeking information on measuring athlete movements in field settings are referred to Part 5 on field testing, sensors and instrumentation). Therefore, these tests are inherently associated with biological aspects of the human body, such as its shape and movement. Testing can include simulations, such as computational fluid dynamics and finite element analysis, as well as physical testing. Sometimes users are also directly asked to test and provide feedback on sporting goods, and sometimes engineering mimics are used instead to provide repeatable and safe physical testing of sporting goods. This includes the use of bionic surrogates and robots to mimic human form, movement, or both.

Surrogates are especially useful for measuring the efficacy of sports safety products, such as helmets, knee pads, wrist protectors, and padded clothing. Payne et al. [34] conducted a review of human surrogates for the study of sports safety products. They made a shortlist of requirements that surrogates should have, many of which need to mimic the biological form, including having biofidelic: (i) external geometries, (ii) inertial properties, (iii) tissue structure, and (iv) tissue impact responses. Therefore, both the shape and the material of the surrogate will often be inspired by biology, especially in providing realistic responses of the conformability of materials to mimic skin, muscle, tendons, and bone. Indeed, one area that has garnered interest in sports technology and engineering is that of developing realistic skin simulants [35,36]. In other areas of engineering, surrogates are usually associated with serious or life-threatening injuries, such as penetrating bullet impacts or vehicle impacts. In sports, however, injuries such as lacerations and contusions are common and debilitating [34]. They may also deter people from participating in the sport. Therefore, realistic skin simulants have been developed for surrogates. They can be used to test sporting goods and assess the prevalence of cuts, bruising, and grazing by examining the skin simulant after testing [37]. Having a surrogate with a similar coefficient of friction and compliance to skin [38] may also better reproduce the fit and movement of protective products on the body.

While having the most realistic surrogate possible is useful for research testing, once these surrogates are incorporated into standards (e.g., ISO) and sporting governing body regulations, there is pressure for them to be easy to reproduce, with limited manual steps. For example, manually dipping a surrogate into a silicone skin simulant might add variability to the process, causing testing to not be repeatable and equivalent between surrogates. Therefore, while a research surrogate may be largely biomimetic, surrogates in standards and regulations are more bioinspired, and future developments must balance realistic and reliable testing.

Testing of sporting goods often needs to replicate use and have relatively realistic biological motion. Bio-inspired robots can be used for testing sporting goods, and many types of sports technology and engineering robots exist. Perhaps the most common are robot arms, which have been demonstrated to play racquetball games and volleyball [39]. Bespoke robots can also be produced

to mimic aspects of use, such as a tennis serve or golf swing [39]. Indeed, most sports engineering tests are somewhat inspired by the product in use. Robots can also be used as training aids and even participate in sports themselves (such as the robot football tournament) [39].

These robots can be thought of as Artificial Intelligence (AI). AI is a useful tool for sports technology and engineering, for generative design (as we briefly discussed above), as well as for data analysis and testing (Readers wishing to learn more about AI applications in sports are referred to Chapter 33, "Artificial intelligence approaches in sport"). AI is inspired by biology, specifically in the fields of neuroscience and cognitive science. AI is inspired by neuroscience in two ways. Firstly, AI can mimic the behaviour or intelligence of biology. For example, the sports robots above all have control mechanisms to reproduce simple behaviours, such as a tennis serve. The second way is that the structure of the AI algorithms mimics brain structures and function. Sakai [40] built a ball-pitching robot that has three rollers and two actuators; the motion of these components is controlled by a hierarchical neural network, inspired by the architecture of the brain, so the machine can repeatedly pitch balls reliably to any specification of ball speed, direction, and rotation. Many algorithms have these hierarchies—multiple layers that have different functions, with feed-forward and feedback loops, just like we can see in the brain. Indeed, the first models of AI (neural networks and deep learning) were all generated by cognitive scientists and psychologists with an understanding of neuroscience [41]. While the two fields have separated somewhat recently, AI algorithms are still loosely inspired by the brain. Research in neuroscience is continually discovering new things, and many researchers believe that neuroscience still has an important role in generating ideas that will continue to accelerate and guide AI research [41].

Where does biological engineering fit into sports technology and engineering?

We adopt here the terms biological engineering and bioinspiration as the loosest possible association with biology, and with these definitions, we suggest that biological engineering is useful for sports technology and engineering design. Using a range of examples, which are in no way exhaustive, we demonstrate that biological engineering can be adopted at many different stages throughout the sporting goods design process (Figure 21.4a). It can be used to generate and select concepts, as well as for testing and data analysis. Indeed, biology can be used to inspire new design concepts for sporting goods in the process of bio-inspired design (Figure 21.4b). Theories and techniques in evolutionary biology and neuroscience can be used for complex shape analyses, and AI can be adopted to generate designs, control robots, and analyse complex data. Bionics can also be adopted to develop life-like surrogates (Figure 21.4c) and robots (Figure 21.4d) for product testing. Therefore, we see the possibility of embedding biology throughout the design process.

It is worth bearing in mind that, as their title suggests, sports engineers are primarily engineers. While sports engineers may work closely with sports scientists, coaches, biomechanists, and physiologists, it is rare for them to collaborate with evolutionary biologists, comparative biologists, or neuroscientists. Hassabis et al. [41] make a compelling argument for neuroscientists to be better integrated with AI research to develop new ideas and applications. Furthermore, Graeff et al. [42] also identify that biologists should be incorporated into teams developing biomimetic or bioinspired designs. Both papers identify biologists as an asset for the future development of these fields. We would also support this and suggest that transdisciplinary working would be beneficial for future innovations in sports technology and engineering.

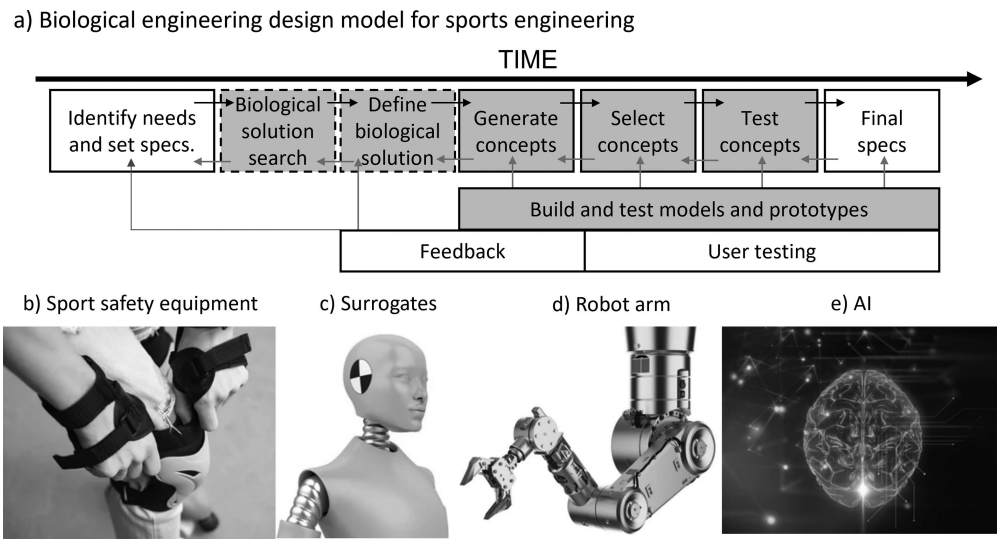

Figure 21.4 (a) Example of biological engineering design process model for sports technology and engineering feeding into biological solution searches and solution, generation and selection of concepts and testing concepts (all in grey boxes). Black lines show a standard timeline, and grey lines indicate required feedback loops that need to be considered at the end of each stage. Testing can take place for (b) sport safety products, for example, here with knee pads and wrist guards; (c) surrogates can be used in testing; (d) robot arms can make realistic movements to simulate user movement for testing; (e) AI can be used for testing control and analysis.

Conclusions

As well as the study of the user, or athlete, we posit here that inspiration from the wider field of biology can have useful and important implications for sports technology and engineering design. We have presented a broad range of examples and summarise that biological engineering is useful for sports technology and engineering design, especially in the research of concepts, the generation of design concepts, and the testing and analysis of the designs.

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22
DESIGN FOR ATHLETES

Andrey Koptyug and Mikael Bäckström

Design in general and design for athletes

Multiple definitions of 'design' can be found on the web and in dictionaries in relation to technical aspects. These definitions refer to 'plan', 'underlying scheme', 'protocol for carrying out or accomplishing something', 'arrangement of elements or details in a product' [1], and a set of descriptions of 'how a product is to be made and how it will work and look' [2]. The majority of books and courses for technical designers (many sources are available; see, for example, [3]) refer to major stages of a process flow, such as: (i) defining targets and constraints, (ii) generating concepts, (iii) selecting adequate concepts, (iv) conceiving a prototype, (v) testing the prototype against the constraints and targets, and, if necessary, returning to (i) or (ii) until an adequate final design is achieved and transferred for further realisation (Figure 22.1).

It is also possible to outline numerous intermediate stages, but the process is always entered via formulating demands and constraints, and during all stages of the process designers analyse the results against them. Often, more than one iteration is needed before an adequate design is achieved. In some cases, one would even need to change initial settings, realising that with the existing technology level, it is not possible to achieve the demands within the chosen constraints and repeat the whole process again. Even when the designer achieves what is assumed to be a 'final' version suggested for manufacturing or in-depth testing of the corresponding prototype, it may happen that some additional demands or constraints will appear and new iterations will be needed. Note that although older versions of a design process flow may not assume prototyping, in modern settings, it is always advisable. This is determined by developments in computer technology, allowing for 'virtual prototyping and testing' and fast physical prototyping with 3D printing, allowing for prompt real testing. This discussion leads us towards the question: what are the most critical demands and constraints one encounters when entering the process of designing for athletes?

It is possible to analyse it from the point of view of the athlete (here we can refer to user-centred design) and everybody else (societal, environmental, economic aspects, etc.). There is a lack of publications related to general aspects of user-centred design (UCD) in relation to technology, sports technology in particular (e.g. [4]). Nevertheless, a lot of information can be found in standards and governing body regulations (e.g. [5]) and in the introductory parts of publications

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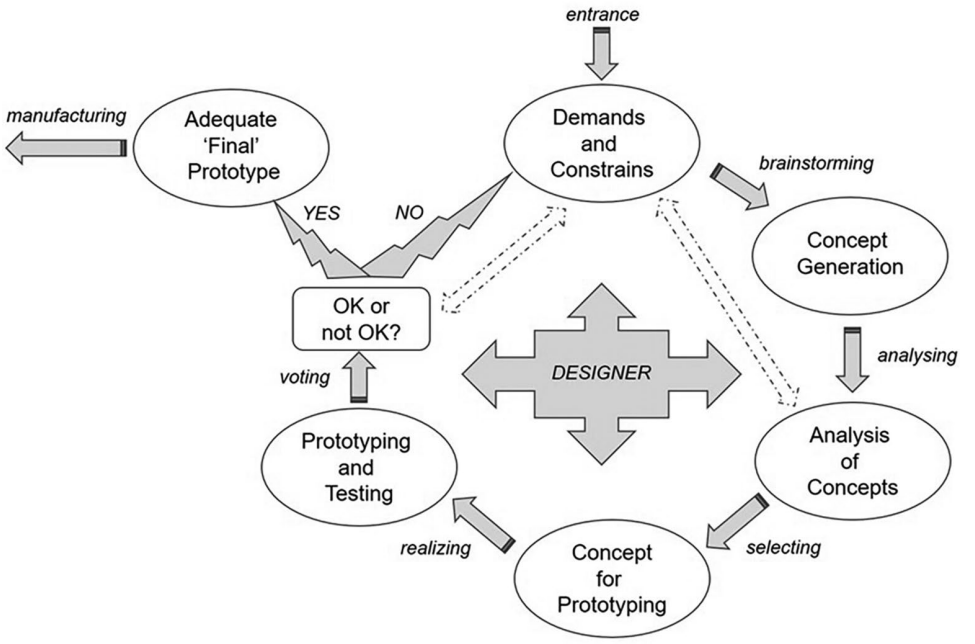

Figure 22.1 Major steps of technical design cycle.

focusing on other subjects and disciplines (e.g. [6–9]). Starting from the user's point of view, many athletes and coaches would select 'performance' and 'safety' as the most critical demands, most probably also adding 'comfort' and 'image' as important factors. Clearly, a training and competing athlete is inseparable from apparel, footwear, and technical equipment, and designers should consider these aspects in their interaction with the 'outside world'. Now, the list of demands and constraints starts to expand, adding manufacturability, economic aspects, testability, reliability, sustainability, societal factors, general legislation, and sports discipline-related legislation, etc. (Figure 22.2).

Many of the mentioned aspects are interdependent, and changes in design affecting one of them could affect others. Thus, choosing the proper design path demands a broad background knowledge to break the demands/constraints into critical (must be fulfilled), necessary (should be fulfilled), and optional (good if they can be fulfilled). In many cases, such knowledge is broader than that acquired through studies of purely technical design.

An additional challenge is represented by the rapid penetration of newly emerging technologies and ideas into sports. The widespread presence of digital and computer-based technologies is today common and widely appreciated. The value of different Artificial Intelligence (AI) methods for sports-related applications is already recognised, and AI is beginning to influence the design process in many aspects, for example, by providing advanced analytical possibilities and strategies for improving performance and safety features (e.g. [10–12]). However, the rapid development of technologies in areas previously having no connections to sports is starting to generate exciting new ideas. For example, recent developments in space exploration related to space tourism have already contributed to the concepts of space sports. Such ideas have appeared in science fiction novels for decades, and now sport scientists are beginning serious discussions on emerging realistic possibilities (e.g. [13]).

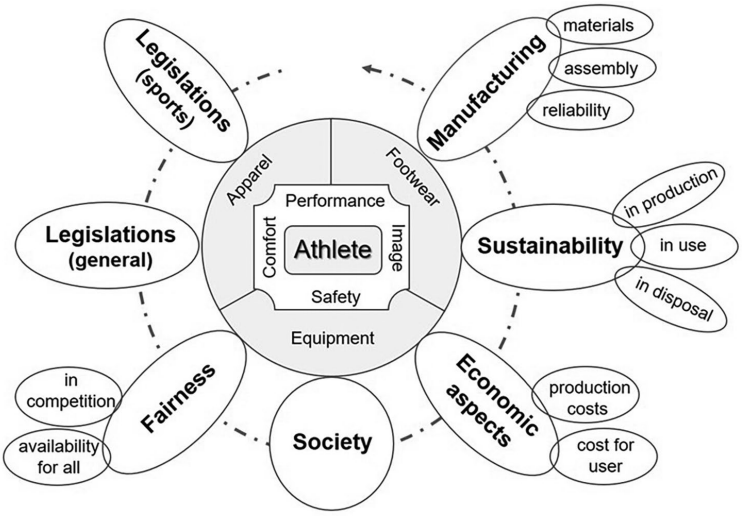

Figure 22.2 Some of the major issues defining strategy and workflow of designing for athletes.

This chapter discusses some critical concepts essential to the process of designing for athletes, navigating through multiple relevant issues. It should be noted that designing for disability sports involves additional aspects that should be considered, which are discussed in Chapter 28, Biomechanics of athletes with impairment and adaptive equipment.

Design for performance and functionality

Performance is one of the major factors driving athletes, coaches, and sports in general (see, for example [14]). The performance of competing athletes is also a factor that drives increasing interest from spectators and followers. Top athletes become role models, stimulating and inspiring others, helping to increase their confidence and build better performance, and influencing the decisions of many young people choosing sports as both a recreational activity and a career pathway (e.g. [15–18]). From this point of view, technical design should be integrally coupled with the artistic design, generating an attractive image of the athlete and their apparel, footwear, and equipment.

During the design process, targeting the best performance is deeply connected to the concept of design for functionality (e.g. [19–21]). Indeed, when stating that something should *perform* (thinking of the best achievements), it is assumed that all components should work properly and have optimal *functionality*. In many cases, functionality becomes a major driving consideration, even leading to a minimalistic approach where "item equals function" (e.g. [22]). In most cases in design for sports, one deals with multifunctional systems that cannot be wrapped or separated into single-function objects. However, functionality-driven analysis often helps in better conceptualisation and structuring of the design process.

Design for performance, ethical issues, and legislations

Modern advances in design, materials, and technology have led to a continuing race for increased performance, raising an important question: what is the contribution of the technology versus the athlete in the achieved results? This dilemma has attracted the attention of the research community

for decades (see, for example, [23–27]), but three specific cases have caused many publications in both research journals and the popular press. Introducing so-called 'sharkskin' swimming suits initially caused controversy, but corresponding regulations were timely put in place (e.g. [28,29]). The so-called Nike 'Vapor Fly' racing shoes case (now often referred to as super shoes) and Oscar Pistorius' 'running blade' still stir the media (see, for example, [30,31] and [32–35], respectively). Some concerns are also related to the introduction of active elements into sports equipment, for example, sensors (e.g. [36]) or active dampers in snow sports [37,38]. So far, such active elements are forbidden in competitive sports and are used mainly for research purposes.

Another case representing the wide penetration of new materials and associated technologies into sports equipment is related to carbon fibre (e.g. [39–44]). Carbon fibre-based composite materials, with their exceptional weight-to-strength ratio and ability to be formed into complex, optimally designed shapes, have already revolutionised the manufacturing of high-performance equipment in badminton and tennis, cycling, golf, skis and snowboards, and water sports, and continue to impact all sports disciplines. One of the interesting recent trends is the use of carbon fibre in sports footwear (e.g. [43,44]). These examples clearly illustrate the need for modern designers to be proficient in material science and manufacturing technology. Readers seeking further information on composite materials are referred to Chapter 11.

An interesting case is related to the introduction of bulk metallic glass (amorphous) metallic materials into sports equipment. Bulk metallic glass (BMG) is an emerging engineering material without a regular crystalline structure. One of its unique properties is high resilience, or the ability to return most of the energy it stores elastically (e.g. [45–47]). When introduced into the construction of a tennis racket, it is claimed to provide up to 30% more energy delivered to the ball. As claimed by the popular press and some of the manufacturers, BMG introduced into golf club heads and tennis rackets provides notably longer shots. Although BMG-enhanced equipment is not allowed in competitions, a few companies continue the application of amorphous metals and the development and manufacturing of BMG-enhanced sports equipment such as tennis rackets, golf clubs, skis, and snowboards. Although rather complicated to manufacture, BMGs are becoming increasingly affordable. Moreover, it is just a question of time before BMG-enhanced running blades or poles for pole-vaulting will be introduced, further changing the capacity for achievable results.

Technological progress never stops, and it inevitably penetrates sports-related areas. Moreover, although ethics and fairness in competitions are intensely discussed mainly at the societal level, it is essential to cover such aspects in scientific and technical publications (e.g. [26,48–52]). The availability of solid research data related to the impact of novel technologies and solutions in sports together with data-supported discussions related to competition fairness, provides essential grounds for the work of various governing bodies. Because of the fast changes, existing rules and regulations related to sports competitions need constant amendments and updates. Corresponding amendments in rules and regulations in a way initiated through novel designs, return as restrictions to new designs. There always exists a significant risk if designers and manufacturers of sports equipment, apparel, or footwear are becoming aware of certain restrictive regulation too late, when introducing changes could be expensive or impossible. Thus, to be successful, novel solutions and designs should be verified against rules and regulations within specific sports and disciplines. It should also be noted that achievements in automatic pattern recognition, virtual reality, and artificial intelligence are starting to penetrate the decision-making and judging processes in different sports disciplines, helping to decrease, if not eliminate, subjective bias (e.g. [53–55]). Today, developing and implementing such systems has become an important application area within sports technology.

Although equality aspects are high on the agenda for modern design in sports, one should not dismiss aspects related to human anthropology (e.g. [56–63]). Corresponding issues are most pronounced in garments and footwear for different age and sex groups of able-bodied athletes [59–62] and in designing for disability sports (e.g. [64–66]). Those who are interested in the history of the subject are referred to a recent research paper on the history of female sports and activewear [63].

Design for affordability and manufacturability

In many cases, the issues of ethics and legislation have a direct link to the issue of technology affordability. One example is user-centred design within technology for inclusive physical activity and disability sport (e.g. [64–66]). Here, the aspects of individualisation are intertwined with affordability or, in simpler words, fairness of access (e.g. [67]). More details on design for inclusive physical activity and disability sport are given in Chapter 28.

It is hard to foresee all aspects related to affordability during the design stages, as proper systematic analysis demands active knowledge in business and administration, as well as production and sales, among other things (e.g. [68]). However, general knowledge of cost analysis and the methods and systems involved is essential (for an overview, see, for example, [69,70]). Unfortunately, in some cases, education programmes developed for designers are not covering this knowledge area, as the focus is commonly shifted towards the technical and technological aspects, including manufacturing.

Main aspects of manufacturing and manufacturability issues are commonly well covered by handbooks for designers and different research articles (recent examples can be found in [71–75]). In addition, a general understanding of concepts like design for manufacturability and assembly/disassembly could be effectively used in relation to design for sports, especially when complex systems are concerned. One good example is the modularisation of equipment for assistive/inclusive physical activity and disability sport helping to resolve the contradiction between availability for all users and individualisation. The main idea is to break the system into functional modules, individualising only necessary items and incorporating possibilities for adjustments into others. One typical example can be represented by developing affordable sitski constructions (see, for example, [76,77] and references therein). Sitskis (sometimes also referred to as a sit-ski) are used by people with limited (or absent) use of their lower body. It consists of an individualised seat (bucket), often moulded or made from composite materials, exactly fitting the user, and a frame mounted on one ski. Certain elements of the sitski frames from different manufacturers or research groups are common or universal (for example, the shock absorber or the element binding the frame to the ski). However, users commonly need to have different adjustments, such as the height of the seat under static loading, the offset for the centre of gravity, the stiffness of the shock absorber, and the limit of its extension in dynamic situations, etc. Although individualising all elements of the sitski may seem like an optimal solution for the best performance, it may be prohibitively expensive for many users. Thus, full individualisation of the bucket (essential not only for performance and comfort but also for safety) could be combined with a modularised frame, allowing interchangeable elements and adjustments, resulting in more affordable solutions.

Design for safety and reliability

Safety in recreational or competitive sports is one of the most important factors, along with performance. Injury prevention is an important consideration when designing for people participating in sports. During the design phase, it is important to become aware of the injury risk factors common

in specific disciplines. One can refer to handbooks (e.g. [78,79]), review papers (e.g. [80–82]), and guidelines (e.g. [83–85]), focusing on this subject. Extra attention should be put in place when designing for athletes with limited functionality or disabilities. One should remember that such athletes, in many cases, are more vulnerable to additional injuries. For example, able-bodied alpine skiers have a certain shock-absorbing capacity provided by the feet. For athletes in sitskis, all shocks transferred to the bucket seat are transferred to their spine almost without any attenuation, thus the equipment should provide the needed shock absorption. In many cases, amputation disrupts nerve pathways, which can lead to partial or complete loss of sensitivity in the remaining part of the limb or hand. A sitskier tightly strapped into the seat may have additional restrictions in blood circulation, which also increases the risk of severe hypothermia or frostbite. Consequently, in winter sports, disabled athletes can have a higher risk of frostbite, even in supposedly mild outdoor conditions. Thus, extra measures should be taken when designing equipment and apparel, focusing more on safety rather than just comfort. Publications related to the subject of safety in disability sports are available (e.g. [86–88]).

Static and dynamic testing are another aspect of injury prevention strategies (e.g. [89]). An interesting example is provided by studies on the design of wrist protection devices. Corresponding publications illustrate the process of analysing risk factors and potential injuries [90,91], the design process, and the assessment of protection efficiency [92], as well as objective testing methods [93]. It should be noted that testing safety devices can result in a 'Catch 22' situation: equipment should be safe before it is allowed for use, but new protection equipment cannot be tested by people as it is not yet shown to be safe.

Computer simulation and advanced functionality assessment using computer models have, for some time, been almost the only way around. Modern models have reached complexity and fidelity levels, allowing them to be referred to as 'digital twins' of real processes, bodies, and body parts [94,95]. Together with the digital twins of safety devices, such systems allow the modelling of real-life scenarios and events allowing for virtual testing and assessment of safety devices. Corresponding virtual tests are performed without endangering human participants, which is one of the main advantages of digital twin systems. Further development of twin-based testing has led to the availability of 'physical twins' or realistic models made using additive manufacturing technologies (e.g. [96–100]). These physical models, often supplemented with embedded sensors, allow for real static and dynamic testing and assessment of safety devices and equipment. Complementing digital and physical twins, these technologies are becoming valuable tools in the assessment of equipment and systems without endangering users. Readers seeking more information on physical twins are referred to Chapter 27, sports injury and trauma biomechanics: surrogates, human data and modelling.

Design for sustainability

Sustainable development is one of the critical issues that define the progress and survival of humanity. Discussions on issues such as saving energy and other resources, moving towards more environmentally friendly technologies, and recycling are becoming everyday realities. Corresponding concerns penetrate all aspects of our lives, including sports, and demand that certain actions are needed (see, for example, [101–105]). United Nations commissions and other international organisations and forums take the lead in developing corresponding "New Economics for Sustainable Development" (NESD) concepts and science-based Sustainable Development Goals (SDGs) [105,106]. Table 22.1 shows the NESD developed by the United Nations and their relevance to sports and sports equipment design and manufacturing. Several publications provide

Table 22.1 "New Economics for Sustainable Development" (NESD) concepts developed by UN [5, 105, 106]

NESD acronym	*Key actions*	*Relevance to Design for Sports*
Blue Economy	Reduction of dispersion of pollutants to waterways and oceans	Indirect
Green Economy	Less carbon intensity through higher reuse and recovery of secondary materials	Very high
Orange/Creative Economy	Design-thinking for resource-efficient production and processes, and use of resource-efficient digital technologies	Very high
Purple/Care Economy	Valuing the innovation and productive capacities of labour and improved living conditions (environmental, health)	High
Social and Solidarity Economy	Opportunity for using human innovation capacities in recycling, recovery, and repair activities, including solidarity economy models conducive to upgrade working conditions from informal to formal sectors	High
Yellow/Attention Economy	Norms setting to shift from goods-ownership to delivery-as-a-service models, highly driven by digital infrastructure and sustainable consumption behavior	High
Frugal/Innovation Economy	Reduction of inputs and natural resources; higher traditional/natural material usage; and emphasis on product and systems design to minimize waste. A circular economy also improves efficiency by aggregating resources through asset sharing business models	Very high

detailed descriptions of the Sustainable Development Goals (e.g. [105–111]), some of which are specifically related to sports (e.g. [107–118]). Among the goals most relevant to sports and the design of sports equipment are those of ensuring healthy lives and promoting wellbeing for all at all ages (SDG 3), ensuring equality and inclusivity (SDG 4, 5, 10), and ensuring sustainable growth (SDG 8, 9, 12, 14, 15, 16). However, only one SDG formulation contains the word 'urgent': "Take urgent action to combat climate change and its impacts" (SDG 13). Table 22.2 provides a list of SDGs indicating their relevance to sports in general and design for sports and athletes in particular.

Several reports describing the status and progress in achieving sustainability goals are compiled by different projects and initiatives (e.g. [113,116–122]). So far, there are no universally accepted sets of indicators allowing quantitative assessment of how organisations progress in implementing sustainability goals in relation to sports and sports equipment, although some organisations and initiatives are working towards achieving this (e.g. [120–122]). There is currently a lack of research publications assessing the progress in achieving sustainability goals by sports goods and equipment manufacturers, with the main information scattered across different public websites and social networks. However, publications by Heidi Niskanen provide interesting insights into how Nike, Adidas, and Puma are implementing SDGs (see, for example, [123] and references therein).

Table 22.2 List of Sustainable Development Goals according to [S5], and their relevance to sports in general (S), and design for sports and athletes (D) in particular

Goal #	*Sustainable Development Goal description*	*Relevance to Sports in general (S) Design for Sports (D)*
Goal 1.	End poverty in all its forms everywhere	S+
Goal 2.	End hunger, achieve food security and improved nutrition and promote sustainable agriculture	
Goal 3.	Ensure healthy lives and promote wellbeing for all at all ages	S++, D+
Goal 4.	Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all	S+, D+
Goal 5.	Achieve gender equality and empower all women and girls	S++, D++
Goal 6.	Ensure availability and sustainable management of water and sanitation for all	
Goal 7.	Ensure access to affordable, reliable, sustainable and modern energy for all	S+, D+ (indirect)
Goal 8.	Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	S+, D+
Goal 9.	Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation	S+, D++
Goal 10.	Reduce inequality within and among countries	S++, D++
Goal 11.	Make cities and human settlements inclusive, safe, resilient and sustainable	
Goal 12.	Ensure sustainable consumption and production patterns	S+, D+ (indirect)
Goal 13.	Take urgent action to combat climate change and its impacts	S+++, D+++
Goal 14.	Conserve and sustainably use the oceans, seas and marine resources for sustainable development	S+, D+ (indirect)
Goal 15.	Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation, and halt biodiversity loss	S+, D++
Goal 16.	Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels	S++, D+
Goal 17.	Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development	S+, D+

As mentioned above, the implementation of the targets accepted in the Climate Action agenda is one of the most urgent issues. The most critical initiatives and actions in this respect are related to so-called decarbonisation (decreasing the CO_2 footprint), moving away from fossil-based fuels, energy generation, and fossil-based chemistry, increasing efficiency of using energy, and sparing resources through the wider use of reuse and recycling. One should remember that decarbonisation is part of the wider goal of reducing emissions of all greenhouse gases. This is relevant to

design-centred thinking, as many of the production processes used in manufacturing sports goods are directly or indirectly involved in producing greenhouse gases. According to the Paris Agreement of 2015, global warming should be kept below 1.5°C. Thus, greenhouse gas emissions need to be reduced by 45% by 2030 and reach net zero by 2050. Several initiatives are currently developing science-based emission reduction targets and monitoring progress in this process (see, for example, the documents generated by the Science Based Targets Initiative [124] and the Sports for Climate Action Network [125]).

One of the important targets in reaching sustainability in production, including all sports-related goods, is moving towards environmentally friendly materials and energy-saving technologies. A significant role here belongs to research and education, generating the necessary new knowledge and providing designers with a deep understanding of existing and emerging technologies and materials. As in many other cases, preventive approaches to reaching sustainability goals are important, and thinking of sustainability during the design process is essential. Many general design-oriented handbooks and research papers focusing on sustainability and 'green technologies' are available (e.g. [126–130]). Sports technologies and sports equipment are not exempt, and some corresponding guideline documents (e.g. [131]), research papers (e.g. [126,131–138]), and books (e.g. [139]) related to sustainability and lifecycle analysis are available. Today, many items, including apparel, footwear, and sports equipment contain fossil-based polymer materials that should first be partially and, in near future, completely substituted by non-fossil alternatives. This task becomes an integral part of the modern design process, defining the need for continuously expanding knowledge of novel 'green' materials and their properties (e.g. [139,140]). The rapidly changing situation dictates changing the common strategy of following the corresponding regulations within this area towards a more proactive one, generating preparedness for inevitably changing legislation.

A few main pathways to reaching the sustainability of major materials are continuously being developed and used. Energy-efficient material production (from mining to material through all stages) and manufacturing methods, recycling (complete or partial), and biodegradability should be mentioned as the main targets. With metallic materials, the focus is on energy efficiency and recycling, with an unfortunate lack of research on the potential of biodegradability. With polymers, attention to substituting them with their non-fossil analogues, recycling, and biodegradability has become a dominant issue. The concept of biodegradable materials is not new, but wide introduction of such materials needs carefully balancing the operating-life stability of the products and their biodegradation in the 'afterlife'. In the previous technological cycle, the concept of lifelong stability was dominant, and changing this paradigm needs time and efforts. At the present stage, combining a certain degree of biodegradability with recycling seems to be most promising. Progress in this direction will depend on the research into the properties of new biodegradable materials (see, for example, reviews on biodegradable polymers [137,141]).

The concept of recycling in a broader sense is continuously developing and generating national and international initiatives. One can suggest here a slogan, "From Waste to Value". It is stressing that innovative approaches to the items commonly regarded as waste (and thus assuming additional expenses for their treatment) could be returned into the production chain as a whole or partially, generating additional profit and reducing the load on the environment. The intense development of the concept of recycling, carried out by initiatives and independent researchers, has given birth to the concepts of *upcycling* and *downcycling* (see, for example, [142–148] and references therein). The meaning of downcycling is close to that is commonly associated with 'traditional recycling' and mainly refers to the reuse of materials from damaged or degraded items, for example, melting plastic bags and bottles and using polymer for forming of other items. 'Down'

here mostly refers to the fact that polymers recycled in such way can be used in similar ways as virgin materials, although not with the same (commonly lower) properties. Upcycling, on the other hand, targets reclamation or reuse of the same materials for new products. Here, 'Up' mostly refers to increasing value through the clever reuse of materials or parts of products. It should be noted that the boundary between upcycling and downcycling is rather diffuse. In both cases, the focus is on extending the lifecycle of initial materials, preventing polluting the environment, and generating new value from something that was assumed to be 'waste' and a source of loss.

Modern designers should understand the importance of proactive approaches to sustainability issues in contrast to the more commonly used reactive ones. This means that throughout the design process, not only current legislation and regulations but also sustainability targets must be considered. One of the latest illustrations of sustainability issues and new legislation becoming acute drivers for reactive measures in designing and developing new products and technologies for athletes is linked to the upcoming EU ban on certain perfluorocarbons (polyfluoroalkyl substances, PFAS). According to the plan, the use of such substances must be phased out by 2025 [149,150]. Two specific applications are still common in sportswear (water-repellent agents and impregnating technologies) and sports equipment (fluorine-containing ski waxes). Research has shown that PFAS from water-repellent agents and ski waxes are not only harmful to the environment but also present a direct threat to the health of those exposed to contact with them (see, for example, [151–154]). As a result, the International Ski and Snowboard Federation, also known as FIS, started implementing a PFAS ban in competitions in the 2021 events and plans to fully implement the ban in the 2023–2024 season [154]. It should be mentioned that the industry was not well prepared for the ban on PFAS-containing ski waxes, and for some time, PFAS-free alternatives did not provide as good gliding performance. Consequently, significant and rather urgent efforts were needed to develop, manufacture, and market 'greener' ski wax alternatives with adequate performance. This story conveys an important message related to the design for athletes: sustainability and environment-related issues are of high importance, they can be directly linked to safety issues, and they should be considered at all stages of the design process, including the early ones related to material choice. Readers seeking more detail on sustainability in relation to sporting goods are referred to Chapter 17, Sustainable design and manufacturing of sport equipment.

Design for comfort

Ideas that comfort in active sports is important are not new, as comfort affects both performance and safety aspects. For the best athlete performance, it is essential that all sports equipment and sportswear work perfectly together with the athlete. Best athlete performance with poor functionality of equipment or improper 'matching' between equipment and the athlete's body rarely allows for top performance results. This means that both technological and physiological factors should be considered together during the design stage. Various books and research papers on general concepts of comfort in sports are available (e.g. [155–158]). Discussions on the perception of comfort being influenced by an interrelated set of parameters can be found, for example, in [159,160]. Several comfort-related factors are commonly considered during design and manufacturing, including dimensioning (best fit), thermoregulation, when it refers to the apparel and footwear, and confidence of athletes (e.g. [161,162]).

In dimensioning, there is an intricate balance between a universal and an individualised approach. Although truly individualised adjustment and even manufacturing of equipment, garments, and footwear are assumed to be most advantageous, they can contradict the criteria of fairness and may be prohibitively complicated and expensive. In most cases, 'next best' options

satisfying all formal regulations and criteria would be chosen. It should also be noted that there is a specific issue in designing sports equipment for junior athletes. This is a specific undertaking, as not only differences in body part dimensions, but also differences in physiology and body part biomechanics should be considered. Discussions on this subject can be found, for example in [56–63,163,164] and references therein.

One interesting example of the comfort-performance relationship is application of so-called compression clothing (e.g. [165–167]). Although we still lack a complete understanding of the underlying physiological mechanisms, athletes in training, competition, and sometimes during recovery from injuries often use compression clothing. At the same time, the tight fit of such clothing can prevent good thermoregulation, making the overall benefit of its use for athletic performance questionable. Thus, the design process of such clothing demands knowledge from engineering (functionality, pressure distribution, etc.) together with that in materials (air and moisture permeability of the fabrics) and physiology (perspiration, cooling/heating in different parts of human body), and certain ingenuity (combining different materials in several segments, etc.).

The importance of thermoregulation aspects goes far beyond comfort. Improper fit or choice of materials for clothing or footwear can lead to body overheating or overcooling, with the risk of serious consequences for the athlete's health. Here, one should clearly understand the difference in the environment (e.g. [168–171]). Indeed, there would be a significant difference in heat and moisture transport from the athlete's body towards the environment in water sports and in hot and cold climate (e.g. [172–176]). It should also be considered that comfort, like other factors deeply rooted in human physiology, is largely individualised and varies for different sub-groups of athletes (e.g. [177–180]). Thus, studying and assessing comfort demands combining subjective and objective evaluation methods [181].

A recent example of the ingenuity of researchers and designers working on improving comfort is related to backpacks for cyclists. It is known that a tightly fitted backpack prevents proper cooling of the cyclist's back during warm or hot weather. It was suggested that the natural flow of air around the cyclist's body could be utilised for active ventilation, and thus, increased heat production during more active pedalling could be compensated for by the increased cooling effect of the specially channelled airflow. A backpack for testing this concept was designed, manufactured, and tested, showing improved thermoregulation for the cyclist [182].

Continuous feedback in design process

One of the modern trends in the design flow is the deep integration of evaluation and assessment methodologies at all process stages (e.g. [183–187]). Historically, experimental testing of prototypes and final products was the only available option for assessing design results. Today, as the number of accepted sports disciplines continuously increases, as does the number of people participating in amateur, recreational, and professional sports, and the equipment used is widening and becoming increasingly complex. Corresponding time-to-product (from idea through design and prototyping) and time-to-market (up to the final item in the hands of a user) in a highly competitive environment have become critical, demanding significant investments of finances, labour, and hardware. Thus, failure at the final stages of the design process or in the process of using footwear, apparel, or equipment is, in most cases, becoming unacceptable. Thus, the increasing value of assessment continuity is in the provision of feedback, allowing for the implementation of corrections in the design itself and in the design flow structure (e.g. [182–192]). In modern product development, "digitalisation", computer support, and virtualisation have become essential parts of the design flow process. Computer assistance in research and engineering has progressed

in the last decades from supporting computations, through advanced dynamic visualisation of 3D objects, towards advanced modelling at the level of virtual experimentation and generating digital and physical twins [94–100]. Although modelling and simulations cannot entirely substitute experimental testing, they allow for discarding faulty or unsatisfactory solutions at early design stages and provide support in making correct decisions. Computer simulations also allow for virtual testing of many different "what if" scenarios, which helps decrease the probability of failures during the product lifetime.

Additional advantages are provided by the development of complex design-oriented systems integrating databases, expert systems, and advanced computer models that link together the shapes and dynamics of 3D objects and the properties of various design elements and materials. Initially, developers of such systems even hoped that they would be able to 'invent' new solutions. Although this did not happen, such systems provide effective support for designers in the search for better solutions (e.g. [193–197]). With the wide availability of such systems, one of the tasks for the modern designer is the proper choice of the corresponding tool for specific tasks. This demands a proper understanding of the current tasks and the ability to correctly formulate the specific problem (e.g. [194–197]). The development of artificial intelligence approaches generates significant interest in exploring the possibilities of its applications, including in sports and sports equipment manufacturing (additional examples can be found in references [11,12,198]).

The complicated history of developing testing methods in sports and outdoor equipment is still awaiting deep study. Several research publications providing a glimpse into the historical perspective of sports industry progress and related material and technology development can be recommended in this respect [199–203].

Impact of emerging technologies

It should be noted that developments in new materials and different technologies in areas not focusing on sports-related applications provide interesting opportunities for the design and manufacturing of sports-related equipment, clothing, and footwear. Some of the relevant examples were given above, and here, we outline only a few of the existing exciting opportunities.

In relation to comfort-driven properties, one should mention the so-called 'active fabrics' containing various microporous membrane materials and fabrics incorporating phase change materials. Semi-permeable membranes that help control moisture transport through multilayer fabric composites were pioneered in large-scale garment manufacturing by W. L. Gore & Associates, Inc. (Gore) and are known to the public as Gore-Tex (e.g. [204]). Today, materials with similar properties are manufactured by several companies and are widely used. However, the fabrics incorporating such membranes may not be equally effective in supporting moisture transfer from the human body under all environmental conditions, as their efficiency depends on the difference between temperature and humidity on either side of the membrane (e.g. [205–207]). Thus, designing garments and footwear incorporating microporous membranes demands a good knowledge of the properties of the membrane in combination with other fabric layers surrounding the human body, the physiology of heat and sweat production, and the service environment conditions. Failure to consider such factors could lead to the situation where the microporous membrane actively transports humidity from the environment towards the skin [208].

The application of fabrics with phase change materials in the yarn is not yet common, but corresponding applications are growing rapidly. The idea of the phase change material application for cooling and heating is not new. It is known that during a phase transition (most commonly between melting and solidification), some materials consume heat energy, while others produce

it. Although applications with reasonably large volumes of different materials have been known for many years, progress in yarn production incorporating phase change materials with the necessary physical properties is recent (e.g. [209–212]). Cooling and heating-support fabrics belong to a wider modern concept of 'intelligent' or 'smart' materials in clothing and footwear design and manufacturing (e.g. [213–215]), and other concepts developed in this approach could also be successfully used in designing for athletes.

Another revolutionising technology, additive manufacturing, was already mentioned earlier in the discussion of the body part surrogate applications in safety research and applications [96–100]. Being a representative of modern digitally-enabled technology, it is, of course, useful in all aspects of digital modelling and physical prototyping (e.g. [216,217]). The serious impact of additive manufacturing on the landscape of modern industrial manufacturing is due to the unprecedented freedom of component shapes and the wide variety of materials it already allows for. Sports-related technology is also utilising the advantages of additive manufacturing, primarily within metallic and polymer materials (e.g. [218–220]). The extra value of additive manufacturing is in allowing better product functionality and enabling more sustainable production (e.g. [146,147]). There are examples of utilising additive manufacturing in sports-related design and manufacturing (e.g. [220–224]). Among the most interesting applications, one can refer to feature articles related to revolutionising tennis racket manufacturing (e.g. [225,226]). Readers seeking more information on the application of additive manufacturing to sporting goods are referred to Chapter 16, 3D printing in sports.

The cited examples stress the need to pay careful attention to the new emerging technologies in science, engineering, and industrial applications. With modern design for athletes being so multi-dimensional, it becomes critical that design teams include specialists not only with deep knowledge in specific areas but also with broad knowledge in different neighbouring disciplines.

Concluding remarks

It is hardly possible to give a comprehensive guide to designing for athletes in a short text, as so many aspects are relevant to the process. Although specialists involved in the process should have a wide background knowledge in many specific subjects, it is hardly feasible to be simultaneously proficient in all of them. Modern media technology allows for finding information on many methods, materials, applications, solutions, and examples that could be useful in the design process. The key to being effective in harvesting the needed information is a good knowledge of basic concepts, the ability to set up correct questions, and an understanding of where to find corresponding answers. Thus, in the present chapter, the most important issues in designing for athletes are outlined, and references to detailed information on several specific issues are given, helping the challenging work of providing athletes with better solutions. This chapter provides an extensive list of references to help navigate the complex and multidisciplinary process of designing for athletes.

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PART IV

Sports biomechanics and human-equipment interaction

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23

INTRODUCTION TO SPORTS BIOMECHANICS AND HUMAN-EQUIPMENT INTERACTION

Duane Knudson

Introduction

Advances in sports engineering and equipment technology are particularly challenging, given that most sports involve interaction with living actors (e.g., Figure 23.1) with considerable agency and biomechanical complexity in the use of that equipment [1]. Sports engineers benefit from interdisciplinary cooperation with colleagues in anthropometrics, biomechanics, and ergonomics/human factors. This interdisciplinary collaboration is particularly important because the two primary outcome goals of sport driving engineering innovations, improving performance [2] and reducing the risk of injury [3], are often inversely related.

While considerable progress has been made in improving sports equipment from early hand-crafted adaptions of common items (e.g., gloves, shoes), improved training, nutrition, and specialized sports equipment have greatly increased the demands on high-level athletes. Early in the 21st century, we are into the fourth generation (4G) of sports equipment development [4]. This section presents five chapters exploring the current techniques and collaborative research integrating biomechanics with sports engineering and sports equipment development. These chapters summarize and illustrate important research methods and results in a variety of sports, addressing both performance and injury risk issues in both able-bodied and disabled participants. This extends previous scientific monograph summaries of the synthesis of sports engineering and biomechanics [5,6].

In Chapter 24 "Biomechanical interactions with footwear," Kong and colleagues summarize footwear features, research testing protocols, important variables, and illustrate accommodating unique biomechanical factors in athletic shoes for persons with disabilities. This highlights unique biomechanical and functional characteristics of the athlete that influence successful interaction with sports equipment. MacKenzie and Henrikson illustrate the interaction of the golfer and their driver in the performance of this dynamic and important skill in the sport in Chapter 25 "How the golfer and driver interact to determine performance." Customized changes in club properties may be matched to individual player properties and swing style tendencies of this important golf technique. In Chapter 26 "The quantified tennis athlete: The performer, the ball, the racquet and the technology" Reid et al. explore how the latest advancements in technology are being

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Figure 23.1 Examples of human-equipment interaction in sport: (left) field hockey and (right) wheelchair tennis.

used to measure the sports' demands. Particular attention is paid to the use of non-invasive data collection to the physical responses of athletes in actual training and competition. Petrone and colleagues provide an extensive overview of research in sports injury and trauma related to developing protective equipment in Chapter 27 "Sports injury and trauma biomechanics: Surrogates, human data and modelling." The use of surrogates, biomechanical measurements, and simulation for all body regions is summarized. In Chapter 28 "Biomechanics of athletes with impairment and adaptive equipment," Linzenberger et al. explore the challenges from both different biomechanical issues and the ethical principles of the International Paralympic Committee's (IPC) Policy on Sports Equipment. Advanced data collection and simulation are summarized in successfully adapting sports equipment to people with specific impairments. The chapters in this section all illustrate the complexity and exciting challenge of developing effective and safe sports equipment.

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24

BIOMECHANICAL INTERACTIONS WITH FOOTWEAR

Pui Wah Kong, Wing-Kai Lam, Phillis Soek Po Teng, and Jing Wen Pan

Introduction

Wearing appropriate shoes for sport participation is important, as athletic footwear plays a role in enhancing physical performance and reducing the risk of injury. The shoe acts as an interface between the human foot and the surrounding environment. The physical features and mechanical properties of shoes can affect the way our body moves and interacts with the environment. Thus, understanding biomechanical interactions among footwear, the human body, and the environment is a prerequisite for footwear innovation and prescription [1].

This chapter focuses on the biomechanical interactions of the human body with athletic footwear when engaging in sport activities. First, common footwear features and testing protocols employed in the design and development of various sport shoes are described. Next, an overview of the biomechanical effects of footwear pertaining to sport performance and injuries is provided. As individuals with disabilities often experience some degree of structural deformities or physical pain, or both, this chapter also discusses the current status and future potential of developing athletic footwear for individuals with disabilities to support their full and safe participation in sports.

Footwear features

There are several footwear features regarding physical, mechanical, and perceptual aspects to consider in the design of athletic footwear. Examples include shoe mass, heel-to-toe drop, flexibility, forefoot bending stiffness, cushioning, and traction (Figure 24.1). Several features and modifications can be made to match the functional requirements of athletes and sports, for example, improving comfort, enhancing performance, and reducing the risk of overuse injuries. Among all sport footwear, running shoes and general court sport shoes may be the most common ones used by sport participants. In the literature, biomechanical studies have extensively investigated the effects of footwear interventions on runners of varying levels. Compared with running shoes, there are obvious differences in the functionalities of general court sport shoes [2]. The specific key features of court sport shoes are the abilities to reduce excessive foot pronation during lateral/cutting movements, provide adequate heel and forefoot cushioning for comfort and shock attenuation, provide appropriate bending and torsional stiffness, and offer optimal traction to avoid foot interlocking and slippage during sport-specific movements.

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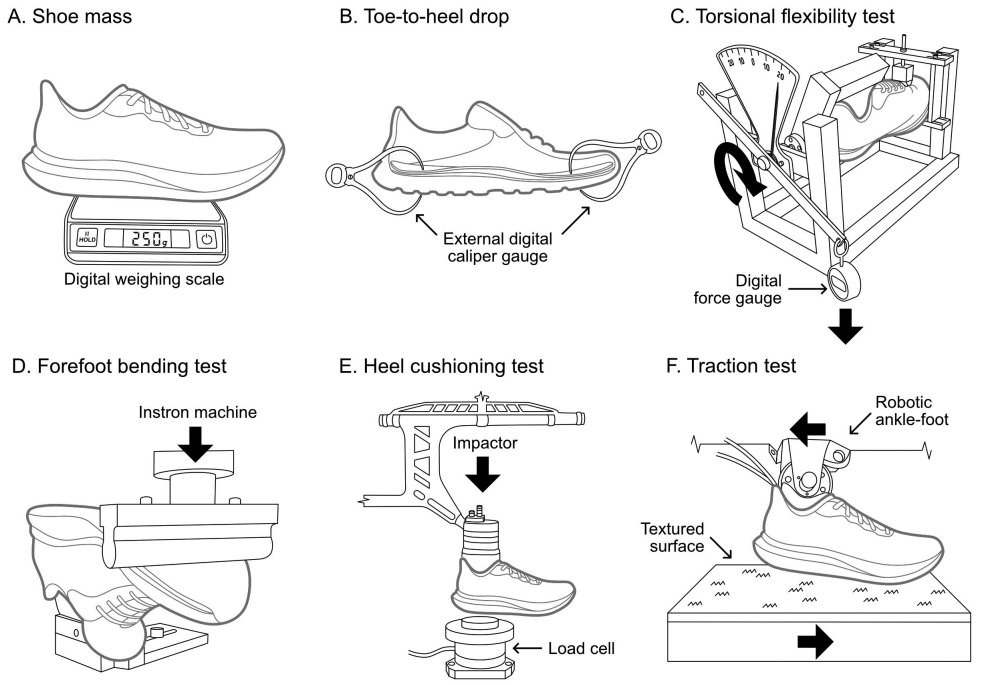

Figure 24.1 Examples of experimental set-up for testing key physical and mechanical features of athletic footwear. These tests are modified from existing practices or following standards such as the ASTM F2333-04 (2017) for traction characteristics.

Table 24.1 lists the common shoe features with respect to running and a few court sports, including footwear for running [3], badminton [4], basketball [5,6], gym training [7], and tennis [8]. As well as physical and mechanical shoe features, it is also important to consider perceptual factors such as comfort and fit. There are clear differences among footwear used in various sports to match sport-specific movement characteristics, such as lateral movements for court sport shoes, lunge movements for badminton shoes, and stability for gym training shoes. We believe that multiple determining factors, including sport type, playing level, and population, must be considered for footwear selection and modification to optimise the interaction among footwear, the human body, and the environment.

Test protocols for human-footwear interaction

It is essential that the functions of athletic footwear match the characteristics of sport movements and the intensity of the sports. While the physical properties of a shoe can be accurately tested using mechanical tests (Figure 24.1), these test results do not necessarily reflect the biomechanical loading on the wearer during sport movements. It is, therefore, critical to carefully consider the potential interaction between footwear and the human body. As such, various wearer testing protocols are designed to assess the biomechanical effects of athletic footwear (Figure 24.2). Common biomechanical footwear testing protocols and variables are summarised in Table 24.2.

Table 24.1 Key footwear features for consideration in sport shoes

Footwear types	*Key features to consider in the design of sport shoes*	
	Physical features	*Mechanical features*
Running shoes	Crash pad, forefoot flares, heel counter, heel flares, heel (stack) height, heel-to-toe drop, insole shape, lacing system, medial post, midsole thickness, rocker, shoe mass, toe spring, and upper breathability	Forefoot bending stiffness, forefoot midsole hardness, midfoot bending stiffness, outsole traction, rearfoot midsole hardness, torsional bending stiffness, and upper elasticity
Badminton shoes	Arch support, breathability, injury protection features (heel curvature, counter), shoe mass, shoe sole durability, and shoe upper durability	Forefoot bending stiffness, forefoot cushioning, forefoot stability, forefoot traction/grip, rearfoot cushioning, rearfoot stability, and rearfoot traction
Basketball shoes	Breathability, durability, injury protection features (shoe collar, counter), and shoe mass	Cushioning, flexibility, forefoot bending stiffness, midsole hardness, stability, and outsole traction
Gym training shoes	Arch support, breathability, heel-to-toe drop, outward appearance, outsole durability, shoe mass, and upper durability	Forefoot cushioning, forefoot outsole flexibility, forefoot stability, rearfoot cushioning, rearfoot stability, rearfoot outsole flexibility, traction, upper flexibility, and other injury protection features (counter)
Tennis shoes	Arch support, breathability, colour, easy to clean, outward appearance, outsole durability, and shoe mass	Forefoot cushioning, forefoot flexibility, rearfoot cushioning, rearfoot flexibility, rearfoot stability, and traction

Perceptual variables

When wearing shoes to perform sport movements, it is important that participants feel comfortable with the shoes. Subjective perception of athletic footwear is often assessed with Likert and Visual Analogue Scales in both industry and academic research [9–11]. Fit perception is the prerequisite for overall comfort in footwear, alongside other factors such as perceived cushioning, arch support, and stability [11], [12]. Increased comfort perception has been found to be related to a lower incidence of injuries [13], better performance [14], and lower energy expenditure [15].

There are mixed findings on whether perceptual variables can predict biomechanical responses to sport shoes. One study on football boots reported a weak and negative correlation between peak plantar pressure and comfort ratings whilst executing football-specific movements [16]. Another study on basketball shoes found no relationship between subjective perception of stability and objective centre of pressure measurement when performing 45° cutting and layup manoeuvres [9]. In summary, perceptual outcomes may be influenced by multiple factors involved in the complex interaction between the foot and shoe, rather than a specific footwear feature. As such, subjective perception remains an essential assessment tool for the comprehensive evaluation of footwear alongside mechanical and biomechanical tests. Therefore, future studies on perception and its interaction with footwear mechanical and biomechanical characteristics in different sports are encouraged.

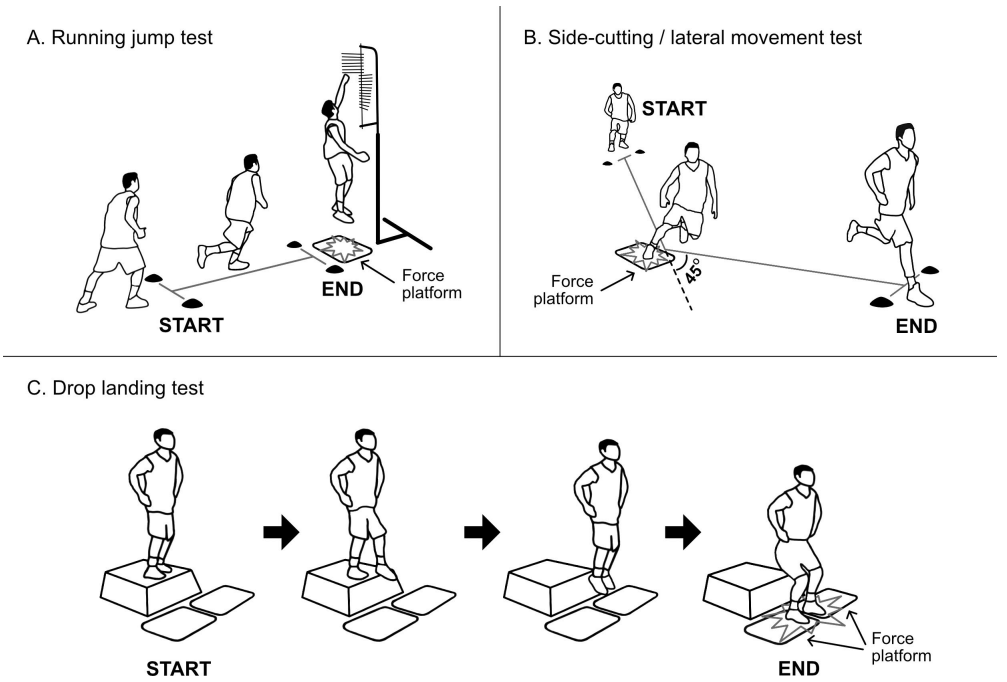

Figure 24.2 Common biomechanical test protocols for athletic footwear: (a) running jump, (b) side-cutting/lateral movement, and (c) drop landing.

Biomechanical variables

Ground reaction force (GRF) is the force exerted at the interface between the human body and the ground. When the foot/shoe pushes off the ground, GRF increases to accelerate the sport participant, and larger GRF is often associated with better athletic performance. From the injury prevention perspective, however, there is a general agreement that excessive peak GRF may cause high tissue stress and acute or chronic damage to the human musculoskeletal system. There is likely a window of loading in which biological tissues react positively to the applied GRFs. Hence, balancing stiffness and shock absorption are important competing shoe design criteria that must be integrated with athlete biomechanical response. Even a small change in midsole hardness of basketball shoes can trigger a different prelanding muscle activation strategy and, hence, alter GRF and joint loadings during landing [17]. To this end, footwear cushioning is of concern. We should consider a shoe's ability to attenuate impacts (magnitude – peak impact force, time – loading rates) during the initial phases (~20–50 ms) of landing movements, such as running, landing, side-cutting, and lunging. This damping/shock absorption, however, cannot be so excessive as to impede the progressive force transmission through the shoe to the human body. Readers seeking details on how material selection influences such properties of shoes are referred to Chapter 5 "Soft materials in sport" and Chapters 9 and 10 "Cellular solids for sports applications".

Shoe frictional properties/traction is one of the major factors for stopping, cutting, turning, and accelerating in most court sports. Footwear scientists have optimised the traction properties by changing the outsole material, hardness, and tread pattern to minimise slippage on the respective sport surfaces, leading to better sport performance. Traction must match the playing characteristics

Table 24.2 Common footwear testing protocols and variables for different sports.

Footwear types	*Testing movements*	*Biomechanical variables*	*Perceptual variables*
Running shoes	1. Overground running 2. Treadmill running	1. GRF[#] variables (impact force, loading rate) 2. Rearfoot mechanics (initial contact angle, pronation excursion, pronation velocity) 3. Knee mechanics (varus-valgus movement, varus-valgus moments/loading) 4. Neuromuscular mechanics and physiology (lower leg muscle activation, maximal oxygen consumption)	1. Comfort 2. Fit 3. Perceived shoe functions*
Basketball shoes	1. Jumping 2. Landing 3. Cutting / lateral movements 4. Agility movements	1. GRF variables (impact forces associated with forefoot and rearfoot contacts, loading phase, maximum take-off force) 2. Rearfoot mechanics (initial contact angle, inversion angle, inversion velocity) 3. Knee mechanics (flexion-extension and varus-valgus movement and associated joint moments) 4. Functional performance (jump height, sprint time, task completion time)	1. Comfort 2. Fit 3. Perceived shoe functions
Tennis shoes	1. Cutting/lateral movements 2. Agility movements	1. GRF variables (traction) 2. Rearfoot mechanics (initial contact angle, inversion angle and velocity) 3. Knee mechanics (flexion-extension and varus-valgus movement and associated joint moments) 4. Functional performance (task completion time)	1. Comfort 2. Fit 3. Perceived shoe functions
Badminton shoes	1. Lateral movements 2. Lunge 3. Agility movements	1. GRF variables (heel landing impact, loading rate, traction) 2. Rearfoot mechanics (initial contact angle and velocity) 3. Knee mechanics (flexion-extension movement and associated moments) 4. Functional performance (competition time)	1. Comfort 2. Fit 3. Perceived shoe functions

Note. [#]GRF denotes ground reaction force. *Perceived shoe functions include the subjective rating of shoe functions such as cushioning, hardness, arch support, stability, bending stiffness, traction, agility, and other functions of interests by manufacturers or investigators.

to prevent excessive joint loading. Taking tennis as an example, players often experience greater slippage on clay courts than on hard courts when executing forehand and backhand strokes. This is due to the relatively slippery surface of clay, which can make it more challenging to maintain proper footing during these types of strokes. It is not straightforward to simply increase shoe friction to prevent slippage because increased footwear traction can lead to higher joint loading, and elevated risks of ankle and knee injuries [18–20]. Studies have compared the effects of sliding and non-sliding

surfaces on tennis strokes and injury rates [19,20]. The results showed that tennis players had fewer injuries and experienced smaller frictional forces when using sliding surfaces than on the non-sliding condition. Another study on basketball v-cut and side-shuffle tasks found that ankle joint moments were lower and resultant knee moments were higher on the sliding surfaces than the conventional surfaces [21]. The authors concluded that sport surfaces that allow a few millimetres of horizontal movement during ground contact may reduce joint loading at the ankle joint but increase the risk of knee injuries. Recent developments in shoes with shear reduction or sliding constructions, such as multi-layer insoles or groove-type outsoles, have shown promising results. These shoes delay the time to peak force, restrict ankle motion, and reduce knee joint loading, all without compromising a player's cutting performance on basketball court surfaces [22].

In summary, it is important to consider the human-footwear interaction to holistically evaluate the performance of athletic footwear. Sport-specific testing protocols should be employed where possible, together with a wide range of measurements, including perceptual and biomechanical variables. As movement characteristics and intensity can vary between populations (e.g., elite versus recreational players, males versus females) [3,23], footwear recommendations should also take into account performance level, training volume, and individual characteristics.

Footwear and sport performance

Sport scientists and engineers have studied several contributing factors and underlying mechanisms to improve footwear functionality for enhancing sport performance. Lower shoe mass has been suggested to positively affect running economy and jump or sprint performance, or both [23–25]. The energy cost of locomotion is reduced with lighter footwear [24,25]. It has been reported that adding 100 g (about 40–55% of the mass of a typical running shoe) to a shoe can impair running economy by up to 10.2% in trained runners [24]. Another study found that heavier rocker shoes (858 ± 96 g per pair), while providing protection to lower limbs, resulted in 4.5% and 5.5% poorer running economy than lighter standard shoes (541 ± 44 g) and minimalist shoes (321 ± 25 g), respectively [25]. The research findings illustrate the complex interaction between human and footwear features. As a change in shoe mass can alter the energy requirement during locomotion and because not all athletes will respond in the same way, one cannot easily predict proportional changes in athletes' response from a single shoe design feature.

Higher forefoot bending stiffness is associated with lower energy cost during running [26]. The concept of optimal bending stiffness has also been applied in other types of sport footwear such as basketball [27], badminton [28], and soccer shoes [29]. To date, the working mechanism of the bending stiffness features, their interaction with athletes' biomechanics, and injury rates remain uncertain. Theoretically, increased bending stiffness would provide a stiffer/longer moment arm which, in turn, would induce a teeter-totter effect or minimise energy dissipation at the metatarsophalangeal (MTP) joint during the push-off phase of jumping and running tasks [30]. It is interesting to note that higher MTP joint stiffness of an athlete is related to improved jump performance in consecutive jumps but not in single jumps or running jumps [31]. This finding suggests that the optimal forefoot bending stiffness of a shoe must be tailored to specific movement tasks and individual MTP joint stiffness to maximise athletic performance.

Footwear and sport injuries

Sport participation can promote good health and well-being, but there are also acute and chronic risks of injury associated with vigorous sports and exercise. Running is a popular sport enjoyed by

a huge population across the world. Running footwear has moved from minimal to well cushioned shoes, with a repeat of this trend in the last 50 years [32]. Since the foot is the first body segment to contact the ground, wearing appropriate footwear is important to attenuate the forces transmitted up the lower extremity that may contribute to injury. Biomechanically, sport scientists have identified injury patterns and potential risk factors to guide how footwear features should be developed, tested, and modified. Injuries are multifaceted, requiring substantial evidence to make meaningful links between shoe features, athlete biomechanics, and injury rates.

Some features are integrated into marketing copy and have become well known to athletes before solid injury rate evidence can be obtained. For example, pronation-reducing features (crash pad, arch support, heel counter, dual-material midsole) are well-known within the recreational running population to control "over-pronation". This concept is widely accepted despite there being no direct prospective evidence between pronation and injuries [33] or a clear definition of overpronation. In addition, preliminary research has begun linking injury risk at the knee joint with running footwear with thinner midsoles that promote greater movement stability [23]. For basketball footwear, a higher shoe collar is hypothesised to enhance ankle proprioception and control ankle inversion movement and velocity [6]. In the sport of badminton, heel design (curvature, thickness, and material) of a shoe has been shown to be important for modulating shock attenuation and heel-to-forefoot transition during lunging [34]. The following sections describe the influence of footwear on biomechanical parameters associated with injury risk, mainly using running as an example.

Influence of footwear on loading rate

Increased loading rate has been commonly associated with a higher injury rate in runners [35,36]. Loading rate refers to the gradient of the first peak of the vertical GRF – time curve (Figure 24.3a) [35,36]. As the first peak is applied rapidly, the viscoelastic body tissue may have insufficient time to react to resist/dampen the force, thus resulting in higher tissue stress, strain, and ultimately injury. Acute injuries also occur when force exceeds the ultimate strength that tissues can withstand.

Cushioned midsole materials, typically made of ethylene-vinyl acetate (EVA) or thermoplastic polyurethane (TPU), have been used to attenuate the vertical GRF experienced by athletes. Readers seeking details on such materials are referred to Chapter 5 "Soft materials in sport" and Chapters 9 and 10 "Cellular solids for sports applications". These materials are often applied to the heel region of running shoes since most runners are habitual heel strikers [37]. As cushioning materials are compressed under load, time is increased to decelerate the foot, thereby reducing the loading rate. For the same impulse, an extended time effectively reduces peak force. Since shoe impact with the ground is inelastic, force is also partially absorbed through the compression of the cushioning materials. However, recent reviews indicate inconsistent evidence regarding the influence of cushioned midsoles on GRF measures [35,38]. Maximalist shoes with midsole thicker than 20 mm do not seem to be associated with lower vertical GRF [35]. This lack of consistent evidence could be because midsoles with a fixed cushioning property might not attenuate forces sufficiently for every runner. In fact, research indicates that only runners of lower weights (median cut-off of 78.2 and 62.8 kg for male and female runners, respectively) benefit from the cushioning midsoles [39]. This is probably because heavier runners require greater compressive properties from the midsoles for a protective damping effect. Designing optimal midsoles for the right differentiated groups of runners may thus be required [38]. As the interaction between athletes and shoes is multifactorial and can vary greatly, changes in material mechanics do not always directly elicit the expected biomechanical response. For instance,

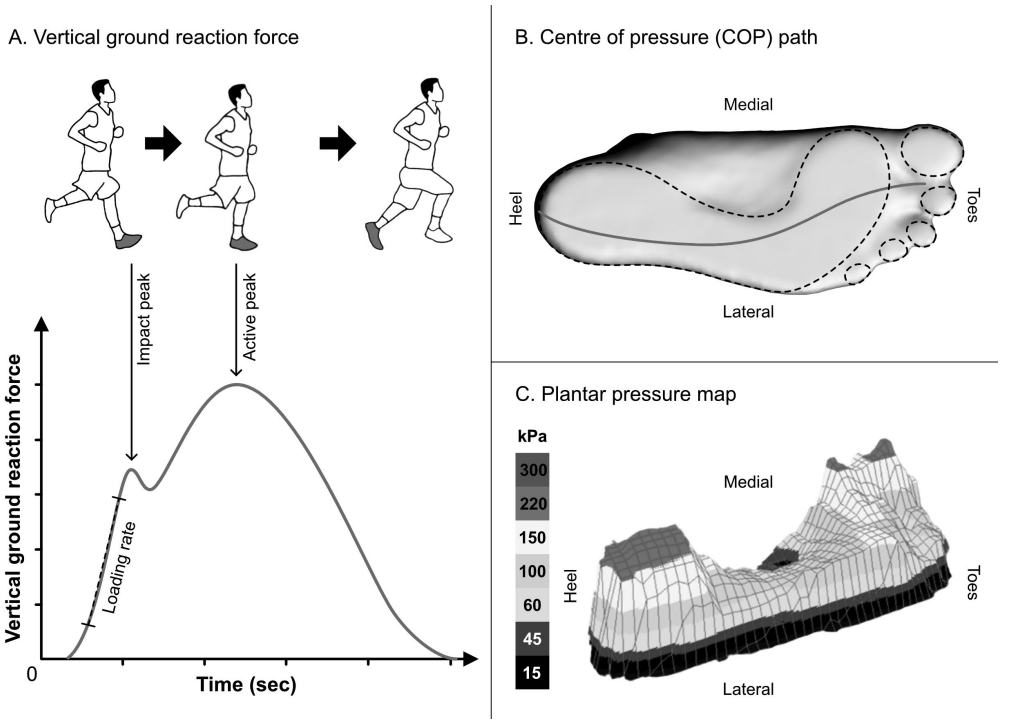

Figure 24.3 Representative biomechanical profiles during running: (a) vertical ground reaction force, (b) centre of pressure (COP) path, and (c) plantar pressure map.

runners adopted a landing strategy that required higher apparent knee joint stiffness when running in softer shoes [40]. Thus, we can expect athletes to adjust their movement strategies in response to the footwear they wear.

Arguments have also been made against the use of current peak GRF and loading rate variables to assess the effectiveness of footwear cushioning [35,41]. Shorten and colleagues [42] determined that lower and delayed impact maximum loads of cushioned footwear could have been artificially inflated by the higher components of low-frequency movements of the lower extremity. In fact, Malisoux and colleagues [41] found that softer cushioning footwear resulted in a reduction in high-frequency vertical loading rates compared with harder versions, extracted with a 10 Hz cut-off frequency. Furthermore, in a 6-month follow-up of more than 800 recreational runners, the softer footwear cushioning (61.3 ± 2.7 N/mm) was indeed still more protective against injuries than the harder ones (94.9 ± 5.9 N/mm) [39]. Higher-frequency loading rates could therefore be a better measure for assessing future cushioning effectiveness. Although a cushioned midsole (e.g., thickness, hardness, viscoelastic response) can reduce the impact force by interacting with an athlete's impact attenuation system [42], one should also note that athletes can be influenced by factors such as advertising descriptions [43] and their own beliefs and expectations about the function of their footwear [44,45]. Studies have shown that even when participants are prescribed physically identical shoes, manipulating their perception of the footwear (e.g., higher price, superior function) can lead to biomechanical changes related to stability and jump performance [44,45]. Therefore, it is important to consider not only the physical properties of the shoes but also the psychological factors that can affect how athletes use them.

Influence of footwear on increased joint stiffness

While better-cushioned footwear may help attenuate high-frequency loading rates, biomechanical studies have correspondingly found increases in leg, ankle, and knee joint stiffness, thereby possibly resulting in higher impact forces in running [40,46,47]. It is proposed that to keep a similar preferred centre of mass path, individuals tend to stiffen the ankle and knee joints when wearing footwear with a softer midsole, especially when running faster [40,47]. Apparent ankle joint stiffness was generally more sensitive to changes in the midsole, but the effect of midsole stiffness seemed more pronounced in females than in males in the knee joint [40]. It is likely that this joint stiffening is also related to the possibility of perceiving differences in cushioning levels. Studies have shown that runners could possibly perceive differences in the cushioning levels of footwear [48,49], especially when the difference was at least 1.4 g using the ASTM F1976 impact test [48]. Thus, an interplay of runner perception of cushioning and mechanical midsole properties, as well as corresponding changes to joint stiffness, should be taken into consideration when determining target levels of cushioning for runners.

Influence of footwear on plantar pressure and centre of pressure

Plantar pressure, commonly measured using pressure insole sensors, could provide an indication of where high localised loads on the feet are present during sport movements (Figure 24.3c). Repeated high plantar pressure could result in injury and decreased perceived comfort levels [50]. Typically, the highest-pressure points are experienced at the heel, as well as the first and second metatarsal head regions in locomotion [51]. Individuals with high arches tend to have less pressure at the midfoot, while those with flat feet tend to have pressure across the entire feet [51]. In general, athletes with high-arched feet are recommended to wear neutral-cushioned running shoes, which are manufactured on a curved last, have additional shock absorption properties, and lack features to reduce pronation of the foot. Previous research reported that wearing neutral-cushioned shoes was effective in reducing plantar pressures in recreational runners with high-arched feet [52]. Maximum pressure at the hallux, lesser toes, lateral midfoot, medial forefoot, middle forefoot, and lateral forefoot was found to be significantly higher in racing flats compared with training shoes, which had better cushioning [53]. Additionally, maximum plantar pressure and associated injury risks were found to be higher in minimalist shoes than in regular running shoes [38,54]. While the lack of cushioning in minimalist shoes could lead to injuries, a gradual and proper transition to minimalist shoes is essential to minimise injury risk [54]. A softer midsole (50 Shore C) in the forefoot region was found to be effective in alleviating plantar pressure experienced during basketball movements involving running, maximal forward sprinting, maximal 45° cutting, and layups [55]. Collectively, the aforementioned studies suggest the importance of cushioning in reducing localised plantar pressure. However, taking running as an example, a review study did not find an association between plantar pressure measures and specific running-related injuries [56]. The extent and mechanism by which reduced plantar pressure in footwear affects injury rates are therefore still unknown.

Footwear also influences the centre of pressure (COP) of GRF [51]. COP is a misnomer in that it is the location of the resultant GRF, rather than a pressure variable. The typical COP of a normal step starts marginally to the lateral side of the mid-heel, alongside the middle of the foot, through the metatarsal heads and toe-off at the first or second toe regions (Figure 24.3b). The transition of the COP location can effectively affect the moment arms at the knee and hip, and hence influence the moments at those joints. The COP path is particularly important, especially in the delay of the

onset of knee osteoarthritis (OA). Although knee OA is a degenerative disease, studies have shown that knee OA risk is associated with certain sports, pre-existing joint injury, and lack of muscle strength [57,58]. Free-flexing footwear, mimicking barefoot walking, has been found effective in reducing external knee adduction moments (KAM), which are associated with knee OA risk [59]. Barefoot walking, a target for footwear knee OA intervention, likely positions the COP such that the moment arm at the knee is reduced, thus lowering KAM. In addition, COP was found to influence rearfoot eversion, which indirectly affects the risk of sustaining running-related injuries [60]. Pertaining to the use of motion control or stability footwear in reducing running-related injury risk, current research findings remain inconclusive [61]. In the future, individual foot profiles and movement patterns should be taken into consideration, as standardised footwear may not be effective for every individual and future work might clarify customised shoe features contributing to lower injury rates. The challenge for engineers and the footwear industry is to develop a process for offering customised footwear at a reasonable cost.

Athletic footwear for individuals with disabilities

Individuals with physical and intellectual disabilities can excel in sports and compete at high-level events such as the Paralympic Games and Special Olympics World Games. Wearing appropriate footwear is important for athletes to play well in a sport, regardless of whether they are able-bodied or para-athletes (see Figure 24.4). Many athletes with disabilities experience some degree of foot/ankle pain or have limb deficiencies, structural deformities, and dermatological conditions of their feet [62,63]. In the literature, footwear research has often focused on able-bodied athletes [64–67]. Promising advancements can be seen via customised orthoses, shoe gear, and shoe modifications for the purpose of improving sport performance and preventing sport-related injuries [6]. In contrast, little attention has been paid to footwear development for athletes with disabilities who aspire to compete in sports. Shoe modifications and therapeutic footwear have been shown to be effective in improving the daily physical activities of individuals with foot deformities or ulcers [68–70], but their contributions to sport performance are still unclear. Concerning well-trained amputee athletes, Hobara and colleagues extensively investigated the biomechanics of running gait in amputees and able-bodied runners and reported that amputees had a higher injury risk to their unaffected limbs [71,72]. At present, the scientific community has devoted little research to developing desirable athletic footwear for individuals with different types of disabilities. Insights into the potential for this research can be gained from media coverage of a few para-athletes who received customised shoes to boost their sport performance.

Current status

From a social perspective, people with disabilities can be frequently prevented from engaging in activities they would like to undertake due to the lack of appropriate clothing and footwear options [73]. One example recently reported in the media highlighted the struggles of a world-class para-badminton player, Nilesh Gaikwad, who has been unable to find the right shoes to pursue high-level sport performance [74]. He has been playing barefoot since he perceived that most regular shoes made him slow and restricted his movement. Through personal contact, he was fortunate to obtain customised badminton shoes based on his leg conditions and measurements offered by Nivia Sports. After a few rounds of prototyping and wearing experiments, Nilesh was delighted to finally experience comfortable shoes that he perceived to have increased his speed and agility while playing badminton. While this example is only a case report with no

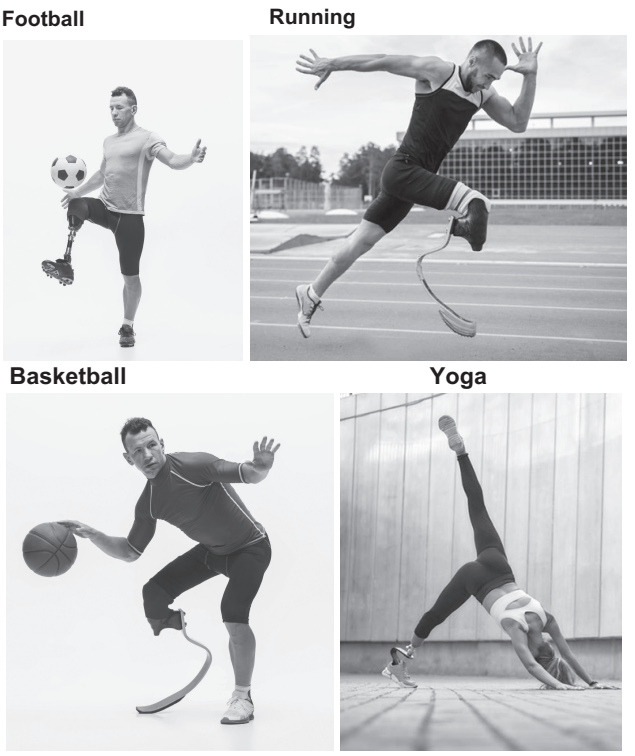

Figure 24.4 Examples of individuals with disabilities engaging in a variety of sports which require appropriate athletic footwear.

scientific data on the modified footwear or sport performance, it is concerning that our society currently lacks research and a systematic approach to support footwear development for most individuals with disabilities.

From a health perspective, it is of interest to examine the possible role of footwear in athletes' injury risk and musculoskeletal conditions. Footwear can impact one's daily activities as well as tasks related to sports and exercise, for example, working out in a gym, running on a treadmill, and cycling. In a recent systematic review on athletes with physical, visual, or intellectual disabilities, Santos Silva Lopes et al. [75] found a high prevalence of muscle injuries. Interestingly, the pattern of injury did not seem to specifically relate to the type of disability but was more related to the biomechanics of the sports involved. For example, para-athletes engaging in sports that were practised in a standing position presented a high rate of injuries to the thigh region. Another large-scale study examining foot conditions among 1,580 Special Olympics athletes also reported biomechanical concerns, including over-pronated gait, abducted gait, and restrictions in the ankle and first MTP joints [62]. Given the high prevalence of various foot conditions observed, the authors advocated the great importance of properly fitted shoe gear for athletes with intellectual disabilities. Stemming from research on the able-bodied population, one possible solution is to modify the shoes with pronation-control features (e.g., harder medial midsole, arch support) to help correct the over-pronated gait. In addition, shoes with higher forefoot stiffness, together with a forefoot rocker, can be considered for athletes with ankle and first MTP restrictions.

Individuals with disabilities are less homogeneous than non-disabled athletes and are often characterised by unique physical characteristics and functional limitations. Their diverse needs present a great challenge for scientific research and the development of para-sport footwear. Economically, it is costly to customise footwear for individuals. Researchers have recognised these challenges and urged greater, continuous efforts to guarantee technical, theoretical, and practical support for this population [75]. It is likely that knowledge gained from the study of disabled athletes may also inform shoe interactions with injured and able-bodied athletes.

Opportunities

Footwear can be deemed as important as food and housing, playing a crucial role in ensuring social and economic inclusion, as well as equal opportunity for healthy sports and exercise [73]. Athletes with intellectual disabilities frequently encounter structural foot conditions such as pes planus (flat foot) and hallux valgus (bunion) [62]. Substantial efforts have been made to customise orthoses, shoe gear, and shoe modifications for mentally and physically challenged athletes, with the hope of improving their physical activity and wellness [76]. The general concepts in footwear design can be applied to para-sports with additional considerations of individual differences in anatomical and functional limitations. For example, footwear can be tailored to provide comfort and reduce plantar pressure [68,69]. Regarding sport performance, depending on individual needs, footwear can be designed to improve running economy [24,25], lower sprint time [6,77], or enhance jump performance [6,78]. There is no doubt that the current technology, research, and development infrastructure are capable of designing desirable athletic footwear to cater to specific disabilities. However, the niche market unlikely attracts major investment from the industry. To accommodate different types of disabilities, it is difficult to lower production costs substantially because the modified footwear will not be produced at a large scale. Governments, non-governmental organisations (NGOs), and national sports associations (NSAs) can implement strategies and provide incentives to encourage more efforts towards the well-being of individuals with disabilities, including the provision of appropriate footwear for sports.

There are individual para-athletes who have benefitted from major sport companies that customise footwear to fulfil their sporting needs. In addition to the para-badminton shuttler noted earlier, Paralympian Jessica Heims was grateful to receive two pairs of adaptive shoes (slides and sneakers) from Nike [79]. She explained that the added strap at the back of the shoe allowed her prosthetic to stay in place during competition. In addition, she pointed out that the "step, lock, and release" laceless features were beneficial to people who have limited arm mobility, as they might have difficulty tying their shoes. These examples of the sport industry stepping up to support para-athletes are promising, sending a positive message to the greater community. In the future, it will be exciting to expand and scale up such efforts to benefit more individuals who wish to participate in sports but have yet to have access to appropriate footwear that accommodates their disabilities.

Students and educational institutions can also play an important role in nurturing academic research and the footwear development landscape across the world. Student projects, such as those in the engineering and design disciplines, can be used as a platform to innovate ideas and generate preliminary data, which are key to obtaining research funding and wider recognition in the industry. One university student collaborated with the Ecuadorian shoe brand *Pancos* to develop smart footwear incorporating satellite-based navigation system technology to prevent blind people from getting lost through a vibration system [73]. Although this application is not directly related to sports, it illustrates the potential of ground-up innovation from students that may stimulate industrial research and commercialisation, which could be of interest to many people and society.

Summary and future innovation

Understanding biomechanical interactions among footwear, the human body, and the environment is a prerequisite for footwear innovation and prescription. The footwear innovation process should incorporate functional requirements, available technologies (e.g., advanced materials), structural designs, and testing of biomechanical responses. In the industry setting, sport scientists, designers, material specialists, tooling engineers, and product managers work closely as a team to explore innovation concepts and prioritise product launch agendas that fulfil the company's needs. In addition, Sterzing et al. [80] described a general model that illustrated how athletic footwear research can be initiated by industry and academia. The authors also determined the general goals of companies for making use of the external infrastructure, intellectual inputs, and reputation of academic research institutions for concept validation and marketing purposes. Some commercially available footwear concepts are highlighted below:

- Minimalist/barefoot concept (e.g., Five Fingers)
- Pronation control concept (general features for several companies such as crash pad, dual-density, etc.)
- Instability shoe concept (e.g., MBT)
- Maximalist concept (e.g., Hoka)
- Racing shoe concept (e.g., Nike 4%, Li Ning Boom)

At the initial phase, innovators or sport scientists should familiarise themselves with sport characteristics and refer to injury incidences and the associated risk factors for injury prevention features. Likewise, understanding performance-enhancing factors can provide insightful directions to create athletic footwear with positive performance features. Innovators may be particularly interested in looking up new scientific findings to cultivate innovation concepts. For example, to create minimalist/barefoot shoe concepts, footwear scientists and innovators could build shoes that allow the feet to move naturally and close to barefoot walking and running conditions. The typical footwear characteristics, including zero-drop, flexibility, thin sole, wide toe box, minimal support, no stiffeners, and very light weight, are then defined as the company silos and innovation directions. Current research findings suggest that taping the foot can be a potential method to reduce plantar fascia strain loading in runners [81]. Footwear scientists and innovators may consider applying this information on taping and the underlying mechanism to brainstorm ideas for footwear innovation for runners who are experiencing or at high risk of developing plantar fasciitis.

Generally, most of the athletic footwear research has focused on running, while less scientific evidence is published for other athletic shoes (e.g., basketball, badminton, table tennis, soccer, squash, volleyball, etc.). In these sports, some typical shoe features/modifications, including (but not limited to) anti-torsion, heel counter, shoe collar, and midfoot straps, have been used for years, but the associated functionality, underlying mechanisms, and biomechanical interactions are not fully addressed. Apart from sport category, there is a lack of emphasis on athletic footwear for females, children, and individuals with disabilities. For instance, few manufacturers have produced female-specific footwear for sports [82]. In other words, female athletes can only select their sports shoes from male footwear. There are several etiological issues for foot and footwear interactions that could differ between females and males. The foot shape of females is characterised by a wider forefoot, higher instep height, narrower heel, more medial curve, and higher foot arch than males' [83]. The possible shoe ideations are to consider functional grading of female-specific flex

grooves, toe spring, and/or rolling bars to address the gender difference in toe length or to take the redesign of female shoe lasts into consideration to address the foot shape differences [84]. Moreover, female athletes are reported to have a larger Q-angle, weaker hip abductors, larger joint laxity, and greater knee joint loading during sports than males, which may lead to different landing angles and rearfoot pronation with higher injury risk potentials [85,86]. Confirmed with X-ray imaging of the talus joint, female athletes have been reported to have a larger range of subtalar joint movement and thus higher injury risks in various cutting and landing tasks compared with their male counterparts [87]. Since knee joint laxity varies over the course of the menstrual cycle [88], some female-specific footwear designs are recommended to promote healthy sports. One shoe ideation could be increased torsional stiffness or mediolateral stiffness or both, to support joint laxity in the mediolateral direction by reducing pronation (eversion).

We recommend that future investigations and innovations in athletic footwear consider (but are not limited to) the following factors:

- Footwear features/modifications for a wide variety of sports and their interaction with biomechanical responses
- Footwear requirements for different sport levels (e.g., elite, recreational) and environments (different surfaces and conditions)
- Identifying mechanisms of injury and performance effects of footwear features and biomechanical responses
- Female-specific footwear for sports
- Children-specific footwear for sports
- Footwear for individuals with various types of disabilities

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25

HOW THE GOLFER AND DRIVER INTERACT TO DETERMINE PERFORMANCE

Sasho J. MacKenzie and Erik M. Henrikson

Methods to study the kinetics of the golf swing

From a mechanical perspective, the external kinetics a player applies to the club, along with its inertial properties (mass, center of mass, and the moment of inertia matrix), determine its change in motion. There are three sets of external forces that act on the club: gravity, air resistance, and the forces a player applies to the grip with their hands. Relative to the forces applied by the player, gravity and air resistance play minor roles in changing the club's motion during a full-effort swing [1]. Different approaches to analyzing the forces applied by the player can be used, and are described in the following two sections.

Forward dynamics vs. inverse dynamics

There is extensive biomechanics research on the golf swing [2,3]. The majority of this research investigating the kinetics applied to golf clubs has been from an inverse dynamics perspective [1,4–8]. Inverse dynamics means that the motion of the golfer and club are measured, typically with high-speed imaging, and Newton's Laws of Motion are inversely applied to the captured kinematic data using a rigid-body linked segment model to determine the net forces and moments that a player must have applied to generate the observed motion.

All parts of a player's hands that are in contact with the grip are applying forces. Currently, it is neither possible, nor probably helpful, to characterize all the separate forces within this overall interaction, although several studies have documented hand kinetics in an aggregate form. To determine the kinetics a player applies to the club, a golf swing researcher would typically use a Newton-Euler approach [6]. The myriad of individual forces a player applies to the grip (Figure 25.1a) would be represented by an equivalent force-couple system [9,10]. This provides the researcher with a net force and a net moment (due to a theoretical couple) applied to the grip by the player (Figure 25.1b).

Another method, known as forward dynamics, utilizes the same governing equations to understand how a player or club's properties impact performance, but it employs a different approach for gaining insights. Forward dynamics assumes the applied kinetics are known, or defined, and determines the resulting motion of the club [11]. One of the novel ways this method has been used

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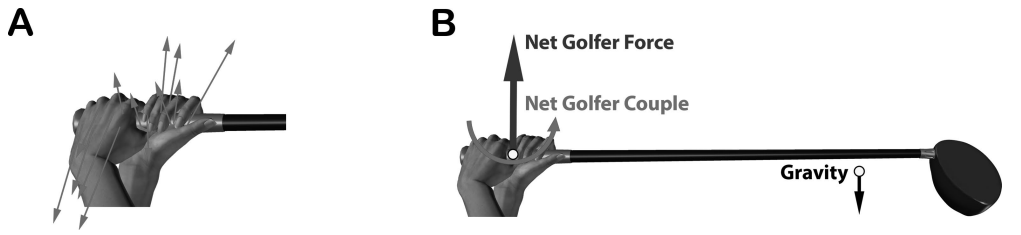

Figure 25.1 (a) Visual representation of a theoretical system of forces applied to the grip by the golfer. (b) The system of forces acting on the club can be replaced by an equivalent arrangement consisting of one force and one couple as well as gravity acting at the center of mass.

is to couple a forward dynamics model with a non-linear optimization method to determine how a player's torque generators (muscles) can be coordinated to maximize or minimize a given objective, such as clubhead speed [12]. This method has also been used to examine how varying different club properties, such as shaft flex, may influence clubhead delivery and speed [13].

Work-energy vs. impulse-momentum methods

The study of the kinetics of dynamic motions like the golf swing follow two forms of expression: The Newtonian vector approach of the impulse-momentum relationship or scalar work-energy theorem. The work-energy approach allows for the calculation of the linear work done by a player through the application of their net force and angular work done by their applied couple [14]. Specifically, linear work is a function of the length of a player's hand path and the portion of force they apply in the instantaneous direction of the hand path. The path of the hands refers to the trajectory traced out by a point on the grip in the middle of a player's hands (Figure 25.1b). This represents a single point where the researcher can consider a player's net force to act during the swing. Similarly, angular work is a function of the rotation of the club about its axis of rotation and the net couple a player applies about that axis.

It is also possible to consider the golfer-club interaction from an impulse-momentum perspective. From this perspective, a player's kinetics act over a period of time to create an impulse, which changes the motion (momentum) of the club. Both perspectives have value. One benefit of explaining the full swing using work-energy is that handpath length is not explicitly influenced by the magnitude of force applied to the grip. A golfer can imagine applying more force along the handpath while simultaneously maintaining, or even increasing, handpath length. Conversely, with impulse-momentum, it seems clear that increasing the force applied to the grip will result in a downswing that is shorter in duration. More time would allow a player to apply more impulse, yet higher average kinetics would result in a downswing that takes less time [1]. Stated another way, players with higher clubhead speeds tend to have longer handpaths, but downswings that are shorter in duration.

Changing a player's swing to improve performance

To illustrate key principles, we introduce Paul—a 17-handicap golfer hoping to break 80 this season. Currently, Paul's clubhead speed at impact hovers around 95 mph (42.5 m/s), and he has a tendency to hit his drives with a moderate to extreme amount of left-to-right shot bend. Since Paul

plays right-handed, this bend is known as a 'slice' and results when a vector normal to the impact point on the face points to the right of the path of this point.

He has a great setup at address; his body positioning is excellent and so is the orientation of the club relative to his hands. Knowing this, assume Paul wants more clubhead speed, a different clubhead path, or an altered timing of the clubface closing (to square with the target). His options are to apply different kinetics during the swing, alter the properties of the club, or both. This section will examine the different ways Paul may choose to alter his swing, while the following section will look at ways a club fitter may choose to alter the properties of his driver to optimize performance.

Work-energy analysis of amateur golfers

Paul spends much of his time on the range trying to increase the distance he hits his driver, which makes sense since the goal of most golf drives is to maximize the displacement of the golf ball down the fairway. Golf ball displacement is primarily determined by ball launch speed and can be further optimized by creating a launch angle and spin rate that are best suited to the ball speed. Ball launch speed for a given club is primarily determined by clubhead speed. For a given clubhead speed at impact, the delivered loft, angle of attack, and the vertical impact location of the ball will all meaningfully influence ball launch angle and spin. As previously stated, the forces applied by Paul to the grip primarily determine the clubhead's impact parameters [1].

MacKenzie et al. [1] investigated how amateur golfers delivered energy to the driver. The primary objective was to determine the kinetic factors that determined the differences in clubhead speed among a large group of golfers. The participants did ~4.5x more linear work on the club (174 ± 40 J [mean ± standard deviation]) relative to the amount of angular work (39 ± 14 J) throughout the downswing. Linear work also accounted for 90% of the differences in clubhead speed between golfers. They concluded that methods of training and practice that increase the average force applied in the direction of the handpath during the downswing would have the greatest probability of generating increases in clubhead speed. According to Nesbit and Serrano [15], angular work peaks prior to impact because the wrist joints are not able to keep up with the angular velocity of the grip prior to impact. This is in agreement with MacKenzie et al. [1], who reported that the couple applied by the golfer became negative over the final 30 ms of the downswing. Further, results from MacKenzie et al. [1] suggested that for amateur golfers, increasing the length of the handpath was more likely to increase clubhead speed than rotating the shaft through a larger angle.

For Paul, this means focusing on more hip and shoulder turn in the backswing, which would increase handpath length compared to increasing the amount of wrist cock at the top of the swing, which would increase angular displacement of the club but have essentially no influence on hand path length.

Optimized forward dynamics and inverse dynamics analysis of amateur and pro golfers

Squaring the face is an extremely important skill in golf, and the instruction literature is full of coaching tips that address the tendency of high handicappers to leave the face open to the path, which results in a slice [16–18]. While most studies focus on clubhead speed generation, the kinetics behind face squaring has been investigated using a forward dynamics model of the golf swing [19]. In one simulation, MacKenzie [19] demonstrated that if the club's center of mass is below the force vector applied by the golfer early in the downswing, then a torque is produced on the club about the longitudinal axis of the lead arm. This torque will act until the

center of mass of the club moves in-line with the force vector. The angular impulse generated by the torque creates angular momentum about the longitudinal axis of the lead arm. This angular momentum about the longitudinal axis of the lead arm facilitates face squaring as the clubhead approaches the ball. In a different simulation, the center of mass was positioned above the golfer-applied force vector early in the downswing, and the opposite effect was observed: face squaring was strongly inhibited.

Several years after this research was published, MacKenzie started working with PGA Tour players using systems such as AMM3D (Advanced Motion Measurement 3D, Phoenix, AZ, USA) and Ping's ENSO system to capture club kinematics and perform inverse dynamics analyses. A common characteristic among the world's best golfers was the creation of the face squaring mechanics shown by the forward dynamics model. Namely, that the center of mass of the club would fall below the net force vector applied by the golfer early in the downswing (Figure 25.2a). The torque generated about the longitudinal axis of the lead forearm during the downswing of a PGA Tour player who has won multiple majors follows a very different pattern in comparison to the typical high handicapper who has a strong tendency to slice the golf ball (Figure 25.2b). The primarily positive torque generated by the PGA Tour player in the first 80% of the downswing facilitates supination of the lead forearm and, as such, promotes face squaring. In contrast, the negative torque generated by the high handicap player makes it more challenging for them to square the face heading into impact with the ball.

For Paul, this means having less tension in his hands and forearms at the top of the backswing and allowing the clubhead to move further behind him as his lower body transitions in the downswing direction. If Paul resists the urge to supinate his lead forearm early in the downswing – which seems like a face-squaring mechanism – , then he can more easily square the face for impact. This results in a straighter ball flight.

When adjusting swing mechanics, it is important to monitor performance changes in a reliable manner both in practice and on the course. Tracking the strokes gained off-the-tee with the driver is a suggested method as it incorporates both distance and accuracy [20].

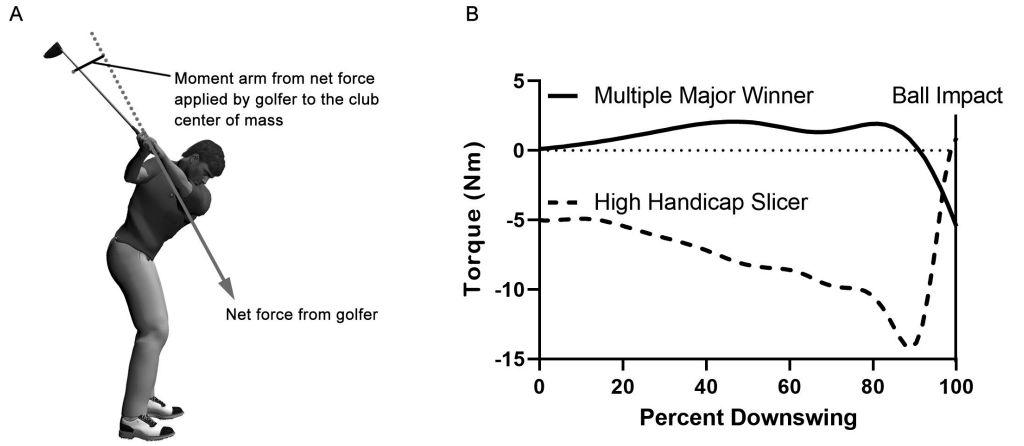

Figure 25.2 (a) Visualization of how a PGA Tour player has the center of mass of the driver "below" the net force vector he is applying to the grip early in the downswing. (b) Torque acting about the longitudinal axis of the lead forearm during the downswing for both a PGA Tour player that has won multiple majors and a less-skilled amateur that tends to slice the golf ball. Positive indicates a torque that facilitates supination of the lead forearm, which promotes face squaring.

For example, it is possible that accuracy may suffer as Paul begins to incorporate swing changes that promote hitting longer drives with a right-to-left ball flight. It's also important to consider the challenges in manipulating the mechanical causes of clubhead speed independently. For example, increasing both the angular distance the club rotates through on the downswing as well as applying a higher average couple will increase clubhead speed. However, an increase in club angular distance may come at the expense of a decrease in the couple. Paul should eventually expect to see an improvement in strokes gained off-the-tee; otherwise, the swing changes should be reconsidered. Paul should also monitor how his evolving swing changes are interacting with his driver.

Changing equipment to improve performance

Paul has been spending some time working on his swing, and by applying some of the principles outlined in the previous section, he has been able to gain some speed and reduce the amount of left-to-right curvature of his ball flight. With those improvements, Paul has made an appointment with an experienced local club fitter, Bill, to see what additional improvements a new driver could provide. As part of the fitting process, Paul will have the opportunity to explore how the mass properties, length, and shaft influence his performance. To kick off the driver fitting process, Bill has Paul warmup and then hit several drives with his current driver to serve as a baseline. With the work that Paul has put into his swing, his clubhead speed is now averaging 99 mph and generates a ball speed around 147 mph. The ball is launching at 10.5 degrees, spinning at 2,350 rpm of backspin, and curves with a slight fade. Bill knows right away that there are some gains on the table if he can increase Paul's launch angle and spin rate and feels like he may be able to increase ball speed using his knowledge of how different fitting variables influence performance. Bill starts by settling on a driver head model, backweight position, and loft, which immediately straightens out Paul's ball flight and provides 1 mph of ball speed. From here, Bill feels he can continue to maximize ball speed while optimizing launch conditions by varying the weight and balance of the driver, length, and shaft properties.

Effect of driver mass properties on performance

Decreasing any of the inertial parameters of Paul's driver will increase his ability to generate maximum clubhead speed, something that may help him hit the ball further and ultimately lower his scores. The phrase "inertial properties" is a short form representation of the following club parameters: mass, center of mass location, and the moment of inertia (MOI) tensor. The center of mass is the mass centroid of an object (Figure 25.1b for a golf driver). When understanding the mechanical influence of the center of mass, it is also critical to realize that a player will always be applying forces at the grip end of the club. The distance between the point of a player's force application and the club's center of mass is an important mechanical variable. The MOI tensor (also referred to as the second moment) is a collection of values that represent the club's resistance to angular acceleration or change in state of rotation. There is an additional parameter related to a club's mass properties that is specific to golf referred to as the club's swing-weight. Clubs with the same swing weight share the same first moment about a point, typically measured 14 inches from the grip end. This distance is arbitrary, and swing weight is expressed on an alphanumeric scale. For instance, a measurement of D-0 corresponds to a first moment at 14 inches of 213.5 oz-in. Of the components that comprise a driver, the grip and the head are the two that can influence these mass properties the most, both due to their location on a built driver and the practicality of adding or removing weight from these components. How would altering the mass of these components interact with

Paul's golf swing and the impact dynamics of the clubhead and ball, ultimately influencing overall performance?

The most straightforward manner of increasing or decreasing all parameters simultaneously is to add or remove mass from the clubhead of Paul's driver. This will not only influence overall club mass, but it will move the center of mass position relative to the hands and increase or decrease the moment of inertia of the club. MacKenzie et al. [21] investigated the influence of clubhead mass on driving performance with 18 low-handicap golfers. On average, participants generated 2.6 mph more clubhead speed with the lightest head (174 g) relative to the heaviest head (200 g). Burritt and Henrikson [22] also examined the effect of clubhead mass on performance and found that for every 10 g added to the head, a player's clubhead speed was reduced by between 0.6 and 1 mph.

Based on Bill's experience, he decides to have Paul try a clubhead that is 5 g heavier, and as expected, his speed decreases by 0.5 mph. However, Paul really likes the feel of the heavier head, and as a bonus, Bill reports a 0.5 mph increase in ball speed. It is possible for Paul to have a higher ball speed – the speed that matters – with a heavier head. Five of the participants tested by MacKenzie et al. [21] generated higher ball speeds with the heaviest clubhead. For a given set of impact kinematics, more clubhead mass will result in more ball speed. In addition to the player's response to added mass, the moment of inertia of the driver head itself will increase with the added mass, leading to less ball speed loss on off-center impacts. The optimal clubhead mass for generating ball speed will depend on Paul's speed sensitivity and response to the added clubhead mass. Players with less sensitivity to head weight changes can generate higher ball speeds with heavier heads than players who are more sensitive to changes in head mass [22]. This is in part due to how clubhead mass and ball speed are related when evaluating conservation laws, as well as the inherent increase the moment of inertia of the driver head as mass increases. This increase in driver head MOI helps preserve ball speed on off-center hits.

Bill notes that the move to a heavier clubhead has resulted in Paul's drives having a slightly more fade-biased trajectory, something he has been working to straighten out. Adding clubhead mass interacted with Paul's kinetics in such a way as to result in the face being more open to the path at impact, which will tend to promote a fade and move the center of Paul's driver dispersion to the right. This is in line with MacKenzie et al. [21], who found that the heaviest condition tended to launch the ball the farthest right with a fade spin, while the lighter conditions were associated with draw spin. The result of these interactions gave the heaviest condition an average finish location approximately 20 yards right of the lightest condition. Bill feels that by altering some of the other properties of the club, he can straighten the ball flight out with the heavier head.

Paul's positive psychophysical reaction to a heavier club prompts Bill to test his response to a driver with increased grip mass. Bill secures a 25 g cylinder of mass inside the shaft at the grip end of Paul's driver. Adding mass to the grip (or upper portion of the shaft) will influence swing mechanics and clubhead delivery, but not impact dynamics [23]. Given the small duration of impact (~0.00045 s), only the mass of the clubhead and bottom portion of the shaft play a role during impact [24]. Bill is familiar with a study – MacKenzie et al. [23] – that investigated the influence of three grip mass conditions (0 g added, 50 g added, and 100 g added) on driving performance. These grip mass variations are quite large relative to typical adjustments made in a fitting, yet the differences found between conditions were quite small. For example, clubhead speed with the lightest condition was only 0.2 m/s faster, on average, relative to the heaviest condition. MacKenzie et al. [23] concluded that large changes in grip mass should not be expected to have meaningful or predictable changes to performance with the driver. Paul likes the feeling of the additional grip mass in his driver, and Bill cannot detect any systematic changes to driving performance, so Paul's final driver build will have this counterweight installed.

Effect of driver length on performance

Another property of the driver that can influence clubhead delivery kinematics is shaft length. The current limit on club length imposed by golf's ruling bodies is 48 inches, with a model local rule, which may be implemented for tournament play, of 46 inches [25]. Based on kinematic constraint equations alone, if Paul used a longer driver shaft, his clubhead speed should increase. The kinematic equation linking linear velocity with angular velocity in pure rotation, $v = r \times \omega$, implies that the speed at the distal end of the club (v) is proportional to the length of the club (r). However, this assumes Paul can create the same angular velocity (ω) with the longer shaft. In practice, longer shafts are typically associated with reduced angular velocity, since increasing length will increase the club's inertial properties. As such, when assessing the relationship between clubhead speed and length, the inertial properties must be considered. Multiple empirical studies have been conducted to examine the effect of driver length on performance; however, each study took a different approach to controlling the inertial properties. Kenny et al. [26] kept the mass of the head constant as length changed from 46 to 48 inches, which led to an increase in total club MOI and swingweight (ranging from D9/229.6 in-oz to F4/267.5 in-oz). Conversely, Lacy et al. [27] used club builds where the swingweight was within a closer range (D2 to D6), but the total weights were up to 70 g different for lengths ranging from 44.5 to 46.25 inches. Both studies did report increases in clubhead speed and ball speed as length increased, with varied results at the individual level. The Kenny et al. [26] study saw a club head speed increase of 3 mph and an average ball speed increase of 4 mph from the shortest to longest condition, while the Lacy et al. [27] study saw an average ball speed increase of 3 mph from the shortest to longest condition.

One way to evaluate how changes in length and the resulting mass properties influence performance in a theoretical fashion is to use a forward dynamics model with a prescribed set of kinetics to look at how varying length, while keeping head weight or swingweight constant, influences clubhead speed. A previously presented two-armed forward dynamics model [28] was used for this purpose. Figure 25.3 shows the results of simulations in which the muscular activation patterns of the golfer were kept constant while systematic changes were made to shaft length and the inertial properties. Keeping *head weight* constant, the model demonstrated that clubhead speed

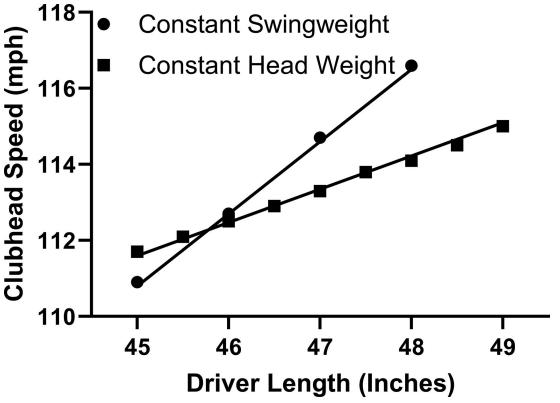

Figure 25.3 Clubhead speed versus driver length results from a forward dynamics model. When swingweight was kept constant, clubhead speed increased by 1.9 mph per 1" increase in length. When head weight was kept constant, clubhead speed increased by 0.8 mph per 1" increase in length.

will increase by 0.8 mph for every 1″ increase in driver length. This rate of change is within the range seen by Kenny et al. [26] in their study, where head weight was kept constant as length changed. When *swingweight* was kept constant, by using a lighter head as length increased, the model predicted a 1.9 mph increase in clubhead speed for every 1″ increase in driver length.

To explore this behavior empirically, two separate experiments were conducted using drivers where length was varied but swingweight remained constant. One experiment was conducted indoors where golf club kinematics were collected using an eight-camera optical motion capture system (T40S, Vicon Motion Systems Ltd, UK) operating at 720 frames per second, and reported ball initial conditions were recorded with a Foresight GC2 launch monitor (Foresight Sports, San Diego, CA, USA).

A second experiment was conducted outdoors using a Trackman launch monitor to record both club and ball data. Twenty players of varying playing abilities (Motion Capture Test Mean Handicap = 3.6 ± 4.7, Range Test Mean Handicap = 14.3 ± 7.7) participated in each of the experiments. The indoor experiment compared three different drivers of lengths 44.25″, 45″, and 45.75″ with a swingweight of D3, while the outdoor experiment compared four different drivers of lengths 43″, 44″, 45″, and 46″ with a swingweight of D3.

Figure 25.4 shows the results of these two experiments. The clubhead speed in the indoor test increased by 1.87 mph for every inch added while the experiment on the range showed a

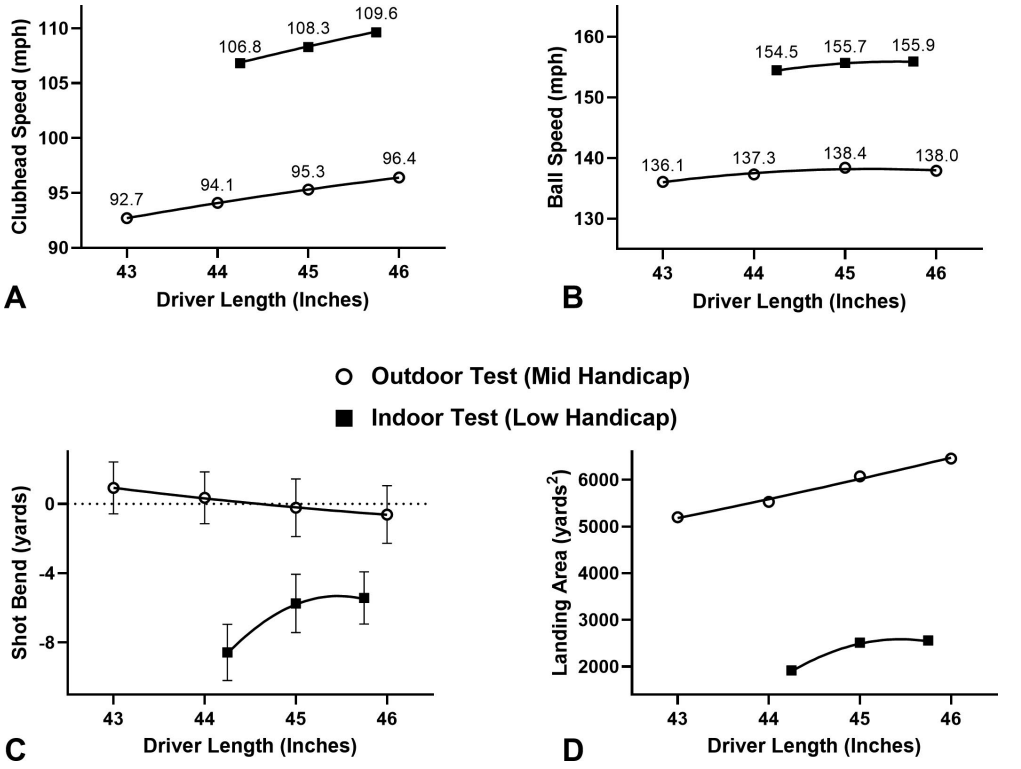

Figure 25.4 Results of two experiments regarding the influence of driver shaft length on (a) clubhead speed, (b) ball speed, (c) shot bend, and (d) landing area. One test was conducted indoors on low handicap (3.6 ± 4.7) golfers, while the other test was conducted outdoors using mid handicap (14.3 ± 7.7) golfers.

lesser increase in clubhead speed for every inch added of 1.23 mph for every inch. The indoor experiment with higher-skilled players was in reasonable agreement with the forward dynamic simulations. Conducting these experiments is crucial to confirm whether the anticipated improvements take place or if the intervention triggers unintended biomechanical adjustments. Players of varying skill levels often exhibit variability in their swing techniques, and they may unconsciously adapt their approach based on club properties or environmental conditions.

Ball speed demonstrated quadratic behavior, decreasing or minimally increasing once the length increased past 45″. Note that lighter driver heads were used to keep swingweight constant as shaft length increased. It seems that, at these lengths, the positive influence of clubhead speed on ball speed can be meaningfully negated by the retarding influence of reduced driver head inertia on ball speed. Less consistent impact patterns on the clubface as shaft length increased likely also played a role in diminishing the positive influence of clubhead speed on ball speed [29]. Lastly, the shot bend (or curve) for both test groups did not vary in a statistically significant way, but offline and downline variability did increase for the longer drivers.

Paul has been testing drivers at a length of 45.25″ up to this point during the fitting, and Bill wants to see if maybe adding 0.5 inches to the driver could provide a little extra clubhead speed, leading to more ball speed. Bill is familiar with the results of the experiments dealing with driver length and its effect on performance, but is still curious to see how Paul responds to a little extra length. Since Paul has arrived at a weight that feels comfortable, Bill elects to build a driver at 45.75″ and a head weight that is 3.5g lighter for Paul to try. This will keep swingweight approximately the same. Paul hits 10 drives with the longer driver and sees an increase in clubhead speed of 0.75 mph but sees a loss of ball speed. This seems to be a result of a more dispersed impact pattern coupled with the lighter, less forgiving driver head. Bill and Paul both decide to stay at a length of 45.25″ with the driver.

Effect of driver shaft properties on performance

Shaft flexibility has perhaps been the most researched property that will interact with Paul's swing kinetics [11,13,30–35]. Although a universally perfect shaft for all golfers does not exist, the stiffness of the shaft is heavily emphasized as a key factor in the club fitting process. Previous research suggests that clubhead speed and orientation at impact can be influenced by shaft stiffness at the individual level, but overall performance differences across test participants for shafts of different stiffness values averaged out and were not significant [13,30,32,35]. MacKenzie and Boucher [32] showed that some players gained over 2 mph of clubhead speed with a particular shaft stiffness while others experienced no change in clubhead speed.

Bill has Paul try shafts with varying levels of stiffness to alter the amount of loft delivered to the ball at impact. The delivered loft will primarily influence ball launch and spin. A more flexible shaft will have more 'lead' deflection (or bend loft) at impact [13,32] (Figure 25.5). More lead deflection will tend to increase the delivered loft, which will increase both the amount of ball spin and vertical launch angle. Paul's vertical launch angle (10.5 deg) and backspin (2350 rpm) numbers are both low for his ball speed, so Bill initially tries a more flexible shaft. As expected, Paul increases his delivered loft, resulting in the ball now launching at 12.5 deg with 2600 rpm of backspin, yielding a carry distance of 250 yards (an increase of 10 yards). The explanation in this scenario is straightforward. Paul's downswing club kinetics are similar with different shafts. If the same kinetics are applied to a more flexible shaft, that shaft will experience a greater magnitude of deflection during the swing [33].

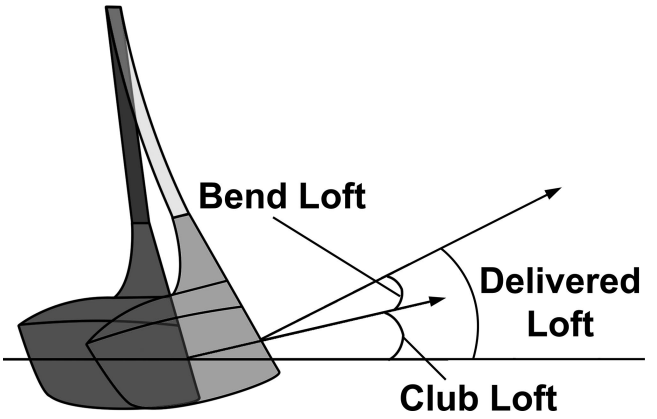

Figure 25.5 Graphical definition of club loft, bend loft, and delivered loft. The solid head represents the position if shaft did not bend. The transparent head shows the position of the actual clubhead at impact.

While Paul's performance with the more flexible shaft is likely, it would not be surprising to see the opposite response [36]. This is because shaft stiffness has a strong influence on grip kinematics as well as clubhead kinematics. While a more flexible shaft will have more lead deflection at impact, it will typically be associated with a grip orientation that will reduce delivered loft. For example, Paul's increased lead deflection might tend to increase loft by 2 deg, but it would not be surprising if he also leaned the grip towards the target by an extra 2 deg at impact. This would result in no change in delivered loft. Two potential explanations exist concerning the impact of shaft stiffness on grip kinematics. One scenario suggests that golfers might employ a consistent motor pattern with every shaft. However, due to the differing bending characteristics of each shaft, distinct patterns of reaction forces may be exerted onto the golfer, subsequently leading to changes in grip kinematics. Another possibility is that certain golfers could perceive distinctions between the shafts, prompting the utilization of a motor pattern specific to the stiffness of the shaft during their swings [32].

As an experienced fitter, Bill is not likely to try and find a shaft stiffness that maximizes Paul's clubhead speed. A more flexible shaft results in higher kick velocity (clubhead speed relative to grip speed). However, the hands will typically slow down approaching impact due to the recoil of the shaft, which will have a retarding influence on clubhead speed relative to the ball. These findings may be the result of the golfer changing how they swing the club due to the feel differences in the shaft [32]. This research suggests that shaft flexibility can influence clubhead speed at the individual golfer level, but not in a readily predictable manner. It is possible that Paul could see a 2 mph gain in clubhead speed with just the right shaft; however, there is no guarantee that this shaft will also produce the desired ball launch angle and spin rate.

Up to this point, Paul has improved his launch and spin, increased ball speed, and is comfortable with the weight and balance of the test driver. The remaining element of performance that Bill and Paul would like to try and address is the slight fade that came along with the heavier driver head Paul preferred. Another property of golf shafts that is often referenced when looking at the specifications of a golf shaft is torque. Torque, in the golf industry, is a shaft's resistance to twisting about its long axis (torsional rigidity). Unlike shaft flexibility/stiffness, there are few published studies on the effect of varying shaft torque on performance. One study conducted to determine

the influence of golf shaft torque on clubhead kinematics and ball flight compared two shafts with differing torque properties but the same longitudinal stiffness [20]. This study had 40 participants (age: 33.4 ± 8.1, handicap: 10.7 ± 4.5) and found that at impact, a shaft with a higher torque value (one that is more torsionally flexible) was associated with increased delivered loft and a more open face relative to a shaft with lower torque. This resulted in the higher torque shaft producing an average result of the ball finishing significantly further right (~5 yards) than the lower torque shaft. With this knowledge, Bill finds a driver shaft that has a similar weight and overall stiffness, but a lower torque value for Paul to try to see if this will help mitigate the fade. Indeed, the new shaft with lower torque does help straighten out Paul's ball flight, and as a bonus, Paul loves the feel of this new shaft.

The complete fitting process has resulted in Paul gaining 1.5 mph of ball speed and ten yards of carry over his old driver. This is on top of the extra club head speed and reduced shot bend that he has achieved by working on his swing mechanics.

Summary

A general understanding of the biomechanics involved in generating clubhead speed, other desirable impact kinematics, and a novel mechanism governing face squaring has been provided. Equally, a solid comprehension exists regarding how different properties of a driver can interact with and influence the golfer's kinetics, resulting in predictable changes to ball flight outcomes. Consequently, it is possible, and recommended, for a golfer to get fitted for a driver that will maximize their performance. Similar to monitoring the influence of swing changes, Paul should expect to see an improvement in strokes gained off the tee; otherwise, the fitting was not effective. A thorough understanding of the principles presented can help a player obtain meaningful performance gains. Paul's example of gaining 10–20 yards through swing and equipment improvements is not only realistic but can be quantified in terms of on-course performance. Broadie, in Every Shot Counts, shows that a 20-yard increase off the tee will lead to a 1.6-stroke improvement per round for a 17-handicap golfer [37].

One area for future research centers around understanding which golfer movement patterns would lead to more consistent performance outcomes. To be more explicit, are there certain movement strategies that would increase a golfer's probability of delivering the clubhead to the ball with the intended kinematics to produce the desired ball flight? There are also specific elements of modified swings in iron and wedge play, as well as putting, that warrant further investigation to understand how golfers interact with the clubs used in those aspects of the game.

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26

THE QUANTIFIED TENNIS ATHLETE

The performer, the ball, the racquet and the technology

Machar Reid, Thomas Perri, Olivia Cant, Adrian Eassom, Cameron Armstrong, and Lyndon Krause

Part 1: How tennis arrived at the quantified athlete

Ever since tennis turned professional in the second half of the last century, it has not just captured the imagination of audiences around the world but also that of sports scientists and engineers. One focus of sports scientists has generally been on describing what the eye cannot see, given the speed of the movement, to help explain how players can move and hit in the way that they do. Armed with this knowledge, sports scientists then help coaches plan or tailor training to best suit the individual. From an engineering perspective, the lines of inquiry have tended to centre on the racquet, string design, as well as the aerodynamics of ball flight and court-player interaction [1–3]. While this has given rise to much quality research over time, its ecological validity has often been constrained, with many experiments conducted away from the court, in laboratories, or in simulated training or competition settings.

Over the last two decades, though, this gap between what happens during match play on the court and what the scientist/engineer can quantify has been bridged. Advances in technology (both hardware and software) have enabled the timely collection, storage, and analysis of biomechanical data on player performance. The introduction of ball and player tracking technologies, powered by computer vision, as well as wearables that can be affixed to the player, footwear, or racquet, helps to quantify the game like never before. While the industry may be at the beginning of its journey in interpreting all this data coming its way, the era of the Quantified Athlete will undoubtedly create opportunities for sports engineers, biomechanists, and sports analytics experts to impact the design and use of tennis equipment at a scale not previously thought possible.

Part 2a: Quantification of ball flight and hitting characteristics of tennis shots

The characteristics of ball flight in match play

Ball and player-tracking technologies are commonplace at professional tennis tournaments, with Hawk-Eye being the sport's seminal tracking technology. Hawk-Eye [4] is composed of

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multiple cameras stationed around the court, producing three-dimensional coordinates of the ball and two-dimensional coordinates of players as a function of time. Its primary use is to call lines electronically and instantaneously, with the integration of its data into broadcast media. Increasingly, the use of these data in player analysis is also becoming mainstream in professional tennis, and readers are encouraged to refer to Chapter 35, Big Data in Sport, for a deeper analytical dive. When compared to high-speed quantitative videography, Hawk-Eye is estimated to carry a mean error of 2.6 mm for the ball's bounce location [5] as part of its compliance with the governing body's electronic line-calling system guidelines. To the knowledge of the authors, however, there has been no independent review of the accuracy of ball flight or player movement in the peer-reviewed literature.

The proprietary spin measure of Hawk-Eye is one output that has been investigated by researchers. Specifically, ball spin rate (revolutions per minute, RPM) and spin direction (topspin/backspin) have been compared to high-speed videography in lab conditions [6]. It was found that the spin measure implemented by Hawk-Eye correctly classified the spin direction (topspin/backspin) for 97.6% of the ball trajectories analysed [6]. Estimation of spin rates however proved more challenging, with root mean square errors of 550 RPM (across trajectories with up to 4,500 RPM of spin) recorded, alongside evidence of spin being underestimated at higher magnitudes and overestimated at lower magnitudes [6]. As these technologies become more available to the research community, we can expect the accuracy of raw tracking and derived variables to be scrutinised.

Data-driven insights into tennis stroke production

With the widespread use of contemporary camera-based player and ball-tracking technology, there is scope to examine aspects of match play and stroke production *in situ*. To date, these data have been used in various research studies, including; to describe match play characteristics and demands [7–9], predict the outcome of shots and points [10,11], develop shot dictionaries [12,13], as well as investigate the relationship between serve characteristics and success [14,15]. This sports analytics use of motion data builds on more qualitative, notational efforts to understand tactical and strategic issues in tennis [16].

With a focus on stroke production, prior research has detailed the characteristics of shots produced by players during Grand Slam singles matches, especially at the Australian Open. For example, compared to their female counterparts, male players at the Australian Open were reported to generate higher average serve (1st serve: 184 kph males, 156 kph females; 2nd serve: 152 kph males, 131 kph females) and groundstroke speeds (111 kph males, 106 kph females), but not serve-return speeds [9]. Shot-specific gender differences were also observed in the same cohort of players for contact point (depth and vertical) and net clearance [9]. Some of these effects can be observed through Figures 26.1 and 26.2, again with data taken from the Australian Open. Specifically, in Figure 26.1, there is some suggestion of topspin forehand speed but not necessarily topspin backhand speed trending higher. While this needs to be interpreted cautiously, given the number of interacting and uncontrolled variables, including court speed, the difference in hitting speeds – especially on the forehand side – of the top 10 players versus the field is marked. When the spin that players apply to the ball is combined with its speed (Figure 26.2), the top 10 players can be seen to separate themselves in a similar fashion. Their forehands are particularly characterised by a combination of higher ball speed and spin than typical top 100 players, which might reasonably be considered to produce the so-called "heavier" ball of top players. The effect of age or maturity has also been examined at the Grand Slam level, with shot speed differences observed

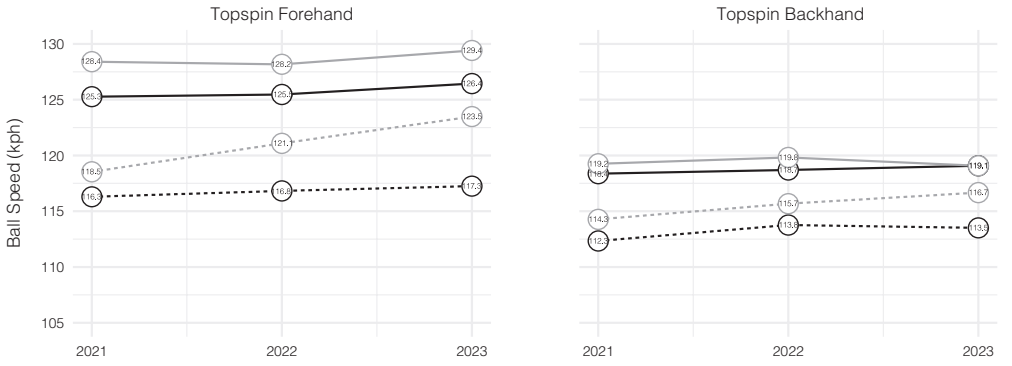

Figure 26.1 Median ball speed (kilometers per hour, kph) of topspin groundstrokes hit during 2021–2023 Australian Open main draw singles matches (Source: Australian Open data, Tennis Australia).

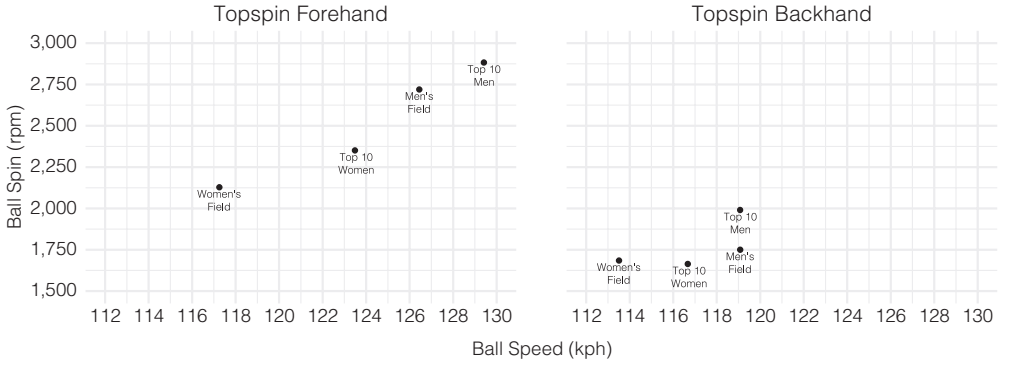

Figure 26.2 Median ball speed (kilometers per hour, kph) and spin (revolutions per minute rpm) of topspin groundstrokes hit during 2023 Australian Open Main Draw Singles matches (Source: Australian Open data, Tennis Australia).

between junior and professional players (median serve speeds: 179 kph males, 158 kph boys; 153 kph females, 146 kph girls) [8].

By considering additional contextual features of matches, such as shot or point outcome, researchers have also provided insight into the effectiveness of stroke production. For instance, as may be expected, faster-paced serves in both the men's and women's games have been associated with higher winning probabilities [13,14]. Further, in the men's game, first-serve speed has also been linked to aces, although it was found that serve placement and angle were the main factors discriminating aces from serves that were returned into play [15]. Such technical insight may help identify key aspects of stroke production that players might focus on to maximise their chance of ultimately winning a point. Perhaps future applications may lie in evaluating equipment (racquet and strings) adjustments proposed for professional players.

Approaches to understanding the types of shots played in tennis have also been informed by multiple shot-specific contextual features. In other words, rather than simply describing shots as being cross-court or down-the-line, short or deep, angled or not, researchers have used clustering techniques to create shot dictionaries based on the trajectory data collected by line-calling technologies [12]. For instance, Kovalchik and Reid [13] used data detailing impact locations and each shot's

trajectory information (e.g., ball speed, net clearance, landing location) to develop a dictionary of serves, serve-returns, and groundstrokes. Such an approach allowed for common features across shots to be identified and for the frequency as well as the effectiveness of distinct shots to be analysed. By way of example, in the women's game, 17 unique serves were identified to the ad court [13]. This contrasts with more traditional tennis analytics approaches, which classify the serve by target (centre "T" in the service box, an opponent's "body", or the "wide" corner of the service box) or spin/speed (minimal spin "flat," side and topspin "kick," or primarily sidespin "slice").

Part 2b: Our technology powered understanding of on-court movement

Today, technology plays an increasingly critical role in informing the physical preparation strategy of elite-level tennis players. In what is an interdisciplinary process, the emphasis is on the delivery of timely yet precise information to the coach and player. Sources of these complex insights continue to evolve from manual or semi-automated methods using video data [17,18] to applications of wearable microsensors (e.g., satellite-based navigation systems [SBNS], micro-electromechanical sensors) [19–33] that complement the above-mentioned camera-based player-tracking systems at the Grand Slam level [7–9]. Logistics, cost and the rules of the sport have constrained the harmonisation of these technologies across match play and training, meaning that many practitioners use a mix of technologies to inform player preparation [34].

The use of satellite-based navigation systems to quantify movement demands in training and match-play

The nuanced movement patterns and small space of tennis have rendered mostly unsuccessful outcomes across initial efforts to use SBNS for tennis-specific information. Indeed, the early work of Duffield et al. [35] identified 1 Hz and 5 Hz SBNS to significantly underestimate distance covered—by up to 30%—amongst tennis-specific movement patterns when compared to motion-tracking technologies. In an attempt to solve this inaccuracy, Vickery et al. [36] replicated these movement activities with SBNS units sampling at 10 Hz and 15 Hz. Despite finding lower underestimations of distance covered, significant differences from the criterion video motion-tracking system were still observed, thus determining the poor accuracy of SBNS metrics for use in tennis. In combination, these findings highlight limitations for practitioners to be aware of when relying on two-dimensional SBNS metrics to quantify tennis-specific movement actions. Similarly, the reliability of distance covered, velocity, and acceleration from SBNS devices has been questioned during point-play activities [30], further contributing to the conjecture regarding SBNS metrics and their application to tennis.

Nevertheless, the use of SBNS as an athlete management input in tennis has followed a similar trajectory as in other sports, where the distance covered acts as a key metric to monitor lower-body competition/training "load." The conditioning and sports medicine literature often uses the term load for the overall running volume, physical requirements of the competition, or training/practice. Indeed, SBNS applications in junior-elite tennis report distances covered approximately 2000–4500 m across both hard and clay courts [22,37], with clay court running distances generally higher [32] in matches and training. As SBNS studies commonly lack the ability to report distance covered at the point level, normalising distance covered to match time can be instructive. On this basis, both surface (47.1 m/min on hard versus 41.5 m/min on clay [13]) and player age [6,17] can can influence the distance and average running speed demands of the sport. Clearly, these findings, as summarised in Table 26.1, are best considered a guide given the reported limitations of SBNS in quantifying this aspect of court coverage in tennis.

Table 26.1 Tennis "load" studies using wearable technology

Reference	*Population (Mean Age ± SD)*	*Tennis context (court surface)*	*Wearable technology (load metrics)*	*Key findings*
[24]	20 junior males (13 ±1y)	Simulated match play (clay)	SBNS (distance covered, velocity, acceleration)	Distance covered = 3362 ±869 m. Peak velocity = 4.4 ±0.8 $m \cdot s^{-1}$. NS differences between winners and losers.
[38]	7 professional males (21.4 ±2.2y)	Simulated match play (hard)	Accelerometery (PlayerLoad)	↓PL (AU) on days 2 and 4 vs. day 1 during consecutive match-play.
[33]	12 national and regional junior males (13.0 ± 0.7y; 15.7 ± 0.8y)	Simulated match play (clay)	SBNS and Accelerometery (distance covered, body load, *g* force)	↑distance covered (m) and actions >1.5 g in national players.
[32]	8 junior males (15.5 ±1.2y)	Simulated match play (clay)	SBNS and Accelerometery (distance covered, body load, g force)	↑distance covered (m) and actions at >1.5 g on clay courts.
[23]	40 junior and adult males (13.0 ±1.0y; 25.0 ±4.0y)	Simulated match play (clay)	SBNS and Accelerometery (distance covered, velocity, acceleration/ decelerations, PlayerLoad)	NS differences in PL (AU) between winning and losing players.
[29]	14 junior males and females (12.9 ±1.3y)	Simulated match play (clay and hard)	SBNS and Accelerometery (distance covered, velocity, acceleration)	↑m/min on hard courts vs. clay courts (47.1 m v 41.5 m).
[27]	26 junior females (13.1 ±1.8y)	Official match play (hard)	SBNS and Accelerometery (distance covered, velocity, acceleration)	Total distance 3154 ±1204 m. Peak velocity 4.1 ±0.5 $m \cdot s^{-1}$. Mean velocity = 2.8 ±0.4 $m \cdot s^{-1}$. Relative distance = 44.4 ±7.8 m/min.
[28]	29 junior males and females (14.0 ±2.1y)	Official match play (hard)	SBNS and Accelerometery (distance covered, velocity, acceleration)	Mean distance covered = 3651 ±1572. Relative distance = 46.01 ±7.01 m/min. PL = 7.65 ±3.85 AU/ min. ↑m/min in higher-ranked players.

(*Continued*)

Table 26.1 (Continued)

Reference	*Population (Mean Age ± SD)*	*Tennis context (court surface)*	*Wearable technology (load metrics)*	*Key findings*
[31]	28 junior males (12.2 ±0.3y)	Simulated match play (clay)	Accelerometery (*g* force)	NS differences in average acceleration (*g*) between winners and losers.
[22]	39 junior males and females (age range = 11–18y)	Official match play and training (clay and hard)	SBNS and Accelerometery (distance covered, player load)	↑m/min during matches for older adolescent males. ↑PL/min during training for younger aged females.

SBNS = global positioning systems. **NS** = not significant. **PL** = PlayerLoad. **AU** = arbitrary units.

The use of inertial sensors to quantify movement demands in training and match-play

Commercially available SBNS devices house piezoelectric accelerometers, which detect linear accelerations occurring in three movement planes (antero-posterior, vertical, medio-lateral) [39]. In sports, wearable sensors often house accelerometers which integrate with gyroscope and magnetometer signals, which respectively detect angular acceleration and unit orientation [40]. Collectively, these devices are referred to as micro-electromechanical sensors (MEMS) in the literature and contribute unique, less-invasive insights into body movements in sports [41].

With a focus on accelerometry, one commonly reported metric is termed 'player load' (PL), which calculates the square root of the sum of individual accelerations captured in each of the three movement planes and divides it by 100 [42]. While an arbitrary measure of resultant acceleration of the body, the PL metric may have increased suitability for court-based sports that are characterised by high acceleration actions yet incur a lower velocity or involve changes in direction [43–46]. Indeed, Gescheit et al. [38] used PL to identify the presence of fatigue in male players during consecutive match-play days and revealed mean values after two hours to be between 400–600 AU. These values significantly reduced following the initial match and point to the sensitivity of PL in possibly detecting changes in physical output, although whether those acceleration changes are distributed between running and hitting actions is unknown. Other literature reporting PL measures in tennis has provided contrasting perspectives on the physicality of the sport, which may be related to variability in the playing cohorts, match times, or data processing methods [27,28]. More research is needed on what PL is actually measuring and whether more detail with calibration and full MEMS provides more relevant tennis intensity data.

Inertial measurement analysis (IMA) is also possible with MEMS, as the acceleration signal is integrated with the direction from the magnetometer [25,47]. Its use in research is in its infancy, but early work has revealed that the greatest proportion (39%) of accelerations occur in the vertical axis, likely reflecting tennis's repeated running actions with possible contributions from trunk rotation [25].

Machine learning capabilities have increasingly been applied to inertial sensor data in tennis to categorise accelerometer load from hitting or movement actions [21,48]. Whilst technologically

promising, data from junior-elite match play has shown that accelerometer load from movement actions (i.e., movement player load [mPL]) remains largely unchanged across court surfaces or between winners and losers [19]. Separately, compartmentalised player load measures have been somewhat useful in establishing a compendium of drill categories that impose desired relative PL [20].

The use of Hawk-Eye to quantify movement demands in training and match-play

As discussed, at the Grand Slam level, player tracking data is captured via object-tracking technology and typically reports distance travelled and movement speeds during broadcasts [49]. Whilst data on the precision of these systems remains unpublished, the mean errors (55 mm) of single-camera analysis systems are encouraging [50]. Accordingly, running distances from recent Grand Slam competitions can provide useful benchmarks for coaches.

Indeed, female and male players at the Australian Open have been reported to cover 5,656 ± 947 m and 9,647 ± 1,935 m (mean ± standard deviation), respectively, over the tournament's first four rounds [51], at an average match distance of 2,110 m and 1,232 m [9]. At the set level, distances of 572 ± 152 m for men and 553 ± 172 m for women were traversed, highlighting how extended match lengths in the men's game influence running volumes [7,9]. This observation is further supported by a near-perfect correlation between total match shots played and total distance covered, which is almost identical between the sexes [52]. Further granularity afforded by player tracking systems reveals approximately 5–8 m covered per point, which is perhaps more instructive for training drill planning [8,9,18,52]. Additionally, recent work has detailed similar distance requirements for the movement cycle between the sexes, which stands at ~4.5m [52]. Furthermore, these precise descriptions of tennis movement suggest an increase of approximately 2 m during winning points and support previous research using video tracking in men's tennis that reports an additional 4 m of distance covered per point [9,53]. It is worth noting that these insights are from play on hard courts and that professional players can be expected to engage in more extended rallies and traverse longer distances on clay courts.

The stop-start nature of tennis movement introduces a ceiling on the speed of player movement, with mean resultant movement velocities reported to be in the range of ~3.4–3.7 $m \cdot s^{-1}$ for men's and women's tennis [9]. Practically speaking, descriptions of movement velocities (speed and direction) are pertinent in understanding the interplay of locomotor actions with stroke performance [17], especially in relation to lower limb joint kinematics which are known to vary with increases in movement velocity into a shot [54]. Similarly, the use of advanced analytics to classify change-of-direction kinematics can be helpful in profiling elite tennis players and providing a coaching framework for planning training [55]. Chapter 32 provides additional context for readers interested in learning more about the application of wearables in sports.

Part 2c: Quantifying racquet performance in practice

Extensive research by biomechanists, sports engineers, and physicists pioneers like Howard Brody and Rod Cross has laid the foundation that informs decision-making regarding contemporary racquet and string design [2,56,57]. This large body of work is less well known among coaches, which is an interesting paradox given the suggested importance of equipment to a player's performance and injury potential. Part of this can be explained by the nature and focus of the research (often lab-based and highly technical). Equally important in this theory-to-practice gap are the

unclear marketing of actual equipment innovations in tennis and the funding emphasis of federations on applied sports science and medicine rather than engineering.

It is with this backdrop in mind that this section has chosen to focus on the outcomes of the co-authors' applied equipment work with players and coaches. These learnings have been translated into practice as part of an effort to highlight the potential influences of equipment on high-level tennis play.

The importance of swingweight

Of all the racquet characteristics of interest to players and coaches, swingweight is perhaps the most investigated. Swingweight is the swing plane moment of inertia of the racquet about the typical axis of rotation at the grip. A racquet with a higher swingweight would make it harder for a player to accelerate throughout the stroke. Still, beyond the time-honoured practice of players adding lead tape to various locations on the racquet head, practical insights are sparse.

Conversations with manufacturers indicate that there is a generational change in the specifications of players on tour. In the search for more spin and racquet acceleration, a growing number of elite players are opting to use racquets with lower-strung swingweights <310 kg·cm^2. Anecdotally, the authors have also observed positive effects for the management of acute overuse injuries, such as distal-humeral bone stress, through the introduction of lower swingweights (beyond changing racquet characteristics).

Furthermore, no equipment change in the history of the sport was more publicised than that made by Roger Federer in 2014. Through some fine-tuning and a redesign of his classic Pro-Staff Wilson, he created a racquet specifically for his game and performance priorities. The result; a marginally bigger sweet spot, stiffer frame, and lower swingweight. The sweet spot of a tennis racquet refers to the contact point near the centre of the string bed that imparts maximum speed to the ball but can also refer to the contact point that minimises the impulsive forces transmitted to the player's hand [58]. While the size of the sweet spot refers to the size of the region around these contact points where the shock and vibration from impact with the ball remain low and the outbound ball speed remains near the maximum, the cumulative outcome for Federer was therefore a racquet that is easier to

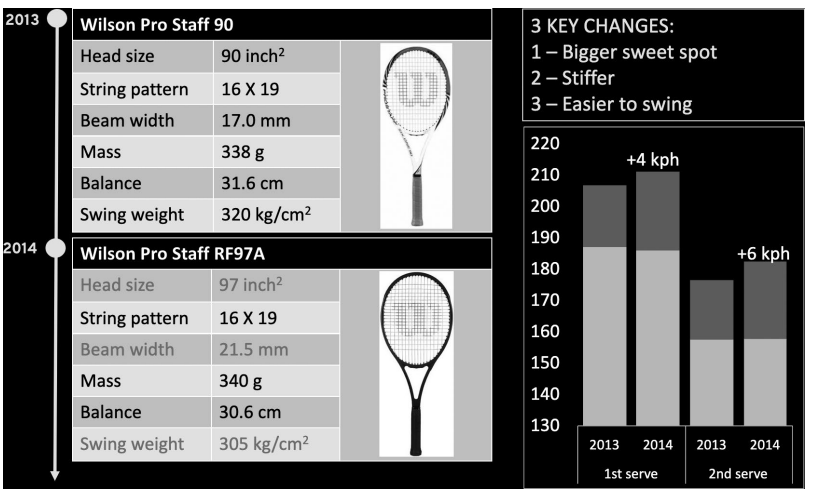

Figure 26.3 Example of how racquet changes can contribute to improvements in serve speed in the professional game.

manoeuvre and has higher potential for power/ball speed. The comparative profile, created in Figure 26.3 using Australian Open data, is clear: while average serve speeds remained consistent, his top-end speeds improved by 4 kph on the first serve and 6 kph on the second.

Guidelines for the youth market are slowly becoming less arbitrary

Research guiding racquet selection for young players is rare, with only a handful of peer-reviewed articles in the Open era [59–63]. In general, the work provides broad guidance that racquets should be modified to the player's physical maturity to enhance racquet control [62,64] and skill development. Even then, the recommendations for age-appropriate racquets offered by stakeholders in the tennis industry are confusing and inconsistent. For example, some groups like the Lawn Tennis Association [65] rely on age to recommend racquets, while others use height [66] or a mix of height, court size, and skill level [67] (Table 26.2).

Given this ambiguity, a team at Tennis Australia reviewed the equipment set-ups of hundreds of elite 10 and under tennis players (51% male, age = 10.7 ± 0.4 years, height = 148.1 ± 6.4 cm)

Table 26.2 Guidelines for racquet selection from different tennis industry sources

Organisation	*Criteria*				*Recommended length (inch)*
	Age (yrs)	*Height (cm)*	*Playing level*	*Court size (m)*	
Major retail store					
Tennis Warehouse	3–5	-	-	-	19
USA	6–8	-	-	-	21
	9–10	-	-	-	25
	>11	-	-	-	26
	Adult	-	-	-	27
Tennis Warehouse	-	0–100	-	-	19
AUS	-	101–113	-	-	21
	-	114–126	-	-	23
	-	127–141	-	-	25
	-	142–158	-	-	26
	-	158+	-	-	27
Racquet Manufacturer					
Head	2–4	-	-	-	19
	4–6	-	Beginner/Advance/Tournament	-	19–21
	6–8	-	Beginner/Advance/Tournament	-	23
	8–9	-	Beginner/Advance/Tournament	-	25
	9–11	-	Beginner/Advance/Tournament	-	26
	>11	-	Beginner/Advance/Tournament	-	27
Babolat	-	<80	-	-	17
	-	81–100	-	-	19
	-	101–110	Future/challenger	-	21
	-	111–125	-	-	23
	-	<140	Future/challenger/master	-	23
	-	<145	-	-	27

Table 26.3 Mean and range [min, max] of racquet specifications used by elite 10 and under players

	Length (inch)	*Mass (g)*	*Balance point (cm)*	*Swingweight (kg/cm²)*	*Twistweight (kg/cm²)*	*Head size (inch)*
Male	26.6 [25.0, 27.5]	288.6 [259.8, 323.7]	33.0 [31.2, 35.4]	277.2 [215.5, 321.5]	15.5 [7.0, 19.0]	99.7 [97.0, 102.0]
Female	26.5 [25.0, 27.0]	279.2 [229.9, 327.5]	33.3 [31.0, 35.6]	271.3 [216.0, 326.5]	15.3 [5.5, 24.0]	101.2 [98, 105.0]

across multiple years to better understand the equipment choices of the modern youth player. The specifications of each player's racquet were recorded, including mass, balance point, swingweight, length, grip size, head size, and string choice. Results revealed that a large proportion of players used full-length adult racquets, which is in deference to the age-based guidelines in Table 26.3. These decisions were often influenced by their favourite player and not necessarily what was best for their game. Head sizes of 100 inch^2 were preferred, albeit young male players tended to use heavier racquets (>300 g) with smaller head sizes than young females. In a testament to the range of racquet specifications among this category of player, the mass of rackets varied by 43%. In summary, the incongruence between the guidelines of the tennis industry and the equipment choices of players was stark. Rather, the diverse range of racket specifications highlights the need to reimagine the existing arbitrary guidelines, perhaps based on subsequent longitudinal research to determine what racquet properties best match skill advancement in typically developing young tennis players.

Performance data is now powering racket selection

Another way in which electronic line-calling technology can be used in training contexts is to assist with the equipment selection of players. Rather than players simply relying on the advice of a retailer, stringer, coach, or their intuition when trialing a demo racquet on-court, they can now participate in experiences that allow them to systematically manipulate features of their racket and string set-ups and receive real-time performance feedback (Figure 26.4). This allows players to reconcile (or challenge) their subjective views with objective data to inform equipment selection in much the same way as golf has done for decades. Figure 26.4 provides an example report generated by Tennis Lab, Melbourne, Australia. Outlined in the report are forehand speed, spin, depth, accuracy, and shot heaviness (a combination of ball speed and ball spin) for three racquet setups. Despite the proficiency with which the current racquet produces spin, it ranks as the least powerful. In this example, if the priority for this player is to improve forehand ball speed, a racquet change is recommended. While science has previously recommended equipment changes to improve tennis shot performance, bridging the gap between theory and practice has often proved a challenge. For instance, players in the mid-twentieth century believed higher string tensions created higher shot speeds based on the 'feel' of the impact. Once research identified this as a myth, it took a decade or more for that knowledge to penetrate the coaching and player populations.

Accordingly, by quantifying the interaction between racquet-string set-ups and hitting performance in this way, professional players can scrutinise their games in more detail. Recreational or competitive players can also receive data-driven recommendations to better guide their equipment selection (i.e., www.tennislab.com.au).

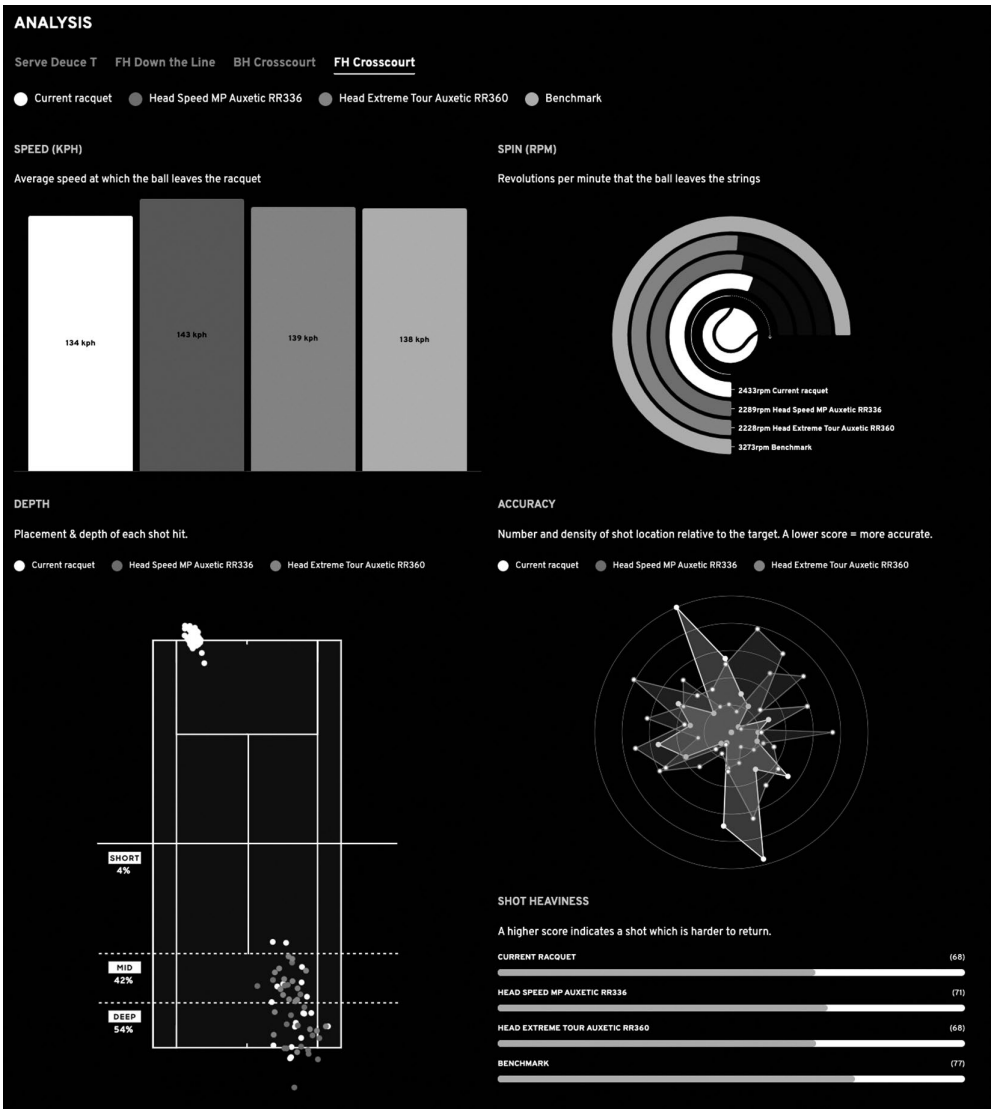

Figure 26.4 An example report comparing the on-court performance characteristics (speed, spin, heaviness, depth of landing location, and accuracy of shot), quantified by HawkEye, of various racquets by an individual player.

Summary

Technology that quantifies tennis play in new ways can create new insights for the sport. These insights can inform improvements in player performance, injury rate reduction, and equipment design. Importantly, these advances can benefit both professional and amateur athletes.

The measurement precision and reliability of the tracking and sensor technologies will continue to evolve, presenting opportunities for governing bodies to simultaneously promote innovation

and uphold the integrity of the sport. On a practical level, the sports sensors and performance industry are still in their infancy in understanding how to best synthesise and leverage the various technologies and their data outputs. What is clear though is that *quantified athletes* are here to stay. Biomechanists, engineers, and other sports scientists have the opportunity to collaborate to validate the impact of these technologies and data on the tennis ecosystem.

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27

SPORTS INJURY AND TRAUMA BIOMECHANICS

Surrogates, human data and modeling

Nicola Petrone, Gemma Leslie, and Giuseppe Zullo

Introduction

Sports injury or trauma [1,2] represent one of the most frequent and long-term causes of suspension or retirement from competitive and recreational sports. Injuries in sports are defined in many ways, but most often are classified as either acute or overuse. Overuse injuries result from the accumulation of microtrauma in tissues over many training and competition cycles without adequate rest, eventually leading to pain and disability. This chapter focuses on acute or traumatic sports injuries where the loading and energy transfer to the body in a single event are enough to cause damage to body tissue and function. Acute sports injuries can result in expensive treatment, reduced performance, and long-term disability that goes beyond elimination from the sport to impaired health through decreased physical activity. Consequently, there is considerable interest in understanding traumatic sports injuries, protective equipment that works with biomechanical and environmental factors, and how the risk of these injuries can be reduced.

Types of injury and trauma occurring in sports vary in terms of body segments and joints, degree of severity, and the level of competition. Risk factors for sports injuries are classified as internal to the athlete (biomechanical-neuromuscular state determined by age, strength, flexibility, fatigue, psychological conditions, etc.) and external (such as type of sport discipline, workload, quality of sport gear, presence/quality of personal protective equipment or environmental/ground conditions etc.) [3–5]. Epidemiology, engineering, and biomechanical research have focused on identifying risk factors in sports injuries, to work toward changes in equipment, training, and rules that would reduce the risk of these injuries in different athletes.

This chapter addresses how engineering and biomechanics interact to create the most accurate physical and theoretical models for the study of traumatic sports injuries. First, the methods and surrogate models for the biomechanical study of injuries are summarized. The subsequent sections illustrate the utility of physical surrogates of the human body, starting from full-body surrogates to various regions of the body. These regions are the head and neck, upper limb and hand, lower limb and knee, and ankle and foot.

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Surrogates and models for sports injuries and trauma research

Decades of research in biomechanics and engineering have focused on the mechanisms and potential reduction of traumatic sports injuries. The use of several different measurements provides a synergy and synthesis of results to understand these complex problems and potential solutions. Some of these research techniques include epidemiological surveys, video-based analyses, experimental in-vivo biomechanical acquisitions and analyses, in-vitro cadaver or Postmortem Human Subjects (PMHS) studies, multibody simulations, finite element (FE) simulations, musculoskeletal modeling, and their combination. The recent development of big-data analysis and Artificial Intelligence can also open new perspectives in the exploitation of large amounts of data collected from in-vivo and in-vitro human studies, large databases of measurements and video collected by wearable smart devices, and the availability and refinement of modeling techniques. This section summarizes a van Mechelen and colleagues' [6] model of the study of injury and its potential prevention and then focuses on the use of surrogate models.

Sports engineering and biomechanics play a fundamental role in the five-stage injury and trauma prevention model proposed by van Mechelen et al. [6]. These stages are: (i) the **interpretation** of injury and trauma mechanisms, (ii) the evaluation and **modeling** of external/internal loading conditions, (iii) the **characterization** of biological tissue response and limit loading conditions, (iv) the **identification** of preventing or protecting strategies and (v) the **assessment** of personal protective equipment or behavioral/physical training for the reduction of injury/trauma risk levels.

In the **interpretation** and **modeling** of sport injuries and trauma mechanisms, in-vivo simulations and video-based analysis play a dominant role (supported by 3D motion capture (MOCAP) and virtual simulation). Whereas in-vitro testing is fundamental for the characterization of structural response and strength of biological tissues/segments/organs, and in the assessment of the effectiveness of countermeasures adopted for the prevention of injuries and traumas, mainly in the case of personal protective equipment (PPE) and protective clothing [2].

The mechanical **characterization** of body segments/joints/tissues requires the measurement of many mechanical properties of human subjects, striving for primarily in-vivo experiments. Common measures include joint range of motion (ROM) and stiffness properties in active and passive movement, skin/fat/muscle surface properties via mechanical indentation or ultrasound methods, and body segment inertial properties via quick release or oscillating tests. Safety limits or ultimate loads associated with muscles, tendons, ligaments, bone, and all internal vital organs, however, cannot be retrieved for ethical reasons. In this case, full body cadaver studies on PMHS or body segment samples from donors are the only way to collect limit values for the strength/integrity of soft and hard tissues composing the human body. Researchers in trauma biomechanics in this case need to liaise with medical doctors and anatomists to ethically access cadaver/segment or tissue/organ specimens to estimate the effect of simulated overload/impacts/crashes in the case of protected or unprotected events.

On the other hand, the **assessment** of the effectiveness of PPE in preventing sport trauma and injuries ideally requires the reproduction of critical loading conditions in realistic game/accident scenarios and postures. This testing is typically achieved using direct drop tests, impactors (from pendulum arrangements, drop towers, or air/electro/mechanical cannons) or multi-actuator controlled test benches that are designed to apply representative loading conditions to a surrogate body segment wearing the protector. The correct assessment of a protective device depends on (a) the **biofidelity** of the surrogate body segment/joint assembly, (b) the **compatibility** of the surrogate shape/size with the wearable protection under evaluation, (c) the **representativeness** of

the loading/restraint conditions of the test methods, and (d) the adoption of appropriate **Injury Risk Functions** [7] or **Injury Criteria** [7] associated with the surrogate test method and validated against human data.

Surrogates, also called phantoms, replicas, or dummies (full-body regions, or segments), are essential tools for the qualification process of any wearable equipment for prevention and protection. The International and National Standards bodies are the official regulatory organizations in charge of defining performance and safety minimum requirements and test methods. These standards shall be followed by manufacturers, users, athletes, and regulatory bodies of sanctioned and unsanctioned events. Being the result of a long and qualified consensus process based on scientific evidence, industrial experience and customer surveillance, International Standards shall be considered the state of the art of knowledge in the field of product safety and appropriateness for PPE of any devices/vehicle/assistive technology interacting intentionally or potentially with the human body.

Surrogates find large application in all fields of safety standards where there is a need to protect the human body from harmful situations encountered in daily life occupations, ranging from transportation, medical and personal care, to various occupational risks (fire, police, and military). Their appropriateness can be summarized in the concept of **biofidelity**, which is the capability of the surrogate to represent the response of the human form to input actions. The field of injury and trauma biomechanics uses both **static** and **dynamic** actions or conditions. Depending on the specific requirement of the undergoing test, biofidelity is typically assessed with respect to the physical phenomenon being simulated by the standard test method. In road vehicle safety, for instance, when assessing the crashworthiness of front bumpers against side impacts to pedestrians, biofidelity of a lower leg surrogate is defined in ISO as the "capability to represent impact responses of human subjects" [8]. Historically speaking, surrogates of the human body first appeared in the field of daily life apparel, such as a wooden foot used for shoemaking or dress forms (mannequins) used since the 15th century to design and show dress fitting. In these cases, the fidelity to the human body was only limited to the external shape of the form, with no distinction initially even from right/left side, human gender, foot size, or any biomechanical motion of the foot.

Current biofidelity of a surrogate is evaluated by a sequence of hierarchical equivalences of the surrogate to the corresponding body segment/complex properties. The subsequent additional requirements that follow (R1–R6), a concept being introduced by the authors in this chapter, will qualify the increasing biofidelity of a human body segment/complex surrogate.

R1 **Dimensions**

Dimensions of the surrogate segment shall adapt to Human Anthropometry regarding segment length, width, breadth, girth, as measurable from standardized methods with respect to standardized body landmarks [9]. Also, depending on the subject's gender, age, and ethnicity peculiarities that body dimensions show, as documented by several international Standards and Databases. Head circumference is typically a sizing body measure to which an appropriate headform shall conform. Foot surrogate length as well should reproduce the foot longitudinal dimension of the corresponding size foot.

R2 **Shape**

The shape of the surrogate segment shall correspond even in localized contour or surface curvature to the shape of an average surface/volume conformation of the corresponding limb/segment: not only the overall dimensions or dimensional ratios, but ideally, the whole surrogate volume shall be consistent with the statistically meaningful percentile shape of the biological counterpart. Here, refined statistical methods for the parametrization of a large database of human samples and the retrieval of mean or extreme percentiles of the parameters shall allow

generation of a virtual average volume, typically after simplification of irrelevant fine features. Given the same head circumference, the same helmet can conform reasonably comfortably to participants with differences in head frontal bone shape; shoes of the same nominal size may fit perfectly or be totally uncomfortable to participants with large foot anatomical differences.

R3 **Inertial properties**

The further step for a segment surrogate towards biofidelity is the reproduction of segment Inertial Properties. These are the overall mass, location of the overall Centre of Mass (COM), and correspondence in terms of Inertial Ellipsoid, mainly in the three principal inertial axes. In this case, the availability of quantitative rigorous databases is less frequent and typically, researchers shall refer back to cadaver studies from military or vehicle safety studies or to more recent in-vivo studies where inertial properties are estimated on the basis of body composition, volume distributions, and tissue-specific weight. The recognition of human musculoskeletal landmarks is fundamental for the introduction of appropriate systems of references and for the correct localization of segment COM and radii of gyration in the three different axes. For instance, localization of head average COM can be carried out only with respect to the anatomical frame of reference associated with the Frankfurt plane and the acoustic meatus. Given a head circumference, failing to reproduce the head's total mass, the head COM location and the head inertial properties in the three anatomical head axes means that the surrogate headform is not behaving correctly during a simulated drop test for impact assessment of helmets or during a full-body crash test simulation.

R4 **External mechanical properties**

External mechanical properties of a body segment surrogate may influence the way the surrogate interacts structurally or in dynamic conditions with a PPE or a protective garment during the simulated reproduction of a fall/impact/overload event. The surface compliance of limb soft tissues, and the friction coefficient between skin and device, is typically mimicked by the artificial cover of the surrogate to reproduce a correct biofidelic behavior. For this reason, dummy segments are typically externally covered by soft silicon/polymer covers of not always appropriate thickness or response: Shore Hardness, for instance, is quite specific, and does not necessarily capture all required properties of soft tissues. When it comes to helmets, discussion on the outer friction coefficient of a surrogate headform or its virtual model in rotational impacts is still challenging researchers. In ski-boot testing, the adoption of a steel/wooden shank element when testing boot flexibility can give different results with respect to the viscoelastic behavior of a silicon calf interacting with the soft liners of modern ski-boots.

R5 **Internal structure mechanical properties**

In most cases, the correspondence of the surrogate to the human limb in terms of dimensions, shape, inertial properties, and external properties is sufficient for the purpose of the test, especially when the structural integrity of PPE or equipment is under static, dynamic or fatigue testing. However, there are some cases when the behavior of internal structures of the segment/limb is also under investigation, or even the main object of investigation. The head, lower limb segment, and knee may be such examples. In this case, the surrogate shall not only reproduce the overall outer appearance or dynamic behavior of the biological counterpart but shall also include some internal structures such as the brain, the frangible bones or the knee ligaments, which must correspond to the average known behavior in terms of stiffness, damping, and ultimate load. In this way, a failure/damage in the surrogate element shall indicate a known probable failure in a human participant hypothetically undergoing the same test conditions.

R6 **Interaction with internal fluid/soft tissues**

Finally, when segments like the head, chest, or abdomen consist internally of vital organs containing fluid influencing the soft-tissue behavior, the biofidelity of the surrogate shall

require that the pressure/compression/tensile/shear strain/stress conditions in the surrogate internal simulant correspond to available data collected in-vivo or on PMHS during similar loading conditions. The presence of a Cerebrospinal Fluid (CSF) simulant in a head surrogate is an improvement in the Bio-Fidelity Level (BFL), as it can be used to measure the pressure waves during frontal bone high-energy or displacement fields. Indeed, data recorded on PMHS impacts [10,11] still represent reference validation tests for advanced FE head models.

Depending on the complexity of the body segment (for instance, Head vs. Foot), as well as the distinction between segments (Trunk vs. Thigh, Neck vs. Arm) and joints (Knee vs. Hip), or body segment complexes (head and neck, lower leg, foot/ankle…), these requirements can be refined and detailed appropriately. Furthermore, they can be seen as an additive checklist for the evaluation of the overall biofidelity of a surrogate and a possible metric for their quantitative comparative evaluation. For instance, a surrogate satisfying the first three requirements (R1–R3) can be associated with a BFL value of 3, while a surrogate satisfying all six requirements (R1–R6) can achieve a BFL-6 value. Surrogates with high levels of biofidelity, from BFL-5 up, for instance, rather than simply dummies (dummy heads or headforms), are considered **replicas** of the segment. Surrogates eventually embody the trade-off between realism and robustness, as well as between accuracy and reliability.

A surrogate body segment can further be qualified by the type and amount of information that can be collected during its application in testing. Surrogates that simulate the presence of a biofidelic body part without any inbuilt sensor system are **Passive Surrogates** (PS). A PS ensures that the physical interaction between the equipment under testing and the surrogate correctly mimics an in-vivo test. On the other hand, a surrogate equipped with inertial motion sensors at its COM can collect kinematic data during an impact or a crash simulation and is an example of an **Instrumented Surrogate** (IS). Some IS may also contain further sensor systems able to retrieve information at deeper internal levels, such as kinetic load components at the joint/interfaces, pressure values at the contact surfaces or in the inner fluid components, or even stress/strain values on bone/tissue simulants [12,13]. Data collected from an IS are important for the validation of the physical surrogate itself in case similar data are available from in-vivo or PMHS tests. Further to this, these IS may support the validation of FE models of increasing biofidelic detail [14–18], given the ethical and complexity issues connected to in-vivo and PMHS test campaigns. Cameras and digital image correlation (DIC) can also be used to obtain information on the movement/deformation of a surrogate.

Full body surrogates

The need for full-scale physical surrogates of the human body first arose with the development of safety requirements in civil transportation, with the diffusion of car traffic and the consequent increase in car accidents beyond acceptable limits of statistical incidence. Driven by experiences in the field of military ejection seats, the first dummies made available to the car industry were developed in the late 1940s. The commitment of large car manufacturers like General Motors, the involvement of the National Highway Traffic Safety Administration (NHTSA), and the biomechanics committees of the Society of Automotive Engineers (SAE), led to the development of the Anthropomorphic Test Devices (ATD) of the Hybrid family. The Hybrid III (Humanetics, Michigan, USA) is the most widespread crash test dummy, delivered in 1976 and specified in a set of widely available documents [19].

The Hybrid III 50th percentile male and the THOR (Test device for Human Occupant Restraint) dummies, a more biofidelic dummy recently developed and adopted by the European New Car

Assessment Program [20], are commonly used in the automotive sector to assess the severity of injuries resulting from car crashes. The THOR dummy is an advanced ATD with a more biofidelic reproduction of the human spine and pelvis, equipped with a wider range of sensors that are more sensitive than those of the Hybrid III. These include face sensors which allow analysis of facial impacts. Despite the enhanced accuracy of THOR, its higher cost and production issues limited its diffusion and Hybrid III is still more available and more commonly used in crash testing [21].

The Hybrid III family also includes a 95th percentile male, a 5th percentile female (representing ideally the lower extremity of the possible adult car driver population) and three child dummies representing a ten-year-old, a six-year-old and a three-year-old. Hybrid III pedestrian (Figure 27.1a) dummies are also available for the investigation of passive impacts suffered by standing or walking pedestrians in car crashes or in the simulation of other vehicle crashes such as motorcycles and bicycles (Figure 27.1b–d). Pedestrian dummies allow adjustment from the initial dummy posture of a car-seated posture due to an adjustable hip with three degrees of freedom (flexion-extension, ab-adduction, intra-extra rotation). In general, the friction of all body joints can be set at different values by tightening bolts acting on polymeric washers, and furthermore, the neck base orientation can also be adjusted to a more flexed or extended position by acting on a toothed clamp. The anthropomorphic biofidelity refers to the US population at the time of development (1970s); this, together with the lack of other representatives, the gender simplifications, criticism in the female presented, and the improvable biofidelity of the child dummies, represent the major limitations implicit in the adoption of ATDs for full-scale injury physical simulations.

ATDs are currently designed to host various sensors across the body [19]. Full kinematics can be recorded by triaxial accelerometers and gyrometers at the head COM, at the chest, and at the pelvis housings. Displacement sensors are inbuilt for measuring chest deflection under compression and for the tibia drawer displacement at the knee (for instance under frontal impact of the dummy against a car dashboard). Kinetic data consisting of triaxial forces and triaxial moment load cells can be retrieved at the skull and neck bases, at the spine/pelvis interface, on the thigh/shank bones and at the knee joint, just to mention some examples. Dummy manufacturers have developed calibrated sensors and calibration test apparatus to ensure that the dynamic response of the ATD corresponds to the human data available from cadaver tests.

Application of full-scale impact or fall tests in sport accident analysis is less common than in the car or motorcycle industry. A car crash simulation can cost up to 25 k€ per impact, depending on the combination of moving conditions of the vehicle under examination, the opponent vehicle, and the number and type of sensors applied to the dummy. However, some examples of sport accident analysis can be found in the literature, performed by research teams in conjunction with manufacturers of PPE, safety barriers, or active protection systems against crashes and falls. Further examples of applications of ATD pedestrian dummies can include a pedestrian Hybrid III dummy seated on a full suspension e-bike for indoor passive vibrational analysis (Figure 27.1b), a suspended Hybrid III 50th Male Pedestrian dummy used for drop tests for indoor helmet and airbag interaction tests (Figure 27.1e), and a seated Hybrid III 50th Male Pedestrian dummy restrained at the chest (Figure 27.1f), ready for impact tests for helmet and neck-brace compatibility tests.

A noticeable application of ATDs in the field of sport injury prevention is the study of backward snowboarder falls after an edge-catching event. Scher et al. [22,23] developed a cableway support system to guide an ATD fully equipped with snowboard, boots, suit, and helmet against a snow slope at a realistic speed (~30 km/h) to replicate the effect of a back edge catching and to collect the 2D motion kinematics (normal and tangential to the snow slope) and the triaxial impact accelerations at the head during the resulting backward fall against the snow. This work showed how challenging the design and construction of a safe, reliable, and repeatable full-scale

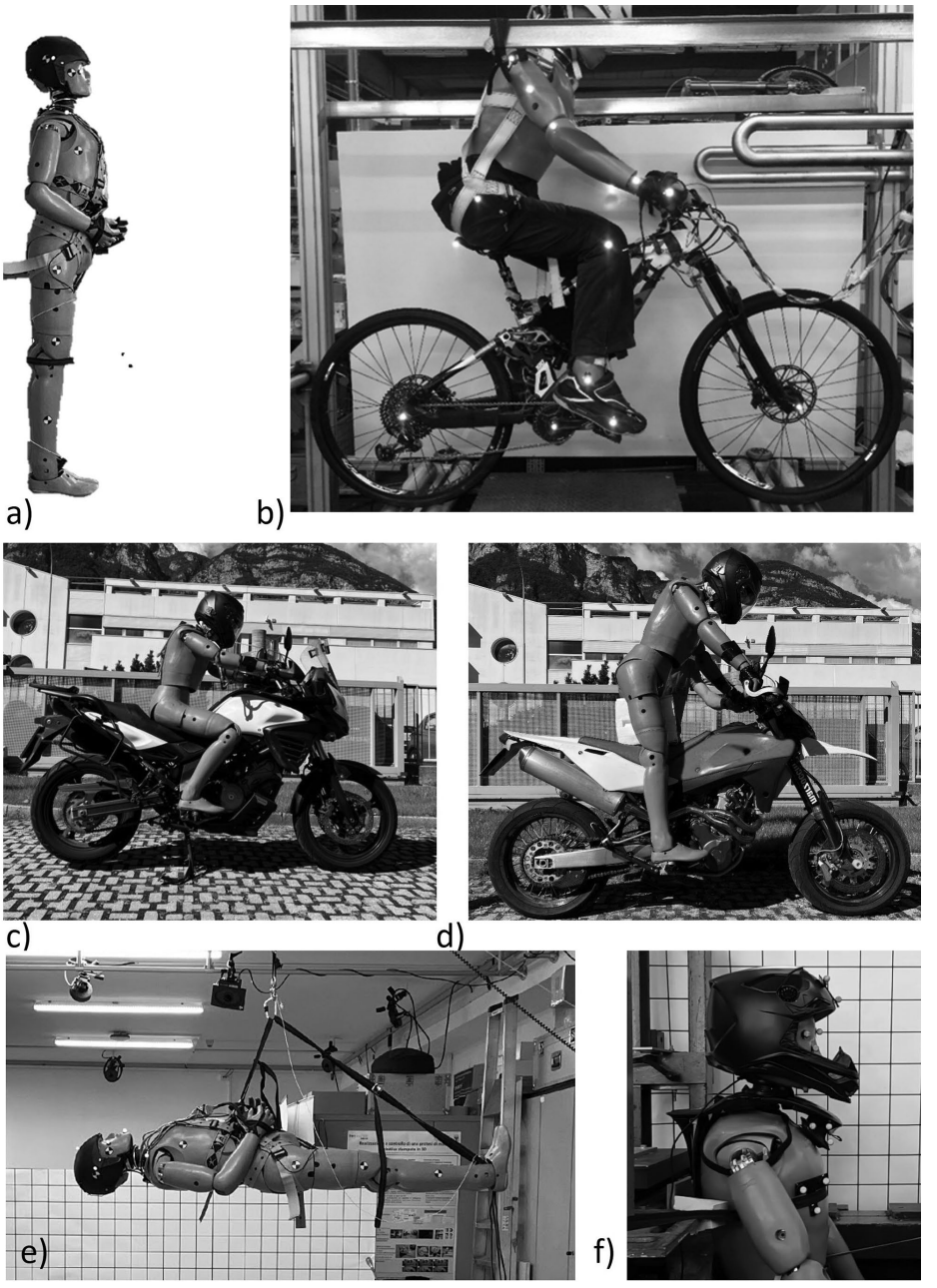

Figure 27.1 Full body surrogates and applications, Anthropomorphic Test Devices. (a) Hybrid III 50th male standing pedestrian dummy. (b) Pedestrian Hybrid III dummy seated on full suspension e-bike for vibrational analysis. (c-d) Seated and standing pedestrian Hybrid III 50th male for road and offroad crash simulations in airbag functional evaluation tests. (e) Suspended Hybrid III 50th male pedestrian dummy ready for drop tests for indoor helmet and airbag interaction tests. (f) Seated and restrained Hybrid III 50th male pedestrian dummy ready for impacts tests for helmet and neck-brace compatibility tests.

test setup for the reproduction of falls in real/simulated winter environments can be, given the need for a large indoor test lab, the complexity of preparing a large snow-covered ramp, and the need for high-speed cameras and rugged portable data loggers for unrestrained falls of the dummy. Results of such experiments still act as validation tools for the multibody simulation of the same backward falls, allowing virtual investigations of changes in snow properties, helmet construction, joint stiffness tuning, and other influencing parameters [24,25], as well as the investigation of risk factors inside more detailed models including the spine. This example highlights the interaction between physical experiments and models in the sports sector: due to the cost and complexity of physical experiments, one good experiment can serve to validate models, which can then be used for more efficient investigations. The approach is similar for cadavers, where it may be hard to justify continuing to use cadavers, so old experiments with cadavers are used to inform surrogates and models.

Testing involving solid and anthropomorphic dummies also finds application in the study of safety nets and safety mattresses (air or foam) used in ski racing competitions to ensure the containment of skiers in the case of high-speed falls and the mitigation of collisions against rocks, trees, chairlift pillars and other fixed obstacles [26–28]. In the case of safety nets, the use of solid or anthropomorphic dummies launched against the nets by a snow toboggan, tower pendulum, or an elastic catapult allows the quantitative evaluation of acceleration peaks and penetration depths at different speeds [27,29,30] (Figure 27.2a–c). The use of anthropomorphic dummies allows the inclusion of ski equipment that in turn can influence net damage and dummy segment entrapments or joint loads (Figure 27.2d–g); moreover, testing in the ambient, despite the increased complexity, ensures that event weather effects on materials such as temperature and humidity are accounted for in the most realistic way.

Recent extensions of such an approach have led to the use of fully equipped ATDs in the investigation of jump kinematics and landing consequences in downhill mountain biking. The introduction of a portable elastic catapult applied to a jump, set at various take-off angles (−4°, 0°, +4°) and realistic jump speeds (~45 km/h), allowed the collection of flight trajectory and crash-landing accelerations at the head and chest (Figure 27.2h–i). These types of evaluations, impossible in-vivo unless by video analysis, justify the use of full-body human surrogates, despite the complexity and cost issues associated. The logistical complexity of taking a heavy dummy up to a mountain/hill for snow sports/bicycle crash scenarios is greater than typical applications in an automotive crash lab, particularly for ensuring the appropriate dummy dynamic posture during the crash.

Head

While instrumented ATDs are generally designed for kinematic and kinetic analyses of the whole body, accurate testing on specific body parts like the head and neck requires specific instrumented surrogates. Such surrogates can be used for analyzing the effects of localized impacts on the head and the effectiveness of PPE such as helmets, mouthguards, neck braces or airbags.

The drive to study biofidelic head surrogates can be found first in civil transportation (car, motorcycle, and bicycle) and military applications, and secondly in the development of safety regulations for impact sports such as hockey, American football, skiing and other disciplines requiring protective headgear. As well as protection from penetrating traumas (climbing and ski mountaineering), attention to blunt trauma has increased in the last decades, in parallel with the development of accurate modeling of the head and the attention to rotational kinematics as a traumatic mechanism responsible for severe focal or diffuse brain injuries [7].

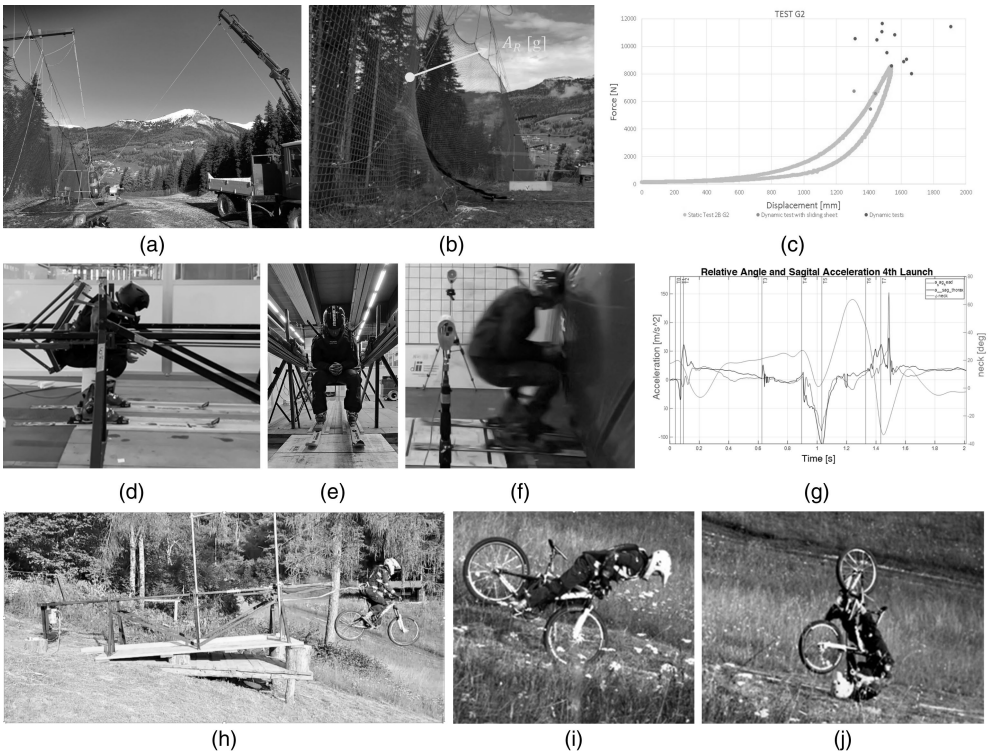

Figure 27.2 Full-scale testing involving full body surrogates, like solid and ATDs at the University of Padova, Italy [27,30]. (a) Pendulum testing of A-nets with a solid dummy at Saslong, Val Gardena, Italy ski downhill course. (b) Peak resultant deceleration A_R and penetration depth during the impact. (c) Comparison of force-displacement curves of A net during static (solid curve) and impact tests (dots). (d-e-f) Fully equipped ATD in skiing egg posture during indoor tests against air-foam mattresses, launched at 45 km/h. (g) ATD longitudinal acceleration at head and trunk during a test, neck flexion (+) extension (−) angle. (h) Elastic Catapult applied to a Downhill jump feature in the +4° take off configuration angle at the ATD release instant. (i-j) Crash landing photograms collected during the tests.

One of the first human head surrogates developed with the intention of reproducing dimensional (R1), shape (R2), and inertial properties (R3) of the human head is the Hybrid III head [31]. It is still one of the most widespread head surrogates used in testing laboratories. The head of the Hybrid III 50th Percentile male (Figure 27.3a) is a hollow aluminum body including the face and chin, accessible by an occipital cover to place sensors (triaxial accelerometers and gyrometers) at its COM. The skull base allows combination with the Hybrid III neck, with a reference plane parallel to the Frankfurt plane. The skull is covered with a removable silicone rubber layer to mimic skin, without any simulant for the brain wobble and viscous behavior. The Hybrid III head/neck assembly finds wide application in sport gear assessment and comparative evaluation of several PPE in different disciplines [32–36]. Another rigid headform is the EN960 (Figure 27.3b), which consists of a hollow aluminum or magnesium skull, with a triaxial accelerometer at the COM. This headform is commonly adopted for helmet standard certification but finds less sport application than the Hybrid III as it is not compatible with any neck surrogate. With the introduction of triaxial gyros at the EN960 COM, it is adopted for oblique impacts in the present motorcycle

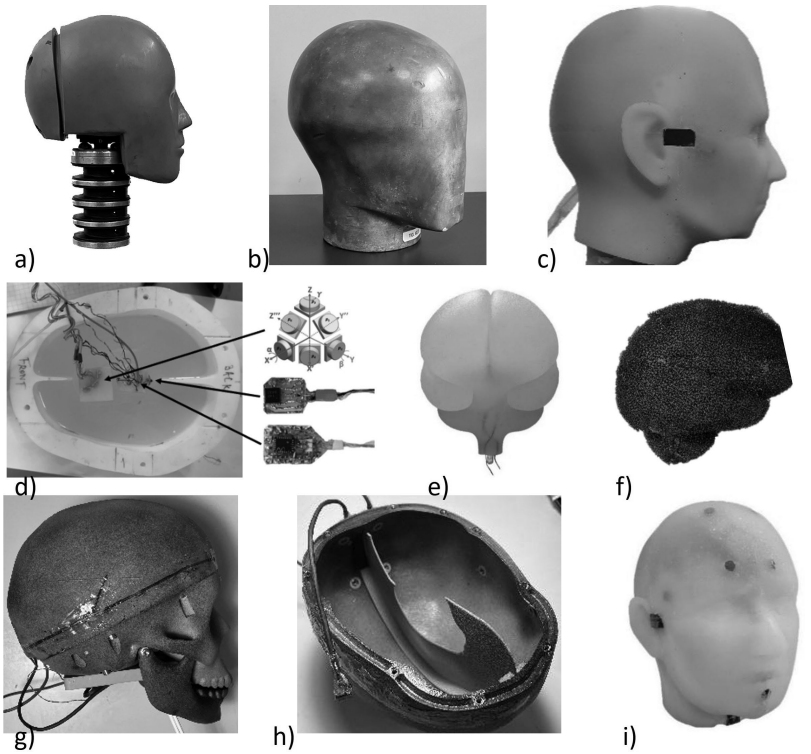

Figure 27.3 Headforms, head surrogates, and replicas. (a) Hybrid III headform with Hybrid III neck: nodding joint between upper neck and head is noticeable. (b) EN 960 headform adopted for helmet standard impact testing. (c) IHHR 2.0 external appearance. (d) IHHR at UniPD-MIUN: brain surrogate with embedded triaxial acceleration, gyros and stress MAPS sensors [13,47,48] (e) Frontal view of the surrogate brain. (f) Side view of the surrogate brain with arachnoid simulant (g) Side view of the closed PA12 IHHR 4.0 skull with screw closure and pressure sensor cabling. (h) Inner view of the upper IHHS 4.0 skull with CSF pressure sensors and surrogate falx. (i) IHHR 4.0 with surrogate skin ready for impact testing [47].

UNECE Regulation 22.06. The NOCSAE headform, which consists of a nylon skull covered with a urethane layer simulating the external soft tissue along with a gelatine mass representing the brain, is claimed to be a more biofidelic surrogate [33]. This headform is instrumented with triaxial accelerometers, but it still does not give information about the motion/stress state at the brain simulant, which is indeed the aim of developing more biofidelic instrumented head replicas (Figure 27.3c) [13].

In the evolution of advanced headforms, several attempts to introduce a brain simulant have been conducted. Much of this work is related to military blast-induced brain trauma and some to sport/vehicle impact research focusing on helmet design. These attempts to introduce a brain simulant are briefly listed in terms of type, number, and location of embedded sensors.

Regarding the early analyses of blast-induced trauma, in 2009, Zhang et al. [37] proposed a physical headform composed of an ellipsoid transparent polycarbonate deformable skull. This outer shell was filled with silicone gel (SYLGARD™ 527, Dow Corning, USA), with a flexible neutral density grid inserted at the mid-sagittal plane to measure brain strain. Similar approaches were adopted by Zhu et al. [38] with an egg-shaped polyethylene 3 mm thick shell filled with

silicone gel (SYLGARD™ 527) to simulate the brain and by Varas et al. [39] with a 7 mm thick polyurethane spherical shell, filled with ballistic gelatine as a brain simulant, and equipped with four pressure sensors to measure the brain simulant pressure values and four strain gauges applied to the exterior surface of the skull shell. In these cases, dimensions (R1) and shape (R2) biofidelity requirements were clearly not fulfilled. In 2009, Merkle et al. [40] developed a Human Surrogate Head Model (HSHM) to examine brain behavior while subjected to overpressure loading, along with the effects of the helmets. The aim was to measure the relative displacement at the brain/skull interface due to a physical force occurring in explosive, ballistic, or automotive crash events. The prototype consists of a glass-epoxy mixture skull, a brain simulant made of silicone gel (SYLGARD™ 527), a synthetic foam to mimic the facial features, and the Hybrid III dummy flexible neck. The surrogate was equipped with two pressure sensors located at the front and behind the headform along the mid-sagittal plane. Additionally, four custom-designed displacement sensors based on an electromagnetic field were attached in the same plane to assess both linear and angular head displacements along the three orthogonal directions.

In 2014, Freitas et al. used an advanced human head surrogate for investigations on blast effects on military helmets [12]. This headform was made of a re-hydrated human cranium and gels to reproduce the soft tissues like the skin (5–7 mm Perma-gel layer), the dura (0.5 mm silicone membrane), and the brain (Perma-gel with iron). This headform was equipped with four pressure sensors in the surrogate brain to measure intracranial pressure, 12 strain rosettes to detect cranial bone strain, and a triaxial accelerometer placed at the hard palate to acquire skull and helmet acceleration.

In 2018, Ouellet et al. with experiences again deriving from blast trauma analyses, proposed an instrumented headform having general anthropomorphic features and the required anatomical structures to capture global and local head response when subjected to blast loading, denominated BIPED [41]. In particular, the prototype was composed of districts simulating skin, skull, brain, cerebellum, CSF, falx membrane, and tentorium membrane. The choice of the materials was based on the biological tissues' elastic or visco-elastic response. In particular, the polymeric skull was molded in two parts, thus allowing the insertion of the brain, which was cast with silicone gel (SYLGARD™ 527). A saline water mixture was used to simulate the CSF, thus ensuring the brain buoyancy. The brain surrogate was suspended in the CSF simulant and its movement was restrained just by the falx and tentorium membranes. The headform was mounted on a Hybrid III neck [19], which has higher flexural stiffness and less damping than the human neck. The prototype headform can monitor external pressure field history at the forehead, right side, left side, top, and back locations, along with intracranial pressures at the frontal, temporal, and parietal regions, and global linear and rotational accelerations. The BIPED showed excellent reproducibility of the tests and it weighs about 3.8 kg [42].

In 2017, Miyazaki et al. [43] introduced a new anthropometric test dummy head capable of direct brain motion evaluation and visualization, for intracranial brain motion measurement in car frontal sled tests. It was composed of a transparent polycarbonate skull equipped with triaxial accelerometers and gyros (6DX pro, Diversified Technical Solutions [DTS], USA) at the face part, a silicone gel (SYLGARD™ 527 A/B, Dow Corning mixed with black dye) brain with optical markers for video capture of brain motion during impacts, and water as CSF simulant [44]. The skin cover (silicon rubber, SLJ3266, Wacker Asahikasei Silicone, Japan) presented a window at the parietal top part for high-speed visualization of brain sagittal motion in frontal car crashes. Despite the absence of sensors in the brain and CSF simulants, these tests showed the possibility of detecting the sagittal displacement of the brain with respect to the skull, even if the lack of meninges and bridging veins caused 3–4 times larger displacements of the physical surrogate than of the corresponding FE model simulations.

Stone et al. [42] recently developed a modular and destructible headform with the aim of addressing limitations of existing headforms, such as the Hybrid-III and EN 960 headforms. In some unprotected impact scenarios, these existing headforms were claimed to produce accelerations possibly unrepresentative of a human head, particularly when impacts produce areas of high pressure where skull deformation and resonance excitation may influence the dynamic response. Stone et al. analyzed the NOCSAE headform, which can produce a more suitable response during these types of impacts but is at risk of permanent deformation in some unprotected impact scenarios, resulting in the entire headform needing to be replaced. The intent was to produce a modular and destructible headform (LU headform) that can be used in potentially destructive testing, where individual components can be replaced. The headform represents a UK 50th percentile male, with inertial properties within 6% of those observed in humans. The additively manufactured skull simulant properties were within the range of values reported for human bone tissue in two out of three build orientations. The lowest and highest resonance frequencies observed in the headform model were within 5% of those observed in humans. The headform was validated against drop and projectile cadaver tests with accelerations within the range reported for PMHS, making it suitable for simulating head dynamics during localized unprotected impacts or in protected impacts where local deformation or resonance frequency excitation or both are of interest.

One of the most biofidelic replicas of the human head has been developed by Potts et al. [44], with a motivation that pertains more to the clinical preparation and conduction of brain surgery rather than to sport head traumas. Indeed, the replica intends to give insight into the positional static shift of the brain due to intrinsic brain tissue deformation under gravitational loads at different head absolute orientations. Technological solutions adopted for the construction of the brain's inner structure, such as the ventricles, the placement of embedded magnetic resonance imaging (MRI) markers in the brain, the replication of the sub-arachnoid space, and the CSF simulant, represent noticeable achievements in the biofidelity level of a human head replica. Care in the selection of the brain simulant, a frozen intermediate hydrogel, and its comparison with data from human cadavers is also worth noting [45]. A major limitation of such a biofidelic replica is the almost complete absence of sensors embedded in the head. Information retrievable is only related to MRI displacement fields associated with different static postures of the head, with no possible application to simulated protected or unprotected head impacts.

In this direction, the effort towards such a high level of biofidelity (allowing the shift from "headform" to "replica"), combined with embedded sensors correlated with brain injury criteria, is worth mentioning in the experimental activity carried out at the University of Padova and Mid Sweden University [13]. The result of a long-term collaboration project, which aimed to design and build the Instrumented Human Head Surrogate (IHHS), an instrumented biofidelic replica of the human head, equipped with different types of sensors (accelerometers, gyroscopes, and pressure sensors) for the acquisition of data from protected and unprotected impact tests, specifically intended for sport applications. The project has led to several prototypes with progressive levels of biofidelity, as described later.

In the first version of IHHS (IHHS 1.0) the skull was made with ABSplus-P430 (acrylonitrile butadiene styrene thermoplastic), additively manufactured with FDM technology, and permanently assembled with a sawtooth suture. The skull was fitted with sensors placed outside (3 triaxial accelerometers and 2 biaxial gyroscopes) and inside (10 pressure sensors). CSF was simulated with siliconic oil, and meninges were manufactured using a 0.7 mm thick silicone coating (Dura mater) and nonwoven polyester (arachnoid trabeculae). The brain was

made with silicone rubber (PlatSil® Gel-10, Polytek® Development Corp, USA), embedding eight triaxial accelerometers and two biaxial gyroscopes during casting [13]. The model was then downscaled by 10% to allow it to fit a helmet for further testing. The IHHS 2.0 skull was made of Polyamide 12 (Nylon 12), and brain tissue was surrogated with a softer (PlatSil® Gel-OO20, Polytek®). The sensor equipment was updated with seven triaxial accelerometers and two biaxial gyrometers [13]. The present version, IHHS 4.0, adopts the same skin simulant material (PlatSil® Gel-10), with the skull additively manufactured in Polyamide 12 in two parts connected with 10 screws (M3) and brass inserts to allow a solid, sealed, and removable closure [46]. The CSF simulant was distilled water. The arachnoid trabeculae meninges were simulated by a 5 mm thick layer of open-cell polymer foam (30 PPI, pores per inch). Falx cerebri and tentorium cerebelli were designed in computer-aided design (CAD) software and additively manufactured in thermoplastic polyurethane (TPU Shore 95A) using the selective laser sintering (SLS) technique. The brain was cast in silicone rubber PlatsilGel OO30 softened by the addition of Smith's Theatrical Prosthetic Deadener (1:1 weight ratio with the standard material). Further refinements were made to improve force transmission, the assembly, and the sensor equipment. In detail, a triaxial accelerometer and gyrometer (DTS 6DX PRO-A) was placed under the mouth floor of the skull, 12 piezoresistive pressure sensors were mounted on the upper half of the internal skull to measure CSF pressure, and the MAPS (Multi-Axial Pressure Sensor) system [47] allows the evaluation of the instantaneous triaxial stress state together with a 200 g triaxial accelerometer and a 2000 dps triaxial gyrometer, all located at the brain COM (Figure 27.3d–i).

Due to the complexity, inhomogeneity, and interconnection of the head's anatomical structures, as well as variability in the mechanical characteristics of biological tissues, obtaining a biofidelic instrumented head surrogate or replica is challenging. The first issue is the choice of simulant materials. Biological soft tissues have peculiar physical and mechanical properties that are difficult to replicate using common polymeric materials. In particular, brain tissue is visco-hyperelastic, so it is subjected both to time-independent hyperelasticity and time-dependent viscoelasticity. Therefore, quasi-static tests and dynamic mechanical analysis need to be performed to compare the mechanical response of brain simulant materials with literature data from human or animal model brain tissue. The amount of data in the literature is still limited, and collecting more data presents ethical challenges.

Additive manufacturing has opened new perspectives in this field, first by the possibility of correctly reproducing a specific human skull geometry retrieved from 3D computed tomography (CT) scans, and secondly for the possibility of choosing equivalent materials and tailoring the density of lattice and filling to mimic cortical and trabecular/diploe bone. Finally, the possibility of producing modular molds for casting soft tissue bulky organs, such as the brain, has given a boost to the exploration of several simulant polymeric gel-based materials and their comparison with human or animal model available data [48]. Readers seeking more information on additive manufacturing are referred to Chapter 16 3D printing in sports.

In any case, the fabrication of biofidelic surrogates is high-cost and time-consuming. The embedding of sensors in the surrogates is challenging due to the need for miniaturized or distributed field sensors, collecting large amounts of inertial, contact, pressure, and external/internal strain/stress values at high sampling rates.

A recent trend in the assessment of the effectiveness of helmets and other head protective gear (airbags) is coupling tests from standard headforms (Hybrid III, EN 960, NOCSAE) with FE models of the head [36]. First, the linear and rotational kinematics of the headform COM are

collected during protected impacts on flat horizontal and inclined anvils. This is then applied to the FE model to estimate the stress distribution in the brain [35]. These models would typically have been validated against available pressure and motion data from testing cadavers. It is worth noting that modeling the brain is challenging, and care should be taken with the interpretation/reliance on such results [49].

As a contribution to the discussion and the comparative evaluation of head surrogates and replicas, Figure 27.4 shows a possible bi-dimensional map for surrogate collocation, as a combination of biofidelity and quantitative information retrievable for research and testing. On the horizontal axis, biofidelity can be seen increasing from poor to perfect, with perfect being a living human head. As a metric, the BFL scale from 1 to 6 may be associated with the increasing fulfillment of requirements R1 to R6. On the other axis, if tests/investigations are performed only on passive surrogates, typically only a small amount of information can be retrieved. As the number of sensors embedded in the surrogate, or diagnostic tools used in the tests (X-rays, MRI, High-speed video…), increases, so does the amount of information retrieved. The hypothetical top-right corner occupies a fully biofidelic detailed and validated FE model able to reproduce the hyper-visco-elastic behavior of the brain under impacts. Surrogates that can reach a BFL corresponding to 6 may reach the status of "replicas" and are more suitable for research in impact biomechanics than in product standard assessment (i.e., within commercial test houses). Headforms with BFL2 or BFL3, are suitable for product standard assessment and are preferably used for their construction simplicity and robustness. The potential of head replicas to become standard assessment tools is high. These replicas could also act as an indirect validation stage for FE models as the mechanical modeling of simulant materials is well known and does not suffer from the ethical and statistical issues associated with PMHS testing.

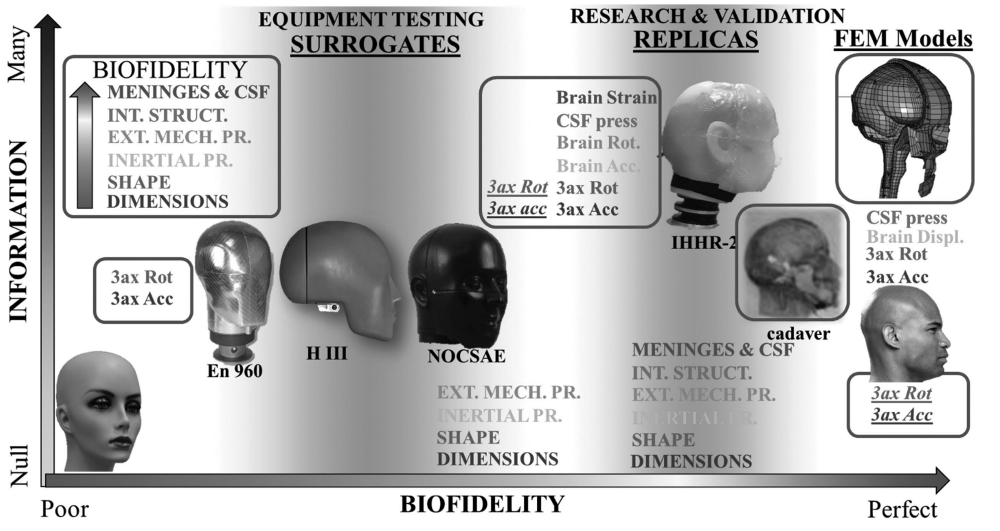

Figure 27.4 Head surrogates, headforms, and replicas. Possible map for surrogate qualification. Biofidelity increases from poor to perfect (living human head). Information increase at increasing number of sensors embedded in the surrogate or adopted diagnostic tools. Surrogates reaching BFL6 can be "replicas," more suitable to research, headforms (BFL2 or BFL3) are suitable and used for product standard assessment due to simplicity and robustness.

Neck

Attention to the neck surrogate has improved considerably in recent years. This attention is in combination with the ongoing discussion focused on helmet test methods regarding the possible inclusion of a surrogate neck. The lack of standardized test methods covering neck protective equipment, used for instance in motocross or downhill mountain biking, has raised demand for a representative biofidelic neck and a standardized test method. In this direction, the increasing diffusion of sophisticated techniques for the design, prototyping, manufacturing, and testing of 3D biofidelic components has favored several solutions for the preparation of a biofidelic surrogate neck.

The neck is a complex body segment presenting a compromise between mobility, stability, and protection of the spinal cord. Indeed, thanks to the multi-vertebral cervical joints, the head has large combined mobility in flexion/extension, lateral bending and rotation, while preserving fundamental stability thanks to passive cervical fundamental spinal units (FSU). The disc and articular facets support gravitational, inertial, and muscular loads surrounded by robust ligaments. The neck's upper extremity shows active mobility and tunable stiffness due to the intervention of several deep and superficial muscles connecting the head/vertebrae/trunk.

The behavior of this special body segment, optimized by human evolution and deeply anatomically studied and characterized by White and Panjabi [50], as well as by several anatomists and biomechanists, is characterized by the preparation of a set of "corridor" curves expressing the Moment/Angle curve of the upper extremity with respect to the lower base obtained by in-vivo and in-vitro PMHS studies, in the different planes. Together with literature sources [51–53] this neck biomechanical corridor finds recent synthesis also in the ISO 13232-3 standard for "Motorcyclist anthropometric impact dummy" [54] prescribing the specific behavior that a biofidelic neck shall provide.

From an engineering perspective, the approach to the development of a surrogate suitable for head/neck complex biomechanical testing has followed three main construction philosophies. Type-I involves the development of an equivalent **beam-like** segment, with the base on the upper trunk and the extremity compatible with a surrogate head base (typically the Hybrid III), of suitable elastomeric material, geometry, and structure to reproduce the kinematic behavior and the stiffness biomechanical corridor. Type-II concepts reproduce the **multiple spinal units of the neck**, with the intention of reproducing the FSU arrangement and overall behavior. Type-III solutions involve the development of a **functionally equivalent neck segment,** without morphological correspondence to the anatomical neck but designed to reproduce its kinematic and kinetic behavior in all directions.

The first and most used surrogate neck is again the one developed for the Hybrid III project [19]. This neck was further developed with the THOR evolution and the BioRID-1. The Hybrid III neck is a Type-I surrogate. It is made of an elastomeric single cylindrical body with a metal plate at each extremity and an axial steel inner cable with a threaded extremity nut enabling its pretensioning for the elastomer precompression. Transverse cuts in the anterior portion of the elastomeric cylinder make the Hybrid III neck stiffer in flexion than in extension. Tests on sport-related protections requiring a head/neck assembly often use the Hybrid III neck, despite its limitation of excessive stiffness with respect to the biological corridor.

To overcome the different behavior of the Hybrid III neck in the forward, backward, and lateral flexion, Walsh et al. [55] developed an unbiased neck surrogate. They intended to replicate the overall dimensions and connections of the Hybrid III neck, presenting a homogeneous unbiased behavior in the four flexion directions. The two neck surrogates were tested both in

a quasi-static flexion test (40 mm displacement of the distal neck nodding joint) in the four flexion planes (front, back, right, left). The "unbiased" neck surrogate was insensitive to flexion direction, showing a stiffness similar to the Hybrid III backward bending. However, when tested under impact (duration ~5-10 ms), using the Hybrid III head surrogate, the response of the two neck surrogates was more similar. The authors attributed the difference between the dynamic and quasi-static test results to the short duration of the impact tests, during which the two neck surrogates had only low flexion angles. The authors also noted that the primary purpose of their prototype was removing the bias due to flexion direction, rather than achieving a biofidelic neck behavior.

Inverted falls, such as in snowboarding freestyle tricks or motocross, can induce large neck angular deflections. In that case, the neck surrogate should mimic the human passive neck stiffness across the full range of combined flexion/torsion deflection. Axial loading behavior should also be included in the validation process to determine if the surrogate can be used for simulating inverted falls.

Type-II neck surrogates may be more conceptually biofidelic as they try to mimic the periodic construction of the human cervical spine, by the development of bodies separated by elastic elements simulating the disc and ligament behavior. Recently, Macgillivray et al. [56] and Rueda-Arreguin et al. [57] developed articulated neck surrogates intentionally designed to assess impact-induced head injuries. In both cases, the solution was based on the stack-up of vertebral bodies connected by elastomeric [56] or elastic [57] elements and stabilized in the sagittal and lateral planes by other cable/elastic elements. In reference [56] the neck successfully sustains the high-energy impacts and shows interesting correspondence with in-vivo volunteer available data [58]. In reference [57], despite the interesting concept, the work seems limited to a desk prototype claimed to correspond to the biological behavior of a human neck. The prototype was far from the rugged construction needed for withstanding the loads recorded during crash or fall tests with dummies or helmet impact tests.

A similar concept has been developed in parallel by Zullo et al. [59] with a more robust construction. The surrogate consists of six independent FSUs, each composed of two additively manufactured rigid plastic disks. These disks were joined together by a compound made of silicone rubber poured and anchored to them with a grid. This design gave a similar neck bending response to the cadaveric experiments of Nightingale et al. [53]. The stiffness of the surrogate could be modified independently in the four directions (flexion, extension, and left/right lateral bending) using four hooks carved on the front, back, and lateral sides of the outer circumference, which allowed the positioning of visco-elastic elements such as O-rings. In a central hole, there was a cable to control stiffness and range of motion like in the Hybrid III surrogate, in addition to a set of sensors giving direct measurement of the sagittal and lateral bending angles.

A recent achievement from Farmer et al. [60] is a representative example of a Type-III functional equivalent neck surrogate. The authors analyzed the kinematic behavior of the head base relative to the neck base during sagittal flexion-extension, lateral bending, and rotation. They then adopted three ball-type joints with superior and inferior bodies, in combination with a plain bearing axial joint to provide a high degree of axial rotation. The stiffness of the system was achieved via a set of four elastic bands surrounding each ball joint, designed to reproduce the available cadaver or passive behavior of the human neck [51–53, 61]. The compactness and robustness of the surrogate make it suitable for impact loads.

The comparison between conceptual solutions of neck surrogates suggests that, in the case of the neck, the most relevant biofidelity requirements can be R1, R2, and R4, namely dimensions, shape, and external mechanical properties. Ruggedness is a key requirement if surrogates

are intended to evaluate the upper trunk-neck-head complex behavior under impact. At present, Type-I surrogates are still most commonly used in crash/fall/impact full-scale simulations, despite showing higher stiffness than the human neck. Promising developments of Type-II and Type-III surrogates are foreseen, provided their robustness improves.

Upper limb and hand

The adoption of upper limb and hand surrogates has been for the assessment of sports PPE, specifically for the shoulder, elbow, and wrist. Sports injuries in this body segment are often related to cuts and lacerations due to player collisions, and fractures due to falls.

Shoulder

Laceration injuries are common in rugby, and hence padding is typically incorporated into the clothing, specifically shoulder padding. World Rugby (Rugby Union's governing body) has regulations and tests for such padding, which uses a shoulder surrogate [62]. The shoulder surrogate is a Ø90 mm steel cylinder, typical of sports PPE standards. Imam et al. [63] developed the external (R4) biofidelity of this surrogate by adding muscle and skin simulants (Figure 27.5). A 14 mm thick muscle simulant layer and a 1 mm thick skin layer were molded around a Ø75 mm steel hemisphere. Silastic 3481 (Dow Corning, UK), with the addition of a catalyst and deadener, was used for the muscle simulant, and synthetic chamois cross-woven polyvinyl acetate for the skin simulant. These simulants have previously been used by Hughes et al. [64] for a generic soft tissue surrogate (200 × 150 × 8 mm) specifically designed for a rugby stud raking test. The catalyst and deadener were added to the Silastic 3481 to reduce its stiffness to better match porcine muscle tissue. Readers seeking more information on rugby padding are referred to Chapter 19.

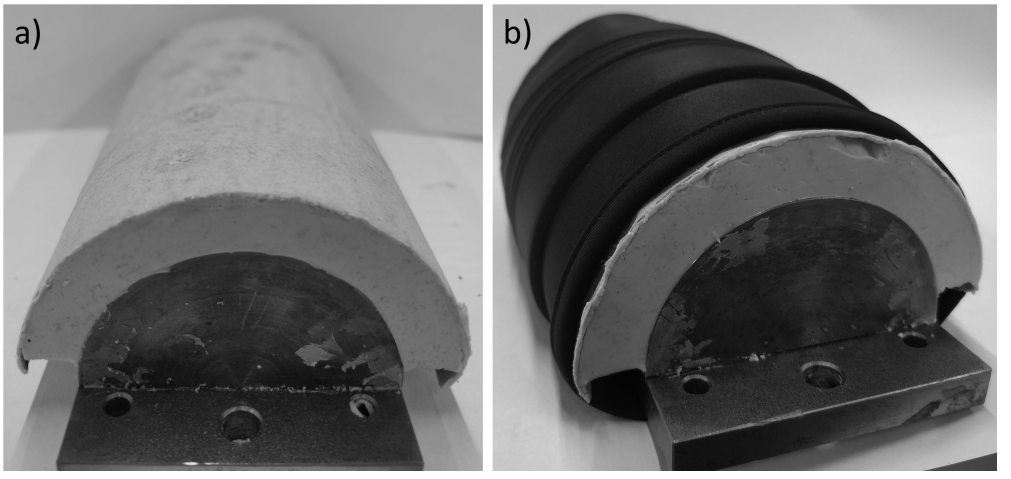

Figure 27.5 Photos of the shoulder surrogate developed in reference [64] for assessing rugby shoulder padding showing (a) the cross-sectional view and (b) the surrogate covered in a sample of rugby padding. Image courtesy of Dr Adil Imam.

Wrist

Attention to wrist surrogates and the ongoing discussion on the effectiveness of wrist protection heightened in 2013 following a call from the snow sports safety community to develop a standard for snowboarding wrist protectors [65]. This call was due to the high number of wrist injuries among snowboarders. Wrist injuries are the most common snowboarding injury due to participants instinctively reaching out their hands to try to cushion a fall [65]. The development and adoption of wrist surrogates, alongside numerical models, have aimed to determine injury biomechanics and assess wrist protector effectiveness.

It is hard to reproduce the anatomical structure of the wrist due to the number of bones and the complexity of the joint. The wrist complex is made up of 15 bones and has five major muscles acting across the joint. The joint is considered biaxial, with motions of extension/flexion and ulnar/radial deviation, in which motion in one direction affects the motion capability in other directions [66].

As previously mentioned, it is unethical to use participants to assess injury-inducing scenarios. As such, cadaveric arms have been used to gain knowledge of wrist injury mechanisms and fracture thresholds to inform the testing of wrist protectors. Greenwald et al. [67] was one of the first to assess the impact response of braced and unbraced forearm cadavers relating to snowboarding fall scenarios. The cadavers were sectioned below the elbow and the wrist joint was set to 40° dorsiflexion and 10° pronation for drop rig testing. The loading period of the braced and unbraced cadavers were compared, and fractures occurred in both, implying that the loads were beyond the protective capabilities of the wrist protector. The fracture patterns varied, including fractures of the ulnar shaft, distal radius, scaphoid, lunate, ulnar styloid and midradial shaft.

The approach to developing a wrist surrogate has focused on suitable dimensions (R1 biofidelity) and shape (R2 biofidelity) of the hand and forearm sections, with the joint simplified to a hinge. One method has been through casting a model of a ~50th percentile male's hand and forearm. Maurel et al.'s [68] surrogate was cast from body filler (resin and hardener), with the wrist at 80° extension. Although there was no joint, as the wrist was fixed at the angle it was molded, there were efforts to develop the compliance of the hand (R4 biofidelity), which was found to be too hard during preliminary tests. Through analysis of peak impact forces, a 3 mm thick rubber layer was found to be the most suitable for simulating the padding of the human hand, and thus was added to the palm of the surrogate. Greenwald et al.'s [69] surrogate was cast from 80 Shore A hardness polyurethane. The hand and forearm sections were connected via a hinge joint at the wrist, which was set to 40° dorsiflexion during testing. Adopting a different approach, Kim et al. [70] utilized the forearm/hand complex of the enhanced airbag interaction Hybrid III female dummy. This dummy was developed to study arm accidents due to the inflation of steering wheel-mounted airbags. As such, the wrist was fitted with a load cell and rotatory potentiometer, and the forearm also had a load cell. Efforts of external (R4) biofidelity were made by adding a polymer covering to simulate arm flesh. These surrogates facilitated external force measurement during testing of wrist protectors. The Hybrid III dummy surrogate also enabled measurement of the transmitted impact forces to the forearm bones.

Multibody simulations and FE models have also been used to acquire knowledge about the functional requirements of wrist protectors. The human (50th percentile adult), ski slope, and snowboard have been modeled, with a detailed upper limb. Lehner et al.'s [71] model was developed to evaluate worst-case fall scenarios in snowboarding and calculate the resulting loads in the upper extremity. The bone geometries of the humerus, radius and ulna were based on CT scan data, with soft tissue modeled as wobbling masses. The skeleton will decelerate in the first instant of

time, and then, with some delay, the soft tissue follows. The use of wobbling masses enables this response and results in more realistic maximum loads [72]. The contacts in the wrist joint were modeled as areal contacts, and the elbow was modeled as a hinge joint. The stiffness of the joints was modeled with rotational spring-damper elements. The model simulated forward and backward falls, comparing joint angles of the shoulder, elbow, wrist, and fingers to experimental data. This data was from Schmitt et al. [73], who conducted fall experiments on participants at heights from 12 to 80 cm. Ground reaction forces were compared to Degoede et al.'s [74] model, the parameters of which were based on muscle strengths and ballistics tests from participants.

Senner et al.'s [72] model focused on the effect of bone strength changes with age and gender in children and adolescents on the effectiveness of wrist protectors in backward snowboarding falls. The scapula, humerus, radius, ulna, carpals, and phalanges were modeled within the full-body model. The soft tissue connecting them was modeled as wobbling masses via spring-damper elements, and the ranges of motion and joint stiffness were described with torque elements. Age-specific values of compact bone and trabecular structures (Young's modulus, Poisson's ratio, density) were applied, with emphasis put on the radius bone (shaft, proximal sector, distal sector) and the epiphyseal plate in children (9 years old). The model was validated by remodeling quasi-static compression tests of cadaver bones [75], and fracture forces were determined from ultimate stresses for cortical bone [76]. Models of different age groups (9, 13 to 14, and 20 years old) were created, and the effect of wearing wrist protectors with different combinations of protective elements (dorsal splints, palmar splints, 5 mm padding, 10 mm padding) were simulated. Axial loading of the radius and wrist extension angles were compared between simulations, and joint angles for the elbow and wrist were compared to Lehner et al.'s [71] work.

In the interest of developing a new standard for snowboarding wrist protectors, EN 14120 (roller sports equipment) was identified as a starting point [65]. A wrist surrogate is prescribed to test the bending stiffness of wrist protectors. This wrist surrogate consists of a paddle-like hand (made from wood or any light and stiff material) and a cuboid forearm, connected via a hinge joint. It is unclear how the size of the hand and forearm correspond to anthropometric data [77]. Schmitt et al. [78] used this surrogate design to test the bending stiffness of snowboarding wrist protectors. They modified the hand, introducing fingers, to enable the testing of glove-integrated wrist protectors, as found in snowboarding.

Adams et al. [79] developed two wrist surrogates, specifically for testing snowboarding wrist protectors. The first was based on eleven dimensions from anthropometric data, improving the dimension (R1) and shape (R2) biofidelity, creating a geometric design. The second surrogate was created from a 3D scan of a ~50th percentile male's hand and forearm, further improving the shape (R2) biofidelity. Both surrogates were additively manufactured in polyamide, connected via a low-friction hinge joint, and had two steel fingers. Following bend testing of snowboarding wrist protectors using the two surrogates, the first one (geometric) was implemented into ISO 20320:2020, the new standard for snowboarding wrist protectors [80]. This surrogate was selected for the standard as it produced more repeatable measurements in the bend test, and furthermore, the geometric design could be recreated in CAD software and scaled to represent different-sized people. The 3D scan surrogate design approach was taken forward for impact testing snowboarding wrist protectors [81], and furthermore in a corresponding FE model [82]. In contrast to a multibody simulation, Newton-Mann modeled the wrist protectors on the wrist surrogate (Figure 27.6a). The model was used to determine the effectiveness of the wrist protectors' protective elements (palmar padding, splints) in reducing impact force and wrist extension. This was achieved via simulations with different padding and splint dimensions and materials. For the experimental impact testing (Figure 27.6b), a potentiometer was fitted to a steel core within the forearm and linked to the wrist

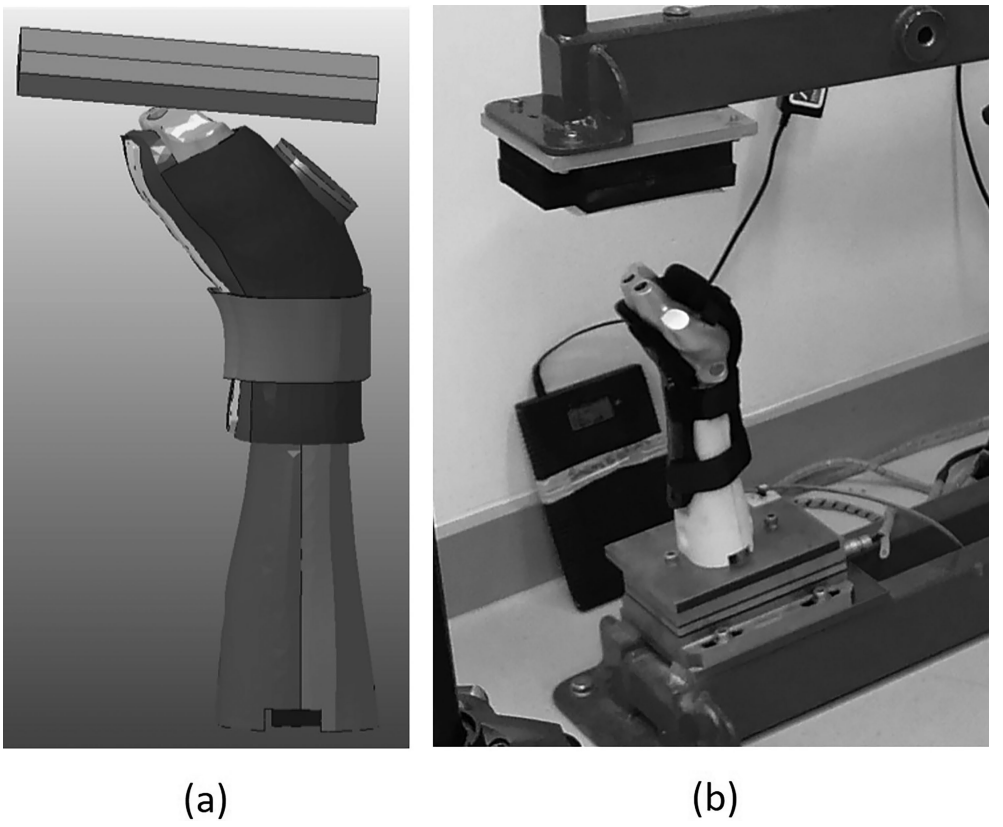

Figure 27.6 FE model of snowboarding wrist protectors (a) and impact testing (b) using the 3D scan surrogate [82, 83]. Authors' own image.

joint via a toothed timing belt, enabling the measurement of wrist angle during testing [81]. The hand was made from aluminum to withstand the high-impact forces, and the fingers were removed.

Following the publication of ISO 20320:2020, Leslie et al. [83] developed the external (R4) biofidelity of the corresponding surrogate by adding a 3 mm thick silicone outer layer to represent a skin/soft tissue (Figure 27.7a). The modified surrogate was created by first reducing the periphery of the original geometry by 3 mm, leaving a stiff core for the hand and forearm. A mold encapsulating the original geometry was then additively manufactured, which fitted around these cores, creating a 3 mm cavity for silicone to fill. The core of the hand and forearm were additively manufactured in Onyx (Markforged, Massachusetts, US), and the skin/soft tissue simulant was maxillofacial M511 silicone (Technovent, Bridgend, UK). Maxillofacial silicone (M511) is typically used for facial and finger prosthesis [84]. The hand and forearm parts were connected via a low-friction hinge joint, and the hand had two steel fingers, as per ISO 20320:2020. Snowboarding wrist protectors were fitted to the developed surrogate (rigid core and 3 mm thick silicone outer layer) and an equivalent rigid surrogate (i.e., no silicone outer layer) followed by ISO 20320:2020 bending stiffness tests. Although the results indicated the silicone outer layer increased the friction between the protector and surrogate, and hence measured stiffness, no relation between the surface friction of the silicone and that of human skin was made.

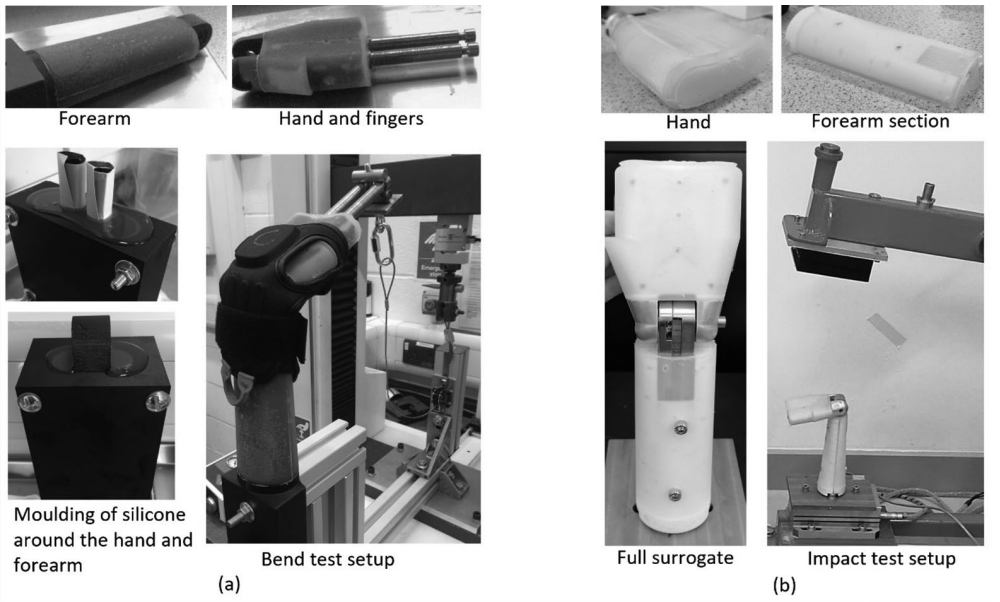

Figure 27.7 Wrist surrogates consisting of a rigid core and silicone outer layer, used for testing snowboarding wrist protectors in a (a) bend test [84] and (b) impact test [86]. Authors' own image.

In 2023, Leslie et al. [85] developed a wrist surrogate for impact testing on Adams et al.'s [81] test rig (Figure 27.7b). The forearm's casing was made from polyamide with a 3 mm thick silicone outer layer and fit around a steel central core connected to the test rig. The hand core was made from aluminum to withstand impact forces. It had a respective 3 and 7 mm thick silicone layer on the dorsal and palmar side. The thicker silicone layer over the palm of the surrogate simulated the soft tissue thickness of the human palm [86], further improving the external (R4) biofidelity of the surrogate. The hand and forearm parts were connected via a hinge joint, and a potentiometer was fitted to measure wrist angle during testing, as per Adams et al. [81]. Snowboarding wrist protectors were fitted to the developed surrogate (rigid core and silicone outer layer) and an equivalent rigid surrogate (i.e., no silicone layer), and were impact tested at ~31 J.

Although upper limb surrogates are less developed in terms of internal structure and instrumentation than those for the head and neck, they have focused on the external geometry to comply with PPE testing. This fulfills their purpose and makes them more repeatable for both manufacture and testing.

Lower limb

Lower limb injuries are among the most frequent types in sport practice. The injury mechanism can be associated with functional overloads or collisions/impacts overloads, depending on the discipline. Lower limb injuries are typically less severe than head or neck injuries, but their higher frequency of occurrence and the long recovery time for athletes have led biomechanists, trainers and engineers to study them. Technological progress and sport equipment advancements have an influence on the evolution of the types of injuries over time in several sports. For example, the introduction of high-rise plastic boots shifted the predominance of alpine ski injuries from tibia

fractures to knee ligament injuries [87]. In this context, the development and adoption of limb segment surrogates, as well as the development of numerical models of the segment/joints under investigation, have favored the interpretation and simulation of injury biomechanics and the development of methods for assessing PPE effectiveness.

In the case of impact overloads, the challenge for a physical or numerical simulation is again the ability to reproduce in the model the six BFL requirements previously proposed, expressing the increasing degree of biofidelity of a model, either physical or numerical. In addition to that, when it comes to dynamic events such as impacts or collisions, the mechanical properties of surrogate materials should correspond to the human model at high strain rates. The anatomical structure of a lower limb segment, such as the thigh and shank, requires the introduction of appropriate simulant materials for skin, muscle/fat tissues and bones, which are indeed a frangible structure, whereas ligaments are non-linear fibrous/collagen structures. Correct biomechanical behavior also requires appropriate joint kinematics, involving a spherical hip joint and correct knee and ankle joints.

Several surrogates have been developed and proposed to address lower limb injury prevention. In the present section, full limb crash surrogates, thigh impact surrogates, knee joint surrogates and ankle-foot surrogates will be mentioned.

Impact surrogates

Among the most advanced and well-documented full-leg impact surrogates, the Advanced Pedestrian Legform Impactor (aPLI) should be mentioned [88]. This legform has been recently developed by the automotive industry to mimic the impact behavior of a pedestrian crossing the street and being impacted by a car at the front bumper. Despite being far from sports injury applications, this surrogate is worth mentioning as it represents the state of the art in the production and calibration of a full-leg surrogate, showing the workflow and requirements of a biofidelic industrial surrogate oriented toward repeatability and reproducibility. This device can also be taken as an example of the evolution of a bespoke/prototype surrogate used in a research lab toward a validated, approved, and standardized surrogate that can then be produced commercially and adopted worldwide. The extensive appendix regarding the rational background of the surrogate development describes the methods followed in the identification of simplified front car models, the introduction of four different FE models for the validation of the surrogate, and the comparison with PMHS data. The stress-strain curves for the selection of appropriate foams mimicking the skin and flesh are reported in the standard. The surrogate leg is equipped with strain gauges at different sections of the fiberglass surrogate bones, inertial sensors at the hip and knee, and linear potentiometers on the medial collateral ligament (MCL), anterior cruciate ligament (ACL) and posterior cruciate ligament (PCL) surrogates (whose stiffness is specified to mimic the real behavior of the knee when struck from the lateral side).

The aPLI has the limitation of being designed for assessing frontal car bumper compliance to a pedestrian impact. Its morphological shape biofidelity is limited to the lateral side, as it is not intended to wear a specific garment or PPE. However, such types of collisions between players, frequent in rugby, football, or soccer tackles, can find such surrogates useful as a tool for evaluating protective equipment like padding garments, kneepads, or knee braces.

A noticeable improvement in biofidelity has been achieved through the introduction of additive manufacturing in the development of human limb surrogates. Payne et al. [89] developed a multi-material human thigh surrogate, consisting of skin, subcutaneous adipose, and muscle tissue simulants molded around a synthetic femur. Each tissue structure was analyzed, and simulants were selected following material testing and comparison to available literature data [90]. The

development stages followed to fabricate this multi-material synthetic soft tissue surrogate are described with quantitative details and images. The potential of additive manufacturing in the preparation of molds and casting is appreciable, and similar techniques were later used by Leslie et al. [83] to develop a wrist surrogate. The resultant surrogate clearly showed the use of sequential molding techniques to construct a full-scale anatomical human impact surrogate suitable for testing PPE.

Knee surrogates

The creation of a knee surrogate has been studied extensively over the years with different approaches and degrees of biofidelity. This has led to various devices typically dependent on available technologies and the focus of the study. The main focuses were to reproduce typical dangerous loads acting on the knee, such as valgus loading and anterior-posterior shear loading, or their combination. Surrogate development was based on available in-vivo and cadaver studies, as well as on the implementation of numerical models of the knee, typically validated against available cadaver studies. The challenge for such surrogates lies in the need to reproduce the complex combination of kinematic and kinetic actions loading the knee during risky situations occurring in sports practice. This implies the development of a multi-component controlled test bench, ideally including all muscle bundles crossing the knee. In addition to that, passive structures such as the four main ligaments must be reproduced and instrumented to capture the instantaneous combination of loads to be compared with available cadaver measurements or model simulations.

In early works, France et al. [91,92] developed an instrumented knee simulating the body's weight by integrating a contralateral limb and a pelvis/torso assembly. Aluminum femur, tibia, and fibula components of the instrumented limb, which were modeled after cadaveric specimens, were used (neglecting the patella). Instrumented ligaments (ACL, PCL, MCL, LCL) and tendons (quadriceps and hamstrings) were included in this surrogate. Lateral impact tests were performed with and without different types of knee braces on the surrogate knee to measure the force acting on the MCL ligament. Various external conditions, such as the impactor's weight and velocity, were shown to affect the knee braces' protection ability, where an increase in knee flexion angle clearly led to an increase in MCL tension. The effectiveness of braces increased with lower velocity and larger impact masses when hips and ankles were fixed. In further development works, the patella-quadriceps was introduced. Like other researchers who included the hamstrings**,** the device was again improved to simulate the response of a real knee under valgus loading and anterior/posterior loading [93].

This experiment evaluated the effectiveness of various knee braces in limiting displacement and, therefore, stress on the knee's inner structures. The leg surrogate was equipped with a vertical torque actuator as well as a vertical linear actuator that simulated body weight. Two pneumatic actuators were used to mimic the external force acting on the leg—one for anterior-posterior tibial translation and the other for valgus movements, positioned laterally on the ankle or knee. A pneumatic actuator was used to simulate muscles, such as the quadriceps and hamstrings, as well as ligament tensioning. Tibial plateaus were equipped with load sensors. Knee stress is the sum of the internal loads imposed by the quadriceps and the external loads imposed by forces acting vertically and laterally. Tests were conducted using different types of knee braces. The lateral actuator was used to apply a valgus displacement while the frontal actuator was used to apply an anterior/posterior displacement.

Adopting a different actuation approach**,** Hermann et al. [94] developed a detailed mechanical right knee surrogate to study its behavior during simulated alpine skiing loading conditions

(Figure 27.8a), as the knee is the most commonly injured body part in this activity [95,96]. The aluminum tibia and femur were obtained from CT scans, along with four knee ligaments (MCL, lateral collateral ligament (LCL), ACL, and PCL), 10 active muscles, and the thigh/shank external geometry to accommodate a knee brace. The patellofemoral joint (using commercial implant components) and tibofemoral joint formed the articulating surface. An additional joint in the hip was pivot-mounted, allowing a ±20° range of motion. The muscles were simulated with steel ropes—eight in the thigh (including the quadriceps) and two in the calf. To measure each muscle's load, they were equipped with an independent tensioning unit and load cells. Artificial polyester ligaments (LARS TM) were used to make the ACL, PCL, LCL, and MCL knee ligaments. Ligament displacement was measured with a special displacement sensor. ACL and MCL loads could be measured with customized load cells. Externally, a plastic skin compatible with knee braces could be applied. By using instrumented ropes, force and moment could be statically applied to the ski, located under the artificial foot, and with the help of a marker-based video system, the movements of the leg could be captured. Hermann et al. performed quasi-static movements (by hand) of the tibia in different conditions of muscle contraction (without muscles, quadriceps dominant, hamstring dominant, and their combinations). ACL loads were recorded, and results showed that hamstring activation could reduce load in the ACL, whereas dominant quadriceps activation could increase ACL loads for knee flexions smaller than 40°. The ACL is loaded differently in flexion or extension movements, with flexion movements mainly resulting in higher ACL loads, in agreement with findings for cadavers [97].

In parallel to experimental tests, Hermann et al. [94] developed a multi-body model of the surrogate knee incorporating all the actuated muscles and the passive ligaments of the physical surrogate, with known mechanical properties (Figure 27.8b). After validating the multi-body model against the surrogate knee experimental test results, the simulation explored a wider range of load conditions and several realistic risky skiing load conditions, such as internal torque combined with valgus/varus/anterior tibial load. Results confirmed several findings from the literature, but the role of quadriceps vs. hamstring contractions was clarified by the simulations. A limitation of Hermann et al.'s bench test is the limited loading conditions that can be simulated, which are typically quasi-static.

In this regard, a new test bench incorporating a knee surrogate has been developed at the Department of Industrial Engineering, University of Padova, with the aim of allowing high-speed angular loading of the knee in flexion/extension, abduction/adduction, and internal/external rotation (Figure 27.8c-e). The bench can be used to test the stiffness of commercial knee braces due to the presence of a leg surrogate equipped with an instrumented cardanic knee surrogate. Stiffness in abduction/adduction or internal/external rotation loading can be measured at different speeds and knee flexion angles. In addition to that, the development of a biofidelic knee surrogate with seven sensor channels—ACL, PCL, LCL, MCL, two tibial plateaus, and the patellar tendon—allows for the evaluation of internal loads acting on the ACL or MCL in the presence or absence of the knee brace. The biofidelic knee is undergoing validation against results from the literature such as the work of Markolf et al. [98].

Ankle and foot

When it comes to the ankle/foot complex, sports traumas and injuries again represent frequent causes of sports practice interruption at all skill levels. Ankle sprain incidences, for instance, are higher in (i) females than males, (ii) children than adolescents, and (iii) adolescents than adults.

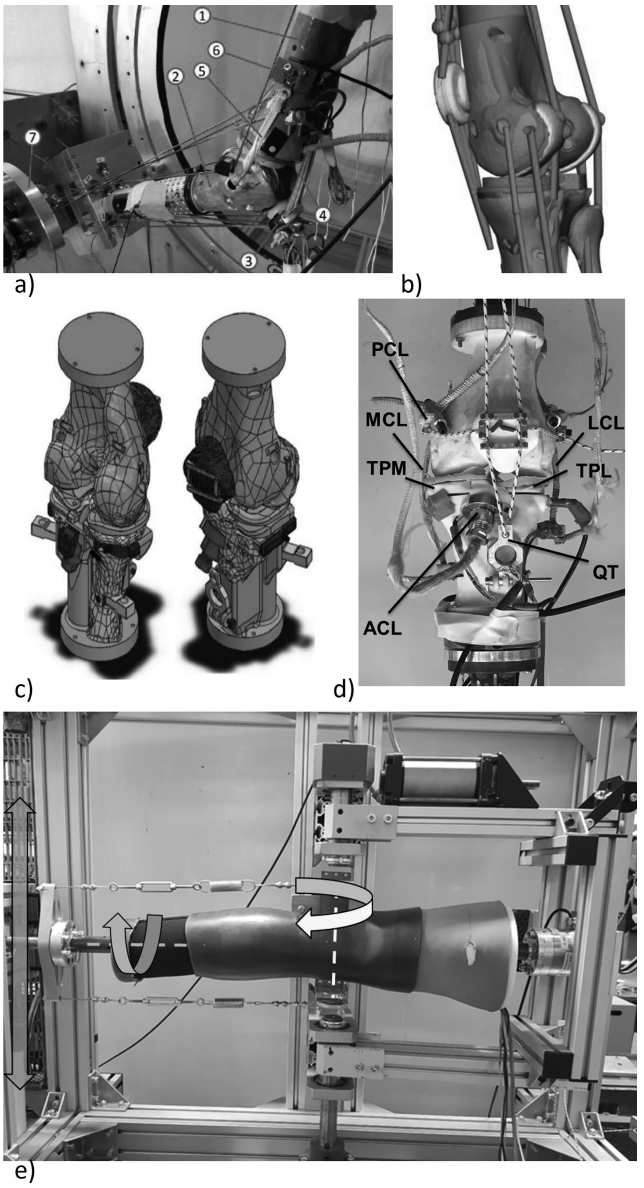

Figure 27.8 Knee surrogates and models. (a) Knee surrogate developed by Herman et al. [95] assembled in the bench for static testing of knee biomechanics under static external loads and internal muscle contractions (1-tibia, 2-femur, 3-patella, 4-patellar tendon, 5-collateral ligament, 6-load cell for collateral ligament tension, 7-block of muscle tension actuators); (b) multi-body model of the surrogate knee developed by Herman for the simulation of loads crossing the knee. Images (a) and (b) courtesy of Dr Aljoscha Hermann, TUM. (c) 3D model of the surrogate developed at UNI PD for the study of knee-braces effectiveness. (d) Additively manufactured instrumented surrogate knee with sensors at ACL, PCL, MCL, LCL, QT, and medial/lateral tibial plateau (TPM, TPL). (e) Dynamic multiaxial test bench: servomotors can apply flexion (right arrow), ab/adduction (left arrow), and int/ext rotation (middle arrow), with/without knee braces.

The sports category with the highest incidence of ankle sprain is indoor/court sports (basketball, futsal), with lateral ankle sprain being the most observed type [99]. Ankle injury prevention studies have not yet provided recommendations for the adoption of high-rise footwear, but, for instance, outdoor hiking with a high-shaft boot can be less efficient [100] in the sagittal plane but safer with respect to lateral stability.

The ankle joint is not as simple as a basic cylindrical hinge, as some may suggest. The dorsi/plantar flexion axis is inclined in the frontal plane, and the subtalar joint axis is also inclined in the sagittal plane in a way that a simple Cardan joint is not representative of the dorsi/plantar flexion and the inversion/eversion movement of the foot.

Many studies on footwear, boots, ski boots, or snowboard boots have involved the development of ankle-foot surrogates [101–105]. Foot surrogates can be recognized as a consolidated tool for footwear construction and testing. Shoe lasts are possibly the first historical example of a body segment surrogate, as shoemakers first experienced the need to replicate the size and shape of human feet to sew and glue the shoe upper to a sole around an appropriate wooden surrogate. In that case, a shoe last is an example of a surrogate reaching no more than the first two levels of biofidelity requirements (R1 and R2). The foot/ankle surrogate adopted in forefoot flexibility studies introduces the metatarsal-phalangeal joint [101,106] for the independent evaluation of the effect of shoe upper consistency on overall stiffness. In this case, the external stiffness of the foot is not included in the surrogate.

In studies related to ski boot flexural stiffness, the use of appropriate shank/foot surrogates is crucial. This is because the flexural behavior of the boots is not only influenced by buckle closure conditions [107] but also by surrogate ankle joint stiffness. Indeed, the size, shape, and hardness of the surrogate surface, as well as the ankle joint kinematics and the ankle stiffness of surrogates, can influence stiffness hysteresis curves resulting from bench tests [108]. There is no standard defining the characteristics of a shank/foot surrogate, nor a standard test method for measuring and quantifying the flexural stiffness of a boot (namely, the "Flex Index" commercially used). Thus, researchers in this field have used either wooden surrogates with free ankle joints [109,110] or surrogates with a Cardanic ankle joint embedded in a silicon-casted form [108,111,112]. In some cases, attention has focused on internal biomechanical biofidelity (involving Achilles tendon stretching) rather than external shape/tissue properties [113]. This implies that each surrogate must be appropriate for the bench and the test method for which it was designed. The fact that each ski boot manufacturer has developed a shank/foot surrogate ("prosthetic leg") and test method independently does not help in defining a common set of standard requirements and test methods for ski boots.

Similar experiences in the development and usage of a surrogate shank/foot complex can be found in the development and validation of Ankle-Foot Orthoses (AFO), particularly in the quantification of stiffness properties. Early studies focused on the introduction of a surrogate shank and applied the loads to the AFO sole [114]. More recently [115], the surrogate design adopts components from orthopedic prosthetics, such as a pylon, a foam shank casting covered by a silicon liner, a single-axis hinge, and a rigid foot. Despite the attention to surface/shape biofidelity in the shank, ankle-foot simulation is limited. A more biofidelic foot mechanical behavior is sought with the development of a surrogate lower limb [116], where the ankle joint is a ball socket associated with tunable elastic elements crossing the ankle. In this surrogate, the foot is divided into four segments (hindfoot, midfoot, forefoot, and toes) connected by ball-socket joints embedded in a silicon-casted cover.

Summary

The development of physical surrogates of the full body or portions of it, despite being complex, challenging, and expensive, is justified by the possibility of using them to reproduce harsh and destructive tests. As presented in this chapter, challenges consist of the identification of biological corridors for surrogate validation, the quantification of its biofidelity, and the ability to extend the amount of information it can retrieve via embedded sensors. The cited examples are representative of various surrogates with varying biofidelity requirements, following the six proposed requirements. Recent advances in additive manufacturing technologies, in combination with the progress of miniaturized sensors and wireless transmission for data logging, facilitate the diffusion of instrumented surrogates.

Surrogates of body segments and joints present in the scientific literature do not yet correspond to those commercially available for research and certification testing. Full-body surrogates are the most widespread and accessible surrogates, with BFL values not exceeding BFL 4. Foot and limb segment surrogates are the most established and widespread. Neck and head segments still challenge researchers and will possibly reach BFL6 in the next few years, given the demand for more sophisticated protective devices. Joint surrogates aiming to reach BFL6, which include internal biomechanical structures like ligaments at the knee, still require scientific and industrial efforts to achieve the highest level of biofidelity, combined with robustness and embedded sensing technologies.

Numerical models, in parallel with physical surrogates, are powerful tools for investigating the response of body segments, organs, and joints to overloads and impacts. Their development also requires knowledge of the geometrical and mechanical characteristics of human tissues and organs. Their validation requires the availability of in vivo or in vitro PMHS data. The advantage of such models, once validated, is the depth of analysis that can be performed at many scales and of the ability to investigate virtual loading conditions that are possible but not yet recorded. Sensors provide discrete localized information, whereas numerical models can show the field of quantities.

The examples cited in the chapter cannot cover all peer-reviewed studies available in this field but illustrate key factors for distinguishing among different approaches and tools for conducting such research. Care should be taken to remain aware of complexity and the levels of biofidelity achieved when evaluating and integrating the results of models in predicting responses of biomechanical systems in acute injury situations.

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28

BIOMECHANICS OF ATHLETES WITH IMPAIRMENT AND ADAPTIVE EQUIPMENT

Stefan Litzenberger, Marie Lund Ohlsson, Jonas Danvind, and Nicola Petrone

Introduction

This chapter discusses the special requirements of sports engineering for athletes with impairments. It is intended to give the reader an idea of the possible approaches and parameters that need to be considered when working in sports for athletes with impairments, showing examples of how the equipment development process can consider the individual adaptations regarding the athlete's function and impairment, as well as the sport-specific rules and classification when enhancing performance, training, or reducing the risk of injury.

From a sports engineering perspective, sports equipment for athletes with impairments appears to offer a wide range of possible improvements, as the interaction of equipment and athlete biomechanics plays an even greater role in sports for athletes with impairments than in able-bodied sports. However, care must be taken not to violate the International Paralympic Committee's (IPC) Policy on Sports Equipment [1, Ch. 3.10], which identifies four basic principles for equipment: Safety, Fairness, Universality, and Physical Prowess. These principles are further explained in the sub-section Equipment Policy in IPC Sports.

Competitive sports for people with impairments

There are many sports that people with impairments can participate in today. The Paralympic Games are the largest sports event for athletes with impairments. It is organised by the International Paralympic Committee (IPC) every second year, alternating between the summer and winter games. The Games have grown from "sport as rehabilitation" to an elite performance competition. The Summer Paralympic Games in Tokyo in 2020 engaged over 4,400 athletes and millions of spectators throughout the world.

The Paralympic Games include sports for athletes with physical impairments (comprising the eight impairments that cause physical activity limitations that are biomechanical in nature—impaired muscle power, impaired range of movement, limb deficiency, leg length difference, hypertonia, ataxia, athetosis, and short stature), visual impairment and intellectual impairment. In the Paralympics, most of the sports (25 out of 27) are open to athletes with physical impairments, 11 are open to athletes with visual impairments, and three for athletes with intellectual impairments.

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Not all sports are open to athletes with all kinds of physical impairments. For example, an athlete with short stature cannot compete in the skiing events in the Paralympics. This is determined through the classification for each specific sport (IPC Classification and sometimes classification governed by an International Sports Federation).

There are also other organisations hosting sports for people with impairments. For example, the International Committee of Sports for the Deaf (ICSD) currently hosts competitions in 30 sports. For people with intellectual impairments, VIRTUS, the international federation for athletes with intellectual impairments, hosts 12 sports and has partnerships with several international sports federations, and Special Olympics for athletes with intellectual impairments, with the purpose of increasing health and inclusion in society through sports participation. There are also international federations hosting sports that are not on the Paralympics programme such as taekwondo for athletes with intellectual impairments.

Adapted sports equipment

Athletes with intellectual impairments generally do not use any adapted sports equipment. They find it harder than others to learn, understand and communicate [2]. Also, proprioception and sport strategies (for example pacing) may be more difficult. To make sports accessible, adapted pedagogical support, such as simplified language, image support, or sign language, should be used.

Athletes with visual impairments often use a personal guide during training and competition, such as in running, cross-country skiing and alpine skiing. In some sports for athletes with visual impairments special equipment is used. The cyclist uses a tandem cycle, and the swim coach uses a tapper to tap on the athlete's head to inform them that the end of the lane is approaching. For training purposes, digital guide systems are being developed [3].

People with physical impairments use equipment to a large extent in their daily lives, which also implies the need for adapted equipment when performing sports. The need for equipment and adapted equipment is different for athletes with different impairments and different sports disciplines. For example, powerlifting, sitting volleyball, soccer, swimming, and taekwondo do not require almost any adapted equipment at all. In contrast, sports such as Nordic skiing, alpine skiing, cycling, and canoeing, require more equipment both in the able-bodied sport and individual-specific adapted equipment for athletes with impairments.

From here on, this chapter focuses on physical impairments.

Equipment policy in IPC Sports

The rules for each specific sport determine which equipment is allowed to be used. The *IPC policy on sport equipment* [1, Ch. 3.10] describes the sports equipment policy where it says:

> Equipment that results in sport performance not primarily being generated by the athlete's own physical prowess but being generated by automated, computer aided, or robotic devices is prohibited in IPC Sanctioned Competitions and Events, and at Paralympic Games.

The rules regarding which equipment is allowed, and the classification system, are also tightly related. The IPC classification code has two critical functions: (1) to define eligibility to compete and (2) to group athletes into classes, which aim to ensure that the impact of impairment is minimised, and sporting excellence determines performance and potential victory.

However, how the impairment impacts performance is a challenging question to answer. In some sports and impairment types, the equipment also plays a role. For the cross-country skier with paralysis in the legs, the sit-ski design will impact performance [4]. The complexity of the athlete (with an impairment), the sport-specific equipment, the adapted equipment, and the performance, together with the sport technical rules and the classification system, has been illustrated as the "Bermuda Triangle" of evidence-based classification [5].

This "Bermuda Triangle" describes the sometimes conflicting, but inextricably linked, complex five elements that must be considered when designing sports equipment for athletes with impairments:

- athlete with specific impairments
- sport-specific equipment
- IPC classification
- IPC technical rules
- athlete sport-specific performance

and all influence each other to a certain degree. Changes in one of the five factors immediately influence the other factors, causing a need for adaptations to ensure fairness, safety, health, and performance.

The reason for this sometimes troublesome complexity is that the three factors—the classification, the sports technical rules, and the athlete (with the impairment, equipment, and performance)—are all related. If one of the factors is changed, for example, an innovative new piece of equipment, it can impact the "impairment-determinants of performance" relationship on which the classification system is built.

Ethical considerations

In many sports, there is a need for individual adjustments of equipment, and sometimes there is also a need to create new sports equipment. This is often made by an orthopaedic or mechanical technician, and sometimes by a specialist sports engineering and technology entity. Many ethical questions are raised about technology in para sports, such as "Does this technology enhance performance, or is it essential for performance?" [6,7]. The IPC wants to ensure that the gold medal is given to the athlete because of human performance and not an "arms race" of scientific and technical advancement in equipment.

When making adaptations to equipment or creating new equipment, it is important to be aware of the rules of the specific sport, especially what is allowed and how and when the equipment should be examined by the technical committee. Also, according to the IPC policy, the equipment should be safe, fair, universal, and promote physical prowess.

The IPC policy on sports equipment [1] says: "The International Paralympic Committee recognises the important role of sports equipment in enabling Paralympic competition and is committed to a sports environment where there are fair and clear rules governing the use of sports equipment for each sport."

This policy is based on four fundamental principles:

- **3.1.1 Safety.** All equipment in use must protect the health and safety of the user, other competitors, officials, spectators, and may not cause damage to the environment (e.g. Field of Play).
- **3.1.2 Fairness.** Equipment needs to be regulated in sports rules in sufficient detail.

- **3.1.3 Universality.** The cost and large-scale availability of (principal components of) equipment should be considered to guarantee access to a sufficiently large number of athletes in the sport.
- **3.1.4 Physical Prowess.** Human performance is the critical endeavour for sports performance, not the impact of technology and equipment.

Nevertheless, it happens time and again that development is only pursued from the point of view of optimising performance, and the other principles are overlooked. Many examples can be given of how technical advantages are achieved by developing equipment with advanced technology that not everyone has access to or can afford [8]. The intentions have been good to assist the athlete in achieving success, but along the way, the other aspects have been forgotten. Performance improvement through technical development is desirable because it moves the Paralympic movement forward and provides more opportunities to participate, but it needs to be done with good ethical considerations by all parties.

Design of new sports equipment, recommendations, and economy

Sports equipment development for athletes with impairments is not just about considering the four IPC principles in equipment development; there are also many steps in the product design process that are influenced specifically by adapted equipment. However, in this chapter, the focus is not on all aspects of the design process. For further reading on the inclusive and universal equipment design process, Torrens and Black [9] present a methodology based on the synthesis-analysis loop for inclusive physical activity and disability sports that forms the basis of recommendations to developers of adapted equipment. It is recommended that new methods for equipment design, using biomechanics and advanced engineering design, be adaptable to other individuals, as well as scalable both in production cost and volumes to benefit other athletes and persons with disabilities.

One crucial factor, which is delimited in this chapter, is the economy of adapted sports equipment. In short, the cost is too high for many users, leading to non-universal products and unfair sporting conditions [10]. Therefore, economics is likely to be the strongest driver of unfair competition at the elite level and access to equipment at the grassroots level, impeding health-promoting physical activity and the additional benefits of sports. In connection with these ethical and financial factors in the development of adapted sports equipment is the use of biomechanical testing to analyse the interaction of the new equipment with the complex biological and neuromuscular determinants of the human body and the variety of impairments.

Challenges for biomechanical measurements

Following the IPC principles (3.1.1–3.1.4), the development of adapted sports equipment should not solely focus on performance optimisation but also the interaction with the athlete. Stefanyshyn and Wannop [11] provide a description of this general principle:

> The purely mechanical system of the piece of equipment becomes a biomechanical system once it is interacting with the athlete. […] A flawlessly engineered mechanical piece of sport equipment can still fail if the athlete's equipment interaction is not properly addressed in the design process.

It is obvious that the analysis of this athlete-equipment interaction requires biomechanical measurements such as optical motion capture, ground reaction force measurement, and others.

In most cases, a combination of several measurement systems is required to fully capture the complexity of the biological and neuromuscular aspects of a particular sport. Furthermore, different impairments or levels of impairment often require different approaches, which often have to be adapted to the special requirements of athletes with impairments. Although the use of one system is often challenging and a combination of systems multiplies the complexity, the conclusions drawn from the data obtained are effective and provide insights that would otherwise be disregarded. On the other hand, sometimes opportunities arise from the possibility of placing sensors directly on the adapted equipment or prostheses, allowing for analyses not possible for able-bodied athletes, thus benefiting both able-bodied and impaired sport and biomechanics.

- **Kinematic analysis** is often the basis of biomechanical analysis and is commonly used for research. The methods most often applied are optical motion analysis (3D infrared video) and the use of inertial measurement devices such as accelerometers, gyroscopes, or a combination of these (Inertial Measurement Units - IMUs). While for optical approaches, marker-placement for non-disabled athletes is well documented, and evaluated models exist [12], this is not the case for athletes with impairments, as missing limbs or marker occlusion by equipment restrict the possibilities and require individual solutions for marker models. Marker placement for optical motion analysis can often be restricted for athletes with physical impairments and pose problems for researchers. However, aids such as prostheses or orthoses can offer a chance for sensor attachment and analysis of forces and limb orientations, which overcome problems usually associated with the measurement of certain parameters on able-bodied athletes. The placement of optical markers, as well as inertial measurement devices on soft tissue, can cause inaccuracies due to the movement of the soft tissue relative to the underlying bony structures, while the placement of these sensors on rigid areas of prosthetic devices can result in less noise in the measurement signal.
- **Force and pressure measurements** are an important source of data for analysing the loads acting on the body of athletes. Again, prosthetic devices can offer the possibility to measure occurring forces on the human body directly via the prostheses. Petrone et al. [13] describe the instrumentation of prosthetic feet for sprinting with strain gauges and gyroscopes to measure the occurring ground reaction forces and the angle orientation during running for every single step (for more information, see the sub-section on Running with Prosthetic Feet). For able-bodied athletes, such measurements require force plates embedded in a running track, only allowing data acquisition for single steps placed directly on the force plate. For the specific analysis of the athlete-equipment interface, mostly pressure measurements are used, making an analysis of contact forces and pressure distribution possible. Especially for athletes using prostheses, they provide an effective way of analysing pressure on contact points, thus not only increasing comfort but also preventing tissue injury.
- **Muscular activity** can be measured using electromyography (EMG) which provides valuable information on neuromuscular coordination and can be used to analyse the impact of adapted sports equipment on coordination strategies and the amount of muscle activation of the athlete. This information can not only aid in assessing the quality of changes in the equipment but can be used as input data for biomechanical models.
- **Inverse dynamic analysis** is a way to calculate internal loads with musculoskeletal models using measured kinematics and kinetics (readers seeking further detail are referred to Chapter 7, Musculoskeletal simulation in sport). However, simulating internal loads may result

in a further problem, as for these simulations anthropometric data (such as segment lengths, segment masses, and inertial properties of segments) must be known to yield meaningful and exact results. For non-disabled athletes, databases and equations to estimate such anthropometric parameters exist, but these can only be applied to athletes with impairments to a certain degree, as missing limbs not only influence the relative distribution of body weight but can also affect the anthropometric properties of adjacent limbs. Additionally, individual differences in impairments generally prevent generalisation. Individual anthropometric data for athletes with impairments are thus often acquired using 2D or 3D body scans, as in reference [14].

Some examples of recent research

This section will highlight examples of recent work done in the field of the biomechanics of adapted equipment. It is structured in terms of the severity of the impairment and, hence, the complexity of assistive equipment being used. Beginning with impairments requiring orthoses, aids, and braces (sub-section Drop-foot orthoses), continuing with technical aids and wheelchairs (sub-section Cross-country Sit-skiing), and ending with examples of equipment where missing limbs are replaced by mechanical substitutes (sub-sections Running with prosthetic feet and Upper limb). More examples of these can also be found in the section on the role of biomechanical simulations in sports for athletes with impairment.

Drop-foot orthoses

Drop-foot is a condition that mostly occurs as a result of neurological problems (e.g., a stroke) or incomplete traumatic spinal cord injury (SCI). The symptoms of drop-foot are weakened dorsiflexion ability of the ankle joint and subsequent problems in the gait pattern of affected persons, such as foot-slap during the loading response and toe-drag during swing [15]. Additionally, a lack of sensory input from the affected limbs can result in excessive pressure and bruises that can lead to ulcers.

To counteract the movement deficiencies, orthoses are worn to support the ankle joint and the adjacent dorsiflexor muscles. Commercially available orthoses are generally designed for everyday use and cannot withstand the forces occurring during sports.

Baur et al. [16] describe a case of a female Paralympic cyclist with paresis of the peroneal muscle after incomplete spinal cord injury for whom an individually designed orthosis for cycling made from carbon-fibre reinforced plastics was developed and tested. From a sports engineering point of view, the development of such an aid needs to be evaluated according to its functional benefits by means of biomechanical measurements (cf. section Challenges for biomechanical measurements). In the case of reference [16], not only the change in power output, but also the muscle activity and plantar pressure were evaluated. While muscle activity helps to assess the functional benefits of the device, measuring plantar pressure was intended to avoid areas of high pressure and, thus, possible negative effects of repetitive high pressures from the orthosis that could lead to overuse injury.

It must be noted, though, that drop-foot is not the only condition requiring the use of orthoses in sports; a wide variety of pathological conditions with different causes can be the reason for using an orthosis in sports [17]. Additionally, the term orthosis covers a wide variety of supportive structures or devices for different pathologies and different purposes, many of which are described in the literature (e.g., [18]).

Running with prosthetic feet

Amputee running is one of the most well-known examples of the complex and controversial issue of fairness in sports technology, garnering worldwide attention due to Paralympic sprinter Oscar Pistorius. The South African double transtibial amputee Oscar Pistorius was a successful runner in the 100, 200, and 400 metres. He won several gold medals in the Paralympics and also reached the 400 metres semi-finals in the athletics world championships for able-bodied athletes in 2011 and in the Olympics 2012. He used prosthetic "feet" (often referred to as "blades") made of carbon-fibre reinforced plastics, which act as a spring storing and returning energy with every gait cycle [14,19].

Human running is a complex motion that engages the whole body: the running distance is covered through cyclic lower limb movements based on continuous acceleration and deceleration phases. Therefore, it is comparable to a rapid sequence of leaps similar to the motion of a bouncing ball, in which each leg acts like a spring [20]. In the bouncing mechanism of running, potential energy and kinetic energy are in phase: this "in-phase oscillation" of the centre of gravity is suggestive of an elastic rebound of the body, as the energy is alternately absorbed and restored by the muscle/tendon complex [21].

In amputee running, passive-elastic carbon-fibre running-specific prostheses (RSPs) are designed to enable athletes with lower limb amputation to run [19], mimicking and enhancing the bouncing mechanism of the athlete's lower extremity. The main components of a prosthetic system for running for transfemoral amputees are the socket, the prosthetic knee, the connecting elements (pylon, pyramidal attack) and the Running Prosthetic Foot (also called a carbon blade foot) including a sole (Figure 28.1a). Transtibial amputees, depending on the length of their stump, only maintain the socket/clamp/foot/sole components (Figure 28.1b).

- **Socket:** a carbon-fibre or fibre-glass negative copy of the residual limb stump, which is the custom element of the prosthesis.
- **Knee:** the cylindrical joint connecting the socket to the RSP in transfemoral amputees. It has a single-axis mechanism that works like a hinge. Hydraulic damping allows the athlete to achieve a more natural swing movement at full extension.
- **Pylon:** a rigid tubular element connecting the socket to the knee unit or directly to the foot, adjusting the appropriate limb length and foot overall alignment.
- **Pyramidal Attack:** allows the foot to be fixed to the pylon and adjusts the angles of alignment of the foot (±7.5° in two perpendicular planes) by acting on four fixing screws.
- **Sole:** provides specialised traction that can be optimised for the foot blade. A spiked sole ensures grip with the surface of an athletics track; a rubber sole could be used on other ground surfaces, such as on a treadmill.
- **Foot:** the prosthetic running foot is frequently defined as "RSP" - "Running Specific Prosthesis." Being part of a more complex prosthetic system, it should more correctly be defined as "RPF" or "Running Prosthetic Foot." It is characterised essentially by shape, category, and height. It is essentially a carbon laminate shaped as a C or J to act as a cantilever spring between the clamp and the tip.

Starting from the 1980s, when the first composite materials (readers seeking more information on composites are referred to Chapter 11 Fibre-Reinforced Composites for Sports Applications) entered the prosthetic industry, the efficiency of lower limb prostheses improved tremendously.

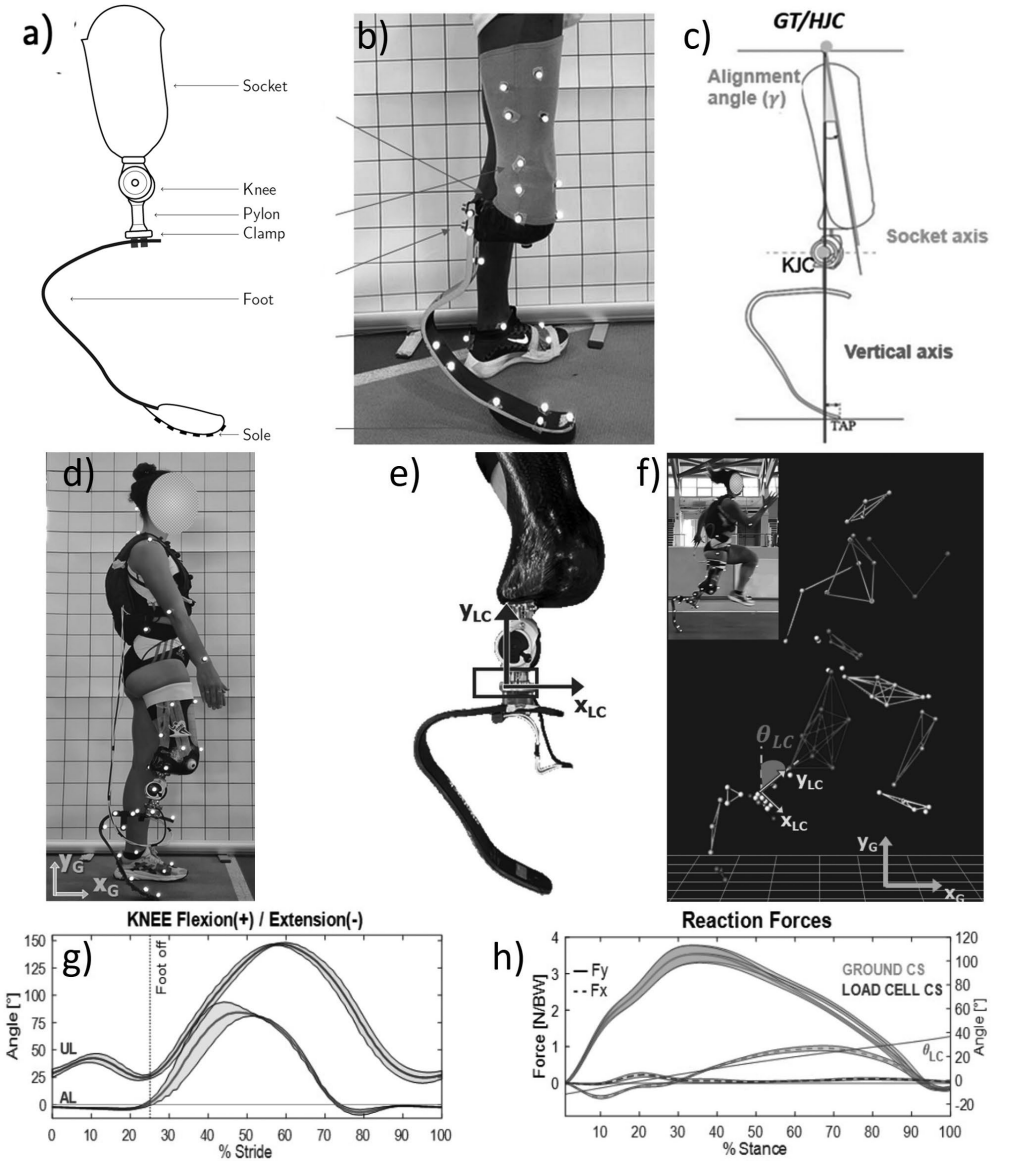

Figure 28.1 (a) Main components of a prosthetic system for transfemoral amputees, with terminology: C-shaped running prosthetic foot (RPF). (b) Transtibial amputee with a J-shaped RPF: marker set adapted for track motion capture biomechanical analyses. (c) Typical alignment axes and parameters for a transfemoral prosthetic system. (d) Transfemoral Olympic sprinter equipped with markers for IR motion capture analyses and wearable load cell applied at the RPF clamp connected to a data logger in a small backpack. (e) Detailed view of the wearable multiaxial load cell applied to the RPF clamp with the corresponding reference system. (f) Dynamic frame captured during a sprint for comparison of the video and the motion capture views: the orientation of the load cell reference frame with respect to the global frame is expressed by angle OLC. (g) Kinematic result of the biomechanical analysis: knee flexion angle throughout the stride of the unaffected limb (UL) and the affected limb (AL). (h) Kinetic result of the biomechanical analysis: Ground reaction forces on the AL captured at the wearable Load Cell and resolved also in the global reference frame.

Carbon-fibre-reinforced materials brought lightness, durability, and strength to the design of prosthetic feet [22].

When analysing the biomechanics of running, two principal phases can be distinguished: the Stance and Swing phases. The Stance phase starts with the *Foot Strike* (FS) and finishes with the *Foot Off* (FO); the *Mid Stance* (MS) instant is typically identified with the passage of the hip joint centre over the vertical line passing through the instantaneous centre of pressure (COP) on the ground surface. The Stance duration is about 40% of the stride cycle, while the swing duration is 60%. In elite sprinting this percentage could reach values close to 20% (stance) and 80% (swing) [23]. The stance phase can be divided into two sub-phases of braking (from FS to MS) and propulsion (from MS to FO) (Figure 28.1g, h). During the braking phase, the blade compresses, converting kinetic/potential energy into elastic, while the hip extensor muscles help stabilise the knee. During the propulsive phase, the hip muscles produce concentric positive work during hip extension; the stored elastic energy is released to the limb, generating a forward/upward push, kinetic/potential energy, and eventually speed.

In transfemoral amputees, hips must generate almost twice the energy of an able-bodied sprinter, as the knee and ankle contributions are missing. Unlike biological ankles, running prosthetic feet cannot generate mechanical power and typically return 63%–95% of the stored elastic energy during the stance phase of running [24].

Research on amputee running biomechanics has documented considerable knowledge about the effect of RPF on Ground Reaction Force (GRF) differences between the Affected Limb (AL) and Unaffected Limb (UL) in unilateral amputee sprinters. It is now widely acknowledged that RPF reduces the peaks of vertical GRF and increases the horizontal GRF impulse at the AL, therefore giving the advantage to double amputees with RPFs over able-bodied sprinters [25,26]. In addition, an increased limb effective length and decreased distal limb mass and inertia can give some inertial and structural advantages in running. Therefore, in competitions, prosthetic height is set at the discretion of the athlete or their prosthetist, or both, in accordance with the IPC guidelines [27]. The height is based on the unaffected leg and residual limb length (of anthropometric proportions before amputation). The total length of the affected limb, when unloaded, is usually set about 5 cm longer than the intact limb to compensate for foot deflection during loading [28,29].

Research regarding the biomechanics of amputee running can be summarised in some main lines that are mutually intersecting but still challenge the scientific and sports community, also for their potential insight into running performance determinants.

a *Development of appropriate instrumented tools for the in-vivo evaluation of GRF and running biomechanics parameters.*

   In vivo evaluation of amputee runners' GRF requires a motion capture system integrated with force platforms that have typically been embedded in the track floor [14,30], or in a treadmill [31]. Recent developments include the direct application of strain gauges to calibrated RPFs [13] and the development of multicomponent dynamometers applied at the clamp (Figure 28.1d, e).

b *Definition, classification, and prescription of RPF stiffness properties.*

   Despite the importance of this mechanical parameter, a clear classification of RPF stiffness properties and test methods is missing. Manufacturers tend to develop their test methods internally and express RPF stiffness by numerical categories that have no standard mechanical corresponding quantity. For sure, load-deflection curves depend on foot/ground orientation angle and on shear/axial load ratio with respect to a clamp axis. Tests based on single actuators and fixed loading axes [24,32,33] have evolved towards variable angles and variable load

ratios [34]. In the future, ISO standards should be developed to address the identification of a common stiffness test method and a common fatigue test method, given the high demand for safety and the high risk of injury associated with an unexpected failure during a sprint or a long jump event.

c *Optimal choice of RPF type and stiffness given an athlete's anthropometry and running technique.*

The first request of an athlete or a trainer to a biomechanist is usually: "What is the most appropriate type of RPF to maximise my performance and minimise my injury risk?" A clear answer to this question is not known due to the complexity of the problem and because often the performance and injury risk criteria are related. Many biomechanical studies of top athletes continue on these topics, but they suffer limitations of low statistical power (due to the low number of elite athletes), individual athlete differences, and the intrinsic adaptive nature of the human neuromuscular system, which typically re-adapts running patterns to the structural/geometric/alignment properties of an assigned device.

d *Identification of an optimal alignment of the full prosthetic system, from the socket to the foot tip.*

The prosthesis alignment has received growing attention in recent years [35,36] as it has been shown that, similarly to able-bodied runners, an increased socket angle during MS with respect to the clinical Thomas' Angle [36], within a certain range and depending on the athlete's response and acceptance, can have positive effects on performance, provided that the knee stability is preserved. New emerging foot shapes require appropriate attention to the socket and foot tip alignment with respect to the Hip Joint Centre (HJC) - Knee Joint Centre (KJC) fundamental leg mechanical axis (Figure 28.1c).

e *Modelling of running biomechanics for the investigation of performance determinants.*

Modelling running in terms of lumped mass/spring/damper elements has proved to be an effective tool to investigate the determinants of performance when varying some factors or across many athletes [37,38]. Leg stiffness of the affected limb is consistently lower than the unaffected limb and is associated with larger values of performance determinants (such as LCOM, the length of COM during stance). Musculoskeletal modelling of running with prosthetic feet is also a challenging but promising direction of investigation, due to the possibility of gaining insight into the residual limb and joint muscle contribution and joint loads.

f *Development of wearable sensors/optical technologies for the monitoring of running activities and the prediction of GRF loads using AI algorithms.*

The use of dynamometric platforms or wearable dynamometers at the clamp, despite their precision and completeness of information, are hardly adoptable for long-term unmonitored data collection regarding the load spectra exerted by a sprinter or a disabled runner on an RPF. Therefore, effort has been put into the development of methods that combine the use of small IMUs with long battery life embedded in the foot with the classification and pattern recognition abilities of AI tools. This line of research seems promising [39] as both IMUs and AI are rapidly developing and becoming more affordable.

Paralympic and prosthetic running is an important and interesting field for the application of principles and methods of mechanics (i.e., the mechanical prosthetic foot) to biological structures (i.e., the residual limb and the sprinter/runner body) during dynamic actions. Progress in understanding these mutual influences and optimisation criteria would also provide insight into able-bodied running biomechanics: indeed, factors affecting prosthetic running can be perturbed or varied to observe their effects more easily than in able-bodied runners.

Upper limb

While the previous section deals with functional prostheses that are built to store energy and generate reaction forces to propel the athlete, in some cases, prostheses may only play the role of restoring equilibrium of the body and passively contribute to overall performance.

The role of the upper limbs in running has been evaluated by a few studies [40–42] and the effect of missing arm swing has been investigated rarely and only with non-disabled participants who suppressed arm swing [43]. Nonetheless, these studies found that the role of arm swing increases with increasing running speed and that the upward and forward momentum as well as the free moment (i.e., moment around the longitudinal axis of the body), changed. Hence, more energy was required for the same running speed, partly to avoid deviation from a straight running direction due to the difference in positive and negative free moments caused by the left and right arm swing.

Mally et al. [44] investigated an elite below-elbow amputee 400 and 800 m Paralympic runner and found that ground reaction forces during the left and right stance phases differed more than for a non-disabled control. Additionally, an extra weight on the affected arm changed the free moment. It was shown that there is a significant influence on the investigated parameters. Eventually, the athlete decided to run with a custom-made prosthesis to equalise the mass moment of inertia of both upper limbs and thus benefit from a reduced left-right difference in forward and upward momentum as well as free moment.

The basic construction of the forearm prosthesis was based on mass moment of inertia calculations of the non-affected and affected limbs. It was finally constructed by a prosthetist and worn during competition by the athlete. This example also shows that when working with athletes with impairments, it is of the utmost importance for sports engineers to take an inclusive approach in the development and adaptation of equipment. The athlete's feedback and acceptance play a major role in this process. Additionally, practitioners such as prosthetists need to be consulted to bring in their experience in the actual manufacturing of prostheses and their adaptation to and interface with the body. Upper-limb prostheses can also fulfil the task of energy storage and recovery for force production. Another example from swimming is also described in the section on the role of biomechanical simulations in sports for athletes with impairments.

Cross-country sit-skiing

Athletes in cross-country sit-skiing have different impairments such as loss of a lower limb or limbs (amputee or dysmelia), loss of muscle function in the legs, hips, trunk, or both, or other conditions. The rules state that the sitting device should have a fixed seat (not adjustable during the race) mounted on a pair of skis with a rigid connection. No springs or flexible articulations are allowed, and the maximum height of the seat is 40 cm. The buttocks must always be in contact with the seat, and the upper thigh must be strapped to the seat.

There are many different sitting positions used in cross-country skiing. Athletes with, for example, a double leg amputation are seated in a kneeling position. Athletes with, for example, a spinal cord injury inducing loss of muscle function in the lower trunk, hips, and legs sit with their knees higher than their hips (KH, Figure 28.2b). Hence, the choice of sitting position depends on several aspects, such as trunk function [45], ability to bend the knees, and the fitness level of the athlete. From an equipment perspective, the question is how to optimise the sitting position for a specific athlete to improve performance and reducing the risk of injuries. Research on non-disabled athletes has shown that a kneeling position compared to the KH position, is

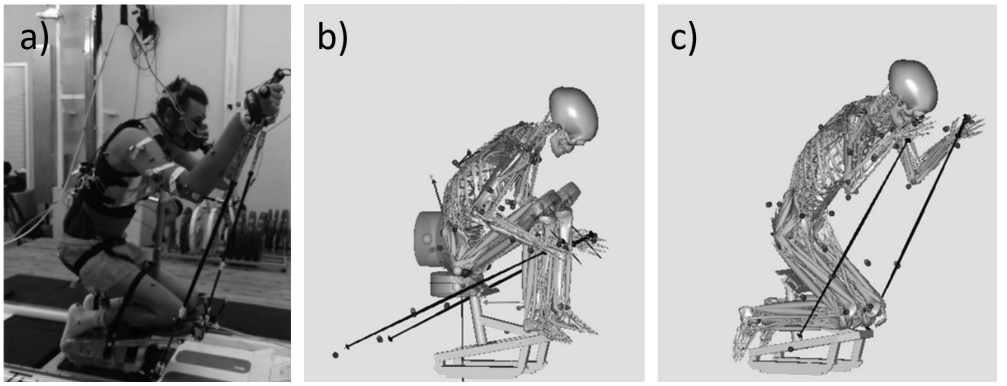

Figure 28.2 (a) Test of athlete with impairment in KLS, (b) simulation of the KH position, and (c) the kneeling position (KLS).

more beneficial both for attaining maximal speed [46] and for being more economical at submaximal intensities [47]. Hence, this position is recommended [48], and many athletes try to adopt this position [4].

However, athletes with loss of muscle function in the lower trunk, hips, and legs cannot sit in a kneeling position. To enable the kneeling position, sit-ski designs also accommodate athletes with paralysis in the legs, hips, and trunk. A sit-ski using trunk support (the chest leaned onto an anterior support) was developed and tested (KLS). The newly developed sit-ski KLS (Figure 28.2a) was tested and compared to the KH sit-ski (the other sit-ski position possible for athletes with loss of trunk function. The evaluation using non-disabled athletes showed that the KH sit-ski, compared to the KLS sit-ski, was better for force and power production and hence better from a performance perspective [49]. Musculoskeletal inverse dynamics simulation (Figure 28.2b, c) also showed that the KH sit-ski, compared to KLS, used muscle power from a larger part of the body [50], had smaller joint reaction forces in the shoulder, and larger joint reaction forces in the lower spine [51]. This can be interpreted as meaning that the new sit-ski KLS design may impede performance and contribute to a higher risk of shoulder pain, but a reduced risk of lower back pain. These hypotheses, however, need to be confirmed by more observational or prospective studies. The two sit-skis, KLS and KH, were also tested and evaluated by an athlete with paralysis in the lower trunk, hips, and legs.

These results showed that the KH position also enhanced performance for the athlete with impairment [52]. In summary, the new sit-ski position KLS was worse than the common sit-ski KH, except for the lower back joint reaction forces.

The seat for a sit-skier is like the boot for a standing skier. To transfer power from the body onto the ground and hence propel the sit-skier forward, this connection is important, as described earlier in, for example, the amputee runner (cf. sub-section on Running with Prosthetic feet). In cross-country sit-skiing, some athletes use custom-made seats of carbon-fibre reinforced composites or plastic, similar to alpine skiing. The use of custom-made seats depends partly on the type of impairment and the need for support. An athlete with a lower leg amputation has full hip and trunk muscle control and often only uses a flat kneeling seat with strapped femurs. An athlete with loss of muscle function around the hips often uses a seat that stabilises the pelvis both in flexion-extension and laterally. Additionally, the athlete and designer need to consider the risk of inducing pressure sores in contact with the seat.

The role of biomechanical simulations in sports for athletes with impairment

As mentioned in the sub-section on Running with Prosthetic Feet, amputee running has drawn attention to Paralympic sprinting. The "Flex-Foot Cheetah" (the carbon fibre foot developed by biomedical engineer Van Phillips and manufactured by Össur) is the piece of equipment in sports for people with impairments that has received the most attention. The "Cheetah" foot was introduced at the beginning of the century and transformed the ability to run for people with transtibial prostheses and also for people with transfemoral prostheses [32].

In 2008, the already mentioned Oscar Pistorius and his prostheses were biomechanically analysed using kinematic and kinetic measurements and inverse-dynamic computations of joint moments and work [14]. The results indicated that running in a straight line with prostheses at full speed requires less mechanical work produced around the knee (11 times lower negative joint work and 8.1 times higher positive work) and 2.8 times higher positive joint work in the ankle joint than able-bodied legs [14]. This thereby indicates that running with prostheses allows for full-speed sprinting at lower metabolic costs [14], which could be judged as "technical boosting" and lead to unfair competition.

There have been several research studies involving biomechanical simulations on the use of both single and double transtibial and transfemoral prostheses. This has been mainly in sprint running but also in the long jump and high jump, both for research questions about performance related to able-bodied athletes and about how different prostheses affect performance [53] and the risk of injuries [54].

Adapted wheelchairs are used in many sports, such as wheelchair racing, tennis, basketball, and rugby [55,56]. There have been several simulation studies of daily-use wheelchairs, and studies of adapted chairs have also been published. For example, the study by Kobayashi and Tanaka [57] on wheelchair rugby shows how lowering the axle position reduced the burden on the subscapularis muscle, which is an important stabiliser of the shoulder girdle and could reduce the risk of shoulder injuries and pain [6].

In cross-country sit-skiing, different sitting positions have been biomechanically simulated to understand joint loads, muscle forces and muscle power distribution (sub-section Cross-Country Sit-Skiing). The intention was to study the risk of shoulder injuries and pain, which is a major risk in seated disability sports. The results showed that sitting in the knee-high position, compared to the kneeling position, utilised more trunk and hip muscle power, and the joint reaction forces in the shoulders were lower.

Simulations of swimming have examined the role of a single transradial prosthesis [58]. These simulations showed how the shoulder joint torque increased in both the intact and the deficient limb when using a transradial paddle (prosthesis). Simulations of different paddle dimensions were performed to design a paddle that balances the load on the left and right shoulders to reduce the risk of injuries.

As mentioned earlier, the results of musculoskeletal simulation partly depend on the choice of underlying mathematical models and biomechanical assumptions. Petrone et al. [59] have shown that two different software packages for biomechanical simulation lead to different results and that both differ from the activity profile in the amplitude and timing of the muscles measured with surface electromyography (sEMG). However, the data still showed reasonable agreement for some muscles examined under some of the conditions tested. While they evaluated the simulation results for isokinetic exercises and hence a controlled condition, Felsner et al. [60] estimated the muscle activity for a hand-cycling motion.

Hand-cycling is a sport in which athletes with impairments of the lower limbs use a special tricycle in which the single front wheel is propelled using a crank that is rotated via handles by the athlete's upper limbs. Unlike in bicycling, where the cranks and thus the legs work asynchronously with an offset of 180 degrees, in hand-cycling, both arms work synchronously.

Felsner et al. [60] recorded the kinematics of an elite hand-cycling athlete along with bilateral sEMG data of four muscles of the upper body and limbs and used the kinematic data to drive a model in the musculoskeletal simulation software AnyBody (AnyBody Technology, Aalborg, DEN). The results of the simulation, in terms of the activation time of the muscles, showed partly large differences from the measured data.

Several reasons for these differences could be identified and might also be applicable to other musculoskeletal simulations.

a Poor accuracy or missing input data (e.g., external forces)
b The model cannot reflect physiological reality due to limitations of the simulation (e.g., muscles with large areas of origin or insertion, which are represented by several single muscles in the simulation)
c The position of sEMG electrodes on the muscle belly and signal changes due to muscle belly movements, which are not represented in the model

This reflects some of the challenges of musculoskeletal simulation, including the complexity of the musculoskeletal system, the choice of muscle recruitment models, the quality of input data, and model validation. On the other hand, there are advantages, such as independence from fitness level, non-invasive predictions of internal and external loads, the ability to model any body composition, and the possibility of separating the influence of equipment and impairment on performance.

In conclusion, biomechanical simulation plays an increasing role in equipment design, injury prevention, muscle recruitment analysis, and performance prediction in disability sports. Consequently, it may also become a supportive tool in classification development in Para Sports, as it has the potential to analyse how physical impairment and equipment influence performance. This could contribute to safe and fair competitions, for example, by ensuring safer equipment and avoiding technical boosting.

Discussion

The success of engineering and technical innovations in sports equipment for persons with physical impairments can be verified through biomechanical testing with athletes. Additionally, biomechanical data (cf. section Challenges for Biomechanical Measurements) and advances in research methodology (cf. sections Some Examples of Recent Research, and The Role of Biomechanical Simulations in Sports for Athletes with Impairment) are important for managing the complexity of the "Bermuda Triangle," with its interactions between athlete impairment, performance and equipment, sport technical rules, and the sport classification system. The simultaneous consideration of the four IPC principles (cf. sub-section Ethical Considerations), as well as the inclusion of human, sport, and neuromuscular factors, adds to these complexities.

Such data are important for understanding and quantifying the forces acting on the athlete and their directions. Currently, many Paralympic sports lack load data and other biomechanical data. Such data could be useful for sizing equipment and engineering new solutions for athletes with impairments. Therefore, more biomechanical data need to be collected in combination with further development of measurement setups, testing protocols, and equipment design procedures in

disability sports to improve the state of knowledge. It is likely that some of these data can also be obtained in collaboration with non-disabled sports. In turn, this knowledge can be used to improve equipment or to calculate and simulate internal loads such as muscle and joint loads (cf. section The role of biomechanical simulations in sports for athletes with impairment), which can help improve performance and reduce the risk of injury.

For over a decade, the IPC, sports, and research communities have focused on evidence-based classification in an effort to improve the validity and equity of the ability classification system [61], supported by frameworks such as Key Elements of Research Design and Measurement Methods [62]. However, such frameworks and collaborative efforts in sports engineering are still lacking for factor-adapted equipment in the "Bermuda Triangle." Therefore, this chapter does not present "standard practices" but rather examples of biomechanics in athletes with impairments and adaptive equipment.

Conclusion

It is anticipated that the body of knowledge regarding the biomechanical interactions of persons using adapted sports equipment will continue to advance. This advancement can be accelerated through more interdisciplinary research networks and collaboration that address adapted equipment in a structured and holistic manner. The result will be an increased availability of design data, technical guidance, sustainable design practices, standards, analysis methods (equipment-impairment performance), and "engineering reinforcement" measures (classification effectiveness). Softer values such as universality (fit, access, and cost) and communication (interdisciplinary platforms), will also improve; the latter will help reduce real and perceived inequalities and "technological controversies" [7,63].

Designers should strive to consider the biomechanics of athletes and develop future adapted sports equipment that performs better while being safe, fair, and universal. Additionally, equipment should ensure that human skills and capabilities play an essential role in competition, benefiting athletes from the grassroots level to the elite.

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PART V

Field testing, sensors and instrumentation for the digital athlete

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INTRODUCTION TO FIELD TESTING, SENSORS AND INSTRUMENTATION FOR THE DIGITAL ATHLETE

Alan Godfrey

Introduction

The use of digital technologies offers scale and high resolution (e.g., millisecond capture) to provide a leap in available data, while computing resources like artificial intelligence (AI) push analytical boundaries. For example, wearable inertial-based technologies and AI-based computing vision provide complementary but contrasting approaches to personalised biomechanical and multi-athlete patterns of movement, respectively. Moreover, associated technologies like wireless connectivity enable data collection in near real-time from an expansive area in nearly any outdoor environment. The use of these types of technologies (albeit a non-exhaustive list) has enabled a paradigm shift in how and where athletes can be studied, moving from bespoke laboratory environments to more realistic and natural settings, i.e., the track, the field, the pool.

Here, this section explores some common technologies (e.g., inertial measurement units) as well as more nuanced (e.g., AI-based computer vision). Although the topic of the digital athlete (e.g., Figure 29.1) is not specifically explored in this edition, the chapters provided allude to the growing trend of harnessing contemporary technologies (like those explored in this section) for performance assessment and digital twinning. To start, in Chapter 30, Mason et al. explore the benefits, trends, and considerations, while providing guidance and opportunities for research regarding the use of technology for longitudinal assessment of athletes. The authors provide a useful overview of some technologies and accompanying methodologies to consider when using these in the field, such as validation and reliability testing. Then, in Chapter 31, Robertson et al. discuss the main types of passive sensing currently used in sport, along with how sporting organisations can navigate this changing landscape to make informed decisions on how they use this technology. Some of the main applications from the literature are presented, along with key developments expected in the future. Next follows a series of chapters that are more technology-focused. In Chapter 32, Stamm et al. examine the use of inertial sensor-based wearables to explore arising kinematics that can be measured and extracted from recordings to help athletes and coaches investigate previously high-resolution parameters, with a case study in swimming. In Chapter 33, Young et al. present AI-based computer vision as an alternative to inertial sensor-based technology. Specifically, the authors adopt an explainable AI (XAI) approach to ensure transparency in how contemporary digital tools can be implemented and developed, avoiding the dreaded black-box concept

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Figure 29.1 Examples highlighting the concept of a digital athlete.

by showcasing a useful tutorial. The use of camera-based technology is further explored by Dunn et al. in Chapter 34, where they present a focus on tennis racket design via a 3-dimensional range of motion and a modelling approach. Lastly, in Chapter 35, Wall et al. help shed light on how data from some of the aforementioned technologies could be processed. Specifically, the authors present the concept of big data and how it can be used in machine learning models to provide predictive insights regarding the tennis serve.

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LONGITUDINAL ASSESSMENT IN SPORT

Injury and performance assessment

Rachel Mason, Gill Barry, Alan Godfrey, and Samuel Stuart

Introduction

Performance is multifactorial and requires adequate physical, mental, technical, and tactical expertise to compete at the elite level [1,2]. The contribution of each element depends on the sport's demands and the characteristics of the individual athlete. Athletes pursue multiple phases over the course of a season, including pre-season, training, competition, and recovery. Considerable time each year is dedicated to their training, and for some sports, athletes' investment amounts to about 17% of their waking time [3]. The training regime of athletes is meticulously scheduled for days (i.e., microcycles) to as long as months (i.e., macrocycles) to ensure progressive adaptation and prevent fatigue, boredom, and injury [4]. Athlete monitoring typically refers to capturing information regarding an athlete's performance or response to training and competition. Professional sports have long assessed and monitored their athletes' physical condition and now have robust programmes for preparing and developing athletes through coaching, nutrition, physiology, biomechanics, and psychology. From this perspective, a fundamental goal of such programmes is to ensure athletes achieve optimal adaptation and peak performance at the correct time whilst simultaneously minimising injury risk. At any given time, 36% of elite athletes have a health problem, with 15% reporting substantial health problems weekly that may negatively impact sporting performance [5]. Accordingly, continuous and longitudinal assessment of athletes is playing an increasingly crucial role, with the emergence of digital technology enabling quantitative measurement in a readily available manner.

Before the adoption of sophisticated technology in the modern era, coaches and practitioners were often reliant on the 'coach's eye' to monitor their athletes' performance, often with some self-report measures. Whilst that broadly subjective approach has its place in sports medicine, the use of more objective monitoring methods can reduce bias from coaching and performance discussions [6]. Traditionally, objective, longitudinal monitoring of athletes has been carried out in large-scale laboratories, with expensive equipment such as 3D motion capture or 2D time-motion videography, where conditions are controlled and data outcomes have been developed and used to improve athlete performance. Although such methodologies offer accurate data, these traditional methods have drawbacks, including lack of ecological validity, the assessment of discrete time

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points, the need for trained personnel, and expensive equipment, thus making them impractical for quick, in-field assessments [7].

Advances in wearable technology represent a departure from previous monitoring methods in breadth and depth. Wearable technology refers to "*devices that can be worn or mated with human skin to continuously and closely monitor an individual's activities, without interrupting or limiting the user's motions*" [8]. Wearable devices can be worn as accessories, embedded in clothing, implanted in the user's body, or even tattooed on the skin [9]. Wearables are powered by microprocessors and enhanced with the ability to send and receive data via the Internet (i.e., Internet of Things, IoT). Wearables include, but are not limited to, heart rate monitors, electrocardiograms (ECG), and inertial measurement units (IMU), with data often transmitted to laptops, tablets, and mobile phones and stored on organisations' and sports teams' central servers [10]. The ease and convenience of data collection enable more varied and extensive outcomes. Wearables have made it possible to monitor a huge range of health- and performance-related data due to their low cost, small size, portability, and mostly unobtrusive manner [11]. This has led to the rise in wearables in the field of sport and exercise medicine. Although body-mounted sensors are capable of delivering precise measurements, comfort is often neglected, leading to potential psychological and performance impacts [12].

In recent decades, significant progress in textile and sensor technologies has been made, whereby sensors are seamlessly integrated into specialised textiles (e.g., aerogels or stimuli-responsive polymers, new finishes in the nanometre range for optimal sweat absorption, and new fabric structures for better aerodynamics [13]) resulting in the emergence of 'smart clothing'. The latter offers a safer approach to monitoring athletic performance, serving the dual purpose of medical and research applications. Smart clothing proves useful in challenging conditions, facilitating data collection and offering vital insights into injury prevention and training support while preserving the athlete's performance. For example, the Maglietta Interattiva Computerizzata (MagIC) was used for the home monitoring of cardiac patients and to evaluate the effects of high-altitude hypoxia on sleep and 24-hour daily life in climbers at 3,500 and 5,400 m above sea level on Mount Everest slopes [14]. The relatively early stage of development in smart clothing, coupled with its limited service life [15] and lower adoption rates [16], leads to a shorter lifecycle compared to conventional clothing. Moreover, that accelerated obsolescence contributes to increased raw material waste and textile accumulation. The sustainability challenges associated with smart clothing have amplified the environmental, social, and economic pressures [17]. Consequently, it is imperative for fashion designers, manufacturers, retailers, and consumers to undertake a comprehensive assessment in order to foster sustainable development [17].

Professional sports organisations integrate wearables for various purposes, from managing training load and optimising performance to preventing injury [18]. Some apply wearables as motivational tools, believing that they offer objective feedback to players to modify behaviour, deterring complacency [18,19]. More recently sports are also exploring the potential of such technologies for longer-term applications, such as assessing the career longevity of current players and talent identification [19,20]. The current conversation around wearables thus far is overwhelmingly enthusiastic, emphasising the potential to predict, prevent or aid recovery from injuries and enhance athletic performance. This chapter will explore the benefits of wearable technology and provide guidance on the implementation of wearables into practice for the longitudinal assessment of athletes, as well as discuss implications and future research opportunities.

Benefits of wearable technology

The adoption of wearable technology in sport is high and has had a profound impact on the evolution of sport, from equipment design to monitoring athlete performance [21]. The incorporation of wearable technology is attractive as it is convenient, inexpensive, and (if validated) clinically useful in decision-making. Furthermore, it enables insight into an athlete's health, well-being, and performance with fewer constraints (e.g., allowing remote or 'on-field' assessment).

There are a myriad of benefits practitioners can expect from successful technological implementation when monitoring athletes. The incorporation of wearable technology into the everyday lives and competition of athletes allows efficient collection and analysis of data from many facets of sports performance and medicine. Variables related to movement patterns [22–24], workloads [25], and biometric markers [26] of athletes can be monitored in various contexts. Furthermore, the incorporation of wearable technology can improve measurement precision, automate processes, and provide more holistic perspectives on athlete performance [19].

Previously, practitioners were limited to snapshots of health and performance outcomes of athletes, and they no longer need to manually record data as they did previously. For example, before wearable technology, which is now commonplace in most elite team sports, understanding match demands (e.g., distance covered, speed) was limited to notational analysis, video, or tape recordings [27]. Such processes were time-consuming, often limiting practitioners' ability to examine only a few key outcomes from certain players in a defined number of matches [28]. Advances in wearable technology provide practitioners with these physical output measures in near real-time [21].

Monitoring a range of health and performance variables may allow for the identification of 'red flags' (e.g., biomechanical fatigue, higher-than-normal resting heart rate (HR)) and enable early intervention in an attempt to prevent injury or illness during training and competition whilst maximising performance [21]. The ability to integrate data streams enables practitioners to better understand the relationships between parameters of performance. Furthermore, it facilitates an integrated, multidisciplinary approach to performance, permitting information to be shared across disciplines in an effective and succinct manner. Where limitations once existed in storing and processing vast amounts of information safely and in a time-efficient manner, technological advancements have reduced many of the challenges involving costs, computing speed, and intelligence tools [29]. Although pitfalls still exist, wearable technology allows practitioners a 360° view of athlete performance when using a strategic, multidisciplinary approach [29]. Improved, individualised training regimes may be deduced from monitoring athlete performance.

Integration into practice

The ability to monitor and quantify an athlete's performance to help minimise the risk of injury has been at the forefront of sports science for the past decade yet remains fully unmet. Wearable technology may serve as a simple and effective solution for sports medicine practitioners to longitudinally assess, monitor, and engage athletes, allowing them to perform optimally while remaining healthy over long, high-intensity seasons. Wearable technologies, such as Catapult, WHOOP™, Zephyr™, and Garmin®, are now almost fundamental to the daily lives of athletes due to their ability to measure various physiological, biomechanical, and well-being outcomes [30]. Wearables have been adopted into many sports, providing previously difficult-to-obtain information [31]. Accordingly, there is a growing need among sports practitioners to identify, utilise and evaluate technologies in a manner that is both effective and efficient. Having a standardised, comprehensive process for implementing technology for athlete monitoring would help mitigate the negative

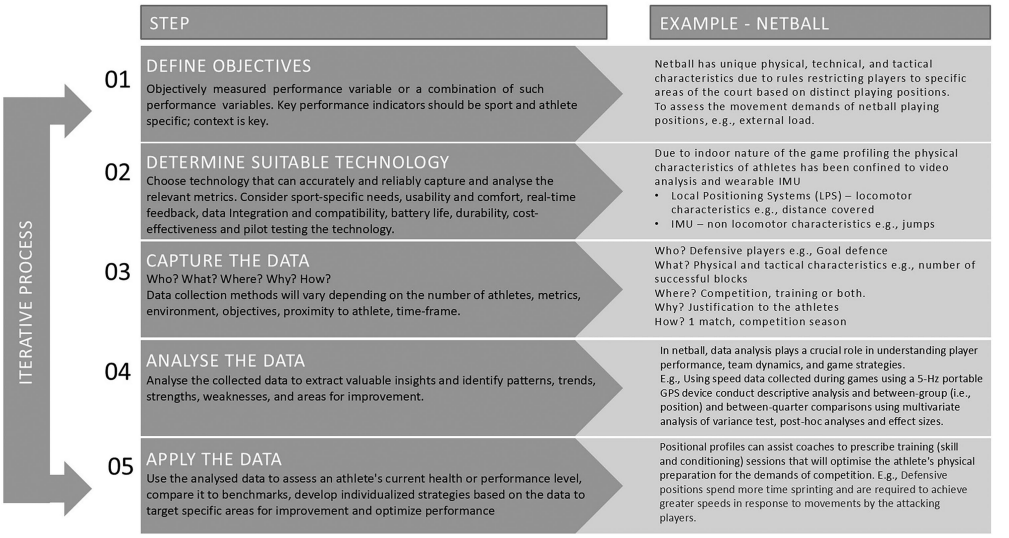

Figure 30.1 An overview of implementing technology into athlete monitoring.

consequences that may arise from poor implementation [32], such as misuse of technology, low athlete engagement, or little practical use [29]. The following steps (Figure 30.1) provide suggested guidance on integrating technology into training and competition efficiently and productively by setting clear objectives, choosing the right technology, and applying good methodology.

Step 1: Define objectives

Defining objectives for sports performance and health requires a systematic and thoughtful approach to ensure that the selected metrics align with the specific goals and objectives of the athletes or individuals involved. In elite sport, objectives may be multifactorial or multidisciplinary, referring to the complex and interconnected nature of the objectives set for athletes' performance and overall well-being. Such objectives should consider various factors and disciplines that contribute to an athlete's success, including physical, psychological, tactical, technical, nutritional, and medical aspects, among others. The integration of multiple factors and disciplines is crucial for achieving optimal performance and maximising an athlete's potential. Examples of objectives may include:

- ***Performance optimisation:*** The primary objective in elite sport is to optimise an athlete's performance. This includes improving physical attributes, technical skills, tactical understanding, and mental resilience to achieve peak performance during competition.
- ***Injury prevention or recovery:*** Reducing the risk of injuries and speeding up recovery from injuries are critical objectives in elite sport. Identifying and addressing factors that contribute to injuries, such as biomechanics, training load management, and individualised injury prevention programmes, can be coupled with baseline characteristics to aid understanding of when an athlete is back to their pre-injury state.
- ***Recovery and regeneration:*** Elite athletes face intense training and competition schedules, making effective recovery and regeneration crucial. This involves developing comprehensive recovery strategies that encompass nutrition, sleep, and active recovery techniques.

- ***Psychological well-being:*** Mental health and psychological well-being play a significant role in an athlete's success. This includes enhancing mental resilience, managing stress, and providing psychological support to athletes.
- ***Nutritional optimisation:*** Nutrition plays a vital role in an athlete's performance, recovery, and overall health.
- ***Tactical and technical development:*** In team sports, multifactorial objectives include improving tactical awareness, decision-making, and technical skills to enhance overall team performance.
- ***Monitoring and performance analysis:*** Multidisciplinary approaches involve utilising advanced technology and sports science to monitor an athlete's progress, analyse performance data, and make informed training and competition decisions.

When defining objectives, it should be recognised that each athlete is unique, and a one-size-fits-all approach may be ineffective. Training, recovery, and nutrition objectives should be tailored to meet the specific needs of each athlete. Furthermore, involving athletes in the decision-making process and empowering them to take ownership of their development is important, as those who understand and buy into the multidisciplinary approach are more likely to adhere to the programme [33]. Athlete objectives require a long-term approach with short-, medium-, and long-term goals embedded. Such strategies allow athletes to progress gradually and avoid burnout or overtraining [34]. Similarly, identifying the unique constraints of the athlete and sport in which the monitoring will take place is vital, ensuring the chosen metrics are relevant and meaningful within the given context, especially due to the abundance of metrics that can potentially be monitored.

Some caution should be given as practitioners should remember that monitoring data serves as a representation of performance, but it is not the actual performance itself [35]. Although many coaches believe that analysing their athletes' performance using technology is beneficial, many acknowledge that it is an investment and that initially no technology is perfect; none of it works all the time, and it can be time-consuming and resource-intensive [29,36]. Establishing benchmark values or targets for each objective will serve as reference points for comparison and allow for the assessment of progress over time.

The establishment of an integrated support team comprising coaches, sports scientists, nutritionists, physiotherapists, psychologists, and medical professionals, allowing for a collaborative approach ensures comprehensive care and optimisation of each athlete's potential. Involving all relevant stakeholders in the early stages of the monitoring process with the most appropriate technology for objectivity will provide valuable insights into the most relevant metrics and help ensure buy-in and commitment to the programme. Subsequently, effective communication and coordination among the support team members are crucial to ensuring that all disciplines are aligned with the overall objectives and strategies. Objectives should be continuously reviewed and refined based on athlete and practitioner feedback, emerging results, research, and changing long-term goals. The monitoring programme can be continually adapted to better suit the needs of the athletes or individuals.

Step 2: Determine the suitable technology

Sport specific analysis

Practitioners are besieged with a multitude of outcomes from wearable technologies. These outcomes may not necessarily be appropriate in all sporting contexts. Selecting the most pertinent outcomes, given each sport's unique constraints, is vital to ensure they are appropriate for the

context of a specific sport. For example, satellite-based navigation system devices can be extrapolated into distance and speed measures, whereas IMUs combine data from multiple sources (e.g., accelerometer, magnetometer, and gyroscope) to measure the acceleration of the body or a body segment. In the applied sport environment, there is a need to distil the myriad of outcomes into concise, meaningful information. Decision-makers (e.g., coaches and performance staff) require simple, accurate, and coherent feedback, including succinct conclusions from data in a timely manner [37]. The onus is on the practitioner to determine the suitable technology, the most relevant outcomes to interpret, and how to communicate with key stakeholders [38]. Although sports performance is often difficult to define and measure (e.g., in team sports) and there is a trade-off between internal and external validity [39], practitioners should outline a clear causal link between their performance or research question. The workflow of selecting outcomes is a continuous process, requiring up-to-date knowledge on the validity and application of wearable technologies, based on research and technological advances. A sound theoretical and practical rationale behind integrating technologies into sporting environments is necessary and greater involvement and improved communication in this process could help to avoid a detrimental gap between information and its impact [40].

The sporting context will drive what technology should be selected to capture the characteristics of athletes during training and competition. For example, in basketball and netball, the use of satellite-based navigation systems is often rendered inoperable, given that at the elite level, both sports are played and trained indoors. Therefore, local positioning systems (LPS), IMUs, and optical tracking may be more appropriate. Additionally, a fundamental consideration when integrating wearable technology is the relevant sport governing bodies. Currently, some sport governing bodies do not permit the use of certain systems within competition (e.g., World Rugby [41] and Federation Internationale de Football Association (FIFA) approval [42]). Thus, practitioners may want to ensure that the technology can be used within competition environments, abiding by regulatory standards and rules within individual sports. The selection of wearable technologies should be dependent upon the context of the sport and needs careful consideration by practitioners.

Technical considerations

Most wearable technologies have been developed based on real-world needs, and despite their widespread use, fewer than 10% of commercially available wearable technologies are validated against an accepted 'gold standard' [43], with reliability less investigated [44]. Even where validity and reliability have been investigated, there is a wide disparity in findings, likely due to various reasons, including differences in technologies tested, experimental design, populations investigated, and outcomes measured. For example, wearable technology has generally been found to be valid for examining most running gait outcomes against laboratory reference measures, particularly spatiotemporal outcomes. However, that appears to be dependent upon factors such as the location of the wearable, the capabilities of the technology, and the testing protocol (e.g., speeds) used, as well as the gait outcomes obtained [44]. To elaborate, IMUs (accelerometers and gyroscopes) on the foot may provide the most accurate derivations of stride measures [45–47], but caution should be taken when using wearables located at the thoracic spine, as outcomes obtained from such placement appear inadequate to robustly predict gait symmetry, peak vertical, and resultant GRF [48–52].

Particular caution is warranted for commercially developed technologies [53]. With the wearable technology market estimated to be about $100 billion in recent years [54], companies are eager to launch as soon as possible to earn a profit [4]. Thus, prior to market launch many manufacturers

often forego independent scientific testing and instead perform internal validity and reliability studies. However, that may limit the robustness of findings and could lead to exaggerated marketing claims by companies [55]. After market launch, independent researchers/institutions may undertake additional investigations, often finding conflicting evidence regarding the accuracy, validity, and reliability of wearable sensors already used in sports medicine settings [56,57]. Thus, although wearable technologies are constantly evolving, practitioners need to balance innovation with a critical consideration of the precision (validity and reliability) of such technology.

With the current landscape of wearable technology being large and heterogeneous, recent research has delved into creating frameworks to align terminology, approaches, and evidentiary standards to progress the innovation and use of wearable technologies. The aim is to prevent the misinterpretation of data, which could lead to serious consequences. Researchers have developed an evaluation framework for monitoring technologies intended for healthcare applications that is applicable to sport. That framework includes verification, analytical validation, and clinical validation (V3) [58]. Verification of wearables entails a systematic evaluation by hardware manufacturers, where evaluation of response to standardised synthetic signals occurs. For example, attaching the accelerometer under verification to a shaker capable of generating known accelerations to see if it produces accurate and precise measures. This step ensures quality assurance by troubleshooting at the most basic level and leads to interoperability of devices by standardising units to facilitate algorithms that are transferable [58–60]. Secondly, analytical validation occurs at the intersection of engineering and clinical expertise [58]. This step translates the evaluation procedure for wearable technology from the bench to in vivo under controlled conditions and is ordinarily performed by the entity that created the algorithm, either the vendor or the clinical trial sponsor [58]. This step introduces human variation but under controlled protocols that are optimal for identifying features of the signal that may be valuable for distinguishing between different movement characteristics (e.g., intensity or type) [60]. Initially performed under laboratory conditions against criterion measures under controlled conditions, this step is still not reflective of real-world conditions. Thus, those results may not reflect how a device/method will perform in free-living applications [60], requiring subsequent analytical validation in 'real-world' settings. The wearable technology outputs should be compared against a reference (i.e., gold) standard device during naturalistic and variable conditions of daily living. That would typically involve human participants wearing both devices to assess metrics such as reliability and time delay from the reference standard. If the wearable device is novel or unique, there might not be an existing wearable system that can be used as the "best" reference. Where that arises, the most optimal reference must be chosen, and an appropriate detailed investigation must occur from the raw data (sample) level to the final outcomes. Upon satisfaction of previous steps, clinical validation can then be performed. The purpose of clinical validation is to validate a measurement that has already undergone verification and analytical validation and to determine its ability to address specific clinical questions [58,60]. This may involve assessing or prognosing certain clinical conditions. It is important to tailor the clinical validation process to the specific context of use. The primary goal of clinical validation is to examine the association between a measurement derived from technology and an outcome (e.g., health condition or performance measure) [58,60]. A technology becomes successfully integrated into practice when it is utilised in various applied contexts, including surveillance, experimental clinical trials, and observational studies that examine the relationship between activity and performance or health outcomes [60].

To provide a holistic view of athlete monitoring, practitioners often combine data from multiple technologies [61]. Practitioners must be aware of between-system interchangeability, as understanding the agreement between technologies is important to ensure meaningful changes in

athletic profiles can be tracked accurately [62]. For example, a comparison of optical tracking to satellite-based navigation systems has shown slight-to-moderate and moderate-to-large differences for total distance and high-speed running distance (> 18 km/hr), respectively [37,61,63]. Such differences highlight the necessity of considering the wider context when using technology (e.g., satellite-based navigation system sampling rate, satellite connectivity, satellite signal strength, and software filtering) when reviewing system comparisons [37]. The subtle differences between technologies need to be critically examined to discern whether they are meaningful regarding their influence on decisions made or the interpretation of data derived [61].

The integration of data from multiple technologies in athlete monitoring is a crucial aspect of modern sports science and performance analysis. However, one significant challenge that practitioners often encounter is the difficulty of synchronising devices from different suppliers, particularly when proprietary software is involved. This is a noteworthy issue in the field, as proprietary systems may not be designed to seamlessly communicate with devices from other manufacturers [64,65]. Different data formats, protocols, and proprietary data structures can create obstacles when attempting to combine information for a holistic view of an athlete's performance. In such cases, the development and adoption of standards can play a pivotal role in addressing these challenges. Standardisation can help establish common protocols and data formats that enable interoperability between devices and software from various suppliers [66]. By adhering to these standards, different technologies can communicate and share data more efficiently. This not only simplifies the process of data integration but also ensures that the information from different sources is accurate and compatible, ultimately streamlining the decision-making process and the interpretation of the data derived [66].

The implementation of wearable technology includes concerns related to data storage and administration (e.g., physical storage limitations, availability, and upkeep), interoperability and the accessibility of diverse resources, as well as security and privacy issues (e.g., permission control, data anonymity, etc.), alongside the need for consistent and universal access. One potential solution to tackle all these challenges is the incorporation of the Cloud Computing paradigm into wearable technologies [67,68]. Cloud Computing offers the capability to access shared resources and a common infrastructure in a ubiquitous and all-encompassing manner, providing on-demand services over the network to execute operations that cater to evolving needs in electronic healthcare applications [67,68]. Further technical considerations include whether the devices require calibration or specialist training to set up and interpret data, the portability and physical range for signal transmission/recording, wireless connectivity between devices (e.g., Bluetooth/ANT+), and real-time data transfer capabilities [43]. An overview of technical considerations is depicted in Figure 30.2.

Step 3: Capture the data

The data capture process will depend on the athlete's individual needs, sport, selection of metrics, the timing and frequency of their monitoring, the scientific basis for measurement, whether the individual is training or competing, and the extent to which they and their coach accept and/or are aware of the benefits and drawbacks of monitoring with wearable technology.

- **Who?** Athletes, coaches, sports scientists, and analysts can be involved in capturing data. Primarily who are the athletes of interest?
- **What?** Various metrics and information related to an athlete's health or performance either during training sessions, competitions, practice, or away from the sporting context.

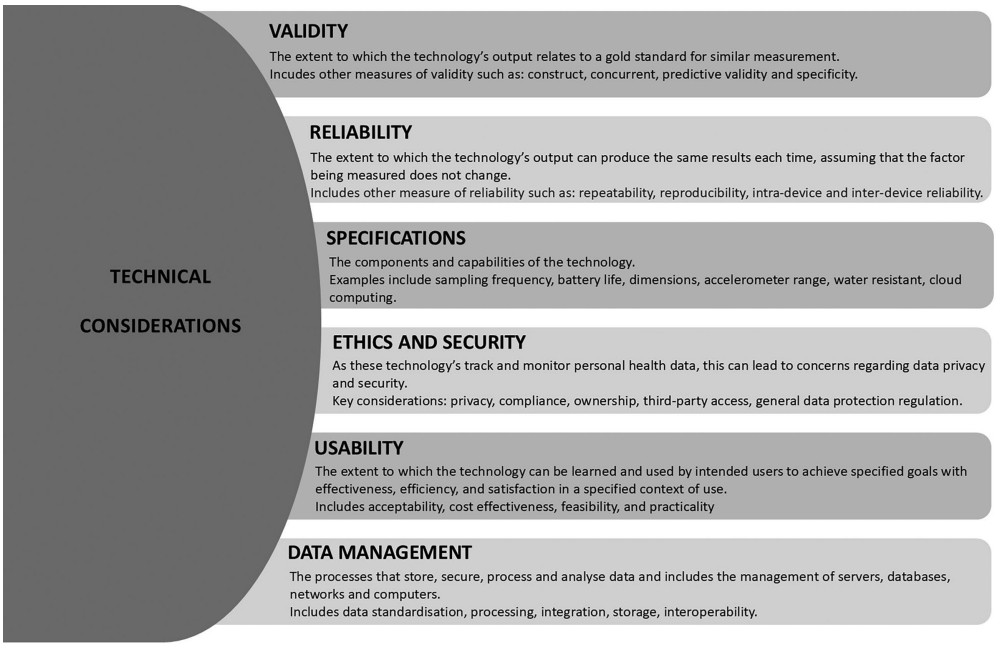

Figure 30.2 Technical considerations for implementing technology into athlete monitoring.

- **Where?** Data can be captured in various locations, including training facilities, competition venues, practice fields, or away from the sporting context (e.g., habitual environments).
- **Why?** Capturing sports performance data is crucial for assessing an athlete's progress, identifying strengths and weaknesses, making data-driven decisions, optimising training strategies, preventing, or aiding recovery from injuries, and enhancing overall performance. This is primarily determined in Step 1 but must be considered throughout the process.
- **How?** How the data is collected depends on numerous factors, some of which will have been addressed in the previous step. This can also refer to more of the intricacies involved in data collection, e.g., real-time monitoring vs. pre- or post-event analysis, integration of multiple data streams (i.e., from on-field, off-field habitual environments, training environments etc.).

Supplementary data to support the primary data obtained through wearable technologies such as race results, video data, and athlete and coach subjective feedback, also need to be considered. Athlete monitoring encompasses two types of measures: 'objective' and 'subjective'. Subjective monitoring delves into the athlete's internal psychosocial and psychobiological factors, such as mental fatigue, effort, perceived stress symptoms, well-being, and motivation [69,70]. These aspects are typically evaluated through self-report tools like the Rating of Perceived Exertion (RPE), wellness items, perceptions of pain, psychological readiness, and more. Such data can provide a more holistic overview of performance and help address the drawbacks of objective monitoring. Objective measures typically fragment observations to allow quantitative precision of metrics in isolation, whereas subjective measures facilitate the merging, integration, and reduction of various information streams that come together to influence the performer-environment system [69,70].

Step 4: Analyse the data

Although there is a growing emphasis on athlete monitoring in both practice and research, it seems that most of the research has concentrated on monitoring specific areas rather than delving into the statistical procedures used [71,72]. Consequently, there is often a risk of accumulating data without a clear plan for its utilisation [35,71]. Although a few studies have touched upon statistical techniques, they do not give significant attention to data preparation and evaluating data quality [35,71,73]. It is important to note that the specific analysis techniques and software tools used may vary depending on the nature of the data, research goals, and the expertise of the analyst. The following steps may be considered:

1 **Data cleaning and preparation**: It is critical that all raw (sample) level data is visually examined (see later). The question should be asked: does all the data make sense? Then a process of cleaning the collected data by checking for errors, missing values, and outliers should be considered. Ensure that the data is in a consistent format and ready for analysis. Prepare the data by organising it into appropriate variables and time points. If not evaluated, the assumptions of normality, homogeneity of variance, and reliability of data collection methods can pose challenges. Violations of these assumptions can undermine the effectiveness of other statistical tests, making data screening essential before conducting further analysis [64]. Although using this data to develop more informed training programmes and potential performance prediction models holds promise, it is crucial for users to address certain statistical considerations when utilising data [71].
2 **Descriptive analysis**: Perform descriptive statistics to gain insights into the data. Calculate measures such as means, standard deviations, ranges, and percentages to summarise the overall trends and characteristics of the data.
3 **Statistical analysis**: Apply appropriate statistical techniques to analyse the longitudinal data. The choice of analysis depends on the research question and the type of data being collected. It is crucial to identify what longitudinal changes in responses (i.e., to training) are meaningful and significant, therefore going above and beyond "normal" or random variability. The standard deviation (SD), effect size, smallest worthwhile change (SWC), coefficient of variation (CV), magnitude-based inferences (MBI), and risk ratio are just some examples of metrics used to determine this meaningful change. However, unsurprisingly, each statistical test will provide different outputs. Common techniques include but are not limited to:

   a ***Trend analysis****:* Examine the direction and magnitude of changes in variables over time using techniques such as linear regression, repeated measures analysis of variance (ANOVA), exponential and rolling averages [74,75].
   b ***Correlation analysis****:* Investigate relationships between variables using correlation coefficients or regression analysis to identify associations between changes in different outcomes.
   c ***Longitudinal modelling****:* Utilise advanced statistical models such as linear and non-linear modelling, mixed-effects models, or growth curve models to account for individual variations, repeated measures, and time dependencies in the data [76,77].
   d ***Group statistical techniques vs single-subject designs:*** The prevailing approach in sports science is to use group statistical techniques, which offer significant benefits, particularly when dealing with large sample sizes [73]. Group statistical methods rely on making inferences based on the mean of data distributions. Consequently, if decisions are solely based on such data, there is a high likelihood of errors occurring, especially for athletes whose data deviates significantly from the mean [73,78]. Single-subject designs may be more appropriate for monitoring and statistically evaluating athletic performance than traditional group

statistical methods [71]. When dealing with the development of elite athletes, practitioners may find themselves working with athletes who fall on the extreme ends of data distributions rather than near the average. It is crucial to acknowledge that the training response of athletes is highly individualistic and unique [35].

4 **Machine learning:** Various machine learning techniques (e.g., deep learning, random forest, artificial neural networks, support vector machines) are applicable in the realm of artificial intelligence and wearables. For instance, deep learning is employed to extract features related to human behaviour, activity recognition, and pattern identification [79]. Modern machine learning methodologies necessitate the incorporation of feature engineering and classification to achieve precision [67,80].

Contrarily, research demonstrates that few who use athlete monitoring methods apply the analytical approaches mentioned previously, with standard deviations and raw scores most commonly used to assess meaningful change [81]. Although it might appear as a lack of adherence to best practices, it is possible that this preference stems from constraints such as limited time, resources, or knowledge, or the need for practitioners to adjust their data analysis approaches to align with coach-friendly requirements [76]. Next steps may include:

- **Visualisation:** Create visual representations of the data using graphs, charts, and plots. Visualisation of the data will help with understanding patterns, identifying trends, and exploring relationships between variables over time. Data should be visualised in a simple and clear manner that does not result in distortion or misrepresentation of the data. This will likely result in an efficient decision-making process and more informed athlete development programmes. Numerous software programs perform data analysis and produce data visualisations and dashboards (single-screen presentations of the most meaningful information) in a time-efficient manner, for example, Tableau [82]. However, many of these come at a cost. More affordable alternatives include spreadsheet software (e.g., Microsoft Excel), yet its capabilities are constrained when it comes to conducting intricate data analysis and generating multiple data visualisations for individual athletes and teams, which can lead to time-consuming and repetitive tasks [71]. Free and open-source programming languages like R and Python can be a viable solution for this issue [83]. Although R and Python provide flexibility in analysis and data visualisation, the initial coding process and learning of the language and packages may require some time investment. One advantage is the ability to loop over or iterate through a sequence of data, which reduces the time needed to complete repetitive tasks for each athlete or test [71].
- **Interpretation and inference**: Interpret the results of the analysis in the context of the research question and objectives. Draw conclusions about the patterns, trends, and relationships observed in the data. Make inferences about the impact of training interventions, performance changes, or any other relevant findings.
- **Reporting and communication**: The findings of the athlete monitoring need to be summarised in a clear and concise manner. Often, much of the data collected, the indicators of performance, and the opportunity to adjust a training programme based on it are time-sensitive [35,71,84]. As a result, it is imperative that data collected be utilised and returned to decision-makers rapidly. Practitioners will be aided by software that can analyse and represent data quickly. Visualisations and data-driven insights can be helpful in conveying key messages. At present, there is a distinct lack of discussion on the return of monitoring data to coaches and the visual display of the information [35,85].

As the amount of data collected increases, more effective systems for analysis, interpretation, and reporting of simple yet meaningful results to athletes and coaches are necessary. With advancements in the analysis of large datasets, suitable contemporary algorithms (e.g., deep learning) may enable better insights into the relationships between the parameters monitored and various aspects of performance or health, or both.

Step 5: Apply the data

The integration of wearable technology for athlete monitoring has allowed coaches and athletes to receive more nuanced feedback regarding performance and health. Athlete monitoring plays a vital role in training programmes [71,86]. By collecting data through monitoring athletes, it becomes possible to either support or challenge the effectiveness of a specific programme and help clarify some of the ambiguity of the training process from a dose-response perspective [35,86]. Data from athlete monitoring programmes go beyond an outcome-driven approach (e.g., win/loss record, season, or personal records), providing the "why" behind athlete performance and health. Coaches and other decision-makers can receive direct, objective feedback on an athlete's progress [48]. Moreover, these data can assist coaches and practitioners in making data-driven decisions to enhance programmes at both the team and individual levels [35,71,86]. Data can only be beneficial to athlete performance and health if practitioners utilise and apply the data correctly; otherwise it is limited to awareness and entertainment.

Applications for athlete monitoring data could be categorised into the following overlapping purposes [27,87]:

- Better understanding of the characteristics of the sport (e.g., physical demands of the sport). Wearable technologies have demonstrated the capacity to determine key kinematic and kinetic measures in snowboarding [88,89], running [44] and rowing [90]. The use of inertial sensors either worn on the hand/wrist/arm of the athlete or integrated into the equipment has proved beneficial in racket and bat sports to assess stroke performance and skill acquisition [91–95].
- Better understanding of the characteristics of the athlete (e.g., internal and external load). For example, in swimming it has been demonstrated that micro-electromechanical system (MEMS) technology assessing swimmers' acceleration profiles has the potential to offer advances in coaching techniques over the traditional video-based approach [90,96–98]. Data from athlete monitoring may also aid in talent identification or the identification of variables that contribute to optimal performance in a particular sport or task [99].
- Helping with decisions pertaining to performance and injury risk (e.g., training load management intended to reduce the risk of injury) or recovery (e.g., assessment of outcome measures towards a pre-injury baseline measure). For example, the incorporation of wearable technology into regular practice would divulge information regarding injury, including normative assessments, injury occurrence and mechanisms, immediate post-injury assessment, monitoring across intervention and rehabilitation, and post-rehabilitation to help inform return-to-play decisions. Wearable technologies have shown promise for concussion detection [100–102], preventing soft-tissue injury [103,104], identifying biomechanical deficits [30,100,105], physiological stress quantification [106–108], and sleep monitoring [109,110].

Regardless of the method or application chosen, the data given back to athletes, coaches, or other decision-makers likely comes in the form of a collection of charts, plots, or dashboards for each athlete. Ensuring athletes receive feedback is essential for their continued engagement with the

technology [81,111] and a step towards developing a supportive athlete monitoring culture [33]. At present, feedback practices within elite sport are underreported [112]. Further establishment of feedback methods is required to enable evidence-based approaches to implementing wearable technology for athlete monitoring and to promote athlete engagement with such technology.

Iterative process

Understanding the demands of a particular sport is an ongoing venture in sport performance. The relationship between competition performance data and collected monitoring data will likely help answer some questions and potentially bring about new questions [86]. Upon completing the implementation of technology into an athlete's training or environment, practitioners must question the effectiveness of the technology, the knowledge gained and its usability. Subsequently practitioners should then be encouraged to make modifications to their implementation of technology to gain more benefit and repeat the process.

Additional considerations

Athlete centred approach

The primary barrier to the successful integration of wearables into sports practice relates to using athlete-centred design in the development of wearable technology [113]. Most wearables have been developed with little to no input from the user (i.e., athletes or practitioners), and as a result, many devices are often not fit for purpose [114]. Technological, design, and aesthetic characteristics are key areas that together affect the adoption of wearable sensors in sport [115,116]. Wearable technology allows for performance monitoring in a cost-effective, accurate and non-invasive manner, yet it is of little use if it lacks practicality, comfort, and aesthetic appeal for the athlete, as this will lead to decreased compliance. Hence, a necessary step is to investigate the opinions of the targeted end-users (i.e., athletes). At present, studies may often lack consideration of the wearer's physical, psychological, and social preferences regarding the technology [115].

Translational research in sports science is dependent on evaluating the efficacy and utility of wearables to derive impact and significance with respect to the athlete [117]. However, a central issue lies in the synthesis and translation of the data. Athletes, coaches, practitioners, and other key stakeholders need to be involved in the design and development of wearables to enable the translation of wearable technology [117].

Although caution is necessary, most wearable technologies are not labelled as medical devices, yet the outputs provided convey explicit or implicit value statements about the user's health or ability [118]. Assuming the accuracy of data provided by wearables can be a risky proposition. Wearables can be susceptible to errors due to factors such as sensor limitations, calibration issues, or variations in data processing algorithms. Inaccurate or inconsistent data can lead to misguided conclusions and potentially inappropriate actions. Thus, sports practitioners and athletes need to be aware of if and how using such technology influences athletes' knowledge and attitudes about their health [118]. For example, self-diagnosis based on data provided by wearable technologies could be inconsistent with clinical, professional diagnoses. Furthermore, the use of technologies may create an unhealthy obsession with health outcomes [118], which could have a detrimental impact on athlete well-being. To address this, independent validation is paramount. Before individuals or healthcare professionals rely on the data produced by wearables, it's prudent to validate their accuracy and consistency through rigorous testing and comparison with established reference systems

or methods [119,120]. Independent validation is critical, as it fosters transparency, credibility, and trust in wearable technologies as valuable tools for health monitoring.

Practitioner role changes

The potential of wearable technology capabilities is often undervalued, and those working in sport may view wearables as a hindrance due to cost and the complexity of integration into an established programme [67]. The integration of wearables into the sporting environment will alter the roles of practitioners and present new challenges. Although wearables offer vast benefits, their use in sport remains largely unregulated and unexamined, with risks compromising athletes' privacy, autonomy, the confidentiality of their data, and ultimately their careers [18].

Professional sport is a high-stakes enterprise, with athletes having lucrative yet extensive contracts. Optimising the performance of athletes and minimising injury is paramount, and professional sports have dedicated laboratories to achieve this [18,121]. It has been calculated that the National Basketball Association (NBA) lost $2.7 billion due to injury over nine seasons, with some teams losing as much as $50 million per season [122]. Thus, the screening and consequent reduction of injuries is of paramount importance. Longitudinal assessment and continuous, remote monitoring of athletes through wearable technology may lead to improved compliance and performance outcomes [18,19], yet it may negatively affect relationships between athletes and practitioners [123].

Wearables facilitate a standardised approach to gathering data in any environment, often under the guidance of an experienced practitioner and usually in conjunction with additional measures in specialist settings [67]. Wearable technology is not designed to replace the human element and practitioner expertise; rather, the technology should provide additional understanding and amplify the voice of practitioners. A balance between technology and the human element is necessary, but there needs to be a strategy that involves all users in the integration of wearable technology into practice and subsequently progresses wearables into the mainstream of sports medicine.

Future directions

Wearable technologies have progressed significantly since their conception, and their potential is growing constantly with technological developments. The opportunities for sports science presented by wearable technology are multifaceted, from improving athlete health to the remote monitoring of key performance outcomes. Despite this, numerous opportunities exist to improve and expand the capabilities and applications of wearable technology. Assessment and research of factors such as, but not limited to, validity, ethical issues, accessibility, privacy, cost, compatibility, acceptability, data interpretation, standardisation, and scientific peer-reviewed evidence are critical to progressing wearable technology [11,124].

Vast amounts of data can be collected using wearable technology, sometimes in real time. Analysing and interpreting such large data sets can pose a significant challenge [125]. These data sets may require the use of machine learning techniques and data mining methods to manage and report wearable technology data in sport. Multidisciplinary competencies and collaboration (i.e., sports scientists, computer scientists, and engineers) will enable the successful integration of wearable technology into the sporting environment and help overcome challenges associated with big data and/or the correct use of technology. Additional challenges such as hardware (e.g., power consumption, connectivity) [67], software [126], ergonomics (e.g., attachment, placement, size) [127,128], and network challenges (e.g., data security) [129,130] would also benefit from such

collaborations. With regard to research, studies should provide a concise, logical flow of information regarding the theoretical underpinning of using wearable technology, the causal chain, and the practical implications arising from the observed data [131].

Area-based initiatives significantly impact the development of wearable technology for sports through funding, regulation, industry collaboration, innovation, standards, data privacy, market access, academic research, professional sports, and technology competitions. As areas continue to emphasise technology and innovation, it is likely to have a significant impact on the future of sustainable wearable technology in sports. For example, the European Union adopted the new Batteries Regulation, aiming to make batteries sustainable throughout their entire life cycle [132].

Conclusion

The increasing accessibility and simplicity of longitudinal assessment of athletes allow for new avenues of sports medicine and performance to be explored. Practitioners and athletes can now gain more insight into various aspects of performance and will either be able to receive immediate feedback during their activity or reflect on it retrospectively. Consideration of the objectives, the athlete, the sporting context, and the technology is necessary when incorporating wearable technology into practice. Furthermore, the integrative possibilities of subjective monitoring and practitioner expertise must not be forgotten.

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31
PASSIVE SENSING IN SPORT

Sam Robertson, Nicolas Evans, Johsan Billingham, and Jade Haycraft

Introduction

Consider a professional golfer struggling with their scoring and missing the cut regularly in tournaments. Out of ideas, the player asks their coach to leave no stone unturned in analysing their swing to help restore their performance to its former glory. The coach organises a comprehensive assessment in a specialised golf lab. Fine-grained detail on the angles and speeds of the hip, shoulder, arm, and wrist during the swing is all captured. The forces imparted through the ground during each phase of the movement are recorded at hundreds of samples every second. Muscle activity and sequencing are obtained to ensure that the body is producing force as it should be. The golfer's heart rate is also measured to ensure that the problem, believed to be technical in nature, does not, in fact, have physiological origins.

All the information outlined in this example is being delivered through passive sensing. Not only is access to this type of technology now accessible to the professional golfer, but also in many other sports. It is increasingly being made available at an affordable rate to the recreational athlete.

Perhaps unsurprisingly then, passive sensing represents the most common method of measuring human and equipment movement in sport. Defined as devices that detect inputs from the physical environment [1,2], passive sensors are often used in the laboratory or training setting; however, many are also now permitted for use in sports competition environments. While the direct impact appears much clearer in individual sports, such as the example above, it is also common in complex team sports (such as all the football codes), where such technology is used to quantify, assess, and prescribe individual drills or particular components of the game. Common examples of sport technology are provided in Figure 31.1 [3–7].

Passive sensors are often used in place of active sensors, although they may also be integrated for various functions, such as in smart devices. In contrast to passive sensors, active sensors use their own energy source to send a signal or wavelength which is reflected off a target, and data is gathered upon receipt of the return information. Examples of active sensors include radar, ultrasound, and Light Detection and Ranging (LiDAR).

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Heart Rate Monitors

Usually worn as a chest strap or on the wrist by an athlete to measure heart rate during physical activity. This information can be used to monitor and prescribe the intensity of training. Newer iterations are being woven into sports apparel and clothing [3].

3-D Motion Capture Systems

Typically utilise the tracking of retro-reflective markers affixed to landmarks on the athlete's body by a multi-camera system. This affords the measurement of kinematic features of human movement such as joint angles, velocities and accelerations. Historically utilised in relatively controlled, indoor research settings, recent hardware improvements have seen such systems also move to outdoor settings capable of capturing larger spaces in different light conditions [4].

Pressure Sensors

Used to measure forces, the distribution of an athlete's weight and balance-related metrics. These sensors are often inserted into shoes or playing surfaces, however, also feature in equipment such as sports racquets or bicycle handlebars [5].

Electromyography (EMG)

Measures the electrical activity in muscles. These devices, often multi-sensor in nature, can facilitate insights into the sequencing of muscle activation during various movements, although recording deeper muscle activity remains a challenge in the field without encumbering the athlete through tethering or sub-dermal sensors.

Global Positioning Systems (GPS)

Can measure the position of an athlete's centre of mass based on aggregated signals received from a network of satellites orbiting the Earth. Whilst most commercial systems report derivative measures of displacement to the end user, many also now provide recordings of location based on latitude and longitude, which are commonly converted to local positional coordinates.

Video Cameras

Used to capture an array of information relating to sporting activities and are used by most practitioners. Video functions by capturing light from the environment which reflects off objects and enters the camera's lens. This incoming light is then converted into the visual representation known as video. Computer vision represents one of the most rapidly growing areas of passive sensing, due to its ability to rapidly take unstructured video and convert it into meaningful, actionable data, such as officiating calls or live tactical positions of athletes on a field or court.

Accelerometers

Used to measure the acceleration of an athlete or body segment and thus providing data around movement speed, and impact forces. They are often used in combination with positional sensors to identify tackles or head contacts in many contact sports by embedding them into equipment such as mouth-guards or helmets [6, 7].

Gyroscopes

Often integrated with accelerometers, serve to measure orientation and rotation of various body segments or the entire body itself. They are used in sports applications to track multi-axis movements and understanding player orientation in team sports.

Temperature Sensors

Monitors and provides feedback during physical activity. They may also help to provide insights into the environmental conditions of training and competition along with heat dissipation of the athlete.

Figure 31.1 Examples of common sports technologies used in industry. Information from references [3–7].

Explaining the popularity of passive sensing in sport

Much of the uptake in passive sensing in sport is due to the advantages such technologies bring above and beyond existing methods of capturing human movement and physiology. Often, existing methods rely on human involvement and predominantly require manual processing to obtain data. In other instances, the existing methods rely on antiquated technologies and warrant upgrading. When considering many processes and challenges in the sporting workplace, now more than ever before the user faces a decision about whether to justify a financial investment in passive sensing to replace existing methods (Figure 31.2). When considered with respect to a specific question or process, this exercise can be termed an 'Information Stocktake' and consists of four main questions:

- **New:** The user wishes to collect new information that, until this point has been previously unmeasurable. For example, although of intuitive interest to coaches historically, muscle activity was not possible based on human visual observation. The development and application of EMG allowed for the actual measurement of this activity, thus creating new lines of scientific inquiry and dispelling myths about how humans generate force in physical activity.
- **Enhanced:** The method provides information to the user in a format of greater quality or resolution than what is currently being used for a given purpose. For example, three-dimensional (3D) motion capture systems allow an athlete's technique to be analysed in higher resolution in three dimensions, which was previously only assessable in two dimensions (2D) using video. Enhancements may emerge not only in terms of accuracy. Passive sensing replacements may also be chosen with respect to their improved reliability, usability, or accessibility, being usable with individuals and in environments where it was not previously possible.
- **Speed:** The technology provides this information to the user in a faster or more efficient manner than what is currently being used. For instance, global positioning system (GPS) and global navigation satellite system (GNSS) tracking provide a more accurate method of tracking the distances run by an athlete without resorting to a tape measure, trundle wheel, or human estimation.

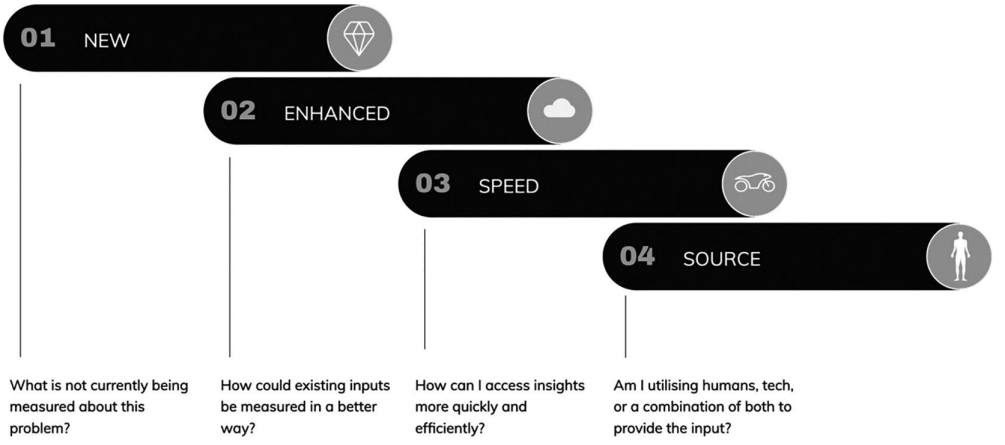

Figure 31.2 The four questions that make up the Information Stocktake.

- **Source:** The technology's ability to detect and input information may automate current processes, thus making it less reliant on human inputs and resources. For example, utilising computer vision to automate event detection, such as kicks, headers, and tackles in football matches, can reduce or remove the time-cost of manually digitising events from match footage [8].

Applications of passive sensing to sport

Passive sensing provides practical value to sport in several different ways. The direct output can be interpreted by a practitioner to intervene with the athlete, for instance, in their technique, movement, or even lifestyle behaviours. Some systems or devices are integrated to create an ensemble output, representing a higher-order response from an athlete, team, or system. An example could be the integration of various physiological functions in an athlete to represent the construct of 'internal load' [9]. In another case, one system may measure a human response, whilst another is used as feedback for the athlete or coach. In such cases, the system does more than simply capture data; it also facilitates the learning process itself. In other cases, the data is not acted on directly at all; it simply forms another part of a broader model.

The applications of passive sensing to sport are seemingly endless (Figure 31.3). Some notable examples from the literature and practice are provided below, categorised by their benefit to different stakeholders.

Athlete performance support

Several reviews have summarised and evaluated the passive sensing literature with respect to performance support. See reference [10] for a review of inertial, force, and EMG sensors, reference [11] for video-based methods, and reference [12] for athlete tracking. Other prominent examples include:

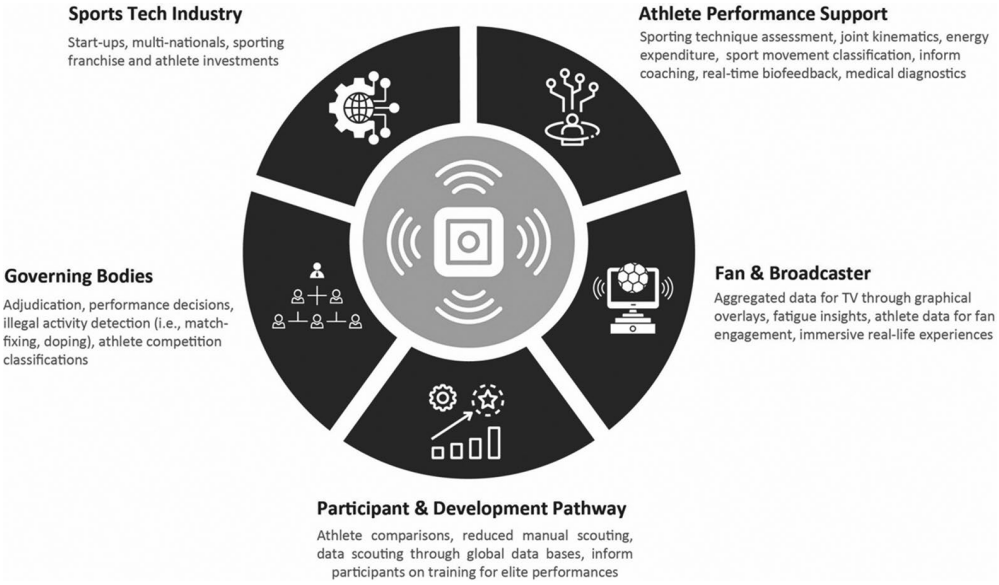

Figure 31.3 Key stakeholders in passive sensing applications to sport.

- Measurement of key features of a sporting technique, such as sensors measuring upper body kinematics [13]
- Predicting a higher-order state (when combined with analytical approaches)
    - Estimating energy expenditure from a GPS tracker [14] or inertial sensor [15]
    - Estimating knee joint forces through leg-mounted inertial measurement units (IMU's) [16]
- Classification of types of training or sporting movements to assist with activity measurement and training prescription [17]
- Relationships between various measures to inform coaching. For example, associations between accelerometer output and running gait kinematics may lead to information that causes a change in technique
- Learning through real-time sensor integration and biofeedback [18]
- Medical diagnostics, such as the detection of critical head impact events with accelerometers [19]

Access to passive sensing outside of the controlled training environment, to supplement current training, is a major benefit as it allows for a more complete picture of the training being undertaken. Another benefit of passive sensing technology is its capacity to monitor physiological and movement measures that can mitigate injury risk or inform the type and intensity of return-to-play training post-injury.

The fan and broadcaster

Sensing methods that allow for the fan or broadcaster to access data from an athlete, particularly if done so in near-real time, can hold real commercial value for the tech provider along with the athletes themselves. Examples include:

- Providing aggregated data for TV or other digital media. For instance, distances covered by players obtained from tracking systems are a popular metric in many football codes. Computer vision tracking combined with graphical overlays is also a visually appealing method for fan consumption and has the advantage of being provided in near real-time. Examples include tracking boat position in sailing or a hockey player's path to shooting the puck into the goal;
- Certain sensors may reveal insights into fatigue in sports such as horse racing, for example, the horse's heart rate or a change in its gait patterns [20]. The lucrative wagering industry is likely to continue to push for access to such insights if they are believed to be valuable information that may enhance the likelihood of success and the overall fan experience;
- There continues to be an appetite amongst fans to gain more intimate details of their favourite athletes. These athletes may wish to sell their data to these fans or third-party organisations as an alternate revenue stream;
- Immersive real-life experiences, such as a treadmill challenge where a fan attempts to maintain Eliud Kipchoge's average marathon running speed for a single minute [21]. Virtual reality and augmented reality will continue to integrate with existing methodology to enhance the sophistication of such experiences.
- E-games and animations, whereby passive sensor output is used to model players for use in the gaming industry, allow fans to mimic their favourite players (e.g., football, basketball, skateboarding). Biofeedback, particularly through vibration in game controllers, features heavily in

such games. New frontiers in exergaming may also emerge, which allow the user to perform sport-specific movements completed by their favourite sporting stars at a level of intensity they are comfortable with.

The participant and development pathway

Data derived from passive sensing now forms part of many talent testing programmes. Utilising such data collected from the same or similar technologies permits interchangeability in the comparison of results between athletes and can be used in large-scale predictive talent models. It also helps to overcome potential issues in reliability between different scouts or talent assessors and can reduce a scout's need for manual means of athlete assessments. The flow-on effect can often mean financial savings due to less travel for scouting individuals, as they no longer need to view athlete performance live around the world, instead having access to global databases of talented athlete data that can inform their selection decisions.

Such information can also serve to inform prescriptive models for development athletes to follow. If the performance or training programmes of elite performers can be profiled, then this can be aspirational to the younger participant and used to inform their own performance pathway.

The sports tech industry

The number of companies now working in sport is staggering. These range from tiny start-ups to huge multi-nationals. The global sports tech industry has grown to a $21.9b industry in 2022, with compound annual growth rate projections of 14% indicating a value of almost $42b USD in 2027 [22]. To help shape the type of passive sensing options available, many sporting franchises and even athletes themselves have acquired or invested in sports tech companies.

Governing bodies

With more sports competitions taking place more regularly all over the globe, governing bodies face an ongoing challenge in areas such as adjudication and the prevention of illegal activities such as doping and match-fixing. Technology can help and may even be reaching the point of being critical to keep up with the increasing complexity and speed of the possibilities (good and bad) these new innovations bring.

Video has long played a role in assisting with measuring timing (i.e., athletics), performance assessment (i.e., figure skating), and the determination of various decisions (i.e., ball in or out of play in various team sports). Video continues to grow in both speed and precision, with FIFA's move to a semi-automated offside technology in football tournaments being one such notable example. Here, computer vision has been integrated with a ball sensor to detect both the position of each player and their limbs at the instance when the ball leaves the passing player's boot. In such difficult human adjudication decisions, such integrated solutions will likely continue to emerge not only in football but in other sports that face similar quandaries. Their introduction, however, requires understanding and mastery of the technology as well as regular management and maintenance of the systems.

Although confirmation of doping often requires rigorous laboratory testing, passive sensing still plays a role in flagging potentially anomalous performances of athletes. For example, if an athlete's movement intensity or running patterns in a team sport appear unusual, this may be used as one of many early warning systems to indicate illegal substance use or potential intentional

match-fixing. In some cases, passive sensing data may be used as evidence to form part of a broader case against an individual. It is also plausible that such data may be used to inform decisions in emerging areas for governing bodies, such as the assignment of athletes into gender categories or para-athlete classifications for competition purposes.

Future challenges and opportunities

Passive sensing will undoubtedly continue to grow. Commercial opportunities combined with industry competition will drive smaller, cheaper, better, and faster solutions. Some of the future key challenges and opportunities likely to present as part of this commercial market growth are discussed below.

Improving technology quality in a saturated commercial market

The importance of establishing a sensor's validity and reliability is well entrenched in the literature and in the field; nonetheless, the push for higher quality products continues to grow. The drivers for this increased emphasis are multi-faceted but are largely driven by competition. Such competition is not only among teams and athletes to use technology to further improvement, but also among the technology providers themselves due to the potential commercial gains that await through product differentiation. As the number and variety of commercial offerings likely expand, companies will need to increase their value proposition to sport. The increased market competition means end-users may be inundated with choice and may find it difficult to differentiate between products.

To assist with this predicament, applied frameworks, such as the Sports Tech Quality Framework developed by the not-for-profit Sports Tech Research Network [23], will be crucial in helping to guide decision-making (Figure 31.4).

This framework lists 23 features under 5 'Pillars', which serve to define the quality of a sports technology product or device. These pillars include Quality Assurance and Measurement, Established Benefit, Ethics and Security, User Experience, and Data Management. Although not all the

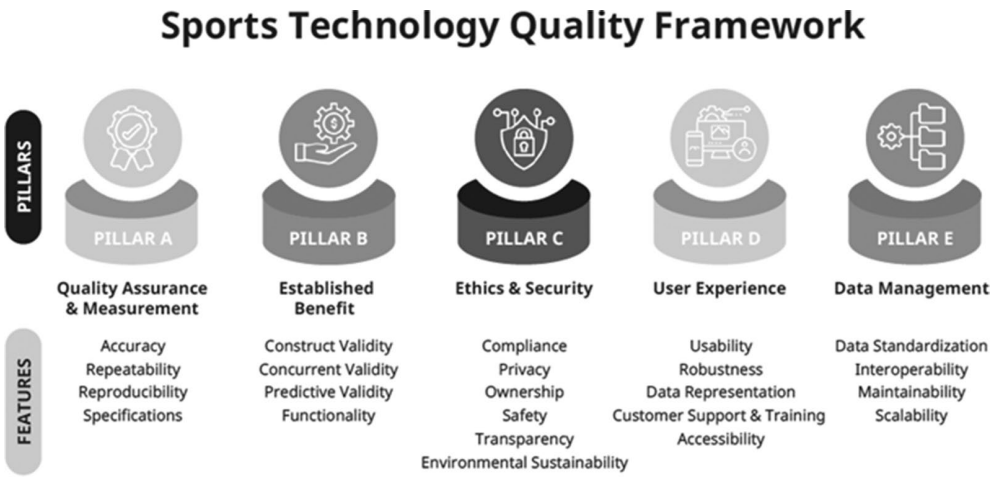

Figure 31.4 The Sports Technology Quality Framework.

23 features are applicable to all passive sensing, the framework does provide a non-prescriptive tool intended to guide decision-making. If the framework becomes ingrained in sporting practice, its value should increase as end-users utilise it to define benchmarks between different tech offerings, guide their own tech design, inform the awarding of lucrative league-wide contracts, advise on purchasing decisions, and implement guidelines to protect the safety and privacy of their athletes. It should also help to hold companies to account for the quality of the product they offer; after all, they benefit heavily from the consumer [24]!

Governance

Unsurprisingly, with such growth, there is a potential for disingenuous salesmen to thrive and an increased availability of unsafe products or devices showing limited to no efficacy. Consequently, the growth in tech must be accompanied by an increase in attention to governance and carefully sought approval, validation, and accreditation in respective sports.

A key consideration around any sports technology is accessibility [25,26]. Inequitable access to a specific technology is typically deemed an unfair advantage, although presently this is loosely governed by most sports at best. Some governing bodies of globally participated sports are taking steps to address inequities through technology democratisation, aiming to ensure that athletes in poorer nations have the same access as those in wealthy ones. However, companies will also face the prospect of needing to provide standardised and potentially scaled-down versions of their products as governing bodies seek to create this technology democratisation environment. Standardisation will help to move the science of sport forward through the ability of data pooling to inform large-scale models, as well as being of use to inform athlete career movement, such as the transfer of a footballer from one league to another. However, governance of technology has the potential to stifle sports tech innovation through the onerous requirements and time demands it presents to founders. Consequently, a balance must be maintained.

Other large-scale challenges exist at the philosophical and ethical levels. In most sports, the ownership of passive sensing data is not well established. As these fundamental personal and legal implications are addressed, there is a risk of further stifling innovation through governance if decisions or agreements are not concluded in a timely fashion. Flow-on effects from this also linger for practical reasons: Who can access an athlete's data? Can they choose for it to remain private? Who can use it for commercial gain? On the latter question, previous situations have arisen whereby the athlete, sensor provider, franchise, governing body, and television broadcaster all claimed ownership of the data. They cannot all be correct!

Individualisation, customisation, and integration

As the current market becomes saturated, smart companies see opportunities in niche areas. Female athletes, underrepresented minorities, and children all provide angles for customisation of existing products. This customisation is important, as it also facilitates the ability for individualisation, which, although time-consuming and expensive for companies, is likely to lead to ongoing adoption.

Customisation and integration also help avoid excessive homogeneity of offerings in certain passive sensing types as has been seen in other technology types such as smartphones, where a relatively small selection of commercial providers dominate the market. Ideally, this customisation also extends to integration, although this likely only emerges when it makes commercial sense to do so. Although many passive sensors have the ability for integration [27], putting this ability in

the hands of the skilled end-user may result in truly innovative results, as they find new applications and unlock new performance frontiers.

Sustainability

Almost no company, sporting-focused or not, can ignore sustainability in 2023. As the race to be the best heats up, it will be tempting for companies to regularly update hardware. Finding an appropriate balance between product evolution, profitability, and environmentally sustainable is a real consideration for many companies, but the responsibility also lies with the end-user to choose their passive sensor providers carefully. This, of course, is contingent on the end-user being informed, with initiatives such as independently audited eco-labels being a potential solution. Sustainability in terms of robustness should also not be ignored. Impacts, laundering durability, and hardware update capacities are just a few examples where gains can be made to increase the longevity of a product. Ensuring battery sustainability across the product lifespan has also received recent attention in strategies and laws passed by the European Union [28]. Readers seeking further information on sustainability considerations are referred to Chapter 17 'Sustainable design and manufacturing of sport equipment'.

Emerging opportunities

It is likely that passive sensing will continue to grow in terms of the varied number of offerings. It is also likely that enhanced and more efficient versions of existing types will emerge. Some of the most prominent examples include:

- **Biometric sensors,** which will likely be embedded onto the skin or into everyday sportswear. Although many examples already exist in this area, the accuracy and reliability in measuring heart rate, hydration, and muscle activity will likely improve, along with modifications such as smaller sizes and lighter masses. New approaches will aim to unlock new frontiers, such as the psychological state of users. The question of data ownership will become even more critical in this context.
- **Smart equipment and apparel,** which will likely incorporate the abovementioned biometric sensors. This will include garments, shoes, and protective equipment. Helmets will measure impact and assist with the diagnosis of concussive episodes, have the ability to adapt settings in real-time, and incorporate active sensing to generate self-cooling and feedback.
- **Augmented and virtual reality (AR & VR),** mixed evidence exists for their benefit in terms of learning and transferability to better sporting performance [29]. Despite this, their point of difference from traditional training and, in general, their high user enjoyment means their use will likely grow regardless of their mixed performance results. Such technology will likely revolutionise fan experiences by allowing them to go 'inside' the locker room or experience a match on a whole new level, either live or from a remote location.
- **Artificial intelligence (AI),** whilst not an example of passive sensing per se, will nonetheless be a crucial element in leveraging its potential, particularly in large-scale, complex device, and system integrations where the amounts and types of data surpass the capacity of humans to process them in reasonable timeframes.
- **Markerless motion capture** should provide more accurate biomechanical analyses and greater detail. The true advantage of such systems will be their ability to capture 2D and 3D information in actual sporting competition. This new frontier will permit technique matching along with the analysis of training transfer from movement interventions into actual competition.

- **Cognitive training:** A range of technologies is already being popularised around the development of cognitive skills [30]. Although evidence is presently is mixed at best [31], in the future, they may help to improve focus, concentration, and decision-making under pressure. Alternatively, these technologies may allow the continuation of cognitive skills training in times when an athlete may not be able to participate physically due to injury.

Conclusion

Many opportunities for passive sensing exist across the sporting landscape. Moving forward, new technologies will emerge, better iterations will be developed, and richer data will become available to the end-user. However, shaping the direction of passive sensing is an important consideration. To access the best technologies, the consumer must demand them. Undoubtedly, to ensure the athlete is protected, regulations and restrictions will be needed. Although it is tempting to leave such matters to governing bodies, it is everyone's responsibility. Working together collectively to help drive the direction of passive sensing has the potential to lead to athletic performances and experiences that nobody thought possible.

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32

INERTIAL SENSING IN SPORTS

Application and technical considerations

Andy Stamm, Conor Wall, and Yunus Celik

Introduction

Investigating the performance of professional athletes to record training progress (or any other form of athletic intervention) is of interest not only to coaches and athletes but also to researchers and amateurs. Investigating performance can be done using different technologies, such as (multi) video cameras with integrated motion tracking systems and wearable inertial sensors. In addition to investigating movements and impacts, there are also devices capable of recording heart rate, blood pressure, oxygen saturation, and other physiological outcomes. Here, we focus on inertial sensing technology, but readers are directed elsewhere for a discussion on other devices [1,2].

Inertial Measurement Units (IMU's) can be attached to athletes in numerous ways. Of critical importance is the selection of the point on the body where the IMU will be placed. This depends on the kind of sports being investigated, as well as finding areas for the IMU that do not restrict or hinder the movement of the athlete. It also relies on the specific movement being recorded since the features of acceleration and angular velocity vary based on the location of the IMU [3]. Where the IMU is attached can also have important effects on the algorithms used, such as setting thresholds [4]. Fixing the IMU can be performed in many ways, such as using medical tape, belts, self-sticking IMUs, or suits with IMUs included. While these attachment methods offer some accuracy in the signals obtained, they may introduce issues such as relative movement between the IMU and the underlying body, caused by soft tissue artifacts or displacement of fixation clothes or straps [5]. Hence, attaching an IMU while considering the location of soft tissues may offer a more stable and dependable signal acquisition [6].

Considering the target parameter, one IMU may be sufficient to investigate the movement or impact. For example, one IMU attached to the sacrum has been used for automatic recognition of swimming activities and assessment of lap times [7]. Similarly, employing multiple IMUs enables the exploration of intricate movements, such as coordination between two body segments, or facilitates the capture of more comprehensive biometric data. For instance, a study utilized six IMUs to create a novel macro-micro approach for analyzing swimming movements [8]. Specifically, the use of IMUs in the referenced study detected swimming bouts, laps, and swimming technique at the macro level and swimming phases at the micro level. Using multiple IMUs, however, will make attachment to the athlete more difficult. It may also lead them to feel uncomfortable

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or distracted during training and may lead to delayed analysis because processing of the data is typically done after the training session. For the purposes of this chapter, swimming will primarily be used as the sport of choice, with IMU application in that context (here extracting velocity). Throughout the chapter, a pragmatic approach is adopted to take the reader through a step-by-step process of using an IMU.

Wearable inertial measurement units (IMUs)

When addressing movements, it is important to mention the different kinds an athlete can experience, such as, but not limited to, those related to acceleration (in all different directions), orientation, rotation, force, and pressure. To measure these kinds of movements and impacts, different sensors are required. To measure acceleration, an accelerometer is needed; to measure rotation or angular velocity a gyroscope is needed; and to measure orientation in relation to the Earth's reference frame, a magnetometer is needed (Figure 32.1).

The acceleration component $a_i(t)$ at time t along the ith axis ($i = x, y$ and z) is given by the equation:

$$a_i(t) = g\cos(\theta(t)) + \underline{i}.\underline{a}_T(t) + \omega(t)^2\,\underline{i}.\underline{r}(t) + \alpha(t)i.r(t) \tag{32.1}$$

where g is the gravitational acceleration, $\theta(t)$ is the time-dependent angle between the axis and the vertical direction, $\underline{i}$ is the unit vector, $\underline{a}_T(t)$ is the translational acceleration, $\omega(t)$ is the angular velocity of rotation, and α(t) is the angular acceleration, both at a distance $\underline{r}(t)$ from the center of rotation.

Previously, those sensors were only available in different packages (accelerometer, gyroscope, magnetometer) and had to be combined on a printed circuit board (PCB) and individually connected to a microcontroller to record movements. Nowadays, all three of these sensors are combined in one Micro Electrical Mechanical System (MEMS) with a miniature form factor that can be included in various apparel. Typically, industry speaks about a 9 Degree of Freedom (9 DOF) MEMS (i.e., 3 degrees per sensor × 3 sensors). These MEMS sometimes already have implemented calibration procedures, which means that the manufacturer has pre-calibrated the sensors, and they can be directly used for recording movements.

Data capture

Recording measurements requires memory to store data (if recorded on the device) or a radio link to transmit the data. Memory can be included on the wearable, which could be an Integrated Circuit (IC) or a Secure Digital (SD) card. While an IC can be more robust, an SD card usually offers more memory and flexibility for less money. Modern SD cards can be connected to a microcontroller as they communicate via Serial Peripheral Interface (SPI), a digital communication protocol used in many electronics and embedded systems. Using these SD cards in IMUs, however, needs careful attention to their physical connection to the electronics. In most modern devices, an integrated SD card slot may not provide a stable connection if the IMU is used in a sport involving high accelerations (e.g., rugby, snowboarding, ski jumping). The high accelerations may cause the SD card to disconnect from the pins of the SD card socket. Soldering the SD card to the electronics would remove that potential failure. Alternatively, edge computing technology has been gaining traction in the field due to several advantages over traditional post-processing. Edge computing

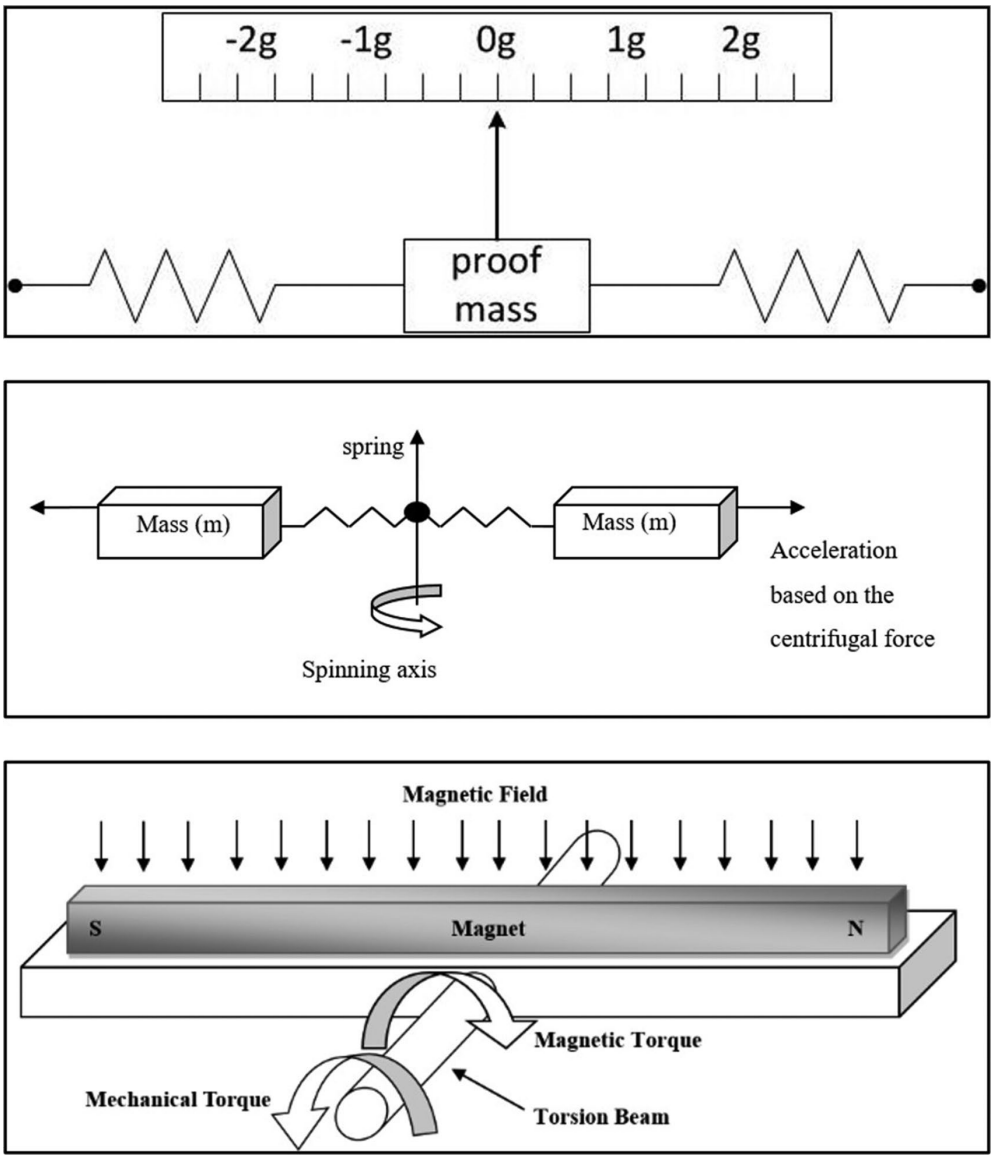

Figure 32.1 Fundamental operation of an (a) accelerometer, (b) gyroscope involving movement of a proof mass attached to springs, and (c) MEMS-based magnetometer.

refers to a decentralized computing paradigm where data processing is performed closer to the sensor technology, rather than saving the data on an SD card for post-processing [9]. This technology facilitates real-time motion analysis without the need for extensive data storage, although it usually comes at a higher financial cost than off-the-shelf wearable IMUs.

We have already elaborated on the devices needed to measure the movements/impacts and how the measured data can be stored. However, the main component in an IMU will be the

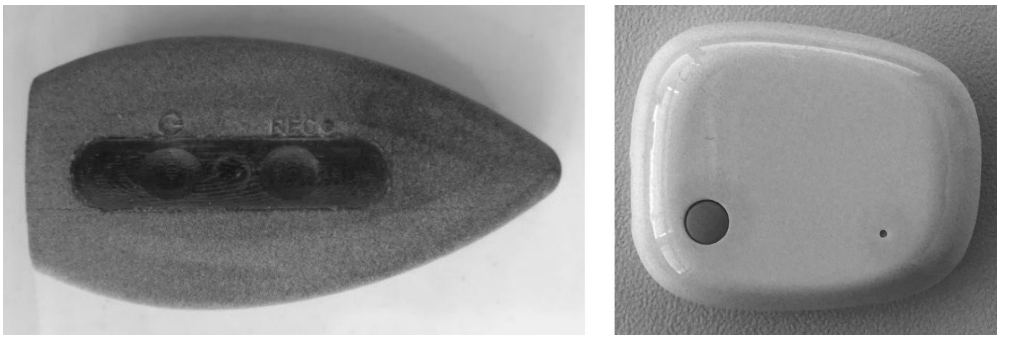

Figure 32.2 Typical IMU-based sensors from MySwimEdge Inc. (Texas, USA, 80 × 41 × 17 mm, left) and MBIENTLAB Inc. (San Jose, USA, 36 × 27 × 10 mm, right).

microcontroller. This microcontroller will connect to the two other main components (memory, MEMS) and will allow the user to extract the measured data. In principle, there are different ways of gaining access to the recorded data: downloading via either a wired or wireless connection. Modern microprocessors, such as those from manufacturers like Atmel (California, USA) or Espressif (Shanghai, China), provide built-in wireless local-area network (WLAN) and Bluetooth connectivity which can be used to transfer the recorded data. Often, microprocessors are powerful enough that preprocessing can be conducted on them.

Every athlete/coach will nowadays have access to a mobile device (e.g., smart phone or tablet) that can be used (with an appropriate Application/App) to connect to the IMU and display the measurements to the athlete in a more streamlined and pragmatic manner. These devices also have multiple connectivity options, including Bluetooth and WLAN. Contemporary software often enables the rapid development of apps (with sufficient domain knowledge) to generate a new level of user experience.

For powering IMUs, a well-established solution for any portable electronic device is the use of a Li-Ion battery in conjunction with inductive charging. This enables manufacturers to make the IMUs waterproof, as the device itself will need no physical connectivity from the inside to the outside. This brings additional benefits for IMUs used in sports where there is a risk of contact with water (e.g., running in the rain or snow sports) or even a must (e.g., swimming). Figure 32.2 shows a typical IMU from MySwimEdge Inc. (Texas, USA) used in swimming analysis. This sensor can provide velocities, distances, and timing parameters to athletes/coaches for each arm separately and therefore also allows symmetry investigation. Another IMU from MBIENTLAB Inc. (San Jose, USA) is also included in Figure 32.2 for comparison.

IMU attachment

The art of data collection is of course mainly driven by the investigated movement. It can be as simple as placing one IMU on a person who will have no contact with water to record the movement and can become more complex when multiple waterproof IMUs are required. When multiple IMUs are included, one needs to think about their synchronization as well, so that the analyzed data from all of them can be related to the athlete's movement [10,11].

A person attaching an IMU to an athlete must consider the investigated sport as well as ensuring that it will not hinder the athlete during movements. Some sports do not allow wearing IMUs

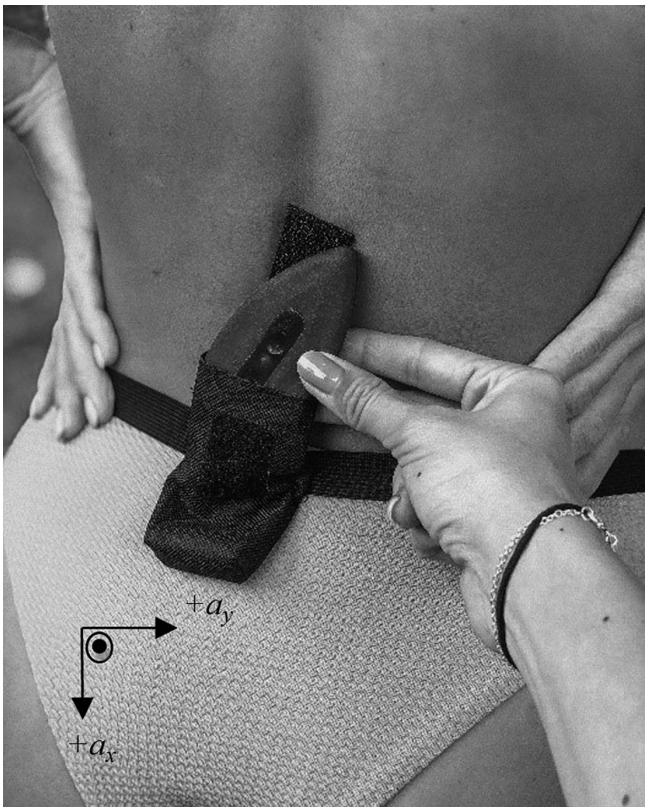

Figure 32.3 Example of a sensor attachment on an athlete.

during competitions which leads to measurements only being possible during training or injury recovery. According to a survey conducted among coaches in racket sports, a notable majority, specifically 96.6%, expressed willingness to permit their athletes to use IMUs during training [12]. It is further important how the IMU is attached to the athlete (see Figure 32.3). Possibilities include taping, belts, Velcro, and suits with embedded IMUs. When attaching an IMU to an athlete the scientist must keep in mind that the IMU is also experiencing some movement, e.g., on the skin in relation to the bones, as already discussed.

As described, an IMU measures the movement of an athlete, usually in three dimensions. Accelerometers, therefore, measure the acceleration in three directions: x, y, and z. Depending on the alignment of the IMU with the athlete or the measured object, a typical attachment is when the vertical direction is aligned with the spine and is represented by a_x, the mediolateral direction by a_y, and the anterior-posterior direction by a_z (see Figure 32.3).

These recorded data, for example shown in Figure 32.4, will then need to be analyzed. This can be done in an analog manner or by automatic analysis, sometimes even cloud-based. While cloud-based analysis usually does not require a scientist for the analysis, manual analysis does and is therefore more time-consuming. Software for analysis is available from various companies. A commonly used analysis tool is MATLAB® (MathWorks, Natick, Massachusetts, United States), and other programs include GNU Octave, Scilab, or Mathematica, to name a few.

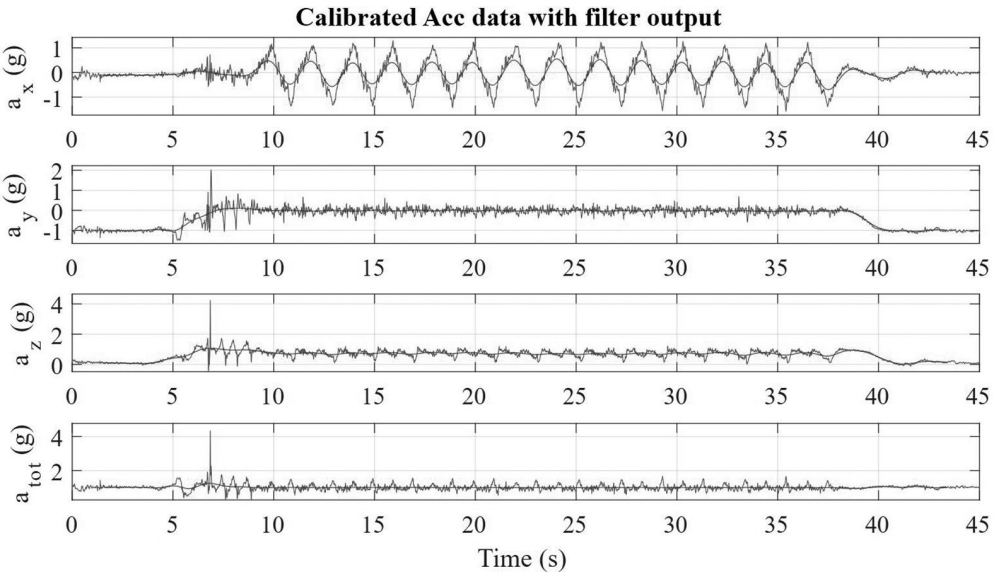

Figure 32.4 Typical acceleration profile of a 50-m freestyle swimming lap swum by an Olympic Gold Medalist (the smooth line shows the Hamming Windowed FIR filter output alongside the unfiltered acceleration data).

Calibration

In general, there are many calibration techniques available [13,14]. Lai et al. [14] use the six-point Newton-Raphson-based calibration technique, which does not require the device to be exactly aligned with gravity. It achieves good results, but the calibration requires six distinct stationary orientations. Another common technique is manual calibration, which requires each axis of the sensor to be aligned with positive and negative gravity to estimate each axis' offset and sensitivity (e.g., ±8g). These offsets and the sensitivity can then be used to calibrate the measured acceleration in relation to gravity, as described by Stamm (Chapter 4.4.3) [15].

Beyond calibration, configuration stands as another crucial consideration preceding data collection employing IMUs. Key parameters in this regard include range (±g or deg/s) and sampling frequency (fs, Hertz, Hz), which necessitate careful selection based on the nature of the activity under consideration. For instance, lower *g* values and degrees per second (deg/s) should be selected for low-intensity movements, whereas higher *g* values (up to 48 g) and degrees per second (up to 2,000) are necessary for high-intensity movements. This selection aims to mitigate the risks of distorting the output signal during the data collection process.

High-pass filtering of the recorded acceleration data removes unwanted interference such as electronic interference. For commercial products, however, this means in most cases that access to raw data is not possible. This needs to be considered for further processing. The sensor placement, close to the center of rotation of a swimmer, for example (lower back), reduces the radius of rotation ($\underline{r} \approx 0$) and therefore minimizes the impact of centripetal and tangential acceleration. Considering this and the effect of the filter, the acceleration component $a_i(t)$ from Equation (32.1) can be rewritten as:

$$a_i(t) \cong \underline{i}.\underline{a}_T(t) \tag{32.2}$$

Data processing

This section focuses on data processing with an accelerometer. The principles mentioned here can also be transferred to measurements with a gyroscope or magnetometer. Data recorded from inertial sensors are usually stored as numerical values between $-2^x/2$ and $+2^x/2$, where x is the value in bits (binary digits) of the sensor data output. This means ±32768 acceleration units for a 16-bit data output. As accelerometers are usually switchable (so that the measurement range suits the use case) between different sensitivities (e.g., 2, 4, and 8 g), these values represent the maximum and minimum values the accelerometer can detect within the selected range.

These numerical values must be converted into gravitational units (g's) with reference to the 1 g acceleration of the Earth's gravity. Calibrations can be conducted in various ways. For example, one is described by Lai et al. [14] and Stamm [15,16]. The acceleration data (see column 2 of Table 32.1) can then be converted to gravitational units (see column 3 of Table 32.1) by applying the aforementioned calibration procedures. This requires subtracting the offset for each axis from each data point of the relevant axis (see Equation 32.3). The result can then be divided by the difference between the positive (+) 1 g and negative (−) 1 g value of each axis, divided by two, which then represents the acceleration in relation to gravity (Equation 32.4).

$$\text{units_dp_off}_x = \text{units_dp}_x - \text{unit_offset}_x \tag{32.3}$$

$$\text{gravity_dp}_x = \frac{\text{units_dp_off}_x}{(\text{sesnitivity_pos_1}g - \text{sensitivity_neg_1}g)/2} \tag{32.4}$$

Figure 32.5 presents the acceleration data in gravitational units (g's). Thus, the converted data can be used for further processing, such as deriving velocity, knowing that 1 *g* represents 9.81 $m \cdot s^{-2}$.

A normal investigation of the gathered acceleration data begins with a visualization (*"eyeballing"*) of the acquired temporal acceleration data. Acceleration signals obtained from accelerometers might contain many other signals, which are not of primary interest for the researcher. These signals could come from imperfectly aligned accelerometers, noise from the accelerometer

Table 32.1 Overview of acceleration units representing the relative gravity units (digits measured by the accelerometer and depending on the resolution of the sensor, e.g. 16 Bit represent a range between +32,768 and −32,768 units for typically ±4g) gained by placing the sensor at rest perfectly aligned to gravity on a flat surface (bench). The sensor is than rotated by 180 degrees on each axis to record positive and negative gravity on each axis applying the methods from Lai et al. [14] and Stamm et al. [16]

Acceleration axis	*Acceleration units*	*Gravitational units*
X	−7973.8	+1 g
	−338.4	0 g
	+8610.5	−1 g
Y	−7906.8	+1 g
	379.3	0 g
	8665.3	−1 g
Z	−8118.2	+1 g
	111.8	0 g
	8341.9	−1 g

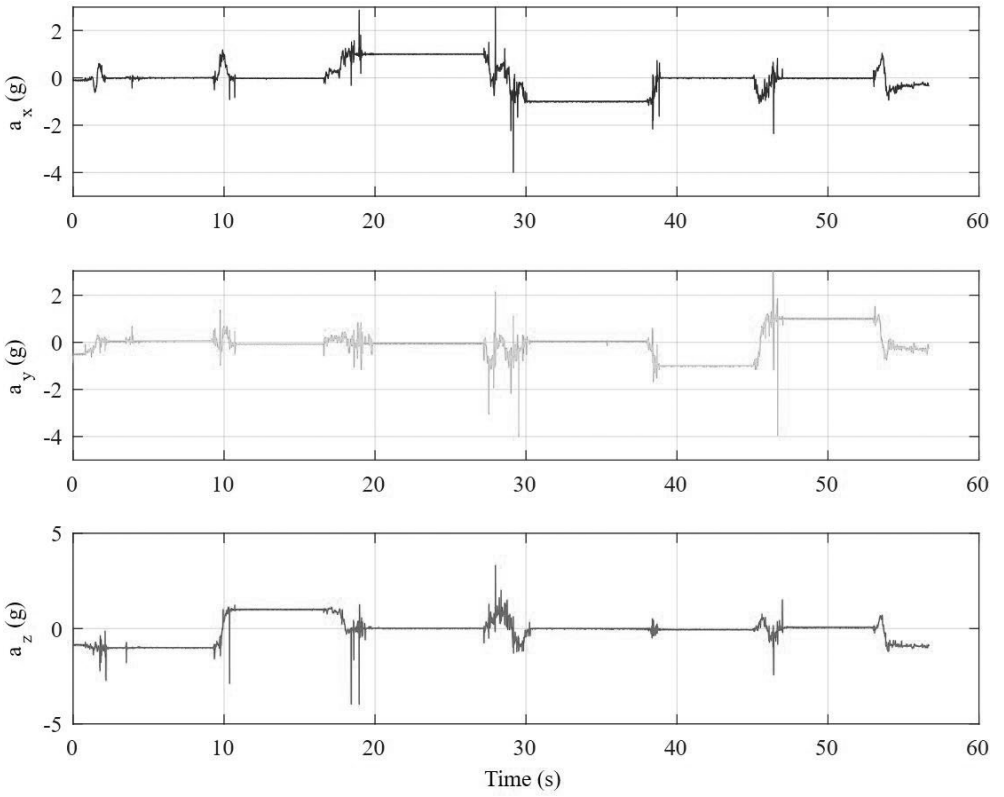

Figure 32.5 Calibration of an MEMS with acceleration shown in gravitational units (g's).

itself, and other noise sources. A way to remove this noise is to apply a filter to the acceleration data. Filtering of inertial sensor data is used for two purposes: (a) to remove the orientation of the sensor and (b) to remove the unwanted signal components from the signal. The ability to remove unwanted signals from acceleration data is determined by the quality and type of the filter, which is dependent on the available processing power. Considering the option of cloud computing, this allows nowadays nearly any filter to be applied to the data with feedback from the cloud within seconds.

Post-processing of data with a desktop computer provides enough processing power and system memory to perform filtering using the best filtering techniques. In comparison, filtering on the inertial sensors is limited by the low processing power and small system memory. Several filtering techniques, such as the Hamming Windowed FIR (Finite Impulse Response) filter, Rectangular Windowed FIR filter, and Moving Average filter, can be used to extract signals from acceleration data and are explained below.

A Hamming Windowed FIR filter, for example, with a length of 500 elements, requires 500 floating-point multiplications and additions per sample and channel. This means, for triaxial acceleration data recorded at 100 Hz, a total of 150,000 floating-point operations per second (100 samples/s × 500 points × 3 channels = 150,000). Other filters, like a Rectangular Windowed FIR filter, need only 600 floating-point operations for the same data.

In sports science, the orientation of an inertial sensor usually refers to its orientation with respect to gravity. Assuming that the sensor is firmly attached to the athlete, this provides an estimation of their orientation according to gravity. However, there might be some differences between the actual orientation of the athlete and that of the sensor due to bone and skin movements. Therefore, the orientation gained from sensors attached to the athlete, even if they are firmly attached, is only an estimation of the athlete's orientation according to gravity. Assuming a good and firm fit of the sensor to the athlete and the investigation of athletes with slow orientation changes such as swimming, the sensor will allow an estimation of the athlete's approximate orientation. Scientists, however, should always compare results against a "gold standard" reference system. This could be a video or tethered system in swimming.

There are several orientation types available which include *Reference Orientation*, *Stationary Orientation*, and *Average Orientation*. In some cases, it might be necessary to use one of them alone or in combination to extract the orientation of interest. Triaxial accelerometers are usually mounted on the athlete in such a way that the human body frame is aligned with the accelerometer axes. This means, for example, that the accelerometer is attached to the swimmer so that the x-axis measures their mediolateral acceleration (represents the body roll), the y-axis their forward acceleration (direction of movement), and the z-axis their anterior-posterior acceleration (up and down movement) (see Figure 32.6). The orientation can be read straight from the calibrated accelerometer recording or extracted using a low-pass filter, as already explained. If the athlete is standing still, the magnitude of the forward acceleration vector of the orientation signal will show about −1 g.

Figure 32.7 provides an example of how sensor orientation can be extracted. The accelerometer shows, in this case, 0.162 g on the mediolateral axis, 0.3 g on the forward axis, and 0.94 g on the anterior-posterior axis. The total acceleration will be 1 g, calculated from the trigonometric sum of

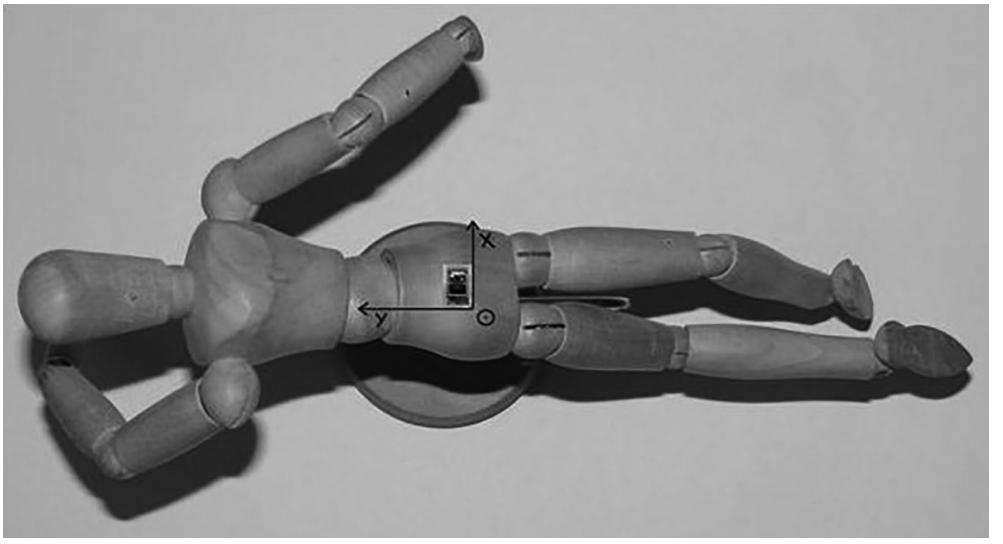

Figure 32.6 Alignment of accelerometer on swimmer used for sensor calibration when stationary.

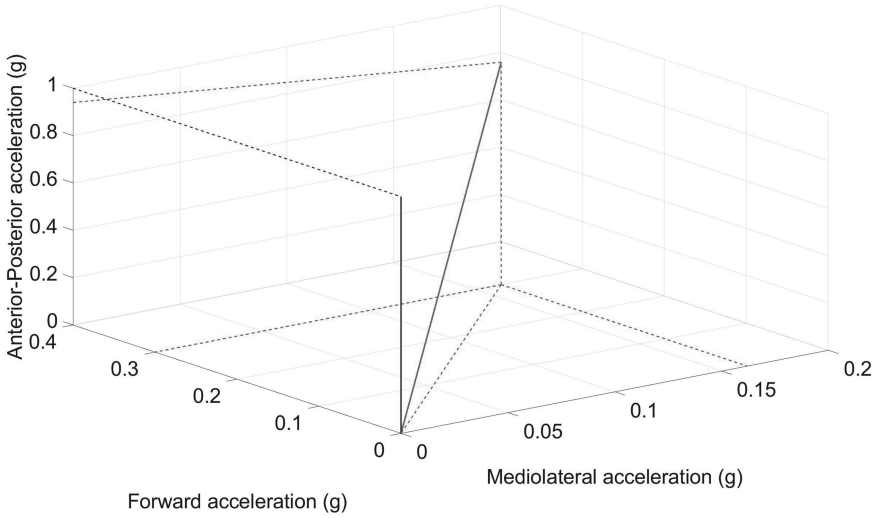

Figure 32.7 Example of sensor orientation extraction. The accelerometer shows 0.162g on the mediolateral, 0.3g on the forward, and 0.94g on the anterior-posterior axis. The trigonometric sum of these values will be the total acceleration and become 1g (vertical solid line).

the three single values. The sensor is therefore 19.93 degree tilted out of the anterior-posterior axis and 28.37 degree out of the forward axis.

- Some sports might require a *Reference Orientation*. This orientation can usually be calculated by performing a calibration of the athlete/sensor combination asking the athlete to perform a couple of stationary positions. For a swimmer, this could be standing upright and still to calibrate the forward direction.
- The *Stationary Orientation* can be calculated using trigonometric analysis of the sensor during periods when the athlete is still. The total dynamic acceleration in these periods will be zero or close to zero, and the total static acceleration is 1 g. This orientation might become a reference orientation if needed.
- The *Average Orientation* can be estimated from the sensor signal during athlete activities. Average orientation will change with the athlete's orientation over time and is therefore representing their position in relation to gravity. The average orientation can be estimated using low-pass filters of appropriate frequencies. The filter frequency is usually determined by the athlete's dominant activity.

For a swimmer with a stroke rate (SR) of about 1 Hz, the sufficient frequency for signal extraction should be at a maximum of 0.5 Hz to correctly determine the orientation component. An increase in the filter frequency causes more "leakage" of the activity signal into the orientation signal, while reducing the filter frequency results in a signal that does not contain fast orientation changes (this is a qualitative assessment). As the swimmer's orientation change in relation to gravity occurs at the lower frequencies, below half of the stroke frequency (e.g., sensor orientation in the body roll signal), the average orientation can be estimated using a low-pass filter with a cut-off frequency of 0.5 Hz. The filtered acceleration data then contains every acceleration component with frequencies below 0.5 Hz (independent of which axis they occur), which includes the body roll component, up-and-down movement of the swimmer, and gravity.

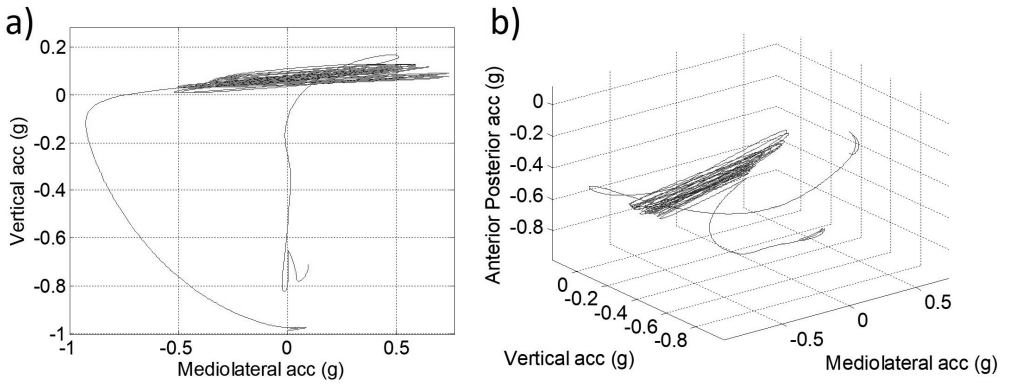

Figure 32.8 Two-dimensional (a) and three-dimensional (b) view of the orientation vector of a freestyle swimming lap.

Figure 32.4 presents the acceleration profile of a low-effort freestyle swimming lap and the extracted orientation (smooth line) using a 0.5 Hz low-pass filter. Figure 32.8 (a) shows a rotated view of the three-dimensional plot of the orientation signal presented in (b) for a freestyle swimming lap. The rotated view presents the two axes of most interest, the forward and mediolateral.

With the sensor firmly attached, the orientation of the athlete with respect to gravity can be determined using trigonometry. The same method can be used to extract angles between the detected orientation and gravity. Orientation can be extracted from static and dynamic data using low-pass filters. This derived orientation may be a sufficient estimation of angles for many sports.

Velocity extraction

One important parameter of interest to researchers and athletes/users is velocity, as it is a good measure of training efficiency and progress. Velocity can be derived by taking the integral of acceleration, which can be further integrated to find displacement.

Inertial sensors can detect acceleration and rotation in three axes if the sensor contains a triaxial accelerometer and gyroscope. The measured acceleration contains different components, such as gravity, sensor orientation, and the actual athlete movement. The athlete's movement can be an oscillatory action, like the body roll in swimming. As the swimmer is constantly turning and twisting, the acceleration signal will be spread across all three channels, making it complex to extract the desired signals. Integrating acceleration data to get velocity is therefore complex and will contain errors. The velocity at a particular time, *v(t)*, can be calculated using an approximation (trapezoidal rule) for the numerical integration of the acceleration *a* (discrete data) using:

$$v(t) = v(t-1) + \Delta t \frac{a(t) + a(t+1)}{2} \tag{32.5}$$

where $v(0) = 0$, $a(t)$, and $a(t + 1)$ are adjacent acceleration values and Δt the time between two samples, which is represented as 1/sampling rate of the sensor.

Figure 32.9 presents the measured accelerations of a resting acceleration sensor in gravitational units (g's) (left) and the low-pass filtered calibrated acceleration data (right). The subtracted result is shown in Figure 32.10.

Using the filtered total acceleration data (see Figure 32.10) to calculate the velocity using Equation (10.5) gives an error of 0.00051 $m \cdot s^{-1}$ for every second of data recording. To remove this

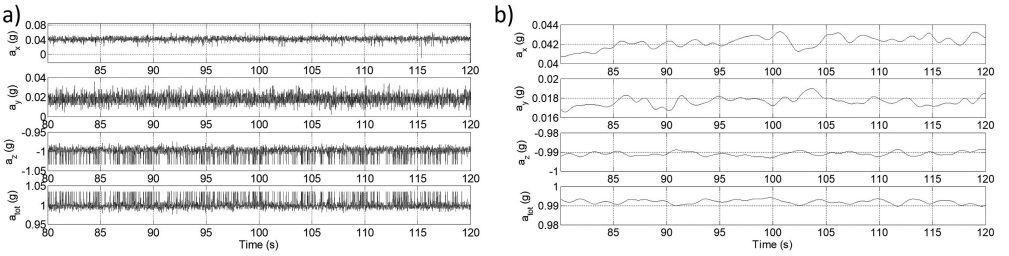

Figure 32.9 Calibrated acceleration in g of a resting accelerometer (left) and low-pass filtered acceleration data of a resting accelerometer (right). The accelerometer was resting with earth gravity recorded on the *z*-axis.

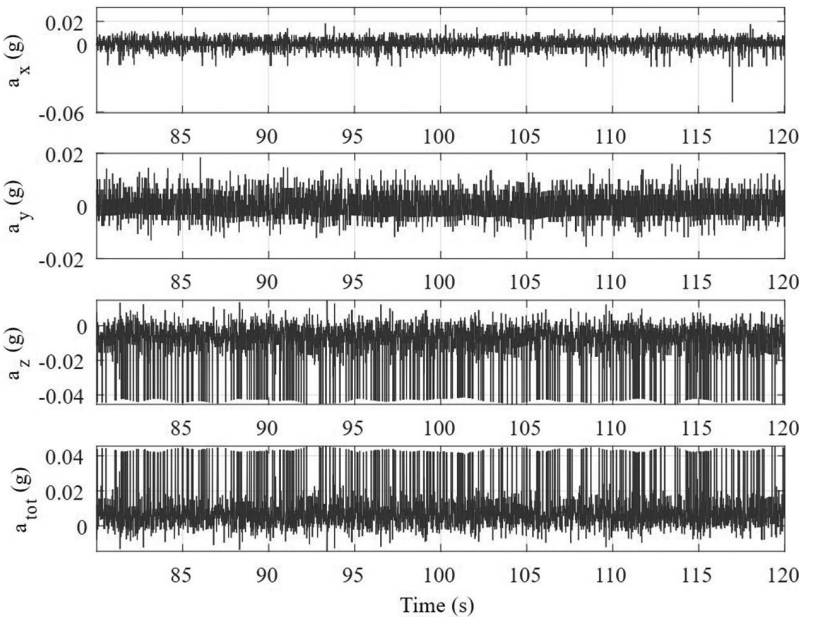

Figure 32.10 Presents the result of the acceleration recorded on all three channels subtracted by the low pass filtered data to generate a high pass filtered acceleration data of a resting accelerometer. Tri-acceleration along each sensing axis (x, y, z) and combined (bottom).

error, one can shift the total acceleration data by the mean acceleration over the time investigated to remove any remaining bias before calculating the velocity. Figure 32.11 shows the velocity calculated from the resting accelerometer over the investigated time frame (40 seconds) after the remaining bias was removed.

Considering the filtered total acceleration shifted by the mean acceleration over the period of interest leads to a velocity calculation for a limited timeframe (period of interest) and therefore to a small integration error. Figure 32.12 presents the calculated velocity profile from recorded IMU data with the methods explained above (smoother line), compared to a tethered velocity meter (SP5000), which can be considered the "gold standard" in measuring velocities in swimming.

A statistical comparison between two different measurement systems can be carried out using regression or Bland-Altman analysis (Figure 32.13). The regression analysis can show how well the measurements (data points) align with the linear fit (red line). The Bland-Altman analysis

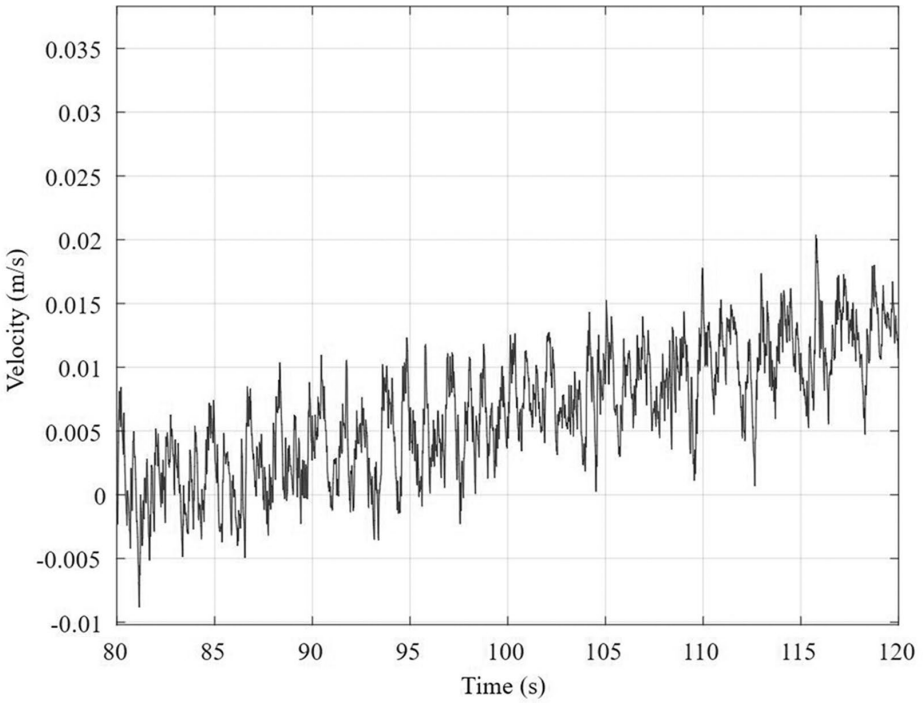

Figure 32.11 Calculated velocity from the gravity subtracted total acceleration data over a 40-s period.

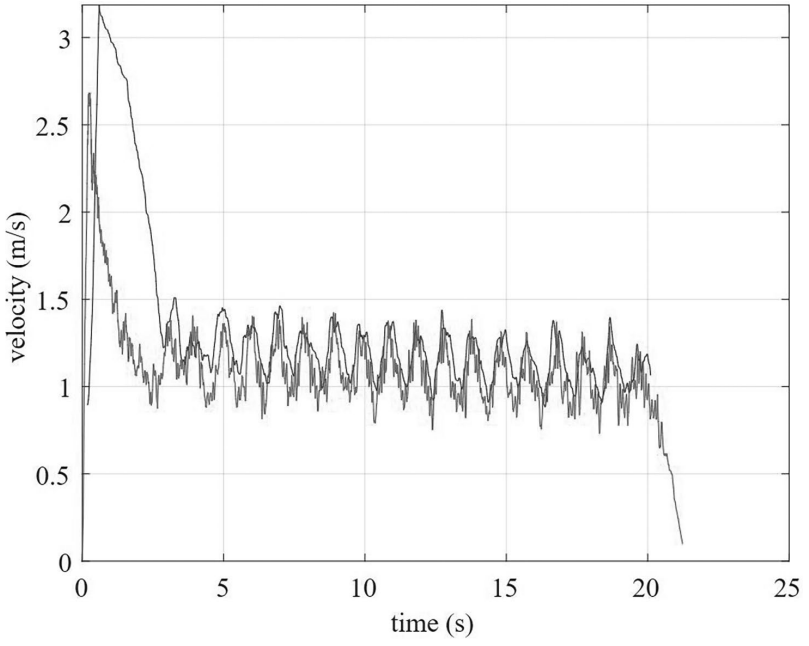

Figure 32.12 Velocity profile calculated from IMU data (smoother line) and a tethered velocity meter over a 25-m pool freestyle swimming session.

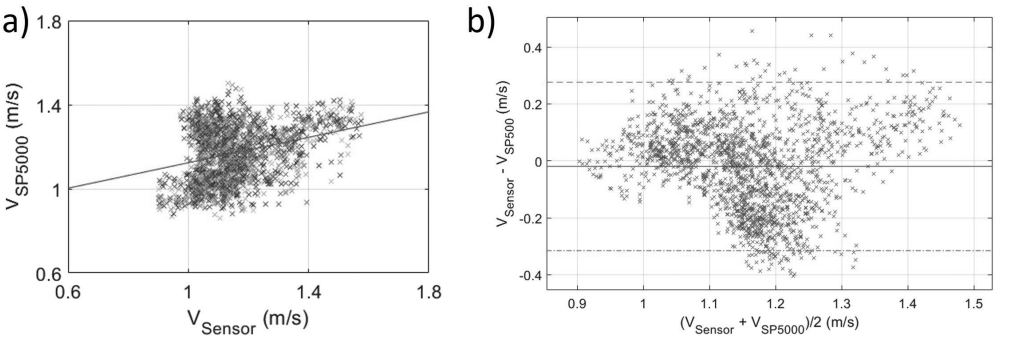

Figure 32.13 Regression analysis (left) and Bland-Altman analysis (right) over a 25-m pool freestyle swimming session.

can determine if there is a systematic measurement error (for uneven scattering around the bias) or if there is no systematic measurement error (if the scattering follows a normal distribution). Figure 32.13 presents an example of each of the methods, considering the velocity profiles presented in Figure 32.12.

Discussion

This chapter discusses the fundamentals and principles of IMUs with an application in freestyle swimming, which is just one scenario where they can be used to find parameters of interest to athletes and coaches. Of course, there are many other possible parameters that could be investigated in swimming, such as identifying stroke type [17,18], finding stroke rate [10,19–22], finding stroke parameters using an IMU at the wrist [17,23,24], and investigating hand movements during the individual swimming styles and phases [25,26], to name a few. In prior reviews [18,27], researchers explored the optimal/preferred sensor locations for extracting various parameters. For instance, wearables were positioned on the lower back (near the center of mass) or head for measuring swim velocity. Upper-limb locations, such as the forearm, wrist, or palm of the hand, were favored for assessing parameters like stroke count, stroke rate, and duration. Additionally, sensors attached to the lower limb were investigated for capturing kicking motions (legs' downbeat) [28].

The type of IMU signal is as important as the IMU location. Earlier studies exclusively employed accelerometers, gyroscopes, or a combination of both sensors to extract the desired outcomes [29–33]. Without magnetometers, a 6-axis system, consisting of accelerometers and gyroscopes measuring in three axes each, can establish relative heading, though it may experience some drift. The inclusion of magnetometers can address this drift by providing an absolute heading reference point based on the Earth's magnetic field, thus offering a global orientation reference [4]. Integrating these three types of sensors enables a more comprehensive and reliable analysis [32].

Wearable sensors can provide real-time feedback to athletes on their performance. In one study, a system was developed comprising a wrist-mounted accelerometer and optical wireless communications. This system was designed to provide real-time feedback to a swimmer via display goggles [29]. Another study utilized multiple wearable sensors (attached to the head, forearms, and shanks) to examine the timing of strokes, breathing, and kicking in front-crawl swimming. The

research specifically employed the angular velocity data from IMUs placed on the head to conduct breathing analysis, a task that often necessitates additional sensor modalities [33].

Beyond competitive swimming, IMUs have also found proficiency in drowning prevention and water safety. IMUs have shown they can be pivotal in detecting swimmers in distress by identifying abnormal or erratic movements, thus enabling timely alerts to lifeguards or automated monitoring systems [34]. This capability is especially critical in complex environments, such as rough seas and crowded swimming areas, where traditional observation methods may fall short.

A recent study involved the development of 'LifeTag', an IMU-based device located on a life-jacket, designed to activate upon immersion in water [35]. The device in the referenced study integrates functionalities for localization, communication, and life-sign detection, collecting accelerometer and gyroscope data while participants simulated drowning scenarios. In processing the data, the research team in that study employed a machine learning framework that selected key features and developed a classifier to distinguish between alive and dead states based on patterns in the inertial data. The results of their experiments indicated the system had nearly 100% prediction accuracy, demonstrating a highly effective classifier for identifying life signs in real-time scenarios and the practical applicability of the device in critical situations.

Another study also emphasized the potential of IMUs in the automatic detection of swimmer's movements, further underscoring their utility in real-time monitoring and safety enhancement in aquatic environments [36]. To further enhance the usability of this technology, the integration of IMUs into wristbands, swimwear, and goggles could be another innovative approach to drowning prevention, where such devices could alert if drowning patterns are detected, contributing to timely rescue and prevention strategies [18].

Conclusion

It is not only swimming that can be investigated using IMUs. There are many other sports, like walking/running [37–41] where researchers investigate strides, step distances, gait analysis, fatigue, and many more. IMUs can also be used to investigate the movement of sports equipment, such as cricket bats [42–44], where they can help to find bat angles, impact times, and so on.

The analyzed measurements can be leveraged to enhance fan engagement and elevate hybrid sports and equipment testing. Equipment testing, especially, benefits during the development cycle, allowing manufacturers to offer better-tested products to customers. Another new way of investigating movements/impacts is to utilize machine vision together with big data. This, of course, requires capturing the movement/impact of interest with a high-quality camera and the ability to store big amounts of video data for later analysis, e.g., with big data algorithms. Readers seeking more information on big data are referred to Chapter 35. Another new way of investigating data is through machine learning algorithms. These allow us to find metrics that cannot be directly measured, e.g., predictions of forces from accelerometer data or similar. Together with wireless IMUs and cloud computing, one could use all these technologies together and provide users with fast feedback on metrics they are interested in. Getting all these technologies to working together, however, requires sophisticated knowledge in hardware design, software programming, and communication protocols for exchanging data.

Conflicts of interest statement

The authors declare that they have no conflict of interest.

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ARTIFICIAL INTELLIGENCE APPROACHES IN SPORT

Fraser Young and Alan Godfrey

Introduction

From match analytics providing high-level performance insights to the most intricate details of injury prevention and management, modern sport relies on data analytics to inform decision-making [1]. Famously, the 'Moneyball' paradigm utilised data analysis to identify undervalued market players in Major League Baseball, leading to one of the most unlikely winning runs in sporting history given the relatively small team budget [2]. The impact of such paradigms has extended beyond baseball, influencing decision-making in various individual and team sports [3,4]. Crucially, the use of data-driven approaches has optimised decision-making, reducing reliance on subjective human experience.

With the perceived success of data-driven decision-making in sport, the scope of data analytics has vastly broadened outside of player purchasing and general management decisions. Prominently, assessing injuries in player populations can inform clinical diagnosis, immediate care plans, treatment, rehabilitation, reconditioning, and performance [5]. For example, one can assess the risk of an athlete carrying a common cruciate ligament injury based on differing features, which is important for contemporary player management, with 3.5% of women and 2.0% of men experiencing such an injury within a seasoned career [6]. Naturally, the utility of such data works to inform relevant parties (e.g., physio, clinician) to assist in decision-making processes is crucial in more modern approaches to player management and welfare.

Recently, a move to understand the underlying mechanisms contributing towards the likes of injury prevention and player management has been realised in conjunction with the prevalence of big data [7]. In utilising the big data paradigm, the sports decision-making process is informed through the identification of micro-trends that could, for example, lead to an injury. That notion has been applied in various sports, with models developed to evaluate the risk of injury and the effects of workload mismanagement [8], leading not only to a lower risk of injury but also to an understanding of the underlying mechanisms contributing towards potential injury. Accordingly, a big data approach to sports analytics has been augmented by a vast uptake in data capture and recording in recent years, with various capture modalities containing the likes of wearables and video [7,8]. However, challenges arise with large datasets often found within the big data context, as useful outcomes become harder to obtain due to high dimensionality, heterogeneous data, and

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computational complexities [9,10]. Consequently, the use of artificial intelligence (AI) has become ubiquitous in providing big data analytics.

AI encapsulates various computations and algorithms, aiming to provide rational, human-like evaluations of data [11]. Recently, AI has experienced a rapid uptake in usage, demonstrating an innate ability to handle, analyse, and provide insight into complex datasets, spanning a range of domains from healthcare and education to robotics [12–14]. Interestingly, the application of AI can provide various personalised outcomes (i.e., micro-level analysis of individuals), providing a new dimensionality of analysis and assessment [15,16]. Of course, the ubiquity of AI has extended to sports-related domains [17], providing smart outcomes to inform all issues faced in modern sports (e.g., player management, injury prevention, decision-making, performance analytics). Consequently, the following chapter will discuss current applications of AI systems, providing insight into the common approaches found in various sporting applications. Insights provided here may help contribute to explainable AI (XAI) in sports, making models and approaches transparent so they can be applied appropriately and effectively.

AI approaches

AI can augment various sporting assessments and analyses. However, when it comes to understanding the mechanisms that power these complex systems, we are often met with a 'black box' of information, i.e., without extensive technical knowledge, one cannot easily understand or implement these systems for pragmatic use. Consequently, the following section will observe varying AI approaches in different sports, providing insight into the approaches used and unpacking the 'black box' for real-world use.

Wearable devices, signal analysis, and classification

Wearable devices are nearly ubiquitous in modern sport, positively impacting various applications. In comparison to video-based computer vision (discussed later), wearable technology can provide more readily available fitness analysis [18] and activity recognition [19], among others, providing a nuanced, micro-level and personalised assessment of athletes during sport. Here, this chapter focuses on the use of technologies in what could be described as the most fundamental action within sports: running.

Gait assessment with wearable technology

Understanding an athlete's gait is paramount to providing injury prevention mechanisms in running-centric sports. For example, adapting a forefoot strike location during impact can reduce gait velocity [20], potentially minimising injury risk, with forefoot strikers reporting fewer injuries than those exhibiting a rearfoot alternative [21]. Additionally, gait is considered a useful biomarker in monitoring the rehabilitation of athletes following an injury. For instance, athletes suffering from a concussion may exhibit lower gait stability and balance in comparison to baseline athlete performance [22,23]. As such, monitoring gait following head trauma could inform return-to-play schedules for injured athletes. Therefore, providing repeatable mechanisms for gait assessment is essential in modern sport, given the ubiquity of running throughout the field.

Recently, wearable technology has positively impacted the field of gait assessment by providing a minimal barrier of entry through low costs in comparison to traditional apparatus (e.g., 3D-motion tracking) and reducing the requirement for bespoke environments, enabling

assessments in real-world scenarios [24–26]. In combination with contemporary AI techniques and signal analysis approaches, the vast quantity of data provided by wearables has enabled a range of gait outcomes that were previously unattainable beyond bespoke research settings. Challengingly, gait assessment with wearable technology can often produce large amounts of data (e.g., 60 Hz data capture during an hour-long training session could produce up to 216,000 data points per sensor), and as such may be unfeasible to analyse in bulk. Consequently, wearable-based gait assessment often relies on intelligent feature extraction layers to inform gait cycle segmentation, reducing potentially large data streams into individual strides for analysis [27].

Often, segmentation relies on the identification of initial and final/terminal contact within the stride [28], providing insight into how the foot makes contact with the ground and for how long contact is maintained. However, the identification of gait features can differ depending on the wearable selected. For example, wearable inertial measurement units (IMUs) typically contain accelerometer or gyroscope units, or both, to provide an understanding of acceleration and rotation of a certain anatomical location during gait [29]. Typically, to extract important initial and final contact timed events from an IMU signal, acceleration signals are searched for peaks (e.g., when the foot makes contact with the ground, there will be a large vertical acceleration peak) and identified utilising, e.g., zero-crossing gradient analysis [30], wherein large peaks are detected by observing gradient changes above a pre-determined threshold. Alternatively, pressure sensor-based devices, such as smart insoles [31], can segment bouts of gait through locating heel strike peaks. Describing a full range of wearable technologies and their respective outcomes is out of the focus of this chapter; however, exhaustive reviews have been covered elsewhere [29,32].

Athlete health and wellbeing assessment

An athlete's wellbeing is a crucial component of their performance in sport, including their mental health [33] and sleep quality [34], among others. However, wellbeing assessment is typically reliant on self-reporting mechanisms between athletes and coaching staff, which may not be truly indicative of the athlete's true wellbeing (e.g., a rugby player with a suspected concussion may minimise symptoms to return to play). Additionally, wellbeing measures are usually subjective [35], and as such, do not scale between athletes (e.g., a good night's sleep may be considered different between individuals). Consequently, there is a need for objective measures of athlete wellbeing.

Recently, wearable technology in combination with AI has provided an interesting impact on monitoring the wellbeing and overall health of athletes outside of training or playing scenarios. Exacerbated by wearable technology's recent uptake in use [36], alongside relatively low costs and a wide range of sensing modalities [37] contained within the devices, wearables are suitable mechanisms for measuring wellbeing in sport. The following section will describe some recent advancements in wellbeing assessment with wearable technology and artificial intelligence.

Cardiovascular health assessment [38,39]: Understanding the cardiovascular health of athletes is paramount, given the obvious impact of poor cardiovascular health on performance and the risk of worsening symptoms or even fatality. Furthermore, there is evidence to suggest that sustained high-intensity exercise over a long period may contribute to the development of cardiovascular diseases [40], which could affect athletes at all levels. Consumer-grade cardiovascular health assessment generally relies on electrocardiography (ECG) sensors, which are commonly found in commercial wearable technology such as the Apple Watch [41]. Traditionally, ECG signals would undergo (PQRST – there are three main components to an ECG signal: (i) P wave, represents depolarisation of the atria; (ii) QRS complex, representing depolarisation of the ventricles and (iii) T wave, repolarisation of the ventricles) wave detection and feature extraction [42], allowing

clinicians to identify temporal abnormalities within the signal that could be indicative of a larger issue. However, more recently, the use of artificial intelligence has automated these assessments, as well as provided predictive outcomes. For example, Raghunath et al., [43] found that by training neural networks based on the ECG signals of 430,000 patients collected over a 35-year span, AI could predict new-onset atrial fibrillation (AF) and alert high-risk individuals upon passing various thresholds. Applied to sport, AF is prevalent in those engaging in endurance sports such as running, cycling, and (cross-country) skiing [44]. As such, by identifying early signs that could lead to AF with wearable technology and artificial intelligence, athletes can change training or participation to minimise their risk of developing a long-term condition.

Sideline concussion assessment [45]: Assessing concussion at the sideline is critical in deciding whether a player is fit to return to play. Currently, should an athlete (e.g., rugby, ice hockey, or American football player) sustain a head impact during play, their return to play is dictated by self-reported symptoms to a trained healthcare professional, which are naturally subjective and could lead to further harm to the athlete. Wearable technology may be able to instrument an objective measure of concussion assessment in near-real time to better inform return to play. For example, a tandem gait test has demonstrated efficacy in identifying concussed players [46,47]. As such, by instrumenting a tandem gait assessment at the sideline with, e.g., a wearable IMU and signal analysis techniques [48], wearable technology could provide an objective measure of concussion [49]. Alternatively, there has been recent interest in the implementation of computer vision-based eye-tracking glasses, wherein devices such as the *Pupil Labs* (https://pupil-labs.com/) or *Tobii* (https://www.tobii.com/) eye-tracker systems use computer vision to track the pupil of an individual [50]. Importantly, the movement of the eye following a suspected concussion can be telling of an affected individual's mental state, with concussed athletes demonstrating higher latency in oculomotor, vestibular, and reaction time tests in comparison to healthy controls [51]. Consequently, by utilising eye-tracking wearable technology, sideline concussion assessment could be better informed.

Considerations

Wearable technology can augment high-resolution athlete assessments when used with AI and appropriate signal processing techniques. However, there are shortcomings to consider when using the technology. Firstly, despite wearable technology's promising application in various sports areas, there is still a need for further validation in a wide range of domains. For example, although a wearable device may demonstrate validity in, e.g., measuring balance control following a concussion in adolescent cohorts [23], the approach's validity will not necessarily scale to other demographics, such as younger children or older adults. Consequently, performing validation studies on diverse applications is critical for wearable technology's uptake as an athlete monitoring tool. This further extends into free-living, out-of-lab testing. For example, currently, most IMU-based running gait assessment studies are conducted indoors, on a treadmill at predetermined speeds, which may not be indicative of overground running gait patterns in uncontrolled environments [52]. To ensure the validity of wearables in different environments, validation studies should move into a free-living context. Furthermore, many wearable technologies are intrinsically 'blind' and cannot establish the context behind the extracted outcomes [53]. For example, should an individual experience an erroneous bout of gait according to an IMU, was it due to intrinsic factors experienced as part of an injury (e.g., concussion), or were they avoiding an obstacle? As such, wearable technology may be most effective in combination with observation, whether by manual observation or autonomous computer vision, to assist in micro-level assessments.

Computer vision

Computer vision encapsulates the field of AI, wherein, using classification methods, one can begin to autonomously classify, segment, and identify a range of objects from video-based cameras (both 2D, e.g., webcam, and 3D, e.g., infrared, thermal) and imaging sources [54]. Perhaps most commonly, the modern world is exposed to computer vision through the likes of autonomous vehicles [55]. However, recent developments have led to its extensive use in sports applications, with prominent applications in athlete tracking augmenting performance analysis, tactic feasibility, and injury assessment [56,57].

Athlete tracking

Traditionally, athlete and player tracking has relied upon satellite-based radio navigation system devices to monitor and track location during training bouts, demonstrating utility in understanding training and performance effectiveness [58]. Naturally, the approach is hindered by the requirement of a wearable sensor, exhibiting a potentially high cost and a moderately intrusive form factor that may cause discomfort in players [59]. Furthermore, high error and low reliability rates have been reported across different satellite-based radio navigation system models, indicating the need for an ensemble of devices or sensing modalities to improve accuracy [58].

In response, the development of markerless tracking systems has been pioneered using computer vision. Generally, a typical system relies upon a two-stage architecture: object detection and object tracking, as shown in Figure 33.1. During the object detection phase, a convolutional neural network (CNN) is often deployed [60], such that:

1 An image is split into searchable regions of interest (RoI), grouped based on similar features (colour, size, shape)
2 Each RoI is reshaped to match the expected input dimensions
3 Each RoI is fed into a classifier to assess the content of a region (e.g., person, ball) [61]

The use of object detection has been shown to be highly suitable for athlete identification [60,62] in image streams, primarily due to highly detailed, large-scale datasets on which ‘gold-standard’

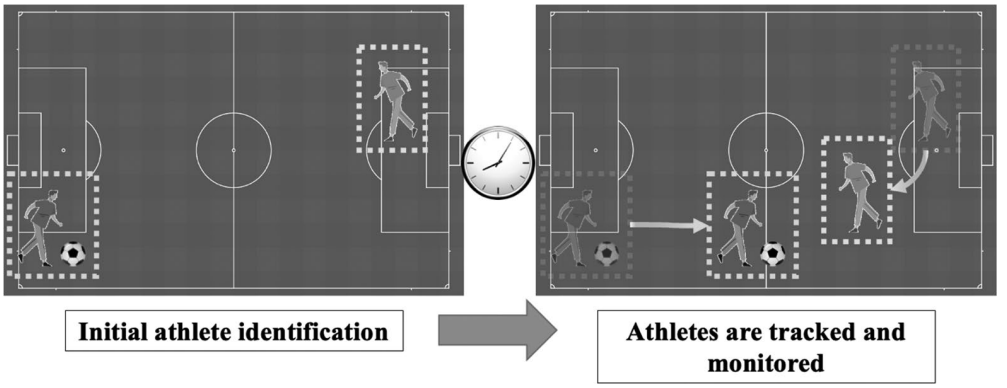

Figure 33.1 Example of athlete tracking in soccer application demonstrating two key stages: initial identification and location tracking.

object detection models are trained, often containing 'human' classes [61]. However, when tracking athletes, the approach is often flawed; as a CNN is unable to distinguish between multiple instances of an object (e.g., during a game of soccer, it may become confused when detecting players with similar uniforms) [60]. As such, the use of a Multiple Object Tracking (MOT) system is warranted, informed by a region of interest (RoI) provided by the object detection stage [63].

Posture and kinematics analysis

Through the successful identification and tracking of an athlete, the provided RoIs work towards informing various sporting assessments. For example, in combination with pose estimation techniques [64], we can begin to understand the joint kinematics of athletes, providing injury assessment and monitoring (e.g., ACL and knee injury risk based on impact and flexion angles [65,66]), as well as performance and training metrics in various sports (e.g., baseball swing kinematics [67], basketball free-throw optimisation [68], and golf swing correction [69]).

Generally, pose estimation techniques aim to locate key anatomical locations on single [70] or multi-person [71] frames in a markerless context (e.g., without the use of neo-reflective markers to identify on-frame or by 3D motion tracking systems), allowing assessments to take place outside of bespoke environments containing, e.g., 3D motion tracking [72]. There are several pose estimation techniques that are commonly used [73], each with its own advantages and disadvantages. For example, some approaches [74] can infer three-dimensional (i.e., depth) anatomical locations from a two-dimensional image, enabling a greater understanding of human motion. However, such approaches may require significant computational complexity, limiting their deployment in real-world settings in comparison to low-powered approaches such as BlazePose (i.e., suitable for use on a smartphone) [75], which prioritises two-dimensional inference.

Despite the differences in approaches, the architecture of pose estimation in sports is fundamentally similar, wherein: (i) athletes are identified within a frame, (ii) the identified athlete's RoI is fed into a pose estimation encoding and decoding approach, and (iii) anatomical locations are returned frame by frame for analysis, as illustrated in Figure 33.2. Following pose estimation,

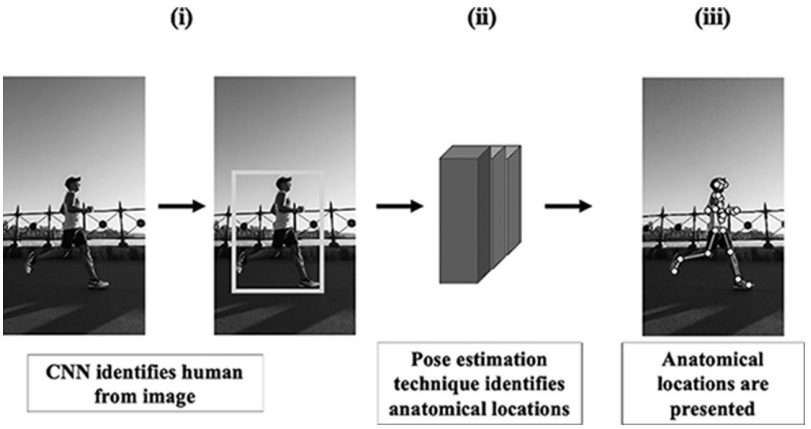

Figure 33.2 Architecture of a typical human pose estimation architecture in sports, showing (a) identification of athlete using a CNN, (b) deployment to a range of pose estimation techniques, and (c) an example of 2D output with extracted anatomical locations.

various analyses can, in turn, be performed. Here, some recent examples of pose estimation in sports are described.

Golf swing performance analysis [76]: Kim et al., proposed a golf swing analysis tool based on PoseNet, a widely-used pose detection method. In the approach, down-range videos are converted into individual frames, wherein the frames are fed into a PoseNet instance, providing 17 anatomical key points for analysis. Following a filtering component to minimise the effect of misclassifications and inaccuracies, the golf swing is then segmented into seven temporal stages based on the intersection of anatomical locations (e.g., wrist intersecting with right shoulder; wrist intersecting hips). With the golf swing segmented into swing-critical temporal stages, the relative positioning of the body and arms can be assessed in comparison to 'gold-standard' athletes (e.g., those competing in professional-level golf) at each stage, informing areas for improvement at each stage of the swing.

Running analysis [77]: The observed study has implemented a custom pose estimation technique utilising convolutional pose machines [64] trained on a custom dataset, which provide a heatmap of likely locations of anatomical locations from two-dimensional images. By extracting the anatomical locations from running videos, the study can extract the following gait and posture outcomes: speed, step frequency, step length, vertical oscillation, trunk angle, arm swing angle and leg swing angle. Interestingly, the observed study performs amicably in comparison to Vicon 3D motion tracking, indicating that pose estimation is valid in comparison to established reference technologies.

Considerations

Computer vision is useful in providing various sports analytics. Particularly, by providing a means of markerless assessment, computer vision can augment low-cost analytics by reducing the reliance on traditional assessment techniques alone (e.g., manual assessment by a professional; instrumenting assessment with expensive equipment such as 3D motion tracking). Additionally, computer vision is physically unobtrusive and, as such, will not affect the performance of an athlete by being mounted with physical sensing modalities (e.g., reflective markers). However, some considerations should be observed before selecting computer vision as a primary means of sports analytics.

Generally, the use of computer vision requires significant computational power, which can lead to both slow execution times (i.e., non-real-time systems) and high costs of machines [78]. As such, the utility of computer vision approaches for sports assessment may currently be limited for real-world applications outside of the research domain, where high-powered computing is commonplace. As a solution, computationally optimised approaches such as Faster-RCNN [79], BlazePose [75], and models included in popular distributions such as TensorFlow Lite [80] should be considered, given their suitability for deployment on, e.g., smartphone technology, allowing the technology to transition into out-of-lab assessments.

Furthermore, there is some debate regarding the validity of computer vision in some assessments in comparison with the likes of wearable technology, which can quantify intricate features in, e.g., gait assessment [81]. As such, computer vision may be more effective at macro-assessments (e.g., player tracking, game analysis), whereas other AI methods (e.g., signal analysis, neural networks, deep learning) may augment micro-level assessments.

Unpacking the black box: Tutorial of a gait assessment tool

There is often a 'black box' description associated with AI in sports, creating a barrier to entry for many who are untrained in a computer science background. Accordingly, there is a focus on XAI,

removing the black box to ensure openness and that all technical approaches are fit for purpose. For example, the 'pose estimation techniques' discussed above provide some insight into the technologies. However, how does one implement such a system? Are these systems feasible to implement without extensive AI knowledge? The following section will provide a demonstration of how to implement a pose estimation approach with minimal experience in the Python programming language. The following example is loosely based on work published on running gait assessment [82].

Environment setup and packages

It is imperative that the programming environment is configured to match that of the given example. It is recommended that a virtual environment tool, such as Anaconda (https://www.anaconda.com/), be used to isolate the project's dependencies so as not to conflict with pre-existing projects. Although outside the scope of this example, a thorough explanation can be found elsewhere [83]. The following example relies on Python version 3.8, which can be obtained from the official Python website. Included with the Python installer, the PIP package installer will reduce the work required to install relevant packages. Here, the main packages are listed with relevant information for installation and purpose.

Package: Mediapipe, *Required*
Install command: pip install mediapipe
Link: https://developers.google.com/mediapipe/solutions/vision/pose_landmarker/
Description: Mediapipe provides several open-source AI solutions for processing image and video streams in near real time [84]. Critically, Mediapipe solutions offer cross-platform, computationally inexpensive computer vision models, enabling inference on virtually any hardware, reducing reliance on, e.g., high-end computing. Here, we will be utilising the included pose estimation model based on BlazePose [75]; however, many other models are available depending on the desired application, including, e.g., iris tracking and hand gesture detection.

Package: OpenCV, *Required*
Install command: pip install opencv-python
Link: https://opencv.org/
Description: OpenCV is an open-source package that handles the loading of images and videos for use with, for example, the Mediapipe package. Images (or video frames) are loaded as NumPy matrices (included), enabling image manipulation with ease.

Package: Pandas, *Optional*
Install command: pip install pandas
Link: https://pandas.pydata.org/
Description: Pandas is an open-source data analysis and manipulation tool. By providing 'dataframes', data can be manipulated as rows and columns, providing a familiar 'Excel-esque' analysis suite to the Python programming language. Here, we use the dataframe structure to generate a CSV with anatomical coordinates for use in additional assessments.

Pose estimation with Mediapipe, OpenCV, and Pandas in Python

With the development environment configured, implementation of a pose estimation system can begin. For our integrated development environment (IDE), VSCode is preferred; however, any

range of modern IDEs or text editors will suffice. Comments highlighted in green will provide further explanation of code extracts.

Step 1: Configuration and loading data

In Figure 33.3, cv2 refers to the OpenCV package. Mediapipe and Pandas are shortened to 'mp' and 'pd', respectively, for brevity (e.g., in code, we can type pd.DataFrame rather than pandas.DataFrame). Once imported, we can begin loading the desired video into the project. Table 33.1 provides some guidance.

Step 2: Initialising pose estimation and reading video files

The 'video_path' (Figure 33.4) variable should store the path of the desired video. Placing the video in the same directory as the Python script will simplify this section, e.g., 'video_path = "video1.mp4"'. Table 33.2 details some guidance.

Firstly, initialising the pose estimation model with 'mp_pose.Pose() as pose' allows configuration of the BlazePose architecture. Here, the minimum detection and tracking confidence have deliberately been set to 20% (0.2, Figure 33.5) to ensure points are detected for illustrative purposes. However, it is recommended that a minimum of 50% be selected for accurate anatomical

```
import cv2
import mediapipe as mp
```

Figure 33.3 Code extract for importing necessary libraries.

Table 33.1 Expected output and common fixes for importing libraries

Expected output	Program should successfully execute, no output expected
Common error	no module named '<package name, e.g., pandas'> found
Troubleshooting	Ensure packages have been installed to the computer using the install commands listed above
Example	pip install pandas **OR** pip3 install pandas

```
# Set video path
video_path = "<DESIRED_PATH, e.g., C://User/Documents/Videos/video1.mp4>"

# Configure mediapipe pose and drawing utilities
mp_drawing = mp.solutions.drawing_utils
mp_drawing_styles = mp.solutions.drawing_styles
mp_pose = mp.solutions.pose

# Load video at path with OpenCV
cap = cv2.VideoCapture(video_path)
```

Figure 33.4 Code extract to load video data and pose estimation model.

Table 33.2. Expected output and common fixes for loading videos into the environment

Expected output	Program should successfully execute, no output expected
Common error	OpenCV: Couldn't read stream from file '<directory>'
Troubleshooting	Ensure the directory is correct, relative to the directory of the project
Example	Given the script being contained within a directory, e.g., 'PoseEstimation', create a directory for images/videos within 'PoseEstimation' named 'data'. You can refer to this directory in code as: video_path = "data/test_video.mp4"

```
# Initialise pose estimation model (may take some time)
with mp_pose.Pose(
    # Pose estimation parameters - see Table 1
    min_detection_confidence=0.2,
    min_tracking_confidence=0.2) as pose:

  # Only run while there are frames remaining in the video stream
  while cap.isOpened():
    row = []
    success, image = cap.read() # Read frame
    if not success: # If frame isn't read, exit loop.
      print("Couldn't read frame")
```

Figure 33.5 Code extract for initialising pose estimation models and cycling through videos frame by frame.

Table 33.3 BlazePose model parameters, default values and their intended purpose

Parameter	*Description*
model_complexity	**Options**: [0,1,2] **Purpose**: There are three models of differing complexity included in Mediapipe pose, with 0 being the simplest and 2 the most complex. There is a trade-off between the three models, with 'model 0' operating the fastest (i.e., most suitable for low-powered machines such as smartphones) and 'model 2' providing the most accurate anatomical locations (at a higher computational cost). **Default**: 1
smooth_landmarks	**Options**: [True, False] **Purpose**: If set to true, landmark locations will be smoothed between subsequent frames, as such, reducing jitter that can be commonly encountered in pose estimation. **Default**: True
enable_segmentation	**Options**: [True, False] **Purpose**: If set to true, a segmentation mask will be provided alongside anatomical locations. **Default**: False
min_detection_confidence	**Options**: [0.0–1.0] **Purpose**: Sets the confidence value of detected anatomical points. **Default**: 0.5
min_tracking_confidence	**Options**: [0.0–1.0] **Purpose**: Sets the confidence value of tracked anatomical points. **Default**: 0.5

tracking. Table 33.3 contains an additional list of configurable parameters that can be adopted depending on the desired application. Table 33.4 details arising guidance.

Step 3: Running inference and displaying results

Finally, with the model configured and the data stream operating, we can process results with the pose estimation model and display them on the screen, as depicted in Figure 33.6. Table 33.5 details arising guidance.

Step 4: Optional: Storing anatomical co-ordinates for analysis

Although displaying the anatomical skeleton over a video stream is interesting, analysing the anatomical coordinates extracted by the pose estimation model can enable interesting assessments such as gait assessment. To augment such assessments, extracting and storing the coordinates is essential. Here, changes will be required to the existing code.

Firstly, proceeding the package imports but outside of the core loop (e.g., next to video_path initialisation), instantiating two arrays to store the location of relevant anatomical locations will allow for data analysis in CSV format (Figure 33.7).

Next, anatomical locations are extracted from the 'results' object. Here, we focus on the extraction of the left foot from the video stream for brevity (Figure 33.8). However, additional locations and their respective positions in the output can be found here [85].

Table 33.4 Expected output and troubleshooting for reading video frames

Expected output	Program should not print: 'Couldn't read frame'
Common error	'Couldn't read frame' is encountered after execution
Troubleshooting	This error is commonly due to an incorrect video path provided, or a corrupted video. Ensure the filename and directory are correct

```
    # Apply pose estimation model to image in video stream
    results = pose.process(image)

    # Draw the pose annotation on the image.
    image.flags.writeable = True
    mp_drawing.draw_landmarks(
        image, # Target image
        results.pose_landmarks, # Landmarks obtained from results
        mp_pose.POSE_CONNECTIONS, # Draw anatomical locations (i.e., skeleton)

landmark_drawing_spec=mp_drawing_styles.get_default_pose_landmarks_style())

    # Show video frame-by-frame
    cv2.imshow('Pose estimation output', image)
```

Figure 33.6 Code extract for running inference with MediaPipe on each frame.

Table 33.5 Expected output and troubleshooting for running inference

Expected output	Window is opened with the supplied video and anatomical skeleton on top of the subject(s)
Common error	Window fails to open/video is black
Troubleshooting	The following examples rely on '.mp4' video encoding or '.png' images. Files may display incorrectly in other formats
Notes	Videos can be cycled through frame by frame by changing cv2.waitKey(5) to cv2.waitKey(0)

```
...
# Set video path
video_path = "<VIDEO_PATH>"

# Build pandas Dataframe output
columns = ["Left Heel Location X",
           "Left Heel Location Y",
           "Left Toe Location X",
           "Left Toe Location Y",
           "Left Ankle Location X",
           "Left Ankle Location Y"
          ]
# Rows will be appended to this array
data = []
```

Figure 33.7 Extended code extract for loading in video and initialising arrays for storing anatomical locations.

Finally, with reference to the coordinates of the left foot, we can then build a CSV file that can be used for supplementary analysis. The following extract (Figure 33.9) should be placed at the bottom of the Python script.

Data analysis and outcomes

Although outside the scope of this chapter, given a structured file containing anatomical coordinates, various assessments can be performed. For example, work in running gait assessment utilising this approach [82] demonstrates how spatial and temporal gait parameters can be extracted from a 2D video stream. In the example, initial and final contact are identified by observing the maximum extension of each leg. In turn, factors such as contact time, swing time, and foot strike location can be extracted.

The dilemma with AI

Generally, within society, there is an acceptance of the use of wearables as tools to help improve health and wellbeing. Wearable watches with various sensors designed to track movement or heart rate are almost ubiquitous, with (bio)feedback used by the wearer to, e.g., monitor training routines. Alternatively, the jury is still out on the use of AI due to uncertainty stemming from the

```
    # Extract left foot X/Y locations
    left_ankle_x = results.pose_landmarks.landmark[27].x
    left_heel_x = results.pose_landmarks.landmark[29].x
    left_toe_x = results.pose_landmarks.landmark[31].x

    left_ankle_y = results.pose_landmarks.landmark[27].y
    left_heel_y = results.pose_landmarks.landmark[29].y
    left_toe_y = results.pose_landmarks.landmark[31].y

    # Build a dataframe row as an array of locations
    # Notice how the structure reflects the column layout above
    row = [left_heel_x, left_heel_y,
           left_toe_x, left_toe_y,
           left_ankle_x, left_ankle_y]

    # Append the row to the data array
```

Figure 33.8 Code extract-to-extract X and Y locations of anatomical locations from a video stream.

```
# Build pandas DataFrame
output_df = pd.DataFrame(columns=columns, data=data)

# Save DataFrame as CSV file for analysis
output_df.to_csv("output.csv")
```

Figure 33.9 Code extract-to-extract X and Y locations of anatomical locations from a video stream. Code extract to convert arrays of anatomical locations into CSV output.

unknown, i.e., the blackbox approach previously described. Fear may also play a role. For example, a commentary debated the use of AI as an electronic line judge in tennis, which would remove or greatly reduce the human factor in sport, thereby curtailing the entertainment, action, and drama that draw one to sporting events [86]. Furthermore, the increasing use of AI (and associated technologies) may lead to deskilling and the devaluation of human experience and clinical intuition [87]. Hence, XAI is an important topic to be considered, and the fields of, e.g., sports science and sports medicine must evolve their exposure to and knowledge of contemporary tools (like wearables and AI) to enhance their work, as findings suggest the use of those tools will become increasingly popular [88].

Conclusion

AI has a vast range of applications within sports, ranging from decision-making to athlete well-being measures and beyond. Here, examples are drawn from recent literature to provide a wide-spanning understanding of what is possible within the field. Currently, AI is inhibited for use within sporting applications, with many studies still existing within the research domain, creating a 'black box' that cannot generally be unpacked by those who would benefit from the technology's

use (e.g., sports therapists, athletes). As such, guidance is provided on AI's use in sports, aiming to enable XAI assessments by interested stakeholders, consequently moving the domain from research into practice.

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34

INTEROPERABLE DIGITAL MEASUREMENT AND MODELLING TO OPTIMISE TENNIS RACKET DESIGN

A case study

Marcus Dunn, Antoine Bacha, and John Hart

Introduction

Since the start of the Open Era in 1968 – the current era of professional tennis – tennis has evolved. In the 1970s, most players would hit the ball with a flat forehand strike or backhand slice [1], so that the ball would reach the net as soon as possible. In the 1990s and early 2000s, topspin groundstrokes were mainly used by clay court players to push their opponents backward with an extremely rounded ball trajectory [1]. By this time, the speed of the game was also increasing, with elite players using a baseline game strategy combined with strong and heavy topspin groundstrokes to win points on clay, acrylic, and grass surfaces [2]. The use of topspin afforded modern elite tennis players the opportunity to play much faster groundstrokes than in the past, whilst still ensuring that the ball remained within court limits [3].

Hitting the ball with topspin induces the Magnus effect, which yields a lift force acting downward on it. This causes the ball to drop sooner than if it were played without topspin, and when combined with greater racket head speed, allows faster strokes to be played within court limits. As such, the ability of players to impart spin on the ball – to increase the distance an opponent must travel or to reduce the time an opponent has to respond to an incoming ball – has become an important skill in the modern game. During flat groundstrokes, the racket follows a predominantly horizontal path during the swing and ball contact phases. For topspin groundstrokes, the racket first moves downward and then upward throughout ball contact. Outbound ball velocity is related to the horizontal velocity of the racket head at impact, whilst its forward rotation (i.e., topspin) is related to the vertical velocity of the racket head at impact [4]. When compared to flat shots, topspin shots comprise a lower horizontal and higher vertical racket head velocity at impact [5]. Therefore, designing the racket so the player can increase the vertical velocity of its head during the swing and ball contact phases of topspin strokes could be a key factor in racket innovation.

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Tennis racket companies constantly evolve their equipment offerings in response to user playing styles. An example of such a company is the family business and Lyon-based brand, Babolat. Babolat's first venture into the world of sport was in 1875 when tennis and badminton were in their infancy. It owed its early success to expertise with a raw material – natural gut – from which the first tennis racket strings were made. After over a century of expertise in the stringing and racket accessories market, Babolat developed and launched its first tennis racket frames in 1994. Since then, the Pure Drive has become an internationally best-selling racket. The Aeropro Drive racket, designed to enhance topspin, was released in 2005. Subsequently in 2012, the first generation of Pure Aero rackets, with a design focussed on improving racket frame aerodynamics, was launched. The world governing body for tennis, the International Tennis Federation (ITF), preserves the nature and integrity of the game by establishing rules and regulations to standardise equipment performance. Material developments have been a major driver of racket innovation in the past. For example, fibre-reinforced composite racket frames are now predominantly used in elite tennis [6] (see Chapter 11 Fibre-Reinforced Composites for Sports Applications for more information on this topic). However, a challenge remains in how to innovate in racket performance while conforming to ITF regulatory constraints [7].

Babolat's traditional racket innovation pathway involved developing a design philosophy and aesthetic based on an existing dataset. Potential racket frame shapes would be assessed from a design perspective, with candidate rackets being prototyped as a first step. Subsequently, prototype rackets would be assessed by players to address key factors such as performance, comfort, and appeal, before a final design was selected. This approach to product development is typical of many sports equipment manufacturers. However, sports equipment manufacturers must continually strive to improve their products owing to the highly competitive marketplace. In understanding that ball spin is an important element of modern tennis, Babolat identified frame aerodynamics as a potential space for racket performance innovation, particularly to increase vertical racket head velocity at impact. It is in this context that Babolat engaged in a research-informed approach to racket innovation. This was achieved by collaborating with the authors of this chapter to re-envisage their racket innovation pathway. A transdisciplinary approach – to measure and analyse motion, model and optimise aerodynamics, and visually communicate performance – was adopted to co-develop a new Pure Aero tennis racket.

Tennis groundstroke racket-ball impacts are high-speed events. The racket undergoes a wide range of accelerations during the swing, impact, and follow-through phases, with impact accelerations propagating differently throughout different sections of the racket. Furthermore, the racket's orientation changes throughout the swing and follow-through phases, and dramatically so during impact, influencing aerodynamic drag experienced by it and how it can be accelerated during the groundstroke. The specific aerodynamic forces acting on the racket during a typical stroke are unknown. It is important to precisely understand racket motion during high-speed impacts, and particularly to understand the influence of technique variation between players. Such understanding will facilitate the aerodynamic optimisation of racket geometry, based on real-world playing data.

In re-envisaging the approach to racket innovation, high-fidelity assessments of racket movement during topspin forehand groundstrokes were first considered. Second, racket models were simulated at precise instants throughout the swing and contact phases using high-fidelity movement data, to allow fluid dynamic optimisation of racket frame cross-sectional geometries. An optimal racket geometry would then be proposed for a future prototype racket design, and subsequently assessed by players for performance, comfort, and appeal factors. Re-envisaging the traditional

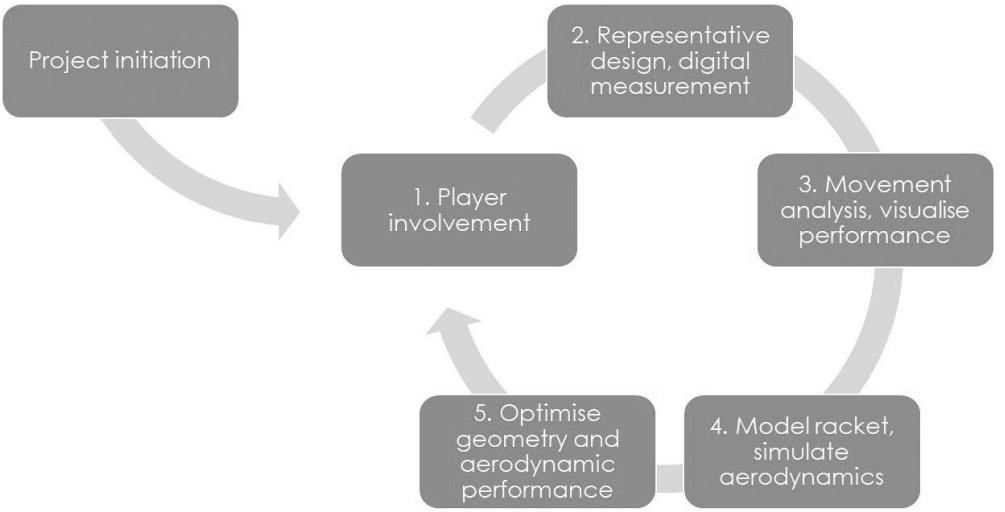

Figure 34.1 Tasks undertaken to achieve project aims: (1) identify different ability and gender tennis players, to capture a range of groundstroke techniques and facilitate boundary conditions for racket optimisation; (2) define representative experimental design and interoperation of digital measurement technologies, to capture high-fidelity racket and ball kinematics; (3) analyse racket kinematics to understand ability and technique, to inform racket optimisation; (4) model and simulate racket aerodynamics to understand groundstroke aerodynamics; and (5) optimise racket cross-sectional geometry to enhance aerodynamic performance, propose candidate geometry for future physical prototype rackets. Authors' own image.

racket innovation pathway as a research-informed approach required the use and interoperation of numerous advanced digital measurement technologies, including high-fidelity motion capture, high-speed videogrammetry, and computational fluid dynamics (CFD). This chapter describes the transdisciplinary co-development of an aerodynamically optimised tennis racket geometry, using multiple interoperable digital measurement technologies. Figure 34.1 outlines the tasks that were undertaken to achieve this.

Player recruitment

It is important to include a range of players (e.g., sex, ability, grip) to characterise different techniques used to perform topspin forehand groundstrokes. Following University Research Ethics Committee approval, male and female players, with a range of player ratings were identified and recruited from local and regional tennis clubs. An initial screening process, whereby player candidates sent videos of their forehand groundstroke playing style via secure internet transfer, was used to aid the recruitment of players with a topspin playing style. Subsequently, four male and two female players of intermediate player rating were identified and recruited. Players were all right-handed, had a Lawn Tennis Association (LTA) rating between 2.2 and 7.2, used Eastern, Full-Western, and Semi-Western grip styles, and used rackets with a mean mass of 321.8 ± 13.0 g (range: 303.1–331.8 g). Players were then invited to a motion capture session, where high-fidelity racket and ball movement data were captured during their topspin forehand groundstrokes.

Digital measurement and interoperable data

Capturing and modelling a full understanding of racket motion and variation in technique across players requires the use of multiple, interoperable digital measurement and modelling systems. High-fidelity three-dimensional motion capture and high-speed camera systems were combined to provide detailed racket and ball information during key groundstroke phases, as well as to provide rich media assets to aid future communication around racket development and innovation. First, a 24-camera motion capture system (Oqus 7+, Qualisys, Sweden), operating at 1,000 Hz and capturing marker positions with sub-millimetre precision, was interfaced with a high-speed monochrome video camera also operating at 1,000 Hz (Miro, Vision Research, USA). Motion capture and high-speed cameras were arranged to focus on the deuce side of an indoor tennis court, inclusive of a full-size net. Figure 34.2 illustrates the experimental setup. To allow capture of representative, cross-court forehand topspin groundstrokes, players were first required to serve from the deuce court. If the ball landed in the opposing deuce court, a ball was projected from a BOLA tennis ball projector back to the player's deuce court. Speed and spin of the BOLA projector were set to allow players to step into the incoming ball and play a cross-court, topspin forehand groundstroke in return. Trials were recorded if the groundstroke landed in the opposing deuce court, and players were permitted as much practice time as needed to accustomise themselves with the experimental setup.

The three-dimensional motion capture system, focussed on the deuce court, was calibrated as described by the supplier, allowing racket markers to be tracked with a precision of 0.2 mm. To calibrate the motion capture and high-speed camera, and to provide interoperable position measurements, intrinsic and extrinsic high-speed camera parameters were first calculated using

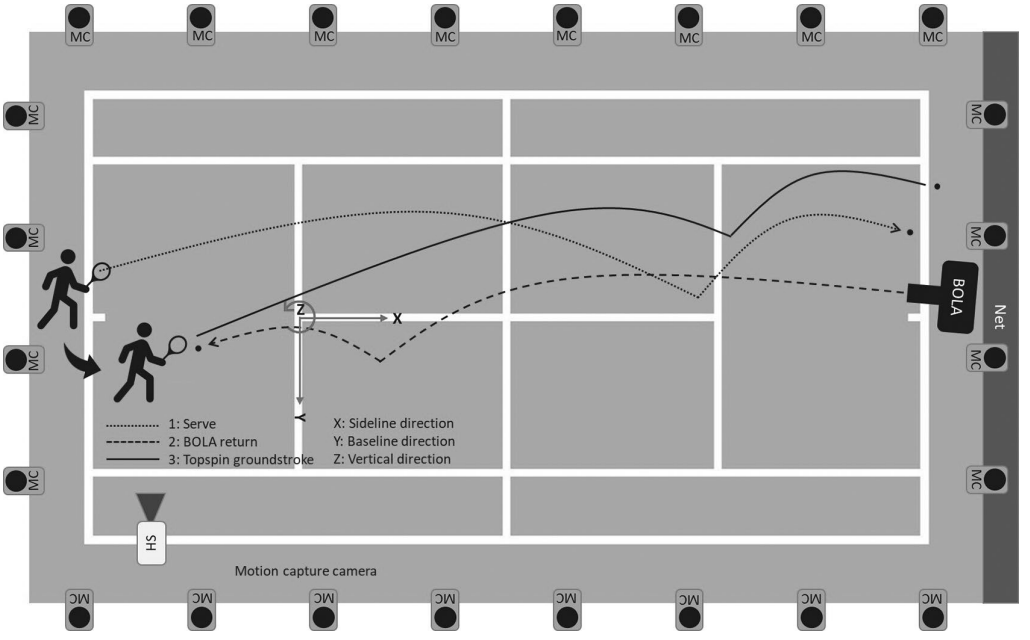

Figure 34.2 Schematic of laboratory setup. Twenty-four motion capture (MC) cameras captured topspin forehand groundstrokes, alongside a single, calibrated, high-speed (HS) camera. Both systems operated at 1,000 Hz. Authors' own image.

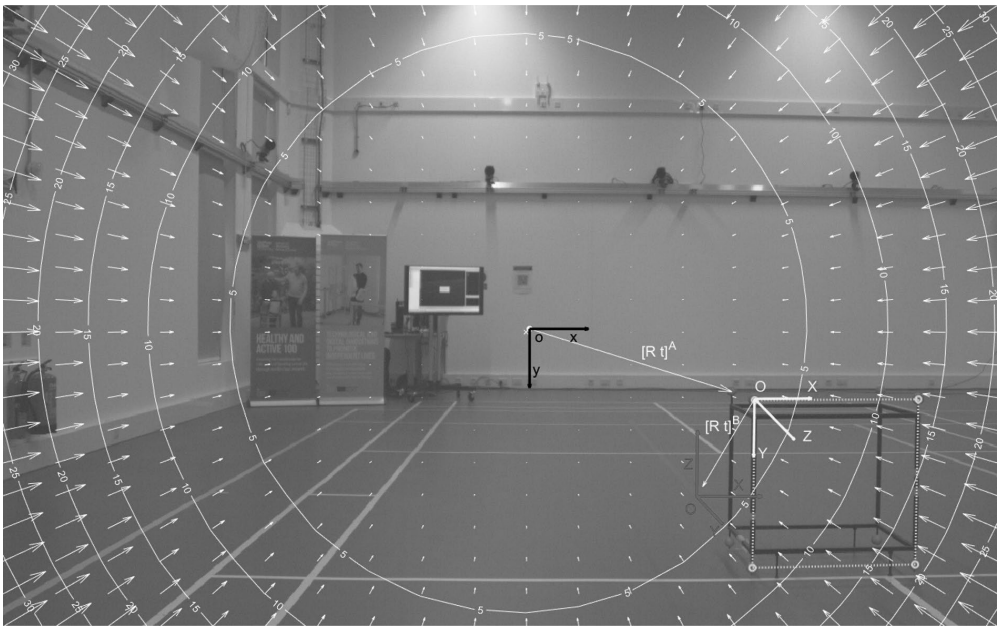

Figure 34.3 High-speed camera image illustrating intrinsic camera parameters (e.g., lens distortion shown as rings and arrows, values in pixels) and the calibration rig used to interface high-speed camera and motion capture data. $[R\ t]^A$ and $[R\ t]^B$ represent homographies between camera (xy), calibration rig (XYZ), and laboratory (XYZ) coordinate systems. Authors' own image.

the planar checkerboard calibration method [8]. When performing the extrinsic calibration of the high-speed camera, a precision-manufactured calibration rig – fitted with large (50 mm) retro-reflective markers – was placed in the motion capture volume (Figure 34.3). Precise, two- and three-dimensional coordinates for each of the retro-reflective markers on the calibration rig were captured by the high-speed camera and motion capture system, respectively, and used to define the extrinsic parameters of the high-speed camera. The use of the calibration rig, which was positioned in front of the high-speed camera, helped to minimise any potential digitising error associated with camera perspective of the motion capture system's calibration frame, which was positioned on the laboratory floor (e.g., Figure 34.3).

The extrinsic high-speed camera parameters (i.e., homography between the calibration rig and high-speed camera coordinate systems) define the transformation matrix $[R\ t]^A$, where R is a 3×3 matrix of direction cosines and t is a 3×1 translation vector (Figure 34.3). This homography allows the transformation of normalised coordinates from the high-speed camera's coordinate system to the calibration rig's coordinate system. In turn, three-dimensional coordinates in the calibration rig coordinate system are transformed into the laboratory coordinate system using $[R\ t]^B$ (Figure 34.3). The definition of transformations $[R\ t]^A$ and $[R\ t]^B$ allows the interoperable exchange of digital information between the high-speed camera and motion capture systems, providing a unique opportunity to measure and visualise tennis racket and ball performance during the swing and impact phases of the groundstroke.

Figure 34.4 demonstrates how digital information, captured by the motion capture system, can be transformed and visualised within a high-speed camera image. Specifically, three-dimensional racket marker positions are used to define the racket bed, and the racket tip marker's resultant

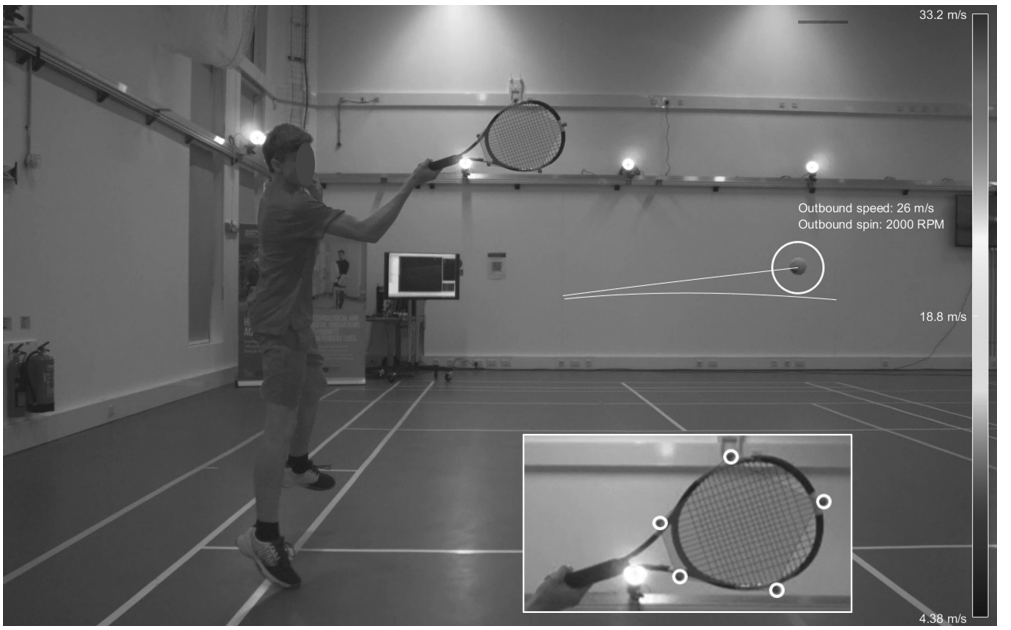

Figure 34.4 High-speed camera visualisation of inbound and outbound ball trajectories and metrics (e.g., speed, spin), and a reprojected, three-dimensional representation of the racket bed, where colour represents racket tip velocity. Racket marker locations are highlighted (white rings) in the zoomed-in image, inset. Authors' own image.

direction velocity is used to indicate its swing speed. The racket bed is then colour-coded accordingly, bounded by maximum and minimum swing-phase velocities. The creation of rich performance visualisations, in both image (e.g., Figure 34.4) and video formats, formed an integral part of player feedback and racket development material.

Upon arrival, players were briefed on the nature of tasks to be completed and were given as much time to warm up and practice as they felt they needed to, to become accustomed to the laboratory setup (e.g., playing balls projected from BOLA, Figure 34.2). Subsequently, each player's racket was weighed using precision scales, and five retro-reflective markers (12 mm diameter) were applied to its tip, throat, and lateral aspects of the head (e.g., Figure 34.4) to define its coordinate system. Following racket preparation, participants undertook groundstroke trials, which followed the task steps previously discussed. In total, ten successful topspin forehand groundstrokes – where strokes played were within the rules of tennis – for six players (n = 60) were recorded. Data for all other strokes were deleted.

Following capture, three-dimensional racket marker trajectories were filtered using a discretised smoothing spline based on generalised cross-validation [9] in MATLAB (2020a, MathWorks, USA). The instant of racket-ball contact was defined by the peak instantaneous velocity (resultant direction) of the racket's tip marker. The start and end of the groundstroke were defined as the racket tip's furthest position from the net (i.e., minimum position in the X direction, Figure 34.2) and highest position during follow-through (i.e., maximum vertical position), respectively. Using these swing phases, relevant racket velocities and orientations were calculated throughout the groundstroke. Inbound and outbound tennis ball trajectories were tracked in high-speed camera footage using a kernelized correlation filter [10] in MATLAB. Pixel-level trajectories were then

fitted with a third-order polynomial to provide a smooth trajectory to extract windowed, ball-centred image sequences. Windowed images were then used to estimate ball centre and radius, using an image differencing technique. Subsequently, localised ball scale information was used to estimate ball velocity, and windowed ball images were used to estimate ball spin (i.e., number of video frames per revolution).

To understand groundstroke technique and playing ability, measures for inbound and outbound ball speed and spin, peak linear and angular racket swing velocity, racket orientation at impact, racket orientation velocity (average racket axial roll) at impact, and racket-ball contact time were extracted. Players were grouped as high, moderate, and low-rated players based on their player rating. A one-way ANOVA with Tukey HSD pairwise comparisons was performed in MATLAB to compare the effects of player ability on dependent variables (alpha set to 0.05). Between-group effect sizes were calculated as $ES_B = \bar{x}_1 - \bar{x}_2 \; / \; S_p$, with effect sizes of 0.2, 0.5 and > 0.8 being considered small, medium, and large effects, respectively [11].

At ball impact, racket orientation was significantly different for different ability players (Table 34.1). *Post-hoc* analysis highlights that this technique difference existed between high and low-rated players (G1–3, Table 34.1). The mean difference in racket orientation at impact (racket's axial roll relative to global vertical) was 8.5°, which corresponds to a large effect size between high and low rated players (Table 34.1). Large and moderate effect sizes for racket orientation at impact were also observed between high- and moderate-rated players (G1–2, Table 34.1), and moderate and low-rated players (G2–3, Table 34.1), respectively. This technique difference highlights that high-level players strike the ball with a more closed racket face (-90° and 0° represent global vertical and horizontal directions, respectively), to facilitate the generation of topspin. Moreover, a moderate effect size for racket orientation velocity (rate of axial roll) during contact was observed between high- and low-rated players (G1–3, Table 34.1), highlighting the importance of racket control when generating topspin during forehand groundstrokes.

Findings demonstrate that players of different level orient and roll the racket differently during topspin forehand groundstrokes, reflecting similar work in this area [12]. Racket orientation will influence the aerodynamic drag it experiences, and in turn, how players move it to generate ball speed and spin during groundstrokes. Understanding the aerodynamic drag experienced by a racket can aid in the optimisation of its design. An optimised racket frame design that minimises the aerodynamic drag experienced by the racket might help players generate greater racket head velocities.

Modelling interoperable data to understand racket aerodynamics

We can understand the aerodynamic performance of an item of sporting equipment through the application of either analytical equations, experimentation, or computational fluid dynamic (CFD) simulation. These methods have individual strengths but also an appropriate time and place for their application. For the aerodynamic analysis and optimisation of a tennis racket frame, the most appropriate approach is the use of CFD for reasons as follows. Although a basic understanding of flow around a racket's sectional profile could be obtained using analytical equations, the flow around a tennis racket is three-dimensional, turbulent, and potentially transitional. This renders analytical approaches impracticable for detailed analysis. Similarly, the use of experimentation was considered unfeasible, as it would have required a complex, time-consuming, and potentially expensive implementation. Experimentation requires the creation of physical prototype models that are suitable for use in a wind tunnel, which can be costly. A greater challenge, however, is affixing a racket to a force measurement system within a wind tunnel in an orientation that is

Table 34.1 Racket and ball kinematics for forehand topspin groundstrokes as a function of playing ability. Authors own table.

Variable	*Group 1 (High)*	*Group 2 (Moderate)*	*Group 3 (Low)*	*P*				*ESB*		
				Main	G1–2	G1–3	G2–3	G1–2	G1–3	G2–3
Inbound ball velocity (m/s)	8.8 ± 0.8	8.7 ± 1.0	8.9 ± 0.5	0.84	0.93	0.99	0.86	0.14	−0.07	−0.21ǂ
Outbound ball velocity (m/s)	26.6 ± 2.8	26.8 ± 3.9	29.0 ± 2.3	0.18	0.98	0.27	0.19	−0.08	−0.95‡	−0.69†
Inbound ball spin (RPM)	2204.3 ± 126.6	2200.4 ± 139.4	2176.9 ± 84.0	0.86	0.99	0.88	0.87	0.03	0.26ǂ	0.20ǂ
Outbound ball spin (RPM)	1699.9 ± 309.7	1619.8 ± 427.4	1696.2 ± 170.7	0.76	0.82	0.99	0.84	0.21ǂ	0.01	−0.23ǂ
Peak linear swing velocity (m/s)	29.1 ± 1.7	29.4 ± 3.0	28.8 ± 1.7	0.77	0.92	0.97	0.78	−0.15	0.15	0.26ǂ
Peak angular swing velocity (°/s)	2263.5 ± 218.9	2652.9 ± 1087.1	2285.7 ± 534.6	0.34	0.47	0.99	0.51	−0.50†	−0.05	0.43ǂ
Racket orientation at impact (°)	−82.6 ± 4.1	−87.6 ± 6.9	−91.1 ± 3.7	0.01*	0.06	0.01*	0.25	0.90‡	2.19‡	0.62†
Racket orientation velocity (°/s)	−1461.5 ± 198.4	−1393.4 ± 334.4	−1331.8 ± 165.2	0.62	0.79	0.59	0.83	−0.25ǂ	−0.71†	−0.23ǂ
Contact time (ms)	5.2 ± 1.5	5.8 ± 2.8	5.4 ± 0.8	0.74	0.77	0.98	0.89	−0.27ǂ	−0.17	0.19

*Significant difference (e.g., $P < 0.05$). ǂ, † and ‡ are small, moderate, and large effect sizes, respectively.

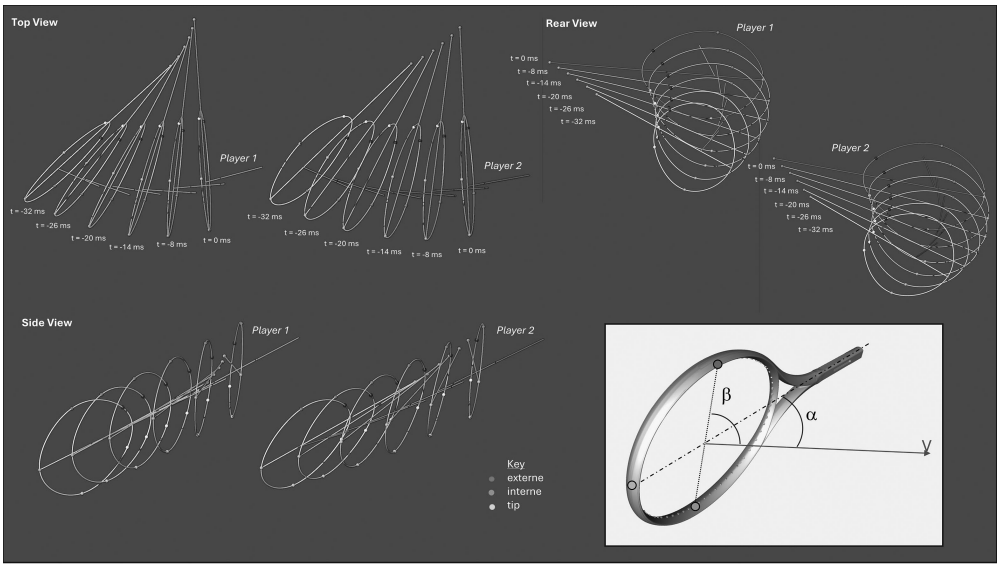

Figure 34.5 Three-dimensional model of the racket used in computational fluid dynamics optimisation of distinct groundstroke techniques. Authors' own image.

aligned with the swung direction. The orientation of the racket changes throughout the swing in a complex oblique manner, with variation in both axial roll (β) and yaw (α). This can be seen in Figure 34.5, which shows player data used here in the aerodynamic optimization of the racket during the 32 ms preceding ball impact. The use of CFD to analyse racket designs does not require the creation of physical models and allows the racket to be placed within a virtual environment in any desired orientation. By using CFD it is possible to gain rapid insight into the flow behaviour and performance of designs, allowing a greater range of designs to be assessed than may be possible through experimentation. The workings and detailed considerations of CFD will not be discussed here but are discussed at length in Chapters 3 and 4.

CFD was therefore applied to aerodynamically optimise the racket frame, but in this optimisation, other design constraints and parameters, beyond aerodynamics, must also be considered. These may include structural (robustness, dynamic response), manufacturing, and aesthetic considerations. Although design for structure and manufacturing may seem obvious considerations, aesthetics are equally of significant importance to sporting goods brands. The aesthetic considerations go beyond imagery and colour but include geometric forms and features within the overall design, which can also be linked to a brand's identity. This was the case here, where the Babolat racket is synonymous with its trapezoidal cross-sectional forms (e.g., Figure 34.6.a), and a requirement of the design optimisation was that the trapezoidal identity should be preserved. For an engineer tasked with aerodynamically optimising geometry, such a constraint can create an interesting challenge! To further complicate the trapezoidal optimisation, as has been shown in Figure 34.5 and discussed, racket orientation varies throughout the swing. This means that the actual cross-sectional form presented to the direction of motion changes throughout the swing, is not a symmetrical trapezoid, and varies between the upper and lower halves of the racket frame. This again is where the use of CFD becomes so effective. Working through multiple design configurations, it is possible to quickly identify the subtle geometric alterations that can result in significant drag reduction.

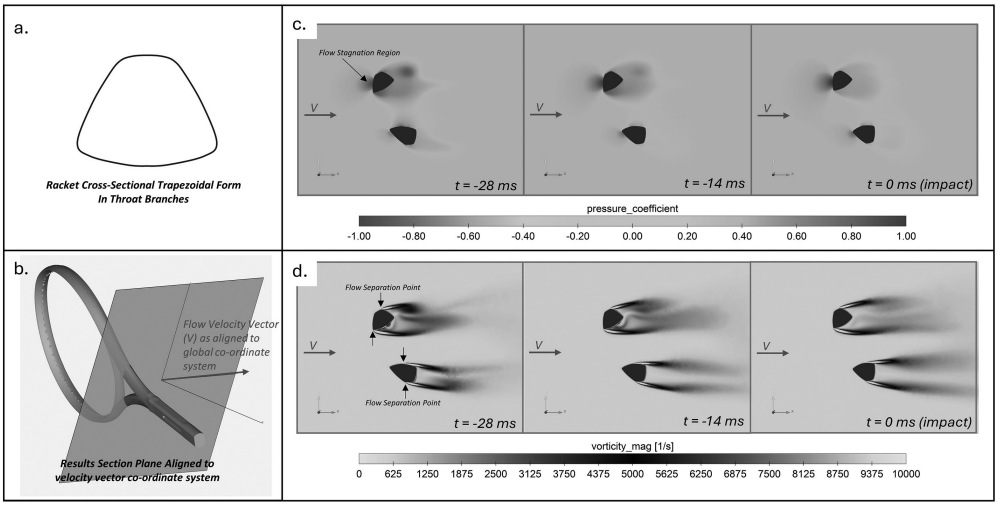

Figure 34.6 (a) Trapezoidal cross-sectional form of the racket throat branch. (b) Pure Aero 2019 tennis racket, showing section plane aligned to racket air flow (*V*) velocity vector used in visualisation of results. (c) Contours of pressure coefficient visualised on section plane at −28 ms, −14 ms, and 0 ms time points. (d) Contours of vorticity magnitude revealing flow separation around racket throat cross-sections. It can be seen in these images how the presented sectional form aligned to the velocity vector changes throughout the swing and differs between the upper and lower branches, due to the racket's axial roll and yaw. Authors' own image.

CFD here was used to understand and optimise the aerodynamic performance of the racket frame in the 32 ms prior to impact, at which point it is moving at over 20 m/s. Five equally spaced time points prior to ball impact were investigated across this time frame, its period of maximum acceleration. The starting point for this optimisation was the analysis of the current Pure Aero 2019 racket. The geometry of this racket was provided in a CAD format by the commercial partner. CFD allowed an understanding to be obtained of not only the aerodynamic drag but also variations in lift and side force. This provided information regarding the aerodynamic stability of the design across the swing.

Typical results obtained of the flow field around a cross-section of the racket frame are shown in Figure 34.6.c. (pressure coefficient) and Figure 34.6.d (vorticity magnitude). Visualisation of the pressure coefficient reveals sectional flow stagnation and potential flow separation both of which cause increased aerodynamic resistance. The visualisation of the vorticity magnitude then allows confirmation of flow separation but also creates an understanding of how wake structures may interact between the upper and lower sections of the racket. The use of interoperable digital data, captured by high-fidelity measurement technologies and modelled using computational fluid dynamics, has allowed the identification of potential geometric improvements that reduce the occurrence of flow stagnation and sudden flow separations around the cross-sections. These were implemented and subsequently analysed in an iterative CAD-CFD cycle. Currently, it is not possible to reveal the geometric alterations, ahead of the racket's release on the market. However, once an acceptable and optimised trapezoidal geometry was obtained that considered both aerodynamics and aesthetics, it was passed back to the commercial partner. They will now proceed with the opening of moulds to create prototype rackets with full functionality. These rackets will then be used in player testing to quantify and analyse performance, capturing racket angle and swing

speed, alongside user feedback. This will bring the cycle of digital measurement and modelling of interoperable data to optimise racket design full circle.

Summary

Generating ball spin is an important skill in modern tennis. This chapter describes how interoperable data, captured by high-fidelity measurement technologies, were analysed, visualised, and modelled to develop a racket geometry intended to enhance aerodynamic performance during topspin forehand groundstrokes. The analysis of topspin forehand groundstrokes revealed that players of different level orient and roll the racket differently, highlighting the need to consider aerodynamic drag during these high-speed impacts. Racket geometry optimisation was necessary because the cross-sectional form presented to the direction of motion varies throughout the swing. As such, aerodynamic drag varies at different locations along the racket's frame as well as throughout the swing. An optimised racket cross-section geometry, intended to reduce drag during groundstrokes, was developed and will proceed to prototype testing, where performance will be quantified alongside user feedback. The use of high-fidelity, interoperable data – captured and exchanged between a range of digital technologies – was crucial to informing and understanding technique, creating rich player feedback visualisations, modelling racket aerodynamics, and optimising racket cross-sectional geometries for use in future prototype rackets. Furthermore, the interoperable data techniques presented here could be applied to the development of other sporting goods.

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35
BIG DATA IN SPORT
Case studies in football and tennis

Conor Wall, Andrew Hext, and Simon Choppin

Introduction

In recent years, big data analytics has become an essential part of many individual and team sports [1]. The Oakland Athletics baseball team, led by General Manager Billy Beane, was one of the first notable examples of data analytics having a significant impact on team sports. In 2001, Billy Beane and Paul DePodesta used a data-driven approach to player acquisition and in-game management to lead the Oakland Athletics to a 20-game win streak, which at the time was the Major League Baseball (MLB) record. The success and potential of this approach led to the adoption of a 'data-driven' philosophy in other team sports [2].

In professional football, big data analytics is being used to reassess and reimagine player recruitment, injury prevention, and performance analysis. This is part of the constant drive to optimise on-field performance [3]. As professional football clubs rely on revenue from league and cup competitions, stadium attendance, and commercial sponsorships, solutions that improve on-field and off-field performance must be explored [4]. In this chapter, we examine technological advancements in data collection and analysis and the significant influence of big data on two prominent sports: football and tennis. These sports were chosen for their global recognition and the detailed complexities inherent in their gameplay. Football, with its emphasis on team dynamics, player recruitment, and performance analysis, provides insight into the application of data-driven decision-making in team sports. In contrast, tennis, utilising technologies like Hawk-Eye, demonstrates the predictive potential that big data offers to individual sports.

Team sports: Technological advancements in professional football analytics

In the highly dynamic arena of team sports, in-game performance measurement is difficult. It is not trivial to determine individual player work rates or to compare team formations for strategic weaknesses. For that reason, team sports have benefited greatly from the introduction of player tracking and big data analytics. In the ever-evolving world of professional football, the integration of big data has revolutionised the way clubs operate, strategise, and perform. Leveraging vast amounts of information has become a game-changer, enabling teams to gain a competitive edge both on

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and off the pitch. Modern football relies on advanced analytical techniques that necessitate the collection of vast amounts of data in a reliable, efficient, and comprehensive manner. Achieving this has been made possible through technological advancements, with sensor-based and video-based systems emerging as the predominant solutions in the professional arena [5].

Innovations in technology for player tracking

Video-based systems

Video-based systems use multiple cameras placed around the stadium to record the game from various angles (e.g., Figure 35.1). Using video data, player tracking and identification are performed using image processing techniques [5]. The recording of positional data during the 2006 World Cup final between Italy and France was one of the first notable examples of this technique being used. The positional data of the 22 players were captured at 25 Hz using multiple cameras, imaging processing techniques, and further processing was used to calculate the XY position for each player every second [6]. This method has grown in popularity among professional football clubs around the world, with the English Premier League adopting it using tools such as Opta Sports, WyScout, PA Sport-Actim Index, and Prozone [7–9].

Figure 35.1 Match Analysis Portable K2 Panoramic Video Camera System. Source: [10].

Figure 35.2 Austrian U21 National Football Team Training with Catapult Wearables. Source: [14].

Wearable-based systems

Various sensors, including satellite-based navigation systems, Inertial Measurement Units (IMUs), and heart rate sensors, are used in wearable-based systems. The player's speed, distance travelled, positional information, body movements, and heart rate are all measured by these low-cost, simple-to-use devices [11]. This data is essential for providing various methods of data analysis to improve different facets of the game, such as training preparation, as coaches can more accurately assess a player's overall fitness and subsequently modify training sessions as a result [12]. Wearable-based systems are now commonly provided in the form of a vest, such as Catapult Sports (South Carolina, USA) (Figure 35.2) and STATSports (Newry, N. Ireland) [13].

Data metrics in football analysis

Key aspects of professional football are examined using a wide range of data metrics. Depending on the service and wearable type, these metrics can change, but they usually revolve around a core set of principles.

Video-based system data metrics

As an example, to demonstrate what data metrics are typically analysed by video-based systems, Opta Sports showcases a total of 67 event-based metrics that are generated, separated into categories of goals and shots, passes, touches, other, duels, goalkeeper, and expected goals and expected assists. Table 35.1 summarises some of the key metrics for each category [7]. These metrics are primarily used for scouting, player recruitment, and performance analysis. To demonstrate the

importance of such metrics, an example to focus on is the 'key pass' metric. It is used to record if a final pass or cross leads to the recipient having an attempt at goal. This is particularly useful as they can be aggregated over a period for any footballer, to provide a means of measuring the overall passing ability and creativity of footballers, without relying on 'end-product' statistics such as assists.

Wearable-based system data metrics

With wearable technology, several metrics can be generated that can be used for the monitoring of a player's fitness and injury prevention. Table 35.2 lists the metrics that are generated by Catapult and STATSports wearable sensors [15,16].

Advanced metrics

Many of the metrics generated by video-based systems existed before this technology was invented. However, because of the invention and use of video-based technology, more advanced metrics have been developed. Expected goals (xG) and expected assists (xA) are two notable examples.

The xG metric is used to estimate the likelihood of a shot being converted to a goal based on various influencing factors, which can differ depending on the xG model used [17]. The Opta

Table 35.1 Opta Sports Metrics by Event Category in Football

Event Category	*Example Metrics*
Goals and shots	Goal/own goal, shot on target, shot off target, blocked shot
Passes	Key pass, pass completion, chances created, cross, pass
Touches	Touches, touch/unsuccessful touch, dribbles/take-ons, missed tackles
Expected goals and expected assists	Expected goals, expected assists
Duels	Duel, successful take-on/dribble, foul won, aerial challenge won
Goalkeeper	Save, clean sheet, catch, penalty faced
Other	Recovery, shield ball out of play, hit woodwork, error

Table 35.2 Catapult and STATSports Wearable Metrics

Wearable Brand	*Metrics*
Catapult	Distance, sprint distance, power plays, energy, impacts, sprints, accelerations, decelerations, HR load, % time in red zone
Statssports	Accelerations and decelerations, average heart rate, average pace, calories, distance, high-speed running, high-intensity distance, intensity, max heart rate, max speed, pro score, sprints, strain, step balance, time in red zone, total distance

Sports xG model is calculated using a logistic regression model with the following regression inputs: passage of play, assist type, post-take-on, rebound, header, distance to goal, visible angle to goal, 1v1, big chance, and competition adjustments [7]. The xG value for each specific shot ranges from 0 to 1, with the closer to 1, the higher the probability of a goal. To calculate the total xG for a team over a specific period, whether that be for an entire match or season, the xG for each shot in that period is summed. Figure 35.3 below illustrates the 2012 Champions League Final between Bayern Munich vs Chelsea, which resulted in Chelsea winning in extra time, despite Bayern Munich outperforming Chelsea in xG during that specific match.

The xA metric is used to calculate the likelihood that a pass will result in a goal assist [18]. Similarly to the xG metric, the xA metric is calculated using various factors. The Opta Sports xA model is calculated using pass type, pass endpoint, and pass length [7]. Like the xG metric, the xA value also ranges from 0 to 1; the closer the value is to 1, the higher the probability of a pass leading to a goal. Both metrics are particularly useful for informing coaches and scouts to assess player performance. For example, the key metric xG allows analysts to make determinations about a player's ability beyond using the goals scored and goals conceded metric. Determining the xG for each shot, and subsequently for each save, can more accurately determine how well the striker and goalkeeper have done for that event [17]. The advanced metric xA has also been used to evaluate a player's ability, as it assigns a fairer level of credit to a player's creativity than using only the goal assist metric, which can be dependent on whether the pass results in a goal, rather than if the quality of the pass was high or not [17].

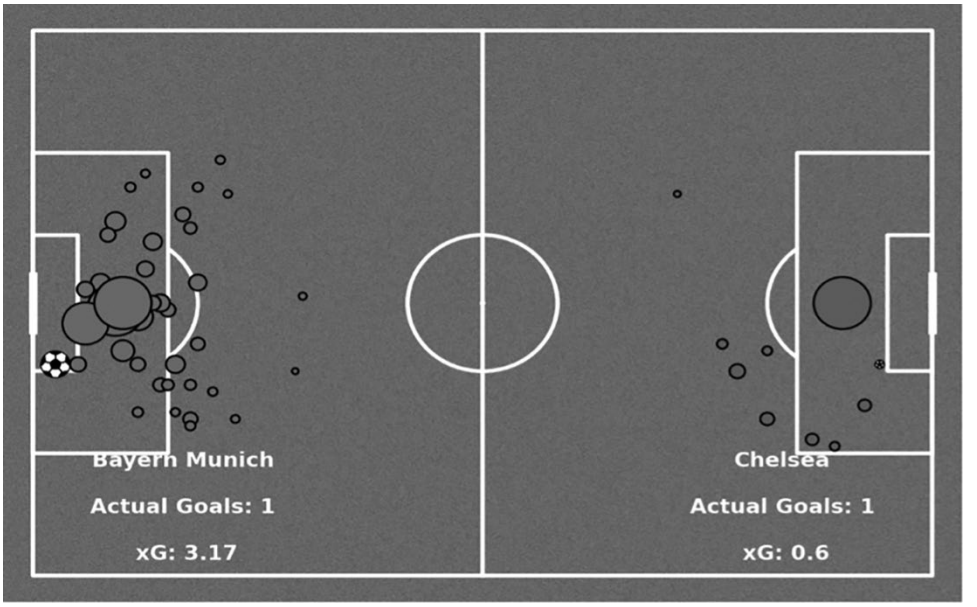

Figure 35.3 xG analysis of the 2012 Champions League Final between Bayern Munich (left) and Chelsea (right). The total xG per team is summed over the match using the xG of each shot a team takes. The circles for the respective team illustrate each shot, with the size of the circle demonstrating the xG value. The larger the circle the higher the xG.

Challenges of data protection and accessibility in UK football

Concerning legal ownership of event data, in the UK there are no laws regarding the protection of 'raw data' unless said data is listed as confidential. Such data is difficult to protect, even under the General Data Protection Regulation (GDPR), as any observer could make notes on any metrics they are interested in. For example, an observer could note how many passes a player makes during a match or how many goals have been scored.

Structured event data is defined as data that is collected and processed, such as being added to a structured database. Even such data and the certain systems and processes that facilitate the structured data are difficult to protect, as there are gaps in the laws that currently exist even under UK intellectual property law. This provides a problem for leagues, professional football clubs, and analysts to keep such data and the associated analysis techniques from being exploited to maintain a competitive advantage [19]. However, concerning wearable data that is collected by professional clubs in the UK, the General Data Protection Regulation (GDPR) requires that any personal data collected from wearables must be protected and processed only for specific purposes. In the case of wearable data in professional football, the primary purpose for collecting and processing the data is to improve the team's performance, unless explicit consent is obtained in advance in writing from the players for other specific purposes, such as research [20].

Regarding data accessibility, any football club can use the aforementioned video-based services like Opta Sports and Prozone, which offer analytical data. However, smaller clubs and countries may not have the funds or resources to access statistical analysis on their teams and players due to the high cost of such systems. To address this challenge, some initiatives are being made to ensure more data availability, with a prominent example being the FIFA football data ecosystem. This initiative by FIFA is a platform that provides event and positional data, which is generated using an optimal tracking system. This platform is made available to all teams, coaches, players, and the media participating in the relevant competition, which so far includes the FIFA Arab Cup 2021, the FIFA Club World Cup in 2019 and 2021, and the FIFA World Cup 2022 [21].

Access to wearable data is dictated by which footballing organisations can afford to purchase and use the wearable technology. Such data generated by a professional football club can only, therefore, be accessed by staff and players of that club and not anyone else, again to maintain competitive advantage.

Applications of Football Analytics and Insights

Scouting and player recruitment

The process of scouting players has changed considerably over time. Historically, a club would hire a scout to attend football matches in person to assess performance and determine if a footballer is worth acquiring. However, with the evolution of technology, scouts adapted, using video recordings of play to make their jobs more efficient [22]. Today, professional scouts employ the use of sophisticated tools such as video footage, machine learning, and human input to create a vast array of metrics to better inform them about a player's performance [23].

FC Copenhagen in the Danish Superliga adopts the use of the scouting tool WyScout to look out for metrics that 'fit' the criteria of the type of player they wish to acquire. For instance, a club may require a centre midfielder who specialises in a certain role, such as being a 'deep-lying playmaker.' As a result, key metrics that show whether a player can effectively fill that role will be assessed in detail. Table 35.1 showcases some of the key metrics that could be used for this

purpose. If a player in the database matches these set parameters, the system would then flag the scout to review the player by reviewing the footage directly. The scout would then watch the player in person and following the approval of the scout, the technical director and manager would then make their assessment [24]. This method of scouting is more efficient and cost-effective as it limits the amount of time spent by scouts identifying potential players to monitor and removes the physical obstacles that were once required 25].

Injury prevention

Players sustaining injuries has been a consistent issue throughout the history of professional football that clubs inevitably must deal with, which typically negatively affects the trajectory of both the team's performance in competitions and the player's performance in general [26]. Until recently in professional football, little could be done regarding analysis to prevent injuries. However, with wearable-based systems, injury prevention analysis using big data has gained usage in professional football. The injury risk for professional footballers is substantially high, which ranges from 2.48 to 9.4 injuries per 1,000 h of exposure [27]. Several key aspects can be analysed to determine the risk of injury. The first is player position, which has been shown to demonstrate a link to injury risk, with forwards having a higher risk and goalkeepers having the lowest risk [28]. Another key aspect can be the type of system that a certain football team plays under. The concept of 'pressing' is a tactic where players work together to regain possession as high up the field as they can. This high-intensity movement is now a commonly used tactic, with teams like Liverpool F.C. in the English Premier League being a key example of its success [29]. Data obtained from Catapult wearables, however, revealed that pressing can increase the risk of injury due to the requirement of high-speed running and sprint running (SR) [30]. With metrics such as those in Table 35.2, coaching staff and analysts can track a player's fitness to help prevent fatigue-related injuries and configure training sessions to suit a player based on their current fitness [31].

To determine a football player's fitness level, various calculations can be utilised, including the VO2 max, heart rate reserve (HRR), maximal lactate steady state (MLSS), and power output. The VO2 max represents the maximum amount of oxygen a player can consume during exercise and is an indicator of their aerobic endurance [32]. The HRR, which is the difference between a player's resting and maximum heart rate, is used to assess the intensity of their workout [33]. The MLSS measures the point at which lactate levels in the blood begin to rapidly increase during exercise [34]. Finally, power output is used to evaluate a player's ability to produce force quickly, an important factor in activities such as jumping, sprinting, and tackling [35].

Performance analysis

Football performance analysis is essential to season-long improvement and involves analysing a team, their players, their opposition, and the opposition's players. For example, some of the underlying metrics that indicate the success of a football team are having the ball in midfield and the offensive zone, making successful passes, scoring successful passes, and having a high passing success percentage [36]. Using this information, coaches can work with the team to focus on various areas of improvement, which, for example, could be improving a low pass success percentage that could indicate a need to work on passing drills in training. Similarly, a player may need to improve their fitness if they frequently cover less ground than their teammates. The fundamental approaches to performance analysis have undergone significant change over time. Traditional

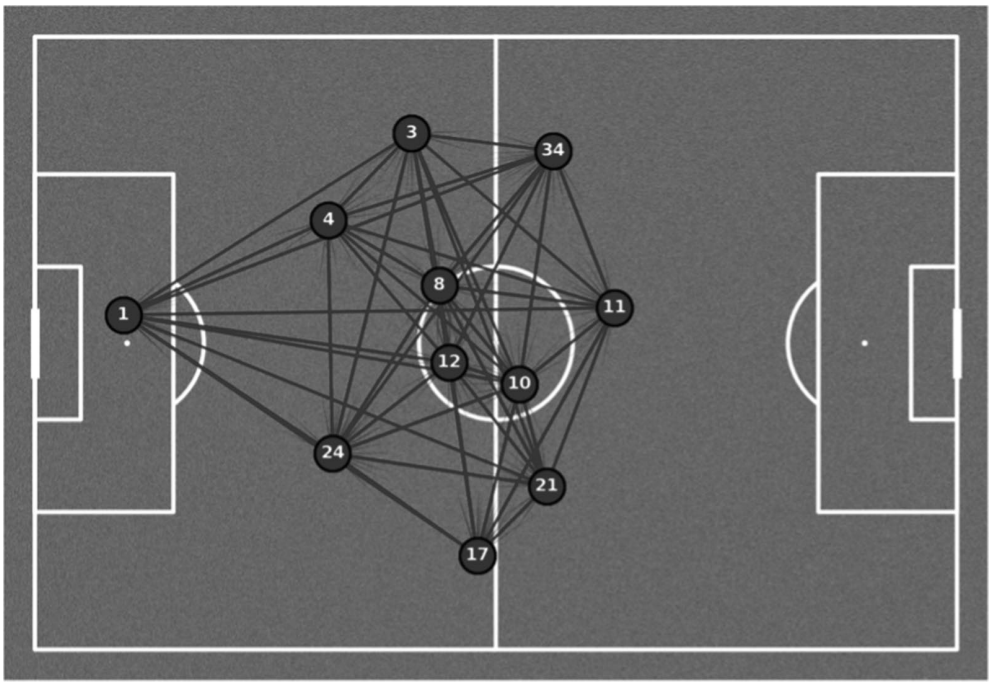

Figure 35.4 Chelsea Passing Network (2012 Champions League Final) [37].

performance analysis techniques involved manual note-taking in person and later progressed to note-taking while watching videos of matches (Figure 35.4).

Today, data gathered and calculated by video-based systems like Prozone are used for performance analysis, and data-driven analysis is subsequently carried out by analysts within the football club. These techniques can be used to analyse performance indicators that show why and how a team is successful or unsuccessful during games. There are several metrics used as performance indicators for both players and overall team performance, of which Table 35.1 showcases those that are analysed by Opta Sports.

Using a series of logistical regressions, a study [38] of 11,032 team-match observations aimed to analyse which performance indicators are indicative of a team being likely to achieve promotion. Regarding defensive statistics, it was found that teams that demonstrated higher levels of defensive contributions such as blocks, clearances, and goalkeeper saves were less likely to achieve promotion. This indicator could, however, be more associated with a team's quality, rather than how it is set up tactically. Set-play statistics, such as converting penalties and scoring goals from corner-kicks, improved promotion probability by 37% and 35% respectively. Regarding open play statistics, penetrative passes and through balls were shown to increase the odds of promotion by 14%. This analysis demonstrates how professional football clubs exploring these types of performance indicators through a series of analyses can put measures into place to implement positive changes to their on-field performance, whether that be through coaching these methods in training or hiring additional coaching staff.

Individual sport: Hawk-Eye data and tennis analytics

Tennis was a relatively early adopter of performance measurement, and large datasets have been available to (a select) number of researchers for many years. As such, this is a good sport to explore how large, rich datasets can be utilised for sophisticated exercises in machine learning, prediction and classification.

Technology's progression has greatly improved our ability to measure the game of tennis. While early scientific work focused on physical interactions of the ball and racket in isolation or on player movement in controlled conditions, the real revolution came from the ability to measure the game during competition. In the modern era, technology allows sophisticated ball and player tracking during competition conditions. This data is collected routinely as part of the Hawk-Eye™ Electronic Line-calling System (ELS), which is used at many competitions and has given players the ability to challenge line-calling decisions. The camera-based system automatically detects the ball and players during play and calculates their position in relation to the court—giving metrics such as velocities, bounce locations and flight trajectories.

While primarily adopted for refereeing the game, the high level of accuracy and large amount of data created by the Hawk-Eye system has provided research opportunities in various applications. Sports engineers have used Hawk-Eye data to characterise ball degradation events [38], investigate the effect of ball wear on ball aerodynamics [39], and measure the speed of tennis court surfaces [40]. While traditionally investigated in controlled test environments, the sheer volume of Hawk-Eye data greatly increases the available statistical power compared to laboratory experiments. This enables subtle effects to be explored in real-world environments – despite a relative lack of precision [39].

Outside the world of rule and equipment governance, Hawk-Eye data has been used for training enhancement, athlete development, and match strategy. Such research can be broadly categorised into three domains: *profiling*, *modelling*, and *prediction*. For example, research into *profiling* has used Hawk-Eye-derived metrics to differentiate between top- and bottom-ranked players [41], understand external match workloads [42], investigate serve effectiveness [43], and differentiate between movement styles [44] by detecting and classifying changes in direction using Hawk-Eye's two-dimensional player tracking data [45]. Similarly, *profiling* research has used Hawk-Eye data to group players based on anthropometric, individual, and match features [46] and identify player styles based on shot selection [47,48]. In the case of the latter, a fundamental stage of such a workflow is based on *modelling* shot types. In this area, researchers have used unsupervised machine learning approaches – such as *k-means* and *multi-stage functional data clustering* – to move away from traditional descriptive shot labels (e.g., flat/slice, down the line/crosscourt, and short/deep) to a data-driven identification of shot types [47,49].

Finally, Hawk-Eye data has been used in various *prediction* tasks. For example, to predict the next serve [47], the next shot location [50,51], -type [52], and -outcome i.e., winner, error, or return [53], the within-point win probability [48], and the overall point outcome [54]. Note that both *profiling* and *modelling* research form a key part of *prediction* tasks as the inclusion of player styles and shot types increases prediction accuracy. Having provided an overview of how Hawk-Eye data has been used in different domains of tennis research, this section provides a more detailed example focusing on its application to understanding the tennis serve: one of the most important elements of tennis [55]. Here, we focus on two different perspectives for analysing the serve: the first uses *point outcome* (win or lose) as the measure of serve effectiveness, and the second uses *shot outcome* (returned or not returned).

Point outcome

The first large-scale study of the tennis serve using Hawk-Eye data was conducted by Mecheri et al. [56]. Their research examined the relationships between various serve attributes (e.g., speed, spin, non-discrete serve location, and serve impacts) and winning-point probabilities. While these questions had previously been investigated using notational analyses – for example, see Gillet et al. [57] and O'Donoghue & Brown [58] – this study provided a more robust methodology – specifically with respect to the sample size and diversity. The dataset, which included 262,596 points from 1,729 matches, was collected from 87 tournaments between 2003 and 2008. Their findings confirmed existing beliefs from notational analyses and demonstrated how Hawk-Eye data could be processed to provide a much higher volume of objective information on tennis performance. Still, by setting out to compare their results with those from notational analyses, the dataset was stratified by first and second service, and each attribute of the serve was investigated independently. An advantage of Hawk-Eye data in tennis is that it affords more sophisticated analyses. Analyses that consider all attributes of the serve simultaneously and thus can move beyond traditional shot categories of first and second serve. For example, we can categorise according to shot speed, placement, and trajectory independently of whether they were first or second serves.

Such an approach is best demonstrated by Kovalchik and Reid [49], who used multi-stage functional data clustering – an unsupervised machine learning approach – in combination with Hawk-Eye data collected at the Australian Open (2015–2017, 385,159 shots) to build a data-driven shot taxonomy. Their approach used location features (ball impact and landing), shape features (the coefficients of 3rd order polynomials), and speed features (impact speed and post-bounce speed) to cluster shot trajectories into shot types. In the case of the serve, Kovalchik and Reid [49] identified 26 serve types for male players (deuce court, n = 13; adv. court, n = 13) and 32 serve types for female players (deuce court, n = 17; adv. court, n = 15). Notably, serve clusters with the same speed label had different shape features, location features, and frequency and win associations, demonstrating how considering the serve's attributes collectively better captures the complexity of the serve and, in turn, enhances the utility of the findings. For example, such a shot taxonomy aids coaches in designing practice that allows players to rehearse shot types according to their frequency, win association, or both [49].

Shot outcome

An alternative perspective for using Hawk-Eye data to investigate the serve is to use serve outcome as a measure of serve effectiveness. One could argue that focusing on the outcome of the serve in isolation is a more theoretically correct approach. When there is more than one shot in the rally, can we exclusively relate the type of serve – and its associated attributes – to the outcome of the point? Especially when we know that the advantage of the serve decreases as the rally length increases [58].

The most notable work in this area was undertaken by Whiteside and Reid [59]. Using spatiotemporal features (impact location, speed, projection angles, landing location, and relative player locations) and contextual features (score) extracted from a Hawk-Eye dataset collected from the 2012 to 2015 Australian Open (25,680 shots), Whiteside and Reid [59] sought to determine the features of an ideal serve in men's professional tennis by constructing a classification tree model that predicted one of two serve outcomes: 'ace' or 'returned in-play'. The classification tree revealed that the serve direction – relative to the receiver, the ball's landing proximity to the nearest service box line, and the serve speed classified aces with an accuracy of 87.0%. Hitting aces appeared more contingent on accuracy than speed, with serves directed >5.9° from the returner and landing <15.3 cm from a service box line most indicative of an ace [59]. While this analysis provides

coaches with empirical data on target locations that are representative of aces, the analysis does not provide a complete picture of how serve characteristics impact whether the shot is returned. Specifically, because the outcome variables used in Whiteside and Reid [59] represent the opposite end of the 'serve return' spectrum. As well as aces, 'not returned' data also includes serves where the receiver makes a shot but the ball lands outside of the court boundary or hits the net. Indeed, when replicating the classification tree method on a dataset where the 'not returned' data included all these scenarios, we found that the same model only had a prediction accuracy of 41% [60]. For this reason, in this chapter, we have explored moving towards a more probabilistic framework for investigating whether a serve is returned or not.

In our approach, we first used the clustering approach described by Wei et al. [47], to assign 39,846 serves[1] to one of 80 clusters (deuce court = 40, ad court = 40). This clustering approach uses *k*-means clustering on ball location and velocity features uniformly sampled from the time when the ball is struck by the racket to the time when the ball bounces. For each cluster, we then calculated the proportion of serves that were returned. The return probabilities of our serve clusters ranged from 21.9% to 88.2% (Figure 35.5). Such a spectrum demonstrates the benefit of moving towards a probabilistic framework, as no cluster had a 0 or 100% return probability. Secondly, we created and refined a multiple regression model of return probability using the serves that represented the centroid of each cluster. Our model showed that the serve speed, serve direction – relative to the receiver, the ball's landing proximity – relative to the service box's centre service line and sideline, and the interaction between the ball speed and serve angle accounted for 92.6% of the variance in the deuce court return probability, and 93.3% in the ad court return probability. In contrast to Whiteside and Reid [59], the probability of all serves being returned was more contingent on speed than accuracy.

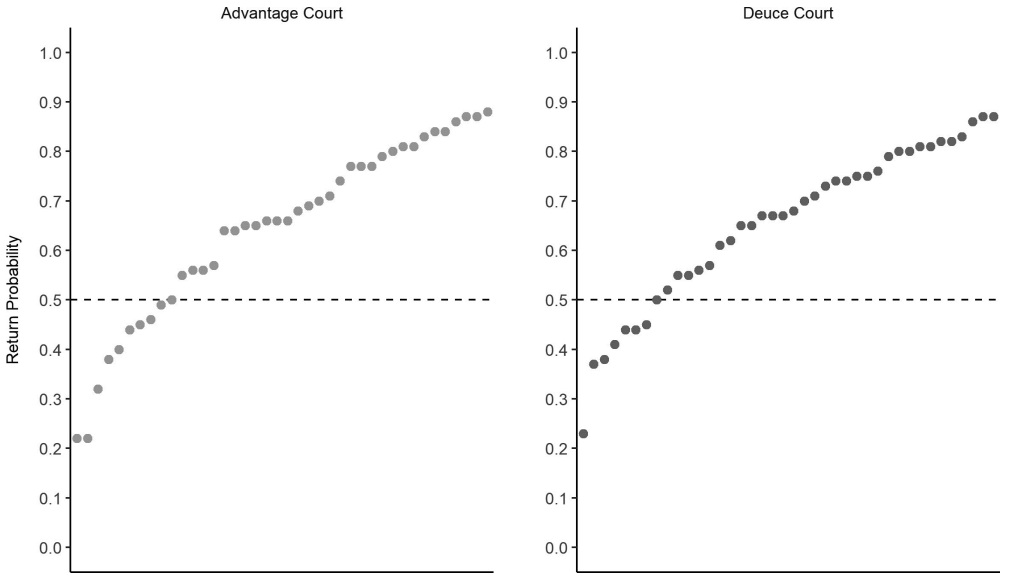

Figure 35.5 The graph illustrates the range of return probabilities for each cluster (represented by a single data point), spanning from 21.9% to 88.2%. The findings emphasise the nuanced role of various serve attributes in determining the likelihood of a return, supporting the shift towards a probabilistic understanding of serve effectiveness. Authors' own image.

While such an analysis approach is more successful for modelling whether any serve is returned or not returned, the approach is not without its limitations. For example, by only using the centroid of each cluster in the model, we lost the granularity of the dataset, i.e., we reduced our dataset from 39,846 to 80 serves. For this reason, we have recently explored using fixed radius nearest neighbour search methods to estimate the probability that each serve in the dataset is returned [61]. Here, for each serve, we find the number of serves that are within a specified distance of the query serve in the serve feature space. As illustrated in Figure 35.6, by not clustering the serve trajectories, our spectrum of return probability retains the granularity of the dataset. In future, these probabilities could be used for modelling.

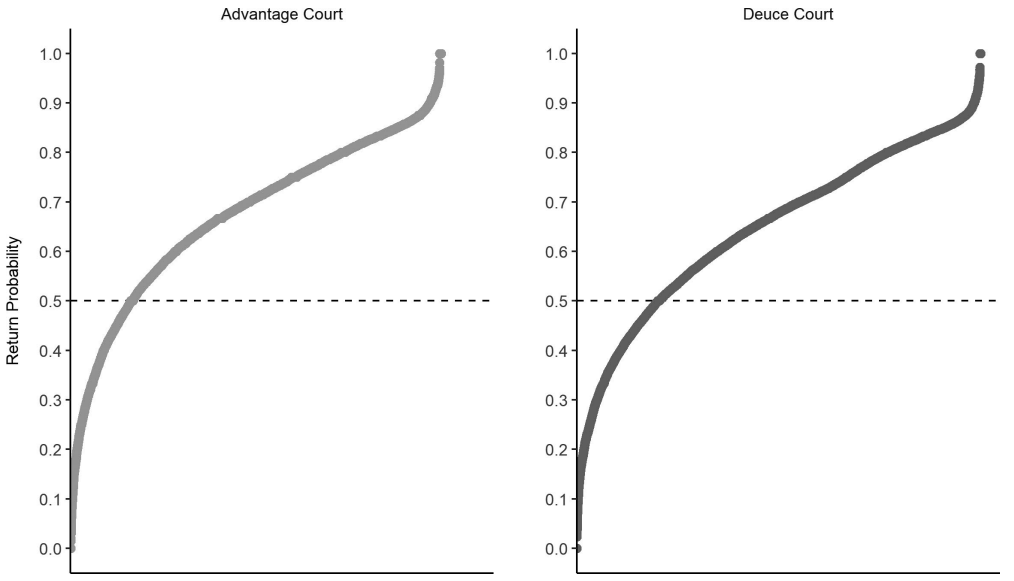

Figure 35.6 The graph demonstrates that the use of nearest neighbour methods retains the granularity of the 39,846-serve dataset, as opposed to reducing it to 80 serves representing cluster centroids. This approach offers a more nuanced perspective on serve return probabilities and serves as the basis for future modelling. Authors' own image.

Challenges

Although access to large datasets of shot-level data offers the potential to advance our knowledge of tennis by using sophisticated analysis techniques, it is important to acknowledge that big data does not automatically solve all our problems. Indeed, as outlined below, it also creates new challenges for the researcher to deal with. As such, there will always be a role for the high levels of control and precision afforded by laboratory experiments.

Existing challenges – Sample size and diversity

A widely accepted limitation with traditional notational analysis is small sample sizes that lack diversity due to the time-expensive process of manual coding. On the surface, Hawk-Eye data remedies such limitations by providing hundreds and thousands of shots for a researcher to analyse.

However, the reality is more complex. Due to the multiple promoters of tennis events – such as the International Tennis Federation, the Grand Slam Board, the ATP Tour, and WTA – each owning their own data, Hawk-Eye data used in studies often comes from one tournament. Accordingly, the generalisability of results is limited to single court surfaces/ tournaments, with the caveat that such findings are likely to be representative of the entire professional game. As elegantly summarised by Kovalchik [62]: "A statistician who wanted to work with the entirety of the data collected at top professional events would have the difficult task of convincing 108 separate tournaments that this research investment was in their interest". A centrally coordinated data-sharing initiative may go some way to remedy this problem, but this may have to be mandated by a governing body, given that ownership lies with individual tournaments.

New challenges – How to process the dataset?

When working with large datasets, researchers must make decisions as to how such data is integrated, transformed, and analysed, and how the results are interpreted and visualised – these challenges are known as *process challenges* [63]. For example, in this case study we have already discussed how a user must decide how to discretise shot trajectories and how various methods can be used to successfully answer seemingly similar questions. Overcoming *process challenges* requires: (1) domain expertise, to understand the problem, the objective of the research, and the fundamental characteristics of the sport; and (2) expertise in sophisticated statistical analysis – to prepare, model, evaluate, and deploy any solution. Therefore, a collaborative approach to conducting research that involves both practitioner and researcher seems best positioned to maximise the utility of such datasets.

Summary

There is little doubt that access to large amounts of data (thanks to developments in technology) has revolutionised practice in team and individual sport. The activities of coaches, scouts, and sports researchers have all changed due to new techniques and access to 'objective' measures of performance. This 'data-driven' approach has reduced the guesswork involved in talent identification and training. It reveals hidden facets of performance which may otherwise have been overlooked and even promises to identify those at risk from injury through overtraining. However, this bonanza of potential upsides must be measured against the risks it brings. Sport has traditionally been (and arguably should always be) a celebration of the unpredictable. The intangible nature of sporting performance creates excitement that brings in an audience. To reduce sport entirely to a set of controllable parameters may extinguish its essential essence. We must also consider that attempting to predict the unpredictable is a futile exercise and that sport will always have room for the underdog and the upset. Sports practitioners must remember that the learned experience of the coach and talent scout still has value. The context and understanding they can provide must not be superseded by the 'objective' decisions that we may be tempted to make using data alone.

Academic researchers have dived into the large datasets coming from professional sport. Just because a researcher has a large dataset at their disposal does not automatically address the challenges of sport research. Indeed, these datasets also pose new challenges for the researcher. It is essential that the outcome of the research is meaningful. As reported by Abt et al. [64], research manuscripts on sports performance routinely conclude that their findings have 'implications for training', without providing details on the 'how', 'why', and the 'likelihood of adoption'. For this reason, researchers and practitioners must work together to ensure the utility of any inferences drawn from big data.

Note

1 This dataset – provided by the International Tennis Federation – comprised of Hawk-Eye data collected from 7 years of Davis Cup Tennis (2012–2018). The dataset included 79 different Davis Cup ties, 284 different matches, 175 different players, and 566 different server-receiver combinations.

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